

# The Weil Conjecture. II

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English translation

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## Introduction

In [1], cited below as I, we proved the Weil conjecture giving the complex absolute value of the eigenvalues of Frobenius acting on the cohomology of a projective smooth variety defined over a finite field. Here we study cohomology with values in a sheaf: the point is to pass from pointwise properties of the sheaf to corresponding properties of its cohomology.

Let therefore  $X_0$  be a scheme of finite type over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a  $\mathbb{Q}_\ell$ -sheaf on  $X_0$ . We choose an algebraic closure  $\overline{\mathbb{F}}_q$  of  $\mathbb{F}_q$ , and we indicate by suppressing the subscript 0 the extension of the base field from  $\mathbb{F}_q$  to  $\overline{\mathbb{F}}_q$ ; see (0.7). For  $x_0 \in |X_0|$  a closed point of  $X_0$ , and  $x \in X(\overline{\mathbb{F}}_q)$  above it, one has the Frobenius automorphism  $F_{x_0}$  of the fibre  $\mathcal{F}_x$  of  $\mathcal{F}$  at  $x$  (I, (1.11)–(1.13), or (1.18)). We shall call its eigenvalues the eigenvalues of  $F_{x_0}$  on  $\mathcal{F}_0$ . We say that  $\mathcal{F}_0$  is pointwise pure of weight  $n$  if, for every  $x_0 \in |X_0|$ , the eigenvalues of  $F_{x_0}$  on  $\mathcal{F}_0$  are algebraic numbers all of whose complex conjugates have absolute value  $N(x_0)^{n/2}$ . We say that  $\mathcal{F}_0$  is mixed if it is an iterated extension of pointwise pure sheaves; the weights of these pure constituents are the weights of  $\mathcal{F}_0$ . Our essential result is the following.

**Theorem 1** (Theorem 1, (3.3.1)). *Let  $f : X_0 \rightarrow S_0$  be a morphism of schemes of finite type over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a mixed sheaf of weights  $\leq n$  on  $X_0$ . Then, for every  $i$ , the sheaf*

$$R^i f_! \mathcal{F}_0$$

*on  $S_0$  is mixed of weights  $\leq n + i$ .*

For  $S_0 = \text{Spec}(\mathbb{F}_q)$ , this theorem says that, for every eigenvalue  $\alpha$  of Frobenius on  $H_c^i(X, \mathcal{F})$ , there is an integer  $m \leq n + i$  (the weight of  $\alpha$ ) such that all complex conjugates of  $\alpha$  have absolute value  $q^{m/2}$ .

Poincaré duality sometimes makes it possible to complement these upper bounds by lower bounds (3.3.5). For example, if  $X_0$  is proper and smooth, and if  $\mathcal{F}_0$  is lisse (“twisted constant”, in another terminology) and pointwise pure of weight  $n$ , then the eigenvalues of Frobenius on  $H^i(X, \mathcal{F})$  are all of weight  $n + i$ ; we shall also express this by saying that  $H^i(X, \mathcal{F})$  is pure of weight  $n + i$ .

For  $\mathcal{F}_0 = \mathbb{Q}_\ell$  (of weight 0), one recovers the main result of I, with the hypothesis “projective and smooth” replaced by “proper and smooth”.

A simple reduction, parallel to the proof of the finiteness theorem for the  $R^i f_!$  (cf. SGA 4, XIV), reduces Theorem 1 to the following theorem, together with a local study at infinity of lisse pointwise pure sheaves on a curve.

**Theorem 2** (Theorem 2, cf. (3.2.3)). *Let  $X_0$  be a proper smooth curve over  $\mathbb{F}_q$ , let  $j : U_0 \hookrightarrow X_0$  be the inclusion of a dense open subset, and let  $\mathcal{F}_0$  be a lisse pointwise pure sheaf of weight  $n$  on  $U_0$ . Then  $H^i(X, j_* \mathcal{F})$  is pure of weight  $n + i$ .*

Here are the main lines of the proof.

### A. Cleanups.

- (i) If  $u : X'_0 \rightarrow X_0$  is a finite surjective morphism whose source is a proper smooth curve, and if primes denote base change by  $u$ , then the groups

$$H^i(X, j_* \mathcal{F})$$

are direct factors of

$$H^i(X', j'_* \mathcal{F}').$$

This argument allows one to reduce to the case where the local monodromy of  $\mathcal{F}$  at the points of  $X - U$  is unipotent.

- (ii) A duality theorem asserts that  $H^i(X, j_* \mathcal{F})$  and

$$H^{2-i}(X, j_*(\mathcal{F}^\vee))$$

are in perfect duality, with values in  $\mathbb{Q}_\ell(-1)$ . This reduces the problem to verifying that the complex conjugates  $\alpha'$  of the eigenvalues  $\alpha$  of Frobenius on  $H^i(X, j_* \mathcal{F})$  have absolute value bounded in the required way. The difficult case is  $H^1$ .

**B. Complex embeddings.** Let  $\alpha \in \overline{\mathbb{Q}_\ell}$  be an eigenvalue of Frobenius on  $H^i(X, j_* \mathcal{F})$ . It is convenient to reformulate the estimates to be checked as follows: for every isomorphism

$$\iota : \overline{\mathbb{Q}_\ell} \xrightarrow{\sim} \mathbb{C},$$

one must prove an inequality of the form

$$|\iota(\alpha)| \leq q^{(i+n)/2}.$$

In the proof, each isomorphism  $\iota$  will be treated separately. This leads one to introduce the notions of pointwise  $\iota$ -pure and  $\iota$ -mixed sheaves: one looks not at all complex conjugates of the eigenvalues, but only at their images under  $\iota$ . It is also convenient to allow the weights to be arbitrary real numbers. See (1.2.6). We shall prove Theorem 2 with “pure” replaced by “ $\iota$ -pure”. I refer to (1.2.11) the reader who, like me, dislikes the axiom of choice implicit in the use of isomorphisms between  $\overline{\mathbb{Q}}_\ell$  and  $\mathbb{C}$ .

**C. Local monodromy of  $\iota$ -pure sheaves.** Put  $S_0 = X_0 - U_0$ . One begins by proving that, if  $\mathcal{F}_0$  is lisse and pointwise  $\iota$ -pure of weight  $\beta \in \mathbb{R}$ , the  $\iota$ -weight

$$w_q(\alpha) = \frac{2 \log |\iota(\alpha)|}{\log q}$$

of an eigenvalue  $\alpha$  of  $F_{x_0}$  on  $j_*\mathcal{F}_0$ , for  $x_0 \in S_0$ , is of the form  $\beta - m$ , with  $m$  an integer  $\geq 0$ , and one determines  $m$  in terms of the local monodromy of  $\mathcal{F}_0$  at  $x_0$  (1.8.4). More generally, one determines the corresponding weights of the eigenvalues of  $F_{x_0}$  on  $\mathcal{F}_0$  in the sense of (1.10.2).

The first step is to prove the necessary upper bound. This is done by using Grothendieck’s formula for the function  $Z(U_0, \mathcal{F}_0, t)$ : after applying  $\iota$ , one finds on the left an infinite product converging in the appropriate half-plane, and on the right a rational function whose numerator is essentially

$$\det(1 - Ft, H_c^1(U, \mathcal{F}));$$

one uses the fact that the fibres  $(j_*\mathcal{F})_x$  for  $x \in S$  contribute to  $H_c^2(U, \mathcal{F})$ . The second step is to apply this result to tensor powers of  $\mathcal{F}_0$  and of its dual, while taking account of the known structure of local monodromy.

Once this is known, it is permissible, and convenient, to study  $j_!\mathcal{F}_0$  rather than  $j_*\mathcal{F}_0$ : if  $\mathcal{F}_0$  is lisse and pointwise  $\iota$ -pure of weight  $\beta \in \mathbb{R}$ , the point is to show that the eigenvalues  $\alpha$  of Frobenius on

$$H_c^1(U, \mathcal{F}) = H^1(X, j_!\mathcal{F})$$

have weight  $w_q(\alpha) \leq \beta + 1$ . To simplify the notation, we assume  $\beta = 0$ ; one reduces to this case by twisting (1.2.7).

**D. Passage to  $X_0 \times X_0$ .** The main geometric idea is to pass from  $(U_0, \mathcal{F}_0)$  to

$$(U_0 \times U_0, \mathcal{F}_0 \boxtimes \mathcal{F}_0)$$

and to analyze

$$H^2(U \times U, \mathcal{F} \boxtimes \mathcal{F}) = H^2(X \times X, j_!\mathcal{F} \boxtimes j_!\mathcal{F})$$

by means of a pencil of hyperplane sections of  $X_0 \times X_0$ , assumed to be in general position, for a suitable projective compactification of  $X_0 \times X_0$ . We assume that the cleanup A(i) has been carried out.

The aim is to prove that the weight of an eigenvalue  $\alpha$  of Frobenius on

$$H^2(X \times X, j_!\mathcal{F} \boxtimes j_!\mathcal{F})$$

satisfies  $w_q(\alpha) \leq 2 + \varepsilon$  in the inductive form of the argument. If  $\alpha$  is now an eigenvalue of Frobenius on  $H^1(X, j_!\mathcal{F})$ , the Kunneth formula asserts that  $\alpha^2$  is an eigenvalue of Frobenius on the preceding  $H^2$ , and one obtains an improved bound for  $H^1(X, j_!\mathcal{F})$ . Iterating this procedure replaces the use of large Cartesian powers in I.

Take a sufficiently general pencil of hyperplane sections of  $X_0 \times X_0$ . The hyperplane sections of the pencil are the fibres of a morphism

$$f : Y_0 \longrightarrow \mathbb{P}_0^1,$$

where  $Y_0$  is obtained from  $X_0 \times X_0$  by blowing up finitely many points. Let  $\mathcal{G}_0$  be the inverse image of  $j_! \mathcal{F}_0 \boxtimes j_! \mathcal{F}_0$  on  $Y_0$ . The Leray spectral sequence for  $f$  reduces, in essence, the study of

$$H^2(X \times X, j_! \mathcal{F} \boxtimes j_! \mathcal{F})$$

to that of

$$H^1(\mathbb{P}^1, R^1 f_* \mathcal{G}).$$

The generalization (1.5.1) of the fundamental estimate I (1.3.2) shows that, over a certain open subset  $V_0 \subset \mathbb{P}_0^1$  on which the sheaf  $R^1 f_* \mathcal{G}_0$  is lisse, it is a successive extension of lisse pointwise  $\iota$ -pure sheaves. The hypotheses left from I (1.3.2) are replaced by the requirement that  $R^1 f_* \mathcal{G}_0$  be, on  $V_0$ , a direct factor of a lisse  $\iota$ -real sheaf, that is, a sheaf  $\mathcal{H}_0$  such that the polynomials

$$\det(1 - F_x t, \mathcal{H}_0)$$

have real coefficients. The results of Section 2 (see E below) let one deduce this from the fact that  $\mathcal{F}_0$  itself is a direct factor of an  $\iota$ -real sheaf, namely  $\mathcal{F}_0 \oplus \mathcal{F}_0^\vee$ : because  $\mathcal{F}_0$  has weight 0, its dual plays the role of a complex conjugate.

How are the weights of  $R^1 f_* \mathcal{G}_0|_{V_0}$  computed? In I (1.3.2), one hypothesis ensured that the sheaf considered had no nontrivial lisse subsheaf, hence was pointwise  $\iota$ -pure by (1.5.1), and another hypothesis made it possible to compute its weight. In the application I (1.6.3), those hypotheses came from the theory of Lefschetz pencils, more precisely from the conjugacy theorem for vanishing cycles. Here  $R^1 f_* \mathcal{G}_0$  has ramification points of three geometrically distinct types, so another argument is needed.

The theory of vanishing cycles makes it possible to control the jump of  $R^1 f_* \mathcal{G}_0$  at a point of  $\mathbb{P}_0^1 - V_0$  from local information on  $X_0 \times X_0$ , namely the groups of vanishing cycles. A local calculation, together with the results in C applied to  $\mathcal{F}_0$ , determines the weights of the Frobenius eigenvalues on those groups of vanishing cycles. These weights turn out to be integers. Applying the results of C to the successive quotients of  $R^1 f_* \mathcal{G}_0|_{V_0}$  for a suitable filtration, one shows that  $R^1 f_* \mathcal{G}_0$  is a successive extension of lisse sheaves on  $V_0$  which either

- (a) are lisse on all of  $\mathbb{P}_0^1$  (with weights not detected by vanishing cycles), or
- (b) have restriction to  $V_0$  pointwise  $\iota$ -pure of integral weight.

Fortunately, the sheaves in case (a) do not contribute to the group

$$H^1(\mathbb{P}^1, R^1 f_* \mathcal{G})$$

which interests us, since  $\mathbb{P}^1$  is simply connected and  $H^1(\mathbb{P}^1, \mathbb{Q}_\ell) = 0$ .

Which integral weights  $m$  appear? If one already knows the theorem with an error  $\leq \delta < 1$  and applies it to the fibres of  $f$ , it gives an estimate  $m < 1 + \delta$ ; since  $m$  is an integer, one even has  $m \leq 1$ . Applying the theorem, with the same error, to  $\mathbb{P}^1$  and to the successive quotients of  $R^1 f_* \mathcal{G}_0$  for the filtration introduced above, one obtains the desired estimate for the eigenvalues of Frobenius on  $H^1(\mathbb{P}^1, R^1 f_* \mathcal{G})$ . At the start, however, one only has the cruder bound, and allowing weight 2 ruins the argument.

**E. The weight of an  $H^1$  is  $< 2$ .** We prove directly that, for  $\mathcal{F}_0$  lisse and pointwise  $\iota$ -pure of weight 0, the eigenvalues  $\alpha$  of Frobenius on  $H^1(X, j_* \mathcal{F})$  have weight  $w_q(\alpha) < 2$ . Applied to the fibre of  $f$  at a point of  $V_0$ , this is enough to start the proof. It also separates the Frobenius eigenvalues on  $H^1(X, j_* \mathcal{F})$ , of weight  $< 2$ , from those on  $H^2(X, j_! \mathcal{F})$ , of weight 2. Applied to the fibres of  $f$ , it is used in proving that  $R^1 f_* \mathcal{G}_0$  is, on  $V_0$ , a direct factor of a lisse  $\iota$ -real sheaf. The proof is inspired by Mertens' proof of the Hadamard–de la Vallée Poussin theorem, namely the nonvanishing of  $\zeta(s)$  on the line  $\Re(s) = 1$ .

From E one can deduce that, if  $\mathcal{F}_0$  is a lisse pointwise  $\iota$ -pure sheaf on a smooth curve  $U_0$ , the fibre  $\mathcal{F}_u$  of  $\mathcal{F}$  at  $u \in |U|$  is a completely reducible representation of the geometric monodromy

group; see (3.4.1)(iii) and (3.4.14). Via the heuristic dictionary of [2], I, this corresponds to the semisimplicity theorem [2], II (4.2.6) in the theory of variations of Hodge structure. A particular case shows that if  $X$  is a nonsingular projective variety over an algebraically closed field  $k$ , if  $(H_t)_{t \in \mathbb{P}^1}$  is a Lefschetz pencil of hyperplane sections of  $X$ , with  $H_t$  singular for  $t \in S$ , and if  $u \in \mathbb{P}^1 - S$ , then the  $H^i(H_u, \mathbb{Q}_\ell)$  are completely reducible representations of  $\pi_1(\mathbb{P}^1 - S, u)$ . A specialization argument reduces one to the case  $k = \overline{\mathbb{F}}_q$ ; then  $X$  and the pencil are defined over a finite field, and the required purity hypothesis follows from the Weil conjecture for the  $H^i$ . As is well known, the original arguments of Lefschetz allow one to deduce from this complete reducibility the hard Lefschetz theorem for iterated cup product with the cohomology class of a hyperplane section (4.1.1). We also give a variant for cohomology with values in a sheaf, even in a complex (6.2.13).

Other purely geometric statements result from the existence of a formalism of weights: the local invariant-cycle theorem (3.6.1), for instance, and the results announced in [3] concerning the existence of a grading by weight on  $\mathbb{Q}_\ell$ -homotopy types (5.3.4). Parallel results are available in Hodge theory, where one can also speak of weights: [12], [9].

Theorem 1 belongs to a formalism that is often far easier to use than I. In (3.7) we show how it simplifies the proofs of the applications given in (1.8). For instructions on its application to estimates of trigonometric sums, I refer to SGA 4 $\frac{1}{2}$  [Sommes trigonometriques], especially nos. 1, 2, 3, and 7.

The essential results of the article concern schemes of finite type over a finite field, with two “buts”. First, they have geometric consequences for a scheme of finite type over the algebraic closure of a finite field, and specialization arguments allow one to pass from there to the case of an arbitrary algebraically closed base field. Second, some results hold for schemes of finite type over  $\mathbb{Z}$ . They follow immediately from the proofs of the analogous results over a finite field, but not from their statements. We shall omit them from the section-by-section review below.

In (1.1) we recall what a constructible  $\mathbb{Q}_\ell$ -sheaf is, with a few variants. We also give a formalism for a corresponding derived category. In (1.2) we define pointwise pure,  $\iota$ -pure, mixed sheaves, and so on, and state in a more precise form a conjecture according to which, on a scheme of finite type over  $\mathbb{F}_q$ , every constructible  $\mathbb{Q}_\ell$ -sheaf is  $\iota$ -mixed. Sections (1.3) to (1.5) are devoted to the announced generalization of I (1.3.2). In (1.3), Grothendieck’s theorem I (1.3.8) is used to define the determinantal weights of a lisse sheaf  $\mathcal{F}_0$  on a smooth curve  $U_0$ . They depend only on the highest exterior powers of the constituents of  $\mathcal{F}_0$ , and control the  $H^1$  (and hence the  $H^2$ ) of every sheaf obtained from  $\mathcal{F}_0$  by passing to a tensor space. Proposition (1.3.13), innocent in appearance, will play the role played in I (1.3.2) by the classical invariant theory of the symplectic group.

Sections (1.6) to (1.8) are devoted to point C above in the proof. Section (1.6) contains algebraic preliminaries: the theory of a nilpotent endomorphism of a vector space. Section (1.7) contains geometric preliminaries: how local monodromy supplies nilpotent endomorphisms. The multidimensional end of (1.7) will be used only in (1.9), which itself is unnecessary for the rest of the article.

Section (1.9) is an application of (1.8) to the local monodromy of a mixed lisse sheaf on the complement of a normal-crossing divisor along which it is tamely ramified. An analogous result for variations of Hodge structure has just been obtained by Cattani and Kaplan.

In (1.10), the methods of (1.6)–(1.8) are applied to non-archimedean absolute values. The semicontinuity result (1.10.7) has made it possible to prove that, for a general complete intersection in characteristic  $p$ , the Newton polygon (defined by the slopes of crystalline cohomology) coincides with the Hodge polygon. That this problem, of  $p$ -adic nature, could be solved only by  $\ell$ -adic methods is due to the present absence of a good theory of vanishing cycles in  $p$ -adic cohomology.

Section (1.11), on specialization of monodromy, is technical and contains no surprises.

Section 2 contains part E above of the proof. In (3.5), by known arguments, one deduces

from it an equidistribution theorem, of which one special case is the analogue, for a function field, of the Sato–Tate conjecture concerning the distribution of Frobenius angles for an elliptic curve over  $K$ .

Section (3.3) contains the proof of Theorem 1 and some corollaries. Section (3.2) gives the proof of Theorem 2, and (3.1) gives the calculation of vanishing cycles used above. In (3.4), Theorem 1 is applied to the study of the successive extensions that define a mixed sheaf. In particular, one proves that a mixed lisse sheaf admits a “weight filtration” by lisse subsheaves, and that the geometric monodromy of a pointwise pure lisse sheaf on a normal scheme is semisimple. Section (3.6) gives the local invariant-cycle theorem; (3.7) gives simplifications in the proofs of (1.8).

Section (4.1) contains the proof of the hard Lefschetz theorem. Sections (4.2) and (4.3) provide complements to SGA 7. In (4.3), the use of an ad hoc topos allows one to transpose the Morse-theoretic-looking arguments used by Lefschetz. In (4.4), one proves that the geometric monodromy group of the vanishing part of the cohomology of the hyperplane sections of a Lefschetz pencil (assumed sufficiently general in the wild characteristic-two case) is Zariski dense in an orthogonal or symplectic group, or is a Weyl group. The last case is studied in detail. In (4.5), one deduces the “gcd theorem” used in [7] to compare the  $\ell$ -adic and crystalline cohomologies of projective smooth varieties.

In Section 5, the yoga of weights is applied to the  $\mathbb{Q}_\ell$ -homotopy type. The definition of the latter makes essential use of the construction, by Miller and Grothendieck, of a graded differential (anti-)commutative algebra attached to a simplicial set and allowing one to compute its integral cohomology.

Finally, Section 6 develops the formalism of mixed sheaves. With proofs inspired by SGA 4 $\frac{1}{2}$  [Th. finitude], it is shown that their category is remarkably stable. One also defines a notion of pure complex, generalizing that of a lisse pointwise pure sheaf on a smooth scheme (6.2.4), (6.2.5). The direct image  $Rf_*$  by a proper morphism carries pure complexes to pure complexes. For them one generalizes the local invariant-cycle theorem and, under dimension hypotheses, the global invariant-cycle theorem and the hard Lefschetz theorem.

## Notations and conventions

**(0.1).** Throughout this article, unless the contrary is explicitly stated, we fix a prime number  $\ell$ , and we consider only noetherian separated algebraic spaces on which  $\ell$  is invertible. We shall simply call them schemes.

The reader who dislikes algebraic spaces may replace, in (0.1), the words “algebraic space” by “scheme”.

**(0.2).** We fix an algebraic closure  $\overline{\mathbb{Q}_\ell}$  of the field  $\mathbb{Q}_\ell$  of  $\ell$ -adic numbers. We denote by  $\iota$  an isomorphism of fields from  $\overline{\mathbb{Q}_\ell}$  to  $\mathbb{C}$ ; see (1.2.11). Exception: in (1.10), the same letter  $\iota$  will denote an isomorphism of fields from  $\overline{\mathbb{Q}_\ell}$  to an algebraic closure  $\overline{\mathbb{Q}_{\ell'}}$  of the field of  $\ell'$ -adic numbers. The word sheaf will mean, according to the section, “sheaf for the étale topology”, “constructible  $\overline{\mathbb{Q}_\ell}$ -sheaf”, or “Weil sheaf”. See (1.1.5), (1.3.2).

If  $f : X \rightarrow Y$  is a morphism, and  $\mathcal{F}$  is a sheaf on  $Y$ , we shall often write, when no ambiguity can arise,

$$H^*(X, \mathcal{F})$$

for  $H^*(X, f^*\mathcal{F})$ . The same convention applies to hypercohomology, with or without supports.

**(0.3).** A geometric point  $\bar{x}$  of  $X$  is a morphism from the spectrum of an algebraically closed field - denoted  $k(\bar{x})$  - to  $X$ . It is said to be localized at  $x \in X$  if  $x$  is its image in  $X$ . It is called algebraic if  $k(\bar{x})$  is an algebraic closure of  $k(x)$ .

Every geometric point defines an algebraic geometric point: replace  $k(\bar{x})$  by the algebraic closure of  $k(x)$  in  $k(\bar{x})$ . We shall sometimes use this construction to extend tacitly to geometric points terminology introduced for algebraic geometric points.

Every point  $x$  of  $X$  whose residue field is separably closed defines a geometric point  $\bar{x}$ : take the perfect closure of  $k(x)$ . We shall often identify  $x$  and  $\bar{x}$ .

If  $f : X \rightarrow Y$  is a morphism, every geometric point  $\bar{x}$  of  $X$  defines by composition a geometric point  $f(\bar{x})$  of  $Y$ , sometimes denoted simply by  $f\bar{x}$ .

**(0.4).** Unless explicitly stated otherwise, locally means locally for the étale topology.

If  $x$  is a point of a scheme  $X$ , we write  $X_{(x)}$  for the henselization of  $X$  at  $x$ , i.e. the spectrum of the henselization of the local ring of  $X$  at  $x$ . If  $\bar{x}$  is a geometric point of  $X$ , we write  $X_{(\bar{x})}$  for the strict localization, or strict henselization, of  $X$  at  $x$ . Its residue field is the separable closure of  $k(x)$  in  $k(\bar{x})$ .

**(0.5).** A smooth curve over a field  $k$  is a smooth scheme purely of dimension one over  $k$ .

Let  $X$  be a scheme of finite type over a field  $k$ . If  $X$  is connected, its structural morphism  $X \rightarrow \operatorname{Spec}(k)$  admits a unique factorization

$$X \longrightarrow \operatorname{Spec}(k') \longrightarrow \operatorname{Spec}(k)$$

with  $k'$  a finite separable extension of  $k$  and  $X/k'$  geometrically connected. In general, each connected component of  $X$  admits such a factorization.

In particular, this gives a mechanical procedure for reducing properties of smooth curves over finite fields to those of absolutely irreducible smooth curves over finite fields. We shall often use it tacitly, passing from one formulation to the other.

**(0.6).** A trait is the spectrum of a discrete valuation ring  $V$ . It is henselian if  $V$  is henselian, and strictly local if it is henselian and its residue field is separably closed. The expression “a trait  $(S, \eta, s)$ ” means a trait  $S$  with closed point  $s$  and generic point  $\eta$ . The expression “a trait  $(S, \bar{\eta}, s, \bar{s})$ ” means a trait  $(S, \eta, s)$ , together with a geometric point  $\bar{s}$  localized at  $s$ , and with a generic geometric point  $\bar{\eta}$  of the strict localization  $S_{(\bar{s})}$ .

**(0.7).** We shall always denote by  $p$  a prime number distinct from  $\ell$ , by  $q$  a power of  $p$ , by  $\mathbb{F}_q$  a finite field with  $q$  elements, and by  $F$  an algebraic closure of  $\mathbb{F}_q$  (or of  $\mathbb{F}_p$ , if  $\mathbb{F}_q$  has not been introduced).

The following conventions will often be in force. The fields  $\mathbb{F}_q$  and  $F$  are fixed. Objects over  $\mathbb{F}_q$ -schemes, or sheaves on  $\mathbb{F}_q$ -schemes - are denoted with a subscript 0. Suppressing this subscript indicates extension of scalars from  $\mathbb{F}_q$  to  $F$ . For instance, if  $\mathcal{F}_0$  is a sheaf on the  $\mathbb{F}_q$ -scheme  $X_0$ , then  $\mathcal{F}$  denotes its inverse image on  $X = X_0 \otimes_{\mathbb{F}_q} F$ .

Geometric points  $\operatorname{Spec}(K) \rightarrow X_0$  of  $X_0$  will always be assumed to be defined by a geometric point of  $X$ ; that is,  $K$  is assumed to be endowed with an  $\mathbb{F}_q$ -morphism  $F \rightarrow K$ .

**(0.8).** If  $X$  is a scheme of finite type over a field or over  $\mathbb{Z}$ , we denote by  $|X|$  the set of its closed points. With the notation of (0.7), if  $X_0$  is of finite type over  $\mathbb{F}_q$ , each point  $x \in |X|$  defines a geometric point, again denoted  $x$ , of  $X_0$ . We shall simply speak of the geometric point  $x \in |X|$ .

**(0.9).** In this article the point of view will be more Galoisian than in I; this will often lead us to write  $F$  where  $F^*$  appeared in I.

**(0.10).** A filtration  $F$  of an object  $V$  of an abelian category is called finite if there exist  $n$  and  $m$  such that  $F^n V = V$  and  $F^m V = 0$ .



(0.11). We write  $\varinjlim$  and  $\varprojlim$  for inductive and projective limits. Quotation marks indicate a limit taken in a category of ind-objects or pro-objects; thus “ $\varinjlim X_i$ ” denotes the ind-object defined by the inductive system of the  $X_i$  (assumed filtered).

We write  $:=$  for an equality whose right-hand side is the definition of the left-hand side.

## 1. Purity

### (1.1) $\ell$ -adic sheaves

(1.1.1). We recall the definition of a constructible  $\overline{\mathbb{Q}}_\ell$ -sheaf on a scheme  $X$  as in (0.1), following SGA 5, VI (1.4.2).

- a) Let  $A$  be a noetherian torsion ring. A sheaf of  $A$ -modules  $\mathcal{F}$  is called constructible if there exists a finite partition of  $X$  into locally closed subsets  $X_i$  such that the restrictions  $\mathcal{F}|_{X_i}$  are locally constant, with fibres finitely generated  $A$ -modules.
- b) Let  $R$  be a noetherian local ring of residual characteristic  $\ell$ , and let  $\mathfrak{m}$  be its maximal ideal. We assume that  $R$  is complete for the  $\mathfrak{m}$ -adic topology. The category of constructible  $R$ -sheaves is the 2-projective limit of the categories of sheaves of  $R/I$ -modules, for  $I$  an open ideal of  $R$ .

More explicitly, a constructible  $R$ -sheaf  $\mathcal{F}$  on  $X$  is a projective system of sheaves of  $R$ -modules, indexed by open ideals  $I$  of  $R$ , such that:

- (i)  $\mathcal{F}_I$  is annihilated by  $I$ , and is constructible as a sheaf of  $R/I$ -modules;
- (ii) for  $I \supset J$ , the transition morphism from  $\mathcal{F}_J$  to  $\mathcal{F}_I$  induces an isomorphism

$$\mathcal{F}_J \otimes_{R/J} R/I \xrightarrow{\sim} \mathcal{F}_I.$$

The map  $n \mapsto \mathfrak{m}^n$  from the positive integers to the set of open ideals of  $R$  is cofinal. Thus one may replace the projective limit over  $I$  by a projective limit over  $n$ , with  $R/I$  replaced by  $R/\mathfrak{m}^n$ .

The functor

$$\mathcal{F} \mapsto “\varprojlim \mathcal{F}_I”$$

from constructible  $R$ -sheaves to pro-sheaves of  $R$ -modules is fully faithful. We shall again call any pro-sheaf in its essential image a constructible  $R$ -sheaf. This leads one to abandon the notation  $\mathcal{F}_I$  and to write instead

$$\mathcal{F} \otimes_R R/I.$$

Justification: for  $\mathcal{F}$  as above, the projective system in  $J$  of the  $\mathcal{F}_J \otimes_R R/I$  is essentially constant, with value  $\mathcal{F}_I$ .

We say that  $\mathcal{F}$  is lisse (in another terminology, twisted constant) if the sheaves  $\mathcal{F} \otimes_R R/I$  are locally constant.

- c) Let  $E$  be a finite extension of  $\mathbb{Q}_\ell$ , and let  $R$  be the integral closure of  $\mathbb{Z}_\ell$  in  $E$ . The category of constructible  $E$ -sheaves is the quotient of the category of constructible  $R$ -sheaves by the thick subcategory of torsion sheaves. Equivalently:
  - (i) There is an essentially surjective functor

$$\mathcal{F} \mapsto \mathcal{F} \otimes_R E$$

from constructible  $R$ -sheaves to constructible  $E$ -sheaves.

(ii) One has

$$\mathrm{Hom}(\mathcal{F} \otimes_R E, \mathcal{G} \otimes_R E) = \mathrm{Hom}(\mathcal{F}, \mathcal{G}) \otimes_R E.$$

A constructible  $E$ -sheaf is lisse if it is locally of the form  $\mathcal{F} \otimes_R E$ , with  $\mathcal{F}$  lisse.

d) For  $F$  a finite extension of  $E$ , there is a functor

$$\mathcal{F} \longmapsto \mathcal{F} \otimes_E F$$

from constructible  $E$ -sheaves to constructible  $F$ -sheaves, and

$$\mathrm{Hom}(\mathcal{F} \otimes_E F, \mathcal{G} \otimes_E F) = \mathrm{Hom}(\mathcal{F}, \mathcal{G}) \otimes_E F.$$

For an iterated extension, one has the canonical isomorphism

$$(\mathcal{F} \otimes_E F) \otimes_F K \simeq \mathcal{F} \otimes_E K. \quad (1.1.1.1)$$

The category of constructible  $\overline{\mathbb{Q}}_\ell$ -sheaves is the  $\overline{\mathbb{Q}}_\ell$ -linear 2-inductive limit of the categories of constructible  $E$ -sheaves, for finite extensions  $E \subset \overline{\mathbb{Q}}_\ell$  of  $\mathbb{Q}_\ell$  becoming larger and larger. In other words:

(i) for  $E \subset \overline{\mathbb{Q}}_\ell$  finite over  $\mathbb{Q}_\ell$ , there is an  $E$ -linear functor

$$\mathcal{F} \longmapsto \mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$$

from constructible  $E$ -sheaves to constructible  $\overline{\mathbb{Q}}_\ell$ -sheaves, inducing isomorphisms on  $\mathrm{Hom}$  after tensoring with  $\overline{\mathbb{Q}}_\ell$ ;

(ii) there are canonical isomorphisms compatible with (1.1.1.1);

(iii) every constructible  $\overline{\mathbb{Q}}_\ell$ -sheaf is of the form  $\mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$  for suitable  $E$  and  $\mathcal{F}$ .

A constructible  $\overline{\mathbb{Q}}_\ell$ -sheaf is lisse if it is locally of the form  $\mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$  with  $\mathcal{F}$  lisse.

For these definitions to be usable, one needs the usual formalism for such sheaves. Such a formalism is partially constructed in SGA 5, V and VI. In those expositions, Jouanolou proves that constructible  $\mathbb{Z}_\ell$ -sheaves on  $X$  form a noetherian abelian category, gives the relation between lisse constructible  $\mathbb{Z}_\ell$ -sheaves and representations of the fundamental group, proves that constructible  $\mathbb{Z}_\ell$ -sheaves are lisse on the strata of a stratification of  $X$ , defines tensor products and inverse images, and defines and studies the direct image functors with proper support  $R^i f_!$  for  $f : X \rightarrow Y$  a morphism of finite type between noetherian schemes.

Thanks to SGA 4 $\frac{1}{2}$  [finiteness], the theory also applies to the functors  $R^i f_*$  and  $R^i f_!$  when  $f$  is a morphism of finite type between schemes of finite type over an excellent regular scheme of dimension  $\leq 1$ . It likewise applies to the sheaves of vanishing cycles  $R^i \Psi$  and  $R^i \Phi$ .

**(1.1.2).** In Section 6 we shall work essentially in the derived category of a category of  $\ell$ -adic sheaves. This theory is not yet fully in place. We shall use the following framework.

Let  $R$  be the ring of integers of a finite extension  $E_\lambda$  of  $\mathbb{Q}_\ell$ , and let  $\mathfrak{m}$  be its maximal ideal. If  $X$  is a scheme, define

$$D^-(X, R) := 2\text{-}\varprojlim_n D^-(X, R/\mathfrak{m}^n).$$

An object  $K$  of  $D^-(X, R)$  is a projective system represented by complexes  $K_n \in \mathrm{Ob} D^-(X, R/\mathfrak{m}^n)$  and by isomorphisms

$$K_{n+1} \overset{L}{\otimes}_{R/\mathfrak{m}^{n+1}} R/\mathfrak{m}^n \xrightarrow{\sim} K_n$$

in  $D^-(X, R/\mathfrak{m}^n)$ . The complex  $K_n$  will rather be denoted

$$K \otimes_R^L R/\mathfrak{m}^n.$$

We shall be especially interested in the subcategory  $D_c^b(X, R)$  consisting of those  $K \in \text{Ob } D^-(X, R)$  such that  $K \otimes_R^L R/\mathfrak{m}$  is bounded with constructible cohomology. Then, for every  $n$ ,  $K \otimes_R^L R/\mathfrak{m}^n$  belongs to  $D_{ctf}^b(X, R/\mathfrak{m}^n)$ : bounded complexes of finite Tor-dimension whose cohomology sheaves are constructible.

The following remarks justify the use we shall make of  $D_c^b(X, R)$ .

- a) If  $K$  is in  $D_c^b(X, R)$ , the sheaves

$$\mathcal{H}^i K := \varprojlim_n \mathcal{H}^i(K \otimes_R^L R/\mathfrak{m}^n)$$

are constructible  $R$ -sheaves. The proof is reduced, by cutting up  $X$ , to the case where the cohomology sheaves of  $K \otimes_R^L R/\mathfrak{m}^n$  are locally constant; then it is enough to examine their fibres at points, where one applies the argument of SGA 5, XV.

- b) Conversely, if  $\mathcal{F}$  is a torsion-free constructible  $R$ -sheaf, the projective system of complexes concentrated in degree zero with term  $\mathcal{F} \otimes_R R/\mathfrak{m}^n$  belongs to  $D_c^b(X, R)$ . We shall not need to consider the case in which  $\mathcal{F}$  has torsion.
- c) Under suitable finiteness hypotheses, for instance when working with schemes of finite type over a regular scheme  $S$  of dimension  $\leq 1$ , the categories  $D_{ctf}^b(X, R/\mathfrak{m}^n)$  are stable under the four operations  $Rf_*$ ,  $f^*$ ,  $Rf_!$ ,  $Rf^!$ , and under the internal operations  $\otimes^L$  and  $R\mathcal{H}om$ . Moreover these operations commute with reduction modulo  $\mathfrak{m}^n$ . They therefore extend trivially to  $D_c^b(X, R)$ . The same applies to the functors  $R\Psi$  and  $R\Phi$  of the theory of vanishing cycles.
- d) If, for all  $K, L$  in  $D_c^b(X, R)$ , the groups

$$\text{Hom}(K \otimes_R^L R/\mathfrak{m}^n, L \otimes_R^L R/\mathfrak{m}^n)$$

are finite, then  $D_c^b(X, R)$  is a triangulated category, with distinguished triangles those whose reductions modulo  $\mathfrak{m}^n$  are distinguished for all  $n$ .

In general, it is better to replace triangles by filtered complexes. One defines  $DF_{ctf}^b(X, R/\mathfrak{m}^n)$  as the derived category of bounded filtered complexes, with finite filtration, whose associated graded object has finite Tor-dimension. Then one puts

$$DF_c^b(X, R) := 2\text{-}\varprojlim_n DF_{ctf}^b(X, R/\mathfrak{m}^n).$$

Whenever one would like to consider a triangle in  $D_c^b(X, R)$ , one can do better: construct an object of  $DF_c^b(X, R)$  whose filtration has only two steps, the two components of the associated graded object being the vertices of the desired triangle and the underlying complex being the third vertex. By abuse of language we shall sometimes call such objects triangles and write them as short exact sequences.

- e) The definition of truncation functors  $\tau_{\leq a}$  is delicate. If  $K$  is a complex of sheaves,  $\tau_{\leq a} K$  is the complex

$$\cdots \rightarrow K^{a-1} \rightarrow \text{Ker}(d^a) \rightarrow 0 \rightarrow \cdots$$

This functor carries quasi-isomorphisms to quasi-isomorphisms and hence passes to the derived category, but it does not preserve finite Tor-dimension. The difficulty is bypassed

by a modified construction using the images of cohomology modulo sufficiently high powers of  $\mathfrak{m}$ . This construction uses in an essential way the fact that  $R$  is regular of dimension one.

**Variante 1** ((1.1.3)). *Let  $D_c^b(X, E_\lambda)$  be the category obtained from  $D_c^b(X, R)$  by extension of scalars from  $R$  to  $E_\lambda$ : there is an essentially surjective functor*

$$D_c^b(X, R) \longrightarrow D_c^b(X, E_\lambda)$$

and

$$\mathrm{Hom}(K, L) \otimes_R E_\lambda \xrightarrow{\sim} \mathrm{Hom}(K \otimes_R E_\lambda, L \otimes_R E_\lambda).$$

One then puts

$$D_c^b(X, \overline{\mathbb{Q}}_\ell) := 2\text{-}\varinjlim_{E_\lambda \subset \overline{\mathbb{Q}}_\ell} D_c^b(X, E_\lambda).$$

(1.1.4). In the sequel we shall freely use, for  $\mathbb{Z}_\ell$ -,  $\mathbb{Q}_\ell$ -, or  $\overline{\mathbb{Q}}_\ell$ -sheaves, the theorems known for constructible sheaves of  $R/\mathfrak{m}^n$ -modules. The expositions SGA 5, V, VI, XV, and the preceding remarks justify our arguments; whenever a difficulty arises, it will be pointed out.

(1.1.5). In this section and the next, we shall simply say sheaf for constructible  $\overline{\mathbb{Q}}_\ell$ -sheaf. This convention will be superseded by (1.3.2).

(1.1.6). An  $\ell$ -adic representation of a profinite group  $\pi$  (or, more generally, of an extension of a finitely generated discrete group by a profinite group) on a  $\overline{\mathbb{Q}}_\ell$ -vector space  $V$  is a homomorphism

$$\rho : \pi \rightarrow \mathrm{GL}(V)$$

such that there exist a finite extension  $E$  of  $\mathbb{Q}_\ell$  in  $\overline{\mathbb{Q}}_\ell$  and an  $E$ -structure  $V_E$  on  $V$  for which  $\rho$  factors through a continuous homomorphism from  $\pi$  to  $\mathrm{GL}(V_E)$ . Representations of  $\pi$  in an algebraic group over  $\overline{\mathbb{Q}}_\ell$  are defined similarly.

For  $X$  connected and pointed by a geometric point  $\bar{x}$ , there is the usual relation between lisse sheaves and continuous representations of the fundamental group  $\pi_1(X, \bar{x})$ : the equivalence of categories sending a lisse sheaf to its fibre. We shall transport to sheaves the familiar terminology for representations. Whether or not  $X$  is connected, a lisse sheaf  $\mathcal{F}$  on  $X$  is called simple or irreducible if it is nonzero and has no nontrivial lisse subsheaf; it is semisimple if it is a sum of simple sheaves. A Jordan–Holder series of  $\mathcal{F}$  is a finite filtration of  $\mathcal{F}$  by lisse subsheaves whose graded pieces are irreducible or zero. The constituents of  $\mathcal{F}$  are the nonzero graded pieces for such a series, and the semisimplification of  $\mathcal{F}$  is the direct sum of its constituents. Constituents and semisimplification are defined up to nonunique isomorphism.

(1.1.7). Let  $k$  be a finite field with  $q$  elements, and let  $\bar{k}$  be an algebraic closure of  $k$ . The Frobenius substitution  $\varphi \in \mathrm{Gal}(\bar{k}/k)$  is the automorphism  $x \mapsto x^q$  of  $\bar{k}$ . The geometric Frobenius  $F \in \mathrm{Gal}(\bar{k}/k)$  is the inverse of  $\varphi$ . The Weil group  $W(\bar{k}/k)$  is the subgroup of  $\mathrm{Gal}(\bar{k}/k)$  formed by the integral powers of  $F$ . It is isomorphic to  $\mathbb{Z}$ , and  $\mathrm{Gal}(\bar{k}/k)$  is its profinite completion. We shall often identify  $W(\bar{k}/k)$  with  $\mathbb{Z}$ , and  $\mathrm{Gal}(\bar{k}/k)$  with  $\widehat{\mathbb{Z}}$ , by the isomorphism sending  $F$  to 1.

Specialize (1.1.6) to  $X = \mathrm{Spec}(k)$ . If  $\mathcal{F}$  is a sheaf on  $\mathrm{Spec}(k)$ , its fibre  $\mathcal{F}_{\bar{k}}$  at the geometric point  $\mathrm{Spec}(\bar{k})$  is a  $\overline{\mathbb{Q}}_\ell$ -vector space on which  $\mathrm{Gal}(\bar{k}/k)$  acts by transport of structure. This action is continuous: the map  $n \mapsto F^n$  extends by continuity to a homomorphism  $\widehat{\mathbb{Z}} \rightarrow \mathrm{GL}(\mathcal{F}_{\bar{k}})$ . Equivalently, the eigenvalues of  $F$  are  $\ell$ -adic units. The functor  $\mathcal{F} \mapsto \mathcal{F}_{\bar{k}}$  is an equivalence from the category of sheaves on  $\mathrm{Spec}(k)$  to the category of  $\overline{\mathbb{Q}}_\ell$ -vector spaces equipped with an automorphism  $F$  whose eigenvalues are  $\ell$ -adic units.

**(1.1.8).** Let  $X$  be a scheme of finite type over  $\mathbb{Z}$ ,  $\mathcal{F}$  a sheaf on  $X$ , and  $y : \text{Spec}(k) \rightarrow X$  a point of  $X$  with values in a finite field  $k$ . Let  $\bar{k}$  be an algebraic closure of  $k$ . The inverse image of  $\mathcal{F}$  on  $\text{Spec}(k)$  falls under (1.1.7), so  $\text{Gal}(\bar{k}/k)$  acts on  $\mathcal{F}_{\bar{y}}$ . By abuse of notation we write

$$\det(1 - F_y t, \mathcal{F})$$

for the corresponding determinant, and we speak similarly of the eigenvalues of  $F_y$  acting on  $\mathcal{F}$ .

If  $X$  is connected and pointed by a geometric point  $\bar{x}$ , one has a conjugacy class of maps

$$\text{Gal}(\bar{k}/k) = \pi_1(\text{Spec}(k), \text{Spec}(\bar{k})) \longrightarrow \pi_1(X, \bar{x}).$$

We write  $F_y$  for the image of geometric Frobenius by any of these maps. For  $\mathcal{F}$  lisse, defined by a representation  $V$  of  $\pi_1(X, \bar{x})$ , one has

$$\det(1 - F_y t, \mathcal{F}) = \det(1 - F_y t, V),$$

which justifies the notation. In the special case where  $y$  is a closed point of  $X$ , this is simply the morphism of inclusion  $\text{Spec}(k(y)) \rightarrow X$ .

**(1.1.9).** When studying schemes of finite type over  $\mathbb{F}_p$ , it is often convenient to replace Galois groups and sheaves by the Weil groups and Weil sheaves defined below. The reader should note that the definition of Weil sheaves is relative to the choice of an algebraic closure  $F$  of  $\mathbb{F}_p$ .

**Definition 1** ((1.1.10)). *With the notation of (0.7):*

- (i) A Weil sheaf  $\mathcal{F}_0$  on  $X_0$  consists of a sheaf  $\mathcal{F}$  on  $X$ , together with an action of  $W(F/\mathbb{F}_q)$  on  $(X, \mathcal{F})$  inducing on  $X = X_0 \otimes_{\mathbb{F}_q} F$  the action deduced from the action of  $W(F/\mathbb{F}_q)$  on  $F$ .
- (ii) If  $\bar{x}$  is a geometric point of  $X$ , the Weil group  $W(X_0, \bar{x})$  is the inverse image of  $W(F/\mathbb{F}_q)$  by the natural map

$$\pi_1(X_0, \bar{x}) \longrightarrow \pi_1(\text{Spec}(\mathbb{F}_q), \bar{x}) = \text{Gal}(F/\mathbb{F}_q).$$

Thus the topology on

$$W(X_0, \bar{x}) = \pi_1(X_0, \bar{x}) \times_{\text{Gal}(F/\mathbb{F}_q)} W(F/\mathbb{F}_q)$$

is the product topology; the subgroup

$$\pi_1(X, \bar{x}) = \text{Ker}(\pi_1(X_0, \bar{x}) \rightarrow \pi_1(\text{Spec}(\mathbb{F}_q), \bar{x}))$$

is open and closed. Replacing  $\mathbb{F}_q$  by  $\mathbb{F}_p$  does not change either the category of Weil sheaves or the Weil group.

**(1.1.11).** The inverse image on  $X$  of a sheaf on  $X_0$  is endowed, by transport of structure, with an action of  $\text{Gal}(F/\mathbb{F}_q)$ . This identifies sheaves on  $X_0$  with particular Weil sheaves: those for which the action of  $W(F/\mathbb{F}_q)$  extends to a continuous action of  $\text{Gal}(F/\mathbb{F}_q)$ .

We shall admit for Weil sheaves the analogue of Grothendieck's results for ordinary sheaves. The generalization is easy, using (1.3.14).

**(1.1.12).** If  $\tilde{X}$  is the universal covering of  $X_0$ , pointed by  $\bar{x}$ , then there is an equivariant morphism, relative to the natural morphism  $W(X_0, \bar{x}) \rightarrow W(F/\mathbb{F}_q)$ ,

$$(\tilde{X}, \text{ with action of } W(X_0, \bar{x})) \longrightarrow (X, \text{ with action of } W(F/\mathbb{F}_q)).$$

The inverse image on  $\tilde{X}$  of a Weil sheaf is therefore endowed with an action of  $W(X_0, \bar{x})$ . If the sheaf is lisse, its fibre at  $\bar{x}$  identifies with its global sections on  $\tilde{X}$ , giving a fibre functor

$$\{\text{lisse Weil sheaves}\} \longrightarrow \{\text{representations of } W(X_0, \bar{x})\}.$$

Here "representation" is defined as in (1.1.6). Standard arguments show that, if  $X_0$  is connected, this functor is an equivalence of categories. As in (1.1.6), it allows us to transfer the language of representations to lisse Weil sheaves.

(1.1.13). Suppose  $X_0$  is connected. For  $y \in |X_0|$  (or more generally for  $y$  a point of  $X_0$  with values in a finite field), the Frobenius elements  $F_y$  lie in  $W(X_0, \bar{x})$ ; the image of  $F_y$  in  $\text{Gal}(F/\mathbb{F}_q)$  is the integral power  $q^{\deg y}$  of the Frobenius substitution. The element  $F_y$  of  $W(X_0, \bar{x})$  is well-defined up to conjugacy.

We shall call degrees the arrows to the right in the standard diagram connecting  $\pi_1(X, \bar{x})$ ,  $W(X_0, \bar{x})$ , and  $W(F/\mathbb{F}_q) \simeq \mathbb{Z}$ . One has  $\deg F_y = \deg y$ . The degree maps are surjective if  $X_0$  is geometrically connected. In general, if the largest finite subfield of  $\Gamma(X_0, \mathcal{O}_{X_0})$  has degree  $n$  over  $\mathbb{F}_q$ , their images are  $n\mathbb{Z}$  and  $n\widehat{\mathbb{Z}}$ .

(1.1.14). Let  $\mathcal{F}$  be a sheaf on  $X$ . If  $F : X \rightarrow X$  is the Frobenius endomorphism deduced by scalar extension from the endomorphism  $x \mapsto x^q$  of  $X_0$ , and if  $\text{id}_{X_0} \times F : X \rightarrow X$  is the action of geometric Frobenius, then, as in SGA 5, XV, there is a canonical isomorphism

$$(\text{id}_{X_0} \times F)^* \mathcal{F} \simeq F^* \mathcal{F}.$$

For a Weil sheaf  $\mathcal{F}_0$  on  $X_0$ , one deduces a Frobenius correspondence  $F^* \mathcal{F} \rightarrow \mathcal{F}$ , and the functor

$$\mathcal{F}_0 \longmapsto (\mathcal{F}, F^*)$$

is an equivalence from the category of Weil sheaves to the category of sheaves  $\mathcal{F}$  on  $X$  endowed with an isomorphism  $F^* \mathcal{F} \xrightarrow{\sim} \mathcal{F}$ .

For constructible sheaves of  $\mathbb{Z}/\ell^n$ -modules, or for constructible  $\mathbb{Z}_\ell$ -sheaves, an analogous result holds. This is not so for  $\mathbb{Q}_\ell$ - or  $\overline{\mathbb{Q}}_\ell$ -sheaves: for example, take  $\mathcal{F} = \overline{\mathbb{Q}}_\ell$  and let  $F^*$  be multiplication by an element of  $\overline{\mathbb{Q}}_\ell^\times$  which is not an  $\ell$ -adic unit.

The induced morphism

$$H^i(X, \mathcal{F}) \xrightarrow{F^*} H^i(X, F^* \mathcal{F}) \rightarrow H^i(X, \mathcal{F})$$

coincides with the action of  $F \in W(F/\mathbb{F}_q)$  on cohomology by transport of structure. The same holds for cohomology with compact supports.

(1.1.15). We call  $\pi_1(X_0, \bar{x})$  the arithmetic fundamental group, and  $\pi_1(X, \bar{x})$  the geometric fundamental group. If  $X_0$  is normal, these are profinite groups. If  $\mathcal{F}_0$  is a lisse Weil sheaf on  $X_0$ , its geometric monodromy group is the image of  $\pi_1(X, \bar{x})$  in  $\text{GL}(\mathcal{F}_{\bar{x}})$ .

## (1.2) Weights

**Definition (1.2.1).** Let  $q$  be a power of a prime number, and let  $n \in \mathbb{Z}$ . A number  $\alpha$  is said to be *pure of weight  $n$ , relative to  $q$* , if it is algebraic and all of its complex conjugates have absolute value  $q^{n/2}$ .

In what follows, “number” will mean “element of  $\overline{\mathbb{Q}}_\ell$ ”.

**Definition (1.2.2).** Let  $X$  be a scheme of finite type over  $\mathbb{Z}$ , and let  $\mathcal{F}$  be a sheaf on  $X$ .

- (i) We say that  $\mathcal{F}$  is *pointwise pure* if there exists an integer  $n$ , the weight of  $\mathcal{F}$ , such that, for every  $x \in |X|$ , the eigenvalues of  $F_x$  are pure of weight  $n$  relative to  $N(x)$ .
- (ii) We say that  $\mathcal{F}$  is *mixed* if it admits a finite filtration whose successive quotients are pointwise pure sheaves. The weights of those quotients which are nonzero are the pointwise weights of  $\mathcal{F}$ .

According to these definitions, the zero sheaf is pointwise pure and, for every  $n$ , has weight  $n$ . It is mixed, with empty set of weights.

**Variant (1.2.3).** If  $k$  is a finite field, sheaves on  $\mathrm{Spec}(k)$  identify with  $\ell$ -adic representations of  $\mathrm{Gal}(\bar{k}/k)$ . We transport the terminology of (1.2.2) to them.

**Variant (1.2.4).** The same terminology applies to Weil sheaves (1.1.10), and also to  $\overline{\mathbb{Q}}_\ell$ -vector spaces endowed with an action of Frobenius, as far as the variant (1.2.3) is concerned.

**Stabilities (1.2.5).**

- (i) The category of pointwise pure sheaves of weight  $n$  is stable under passage to a quotient, to a subsheaf, by extension, by inverse image, and by direct image under a finite morphism.
- (ii) The tensor product of two pointwise pure sheaves of weights  $n$  and  $m$  is pointwise pure of weight  $n + m$ . The dual of a lisse pointwise pure sheaf of weight  $n$  is pointwise pure of weight  $-n$ .
- (iii) The category of mixed sheaves is stable under the operations of (i), and under tensor product.
- (iv) Finally, note that  $\overline{\mathbb{Q}}_\ell(1)$  is pointwise pure of weight  $-2$ .

**(1.2.6).** To show that  $\alpha \in \overline{\mathbb{Q}}_\ell$  is pure of weight  $n$ , it suffices to verify that for every isomorphism  $\iota$  from  $\overline{\mathbb{Q}}_\ell$  to  $\mathbb{C}$ , the complex number  $\iota\alpha$  has absolute value  $q^{n/2}$ . The number  $\alpha$  is then automatically algebraic; otherwise  $\alpha$  could be any transcendental number. Throughout what follows, our arguments will concern each isomorphism  $\iota$  separately. This is why, although the important concepts are those of (1.2.2), the following terminology will be useful to us.

Let  $\iota$  be an isomorphism of fields from  $\overline{\mathbb{Q}}_\ell$  to  $\mathbb{C}$ . If  $q$  is a power of a prime number, and  $\alpha \in \overline{\mathbb{Q}}_\ell$ , put

$$w_q(\alpha) := 2 \log_q |\iota\alpha|; \quad (1.2.6.1)$$

this is the  $\iota$ -weight of  $\alpha$ , relative to  $q$ . For every real number  $\beta$ , a sheaf  $\mathcal{F}$  is called *pointwise  $\iota$ -pure of weight  $\beta$*  if, for every  $x \in |X|$  and every eigenvalue  $\alpha$  of  $F_x$  acting on  $\mathcal{F}$ , one has  $w_{N(x)}(\alpha) = \beta$ . We say that  $\mathcal{F}$  is  *$\iota$ -mixed* if it admits a finite filtration whose successive quotients are pointwise  $\iota$ -pure. The weights of those of these quotients which are nonzero are the pointwise  $\iota$ -weights of  $\mathcal{F}$ . One similarly transposes the variants (1.2.3) and (1.2.4). The analogue of (1.2.5) remains true. Note that, for  $k$  a finite field, a representation  $V$  of  $W(\bar{k}/k)$  is automatically  $\iota$ -mixed.

**(1.2.7) Torsion.** For  $b \in \overline{\mathbb{Q}}_\ell^*$ , we shall denote by  $\overline{\mathbb{Q}}_\ell^{(b)}$  any Weil sheaf on  $\mathrm{Spec}(\mathbb{F}_p)$  of rank one for which  $F$  acts by multiplication by  $b$ . For  $b$  an  $\ell$ -adic unit, this is an ordinary sheaf. If a choice has been made of an algebraic closure  $\bar{F}$  of  $\mathbb{F}_p$ , it will be normalized by the choice of an isomorphism between its geometric fibre at  $\bar{F}$  and  $\overline{\mathbb{Q}}_\ell$ . For  $X$  a scheme of characteristic  $p$  (that is, over  $\mathbb{F}_p$ ),  $\mathcal{F}$  a sheaf on  $X$ , and  $b$  an  $\ell$ -adic unit (respectively,  $X$  of finite type over  $\mathbb{F}_p$ ,  $\mathcal{F}$  a Weil sheaf, and  $b \in \overline{\mathbb{Q}}_\ell^*$ ), we write  $\mathcal{F}^{(b)}$  for the tensor product of  $\mathcal{F}$  by the inverse image of  $\overline{\mathbb{Q}}_\ell^{(b)}$ ; we say that  $\mathcal{F}^{(b)}$  is deduced from  $\mathcal{F}$  by torsion. The sheaf  $\overline{\mathbb{Q}}_\ell^{(b)}$  is pointwise  $\iota$ -pure of weight  $w_p(b)$ ; the torsion  $\mathcal{F} \mapsto \mathcal{F}^{(b)}$  therefore increases weights by  $w_p(b)$ .

**Remark (1.2.8).** In the definitions of pointwise pure sheaves and mixed sheaves, we have required the weights to be integers. In the definitions of pointwise  $\iota$ -pure sheaves and  $\iota$ -mixed sheaves, it is convenient to allow them to be arbitrary real numbers. The deep theorem (3.4.1) shows a posteriori that, for  $X$  of finite type over  $\mathbb{F}_p$ , this generality is largely illusory: every  $\iota$ -mixed sheaf is a direct sum of sheaves deduced by torsion (1.2.7) from  $\iota$ -mixed sheaves of integral weights, and every sheaf which is  $\iota$ -mixed for every  $\iota$  is a direct sum of sheaves deduced by torsion from mixed sheaves.

Every lisse sheaf is a successive extension of irreducible lisse sheaves, and we shall see in (1.3.6) that every lisse sheaf on  $X$  normal, connected, and of finite type over  $\mathbb{F}_p$  is obtained by torsion from a sheaf whose determinant (the maximal exterior power) is defined by a finite-order character of the fundamental group of  $X$ . Therefore the following conjecture

**Conjecture (1.2.9).** On  $X$  of finite type over  $\mathbb{F}_p$ , every sheaf is  $\iota$ -mixed, is a consequence of the more precise conjecture (1.2.10) below. One of our main tools, the theorem (1.5.1), is a partial result in the direction of (1.2.9).

**Conjecture (1.2.10).** Let  $X$  be normal, connected, and of finite type over  $\mathbb{F}_p$ , and let  $\mathcal{F}$  be an irreducible lisse sheaf whose determinant is defined by a finite-order character of the fundamental group.

(i)  $\mathcal{F}$  is pure of weight 0.

(ii) There exists a number field  $E \subset \overline{\mathbb{Q}}_\ell$  such that, for  $x \in |X|$ , the polynomial

$$\det(1 - F_x t, \mathcal{F})$$

has coefficients in  $E$ .

(iii) For  $\lambda$  a non-archimedean place of  $E$  prime to  $p$ , the inverse roots in  $\bar{E}_\lambda$  of the polynomial  $\det(1 - F_x t, \mathcal{F})$ , that is, the eigenvalues of Frobenius, are  $\lambda$ -adic units.

(iv) For  $\lambda$  dividing  $p$ , the valuation of the inverse roots verifies

$$|v(\alpha)/v(Nx)| \leq \frac{1}{2} \operatorname{rg} \mathcal{F}.$$

(v) For a suitable  $E$  (perhaps larger than in (ii)), and for each non-archimedean place  $\lambda$  prime to  $p$ , there exists an  $E_\lambda$ -sheaf compatible with  $\mathcal{F}$ , that is, with the same Frobenius eigenvalues.

(vi) For  $\lambda$  dividing  $p$ , one hopes for small crystalline companions.

The parts (i), (iii), and (iv) of the conjecture follow from the special case in which  $X$  is a curve. For  $\mathcal{F}$  of rank 2 on a curve, Drinfeld's work reduces the conjecture, except for (vi) at present, to control of the constant of the functional equation of the  $L$ -function attached to  $\mathcal{F}$ ; it is a question of  $L(\mathcal{F}, \chi)$ , with  $\chi$  and its twists. In particular, if  $\mathcal{F}$  belongs to an infinite compatible system of  $\lambda$ -adic representations, then (i) through (v) are verified.

**Remark (1.2.11).** I do not claim to believe in the existence of isomorphisms between  $\overline{\mathbb{Q}}_\ell$  and  $\mathbb{C}$ ; they are only a convenience of exposition. Whenever we prove that a number is pure, an easy fragment of the proof would suffice to establish that it is algebraic. For the rest of the arguments, it would then suffice to consider the complex embeddings of the subfield of  $\overline{\mathbb{Q}}_\ell$  formed by the algebraic numbers, and this does not require the axiom of choice.

**Definition (1.2.12).** Let  $X$  be a scheme of finite type over  $\mathbb{Z}$ , and let  $\mathcal{F}$  be a lisse sheaf on  $X$ . We say that  $\mathcal{F}$  is *totally real* if, for every  $x \in |X|$ , the coefficients of the characteristic polynomial of  $F_x$  acting on  $\mathcal{F}$  are totally real algebraic numbers.

The following variant will be useful to us in equal characteristic.

**Variant (1.2.13).** We say that  $\mathcal{F}$  is  $\iota$ -real if, for every  $x \in |X|$ , the polynomial

$$\iota \det(1 - F_x t, \mathcal{F})$$

has real coefficients.



**Remark (1.2.14).** Every lisse pointwise pure sheaf  $\mathcal{F}$  is a direct factor of a totally real lisse pointwise pure sheaf, namely, if it is of weight  $n$ , the sheaf  $\mathcal{F} \otimes \mathcal{F}^\vee(-n)$ . Similarly, every lisse pointwise  $\iota$ -pure sheaf of integral weight is a direct factor of a lisse pointwise  $\iota$ -pure  $\iota$ -real sheaf.

In these variants, one may take  $\mathcal{F}$  to be a Weil sheaf. For Weil sheaves, every lisse pointwise  $\iota$ -pure sheaf of arbitrary weight is a direct factor of a lisse pointwise  $\iota$ -pure  $\iota$ -real sheaf.

### (1.3) Determinantal weights

The conventions of (0.7) are in force. Let  $X_0$  be a normal geometrically connected scheme of finite type over  $\mathbb{F}_q$ , and let  $\bar{x}$  be a geometric point of  $X_0$ . In this section we give consequences of the following theorem.

**Theorem (1.3.1).** The image of the geometric fundamental group  $\pi_1(X, \bar{x})$  in the quotient of  $W(X_0, \bar{x})$  by the closure of its derived group is the product of a finite group of order prime to  $p$  by a pro- $p$  group.

This theorem is well known, but I have not been able to find it in the literature. For our principal results, only the case in which  $X$  is a curve matters. We give below a proof in that case. The general case will be treated in (1.11.4).

Let  $\bar{X}_0$  be the smooth projective completion of  $X_0$ , and let  $S = \bar{X}_0 - X_0$ . Let  $k$  be the function field of  $X_0$ , let  $k_v$  be its completion at a place  $v$  (identified with a closed point  $v \in |\bar{X}_0|$ ), let  $r_v$  be the group of units of  $k_v$ , and let  $k_{\mathbb{A}}^*$  be the idèle group of  $k$ . Class-field theory identifies  $W(X_0, \bar{x})^{\text{ab}}$  with the idèle-class group

$$\left( \prod_{v \notin S} r_v \right) \backslash k_{\mathbb{A}}^* / k^*,$$

the projective limit of the divisor-class groups

$$C_m = \{\text{divisors on } \bar{X}_0\} / \{\text{those of the form } (\varphi), \varphi \equiv 1 \pmod{m}\}$$

for a “modulus”  $m$  which becomes larger and larger and is concentrated on  $S$ . The image of the geometric fundamental group is obtained by replacing  $k_{\mathbb{A}}^*$  by the subgroup  $k_{\mathbb{A}}^1$  of idèles of norm 1, that is, by replacing the  $C_m$  by the subgroups  $C_m^0$  of divisor classes of degree 0. The kernel of the map

$$\left( \prod_{v \notin S} r_v \right) \backslash k_{\mathbb{A}}^1 / k^* \longrightarrow \left( \prod_v r_v \right) \backslash k_{\mathbb{A}}^1 / k^*$$

is a quotient of  $\prod_{v \in S} r_v$ , hence is the product of a pro- $p$  group by a finite group; and the image group

$$\prod_v r_v \backslash k_{\mathbb{A}}^1 / k^*$$

of divisor classes of degree 0, for linear equivalence, is finite (Weil, BNT IV, Th. 7). Indeed the Picard variety  $\text{Pic}^0(X)$  is of finite type, and this divisor-class group is the group of rational points over  $\mathbb{F}_q$  of  $\text{Pic}^0(X)$ .

**(1.3.2).** In the remainder of this article, we shall simply call sheaves the Weil sheaves of (1.1.10). Constructible  $\mathbb{Q}_\ell$ -sheaves will be called étale sheaves.

This convention will sometimes lead us to apply to  $\bar{\mathbb{Q}}_\ell$ -sheaves results proved for  $\mathbb{Q}_\ell$ -sheaves. The justification will always be immediate.

**(1.3.3).** Let  $\mathcal{F}_0$  be a sheaf of rank 1 on  $X_0$ . The group  $W(X_0, \bar{x})$  acts on  $\mathcal{F}_{\bar{x}}$  by homotheties, through a character

$$\chi : W(X_0, \bar{x}) \longrightarrow \overline{\mathbb{Q}}_\ell^*.$$

Conversely, for any such character  $\chi$ , the equivalence (1.1.12) gives us a sheaf  $\mathcal{L}_0(\chi)$  endowed with a trivialization  $\mathcal{L}(\chi)_{\bar{x}} \simeq \overline{\mathbb{Q}}_\ell$ .

The character  $\chi$  factors through the largest abelian quotient of  $W(X_0, \bar{x})$ . By (1.3.1), the geometric monodromy group of  $\mathcal{F}_0$ , that is, the image under  $\chi$  of  $\pi_1(X, \bar{x})$ , is therefore the product of a finite group by a pro- $p$  group. This image is also a compact subgroup of  $E^*$ , for  $E \subset \overline{\mathbb{Q}}_\ell$  a finite extension of  $\mathbb{Q}_\ell$ . It is therefore the product of a finite group by a pro- $\ell$  group and, since  $p \neq \ell$ , it is a finite group. In particular, some power  $\chi^n$  ( $n > 0$ ) of  $\chi$  is trivial on  $\pi_1(X, \bar{x})$ , that is, is of the form

$$w \longmapsto b^{\deg(w)}.$$

If  $c$  is an  $n$ -th root of  $b$ , the character  $\chi$  is the product of the character  $w \mapsto c^{\deg(w)}$  by a character  $z$  of order  $n$ , that is, with values in the cyclic group of  $n$ -th roots of unity of  $\overline{\mathbb{Q}}_\ell$ . In particular, for  $y \in |X_0|$ , one has

$$|\iota\chi(F_y)| = |\iota c|^{\deg(y)};$$

the sheaf  $\mathcal{F}_0$  is pointwise  $\iota$ -pure, of weight the  $\iota$ -weight of  $c$  relative to  $q$  (1.2.6.1). We have proved:

**Proposition (1.3.4).** Let  $\mathcal{F}_0$  be a lisse sheaf of rank 1 on  $X_0$ , and let  $\chi : W(X_0, \bar{x}) \rightarrow \overline{\mathbb{Q}}_\ell^*$  be the corresponding character.

- (i) The geometric monodromy group of  $\mathcal{F}_0$  is finite. More precisely,  $\chi$  is the product of a character  $w \mapsto c^{\deg(w)}$  by a character of finite order.
- (ii)  $\mathcal{F}_0$  is pointwise  $\iota$ -pure, of weight the weight of  $c$  relative to  $q$ .

**Variant.** If  $X_0$  is a curve, the proof of (1.3.1) is complete - independent of (1.11) - and hence already gives (1.3.4). Here, following N. Katz, is how to deduce the general case of (1.3.4) from this particular case.

Let  $E \subset \overline{\mathbb{Q}}_\ell$  be as above, and let  $M$  be the group of elements of finite order in  $E^*$ , that is, the roots of unity. It is a finite group. First one shows that, whatever  $x, y \in |X_0|$  may be, one has

$$\chi(F_x)^{1/\deg(x)} = \chi(F_y)^{1/\deg(y)} \quad \text{in } E^* \otimes \mathbb{Q}$$

(written multiplicatively): by joining  $x$  and  $y$  by a curve, one reduces to the case in which  $X_0$  is a curve and applies (1.3.4). For every zero-cycle of degree 0,  $\sum n(x)x$ , one therefore has  $\chi(\prod F_x^{n(x)}) \in M$ . It follows from Chebotarev's theorem that these products of Frobenius elements are dense in the geometric  $\pi_1$ , which  $\chi$  therefore sends into  $M$ : the restriction of  $\chi$  to the geometric  $\pi_1$  has finite order, and one concludes as above.

**Definition (1.3.5).** Let  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$ . The determinantal weights (relative to  $\iota$ ) of  $\mathcal{F}_0$  are the numbers

$$\frac{1}{d}(\iota\text{-weight of } \wedge^d \mathcal{F}'_0)$$

where  $\mathcal{F}'_0$  runs through the constituents of  $\mathcal{F}_0$ , and  $d$  is its rank.

It is clear that a lisse sheaf pointwise  $\iota$ -pure of weight  $\beta$  is also purely of determinantal weight  $\beta$ .

(1.3.6). If  $\mathcal{F}_0$  is lisse and irreducible of rank  $n$ , and if  $b \in \overline{\mathbb{Q}}_\ell^*$  is such that, for some  $y \in |X_0|$ ,

$$\det(F_y) = b^{n \deg(y) [\mathbb{F}_q : \mathbb{F}_p]},$$

then the sheaf  $\mathcal{F}_0^{(b^{-1})}$  of (1.2.7) is of determinantal weight 0 relative to every  $\iota$ : its maximal exterior power is defined by a character of finite order.

(1.3.7). To study the behaviour of determinantal weights under tensor product and exterior power, it will be convenient to pass from the language of sheaves to that of representations of algebraic monodromy groups. The dictionary is inspired by Serre [11], II (1.3).

Let therefore  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$ . Let  $G^0$  be the algebraic subgroup of  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$  which is the Zariski closure of the geometric monodromy group (1.1.15). By push-out from the exact sequence

$$0 \rightarrow \pi_1(X, \bar{x}) \rightarrow W(X_0, \bar{x}) \rightarrow \mathbb{Z} \rightarrow 0,$$

one defines an algebraic group  $G$  (not of finite type), extension of  $\mathbb{Z}$  by  $G^0$ :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi_1(X, \bar{x}) & \longrightarrow & W(X_0, \bar{x}) & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & G^0 & \longrightarrow & G & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & & & \downarrow & & \\ & & & & \mathrm{GL}(\mathcal{F}_{\bar{x}}) & & \end{array} \quad (1.3.7.1)$$

Here is the exact definition: if  $w$  is an element of  $W(X_0, \bar{x})$  of degree 1, then  $W(X_0, \bar{x})$  is the semidirect product of the group  $w^{\mathbb{Z}}$  generated by  $w$  with  $\pi_1(X, \bar{x})$ . The image of  $w$  in  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$  normalizes that of  $\pi_1(X, \bar{x})$ , that is, the geometric monodromy group. It therefore normalizes its Zariski closure  $G^0$ , and  $G$  is the semidirect product of  $\mathbb{Z}$  by  $G^0$  relative to the action  $\mathrm{int}(w)$  of  $\mathbb{Z}$  on  $G^0$ . The group  $G$  thus obtained fits into diagram (1.3.7.1), and this characterizes it up to a unique isomorphism.

Each linear representation of the algebraic group  $G$  defines an  $\ell$ -adic representation of  $W(X_0, \bar{x})$ , hence a Weil sheaf on  $X_0$ . The original sheaf  $\mathcal{F}_0$  corresponds to the tautological representation (1.3.7.1) of  $G$  in  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ .

If  $\mathcal{F}_0^{(i)}$  is a finite family of lisse sheaves, and one applies the preceding construction to the direct sum  $\mathcal{F}_0$  of the  $\mathcal{F}_0^{(i)}$ , then in (1.3.7.1) one may replace  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$  by the subgroup  $\prod_i \mathrm{GL}(\mathcal{F}_{\bar{x}}^{(i)})$ . Each sheaf  $\mathcal{F}_0^{(i)}$  is determined by the representation of  $G$  in the corresponding factor  $\mathrm{GL}(\mathcal{F}_{\bar{x}}^{(i)})$ .

**Theorem (1.3.8) (Grothendieck).** Let  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$ , let  $G$  and  $G^0$  be as above, and let  $G^{00}$  be the neutral component of  $G^0$ . Then the radical of the algebraic group  $G^{00}$  is unipotent.

We first prove:

**Corollary (1.3.9).** If  $\mathcal{F}_0$  is semisimple, the Zariski closure  $G^0$  of the geometric monodromy group of  $\mathcal{F}_0$  is an extension of a finite group by a semisimple group.

If the representation of  $W(X_0, \bar{x})$  on  $\mathcal{F}_{0\bar{x}}$  is semisimple, the same is true of its restriction to the distinguished subgroup  $\pi_1(X, \bar{x})$ ; the sum of the  $\pi_1(X, \bar{x})$ -subspaces simple for  $\pi_1(X, \bar{x})$  is  $W$ -stable, hence is a supplement to any other. We shall prove (1.3.9) assuming only that the representation of  $\pi_1(X, \bar{x})$  is semisimple. Under this hypothesis,  $G^{00}$  is reductive, and it remains to prove that its largest central torus  $T_1$  is trivial. It is therefore isogenous to the largest torus  $T$  quotient of  $G^{00}$ .

First suppose that  $G^0 = G^{00}$  and that  $G$  is a product  $G \simeq G^0 \times \mathbb{Z}$ . The application  $W(X_0, \bar{x}) \rightarrow G$  furnishes morphisms  $W(X_0, \bar{x}) \rightarrow T$ , such that the image of  $\pi_1(X, \bar{x})$  is Zariski

dense. Since  $T$  is a product of multiplicative groups, it follows from (1.3.4) that this image is finite; hence  $T = \{1\}$ .

We reduce to this case in general by replacing  $X_0$  by a finite cover and  $\mathbb{F}_q$  by a finite extension, hence  $W(X_0, \bar{x})$  and  $G$  by subgroups of finite index. The group  $W(X_0, \bar{x})$  acts on  $T_1$  while respecting the finite set  $F \subset X(T_1)$  of characters whose kernels do not contain  $T_1$ ; this set generates  $X(T_1) \otimes \mathbb{Q}$ . Hence  $G^0$ , and therefore  $T_1$ , acts faithfully; the group  $W(X_0, \bar{x})$  therefore acts on  $T_1$  through a finite quotient, and one is reduced to assuming this action trivial. The group of exterior automorphisms of  $G^{00}$ , finite on  $T_1$ , is finite; one may suppose that  $W(X_0, \bar{x})$  acts on  $G^{00}$  by inner automorphisms. Finally one is reduced to supposing that  $G^0 = G^{00}$ , and then  $G \simeq G^{00} \times \mathbb{Z}$ .

Prove (1.3.8). Let  $G$  be a Jordan–Hölder sequence for  $\mathcal{F}_0$ , let  $P$  be the subgroup of  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$  which respects the filtration  $G$ , let  $N$  be the subgroup of  $P$  which acts trivially on the successive quotients (unipotent radical), and let  $L = P/N$ . The analogue  $G_1^0$  of  $G^0$  for  $\mathrm{Gr}_G(\mathcal{F}_0)$  is the image of  $G^0$  in  $L$ : the algebraic group  $G^0$  is an extension of  $G_1^0$  by a subgroup of  $N$ , automatically unipotent, so that it suffices to prove (1.3.8) for  $\mathrm{Gr}_G(\mathcal{F}_0)$ , where (1.3.9) applies.

**Lemma (1.3.10).** Let  $G$  be a group scheme extension of  $\mathbb{Z}$  by an algebraic group  $G^0$ , whose connected component of the identity  $G^{00}$  is reductive. The following conditions are equivalent:

- (i)  $G$  admits a linear representation whose restriction to  $G^0$  is faithful.
- (ii)  $G$  admits a linear representation whose restriction to  $G^0$  has finite kernel.
- (iii) Let  $Z^{00}$  be the connected component of the centre of  $G^{00}$ . The restriction to  $Z^{00}$  of every inner automorphism of  $G$  is of finite order.
- (iv) The centre of  $G$  maps onto a finite-index subgroup of  $\mathbb{Z}$ .

The implication (i) $\Rightarrow$ (ii) is trivial. Prove that (ii) $\Rightarrow$ (iii): if  $\rho$  is a linear representation as in (ii), the weights of  $Z^{00}$  in this representation form a finite set of characters, rationally generating the character group of  $Z^{00}$ . This finite set is respected by the action of  $G$  on  $Z^{00}$  by inner automorphisms. A finite-index subgroup of  $G$  therefore acts trivially on this set, and hence on  $Z^{00}$ .

Prove that (iii) $\Rightarrow$ (iv). Let  $g$  be an element of  $G$  whose image in  $\mathbb{Z}$  is 1, and let  $\gamma$  be the automorphism  $x \mapsto gxg^{-1}$  of  $G^{00}$ . We know that the group of exterior automorphisms of a reductive group is the semidirect product of the automorphisms respecting a pinning by the adjoint group (inner automorphisms) (SGA 3 XXIV (1.3)), and that an automorphism preserving a pinning is of finite order as soon as its restriction to the connected centre is. Hypothesis (iii) therefore ensures that  $\gamma$  is the product of an automorphism of finite order preserving a pinning by an inner automorphism. Modifying  $g$  by an element of  $G^{00}$ , we may suppose  $\gamma$  of finite order. The restriction of  $\mathrm{Int}(g)$  to  $G^0$  is then also of finite order. If its order is  $n$ , then  $g^n$  is central (because it centralizes  $G^0$  and  $g$ ), and (iv) follows.

Finally, under hypothesis (iv),  $G$  admits a central subgroup  $Z$ , generated by an element  $g$  of nonzero degree in  $\mathbb{Z}$ ; the group  $G/Z$  is algebraic and every faithful representation of  $G/Z$  gives a representation of  $G$  whose restriction to  $G^0$  is faithful.

**Corollary (1.3.11).** Suppose  $\mathcal{F}_0$  semisimple, and let  $Z$  be the centre of  $G$ . Then the kernel and cokernel of  $\deg : Z \rightarrow \mathbb{Z}$  are finite.

The kernel  $Z \cap G^0$  is contained in the centre of  $G^0$ , which is finite by (1.3.9). Again by (1.3.9), condition (1.3.10)(iii) is trivially verified (one has  $Z^{00} = \{e\}$ ), and the cokernel is finite by (1.3.10)(iv).

**Corollary (1.3.12).** Suppose  $\mathcal{F}_0$  semisimple, and let  $g$  be a central element of degree  $m \neq 0$  in  $G$ . For the sheaf defined by a representation  $V$  of  $G$  to be purely of  $\iota$ -determinantal weight  $\beta$ ,

it is necessary and sufficient that all the eigenvalues  $\alpha$  of  $g$  acting on  $V$  verify

$$|\iota\alpha| = q^{m\beta/2}.$$

One reduces to supposing  $V$  simple, in which case  $g$  is scalar. One then replaces  $V$  by its maximal exterior power, to reduce to the case in which  $V$  has rank 1. That case is left to the reader; see (1.3.4).

This interpretation of determinantal weight immediately gives the following proposition, which is the main result of this section.

**Proposition (1.3.13).**

- (i) Let  $f : X'_0 \rightarrow X_0$  be a dominant morphism of normal connected schemes of finite type over  $\mathbb{F}_q$ . For a lisse sheaf on  $X_0$  to be purely of  $\iota$ -determinantal weight  $\beta$ , it is necessary and sufficient that its inverse image on  $X'_0$  be so.
- (ii) If the lisse sheaves  $\mathcal{F}_0$  and  $\mathcal{G}_0$  on  $X_0$  are purely of  $\iota$ -determinantal weights  $\beta$  and  $\gamma$ , then  $\mathcal{F}_0 \otimes \mathcal{G}_0$  is purely of  $\iota$ -determinantal weight  $\beta + \gamma$ .
- (iii) Let  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$ , and let  $n(\beta)$  be the sum of the ranks of the constituents of  $\mathcal{F}_0$  of  $\iota$ -determinantal weight  $\beta$ . Then the determinantal weights of  $\wedge^a \mathcal{F}_0$  are the sums

$$\sum a(\beta)\beta$$

with  $\sum a(\beta) = a$  and  $a(\beta) \leq n(\beta)$ .

For (ii), take  $G$  defined by  $\mathcal{F}_0$  and  $\mathcal{G}_0$ ; see (1.3.7), last paragraph.

**Proposition (1.3.14) (see (1.3.2)).** For an irreducible lisse sheaf  $\mathcal{F}_0$  of rank  $r$  on  $X_0$  to be an étale sheaf, it is necessary and sufficient that  $\wedge^r \mathcal{F}_0$  be one. Every irreducible lisse sheaf is therefore obtained by torsion (1.2.7) from an étale sheaf.

Let  $g$  be central in  $G$  as in (1.3.12). It acts on  $\mathcal{F}_{0\bar{x}}$  by multiplication by a scalar  $u$ . The group  $G$  acts on  $\wedge^r \mathcal{F}_{0\bar{x}}$  through a discrete quotient, which is also a quotient of  $W(X_0, \bar{x})$ ; and for  $\wedge^r \mathcal{F}_0$  to be an étale sheaf, it is necessary and sufficient that  $u$  be an  $\ell$ -adic unit. Suppose this is so, and let  $E$  be a finite extension of  $\mathbb{Q}_\ell$  such that the action of  $W(X_0, \bar{x})$  factors through  $\mathrm{GL}(r, E)$ . It remains to show that the image of  $W(X_0, \bar{x})$  in  $\mathrm{GL}(r, E)$  is bounded; then the inverse image of an open subgroup will be open of finite index, and the representation will extend to the completion  $\pi_1(X_0, \bar{x})$  of  $W(X_0, \bar{x})$ .

Let  $W_1$  be the subgroup of finite index of  $W(X_0, \bar{x})$  which is the inverse image of  $G^{00} \times g^{\mathbb{Z}}$ , let  $W_1^0$  be the inverse image of  $G^{00}$ , and let  $\rho$  be the composite morphism

$$W_1 \longrightarrow G^{00} \times g^{\mathbb{Z}} \longrightarrow G^{00}.$$

Since the  $u^n$  ( $n \in \mathbb{Z}$ ) form a bounded set, it suffices to show that  $\rho(W_1) \subset G^{00}(E)$  is bounded. This subgroup normalizes  $\rho(W_1^0)$ , which is compact and Zariski dense. We conclude by the following lemma.

**Lemma (1.3.15).** In a semisimple group  $H$  over a local field  $E$ , the normalizer of a compact Zariski-dense subgroup  $K \subset H(E)$  is compact.

We treat only the case in which  $E$  has characteristic 0. The logarithm map, first defined on a sufficiently small neighbourhood of  $e$ , is extended to the union of the compact subgroups of  $H(E)$  by the formula

$$\log(x) = \frac{1}{n} \log(x^n).$$

The density of  $K$ , and the fact that  $H$  is semisimple, imply that  $\log(K)$  generates  $\text{Lie } H$  as a vector space over  $E$ . Let  $\mathcal{O}_E$  be the valuation ring of  $E$ , and let  $L$  be the  $\mathcal{O}_E$ -module generated by the compact set  $\log K \subset \text{Lie } H$ . It is stable under the normalizer of  $K$ . Since the adjoint representation has finite kernel, and since

$$\text{ad} : H(E) \longrightarrow \text{GL}(\text{Lie } H)$$

is proper, it follows that this normalizer is compact.

## (1.4) Cohomology of curves and $L$ -functions: reminders

This section gathers together some well-known results that we shall use constantly.

**(1.4.1).** Let  $X$  be a curve over an algebraically closed field  $k$ , and let  $\mathcal{F}$  be a sheaf on  $X$ . The groups

$$H_c^0(X, \mathcal{F}), \quad H^0(X, \mathcal{F}), \quad H_c^2(X, \mathcal{F}), \quad H^2(X, \mathcal{F})$$

are elementary in nature.

- a)  $H^0(X, \mathcal{F})$  is the group of global sections, and  $H_c^0(X, \mathcal{F})$  is the subgroup of those with compact support; if  $X$  has no complete irreducible component, this means finite support. If  $X$  is connected, with base point  $x$ , and if  $\mathcal{F}$  is lisse and defined by a representation  $V$  of  $\pi_1(X, x)$ , then

$$H^0(X, \mathcal{F}) = V^{\pi_1(X, x)}, \quad (1.4.1.1)$$

while

$$H_c^0(X, \mathcal{F}) = \begin{cases} V^{\pi_1(X, x)}, & \text{if } X \text{ is complete,} \\ 0, & \text{otherwise.} \end{cases} \quad (1.4.1.2)$$

- b) If  $U$  is the complement of a finite subset of  $X$ , then

$$H_c^2(U, \mathcal{F}) \simeq H_c^2(X, \mathcal{F}).$$

Taking  $U$  small enough so that  $U_{\text{red}}$  is smooth and  $\mathcal{F}$  is lisse on  $U$ , this makes it possible to compute  $H_c^2$  by Poincaré duality from the calculation of  $H^0$  in a). If  $U_{\text{red}}$  is smooth and connected, with base point  $x$ , and if, on  $U$ , the sheaf  $\mathcal{F}$  is defined by a representation  $V$  of  $\pi_1(U, x)$ , then

$$H_c^2(X, \mathcal{F}) = V_{\pi_1(U, x)}(-1). \quad (1.4.1.3)$$

**(1.4.2).** With the conventions of (0.7), let  $X_0$  be a smooth absolutely irreducible curve over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$ . Let  $\mathcal{F}'_0$  (respectively  $\mathcal{F}''_0$ ) be the largest subsheaf (respectively quotient sheaf) of  $\mathcal{F}_0$  which becomes constant on  $X$ . It is the inverse image of a sheaf  $F'_0$  (respectively  $F''_0$ ) on  $\text{Spec}(\mathbb{F}_q)$ , and the discussion (1.4.1) gives

$$H^0(X, \mathcal{F}) = F', \quad (1.4.2.1)$$

$$H_c^0(X, \mathcal{F}) = F' \text{ or } 0 \text{ according as } X \text{ is complete or not,} \quad (1.4.2.2)$$

$$H_c^2(X, \mathcal{F}) = F''(-1). \quad (1.4.2.3)$$

The sheaves  $\mathcal{F}'_0$  and  $F'_0$  (respectively  $\mathcal{F}''_0$  and  $F''_0$ ) have the same determinantal weights, and these are among the determinantal weights of  $\mathcal{F}_0$ . From (1.4.2) we therefore get:

**Corollary (1.4.3).** If  $\alpha$  is an eigenvalue of  $F^*$  on  $H^0(X, \mathcal{F})$  or  $H_c^0(X, \mathcal{F})$  (respectively on  $H_c^2(X, \mathcal{F})$ ), then there exists a determinantal weight  $\beta$  of  $\mathcal{F}_0$  such that

$$w_q(\alpha) = \beta \quad (\text{respectively } w_q(\alpha) = \beta + 2).$$

More trivially, one also has:

**Corollary (1.4.4).** If  $\alpha$  is an eigenvalue of  $F^*$  on  $H^0(X, \mathcal{F})$  or  $H_c^0(X, \mathcal{F})$  (respectively on  $H_c^2(X, \mathcal{F})$ ), then, for every  $x \in |X_0|$ , the number  $\alpha^{\deg(x)}$  (respectively  $(q^{-1}\alpha)^{\deg(x)}$ ) is an eigenvalue of  $F_x$  on  $\mathcal{F}_0$ .

**(1.4.5).** If  $X_0$  is a scheme of finite type over  $\mathbb{F}_q$ , and if  $\mathcal{F}_0$  is a sheaf on  $X_0$ , then by Grothendieck one has an equality of formal  $t$ -adic series:

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \prod_i \det(1 - Ft, H_c^i(X, \mathcal{F}))^{(-1)^{i+1}}. \quad (1.4.5.1)$$

Applying  $\iota$  to it, one obtains an equality between complex formal series, expressing that the left hand side is the Taylor expansion of the rational function on the right. If the product on the left converges for  $|t| < R$ , one obtains in that disc an equality between analytic functions. Here is a convergence criterion.

**Proposition (1.4.6).** Suppose that, for  $x \in |X_0|$ , the eigenvalues  $\alpha$  of  $F_x$  on  $\mathcal{F}$  satisfy

$$w_{N(x)}(\alpha) < \beta.$$

Then the product

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1}$$

converges for

$$|t| < q^{-\beta/2 - \dim X_0},$$

and hence has neither zero nor pole in that disc.

Set  $d = \dim X_0$ , and decompose  $X_0$  into pieces each of which is quasi-finite over an affine space  $\mathbb{A}^d$ : one obtains an estimate

$$\#\{x \in |X_0| \mid \deg(x) = n\} < \text{Cte} \cdot q^{dn}.$$

This reduces the convergence of the product to that of the geometric series

$$\sum_n q^{dn} q^{n\beta/2} |t|^n.$$

**(1.4.7).** If  $\dim X_0 \leq 1$ , the  $H_c^i$  are zero for  $i \notin \{0, 1, 2\}$ . The formula deduced from (1.4.5.1) by applying  $\iota$  therefore reduces to

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}))}{\det(1 - Ft, H_c^0(X, \mathcal{F})) \det(1 - Ft, H_c^2(X, \mathcal{F}))}. \quad (1.4.7.1)$$

If  $X_0$  is a smooth affine curve and  $\mathcal{F}_0$  is lisse, then  $H_c^0$  is zero and the formula simplifies to

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}))}{\det(1 - Ft, H_c^2(X, \mathcal{F}))}. \quad (1.4.7.2)$$

In the uses we shall make of these formulae, the left hand side will be controlled by (1.4.6), and the poles on the right by (1.4.3).

## (1.5) A criterion for purity

The conventions of (0.7) are in force. Let  $X_0$  be a smooth absolutely irreducible curve over  $\mathbb{F}_q$ . The proof of the following theorem is parallel to that of Theorem (1.3.4), which it generalizes.

**Theorem (1.5.1).** The constituents of any lisse  $\iota$ -real sheaf on  $X_0$  are  $\iota$ -pure.

**Lemma (1.5.2).** Let  $\mathcal{F}_0$  be a lisse  $\iota$ -real sheaf on  $X_0$ , and let  $r$  be the largest of its determinantal weights, relative to  $\iota$  (1.3.5). Then, for every  $x \in |X_0|$ , and every eigenvalue  $\alpha$  of  $F_x$  on  $\mathcal{F}_0$ , one has

$$w_{N(x)}(\alpha) \leq r.$$

After removing from  $X_0$  a point other than the one under consideration, one reduces to the case in which  $X_0$  is affine. For every positive integer  $k$ , one then has the equality (1.4.7.2)

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F}^{\otimes 2k})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}^{\otimes 2k}))}{\det(1 - Ft, H_c^2(X, \mathcal{F}^{\otimes 2k}))}. \quad (1.5.2.1)$$

In this formula:

a) The factors

$$\det(1 - F_x t^{\deg(x)}, \mathcal{F}^{\otimes 2k})^{-1}$$

on the left are formal series with constant term 1 and real non-negative coefficients. The proof is as in (1.3.3), (3.4), with “rational” replaced by “real”.

b) By (1.3.13)(ii), the determinantal weights of the constituents of  $\mathcal{F}_0^{\otimes 2k}$  are at most  $2kr$ . By (1.4.3), the right hand side has no pole for

$$|t| < q^{-(2kr+2)/2}.$$

Because the factors on the left have non-negative coefficients, poles cannot be cancelled. In particular, if  $\alpha$  is an eigenvalue of  $F_x$  on  $\mathcal{F}_0$ , then the contribution of  $\alpha^{2k}$  gives a pole at absolute value

$$|\iota(\alpha)|^{-2k/\deg(x)}.$$

Consequently

$$|\iota(\alpha)|^{-2k/\deg(x)} \geq q^{-(2kr+2)/2},$$

or equivalently

$$w_{N(x)}(\alpha) \leq r + \frac{1}{k}.$$

Letting  $k$  tend to infinity gives the lemma.

**(1.5.3) Proof of (1.5.1).** Let  $\mathcal{F}_0$  be a lisse  $\iota$ -real sheaf on  $X_0$ . For  $\gamma \in \mathbb{R}$ , let  $\mathcal{F}_0(\gamma)$  be the sum of the constituents of  $\mathcal{F}_0$  of determinantal weight  $\gamma$ , relative to  $\iota$ , and let  $n(\gamma)$  be the rank of  $\mathcal{F}_0(\gamma)$ . Let  $x \in |X_0|$ , and let  $\alpha_i^\gamma$  ( $1 \leq i \leq n(\gamma)$ ) be the eigenvalues of  $F_x$  on  $\mathcal{F}_0(\gamma)$ , counted with multiplicity. We have to prove that

$$w_{N(x)}(\alpha_i^\gamma) = \gamma.$$

By the definition of the determinantal weight, one has, for every  $\gamma$ ,

$$\sum_i w_{N(x)}(\alpha_i^\gamma) = n(\gamma)\gamma. \quad (1.5.3.1)$$



Suppose, as we may, that  $\mathcal{F}_0(\beta) \neq 0$ , and let  $N$  be the sum of the  $n(\gamma)$  for  $\gamma > \beta$ . By (1.3.13)(iii), the determinantal weights of the  $(N+1)$ -st exterior power of  $\mathcal{F}_0$  are at most

$$\beta + \sum_{\gamma > \beta} n(\gamma)\gamma.$$

Since each product

$$\alpha_i^\beta \prod_{\gamma > \beta} \prod_{j=1}^{n(\gamma)} \alpha_j^\gamma$$

is an eigenvalue of  $F_x$  on  $\wedge^{N+1} \mathcal{F}_0$ , (1.5.2) gives

$$w_{N(x)}(\alpha_i^\beta) + \sum_{\gamma > \beta} \sum_j w_{N(x)}(\alpha_j^\gamma) \leq \beta + \sum_{\gamma > \beta} n(\gamma)\gamma.$$

Subtracting the equalities (1.5.3.1) for  $\gamma > \beta$ , one obtains

$$w_{N(x)}(\alpha_i^\beta) \leq \beta. \quad (1.5.3.2)$$

To deduce equality from this inequality, one may either pass to the dual sheaf, or observe that summing the inequalities (1.5.3.2) over  $i$  gives the equality (1.5.3.1) for  $\beta$ .

## (1.6) Around Jacobson–Morosov

**Proposition (1.6.1).** Let  $N$  be a nilpotent endomorphism of an object  $V$  of an abelian category. There is then one and only one finite increasing filtration  $M$  of  $V$  such that

$$NM_i \subset M_{i-2}$$

and such that  $N^k$  induces an isomorphism

$$\mathrm{Gr}_k^M V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M V.$$

We prove existence; uniqueness is proved in the same way. We proceed by induction on an integer  $d$  such that  $N^{d+1} = 0$ . For  $d = 0$  one takes the trivial filtration. Next put

$$M_d = V, \quad M_{d-1} = \mathrm{Ker} N^d, \quad M_{-d} = \mathrm{Im} N^d, \quad M_{-d-1} = 0,$$

so that  $\mathrm{Gr}_d^M V$  and  $\mathrm{Gr}_{-d}^M V$  are respectively the coimage and the image of  $N^d$ . On  $\mathrm{Ker} N^d / \mathrm{Im} N^d$ , the  $(d-1)$ -st power of  $N$  is zero. The induction hypothesis therefore defines  $M$  on this quotient, and, for  $d-1 \geq i \geq -d$ ,  $M_i$  on  $V$  is defined as the inverse image in  $\mathrm{Ker} N^d$  of the corresponding subobject of  $\mathrm{Ker} N^d / \mathrm{Im} N^d$ .

**Remark (1.6.2).** The characterization (1.6.1) of  $M$  shows its compatibility with passage to the dual category.

**Definition (1.6.3).** The primitive part  $P_i(V)$ , or simply  $P_i$ , of  $\mathrm{Gr}_i^M V$  is the kernel of the morphism induced by  $N$  from  $\mathrm{Gr}_i^M V$  to  $\mathrm{Gr}_{i-2}^M V$ .

(1.6.4). If  $i > 0$ , then

$$N : \text{Gr}_i^M V \longrightarrow \text{Gr}_{i-2}^M V$$

is injective, since  $N^{i-1}N$  is an isomorphism, and therefore  $P_i = 0$ . If  $i \geq 0$ , writing that

$$N \circ N^{i+1} : \text{Gr}_{i+2}^M V \longrightarrow \text{Gr}_i^M V \longrightarrow \text{Gr}_{i-2}^M V$$

is an isomorphism shows that  $\text{Gr}_{-i}^M V$  is the direct sum of  $P_{-i}$  and of the image of  $\text{Gr}_{i+2}^M V$  by  $N^{i+1}$ ; the morphism  $N$  induces an isomorphism from that image onto  $\text{Gr}_{-i-2}^M V$ . Repeating the construction, one obtains, by descending induction on  $i \geq 0$ , an isomorphism

$$\text{Gr}_i^M V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j}. \quad (1.6.4.1)$$

Through these isomorphisms, the morphism  $N : \text{Gr}_i^M V \rightarrow \text{Gr}_{i-2}^M V$  is the evident inclusion for  $i \geq 1$  and the evident projection for  $i - 2 \leq -1$ .

Schematically, the graded pieces are arranged as follows:

$$\begin{array}{rclcl} \text{Gr}_3^M : & & P_3 & \cdots & \\ \text{Gr}_2^M : & & P_2 & P_4 & \cdots \\ \text{Gr}_1^M : & P_1 & P_3 & & \cdots \\ \text{Gr}_0^M : P_0 & & P_2 & P_4 & \cdots \\ \text{Gr}_{-1}^M : & P_1 & P_3 & & \cdots \\ \text{Gr}_{-2}^M : & & P_2 & P_4 & \cdots \\ \text{Gr}_{-3}^M : & & P_3 & \cdots & \end{array}$$

with arrows induced by  $N$  running downward.

**Lemma (1.6.5).** The morphism

$$N : (V, M) \longrightarrow (V, M \text{ shifted by } 2)$$

is strictly compatible with the filtrations.

It is necessary to prove that  $NM_{i+2} \supset \text{Im}(N) \cap M_i$ . Distinguish two cases. If  $i < 0$ , the morphism  $N : M_{i+2} \rightarrow M_i$  has surjective associated graded, hence is surjective, and  $NM_{i+2} = M_i$ , which proves the assertion a fortiori. If  $i + 2 \geq 0$ , the morphism

$$N : V/M_{i+2} \longrightarrow V/M_i$$

has injective associated graded, hence is injective, and  $N^{-1}M_i \subset M_{i+2}$ , again proving the assertion.

This permits one to exchange the constructions  $\text{Ker } N$  and  $\text{Gr}^M$ .

**Corollary (1.6.6).** The inclusion of  $\text{Ker } N$  in  $V$  induces isomorphisms

$$\text{Gr}_i^M(\text{Ker } N) \xrightarrow{\sim} P_i.$$

(1.6.7). Assume now that  $V$  is a finite-dimensional vector space over a field  $k$ . We first describe  $M$  when, in a basis  $e$ , the matrix of  $N$  is

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{pmatrix}. \quad (1.6.7.1)$$

Put  $\dim V = d + 1$  and index the basis vectors by the integers going by steps of two from  $d$  to  $-d$ , so that  $Ne_i = e_{i-2}$  for  $i \neq -d$  and  $Ne_{-d} = 0$ . Then  $M_i$  is generated by the  $e_j$  with  $j \leq i$ .

In general,  $V$  is a sum of subspaces  $V_a$  stable under  $N$ , with  $N$  of the form (1.6.7.1) on each  $V_a$ , and the filtration  $M$  of  $V$  is the sum of the preceding filtrations of the  $V_a$ .

**(1.6.8).** If  $k$  has characteristic zero, one may interpret the filtration  $M$  in terms of the Jacobson–Morosov theorem: if

$$u : \mathrm{SL}(2) \longrightarrow \mathrm{GL}(V)$$

satisfies

$$du \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = N,$$

and if  $V_j$  is the subspace of  $V$  formed by those vectors  $v$  such that

$$u \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} (v) = \lambda^j v,$$

then  $M_i$  is the sum of the  $V_j$  for  $j \leq i$ . This is reduced to the case where  $V$  is an irreducible representation of  $\mathrm{SL}(2)$ , that is, a symmetric power of the evident representation, and one is then in the situation of (1.6.7).

This interpretation gives the behavior of  $M$  under tensor product and duality.

**Proposition (1.6.9).** Define the tensor product  $(V, N)$  of  $(V', N')$  and  $(V'', N'')$  by

$$V = V' \otimes V'', \quad N = N' \otimes 1 + 1 \otimes N'',$$

and the dual of  $(V', N')$  as  $(V'^\vee, -{}^t N')$ .

- (i) If  $k$  has characteristic zero, the filtration  $M$  of a tensor product is the tensor product of the filtrations  $M$  of the factors:

$$M_i(V' \otimes V'') = \sum_{i'+i''=i} M_{i'}(V') \otimes M_{i''}(V'').$$

- (ii) The filtration  $M$  of a dual is the dual of the filtration  $M$  of the original space:

$$M_i(V'^\vee) = M_{-i-1}(V')^\perp.$$

- (iii) Under these hypotheses, formation of  $\mathrm{Gr}^M$  is therefore compatible with tensor product and passage to the dual.

**(1.6.10).** Assume henceforth that  $V$  is finite-dimensional over a field  $k$  of characteristic zero. On  $\mathrm{Gr}^M(V)$  there is then a unique representation  $v$  of  $\mathrm{SL}(2)$  such that

$$dv \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

is  $N : \mathrm{Gr}_i^M V \rightarrow \mathrm{Gr}_{i-2}^M V$  and such that

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}$$

acts by multiplication by  $\lambda^j$  on  $\mathrm{Gr}_j^M(V)$ . Existence is seen by choosing  $u$  as in (1.6.8), identifying  $\mathrm{Gr}^M(V)$  with  $V$  by means of the splitting  $V_j$  of the filtration, and transporting  $u$  to the graded object by this identification. Uniqueness follows from Bourbaki, Lie VIII, §11, no. 1, Lemma 1. Formation of  $v$  is compatible with tensor product and passage to the dual taken as in (1.6.9). This will allow us to determine the behavior of  $P_i$  under tensor product and duality.

**(1.6.11).** Choose once and for all, for each integer  $d \geq 0$ , an irreducible representation  $S_d$  of  $\mathrm{SL}(2)$  of dimension  $d + 1$ , and a lowest-weight vector  $e_{-d} \in S_d$  such that  $e_{-d} \neq 0$  and

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} e_{-d} = \lambda^{-d} e_{-d}.$$

Put

$$P(d, d') = \{j \mid |d - d'| \leq j \leq d + d', j \equiv d + d' \pmod{2}\}, \quad (1.6.11.1)$$

and choose isomorphisms of representations

$$S_d \otimes S_{d'} \xrightarrow{\sim} \bigoplus_{j \in P(d, d')} S_j, \quad (1.6.11.2)$$

$$S_d^\vee \xrightarrow{\sim} S_d. \quad (1.6.11.3)$$

For  $V, N, M, P$  as above, there exists a unique isomorphism of  $\mathrm{SL}(2)$ -representations

$$\varphi : \bigoplus_j S_j \otimes P_{-j} \xrightarrow{\sim} \mathrm{Gr}^M(V), \quad (1.6.11.4)$$

where  $\mathrm{SL}(2)$  acts trivially on  $P_{-j}$  in the source and as in (1.6.10) on the target, such that the composite

$$P_{-j} \xrightarrow{e_{-j} \otimes -} S_j \otimes P_{-j} \longrightarrow \mathrm{Gr}_{-j}^M(V)$$

is the identity inclusion of  $P_{-j}$ . This yields isomorphisms

$$P_{-j} \simeq \mathrm{Hom}_{\mathrm{SL}(2)}(S_j, \mathrm{Gr}^M(V)). \quad (1.6.11.5)$$

**(1.6.12).** Let now  $V'$  and  $V''$  be as in (1.6.10), let  $V = V' \otimes V''$ , and write  $P_j, P'_j, P''_j$  for  $P_j(V), P_j(V')$ , and  $P_j(V'')$ . The tensor product of the isomorphisms (1.6.11.4), followed by the choices (1.6.11.2), gives an isomorphism, depending only on the choices (1.6.11.2),

$$P_{-j} \xrightarrow{\sim} \bigoplus_{j \in P(j', j'')} P'_{-j'} \otimes P''_{-j''}. \quad (1.6.12.1)$$

For passage to the dual, one similarly obtains isomorphisms, depending only on the choices (1.6.11.3),

$$P_{-j}(V'^\vee) \xrightarrow{\sim} P_{-j}(V')^\vee. \quad (1.6.12.2)$$

**Proposition (1.6.13).** Let  $V$  be an object of an abelian category, endowed with a finite increasing filtration  $W$ , and let  $N$  be a nilpotent endomorphism of  $V$  respecting  $W$ . There is at most one finite increasing filtration  $M$  of  $V$  such that  $NM_i \subset M_{i-2}$  and such that  $N^k$  induces isomorphisms

$$\mathrm{Gr}_k^M \mathrm{Gr}^W V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M \mathrm{Gr}^W V.$$

The proof is by induction on the length of  $W$ . This reduces one to the case where, for an integer  $a$ , one has  $W_a = V$ , and where the assertion is known for  $W_{a-1}$ , endowed with the induced filtration and the endomorphism induced by  $N$ . A simultaneous shift by the same amount on the filtrations  $W$  and  $M$  preserves the condition of (1.6.13), so that we may suppose  $a = 0$ .

Let  $M$  be a filtration satisfying the stated conditions. The filtration induced on  $W_{-1}$  satisfies the same conditions, hence is uniquely determined. The quotient filtration is the filtration (1.6.1) of  $\mathrm{Gr}_0^W V$ . Let  $c$  be such that  $\mathrm{Gr}_i^M \mathrm{Gr}_0^W V$  is nonzero only for  $-c \leq i \leq c$ . For  $i \geq 0$ , the following identities hold:

- 1) for  $i > c$ ,  $M_{-i} = M_{-i}(W_{-1})$ ;
- 2)  $M_{-i} = M_{-i}(W_{-1}) + N^i(M_i)$ ;
- 3)  $M_i = \text{Ker}(N^{i+1} : V \rightarrow V/M_{-i-2})$ .

These identities determine  $M$  uniquely, by descending induction on  $|i|$ , in terms of its restriction to  $W_{-1}$  and of  $N$ .

**(1.6.14).** In the applications, the abelian category is the category of vector spaces over a field  $k$  of characteristic zero, and instead of a single nilpotent endomorphism  $N$  of  $V$  one has a nilpotent action of a one-dimensional Lie algebra  $\mathfrak{N}$ . The preceding theory applies as soon as one chooses  $N \neq 0$  in  $\mathfrak{N}$ . In (1.6.13), existence of  $M$ , and  $M$  itself when it exists, are independent of this choice. The isomorphism  $N^k$  has the canonical avatar

$$\text{Gr}_{i+k}^M \text{Gr}^W V(k) \xrightarrow{\sim} \text{Gr}_{i-k}^M \text{Gr}^W V, \quad (1.6.14.1)$$

where for a vector space  $Y$  one writes  $Y(k) = Y \otimes \mathfrak{N}^{\otimes k}$ .

In the situation of (1.6.1)–(1.6.12), the filtration  $M$  and the  $P_j$  do not depend on the choice of  $N$ , but the isomorphisms (1.6.4.1) and (1.6.12.1–2) do. If  $N$  is replaced by  $\lambda N$ , each of their components, generically denoted  $\alpha : X \rightarrow Y$ , is multiplied by a convenient power  $\lambda^n$  of  $N$ . It therefore defines a map

$$\alpha \otimes N^{\otimes n} : X \longrightarrow Y(n)$$

independent of the choice of  $N$ . In this way the previous isomorphisms become canonical:

$$\text{Gr}_i^M(\text{Ker } N) \simeq P_i, \quad (1.6.14.2)$$

$$\text{Gr}_i^M V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j} \left( \frac{i+j}{2} \right), \quad (1.6.14.3)$$

$$P_{-j} \simeq \bigoplus_{j \in P(j', j'')} P'_{-j'} \otimes P''_{-j''} \left( \frac{j - j' - j''}{2} \right), \quad (1.6.14.4)$$

$$P_{-j}(V'^\vee) \simeq P_{-j}(V')^\vee(j). \quad (1.6.14.5)$$

To remember these formulas, one regards  $\text{Gr}_i$  and  $P_i$  as of weight  $i$ , and  $\mathfrak{N}$  as of weight  $-2$ ; both members are then always isobaric of the same weight. We shall also use a sheaf-theoretic variant of (1.6.14).

## (1.7) Local monodromy

**(1.7.1).** Let  $R$  be a henselian discrete valuation ring,  $K$  its field of fractions,  $k = R/\mathfrak{m}$  its residue field,  $\overline{K}$  an algebraic closure of  $K$ , and  $\overline{k}$  the residue field of the normalization of  $R$  in  $\overline{K}$ . The extension  $\overline{k}/k$  is not algebraic in general. It is often convenient first to choose  $\overline{k}$ , and then  $\overline{K}$ , by taking an algebraic closure  $\overline{k}$  of  $k$ , then an algebraic closure  $\overline{K}$  of the field of fractions of the strict henselization of  $R$ .

The inertia group is defined as the kernel in the exact sequence

$$0 \longrightarrow I \longrightarrow \text{Gal}(\overline{K}/K) \longrightarrow \text{Gal}(\overline{k}/k) \longrightarrow 0. \quad (1.7.1.1)$$

It, or its image in a representation, is also called the local monodromy group. Recall that if  $k$  has exponential characteristic  $p$ , one has an exact sequence

$$0 \longrightarrow P \longrightarrow I \longrightarrow \widehat{\mathbb{Z}}_{(p')}(1) \longrightarrow 0,$$

where  $P$  is a pro- $p$  group and

$$\widehat{\mathbb{Z}}_{(p')}(1) = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$$

is the tame inertia group. Taking the  $\ell$ -component gives

$$t_\ell : I \longrightarrow \mathbb{Z}_\ell(1)$$

whose kernel is a profinite group of order prime to  $\ell$ .

**(1.7.2).** Let  $\rho : I \rightarrow \mathrm{GL}(V)$  be an  $\ell$ -adic representation of  $I$ . Grothendieck proved that the following hypothesis is often verified in practice:

(\*) The restriction of  $\rho$  to a finite-index subgroup  $I_1$  of  $I$  is unipotent.

If this is so, there is a unique representation  $\bar{\rho}$  of  $\mathbb{Q}_\ell(1)$ , viewed as a one-dimensional Lie algebra in  $V$ , such that

$$\rho(\sigma) = \exp(\bar{\rho}(t_\ell(\sigma))) \quad (\sigma \in I_1 \subset I). \quad (1.7.2.1)$$

One calls  $\bar{\rho}$  the logarithm of the unipotent part of the local monodromy. It may be identified with a nilpotent morphism

$$N : V(1) \longrightarrow V.$$

The defining formula is then written

$$\rho(\sigma) = \exp(Nt_\ell(\sigma)) \quad (\sigma \in I_1 \subset I). \quad (1.7.2.2)$$

The theory of (1.6.14) applies to this situation. The twist  $(k)$  of local systems is the usual Tate twist. The filtrations (1.6.1) and (1.6.13) are called, respectively, the local monodromy filtration of  $V$  and the local monodromy filtration of  $V$  relative to  $W$ .

**(1.7.3).** Assume the residue field finite with  $q$  elements. Define the Weil group  $W(\bar{K}/K)$  as the subgroup of  $\mathrm{Gal}(\bar{K}/K)$  inverse image of  $W(\bar{k}/k)$ . If  $V$  is an  $\ell$ -adic representation of  $W(\bar{K}/K)$ , then by Grothendieck (SGA 7 I) condition (\*) of (1.7.2) is automatically verified. The logarithm of the unipotent part of the monodromy

$$N : V(1) \longrightarrow V$$

is therefore defined. It commutes with the action of  $W(\bar{K}/K)$ .

**Lemma (1.7.4).** If  $F'$  and  $F''$  are two liftings in  $W(\bar{K}/K)$  of  $F \in W(\bar{k}/k)$ , then the eigenvalues of  $F'$  and  $F''$  coincide up to multiplication by roots of unity.

To verify this, one may replace  $V$  by its semisimplification, and then suppose that the action of the inertia group factors through a finite quotient. In that case there is  $n > 0$  such that  $F'^n$  and  $F''^n$  have the same image in  $\mathrm{GL}(V)$ , and the lemma follows.

Whatever lift one chooses, the  $\iota$ -weights of the eigenvalues of  $F'$  or of  $F''$ , relative to  $q$ , are therefore the same. Imitating (1.2), one then defines pure, mixed,  $\iota$ -pure representations, and the weights of a representation.

**Proposition-definition (1.7.5).** Suppose that the  $\iota$ -weights of  $V$  are integers. There exists then one and only one finite increasing filtration  $M$  of  $V$ , stable under  $W(\bar{K}/K)$ , such that  $\mathrm{Gr}_i^M(V)$  is  $\iota$ -pure of weight  $i$ . It is called the weight filtration. One has  $NM_i(1) \subset M_{i-2}$ .

Choose a Frobenius lift  $F'$  in  $W(\bar{K}/K)$ . Let  $V'_i$  be the sum of the generalized eigenspaces of  $F'$  in  $V$  corresponding to eigenvalues  $\alpha$  such that  $w_q(\alpha) = i$ , and put  $M'_i = \prod_{j \leq i} V'_j$ . One proves that  $M'$  is independent of  $F'$ . This will make it stable under  $W(\bar{K}/K)$  and gives  $M = M'$ . If

$M''$  is the filtration defined by another lift  $F''$ , let  $I_1$  be as in (1.7.2.1). Since  $I/I_1$  is finite, there is  $n > 0$  such that  $F''^n \equiv F'''^n \pmod{I_1}$ . For a suitable  $\lambda \in \mathbb{Q}_\ell(-1)$ , one then has in  $\mathrm{GL}(V)$

$$F'''^n = \exp(\lambda N) F''^n,$$

hence

$$F'''^n = \exp(\mu N) F''^n \exp(\mu N)^{-1} \quad \text{with } \mu = \lambda/(1 - q^n).$$

Since  $V'_i$  is the sum of the generalized eigenspaces of  $F''^n$  corresponding to eigenvalues of weight  $i$ , the automorphism  $\exp(\mu N)$  sends  $M'$  onto  $M''$ . Since  $N$  respects  $M'$ , this proves the assertion.

**Corollary (1.7.6).** If  $V$  is  $\iota$ -pure, the action of  $I$  factors through a finite quotient.

By twisting, one reduces to the case where  $V$  has integral weights. The filtration  $M$  is then reduced to one step. Since  $NM_i \subset M_{i-2}$ , one has  $N = 0$ , and the corollary follows.

**Remark (1.7.7).** The proof of (1.7.5) shows that any representation  $V$  admits a unique decomposition

$$V = \sum V^\alpha \quad (\alpha \in \overline{\mathbb{Q}_\ell}^*/\{\alpha \sim \alpha' \text{ if } \alpha/\alpha' \text{ is a root of unity}\}),$$

such that the eigenvalues of Frobenius liftings on  $V^\alpha$  lie in the class  $\alpha$ . Each  $V^\alpha$  is obtained from a mixed representation by twisting.

**Variant (1.7.8).** If  $D$  is a smooth divisor in a regular scheme  $X$ , the preceding constructions have an analogue for sheaves on  $X - D$  that are tamely ramified along  $D$ . More generally, let  $D$  be a normal-crossings divisor of  $X$ , the transverse union of smooth divisors  $D_i$  ( $i \in I$ ). We work locally so that each  $D_i$  admits a global equation  $t_i = 0$ . Let  $E$  be the intersection of the  $D_i$ . For  $n$  invertible on  $X$ , put

$$X_n = X[t_i^{1/n}]_{i \in I},$$

the closed subscheme of the affine space over  $X$ , with coordinates  $T_i$ , defined by the equations  $T_i^n = t_i$ . The ramified covering  $\pi : X_n \rightarrow X$  is etale over  $X - D$  and totally ramified over  $E$ ; over  $E$ ,  $\pi$  has the section  $t_i^{1/n} = 0$ . The restriction to  $E$  of a sheaf on  $X_n$  means its inverse image by this section. The group  $(\mu_n)^I$  acts on  $X_n$  by multiplying  $t_i^{1/n}$  by  $\tau_i$ .

Let  $L$  be a set of prime numbers invertible on  $X$ , and let  $\mathcal{F}$  be a sheaf of sets on  $X - D$ , locally constant, tamely ramified along  $D$ , and indeed  $L$ -ramified along  $D$  in the following sense. For  $d$  the generic point of an irreducible component of  $D$ , the henselization  $X_{(d)}$  is a henselian trait; if  $\bar{\eta}$  is a geometric generic point, require the inertia group to act on  $\mathcal{F}_{\bar{\eta}}$  through an  $L$ -group. Abhyankar's lemma then gives an integer  $n$ , all of whose prime factors lie in  $L$ , such that the inverse image of  $\mathcal{F}$  on  $\pi^{-1}(X - D) \subset X_n$  is the restriction to  $\pi^{-1}(X - D)$  of a locally constant sheaf on  $X_n$ . We call this the locally constant extension of  $\mathcal{F}$  to  $X_n$ . We write  $\mathcal{F}[E]$  for its restriction to  $E$ . It carries an action of  $\mu_n^I$  deduced from the action on  $X_n$ . Putting

$$\mathbb{Z}_L(1) = \prod_{\ell \in L} \mathbb{Z}_\ell(1)$$

(a pro-sheaf on  $E$ ), one obtains a functor

$$\mathcal{F} \longmapsto \mathcal{F}[E]$$

from locally constant sheaves of sets on  $X - D$  that are  $L$ -ramified along  $D$  to locally constant sheaves of sets on  $E$  endowed with an action of  $\mathbb{Z}_L(1)^I$ . This functor passes to constructible  $\overline{\mathbb{Q}_\ell}$ -sheaves by passage to the limit, with a parallel variant for Weil sheaves. In many cases,

in particular if  $X$  is of finite type over  $\mathbb{Z}$ , the action of  $\mathbb{Z}_L(1)^I$  is quasi-unipotent: there are mutually commuting nilpotent endomorphisms

$$N_i \in \text{End}(\mathcal{F}[E])(-1)$$

such that, for  $\sigma$  in a suitable open subsheaf  $n\mathbb{Z}_L(1)^I$  of  $\mathbb{Z}_L(1)^I$ , it acts by

$$\exp\left(\sum_i N_i \sigma_i\right).$$

We call  $N_i$  the logarithm of the unipotent part of the monodromy around  $D_i$ . In the case of a smooth divisor, one uses this endomorphism, as above, to define the local monodromy filtration of  $\mathcal{F}[D]$ , and, when a filtration  $W$  is present, the relative local monodromy filtration.

**(1.7.9).** Near an intersection of  $p$  of the  $D_i$ , the divisor  $D$  appears as the union of  $p$  smooth divisors, and the preceding constructions can be repeated. We use the following notation: for  $J \subset I$ ,  $D_J$  is the intersection of the  $D_j$  ( $j \in J$ ),  $X_J$  is the complement of the union of the  $D_i$  with  $i \notin J$ , and  $D'_J = D_J \cap X_J$ . The  $D_i$  with  $i \notin J$  cut out on the regular scheme  $D_J$  a normal-crossings divisor with complement  $D'_J$ , while the  $D_j$  with  $j \in J$  cut out on  $X_J$  a normal-crossings divisor with complement  $X - D$ ; the intersection of their traces is  $D'_J$ .

Let  $\mathcal{F}$  be a locally constant sheaf of sets on  $X - D$ ,  $L$ -ramified along  $D$ . Applying (1.7.8) gives for each  $J$  a sheaf  $\mathcal{F}[D'_J]$  on  $D'_J$  with an action of  $\mathbb{Z}_L(1)^J$ . Repeating the construction yields, for  $K \subset I - J$ , a locally constant sheaf  $\mathcal{F}[D'_J][D'_{J \cup K}]$  on  $D'_{J \cup K}$  with the compatible actions. Transitivity of inverse images gives an isomorphism

$$\mathcal{F}[D'_J][D'_{J \cup K}] \xrightarrow{\sim} \mathcal{F}[D'_{J \cup K}], \quad (1.7.9.1)$$

compatible with the action of  $\mathbb{Z}_L(1)^{J \cup K}$ . For  $J \subset K \subset L$ , the corresponding square is commutative. All this extends to constructible  $\overline{\mathbb{Q}}_\ell$ -sheaves, and to Weil sheaves, by passage to the limit.

**(1.7.10).** The constructions (1.7.8) depend on the choice of the coordinates  $t_i$ . To make them canonical, one introduces the normal bundle  $T$  of  $E$  in  $X$  and the subbundles  $T_i$  tangent to the  $D_i$ . One replaces the construction of  $\mathcal{F}[E]$  by that of a sheaf  $\mathcal{F}[E]'$  on  $T - \bigcup T_i$ . A system of equations  $t_i$  defines a section of  $T - \bigcup T_i$ , and the sheaf  $\mathcal{F}[E]$  defined in terms of the  $t_i$  is identified with the inverse image of  $\mathcal{F}[E]'$  by this section. The construction is also compatible with base change by a smooth morphism  $X' \rightarrow X$ . The  $N_i$  and  $\mathbb{Z}_L(1)^I$  act on  $\mathcal{F}[E]'$ , which plays the role of the restriction of  $\mathcal{F}$  to a tubular neighborhood of  $E$  in  $X$ . It may also be regarded as an analogue of nilpotent orbits in the theory of variations of Hodge structure.

**(1.7.11).** Assume that  $X$  is henselian local,  $X = \text{Spec}(R)$ , and take for  $L$  the set of all primes invertible on  $X$ . We interpret (1.7.8) in Galois terms. Let  $k$  be the residue field of  $R$ ,  $K$  its field of fractions,  $R_1$  a strict henselization of  $R$  with residue field  $k_1$  and fraction field  $K_1$ ,  $\overline{K}$  an algebraic closure of  $K_1$ , and  $K_2 \subset \overline{K}$  the extension of  $K_1$  obtained by adjoining the  $n$ -th roots of the  $t_i$ , for all  $n$  invertible in  $R$ . A locally constant sheaf of sets on  $X - D$  is tamely ramified along  $D$  if and only if it becomes trivial over  $\text{Spec}(K_2)$ ; this identifies the tame fundamental group of  $X - D$  with  $\text{Gal}(K_2/K)$ . It is an extension

$$0 \longrightarrow \mathbb{Z}_L(1)^I \longrightarrow \pi_1^{\text{mod}}(X - D) \longrightarrow \text{Gal}(k_1/k) \longrightarrow 0.$$

The choice of the  $t_i$  and of their roots splits this extension. From a sheaf  $\mathcal{F}$  on  $X - D$ , identified with a set  $F$  on which  $\pi_1^{\text{mod}}(X - D)$  acts, one obtains an action of  $\text{Gal}(k_1/k)$  on  $F$ , and a  $\text{Gal}(k_1/k)$ -equivariant action of  $\mathbb{Z}_L(1)^I$ . This is the sheaf  $\mathcal{F}[E]$  of (1.7.8).



**(1.7.12).** If  $k$  is finite, define  $W^{\text{mod}}(X - D)$  as the inverse image of  $W(k_1/k)$  in the tame fundamental group. Generalizing (1.7.3)–(1.7.7), with the same proof, one has:

- (1.7.12.1) The restriction to  $\mathbb{Z}_L(1)^I$  of an  $\ell$ -adic representation of  $W^{\text{mod}}(X - D)$  is quasi-unipotent.
- (1.7.12.2) If  $F'$  and  $F''$  are two liftings in  $W^{\text{mod}}$  of  $F \in W(k_1/k)$ , the eigenvalues of  $F'$  and  $F''$  coincide up to multiplication by roots of unity. Thus the  $\iota$ -weights are defined as in (1.7.4).
- (1.7.12.3) If the  $\iota$ -weights of  $V$  are integers, there is on  $V$  one and only one weight filtration  $M$ , as in (1.7.5). The analogues of (1.7.6) and (1.7.7) are true.

## (1.8) Local monodromy of pure sheaves

The conventions of (0.7) are in force. Let  $X_0$  be a smooth absolutely irreducible curve over  $\mathbb{F}_q$ , and let  $j : U_0 \hookrightarrow X_0$  be an open subset whose complement is the finite set  $S_0$ .

**Lemma (1.8.1).** If  $\mathcal{F}_0$  is a lisse sheaf on  $U_0$  pointwise  $\iota$ -pure of weight  $\beta$ , then, for every  $x \in |S_0|$  and every eigenvalue  $\alpha$  of  $F_x$  on  $(j_*\mathcal{F}_0)_x$ , one has

$$w_{N(x)}(\alpha) \leq \beta.$$

Removing, if necessary, a point of  $X_0$  lying in  $U_0$ , one reduces to the case where  $X_0$  is affine. One then has the equality (1.4.7.2)

$$\prod_{x \in |U_0|} \det(1 - F_x t^{\deg x}, \mathcal{F}_0)^{-1} \prod_{x \in |S_0|} \det(1 - F_x t^{\deg x}, j_*\mathcal{F}_0)^{-1} = \frac{\det(1 - Ft, H_c^1(X, j_*\mathcal{F}))}{\det(1 - Ft, H^0(X, j_*\mathcal{F}))}. \quad (1.8.1.1)$$

In this formula:

- a) By (1.4.6), the first factor on the left converges for  $|t| < q^{-(\beta+2)/2}$  and has neither zero nor pole in this region.
- b) By (1.4.3), the right-hand side has no pole for  $|t| < q^{-(\beta+2)/2}$ .
- c) Comparing a) and b), one finds that the remaining factors

$$\det(1 - F_x t^{\deg x}, j_*\mathcal{F}_0)^{-1} \quad (x \in |S_0|)$$

are without pole for  $|t| < q^{-(\beta+2)/2}$ .

If  $\alpha$  is an eigenvalue of  $F_x$  on  $(j_*\mathcal{F}_0)_x$ , then  $|\alpha| \leq N(x)^{(\beta+2)/2}$ , that is

$$w_{N(x)}(\alpha) \leq \beta + 2. \quad (1.8.1.2)$$

To conclude, one applies (1.8.1.2) to tensor powers of  $\mathcal{F}_0$ . Since  $(j_*\mathcal{F}_0)^{\otimes k}$  is a subsheaf of  $j_*(\mathcal{F}_0^{\otimes k})$ , the eigenvalues  $\alpha^k$  satisfy  $w_{N(x)}(\alpha^k) \leq k\beta + 2$ , hence  $w_{N(x)}(\alpha) \leq \beta + 2/k$ . Letting  $k \rightarrow \infty$  gives the lemma.

**Remark (1.8.2).** Compare the zeros of the two sides of (1.8.1.1). On the left, there are none for  $|t| < q^{-(\beta+2)/2}$ . On the right, the denominator does not vanish in this same region. Therefore the numerator cannot vanish there. Thus for every eigenvalue  $\alpha$  of  $F$  on  $H_c^1(X, j_*\mathcal{F})$ , one has  $w_q(\alpha) \leq \beta + 2$ .

**(1.8.3).** Let  $y \in S$ , with image  $s$  in  $S_0$ , let  $\eta$  be the generic point of the henselian trait with residue field  $X_0(s)$ , and let  $\bar{\eta}$  be a geometric generic point of its strict henselization. The fibre  $\mathcal{F}_{0\bar{\eta}}$  of  $\mathcal{F}_0$  at  $\bar{\eta}$  is a representation of the Weil group  $W(\bar{\eta}/\eta)$ . We write  $M$  for its local monodromy filtration (1.7.2).

**Theorem (1.8.4).** With the notation of (1.8.1) and (1.8.3), if  $\mathcal{F}_0$  is pointwise  $\iota$ -pure of weight  $\beta$ , the representation

$$\mathrm{Gr}_i^M(\mathcal{F}_{0\bar{\eta}})$$

of  $W(\bar{\eta}/\eta)$  is  $\iota$ -pure of weight  $\beta + i$ .

By twisting (1.2.7), we reduce to the case  $\beta = 0$ . Replace  $X_0$  by a finite covering, and  $U_0$ ,  $\mathcal{F}_0$ , and  $s$  by their inverse images. This changes neither the weights, nor  $\mathcal{F}_{0\bar{\eta}}$ , nor the filtration  $M$ , and reduces us to the case where formula (1.7.2.1) for the representation of  $W(\bar{\eta}/\eta)$  on  $\mathcal{F}_{0\bar{\eta}}$  holds on all of  $I$ :

$$\rho(\sigma) = \exp(t_\ell(\sigma)N) \quad (\sigma \in I).$$

Under these hypotheses,

$$\mathrm{Ker} N = (\mathcal{F}_{0\bar{\eta}})^I,$$

and this vector space is the fibre of  $j_*\mathcal{F}_0$  at  $\bar{s}$ . By (1.8.1), if  $\alpha$  is an eigenvalue of  $F$  on  $\mathrm{Ker} N$ , then  $|\iota(\alpha)| \leq 1$ .

Let  $\alpha$  be an eigenvalue of  $F$  on  $P_{-j}(\mathcal{F}_{0\bar{\eta}})$ . By (1.6.14.2), one first obtains  $|\iota(\alpha)| \leq 1$ . Apply this result to  $\mathcal{F}_0 \otimes \mathcal{F}_0$ . By (1.6.14.4),

$$P_{-j}(\mathcal{F}_{0\bar{\eta}}) \otimes P_{-j}(\mathcal{F}_{0\bar{\eta}})(-j)$$

is a direct factor of  $P_0(\mathcal{F}_{0\bar{\eta}} \otimes \mathcal{F}_{0\bar{\eta}})$ ; hence  $\alpha^2 q^j$  is an eigenvalue of  $F$  on that  $P_0$ , and  $|\iota(\alpha^2 q^j)| \leq 1$ , i.e.  $|\iota(\alpha)| \leq q^{-j/2}$ . Applying this last result to the dual sheaf and using (1.6.14.5), one obtains also  $|\iota(\alpha^{-1} q^{-j})| \leq q^{-j/2}$ , i.e.  $|\iota(\alpha)| \geq q^{-j/2}$ . Thus the eigenvalues  $\alpha$  of  $F$  on  $P_{-j}(\mathcal{F}_{0\bar{\eta}})$  have absolute value  $q^{-j/2}$ , and one concludes by (1.6.14.3).

**Corollary (1.8.5).** Assume that  $\mathcal{F}_0$  is a lisse sheaf on  $U_0$  and that it admits a finite increasing filtration  $W$  by lisse subsheaves such that, for every  $i$ ,  $\mathrm{Gr}_i^W(\mathcal{F}_0)$  is pointwise  $\iota$ -pure of weight  $i$ . Write again  $W$  for the induced filtration on  $\mathcal{F}_{0\bar{\eta}}$ . Then  $\mathcal{F}_{0\bar{\eta}}$  is  $\iota$ -mixed with integral weights; the local monodromy filtration of  $V$  relative to  $W$  exists, and it coincides with the weight filtration of  $\mathcal{F}_{0\bar{\eta}}$  (1.7.5).

By (1.7.5) and (1.8.4) applied to  $\mathrm{Gr}^W(\mathcal{F}_0)$ , the characteristic properties (1.6.13) are satisfied by the weight filtration.

**(1.8.6).** Let  $D_0$  be a smooth divisor in a scheme  $X_0$  smooth over  $\mathbb{F}_q$ , let  $U_0$  be the complementary open, and let  $\mathcal{F}_0$  be a lisse sheaf on  $U_0$  tamely ramified along  $D_0$ . Suppose that  $\mathcal{F}_0$  admits a finite increasing filtration  $W$  by lisse subsheaves such that  $\mathrm{Gr}_i^W \mathcal{F}_0$  is pointwise  $\iota$ -pure of weight  $i$ . Suppose that  $D_0$  is defined by an equation  $t$ , and let  $\mathcal{F}_0[D_0]$  be the sheaf on  $D_0$  endowed with an action of  $\mathbb{Z}_\ell(1)$  obtained from (1.7.8). Restricting  $\mathcal{F}_0$  to curves transverse to  $D_0$ , one deduces:

**Corollary (1.8.7).** With the notation of (1.8.6), the local monodromy filtration around  $D$  of  $\mathcal{F}_0[D_0]$ , relative to  $W$ , exists. For this filtration, denoted  $M$ , the sheaf  $\mathrm{Gr}_i^M \mathcal{F}_0[D_0]$  is pointwise  $\iota$ -pure of weight  $i$ .

**Remark (1.8.8).**

- 1) With the notation of (1.8.1), if  $\mathcal{F}_0$  is lisse on  $U_0$ , then  $(j_*\mathcal{F}_0)_{\bar{s}} = (\mathcal{F}_{0\bar{\eta}})^I$ . This group is deduced from  $\mathrm{Ker} N$  by taking invariants by a finite group  $I/I_1$ . Under the hypotheses of (1.8.4),  $\mathrm{Ker} N \subset M_0$ , and the induced filtration  $M$  on  $(j_*\mathcal{F}_0)_{\bar{s}}$  has associated graded  $\mathrm{Gr}_i^M(j_*\mathcal{F}_0)_{\bar{s}}$   $\iota$ -pure of weight  $\beta + i$ , and zero for  $i > 0$ .

- 2) With the notation of (1.8.6), if  $\mathcal{F}_0$  is lisse on  $U_0$  and tamely ramified along  $D_0$ , the restriction of  $j_*\mathcal{F}_0$  to  $D_0$  is the subsheaf of invariants of a finite group in  $\text{Ker } N \subset \mathcal{F}_0[D_0]$ . If  $\mathcal{F}_0$  is pointwise  $\iota$ -pure of weight  $\beta$ , twisting (1.8.7) gives that, for the monodromy filtration  $M$ ,  $\text{Gr}_i^M(j_*\mathcal{F}_0|D_0)$  is pointwise  $\iota$ -pure of weight  $\beta + i$ , and is zero for  $i > 0$ .
- 3) Under the hypotheses of (1.8.6),  $W$  and the monodromy define on  $j_*\mathcal{F}_0|D_0$  a filtration  $M$  for which  $\text{Gr}_i^M(j_*\mathcal{F}_0|D_0)$  is lisse and  $\iota$ -pure of weight  $i$ .

**Corollary (1.8.9).** Let  $X_0$  be a scheme of finite type over  $\mathbb{F}_q$ , let  $j : U_0 \hookrightarrow X_0$  be an open subset, and let  $\mathcal{F}_0$  be an  $\iota$ -mixed sheaf of pointwise weights  $\leq p$  on  $U_0$ . Then  $j_*\mathcal{F}_0$  is again  $\iota$ -mixed of pointwise weights  $\leq p$ . If the pointwise weights of  $\mathcal{F}_0$  are integral, those of  $j_*\mathcal{F}_0$  are integral as well.

We use the following devissages, justified by (1.2.5): devissage in  $\mathcal{F}_0$  by exact sequences; replacement of the support by a locally closed subscheme; normalization by a finite surjective morphism; replacing the open by a dense smaller open to make infinity a divisor; and finally localization at the generic point at infinity. These reductions bring the proof to (1.8.8), 2).

**Corollary (1.8.10).** Let  $\mathcal{F}_0$  be a lisse sheaf on  $X_0$  of finite type over  $\mathbb{F}_q$ , and let  $j : U_0 \hookrightarrow X_0$  be a dense open subset. If the restriction of  $\mathcal{F}_0$  to  $U_0$  is pointwise  $\iota$ -pure, then  $\mathcal{F}_0$  itself is pointwise  $\iota$ -pure.

Let  $\beta$  be the weight of  $\mathcal{F}_0|U_0$ . Since  $\mathcal{F}_0 \rightarrow j_*j^*\mathcal{F}_0$  and likewise for the dual, (1.8.9) shows that  $\mathcal{F}_0$  is  $\iota$ -mixed of weights  $\leq \beta$ , while its dual is  $\iota$ -mixed of weights  $\leq -\beta$ . Thus the weights are also  $\geq \beta$ , and the assertion follows. Equivalently, one can reduce by curves and normalization to the case of a smooth curve and then use (1.8.1) directly.

**Corollary (1.8.11).** Every lisse  $\iota$ -mixed sheaf  $\mathcal{F}_0$  on a normal scheme  $X_0$  of finite type over  $\mathbb{F}_q$  is a successive extension of lisse pointwise  $\iota$ -pure sheaves.

One may suppose  $X_0$  connected and reduce to the case where the sheaf is irreducible. Since  $X_0$  is normal, its restriction to any nonempty open subset is still irreducible; for a suitable open subset it is pointwise  $\iota$ -pure by the definition of mixed sheaves, and (1.8.10) applies. Later, in (3.4.1), the normality hypothesis will be removed.

**Corollary (1.8.12).** Let  $\mathcal{F}_0$  be a lisse  $\iota$ -mixed sheaf on a connected scheme  $X_0$  of finite type over  $\mathbb{F}_q$ . If  $\mathcal{F}_0$  is  $\iota$ -pure at one point of  $X_0$ , then it is pointwise  $\iota$ -pure.

For  $X_0$  normal this follows from (1.8.11): the weights of  $\mathcal{F}_0$  are read from its fibre at any point. The general case is obtained by applying this to the normalization and then descending the conclusion.

**(1.8.13)–(1.8.14).** The preceding discussion has the expected analogues for variations in Hodge theory. A variation of mixed Hodge structures on a smooth analytic space  $S$  consists of a local system  $V_{\mathbb{Z}}$  of free finitely generated  $\mathbb{Z}$ -modules, an increasing finite filtration  $W$  of  $V_{\mathbb{Q}}$ , and a decreasing finite filtration  $F$  of  $V_{\mathbb{C}}$  by vector subbundles, holomorphically varying and satisfying Griffiths transversality, such that  $W$  and  $F$  define at each point of  $S$  a mixed Hodge structure. The variations occurring in algebraic geometry satisfy additional properties, for instance the polarizability of the variations  $\text{Gr}^W(V)$ .

Let  $(V, W, F)$  be such a variation on the punctured disk  $D^*$ , let  $t \in D^*$ , let  $V$  be the fibre at  $t$ , and let  $T$  be the monodromy. Suppose for simplicity that  $T$  is unipotent, and put  $N = -\log T$ .

**Problem (1.8.15).** Isolate a class of good variations of mixed Hodge structures on  $D^*$  such that, for a good unipotent variation  $(V, W, F)$ , there exists on  $V$  a finite increasing filtration  $M$

with  $NM_a \subset M_{a-2}$  and such that  $N^b$  induces isomorphisms

$$N^b : \mathrm{Gr}_{a+b}^M \mathrm{Gr}_a^W V \xrightarrow{\sim} \mathrm{Gr}_{a-b}^M \mathrm{Gr}_a^W V.$$

One would also like:

- a) the variation  $V$  to be asymptotic, in a suitable sense, to a nilpotent orbit: a good variation for which  $F$  at  $t \exp(u)$  is the transform by  $\exp(-uN)$  of the filtration deduced from  $F_t$  by horizontal transport from  $t$  to  $t \exp(u)$ ;
- b) for a nilpotent orbit,  $(V, M, F)$  to be again a variation of mixed Hodge structures.

### (1.9) Tame local monodromy of mixed sheaves

This subsection, which will not be used later, generalizes (1.8.7) to the case of normal-crossing divisors.

**(1.9.1).** Let  $X$  be a smooth scheme over  $\mathbb{F}_q$ , let  $D$  be a normal-crossing divisor on  $X$ , the transverse union of smooth divisors  $D_i$  ( $i \in I$ ), and let  $E = \bigcap_{i \in I} D_i$ . We assume that each  $D_i$  is globally defined by an equation  $t_i = 0$ , in order to apply the construction of (1.7.8).

Let  $\mathcal{F}$  be a lisse sheaf on  $X - D$ , tamely ramified along  $D$ . Suppose that it has a finite increasing filtration  $W$  by lisse subsheaves such that  $\mathrm{Gr}_i^W \mathcal{F}$  is pointwise  $\iota$ -pure of weight  $i$ . Let  $L$  be the set of primes distinct from  $p$ , and define  $\mathcal{F}[E]$  as in (1.7.8). It is a lisse sheaf on  $E$ , equipped with an action of  $\mathbb{Z}_L(1)^I$ ; this action is quasi-unipotent, hence gives nilpotent endomorphisms

$$N_i : \mathcal{F}[E](1) \longrightarrow \mathcal{F}[E].$$

Given a sheaf  $N$ , two finite increasing filtrations  $W'$  and  $W''$  of  $N$ , and a finite family of endomorphisms

$$N_\alpha : N(1) \longrightarrow N \quad (\alpha \in A),$$

we shall consider the condition

$$L(W', W'', (N_\alpha)_{\alpha \in A}) :$$

the  $N_\alpha$  stabilize  $W'$ , satisfy  $N_\alpha W'_a(1) \subset W'_{a-2}$ , and, for all rational numbers  $c_\alpha > 0$ , the endomorphism

$$N = \sum_{\alpha \in A} c_\alpha N_\alpha$$

induces isomorphisms

$$N^b : (\mathrm{Gr}_{a+b}^{W''} \mathrm{Gr}_a^{W'} N)(b) \xrightarrow{\sim} \mathrm{Gr}_{a-b}^{W''} \mathrm{Gr}_a^{W'} N.$$

By (1.6.13), if  $W'$  and the family  $N_\alpha$  are given, there is at most one filtration  $W''$  satisfying this condition. In its statement one may replace “positive rational number” by “positive integer,” since one can replace the  $c_\alpha$  by  $nc_\alpha$  and  $N$  by  $nN$ .

**Theorem (1.9.2).** With the hypotheses and notation of (1.9.1), there is on  $\mathcal{F}[E]$  a unique system of finite increasing filtrations  $W(J)$ ,  $J \subset I$ , by lisse subsheaves stable under  $\mathbb{Z}_L(1)^I$ , hence under the  $N_i$ , such that:

- a)  $W(\emptyset)$  is the filtration induced from  $W$  by functoriality;
- b) for  $J \subset K$ , the condition

$$L(W(J), W(K), (N_k)_{k \in K-J})$$

is satisfied.

Moreover,  $\mathrm{Gr}_a^{W(I)} \mathcal{F}[E]$  is pointwise  $\iota$ -pure of weight  $a$ .

The uniqueness of the  $W(J)$  follows at once from (1.6.13); it is enough to have the conditions for  $\emptyset \subset K$ . For existence, one first treats the following special case.

**Lemma (1.9.3).** Let  $(c_k)_{k \in I}$  be a system of positive integers. The sheaf  $\mathcal{F}[E]$  admits a unique finite increasing filtration  $W(I)$  by lisse subsheaves such that

$$N = \sum_{k \in I} c_k N_k$$

satisfies the condition  $L(W, W(I), N)$ . Moreover  $\mathrm{Gr}_a^{W(I)} \mathcal{F}[E]$  is pointwise  $\iota$ -pure of weight  $a$ .

By uniqueness, it is enough to prove the existence of the required filtration at one geometric point of each connected component of  $E$ . After a finite extension of the base field, one may suppose the point rational. Complete the equations  $t_i$  to a system of local coordinates  $(t_i, u_j)$  centered at that point. Restrict  $\mathcal{F}$  to the curve defined parametrically by  $u_j = 0$ ,  $t_i = t^{c_i}$ . On the strict localization of this curve, the geometric generic fibre identifies with  $\mathcal{F}[E]$ , and the logarithm of monodromy is precisely  $N = \sum c_i N_i$ . The assertion follows from (1.8.5).

**(1.9.4).** We deduce the theorem from this special case. The filtration  $W(I)$  of (1.9.3) is independent of the choice of the  $c_k$ , since it is characterized independently of them as a weight filtration. For  $J \subset I$ , apply (1.9.3) to the corresponding partial intersection. This gives a weight filtration  $W(J)$ . Reapplying the construction to the residual divisor components gives, for  $J \subset K$ , the condition

$$L(W(J), W(K), (N_k)_{k \in K-J}).$$

The compatibilities with the identifications of (1.7.9) then give the required system on  $E = D_I$ .

**Variant (1.9.5).** Let  $S$  be a scheme of finite type over  $\mathbb{Z}$ ,  $f : X \rightarrow S$  a smooth morphism,  $D \subset X$  a relative normal-crossing divisor, and  $\mathcal{F}$  a lisse sheaf on  $X - D$ , tamely ramified along  $D$ . A variant of (1.9.2) holds in this setting; it is verified fibre by fibre by reduction to (1.9.2). The case of a normal-crossing divisor in a regular scheme of finite type over  $\mathbb{Z}$  appears inaccessible, already for  $S = \mathrm{Spec}(\mathbb{Z}[1/\ell])$ .

**Remark (1.9.6).** As explained in [2], one expects the theory of variations of Hodge structures, always assumed polarizable, often to run parallel to that of lisse pointwise pure sheaves. Cattani and Kaplan have proved an analogue of (1.9.2), restricted to the pure case, for variations of Hodge structures on  $(D^*)^I$ . In their context, the coefficients  $c_\alpha$  in the condition  $L(W', W'', (N_\alpha))$  may even be taken to be real  $> 0$ , not merely rational  $> 0$ .

## (1.10) Non-archimedean absolute values

The conventions of (0.7) are in force. Let  $X_0$  be a smooth absolutely irreducible curve over  $\mathbb{F}_q$ , let  $j : U_0 \hookrightarrow X_0$  be an open subset whose complement is the finite set  $S_0$ , and let  $\mathcal{F}_0$  be a lisse sheaf on  $U_0$ . In (1.8) the trace formula and local monodromy were used to study the archimedean absolute values of the Frobenius eigenvalues at the points of  $S_0$ . Here we study absolute values which need not be archimedean.

Choose either an isomorphism  $\iota : \overline{\mathbb{Q}}_\ell \simeq \mathbb{C}$ , or a prime  $\ell'$ , an algebraic closure  $\overline{\mathbb{Q}}_{\ell'}$ , and an isomorphism  $\overline{\mathbb{Q}}_\ell \simeq \overline{\mathbb{Q}}_{\ell'}$ . In the latter case the natural absolute value is normalized by  $|\ell'| = 1/\ell'$ .

**Lemma (1.10.1).** Let  $b > 0$ . If, for every  $x \in |U_0|$  and every eigenvalue  $\alpha$  of  $F_x$  on  $\mathcal{F}_0$ , one has

$$|\alpha| \leq b^{\deg x},$$

then, for every  $x \in |S_0|$  and every eigenvalue  $\alpha$  of  $F_x$  on  $j_* \mathcal{F}_0$ , one still has

$$|\alpha| \leq b^{\deg x}.$$

One repeats the proof of (1.8.1), treating first the non-archimedean case and indicating the archimedean modifications. In the trace formula, the first factor converges for  $|t| < 1/b$ , while the right hand side has no pole in this disk. Hence the remaining factors

$$\det(1 - F_x t^{\deg x}, j_* \mathcal{F}_0) \quad (x \in |S_0|)$$

have no pole there, which proves the lemma.

**(1.10.2).** Let  $s \in |S_0|$ , and let  $\bar{s}, \eta, \bar{\eta}$  be as in (1.8.3). We call “eigenvalues of  $F_s$  on  $\mathcal{F}_0$ ” the eigenvalues of a lift of  $F_s \in W(\bar{s}/s)$  to  $W(\bar{\eta}/\eta)$  acting on the fibre. This terminology is used only when the choice of the lift is immaterial.

**Theorem (1.10.3).** In the non-archimedean case, let  $0 < b < c < \infty$ . If, for every point  $x \in |U_0|$  and every eigenvalue  $\alpha$  of  $F_x$  on  $\mathcal{F}_0$ ,

$$b^{\deg x} \leq |\alpha| \leq c^{\deg x},$$

then the same inequalities hold for every  $s \in |S_0|$  and every eigenvalue of  $F_s$  on  $\mathcal{F}_0$ .

After replacing  $U_0$  by a finite cover, local monodromy may be assumed unipotent. The action of inertia is then written

$$\sigma \longmapsto \exp(t_\ell(\sigma)N).$$

Lemma (1.10.1) gives the upper bound on the eigenvalues on  $\text{Ker } N$ , and (1.6.14.3) propagates it to the whole representation. The lower bound is obtained by applying the upper bound to the dual sheaf.

**(1.10.4)–(1.10.8).** Applying (1.10.3) to exterior powers gives relations between Newton polygons attached to the Frobenius endomorphisms at points of  $U_0$  and those at the points of  $S_0$ . We record only the form needed later. For  $\ell' \neq p$ , Frobenius eigenvalues are in practice  $\ell'$ -adic units, and one obtains:

**Corollary (1.10.5).** If, for every  $x \in |U_0|$ , the eigenvalues  $\alpha$  of  $F_x$  on  $\mathcal{F}_0$  are  $\ell'$ -adic units, then for every  $s \in |S_0|$  the eigenvalues of  $F_s$  on  $\mathcal{F}_0$  are still  $\ell'$ -adic units.

For  $\ell' = p$  it is useful to use valuations. If  $N$  is a power of  $p$ , let  $v_N$  be the valuation of  $\mathbb{Q}_p$  normalized by  $v_N(N) = 1$ . The preceding argument, applied to exterior powers and determinants, gives corresponding inequalities for the valuations  $v_N(\alpha)$ .

**Corollary (1.10.9).** Suppose that for almost all  $x \in |U_0|$  the eigenvalues  $\alpha$  of  $F_x$  on  $\mathcal{F}_0$  satisfy

$$\gamma \leq v_N(\alpha) \leq \delta.$$

If  $n$  is the integer part of  $\delta - \gamma$ , then the logarithm  $N$  of the unipotent part of the local monodromy at a point  $s \in |S_0|$  satisfies

$$N^{n+1} = 0.$$

Indeed, if  $m$  is minimal with  $N^{m+1} = 0$ , then  $F_s$  has eigenvalues  $\alpha$  and  $q^m \alpha$ , and one applies (1.10.3).

### (1.11) Specialization of monodromy

**Proposition (1.11.1).** Let  $S$  be an irreducible scheme, let  $f : X \rightarrow S$  be a smooth morphism, purely of relative dimension 1, with geometrically connected fibres, let  $g$  be a section of  $f$ , and let  $\mathcal{F}$  be a lisse  $\mathbb{Z}_\ell$ -sheaf on  $X$ . For each geometric point  $\bar{s}$  of  $S$ , let  $M_{\bar{s}}$  be the image of

$$\pi_1(X_{\bar{s}}, g(\bar{s}))$$

in

$$\mathrm{Aut}((\mathcal{F}/\ell^n \mathcal{F})_{g(\bar{s})}).$$

Then there is an open subset  $U \subset S$  such that, for every  $n$  and every geometric point  $\bar{s}$  of  $U$ ,  $M_{\bar{s}}$  is the fibre at  $\bar{s}$  of a lisse subsheaf  $M^n$  of

$$\mathrm{Aut}(g^* \mathcal{F}/\ell^n \mathcal{F})$$

on  $U$ .

One considers the finite étale cover

$$I_n = \mathrm{Isom}(\mathcal{F}/\ell^n \mathcal{F}, f^* g^* \mathcal{F}/\ell^n \mathcal{F})$$

of  $X$ , with the identity section above  $g$ . After shrinking  $S$ , one constructs simultaneously the subsheaves  $M^n$  and a tower of finite étale covers. The construction is compatible in  $n$ , and specialization of the fundamental group gives the desired local constancy.

**Remarks (1.11.2)–(1.11.5).** For the groups  $\pi_1(X_{\bar{s}}, g(\bar{s}))$  to form a local system on  $S$ , it is enough that  $X$  admit a relative compactification

$$X \hookrightarrow \overline{X}$$

where  $\overline{X}$  is proper and smooth over  $S$ , of relative dimension 1, and  $X$  is the complement of a finite étale subscheme  $T$  of  $S$ ; it is enough for such a compactification to exist after a finite radicial surjective base change.

With these notations, if  $X_n \rightarrow X$  is one of the finite étale covers constructed above, one also has compatible compactifications

$$X_n \hookrightarrow \overline{X}_n$$

whose complements  $T_n$  are finite over  $S$ . For a geometric point  $\bar{s}$ , a point  $t \in T_{\bar{s}}$ , and inertia  $I_t$ , the image of  $I_t$  in  $M_{\bar{s}}$  specializes compatibly when  $(\bar{s}, t)$  specializes. This follows after reducing to the strictly local case and using tame coverings of the compactified curves.

The proposition remains valid for any morphism of finite type  $f$ : localization on  $S$  and Mayer–Vietoris reduce to the case where  $f$  is flat with normal fibres; restriction to an open then reduces to the smooth quasi-projective case.

## 2. The Hadamard–de la Vallée-Poussin method

### (2.1) The method

**(2.1.1).** Let  $\Gamma$  be a group isomorphic to  $\mathbb{Z}$  or to  $\mathbb{R}$ , let  $\omega_1 : \Gamma \rightarrow \mathbb{R}_{>0}$  be a non-trivial quasi-character, let  $G$  be a locally compact group extension of  $\Gamma$  by a compact group  $G^0$ , let  $\Sigma$  be an infinite countable set, and let  $(x_v)_{v \in \Sigma}$  be a family of conjugacy classes in  $G$ . We impose hypotheses (A) and (B).

(A1) If  $\Gamma \simeq \mathbb{R}$ , the extension  $G$  is trivial. This hypothesis is in fact automatic.

(A2) If  $\Gamma \simeq \mathbb{Z}$ , the centre of  $G$  maps to a subgroup of finite index in  $\Gamma$ .

For  $s \in \mathbb{C}$ , put  $\omega_s = \omega_1^s$ . If  $\Gamma \simeq \mathbb{Z}$ , we write  $q > 1$  and  $\deg : \Gamma \simeq \mathbb{Z}$  so that

$$\omega_1(\gamma) = q^{-\deg(\gamma)}.$$

Let

$$N_v = \omega_{-1}(x_v).$$

The set of connected components of  $G$  is discrete, and we give  $G$  the corresponding Riemann-surface structure. The hypotheses (B) are:

$$N_v > 1$$

and

$$\prod_{v \in \Sigma} (1 - N_v^{-s})^{-1}$$

converges absolutely for  $\Re(s) > 1$ . For  $\Gamma \simeq \mathbb{Z}$ , this is the same as requiring  $\deg(v) > 0$  and

$$\#\{v \mid \deg(v) = n\} = O(q^{n(1+\epsilon)})$$

for every  $\epsilon > 0$ .

For a finite-dimensional continuous complex representation  $v$  of  $G$ , the product

$$L(v) = \prod_{v \in \Sigma} \det(1 - v(x_v))^{-1}$$

converges absolutely in the region  $\Re(v) > 1$ , and one writes  $L(v, s) = L(v\omega_s)$ .

**Examples (2.1.2)–(2.1.3).** If  $\Gamma \simeq \mathbb{R}$ , the situation is described by a compact group  $G^0$ , conjugacy classes  $x_v^0 \subset G^0$ , and real numbers  $N_v > 1$ ; this is the framework considered by Serre. If  $X_0$  is a smooth absolutely irreducible curve over  $\mathbb{F}_q$ , with function field  $K$ , one takes the Weil group extension

$$1 \longrightarrow \pi_1(X, \bar{x}) \longrightarrow W(X_0, \bar{x}) \longrightarrow \mathbb{Z} \longrightarrow 0,$$

sets  $\Sigma = |X_0|$ , and takes  $x_v$  to be geometric Frobenius.

**(2.1.4).** For  $\Re(s) > 1$ , the logarithmic derivative is

$$-\frac{L'}{L}(v, s) = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-ns} \operatorname{Tr}(v(x_v^n)).$$

Equivalently, for a unitary virtual representation  $v$  and real  $\sigma > 1$ ,

$$-\frac{L'}{L}(\omega_\sigma v) = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-n\sigma} \operatorname{Tr} v(x_v^n).$$

If  $\mu$  is a measure on  $G$ , or on the set of conjugacy classes of  $G$ , its Fourier transform is

$$\hat{\mu}(v) = \int \operatorname{Tr} v(g) d\mu.$$

For positive  $\mu$  of finite total mass one has

$$\hat{\mu}(v \otimes \bar{v}) \geq 0.$$

The last displayed logarithmic derivative is the Fourier transform of the positive measure

$$\mu_\sigma = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-n\sigma} \delta_{x_v^n}.$$



Therefore, for every unitary virtual representation  $v$ ,

$$-\frac{L'}{L}(\omega_\sigma(v \otimes \bar{v})) \geq 0 \quad (\sigma > 1).$$

Let  $r(v)$  denote the order of the pole, the negative of the order of zero, of  $L(v)$  at  $\omega_1$ . If  $r$  extends additively to the Grothendieck group of unitary representations and satisfies the positivity inequalities obtained above, then the Hadamard–de la Vallée-Poussin argument implies that  $r(v) = 0$  for  $v \neq 1$ , except possibly for a one-dimensional character of order 2.

**Variant and proof (2.1.5)–(2.1.8).** It is enough to prove the assertion after replacing  $G$  by the compact group obtained from its unitary representations. In the compact case, let  $[p : \tau]$  denote the multiplicity of an irreducible representation  $\tau$  in  $p$ . For any finite set  $T$  of irreducible representations and any  $\varepsilon > 0$ , there is a representation  $p$  such that

$$\dim \tau [p \otimes \bar{p} : \tau] < (1 + \varepsilon)[p \otimes \bar{p} : 1] \quad (\tau \in T).$$

This follows from density of the algebra of central characters in the algebra of continuous central functions. Using the positivity of  $r(p \otimes \bar{p})$ , letting  $\varepsilon \rightarrow 0$  and increasing  $T$ , one obtains the desired vanishing.

**(2.1.9)–(2.1.12).** For the Riemann zeta function,  $G^0 = 1$ ,  $\Sigma$  is the set of primes, and  $N_p = p$ . The function

$$-\frac{\zeta'}{\zeta}(\sigma + it)$$

is the Fourier transform of the positive measure

$$\sum_{p,n} \log p p^{-n\sigma} \delta_{n \log p},$$

and hence is positive definite. The classical inequality

$$3A_\sigma(0) + 4\Re A_\sigma(t) + 2\Re A_\sigma(2t) \geq 0$$

is the form used by Mertens to derive the theorem of Hadamard and de la Vallée-Poussin that  $\zeta(s)$  has no zero on  $\Re(s) = 1$ .

Assume now that  $\Gamma \simeq \mathbb{Z}$ , that  $L(1)$  is meromorphic for  $\Re(s) > 1$  with a simple pole at  $\omega_1$ , no other pole, and no zero in that domain. For a central element  $z \in G$  of positive degree  $d$ , translate the pieces  $G_{i+nd}$  to  $G_i$ . The measures

$$\mu = \sum_{v,n>0} \deg(v) q^{-n \deg(v)} \delta_{x_v^n}$$

and the Haar comparison measure have Fourier–Laplace transforms whose difference is holomorphic on  $\Re(s) \geq 0$ . Consequently the translated restrictions of  $\mu$  converge vaguely to the corresponding translated Haar measure. This gives the equidistribution theorem (2.1.12).

## (2.2) Purity and compactness

**(2.2.1)–(2.2.4).** Let  $G_{\mathbb{C}}$  be an algebraic group extension

$$1 \longrightarrow G_{\mathbb{C}}^0 \longrightarrow G_{\mathbb{C}} \longrightarrow \mathbb{Z} \longrightarrow 0$$

with  $G_{\mathbb{C}}^0$  reductive, and choose a real form  $G_{\mathbb{R}}$  as in (2.2.1). The exact sequence

$$1 \longrightarrow G_{\mathbb{R}}^0 \longrightarrow G_{\mathbb{R}} \longrightarrow \mathbb{Z} \longrightarrow 0$$

has compact connected part. Two elements of  $G_{\mathbb{R}}$  conjugate in  $G_{\mathbb{C}}$  are already conjugate in  $G_{\mathbb{R}}$ . An element  $g \in G_{\mathbb{C}}$  has a unique Jordan decomposition

$$g = g_s g_u = g_u g_s,$$

with  $g_s$  semisimple and  $g_u$  unipotent.

Let  $X_0$  be a smooth connected scheme over  $\mathbb{F}_q$ , let  $\mathcal{F}_0$  be a lisse  $\mathbb{Q}_{\ell}$ -sheaf, and let  $G$  be the Zariski closure of the image of  $W(X_0, \bar{x})$  in  $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ , with the grading by degree. Assume that  $G^0$  is reductive and that some representation with finite kernel on  $G^0$  gives an  $\ell$ -mixed sheaf. Then every linear representation of  $G$  gives an  $\ell$ -mixed lisse sheaf.

**Example (2.2.5).** If  $\mathcal{F}_0$  is lisse and  $\ell$ -mixed on  $X_0$ , and the restriction of the corresponding representation to  $\pi_1(X, \bar{x})$  is semisimple, then the preceding hypotheses hold.

**Proposition (2.2.6).** Let  $v \in |X_0|$ , and let  $F_v$  be the image of a geometric Frobenius in  $G$ . The semisimple part  $(F_v)_s$  is conjugate to an element of  $G_{\mathbb{R}}$ , uniquely up to  $G_{\mathbb{R}}$ -conjugacy.

The uniqueness follows from the real-conjugacy lemma. The existence follows by applying the weight bounds to a faithful representation and using the compatibility of semisimple parts with representations.

**(2.2.7)–(2.2.8).** For  $X_0$  a smooth connected scheme over  $\mathbb{F}_q$ , set

$$\Sigma = |X_0|, \quad x_v = (F_v)_s \in G_{\mathbb{R}}/\text{conjugacy}, \quad N_v = q^{\deg v}.$$

These data satisfy the hypotheses (A) and (B') of (2.1). If  $\dim X_0 = 1$ , they satisfy (B) as well. If  $\dim X_0 = N$ , the same statements hold after replacing the character by  $q^{-N \deg}$ ; in what follows the character  $q^{-\deg}$  is retained and the statements of (2.1) are adjusted accordingly.

The functors extension of scalars by  $\iota$  and restriction to  $G_{\mathbb{R}}$  give equivalences between linear representations of  $G$ , linear representations of  $G_{\mathbb{C}}$ , and continuous complex representations of  $G_{\mathbb{R}}$ . For a representation  $\tau$  of  $G_{\mathbb{R}}$ , let  $\mathcal{F}_0(\tau_{\ell})$  be the lisse sheaf obtained from the corresponding  $\ell$ -adic representation. If  $\tau$  is irreducible, this sheaf is pointwise pure of weight  $2N(\tau)$ . Moreover the  $L$ -function  $L(\tau, s)$  is expressed by the formal zeta function

$$L(\tau, s) = Z(\mathcal{F}_0(\tau_{\ell}), t), \quad t = q^{-s}.$$

**Proposition (2.2.9).** If  $\dim X_0 = 1$ , the function  $L(\tau)$  satisfies the non-vanishing condition (C).

Grothendieck's formula gives

$$Z(\mathcal{F}_0(\tau_{\ell}), t) = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}(\tau_{\ell})))}{\det(1 - Ft, H_c^0(X, \mathcal{F}(\tau_{\ell}))) \det(1 - Ft, H_c^2(X, \mathcal{F}(\tau_{\ell})))}.$$

If  $\tau$  is unitary and is not a power of the quasi-character, then the restriction of  $\tau$  to  $G^0$  is non-trivial and the denominator is 1. The assertion follows from (2.2.8). For the trivial character, the assertion is the classical assertion for the zeta function of  $X_0$ . A possible zero at a quadratic twist is excluded by passing to the corresponding double cover.

**Corollary (2.2.10).** Let  $X_0$  be a smooth curve over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a lisse pointwise  $\iota$ -pure sheaf of weight  $\beta$  on  $X_0$ . If  $\alpha$  is an eigenvalue of  $F$  on  $H_c^1(X, \mathcal{F})$ , then

$$w_q(\alpha) < \beta + 2.$$

This improves the trivial estimate coming from (1.8.2), and will be improved further in (3.2).

The assertion is invariant under finite extension of the base field. One reduces to the absolutely irreducible case, and then by exact sequences to the case where  $\mathcal{F}_0$  is irreducible. The hypotheses of (2.2.5) are then satisfied. After twisting, one may suppose  $\beta = 0$ . The corresponding representation is unitary on  $G_{\mathbb{R}}$ , and (2.2.9), together with the cohomological expression of  $L$ , yields the stated strict inequality.

### 3. The fundamental theorem

#### (3.1) A calculation of vanishing cycles

The results of this paragraph will be used to compute, modulo integers, the weights of certain groups of vanishing cycles.

**(3.1.1).** Let  $S$  be a smooth projective surface over an algebraically closed field  $k$ , let  $D$  be a normal-crossings divisor on  $S$ , let  $V = S - D$ , let  $j : V \hookrightarrow S$  be the inclusion, and let  $\mathcal{F}$  be a sheaf on  $V$ , tamely ramified along  $D$ . We shall study the cohomology groups  $H_c^*(V, \mathcal{F})$  by the method of Lefschetz pencils.

As in I.5, embed  $S$  in a projective space  $P$  and sweep it out by a pencil of hyperplanes  $(H_t)_{t \in A^*}$ . The pencil has axis  $A$ , and  $\tilde{S}$  denotes the space of pairs  $(x, t)$  with  $x \in H_t$ . Let  $\tilde{V}$  be the inverse image of  $V$  in  $\tilde{S}$ . The two projections give a diagram

$$\begin{array}{ccc} \tilde{V} & \xrightarrow{\tilde{j}} & \tilde{S} \xrightarrow{\pi} S \\ f \downarrow & & \downarrow f \\ A^* & & A^* \end{array}$$

(the repeated letter  $f$  should cause no confusion). The fibers of  $f : \tilde{S} \rightarrow A^*$  are the hyperplane sections  $S_t$ .

We impose the following general-position hypotheses.

- (A) The axis  $A$  is transverse to  $S$  and disjoint from  $D$ ; hence  $\tilde{S}$  is smooth, obtained from  $S$  by blowing up the finite set  $S \cap A$ , and none of these points lies on  $D$ .
- (B) The only singularities of  $f$  are ordinary quadratic points, and none of them lies on  $D$ .
- (C) On the normalization  $D'$  of  $D$ , the only singularities of  $f$  are ordinary quadratic points; none of them lies above a singular point of  $D$ .
- (D) Each exceptional fiber contains only one exceptional point.

An exceptional point is one of the following: a critical point of  $f$  on  $S$ , a critical point of  $f$  on  $D'$ , or a singular point of  $D$ . Thus exceptional fibers are of one of three types: a curve with an ordinary double point, a tangency of  $D$  with  $S_t$ , or two branches of  $D$  meeting on  $S_t$ .

There is a canonical isomorphism

$$H_c^*(\tilde{V}, \pi^* \mathcal{F}) \simeq H_c^*(V, \mathcal{F}) \oplus H^*(S \cap A, \mathcal{F})(-1)[-2]. \quad (3.1.1.3)$$

For what follows it would suffice to know the injectivity

$$\pi^* : H_c^*(V, \mathcal{F}) \hookrightarrow H_c^*(\tilde{V}, \pi^* \mathcal{F}), \quad (3.1.1.4)$$

whose Poincare-dual transpose has a retraction. We use the Leray spectral sequence

$$E_2^{pq} = H^p(A^*, R^q f_! \pi^* \mathcal{F}) \Rightarrow H_c^{p+q}(\tilde{V}, \pi^* \mathcal{F}). \quad (3.1.1.5)$$

The sheaves  $R^q f_! \pi^* \mathcal{F} = R^q f_!(\tilde{j}_! \pi^* \mathcal{F})$  are lisse except at the exceptional values of  $f$ .

Let  $t$  be an exceptional value, let  $A_{(t)}^*$  be the henselization of  $A^*$  at  $t$ , and let  $\bar{\eta}$  be a geometric generic point. Applying the theory of vanishing cycles to  $\tilde{j}_! \pi^* \mathcal{F}$  over  $A_{(t)}^*$ , the sheaves of vanishing cycles  $\Phi^q$  are concentrated at the exceptional point  $x$  of  $S_t$ , and one has a long exact sequence

$$\cdots \rightarrow (R^q f_! \pi^* \mathcal{F})_t \rightarrow (R^q f_! \pi^* \mathcal{F})_{\bar{\eta}} \rightarrow \Phi^q \rightarrow (R^{q+1} f_! \pi^* \mathcal{F})_t \rightarrow \cdots \quad (3.1.1.6)$$

**(3.1.2).** Assume in addition that the local monodromy of  $\mathcal{F}$  around  $D$  is unipotent. If  $d \in D$ , let  $S_{(d)}$  be the strict localization of  $S$  at  $d$ , and  $V_{(d)}$  the inverse image of  $V$ . The hypothesis gives a finite filtration  $F$  of  $\mathcal{F}|_{V_{(d)}}$  by lisse subsheaves whose graded pieces extend as constant sheaves over  $S_{(d)}$ . We denote by  $\mathrm{Gr}_i^F(\mathcal{F})_d$  the fiber at  $d$  of this constant prolongation.

For a two-element set  $B$ , write  $\mathbb{Z}^B/\mathbb{Z}$  for the quotient by the diagonal copy of  $\mathbb{Z}$ ; equivalently it is the free rank-one group endowed with two opposite identifications with  $\mathbb{Z}$ , indexed by  $B$ .

**(3.1.3) Case (a).** Since  $\tilde{j}_! \pi^* \mathcal{F}$  is lisse at the ordinary double point  $x$ , the vanishing cycles are those of the constant sheaf, tensored by the fiber of  $\mathcal{F}$ . Picard-Lefschetz theory in relative dimension one gives

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.3.1)$$

and, if  $B$  is the two-element set of branches of  $S_t$  at  $x$ ,

$$\Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \mathcal{F}_x(-1) \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.3.2)$$

It may be described by the Kummer covering  $T^\ell = uv^{-1}$  when  $u$  and  $v$  are local equations of the two branches.

**(3.1.4) Case (b).** First suppose  $\mathcal{F} = \mathbb{Q}_\ell$ . From the exact sequence

$$0 \rightarrow j_! \mathbb{Q}_\ell \rightarrow \mathbb{Q}_\ell \rightarrow i_* \mathbb{Q}_{\ell D} \rightarrow 0 \quad (3.1.4.1)$$

and the vanishing-cycle long exact sequence one obtains

$$\Phi^q(j_! \mathbb{Q}_\ell) \simeq \Phi^{q-1}(\mathbb{Q}_{\ell D}). \quad (3.1.4.2)$$

If  $B$  is the set of two points of the henselized normalization  $D'_{(x)}$  above the generic point, then

$$\Phi^q(j_! \mathbb{Q}_\ell) = 0 \quad (q \neq 1), \quad (3.1.4.3)$$

and the non-zero group is the corresponding one-dimensional branch quotient. In general, by devissage with the filtration  $F$  of (3.1.2),

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.4.5)$$

and

$$\mathrm{Gr}_i^F \Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \mathrm{Gr}_i^F(\mathcal{F})_x \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.4.6)$$

**(3.1.5) Case (c).** Let  $B$  be the set of the two branches of  $D$  through  $x$ . With  $i : D' \rightarrow S$  the normalization, locally near  $x$  one has

$$0 \rightarrow j_! \mathbb{Q}_\ell \rightarrow \mathbb{Q}_\ell \rightarrow i_* \mathbb{Q}_\ell \rightarrow \mathbb{Q}_{\ell x} \otimes \mathbb{Q}_\ell^B/\mathbb{Q}_\ell \rightarrow 0. \quad (3.1.5.1)$$

Since  $f$  and  $fi$  are smooth at the relevant points, the only contribution to vanishing cycles is again in degree one. Thus

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.5.3)$$

and, for  $F$  as in (3.1.2),

$$\mathrm{Gr}_i^F \Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \mathrm{Gr}_i^F(\mathcal{F})_x \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.5.4)$$

### (3.2) Dimension one

The conventions of (0.7) remain in force.

**Proposition 1** (3.2.1). *Let  $X_0$  be a smooth absolutely irreducible curve over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a lisse sheaf pointwise  $\iota$ -pure and  $\iota$ -real on  $X_0$ . Then the polynomials*

$$\iota \det(1 - Ft, H_c^1(X, \mathcal{F}))$$

*have real coefficients.*

In Grothendieck's formula, the local factors on the left are real polynomials by hypothesis and the rational function on the right is real. If  $\beta$  is the weight of  $\mathcal{F}_0$ , the zeros of the factor attached to  $H_c^2$  are precisely the poles of absolute value  $q^{-(\beta+2)/2}$ , hence are stable under complex conjugation. Poincare duality deals with the proper case and the denominator in the cohomological expression is real; therefore the numerator is real.

*Remark 1* (3.2.2). The arguments of (0.5) remove the absolute irreducibility hypothesis.

**Theorem 3** (3.2.3). *Let  $X_0$  be a smooth projective curve over  $\mathbb{F}_q$ , let  $j : U_0 \hookrightarrow X_0$  be a dense open subset, and let  $\mathcal{F}_0$  be a lisse sheaf on  $U_0$  pointwise  $\iota$ -pure of weight  $\beta$ . Then the eigenvalues  $\alpha$  of  $F$  on  $H^i(X, j_*\mathcal{F})$  satisfy*

$$w_q(\alpha) = \beta + i.$$

For  $i = 1$ ,  $H^1(X, j_*\mathcal{F})$  is the image of  $H_c^1(U, \mathcal{F})$  in  $H^1(U, \mathcal{F})$ . The proof starts from the following induction statement.

**Lemma 1** (3.2.4(k)). *Let  $U_0$  be a smooth curve over  $\mathbb{F}_q$  and let  $\mathcal{F}_0$  be lisse, pointwise  $\iota$ -pure of weight zero on  $U_0$ . Then the eigenvalues  $\alpha$  of  $F$  on  $H_c^1(U, \mathcal{F})$  satisfy*

$$w_q(\alpha) < 1 + 2^{-k}.$$

The assertion for sheaves on a compactification  $j : U_0 \hookrightarrow X_0$  follows from the long exact sequence attached to

$$0 \rightarrow j_!j^*\mathcal{G}_0 \rightarrow \mathcal{G}_0 \rightarrow \mathcal{G}_0/j_!j^*\mathcal{G}_0 \rightarrow 0.$$

Assume (3.2.4)(k), and prove (3.2.4)(k + 1). After finite base extension, choose a compactification  $X_0$  and suppose first that the local monodromy along  $X - U$  is tame and unipotent. Put

$$S_0 = X_0 \times X_0, \text{quad} V_0 = U_0 \times U_0,$$

let  $D_0 = S_0 - V_0$ , and take a Lefschetz pencil satisfying the general-position hypotheses of (3.1). Let

$$\mathcal{G}_0 = \mathcal{F}_0 \boxtimes \mathcal{F}_0 = p_1^*\mathcal{F}_0 \otimes p_2^*\mathcal{F}_0.$$

**Lemma 2** (3.2.7). *The lisse sheaf  $w^*R^1f_!\mathcal{G}_0$  on the non-exceptional locus  $W_0$  is a direct factor of an  $\iota$ -real sheaf.*

Indeed  $\mathcal{G}_0$  is pointwise  $\iota$ -pure and  $\iota$ -real, and the fiber of  $R^1f_!\mathcal{G}_0$  at a point is the  $H_c^1$  of a curve with the restricted coefficient sheaf; apply (3.2.1).

**Lemma 3** (3.2.8). *The sheaf  $R^1f_!\mathcal{G}_0$  has no section supported on the exceptional set  $T$ .*

**Lemma 4** (3.2.9). *If  $\text{Gr}_i^G R^1f_!\mathcal{G}_0$  does not become constant on  $A^*$ , its restriction to  $W_0$  is pointwise  $\iota$ -pure of integral weight.*

The proof uses (1.8.4) at the boundary points and the calculation of vanishing cycles in (3.1). For each exceptional  $t$ , the five-term exact sequence

$$0 \rightarrow (R^1 f_! \mathcal{G}_0)_t \rightarrow (R^1 f_! \mathcal{G}_0)_{\bar{\eta}} \rightarrow \Phi^1 \rightarrow (R^2 f_! \mathcal{G}_0)_t \rightarrow 0 \quad (3.2.9.1)$$

shows both (3.2.8) and the claimed integrality of the weights of the nonconstant graded pieces.

**Lemma 5** (3.2.10). *The constituents of  $w^* R^1 f_! \mathcal{G}_0$  are pointwise pure of weights  $< 2$ .*

This follows from (2.2.10) by testing the weights at one point.

**Lemma 6** (3.2.11). *The eigenvalues  $\alpha$  of  $F$  on*

$$H^1(A^*, R^1 f_! \mathcal{G})$$

*satisfy  $w_q(\alpha) < 2 + 2^{-k}$ .*

The constant graded pieces contribute no  $H^1$ , and the nonconstant ones are handled by (3.2.5)(k).

**Lemma 7** (3.2.12). *The eigenvalues on  $E_2^{0,2}$  and on  $E_2^{2,0}$  of the Leray spectral sequence are of weights  $< 2$ .*

This is a direct consequence of the fiber formula and two applications of (1.4.4).

From the Leray spectral sequence (3.1.1.5), the eigenvalues on  $H_c^2(\tilde{V}, \pi^* \mathcal{G})$  have weight  $< 2 + 2^{-k}$ . Since

$$H_c^1(U, \mathcal{F}) \otimes H_c^1(U, \mathcal{F}) \subset H_c^2(V, \mathcal{G}) \subset H_c^2(\tilde{V}, \pi^* \mathcal{G}),$$

Kunneth's formula gives the induction step. The tame-unipotent hypothesis is removed by a finite cover; pullback injects cohomology. Duality then gives Theorem (3.2.3).

### (3.3) The general case

**Theorem 4** (3.3.1). *Let  $f : X \rightarrow Y$  be a morphism of schemes of finite type over  $\mathbb{Z}$ , and let  $\mathcal{F}$  be a sheaf on  $X$ . If  $\mathcal{F}$  is mixed of weights  $\leq n$ , then, for each  $i$ ,  $R^i f_! \mathcal{F}$  is mixed of weights  $\leq n + i$ .*

The proof uses successive devissages. Exact sequences reduce the assertion in the coefficient sheaf; an open-closed decomposition reduces it in  $X$ ; base change to an open part and its complement reduces it in  $Y$ ; Leray spectral sequences give transitivity. Universal homeomorphisms do not affect étale cohomology. Finally, morphisms of relative dimension zero preserve pointwise purity and have no higher direct images. These reductions reduce to the case where  $f$  has relative dimension one,  $\mathcal{F}$  is lisse and pointwise pure, and  $X$  is the complement in a smooth projective relative curve of a divisor finite étale over  $Y$ , with tame ramification along the divisor.

On each geometric fiber, Theorem (3.2.3) gives purity for  $R^i f_! j_* \mathcal{F}$ . The local-monodromy theorem (1.8.4) gives the weights of the graded pieces of the local monodromy filtration on  $i^* j_* \mathcal{F}$ . The finite part is handled by the relative-dimension-zero case, and the exact sequence

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* i^* j_* \mathcal{F} \rightarrow 0$$

finishes the proof.

**(3.3.2).** Lower bounds for weights are obtained from the  $p$ -adic estimates of SGA 7 XXI, 5. A sheaf is called integral if all Frobenius eigenvalues are algebraic integers. The weights of a mixed integral sheaf are necessarily non-negative, since the norm of a pure algebraic integer of weight  $n$  has absolute value  $q^{dn/2}$ .

**Corollary 1** (3.3.3). *Let  $f : X \rightarrow Y$  be of finite type over  $\mathbb{Z}$ , and let  $\mathcal{F}$  be an integral sheaf on  $X$ . If  $\mathcal{F}$  is mixed of weights  $\leq n$ , then  $R^i f_* \mathcal{F}$  is mixed of weights between 0 and  $n + i$ . If  $i$  is larger than the maximum dimension  $d$  of the fibers, the weights lie between  $2(i - d)$  and  $n + i$ .*

For  $Y = \text{Spec } \mathbb{F}_q$  this becomes:

**Corollary 2** (3.3.4). *Let  $X_0$  be of finite type over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be mixed of weights  $\leq n$  on  $X_0$ . Then  $H_c^i(X, \mathcal{F})$  is mixed of weights  $\leq n + i$ . If  $\mathcal{F}_0$  is integral, the weights are non-negative; if moreover  $i > d = \dim X$ , they are  $\geq 2(i - d)$ .*

**Corollary 3** (3.3.5). *If  $X_0$  is smooth of finite type over  $\mathbb{F}_q$  and  $\mathcal{F}_0$  is lisse mixed of weights  $\geq n$ , then  $H^i(X, \mathcal{F})$  is mixed of weights  $\geq n + i$ .*

This is Poincare duality applied to the dual sheaf. Hence:

**Corollary 4** (3.3.6). *If  $X_0$  is smooth of finite type over  $\mathbb{F}_q$  and  $\mathcal{F}_0$  is lisse pure of weight  $n$ , then the image of*

$$H_c^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})$$

*is pure of weight  $n + i$ .*

**(3.3.7)–(3.3.9).** If an algebraic number  $\alpha$  is pure of weight  $n$  relative to  $q$ , normalize  $p$ -adic valuations by  $v(q) = 1$  and attach to  $\alpha$  the pair  $(v(\alpha), v(q^n \alpha^{-1}))$ . For integral eigenvalues both coordinates are non-negative and sum to  $n$ . For the constant sheaf, the preceding bounds localize these pairs in the corresponding triangles in the  $(r, s)$ -plane. In particular, for  $X_0$  proper and smooth one obtains again the main theorem of I, with projective replaced by proper:

$$\det(1 - Ft, H^i(X, \mathbb{Q}_\ell)) \in \mathbb{Z}[t]$$

is independent of  $\ell \neq p$ , and all complex roots have absolute value  $q^{i/2}$ .

**(3.3.10)–(3.3.11).** The theorem remains valid with “mixed” replaced by “ $\iota$ -mixed” and real-valued weights. The only weights in  $R^i f_* \mathcal{F}$  are those  $\leq \beta + i$  congruent modulo  $\mathbb{Z}$  to a weight already occurring in  $\mathcal{F}$ . The smoothness assumption is used only to invoke Poincare duality, and may therefore be replaced by the corresponding rational-homology-manifold condition

$$Ra^! \mathbb{Q}_\ell = \mathbb{Q}_\ell(n)[2n]$$

for the structural morphism  $a : X_0 \rightarrow \text{Spec } \mathbb{F}_q$ .

### (3.4) Application: the structure of mixed sheaves

**Theorem 5** (3.4.1). *Let  $\mathcal{F}_0$  be an  $\iota$ -mixed sheaf on a scheme  $X_0$  of finite type over  $\mathbb{F}_q$ .*

*(i) There is a unique functorial decomposition*

$$\mathcal{F}_0 = \bigoplus_{b \in \mathbb{R}/\mathbb{Z}} \mathcal{F}_0(b)$$

*according to weights modulo  $\mathbb{Z}$ .*

(ii) If the pointwise weights are integers and  $\mathcal{F}_0$  is lisse, then  $\mathcal{F}_0$  has a unique finite increasing weight filtration  $W$  by lisse subsheaves such that  $\mathrm{Gr}_i^W \mathcal{F}_0$  is pointwise pure of weight  $i$ ; morphisms are strictly compatible with these filtrations.

(iii) If  $\mathcal{F}_0$  is lisse and pointwise pure, and if  $X_0$  is normal, then  $\mathcal{F}$  on  $X$  is semisimple.

**Lemma 8** (3.4.2). *For lisse sheaves  $\mathcal{F}_0$  and  $\mathcal{G}_0$  on  $X_0$  there is an exact sequence*

$$0 \rightarrow H^0(X, \mathrm{Hom}(\mathcal{F}, \mathcal{G}))_F \rightarrow \mathrm{Ext}^1(\mathcal{F}_0, \mathcal{G}_0) \rightarrow H^1(X, \mathrm{Hom}(\mathcal{F}, \mathcal{G}))^F.$$

Here superscripts and subscripts denote invariants and coinvariants for the action of the Weil group.

**Lemma 9** (3.4.3). *If  $\mathcal{F}_0$  and  $\mathcal{G}_0$  are lisse on smooth  $X_0$  and pointwise pure of weights  $\beta$  and  $\gamma$ , then a geometrically non-trivial extension of  $\mathcal{F}_0$  by  $\mathcal{G}_0$  can exist only if  $\beta \equiv \gamma \pmod{\mathbb{Z}}$  and  $\beta > \gamma$ .*

The same proof gives the corresponding vanishing criterion for  $\mathrm{Ext}^1$  with  $\beta \geq \gamma$ .

The proof of semisimplicity takes the largest semisimple lisse subsheaf of the underlying geometric sheaf. It is Frobenius-stable, hence comes from  $X_0$ . The preceding extension criterion forces the quotient extension to be geometrically trivial, contradicting maximality unless the quotient is zero.

The uniqueness in (i) and (ii) follows by looking at generalized eigenspaces of the Frobenius at closed points. Existence is first proved when  $X_0$  is smooth and the sheaf is an iterated extension of lisse pointwise pure sheaves; the extension criterion gives the splitting by weights modulo  $\mathbb{Z}$  and the weight filtration. The general case follows by induction on  $\dim X_0$  using an open-closed decomposition and specialization.

**Variant 2** (3.4.9). *Every lisse mixed sheaf has a filtration by pointwise weights by lisse subsheaves, and every pure sheaf is geometrically semisimple.*

**Potentially pure sheaves.** A sheaf on a scheme  $X$  of finite type over an algebraically closed field is said to have a property potentially if it is obtained by base change from a family over a finite-type base whose closed fibers have the property. If  $f : Y \rightarrow X$  is proper and smooth, the sheaves  $R^i f_* \mathbb{Q}_\ell$  are potentially pointwise pure. Specialization then gives:

**Corollary 5** (3.4.12). *If  $X$  is normal of finite type over an algebraically closed field and  $\mathcal{F}$  is lisse and potentially pointwise  $\iota$ -pure, then  $\mathcal{F}$  is semisimple.*

**Corollary 6** (3.4.13). *For  $S$  normal connected and  $f : X \rightarrow S$  proper and smooth, the sheaves  $R^i f_* \mathbb{Q}_\ell$  are semisimple.*

### (3.5) Application: equidistribution theorems

Return to the notation and hypotheses of (2.2). Thus  $X_0$  is normal geometrically connected of dimension  $N$ , and for every irreducible representation  $\tau$  of  $G_{\mathbb{R}}$  the corresponding sheaf is pointwise pure of weight  $2N(\tau)$ . By Grothendieck's cohomological interpretation of  $L$ -functions, together with (3.3.4) and (3.3.10),  $L(\tau)$  extends meromorphically to  $G$ ; in the half-plane relevant to  $N$  it is holomorphic and invertible except for the simple pole at the trivial character. Therefore the equidistribution results (2.1.10)–(2.1.12) apply.

For  $n > 0$ , let  $X_{(n)}$  be the base change of  $X_0$  to  $\mathbb{F}_{q^n}$ . A rational point  $x \in X_0(\mathbb{F}_{q^n})$  gives a geometric Frobenius element  $F_x$  of degree  $n$ . If a closed point  $x_0$  has degree  $d \mid n$ , then the  $d$  rational points above it contribute conjugates of  $F_{x_0}^{n/d}$ . Thus, on conjugacy classes,

$$\sum_{x_0 \in |X_0|} \deg(x_0) [F_{x_0}^{n/\deg(x_0)}] = \sum_{x \in X_0(\mathbb{F}_{q^n})} [F_x]. \quad (3.5.2.1)$$



After identifying components of  $G_{\mathbb{R}}$  by a positive-degree central element  $z$ , Theorem (2.1.12) gives:

**Theorem 6** (3.5.3). *For every component index  $i$ , the measures obtained from the Frobenius classes of points of  $X_0(\mathbb{F}_{q^{nd+i}})$ , normalized by the expected factor  $q^{-N(nd+i)}$ , converge vaguely to the translated Haar measure on the corresponding component of  $G_{\mathbb{R}}$ .*

**Example: elliptic curves and Sato–Tate.** Let  $f : E_0 \rightarrow X_0$  be a family of elliptic curves and put  $\mathcal{F}_0 = R^1 f_* \mathbb{Q}_{\ell}$ . If the modular invariant  $j$  is nonconstant, the geometric monodromy group is all of  $\mathrm{SL}(V)$ : after imposing level  $n \geq 3$  prime to  $p$ , the associated modular cover maps dominantly to the moduli scheme of elliptic curves with level  $n$  structure, and the known topological monodromy contains a finite-index subgroup of  $\mathrm{SL}(H^1(E_x, \mathbb{Z}_{\ell}))$ .

Then  $G_{\mathbb{R}}$  is  $\mathrm{SU}(2) \times \mathbb{Z}$ , acting on  $\iota V \simeq \mathbb{C}^2$  by  $(g, n) \mapsto q^{n/2} g$ . Identify conjugacy classes in  $\mathrm{SU}(2)$  with  $[0, \pi]$  by eigenvalues  $e^{\pm i\theta}$ . Haar measure pushes forward to

$$\frac{2}{\pi} \sin^2 \theta \, d\theta.$$

For  $x \in X_0(\mathbb{F}_{q^n})$  define the Frobenius angle  $\theta(x)$  by

$$\#E_x(\mathbb{F}_{q^n}) = 1 - 2q^{n/2} \cos \theta(x) + q^n$$

(up to the sign convention of geometric Frobenius). Hence:

**Proposition 2** (3.5.7). *If  $j$  is nonconstant, then as  $n \rightarrow \infty$  the normalized counting measures of the angles  $\theta(x)$ ,  $x \in X_0(\mathbb{F}_{q^n})$ , converge vaguely to the Sato–Tate measure  $\frac{2}{\pi} \sin^2 \theta \, d\theta$ .*

### (3.6) Application: the local invariant-cycle theorem

Let  $k$  be algebraically closed, let  $S$  be the spectrum of the henselization of  $k[T]$  at  $(T)$ , with closed point  $s$ , generic point  $\eta$ , and geometric generic point  $\bar{\eta}$ . For a proper morphism  $f : X \rightarrow S$  there is a specialization map

$$\mathrm{sp}^* : H^*(X_s) \longrightarrow H^*(X_{\bar{\eta}}),$$

whose image lies in the inertia invariants.

**Theorem 7** (3.6.1). *Assume that  $X$  is essentially smooth over  $k$  and that  $X_{\eta}$  is smooth over  $\eta$ . Then, in  $\mathbb{Q}_{\ell}$ -cohomology,*

$$\mathrm{sp}^* : H^*(X_s) \longrightarrow H^*(X_{\bar{\eta}})^{\mathrm{Gal}(\bar{\eta}/\eta)}$$

*is surjective.*

The proof spreads the situation to a family over a smooth curve over a finite field, so that Frobenius weights are available. The Hochschild–Serre spectral sequence for the inertia group gives exact sequences of Wang type

$$0 \rightarrow H^{i-1}(X_{\bar{\eta}})_I(-1) \rightarrow H^i(X_{\eta}) \rightarrow H^i(X_{\bar{\eta}})^I \rightarrow 0. \quad (3.6.1.5)$$

Together with the long exact sequence for cohomology with support and Poincaré duality this yields a cross of exact sequences. The terms above and below have incompatible weights:

**Lemma 10** (3.6.2).  *$H^i(X_{\bar{\eta}})^I$  is mixed of weights  $\leq i$ .*

**Lemma 11** (3.6.3).  *$H^{2N-i-1}(X_{\bar{\eta}})_I(-N)$  is mixed of weights  $> i$ .*

Applying the exact functor  $W_i$  to the cross gives the surjectivity.

In the complex-analytic analogue, for a proper map  $X \rightarrow D$  smooth over the punctured disk, the same result follows by replacing Frobenius weights with the weight filtration of mixed Hodge theory; this is the form related to Steenbrink’s mixed Hodge structure on vanishing cohomology.

### (3.7) Return to (1.8)

The results of (3.3) and (1.8) simplify the proofs of (1.8.2) and (1.8.4). For a family of elliptic curves  $f : E_0 \rightarrow S_0$  over a smooth  $S_0$ , the image

$${}^k W = \operatorname{Im} \left( H_c^1(S, \operatorname{Sym}^k(R^1 f_* \mathbb{Q}_\ell)) \rightarrow H^1(S, \operatorname{Sym}^k(R^1 f_* \mathbb{Q}_\ell)) \right)$$

is pure of weight  $k + 1$ . This replaces the earlier reduction of the corresponding modular-form estimate to the usual Weil conjecture and gives a cleaner input to the Ramanujan–Petersson application.

Finally, let  $\psi : \mathbb{F}_p \rightarrow \mathbb{C}^*$  be a nontrivial additive character. If  $Q \in \mathbb{F}_q[X_1, \dots, X_n]$  has degree  $d$  prime to  $p$  and its homogeneous part  $Q_d$  defines a smooth projective hypersurface, then

$$\left| \sum_{x_1, \dots, x_n \in \mathbb{F}_q} \psi \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p} Q(x_1, \dots, x_n) \right| \leq (d-1)^n q^{n/2}. \quad (3.7.2.1)$$

The sharper statement is

$$H_c^i(\mathbb{A}^n, \mathcal{F}(\psi Q)) = 0 \quad (i \neq n), \quad \dim H_c^n(\mathbb{A}^n, \mathcal{F}(\psi Q)) = (d-1)^n, \quad (3.7.2.3)$$

and  $H_c^n(\mathbb{A}^n, \mathcal{F}(\psi Q))$  is pure of weight  $n$ . One proves this by considering the universal polynomial over the open parameter scheme  $S_0$  where the degree- $d$  part is smooth. The sheaves  $R^i f_* \mathcal{F}(\psi Q_S)$  are lisse by local acyclicity; since  $S_0$  is connected and the sheaves are mixed, it suffices by (1.8.12) to check a single polynomial, namely  $Q = \sum X_i^d$ . Kunneth then reduces to the one-variable case  $Q = X^d$ , already treated in (1.8.11).

## 4. Lefschetz pencils

In this chapter, except in number (4.5), we work over an algebraically closed field  $k$  of characteristic  $p$ . Let  $\ell$  be a prime number distinct from  $p$ . The cohomology groups considered have coefficients in  $\mathbb{Z}_\ell$ ,  $\mathbb{Q}_\ell$ , or  $\mathbb{Z}/m$ , where  $m$  is a power of  $\ell$ .

### (4.1) The hard Lefschetz theorem

In this number we suppose chosen, over  $k$ , an isomorphism between  $\mathbb{Z}_\ell$  and  $\mathbb{Z}_\ell(1)$ . This spares us from writing Tate twists.

**Theorem 8** (4.1.1). *Let  $X$  be a projective smooth variety over  $k$ , let  $\mathcal{L}$  be an ample invertible sheaf on  $X$ , and let  $\xi = c_1(\mathcal{L}) \in H^2(X, \mathbb{Q}_\ell)$ . Suppose that  $X$  is pure of dimension  $n$ . Then, for each  $i > 0$ , cup-product by  $\xi^i$ ,*

$$\xi^i : H^{n-i}(X, \mathbb{Q}_\ell) \longrightarrow H^{n+i}(X, \mathbb{Q}_\ell),$$

*is an isomorphism.*

We prove the theorem by induction on  $n$ . The case  $n = 0$  is trivial; suppose  $n \geq 1$ . Since  $c_1(\mathcal{L}^{\otimes N}) = N c_1(\mathcal{L})$ , the assertion for  $\mathcal{L}$  is equivalent to the assertion for any positive tensor power. We may therefore assume  $\mathcal{L}$  very ample. Let  $Y$  be a smooth hyperplane section of  $X$  for the projective embedding defined by  $\mathcal{L}$ . We first reduce the induction step to the following lemma.

**Lemma 12** (4.1.2). *The intersection form on  $H^{n-1}(Y)$  is nondegenerate on the image of  $H^{n-1}(X)$ .*

For  $i = 0$  the assertion is trivial. For  $i \geq 1$ , cup-product by  $\xi = c_1(Y)$  is the composite of the restriction morphism  $H^*(X) \rightarrow H^*(Y)$  and its transpose under Poincare duality, the Gysin morphism  $H^*(Y) \rightarrow H^{*+2}(X)$ . Thus  $\xi^i$  factors as

$$H^{n-i}(X) \longrightarrow H^{(n-1)-(i-1)}(Y) \xrightarrow{\xi^{i-1}} H^{(n-1)+(i-1)}(Y) \longrightarrow H^{n+i}(X).$$

For  $i > 1$  the two extreme arrows are isomorphisms by weak Lefschetz, and the middle arrow is an isomorphism by induction. For  $i = 1$ , weak Lefschetz gives only the injection  $H^{n-1}(X) \hookrightarrow H^{n-1}(Y)$ , whose transpose  $H^{n-1}(Y) \rightarrow H^{n+1}(X)$  is surjective. The composite is an isomorphism precisely when Lemma (4.1.2) holds.

After replacing  $\mathcal{L}$  by a tensor power, assume that the embedding admits a Lefschetz pencil of hyperplane sections. Let  $Y = Y_t$  be a smooth member of such a pencil  $(Y_t)_{t \in \mathbb{P}^1}$ . We prove (4.1.2) by interpreting the image of  $H^{n-1}(X)$  in  $H^{n-1}(Y)$  in terms of monodromy.

Let  $S$  be the set of parameters  $t$  for which  $Y_t$  is singular. For  $t \in \mathbb{P}^1 - S$ , the groups  $H^i(Y_t)$  form the fibers of a lisse  $\mathbb{Q}_\ell$ -sheaf on  $\mathbb{P}^1 - S$ , and  $\pi_1(\mathbb{P}^1 - S, t_0)$  acts on  $H^i(Y)$ .

**Proposition 3** (4.1.3). *The image of  $H^{n-1}(X)$  in  $H^{n-1}(Y)$  is the subspace  $H^{n-1}(Y)^{\pi_1(\mathbb{P}^1 - S, t_0)}$  of monodromy invariants.*

This is proved in SGA 7, XVIII, (5.6.10), under the hypothesis that  $Y$  satisfies hard Lefschetz, which is our induction hypothesis. Over  $\mathbb{C}$ , in integral cohomology, a direct proof is given in SGA 7, XIX, (4), by combining the corresponding statement with the Picard–Lefschetz formula. In (4.3) below we transpose this direct argument to  $\mathbb{Z}_\ell$ -cohomology over an arbitrary base field.

We also know by (3.4.13) that  $H^{n-1}(Y)$  is a completely reducible representation of  $\pi_1(\mathbb{P}^1 - S, t_0)$ . It remains only to apply the following algebraic lemma.

**Lemma 13** (4.1.4). *Let  $V$  be a completely reducible linear representation of a group  $\pi$ , equipped with a  $\pi$ -invariant nondegenerate bilinear form  $\Phi$ . Then the restriction of  $\Phi$  to  $V^\pi$  is nondegenerate.*

By complete reducibility, write  $V = V^\pi \oplus W$  for a subrepresentation  $W$  having no trivial quotient. Then  $W$  is orthogonal to  $V^\pi$ : indeed  $\Phi$  induces a morphism  $W \rightarrow (V^\pi)^*$ , hence a trivial quotient if it is nonzero. It follows that  $\Phi|_{V^\pi}$  is nondegenerate.

**Corollary 7** (4.1.5). *The odd Betti numbers of a nonsingular projective variety are even.*

It is enough to assume the variety connected and pure of dimension  $n$ . If  $\xi = c_1(\mathcal{L})$  with  $\mathcal{L}$  ample, then the form  $\text{Tr}(\xi^i xy)$  on  $H^{n-i}(X)$  is nondegenerate by hard Lefschetz and Poincare duality. If  $n - i$  is odd, this form is alternating; hence  $b_{n-i} = b_{n+i}$  is even. This result should hold without the projectivity hypothesis, but Deligne does not know how to prove it in characteristic different from zero.

**(4.1.6) Weak Lefschetz.** Let  $X$  and  $Y$  be as above, except that  $X$  is pure of dimension  $n + 1$  and  $Y$  is pure of dimension  $n$ . The key input is the cohomological-dimension theorem for affine varieties: in  $\mathbb{Z}/m$ -cohomology,

$$H^i(X - Y) = 0 \quad (i > n + 1).$$

Since  $X - Y$  is smooth, Poincare duality also gives

$$H_c^i(X - Y) = 0 \quad (i < n + 1).$$

The long exact sequence of relative cohomology then gives

$$H^i(X) \xrightarrow{\sim} H^i(Y) \text{quad}(i < n), \quad H^n(X) \hookrightarrow H^n(Y).$$

The transpose Gysin morphisms satisfy

$$H^i(Y) \xrightarrow{\sim} H^{i+2}(X) \quad (i > n), \quad H^n(Y) \rightarrow H^{n+2}(X).$$

Passing to the projective limit gives the same conclusions in  $\mathbb{Z}_\ell$ - and  $\mathbb{Q}_\ell$ -cohomology. In  $\mathbb{Z}_\ell$ -cohomology the universal-coefficient formula further gives that  $H^{n+1}(X - Y, \mathbb{Z}_\ell)$  is torsion-free, and hence  $H^n(Y, \mathbb{Z}_\ell)/H^n(X, \mathbb{Z}_\ell)$  is torsion-free.

## (4.2) Reminders on Lefschetz pencils

In (1.3) and (1.4) the essential results of the theory of Lefschetz pencils were summarized, excluding the case where  $p = 2$  and the hyperplane sections have even dimension. The aim here is to repair that omission in part and, in doing so, to complete SGA 7, XVIII: in the wild case, the theory holds for general Lefschetz pencils, not only for generic ones.

**(4.2.1) Local theory.** Let  $f : X \rightarrow S$  be as in (1.4.2), of relative dimension  $n$ . Write  $n = 2n'$  or  $2n' + 1$ . Let  $\delta \in H^n(X_{\bar{\eta}})(n')$  be the vanishing cycle, determined up to sign. In the exceptional case  $n$  even and  $p = 2$ , omitted in (1.4.3), the inertia group has several characters of order two. For one of them, depending on the ordinary quadratic singularity of  $f$ , the local monodromy is again given by the Picard–Lefschetz formula of loc. cit. The statements hold in  $\mathbb{Z}/m$ -,  $\mathbb{Z}_\ell$ -, and  $\mathbb{Q}_\ell$ -cohomology.

**Corollary 8** (4.2.2). *The image of  $H^n(X_s)$  in  $H^n(X_{\bar{\eta}})$  is the orthogonal complement of  $\delta$ . In  $\mathbb{Z}_\ell$ - or  $\mathbb{Q}_\ell$ -cohomology it is also the subspace of invariants under inertia.*

The second assertion follows from the Picard–Lefschetz formula: the term  $(x, \delta)\delta$  vanishes if and only if  $(x, \delta) = 0$ . This is clear when  $\delta$  is not torsion; if  $\delta$  is torsion, the pairing  $(x, \delta)$  is identically zero.

**(4.2.3) Global theory.** Retain the notation of (1.5.6). Thus  $X \subset P$  is a connected nonsingular projective variety of dimension  $n + 1$ ,  $(X_t)_{t \in D}$  is a Lefschetz pencil of hyperplane sections,  $S$  is the set of exceptional parameters, and  $j : U \hookrightarrow D$  is the complementary open. Identify  $D$  with a line in the dual projective space  $P^\vee$ , and let  $X^\vee \subset P^\vee$  be the image of  $X$  by the correspondence “the hyperplane  $H$  is tangent to  $X$  at  $x$ .” A Lefschetz pencil is called transverse if  $D$  is transverse to  $X^\vee$ .

A path between two geometric points of a scheme is an isomorphism between the corresponding fiber functors on étale covers. For any path  $c$  on  $U$ , from  $u$  to a geometric generic point of the henselization  $D_{(t)}$ , one obtains an identification

$$H^n(X_{\bar{\eta}}, \mathbb{Z}_\ell) \simeq H^n(X_{\bar{u}}, \mathbb{Z}_\ell),$$

and hence a vanishing cycle  $\pm\delta_c$  in  $H^n(X_{\bar{u}}, \mathbb{Z}_\ell)(n')$ .

**Definition 2** (4.2.4). (i) *The set of vanishing cycles is the set of all cycles  $\pm\delta_c$  obtained in this way, with  $c$  and  $t \in S$  varying.*

(ii) *The vanishing part of  $H^n(X_{\bar{u}})$  is the subspace generated by the  $\lambda\delta$ , where  $\lambda \in \mathbb{Z}_\ell(-n')$  and  $\delta$  is a vanishing cycle.*

Since the projective line  $D$  is simply connected, the group  $\pi_1(U, u)$  is topologically generated by conjugates of local inertia groups  $I_t$ ,  $t \in S$ . Thus the monodromy group is topologically generated by the images of these conjugates. For  $n$  odd these images are transvections  $x \mapsto x + \lambda(x, \delta)\delta$ ; for  $n$  even they are reflections  $x \mapsto x - (-1)^{n'}(x, \delta)\delta$ .

**Proposition 4** (4.2.6). *For a generic Lefschetz pencil the vanishing cycles, taken up to sign, are all conjugate under the monodromy group.*

If  $X^\vee$  is not a hypersurface, the assertion is trivial. Otherwise  $X^\vee$  is irreducible. Let  $\eta$  be the generic point of the Grassmannian of lines in  $P^\vee$ , and let  $D_\eta$  be the corresponding projective line. On the geometric generic line the critical locus is the intersection with  $X^\vee$ , which is connected over the generic point of  $X^\vee$ . The geometric fundamental-group comparison and Bertini's theorem then show that the monodromy action is transitive on the set of critical points, whence the conjugacy of vanishing cycles.

**Corollary 9** (4.2.7). *For  $p = 2$  and  $n$  even, assume that the Lefschetz pencil is sufficiently general. Then:*

- (i) *if  $n$  is odd, the homomorphisms from  $\mathbb{Z}_\ell(1)$  to the monodromy group, given by the Picard–Lefschetz formula  $x \mapsto x + \lambda(x, \delta)\delta$ , are conjugate to one another;*
- (ii) *if  $n$  is even, the reflections  $x \mapsto x - (-1)^{n'}(x, \delta)\delta$ , for  $\delta$  a vanishing cycle, are conjugate to one another.*

For a generic pencil this follows from (4.2.6) and the local Picard–Lefschetz formula; the general sufficiently general case follows by specialization (1.11).

**Corollary 10** (4.2.8). *Under the hypotheses of (4.2.7), the vanishing cycles, taken up to sign, are all conjugate in  $H^n(X_{\bar{u}}, \mathbb{Q}_\ell)$ .*

Modulo torsion,  $\pm\delta$  is determined by the corresponding Picard–Lefschetz transformation.

### (4.3) Complement to SGA 7, XIX, (4)

Choose once and for all an isomorphism over  $k$  between  $\mathbb{Z}_\ell$  and  $\mathbb{Z}_\ell(1)$ , so that Tate twists are suppressed. In this number Deligne gives a direct proof of (4.1.3), generalizing SGA 7, XIX, (4), to an arbitrary base field and to  $\mathbb{Z}_\ell$ -cohomology.

Let  $P$  be a projective space over  $k$ , let  $X \subset P$  be projective, smooth, and pure of dimension  $n + 1$ , let  $A \subset P$  be a codimension-two linear subspace meeting  $X$  transversally, and suppose that the hyperplanes through  $A$  sweep out on  $X$  a Lefschetz pencil. Let  $\tilde{X}$  be the blow-up of  $X$  along  $A \cap X$ , with exceptional divisor  $\tilde{A}$ , and let

$$f : \tilde{X} \longrightarrow \mathbb{P}^1$$

be the morphism whose fibers are the members  $Y_t$  of the pencil. Let  $S$  be the set of critical values, choose  $0 \in \mathbb{P}^1 - S$ , and put  $Y = Y_0$ .

The geometric situation is represented by

$$\begin{array}{ccccc} \tilde{A} & \hookrightarrow & \tilde{X} & \longrightarrow & X \\ \downarrow & & \downarrow \hat{f} & & \\ 0 & \hookrightarrow & \mathbb{P}^1 & \xlongequal{\quad} & \mathbb{P}^1. \end{array} \quad (4.3.1.1)$$

Let  $\text{Ev}(Y)$  be the vanishing part of  $H^n(Y)$  and let  $\text{Ev}(Y)^\perp$  denote its orthogonal complement for  $(x, y) = \text{Tr}(xy)$ .

**Lemma 14** (4.3.2). *In  $\mathbb{Z}/m$ -cohomology,*

$$H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m) \simeq \text{Ev}(Y)^\perp \subset H^n(Y).$$

*In  $\mathbb{Z}_\ell$ - or  $\mathbb{Q}_\ell$ -cohomology this group is also the subspace of  $\pi_1(\mathbb{P}^1 - S, 0)$ -invariants.*

The sheaf  $R^n f_* \mathbb{Z}/m$  has no section with punctual support, and its  $H^0$  injects into the fiber  $H^n(Y)$ . Its image is the fiber at 0 of the largest constant subsheaf. The local theory identifies this largest locally constant subsheaf with  $\text{Ev}(Y)^\perp$ , and since  $\mathbb{P}^1$  is simply connected it is constant.

**Proposition 5** (4.3.3). *The subspace  $\mathrm{Ev}(Y)^\perp$  is the image of  $H^n(X)$  in  $H^n(Y)$ .*

In  $\mathbb{Z}/m$ -cohomology this may be written as the exact cross

$$\begin{array}{ccccccc} & & \mathrm{Ev}(Y)^* & & & & \\ & & \downarrow & & & & \\ 0 & \longrightarrow & \mathrm{Ev}(Y) & \longrightarrow & H^n(Y) & \longrightarrow & H^{n+2}(X) \longrightarrow 0 \\ & & \downarrow & & & & \\ & & H^n(X) & & & & \end{array} \quad (4.3.3.1)$$

where the composite  $H^n(X) \rightarrow H^n(Y) \rightarrow H^{n+2}(X)$  is cup-product with the class of  $Y$ . In  $\mathbb{Z}_\ell$ -cohomology, since the duals  $\mathrm{Ev}(Y)^*$  modulo  $\ell^j$  need not form a projective system, one instead passes to the limit in the exact sequence

$$0 \rightarrow H^n(X) \rightarrow H^n(Y) \rightarrow (\mathbb{Z}/\ell^j)^B,$$

where the vanishing cycles indexed by  $B$  generate the vanishing part. This gives the analogous cross

$$\begin{array}{ccccccc} & & \mathrm{Ev}(Y)^* & & & & \\ & & \downarrow & & & & \\ 0 & \longrightarrow & \mathrm{Ev}(Y) & \longrightarrow & H^n(Y) & \longrightarrow & H^{n+2}(X) \longrightarrow 0 \\ & & \downarrow & & & & \\ & & H^n(X) & & & & \end{array} \quad (4.3.3.2)$$

in  $\mathbb{Z}_\ell$ -cohomology.

**(4.3.4) Relative cohomology.** Let  $u : T' \rightarrow T''$  be a morphism of topoi. Triples  $(\mathcal{F}', \mathcal{F}'', \varphi : u^* \mathcal{F}'' \rightarrow \mathcal{F}')$  form a new topos  $T$ . If  $\mathcal{F} = (\mathcal{F}', \mathcal{F}'', \varphi)$  is an abelian sheaf of  $T$ , set

$$H^0(T'' \text{ mod } T', \mathcal{F}) = \ker(H^0(T'', \mathcal{F}'') \rightarrow H^0(T', \mathcal{F}')).$$

The right-derived functors give the relative cohomology of  $T''$  modulo  $T'$ . Equivalently, for  $K \in D^+(T)$ , there is a triangle

$$R\Gamma(T'' \text{ mod } T', K) \rightarrow R\Gamma(T'', K'') \rightarrow R\Gamma(T', K') \rightarrow,$$

and hence a long exact sequence of cohomology.

**(4.3.5) Sheaves on  $\mathbb{P}^1$ .** Let  $X$  be an irreducible complete variety over  $k$ , let  $x, y$  be two closed points, and let  $K \in D^+(X)$ . A path from a geometric generic point of the henselization at  $x$  to one at  $y$  gives a morphism

$$H^*(X - \{x\}, K) = H^*(X \text{ mod } \{x\}, K) \rightarrow H^*(X_{(y)} \text{ mod } \bar{y}, K). \quad (4.3.5.1)$$

**Proposition 6** (4.3.6). *Let  $t \in \mathbb{P}^1$ , and let  $K \in D^b(\mathbb{P}^1)$  have constructible cohomology sheaves, with finite support in negative degrees. Let  $S$  be a finite subset of  $\mathbb{P}^1$  such that the cohomology sheaves of  $K$  are locally constant outside  $S$ . Then the product of the morphisms (4.3.5.1), for  $X = \mathbb{P}^1$ ,  $x = t$ , and  $y \in S$ , is injective:*

$$H^1(\mathbb{P}^1 \text{ mod } t, K) \rightarrow \prod_{s \in S} H^1(\mathbb{P}^1_{(s)} \text{ mod } \bar{s}, K).$$

The proof first removes a subcomplex supported in negative degrees and reduces to a sheaf  $\mathcal{H}$  without punctual sections, locally constant outside  $S$ . Then  $H^1(\mathbb{P}^1 \text{ mod } t, \mathcal{H})$  classifies  $\mathcal{H}$ -torsors on  $\mathbb{P}^1$  trivialized at  $t$ . Such torsors are determined by their local restrictions because  $\mathbb{P}^1$  is simply connected and the images of the local monodromy groups topologically generate  $\pi_1(\mathbb{P}^1 - S)$ .

**(4.3.7) Plan of the proof of (4.3.3).** The injection of  $H^n(X)$  into  $H^n(Y)$  factors through  $H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m) \subset H^n(Y)$ . It remains to show that, for a class  $\alpha$  in this  $H^0$ , the obstruction  $d\alpha \in H^{n+1}(X - Y)$  to extending  $\alpha$  from  $Y$  to  $X$  vanishes. Since  $X - Y = \tilde{X} - Y - \tilde{A}$ , and if  $\varphi$  is the projection of  $\tilde{X} - \tilde{A}$  to  $\mathbb{P}^1$ , one has

$$H^{n+1}(X - Y) = H^1(\mathbb{P}^1 \bmod 0, R\varphi_* \mathbb{Z}/m[n]).$$

The weak Lefschetz theorem shows that the lower cohomology sheaves have finite support. Proposition (4.3.6) then reduces the obstruction to local obstructions, which are precisely the pairings  $(\alpha, \delta_s)$  with vanishing cycles.

**(4.3.8) Proof of (4.3.3).** Let  $\alpha \in H^n(Y)$  and let  $\tilde{\alpha}$  be its inverse image on the inverse image of  $Y$  in  $\tilde{X}$ . The obstruction to extending  $\alpha$  to  $X$  is the same as the obstruction to extending  $\tilde{\alpha}$  to  $\tilde{X}$ . In the derived-category form of the Leray spectral sequence, take a flasque resolution  $(\mathbb{Z}/m)^*$  of  $\mathbb{Z}/m$  on  $\tilde{X}$ , put  $K = f_*(\mathbb{Z}/m)^*$  and  $L = f_*((\mathbb{Z}/m)^*|_{\tilde{A}})$ . The exact sequence of simple complexes associated with

$$\Gamma(\mathbb{P}^1, K) \rightarrow s(K_0 \rightarrow L_0) \rightarrow L_0$$

identifies the obstruction as an element of

$$H^1(\mathbb{P}^1 \bmod 0, s(K \rightarrow L)[n]),$$

where  $s(K \rightarrow L)$  represents  $R\varphi_* \mathbb{Z}/m$ . Applying (4.3.6), the obstruction is tested after localization at each critical value. These local obstructions vanish exactly for classes coming from  $H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m)$ , and (4.3.3) follows.

**Corollary 11 (4.3.9).** *In  $\mathbb{Q}_\ell$ -cohomology one has an orthogonal direct-sum decomposition, for the intersection form,*

$$H^n(Y) = H^n(X) \oplus \text{Ev}(Y).$$

This reformulation of hard Lefschetz follows from (4.3.3) and (4.1.2).

**(4.3.10)** In  $\mathbb{Z}_\ell$ -cohomology, the cross (4.3.3.2) shows that the intersection of the vanishing part of  $H^n(Y)$  with the fixed part  $H^n(X) = H^n(Y)^{\pi_1}$  is

$$\ker(\xi : H^n(X) \rightarrow H^{n+2}(X)).$$

In  $\mathbb{Q}_\ell$ -cohomology this kernel is zero by hard Lefschetz. Thus one recovers Lefschetz's fundamental lemma: a cycle which is both invariant and vanishing is zero. In  $\mathbb{Z}_\ell$ -cohomology this may fail; after replacing the embedding by a multiple, this kernel may coincide with the whole torsion subgroup of  $H^n(X)$ .

## (4.4) Monodromy of Lefschetz pencils

Retain the notation of (4.3.1). Work in  $\mathbb{Q}_\ell$ -cohomology, and let  $G$  be the group of automorphisms of  $\text{Ev}(Y)$  preserving the nondegenerate intersection form. The group  $\pi_1(\mathbb{P}^1 - S, 0)$  acts on  $\text{Ev}(Y)$  preserving this form; let  $M$  be its image in  $G$ .

**Theorem 9 (4.4.1).** *Assume  $X$  irreducible. If  $p = 2$  and  $n$  is even, assume in addition that the pencil is sufficiently general. Then either:*

- (a)  $M$  is open in  $G$ ;
- (b)  $n$  is even and  $M$  is finite.

Since  $G$  is a semisimple algebraic group over  $\mathbb{Q}_\ell$ , it is enough to prove Zariski density of  $M$  in the nonfinite case. After extending scalars to  $\mathbb{C}$ , this is reduced to two elementary algebraic lemmas applied to the Zariski closure of  $M$  and to the conjugacy class of Picard–Lefschetz transformations.

**Lemma 15** (4.4.2a). *Let  $V$  be a finite-dimensional complex vector space with a nondegenerate alternating form. Let  $M \subset \mathrm{Sp}(V)$  be an algebraic subgroup and let  $R$  be an  $M$ -orbit in  $V$  generating  $V$ . Suppose that  $M$  is the smallest algebraic subgroup containing all transvection one-parameter subgroups*

$$U_\delta = \{x \mapsto x + \lambda(x, \delta)\delta \mid \lambda \in \mathbb{C}\}, \quad \delta \in R.$$

*Then  $M = \mathrm{Sp}(V)$ .*

**Lemma 16** (4.4.3a). *If  $\delta', \delta'' \in R$  are not orthogonal, then every nonzero linear combination of  $\delta'$  and  $\delta''$  belongs to  $R$ .*

The subgroup generated by  $U_{\delta'}$  and  $U_{\delta''}$  is the group  $\mathrm{SL}_2$  of their span; its nonzero vectors form a single orbit. The proof of (4.4.2a) follows by taking a maximal subspace  $W$  for which  $R \cap W$  is dense, using (4.4.3a) to show no vector outside  $W$  can be nonorthogonal to  $W$ , and concluding from nondegeneracy and generation that  $W = V$  and hence  $M = \mathrm{Sp}(V)$ .

**Lemma 17** (4.4.2b). *Let  $V$  be a finite-dimensional complex vector space with a nondegenerate symmetric form. Let  $M \subset O(V)$  be an algebraic subgroup and let  $R$  be an  $M$ -orbit generating  $V$ . Suppose  $(\delta, \delta) = 2$  for all  $\delta \in R$ , and suppose that  $M$  is the smallest algebraic subgroup of  $O(V)$  containing the reflections*

$$s_\delta : x \mapsto x - (x, \delta)\delta, \quad \delta \in R.$$

*Then  $M$  is finite or  $M = O(V)$ .*

**Lemma 18** (4.4.3b). *Let  $\delta', \delta'' \in R$ . If  $(\delta', \delta'')$  is not of the form  $2 \cos \theta$  with  $\theta$  a rational multiple of  $2\pi$ , then the vectors  $\delta$  in the span of  $\delta', \delta''$  satisfying  $(\delta, \delta) = 2$  belong to  $R$ . The same conclusion holds if  $(\delta', \delta'') = \pm 2$ .*

The first case uses the fact that the product  $s_{\delta'} s_{\delta''}$  is a rotation of infinite order in the two-plane, whose algebraic envelope is the connected orthogonal group of that plane. The second case uses the transvection obtained when the plane has a one-dimensional radical. The proof of (4.4.2b) then proceeds as in the alternating case, except that if all maximal subspaces have dimension one, the scalar products in  $R$  take only finitely many values; hence  $R$  is finite and so is  $M$ .

Deligne then studies the finite case.

**Proposition 7** (4.4.5). *If  $M$  is finite, then the analogous group in  $\ell'$ -adic cohomology is finite for every  $\ell' \neq p$ . As a quotient of  $\pi_1(\mathbb{P}^1 - S, 0)$  it is independent of  $\ell$ , and the character of its representation on  $\mathrm{Ev}(Y, \mathbb{Q}_{\ell'})$  has integer values independent of  $\ell'$ .*

The proof removes the trivial factors from the virtual representation

$$\sum_i (-1)^i H^i(Y)$$

and reduces by specialization to the case where the pencil is defined over a finite field. For  $x \in \mathbb{P}_0^1 - S_0$  rational over  $\mathbb{F}_{q^m}$ , the Lefschetz trace formula identifies the trace of Frobenius with the number of  $\mathbb{F}_{q^m}$ -rational points of the fiber. The required independence follows from Chebotarev.



**Lemma 19** (4.4.6). *Let  $U_0$  be normal and geometrically connected over  $\mathbb{F}_q$ , with base point  $u$ . Let  $G$  be the quotient of  $W(U_0, u)$  by a subgroup of finite index in  $\pi_1(U, u)$ , and let  $V$  be a virtual  $\ell$ -adic representation of  $G$ . Let  $V'$  be a virtual  $\ell'$ -adic representation of  $W(U_0, u)$ . If the characters of  $V$  and  $V'$  agree and are rational on powers of Frobenius elements, then  $V'$  is inflated from a virtual representation of  $G$ , and the two characters are equal and rational.*

The proof uses Chebotarev in the form that every element of sufficiently large degree in  $G$  is a Frobenius element. A character of  $G$  is determined by its restriction to elements of sufficiently large degree, via a Vandermonde independence argument.

**Lemma 20** (4.4.7). *If two virtual  $\ell$ -adic representations of  $W(U_0, u)$  have equal characters on powers of Frobenius elements, then their characters are equal.*

One passes to the Zariski closure  $G$  attached to the sum of the representations, cuts according to the  $\ell$ -adic valuations of eigenvalues of a central element of positive degree, twists to reduce to representations of  $\pi_1(U_0, u)$ , and applies Chebotarev density.

(4.4.8) Assume now the hypotheses of (4.4.1), with  $n = 2n'$  even,  $M$  finite, and  $\text{Ev}(Y) \neq 0$ . The intersection form on  $\text{Ev}(Y)$  corresponds, after the twist  $(n')$ , to a  $\mathbb{Q}_\ell$ -valued form; vanishing cycles  $\delta \in \text{Ev}(Y)(n')$  satisfy

$$(\delta, \delta) = (-1)^{n'} 2.$$

**Theorem 10** (4.4.9). *Under the hypotheses of (4.4.8):*

- (a) *the  $\mathbb{Z}$ -module  $L$  generated by the vanishing cycles is such that  $L \otimes \mathbb{Q}_\ell = \text{Ev}(Y)(n')$ , and the form is integral on  $L$ ;*
- (b) *the vanishing cycles  $\pm\delta$  form in  $L$  a root system  $R$  of type  $A$ ,  $D$ , or  $E$ ; the group  $M$  is the corresponding Weyl group.*

The reflections  $s_\delta$  are characterized intrinsically as the elements conjugate to nontrivial elements of local inertia groups. For two such reflections,

$$\text{Tr}(s_\delta s_{\delta'}) = (\dim \text{Ev}(Y) - 2) + (\delta, \delta')^2.$$

Thus  $\text{Tr}(s_\delta s_{\delta'}) - \dim \text{Ev}(Y) + 2$  has a square root in  $\mathbb{Q}_\ell$  for every  $\ell \neq p$ ; hence it is the square of an integer. This gives  $(\delta, \delta') \in \mathbb{Z}$ , proves integrality, and identifies  $R$  as an irreducible simply-laced root system.

## (4.5) Application: the gcd theorem

The conventions (0.7) are in force. The following theorem is used by Katz and Messing to compare  $\ell$ -adic and crystalline cohomology.

**Theorem 11** (4.5.1). *Let  $X_0$  be a smooth projective absolutely irreducible variety of dimension  $n + 1$  over  $\mathbb{F}_q$ , and let  $(X_t)_{t \in \mathbb{P}^1}$  be a Lefschetz pencil of hyperplane sections of  $X$ , defined over  $\mathbb{F}_q$ . If  $p = 2$  and  $n$  is even, assume the pencil sufficiently general. Let  $S \subset \mathbb{P}^1$  be the set of  $t$  for which  $X_t$  is singular. Then*

$$\det(1 - Ft, H^n(X))$$

*is the least common multiple of the polynomials  $f(T) = \prod_i (1 - \alpha_i T)$  having the following property: for every  $t \in \mathbb{P}^1 - S$  defined over  $\mathbb{F}_{q^k}$ , if  $F$  is Frobenius on  $X_t$  relative to its natural  $\mathbb{F}_{q^k}$ -structure, then*

$$\prod_i (1 - \alpha_i^k T) \quad \text{divides} \quad \det(1 - FT, H^n(X_t)).$$

Using the orthogonal decomposition

$$H^n(X_t) = H^n(X) \oplus \text{Ev}(X_t),$$

one sees that  $\det(1 - FT, H^n(X))$  has the required property. To prove minimality Deligne proves the following lemma.

**Lemma 21** (4.5.2). *Let*

$$P(T) = \prod_i (1 - \gamma_i T), \quad Q(T) = \prod_j (1 - \beta_j T),$$

*with all  $\gamma_i, \beta_j$  nonzero. If for every  $t \in \mathbb{P}^1 - S$  defined over  $\mathbb{F}_{q^k}$  one has*

$$P^{(k)}(T) \mid Q^{(k)}(T) \det(1 - FT, \text{Ev}(X_t)),$$

*where  $P^{(k)}(T) = \prod_i (1 - \gamma_i^k T)$  and similarly for  $Q$ , then  $P$  divides  $Q$ .*

It is enough to prove the lemma when  $P = 1 - \gamma T$ . Suppose by contradiction that  $\gamma$  is distinct from the  $\beta_j$ . For each  $j$ , let  $k_j$  generate the ideal of integers  $k$  for which  $\gamma^k = \beta_j^k$ , and let  $N$  be the product of the nonzero  $k_j$ . The hypothesis then gives, for  $k \equiv 1 \pmod{N}$ ,

$$1 - \gamma^k T \mid \det(1 - FT, \text{Ev}(X_t)). \quad (4.5.3)$$

Let  $u$  be a geometric point of  $\mathbb{P}^1 - S$ , and let  $G$  be the image of the arithmetic fundamental group in  $\text{GL}(E) \times \mathbb{Z}$ , where  $E = \text{Ev}(X_u)$  and the projection to  $\mathbb{Z}$  is the degree. For  $t \in (\mathbb{P}^1 - S)_0(\mathbb{F}_{q^k})$  the Frobenius element  $F_t$  has degree  $k$ , and

$$\det(1 - FT, \text{Ev}(X_t)) = \det(1 - p(F_t)T).$$

By Chebotarev, (4.5.3) implies:

$$\gamma^{\deg(g)} \text{ is an eigenvalue of } p(g) \quad (g \in G, \deg(g) \equiv 1 \pmod{N}). \quad (4.5.4)$$

It remains to show that this is impossible even for elements of degree one.

**Lemma 22** (4.5.4). *If  $g \in \text{GL}(E)$  normalizes the geometric monodromy group  $M$ , then the elements  $gm$  for  $m \in M$  have no common eigenvalue.*

One may replace  $M$  by its Zariski closure. This closure is either a symplectic group, an orthogonal group, or a Weyl group of type  $A$ ,  $D$ , or  $E$ . In the symplectic and orthogonal cases, after multiplying by a scalar and by an element of the group, the assertion reduces to the fact that the elements of the symplectic or orthogonal group have no common eigenvalue; already 1 and  $-1$  have none in common. In the Weyl case, one reduces to an automorphism of the Dynkin diagram. If the diagram automorphism is trivial, take a Coxeter transformation. If it is opposition, replace by its negative. The remaining cases are  $D_{2k}$  with an order-two diagram automorphism and  $D_4$  with an order-three automorphism; in the first, two explicit signed permutation matrices have disjoint spectra, and in the second, using the Hurwitz quaternion realization of  $D_4$ ,  $g$  and  $-g$  have no common eigenvalue. This proves the theorem.

## 5. Application to the $\mathbb{Q}_\ell$ -homotopy type

The aim of this section is to justify some of the results announced in [3]. The problem is to attach to each algebraic variety  $X$  over an algebraically closed field  $k$  a differential graded  $\mathbb{Q}_\ell$ -algebra  $A(X)$ , well-defined up to quasi-isomorphism and functorial in  $X$ , which is the  $\mathbb{Q}_\ell$ -homotopy type of  $X$ . In particular, one wants

$$H^*A(X) = H^*(X, \mathbb{Q}_\ell).$$

One also wants that, if  $k = \mathbb{C}$ , the minimal model of  $A(X)$  be obtained from the Sullivan minimal model of the rational homotopy type of the topological space  $X(\mathbb{C})$  by extending scalars from  $\mathbb{Q}$  to  $\mathbb{Q}_\ell$ .

The method which I had in mind for constructing  $A(X)$  when I wrote [3] has evaporated. The method used here depends essentially on a construction, due independently to Grothendieck and to Miller [8], described in (5.1).

### (5.1) The $\mathbb{Z}\{t\}$ -de Rham complex, after Grothendieck and Miller

(5.1.1) The algebra of polynomials with divided powers over  $\mathbb{Z}$  is the subring

$$\mathbb{Z}\{t\} \subset \mathbb{Q}[t]$$

with  $\mathbb{Z}$ -basis

$$\frac{t^n}{n!} \quad (n \geq 0).$$

The  $t$ -grading on  $\mathbb{Z}\{t\}$  is the one for which  $t^n/n!$  has degree  $n$ .

(5.1.2) We shall abbreviate “differential graded” as DG. DG algebras are always assumed commutative in the sense of the Koszul sign rule. The standard example is the DG algebra of exterior differential forms on a variety.

(5.1.3) Let  $A$  be a commutative ring and set

$$A\{t\} = A \otimes_{\mathbb{Z}} \mathbb{Z}\{t\}.$$

For  $A = \mathbb{Q}$ , this is just the polynomial algebra  $\mathbb{Q}[t]$ . A  $t$ -graded DG  $A\{t\}$ -algebra is a DG  $A\{t\}$ -algebra whose homogeneous components are themselves given a grading, the  $t$ -grading, making it a graded  $A\{t\}$ -module and such that the product and the differential  $d$  have  $t$ -degree 0.

(5.1.4) The construction of the  $\mathbb{Z}\{t\}$ -de Rham complex, defined below, is a contravariant functor

$$X \longmapsto \Omega(X)$$

from simplicial sets to  $t$ -graded DG  $\mathbb{Z}\{t\}$ -algebras. We write  $\Omega^p(X)$  for the part of exterior degree  $p$ . For a bihomogeneous element,

$$0 \leq \text{exterior degree} \leq t\text{-degree.} \quad (5.1.4.1)$$

There is moreover a functorial morphism of bigraded  $\mathbb{Z}\{t\}$ -algebras

$$H^*\Omega(X) \longrightarrow \bigoplus_{p \leq n} H^p(X, \mathbb{Z}) \cdot \frac{t^n}{n!}, \quad (5.1.4.2)$$

which identifies the cohomology of  $\Omega(X)$  with the direct sum on the right.

**(5.1.5)** More generally, for every commutative ring  $A$ —the cases needed below are  $A = \mathbb{Z}$ ,  $A = \mathbb{Z}/\ell^n$ , and  $A = \mathbb{Q}$ —there is a functor

$$X \longmapsto \Omega(X, A),$$

the  $A\{t\}$ -de Rham complex, satisfying the same formalism as (5.1.4), with  $\mathbb{Z}$  replaced by  $A$ . It is covariant in  $A$ , and the analogue of (5.1.4.2) is functorial in  $A$ .

For  $A = \mathbb{Q}_\ell$ , there is a morphism from  $\Omega(X, \mathbb{Q})$  to the  $t$ -graded DG  $\mathbb{Q}[t]$ -algebra obtained from the de Rham–Sullivan complex  $\Omega_{\text{DRS}}(X)$  by extension of scalars to  $\mathbb{Q}_\ell[t]$ . This morphism is compatible with (5.1.4.2). It identifies the part of  $t$ -degree  $n$  of  $\Omega(X, \mathbb{Q})$  with

$$\frac{t^n}{n!} \otimes \{\text{forms in } \Omega_{\text{DRS}}(X) \text{ of degree } \leq n \text{ in the barycentric coordinates of each simplex}\}.$$

If  $B$  is a quotient of  $A$ , the natural morphism

$$\Omega^p(X, A) \otimes_A B \longrightarrow \Omega^p(X, B)$$

is an isomorphism except possibly in the component of  $t$ -degree  $p$ .

**(5.1.6)** Some preliminaries are needed before the idea of the construction and then the construction itself are given.

**(5.1.6.1)** If  $a$  is a finite set, write

$$\mathbb{Z}\{(x_i)_{i \in a}\}$$

for the subring of  $\mathbb{Q}[(x_i)_{i \in a}]$  with  $\mathbb{Z}$ -basis the monomials

$$\prod_{i \in a} \frac{x_i^{r_i}}{r_i!}.$$

Let  $\Omega'(a)$  be the corresponding de Rham complex: it is the subcomplex of the de Rham complex of  $\mathbb{Q}[(x_i)_{i \in a}]$  which is freely generated over  $\mathbb{Z}\{(x_i)_{i \in a}\}$  by the products of  $dx_i$ 's. Put

$$t = \sum_{i \in a} x_i,$$

and let  $\Omega(a)$  be the quotient of  $\Omega'(a)$  by the ideal generated by  $dt$ . This is a  $t$ -graded DG  $\mathbb{Z}\{t\}$ -algebra if  $x_i$  and  $dx_i$  are assigned  $t$ -degree 1. It is called the  $t$ -graded DG  $\mathbb{Z}\{t\}$ -algebra of relative differential forms, relative to  $t$ , polynomial and integral with divided powers, on  $\mathbb{R}^a$ .

**(5.1.6.2)** For any ring  $A$ , set

$$\Omega(a, A) = \Omega(a) \otimes_{\mathbb{Z}} A.$$

**(5.1.6.3)** An intrinsic description is as follows.

a) If  $L$  is a free  $A$ -module, consider

$$\Omega(L) = \Gamma^*(L) \otimes \bigwedge^*(L),$$

where  $\Gamma^*$  denotes divided powers. The exterior degree is that of the exterior power, and the  $t$ -degree is the sum of the exterior and divided-power degrees. The differential, compatible with divided powers, is characterized by sending a generator in the divided-power factor to the corresponding generator in the exterior factor.

b) If  $M$  is a free quotient of  $L$ , consider the corresponding quotient

$$\Omega(L, M)$$

of  $\Omega(L)$ .

c) Then

$$\Omega(a, A) \simeq \Omega(A^a, A^a/A\Delta),$$

where  $A\Delta$  is the diagonal copy of  $A$  in  $A^a$ .

**(5.1.6.4)** The  $t$ -graded DG algebra  $\Omega(a)$  is a contravariant functor of  $a$ . This is clear from (5.1.6.3), since the pair

$$(\mathbb{Z}^a, \mathbb{Z}^a/\mathbb{Z}\Delta)$$

is contravariant in  $a$ . More concretely, to a map  $\varphi : v \rightarrow a$  one associates  $\varphi^*$ , sending  $x_i$  to  $\sum_{\varphi(j)=i} x_j$ . For example, if  $v \subset a$  and  $\varphi$  is the inclusion,  $\varphi^*$  is obtained by putting  $x_i = 0$  for  $i \notin v$ .

**(5.1.7)** Here is the idea of the construction in the parallel setting of Godement's "simplicial schemes." Start with a set  $S$ , and a collection  $\mathcal{S}$  of finite nonempty subsets of  $S$ , called simplexes. Suppose that every one-element subset of  $S$  is a simplex and that every nonempty subset of a simplex is a simplex. For  $a \in \mathcal{S}$ , let  $|a|$  be the usual simplex in  $\mathbb{R}^{(S)}$  spanned by the basis vectors  $e_i$ ,  $i \in a$ , and let  $X$  be the union of the  $|a|$ 's. Let  $\tilde{X}$  be the cone on  $X$ , the union of the cones with vertex 0 spanned by the  $|a|$ 's.

It is more convenient to consider the homotopy-equivalent spaces  $X'$ , the union of the affine spaces  $a'$  spanned by the  $|a|$ 's, and  $\tilde{X}'$ , the union of the vector spaces  $\tilde{a}'$  spanned by the  $|a|$ 's. Let

$$t : \mathbb{R}^{(S)} \longrightarrow \mathbb{R}$$

be the function "sum of coordinates." The space  $X'$  is the fiber over  $t = 1$  of the restriction  $\tilde{X}' \rightarrow \mathbb{R}$  of  $t$ .

For each  $a$ , let  $\Omega(a)$  be the  $t$ -graded DG  $\mathbb{Z}\{t\}$ -algebra of relative forms, relative to  $t$ , polynomial and integral with divided powers, on  $\tilde{a}'$ . By definition, a  $p$ -form in  $\Omega^p(X)$  is a family

$$\omega = (\omega_a)_{a \in \mathcal{S}}, \quad \omega_a \in \Omega^p(a),$$

such that, if  $v \subset a$ , then  $\omega_v$  is the restriction of  $\omega_a$ .

**(5.1.8)** Here is the formal definition. Restrict the functor  $a \mapsto \Omega(a)$  of (5.1.6.4) to the finite ordered sets

$$\Delta_n = [0, n]$$

and the increasing maps between them. This gives a simplicial  $t$ -graded DG  $\mathbb{Z}\{t\}$ -algebra  $\Omega_\Delta$ . Put

$$\Omega(X, A) = \text{Hom}(X, \Omega_\Delta \otimes A),$$

where this means the direct sum, over the exterior and  $t$ -degrees, of the groups of morphisms of simplicial sets from  $X$  to the corresponding bihomogeneous component of  $\Omega_\Delta \otimes A$ .

(5.1.9) For the proof of the fundamental properties of this construction, I refer to [8]. Let us only indicate that:

- a) the morphism of (5.1.5) to the de Rham–Sullivan complex is obtained by sending  $x_i$  to the barycentric coordinate  $x_i$  times  $t$ ;
- b) the key point is the calculation of the homology groups—or homotopy groups—of the simplicial group involved: each  $H_i$  is isomorphic to  $\mathbb{Z}$  and is placed in exterior degree and  $t$ -degree  $i$ . This calculation follows easily from (5.1.6.3) if one recalls that the simplicial group  $a \mapsto A^a$  is homotopic to 0: its normalized complex is reduced to  $A^{-1} \rightarrow A$  in homological degrees 0 and 1. One has

$$\Gamma^*(A^a) \simeq \Gamma^*(0) = A,$$

and  $\Lambda(A^a/A\Delta)$  is computed by Quillen’s decalage theorems: its normalized object is the symmetric algebra of the  $A$ -module  $A$ , with  $\text{Sym}^n$  placed in homological degree  $n$ .

## (5.2) The $\mathbb{Q}_\ell$ -homotopy type

(5.2.1) Let  $X$  be a filtered projective system of simplicial sets  $X^\alpha$ . Put

$$H^i(X, \mathbb{Z}/\ell^n) = \varinjlim_\alpha H^i(X^\alpha, \mathbb{Z}/\ell^n). \quad (5.2.1.1)$$

The relation (5.1.4.2) still holds between these groups and the  $t$ -graded DG algebra over  $\mathbb{Z}/\ell^n\{t\}$ :

$$\Omega(X, \mathbb{Z}/\ell^n) = \varinjlim_\alpha \Omega(X^\alpha, \mathbb{Z}/\ell^n), \quad (5.2.1.2)$$

by exactness of filtered inductive limits.

Assume now:

$$(F) \quad \text{the groups } H^i(X, \mathbb{Z}/\ell) \text{ are finite.}$$

The arguments of SGA 5, V, (5.3.1) apply and show that each

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n) \quad (5.2.1.3)$$

is a finitely generated  $\mathbb{Z}_\ell$ -module and that one has the usual exact sequences

$$0 \rightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \rightarrow H^i(X, \mathbb{Z}/\ell^n) \rightarrow \text{Tor}_{\mathbb{Z}_\ell}(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \rightarrow 0.$$

Put

$$\Omega(X, \mathbb{Z}_\ell) = \varprojlim_n \Omega(X, \mathbb{Z}/\ell^n), \quad (5.2.1.4)$$

a  $t$ -graded DG  $\mathbb{Z}_\ell\{t\}$ -algebra. The groups  $H^i(X, \mathbb{Z}/\ell^n)$  are finite, hence their projective system satisfies the Mittag–Leffler condition. In each bidegree  $(i, p)$ , except for  $i = p$ , the transition morphisms in  $\Omega^p(X, \mathbb{Z}/\ell^n)$  are surjective by (5.1.5), and the Mittag–Leffler condition is again satisfied. For  $i = p$ , one has an exact sequence

$$\Omega^{p-1}(X, \mathbb{Z}/\ell^n) \xrightarrow{d} \Omega^p(X, \mathbb{Z}/\ell^n) \longrightarrow H^p(X, \mathbb{Z}/\ell^n) \longrightarrow 0,$$

and hence again Mittag–Leffler, since cohomology satisfies it.

Thus passage to the projective limit in (5.2.1.4) is exact and gives a morphism

$$H^*\Omega(X, \mathbb{Z}_\ell) \longrightarrow H^*(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell\{t\}, \quad (5.2.1.5)$$

which identifies the cohomology of  $\Omega(X, \mathbb{Z}_\ell)$  with the direct sum of the

$$H^p(X, \mathbb{Z}_\ell) \cdot \frac{t^n}{n!} \quad (p \leq n).$$

It is essential in this construction that the inductive limit over  $\alpha$  has preceded the projective limit over  $n$ .

For any  $t$ -graded module  $M$  over  $\mathbb{Q}[t]$ , for instance over  $\mathbb{Q}_\ell[t]$ , write

$$\varinjlim_t M$$

for the inductive limit of the system of homogeneous components  $M_k$ , where the transition morphism from  $M_k$  to  $M_{k+1}$  is multiplication by  $t$ . Put

$$H^*(X, \mathbb{Q}_\ell) = H^*(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \quad (5.2.1.6)$$

and

$$A(X) = \varinjlim_t (\mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \Omega(X, \mathbb{Z}_\ell)). \quad (5.2.1.7)$$

It follows from (5.2.1.5), by passage to the limit, that

$$H^*A(X) = H^*(X, \mathbb{Q}_\ell). \quad (5.2.1.8)$$

**(5.2.2)** Let  $X$  be an algebraic variety over an algebraically closed field  $k$ . It can be defined over a countable algebraically closed subfield  $k_1 \subset k$ . Let  $X_1/k_1$  be such that  $X$  is obtained from  $X_1$  by extension of scalars. Since  $k_1$  is countable, one can find a countable projective system, indexed by the integers, of étale and surjective morphisms

$$X_\alpha \longrightarrow X_1$$

such that, for every étale and surjective morphism  $U \rightarrow X_1$ , there are an  $\alpha$  and a factorization of  $X_\alpha \rightarrow X_1$  through  $U$ .

For each  $\alpha$  and each finite set  $a$ , let

$$(X_\alpha/X_1)^a$$

be the fiber product over  $X_1$  of the family, indexed by  $a$ , of copies of  $X_\alpha$ . When  $a$  runs through the ordered finite sets  $[0, n]$ , these fiber products form a simplicial scheme. We take the associated projective system of simplicial sets and set

$$A(X) = A(\{(X_\alpha/X_1)^\bullet\}_\alpha). \quad (5.2.2.2)$$

It would have been more natural to take, instead of these fiber products, a projective cofinal system of étale hypercoverings of  $X_1$ ; according to Artin, this is essentially equivalent.

By the definition of étale cohomology, if one had used hypercoverings, and by Artin with the definition just chosen, one has

$$H^*(\{(X_\alpha/X_1)^\bullet\}_\alpha, \mathbb{Z}/\ell^n) = H^*(X_1, \mathbb{Z}/\ell^n).$$

On the other hand,

$$H^*(X_1, \mathbb{Z}/\ell^n) \simeq H^*(X, \mathbb{Z}/\ell^n).$$

The finiteness condition  $(F)$  is satisfied by SGA 4 $\frac{1}{2}$ , Théorème de finitude, (1.1). By definition of  $\ell$ -adic cohomology, one therefore obtains an isomorphism

$$H^*A(X) \simeq H^*(X, \mathbb{Q}_\ell).$$

**(5.2.3)** Suppose  $X$  is defined over a countable subfield  $k_0 \subset k$ , and take  $k_1$  to be the algebraic closure of  $k_0$  in  $k$ . One may then choose the projective system of the  $X_\alpha$ 's defined over  $k_0$ . The group  $\text{Gal}(k_1/k_0)$  then acts on the resulting system, hence, by transport of structure, on  $A(X)$ . This action is compatible with the action of Galois on the  $\ell$ -adic cohomology of  $X$ .

One should keep in mind that  $A(X)$  is enormous—this was the price to pay for its depending canonically on  $X$ . Consequently, one can hardly speak of continuity for the Galois action. One can only say that Galois acts continuously on each  $\Omega(X, \mathbb{Z}/\ell^n)$ , given the discrete topology, for every  $n$ .

**(5.2.4)** Now take  $k = \mathbb{C}$ , and let  $X_\alpha$  be a projective system as in (5.2.2). One can then find a projective system of spaces  $X^\alpha$  over  $X(\mathbb{C})$  and maps

$$a_\alpha : X^\alpha \longrightarrow X_\alpha(\mathbb{C})$$

such that each  $X^\alpha$  is the disjoint union of the open sets of an open covering of  $X(\mathbb{C})$ , and every finite intersection of these open sets is either empty or contractible.

Let  $X(\mathbb{C})$  denote the projective system of nerves

$$N^\alpha = \pi_0((X^\alpha/X(\mathbb{C}))^\bullet)$$

of the coverings  $X^\alpha$ . Each of these nerves has the homotopy type of  $X(\mathbb{C})$ . The projective system of the  $a_\alpha$ 's gives a morphism from  $X(\mathbb{C})$  to the étale system  $X$ .

**Construction (5.2.5)** We construct isomorphisms between the minimal models of the DG  $\mathbb{Q}_\ell$ -algebra  $A(X)$  and the tensor product with  $\mathbb{Q}_\ell$  of Sullivan's minimal models  $\mathcal{M}(X(\mathbb{C}))$  of the rational homotopy type of the topological space  $X(\mathbb{C})$ .

By SGA 4, XVI, (4.1), the morphism

$$\Omega(X, \mathbb{Z}/\ell^n) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}/\ell^n)$$

is a quasi-isomorphism: in cohomology it induces the isomorphisms deduced from the comparison isomorphisms

$$H^i(X, \mathbb{Z}/\ell^n) \simeq H^i(X(\mathbb{C}), \mathbb{Z}/\ell^n).$$

There are also quasi-isomorphisms

$$\Omega(N^\alpha, \mathbb{Z}/\ell^n) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}/\ell^n).$$

Passing to limits, one obtains quasi-isomorphisms

$$\Omega(X, \mathbb{Z}_\ell) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}_\ell) \longleftarrow \Omega(N^\alpha, \mathbb{Z}_\ell).$$

There are moreover quasi-isomorphisms

$$\Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Z}_\ell \longrightarrow \Omega(N^\alpha, \mathbb{Z}_\ell)$$

and

$$\varinjlim_t \Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Q} \longrightarrow \Omega_{\text{DRS}}(N^\alpha) \longrightarrow \Omega_{\text{DRS}}(X(\mathbb{C})).$$

Thus there is a chain of quasi-isomorphisms

$$A(X) \simeq A(X(\mathbb{C})) \longleftarrow (\varinjlim_t \Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Q}) \otimes \mathbb{Q}_\ell \longrightarrow \Omega_{\text{DRS}}(N^\alpha) \otimes \mathbb{Q}_\ell \longrightarrow \Omega_{\text{DRS}}(X(\mathbb{C})) \otimes \mathbb{Q}_\ell.$$

Passing to minimal models replaces these quasi-isomorphisms by isomorphisms.



### (5.3) Gradings by the weight

**(5.3.1)** Let  $A$  be a DG algebra over a field  $K$  of characteristic 0. For us  $K$  will be  $\mathbb{Q}$  or  $\mathbb{Q}_\ell$ . Assume  $H^0(A) = K$ , and that the  $H^i(A)$  are finite-dimensional. For the definition of the minimal model  $\mathcal{M} \rightarrow A$  of  $A$ , I refer to [6]. It follows easily from the inductive construction of  $\mathcal{M}$  that there is an increasing exhaustive filtration of  $\mathcal{M}$  by subalgebras of finite type  $\mathcal{M}_n$ ,  $n \geq 0$ , such that:

(\*) every automorphism of  $\mathcal{M}$  is homotopic to an automorphism preserving each  $\mathcal{M}_n$ .

If an automorphism  $\sigma$  of  $\mathcal{M}$  preserves this filtration, it makes sense to speak of its eigenvalues and of the grading of  $\mathcal{M}$  by generalized eigenspaces. After extension of scalars to an algebraic closure  $\bar{K}$ , the algebra  $\mathcal{M} \otimes_K \bar{K}$  is the direct sum of

$$\mathcal{M}(\alpha) = \bigcup_m \ker(\sigma - \alpha)^m, \quad \alpha \in \bar{K}^*,$$

and these subspaces form a grading of type  $\bar{K}^*$ .

Again by the inductive construction of  $\mathcal{M}$ , every eigenvalue of  $\sigma$  on  $\mathcal{M} \otimes \bar{K}$  is a product of eigenvalues of  $\sigma$  on the  $H^i(A) \otimes \bar{K}$ , and even on the spherical quotients of the  $H^i(A) \otimes \bar{K}$ . More precisely, an eigenvalue of  $\sigma$  on  $\mathcal{M}^n \otimes \bar{K}$  is a product

$$\prod_i \alpha_i$$

of eigenvalues  $\alpha_i$  of  $\sigma$  on  $H^{n(i)}(A)$ , with  $\sum_i n(i) > n$ .

Let  $\Gamma$  be the subgroup of  $\bar{K}^*$  generated by the eigenvalues of  $\sigma$ . If

$$v : \Gamma \longrightarrow \mathbb{Z}$$

is a homomorphism, the sums

$$\mathcal{M}(n) = \bigoplus_{v(\alpha)=n} \mathcal{M}(\alpha)$$

form a  $\mathbb{Z}$ -grading of  $\mathcal{M} \otimes \bar{K}$ . If, for every eigenvalue  $\alpha$  of  $\sigma$  on  $H^*(A)$ , one has  $v(\alpha) \geq 0$ , this grading is in nonnegative degrees. If, for every eigenvalue  $\alpha$  of  $\sigma$  on  $H^i(A)$ , one has  $v(\alpha) \geq i$ , then  $\mathcal{M}^n \otimes \bar{K}$  is entirely in degrees  $> n$ . If  $\Gamma$  is contained in a group  $\Gamma'$ , stable under  $\text{Gal}(\bar{K}/K)$ , and  $v$  extends to a Galois-invariant homomorphism

$$v' : \Gamma' \longrightarrow \mathbb{Z},$$

then this grading of  $\mathcal{M} \otimes \bar{K}$  is defined over  $K$ .

**(5.3.2)** The case of interest is the following. Let  $K = \mathbb{Q}_\ell$ . For a suitable integer  $q > 1$ , a power of a prime, one may take  $\Gamma'$  to be the group of algebraic numbers  $\alpha$  all of whose complex conjugates have the same absolute value, this absolute value being of the form  $q^{n(\alpha)/2}$  for an integer  $n(\alpha)$ . Put

$$v'(\alpha) = n(\alpha).$$

The  $\ell$ -adic gradings by weight mentioned in [3] are obtained in this way, taking  $\sigma$  to be a Frobenius. Here are some examples.

**(5.3.3)** Let  $X$  be a complex algebraic variety. It can be defined over a subfield  $k_0 \subset \mathbb{C}$  of finite type over  $\mathbb{Q}$ , i.e. it is obtained by extension of scalars from some  $X_0/k_0$ . This permits  $\text{Gal}(\bar{k}_0/k_0)$  to act on  $A(X)$ , defined as in (5.2). Let  $\mathcal{M}$  be the Sullivan minimal model of the rational homotopy type of the topological space  $X(\mathbb{C})$ . By (5.2.5),  $\mathcal{M} \otimes \mathbb{Q}_\ell$  is a minimal model of  $A(X)$ . More precisely, there is a natural homotopy class of quasi-isomorphisms

$$\varphi : \mathcal{M} \otimes \mathbb{Q}_\ell \longrightarrow A(X),$$

which induce in cohomology the natural isomorphisms

$$H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_\ell \simeq H^*(X, \mathbb{Q}_\ell).$$

For every automorphism  $s$  of  $A(X)$ , one therefore has an automorphism  $\tilde{s}$  of  $\mathcal{M} \otimes \mathbb{Q}_\ell$ , well-determined up to homotopy, such that  $s\varphi$  is homotopic to  $\varphi\tilde{s}$ . It may be chosen to preserve a filtration as in (5.3.1). This applies in particular to  $s \in \text{Gal}(\bar{k}_0/k_0)$ . If  $s$  is chosen to be a Frobenius as in [3], the hypotheses of (5.3.2) are satisfied by  $\tilde{s}$ , and the corresponding grading of  $\mathcal{M} \otimes \mathbb{Q}_\ell$  induces on cohomology a grading which splits the tensor product with  $\mathbb{Q}_\ell$  of the weight filtration of Hodge theory on  $H^*(X(\mathbb{C}), \mathbb{Q})$ . A Sullivan argument then gives a grading of  $\mathcal{M}$  with the same properties.

More precisely:

**Theorem 12** (5.3.4). *There exist gradings of  $\mathcal{M}$  compatible with the weight filtration of*

$$H^*(\mathcal{M}) = H^*(X(\mathbb{C}), \mathbb{Q})$$

*in the sense that*

$$W_n H^*(X(\mathbb{C}), \mathbb{Q}) = \bigoplus_{m \leq n} H^*(X(\mathbb{C}), \mathbb{Q})_m.$$

*Any two such gradings are conjugate by an automorphism of  $\mathcal{M}$  inducing the identity on*

$$\text{Gr}^W H^*(X(\mathbb{C}), \mathbb{Q}).$$

Choose a filtration  $(\mathcal{M}_n)$  of  $\mathcal{M}$  as in (5.3.1). Gradings of  $\mathcal{M}$  compatible with the filtration by the  $\mathcal{M}_n$ 's are the same as homomorphisms from  $\mathbb{G}_m$  into the pro-algebraic group

$$G_1 = \text{Aut}((\mathcal{M}, \mathcal{M}_n)) = \varprojlim_n \text{Aut}((\mathcal{M}_n)),$$

of automorphisms of  $\mathcal{M}$  preserving the filtration. Let  $G_2$  be the subgroup of  $G_1 \times \mathbb{G}_m$  formed by the couples  $(g, \lambda)$  such that  $g$  respects the weight filtration of  $H^*(X(\mathbb{C}), \mathbb{Q})$  and acts on  $\text{Gr}_i^W H^*$  as multiplication by  $\lambda^i$ . One checks that the kernel of the homomorphism

$$G_2 \longrightarrow \mathbb{G}_m, \quad (g, \lambda) \mapsto \lambda,$$

is unipotent:

$$1 \longrightarrow U \longrightarrow G_2 \longrightarrow \mathbb{G}_m \longrightarrow 1.$$

After extension of scalars to  $\mathbb{Q}_\ell$ , this morphism has a section: this is the existence of the gradings of (5.3.3). It is therefore surjective as a morphism of schemes over  $\mathbb{Q}$ . The pro-algebraic group  $G_2$  is thus, over  $\mathbb{Q}$ , an extension of  $\mathbb{G}_m$  by  $U$ . Such an extension is always trivial, and any two sections are  $U$ -conjugate. This proves existence and uniqueness when only gradings inducing a grading on every  $\mathcal{M}_n$  are considered.

To conclude, it remains to prove the following technical lemma, whose somewhat tedious verification is left to the reader.

**Lemma 23** (5.3.5). *For every finite set of automorphisms of  $\mathcal{M}$ , there exists a filtration as in (5.3.1) stable under all of these automorphisms.*

(5.3.6) For  $X$  an algebraic variety over an arbitrary algebraically closed field, one can again define a weight filtration on  $H^*(X, \mathbb{Q}_\ell)$  in terms of generalized eigenspaces of suitable Frobenius automorphisms. The preceding constructions give gradings of the minimal model of  $A(X)$ , inducing on cohomology a grading which splits the weight filtration.

**Corollary 12** (5.3.7). *If  $X$  is proper and smooth, a minimal model of  $A(X)$  is a minimal model of  $H^*(X, \mathbb{Q}_\ell)$ , provided with the DG structure for which  $d = 0$ .*

Let  $W$  be a grading by weights of a minimal model  $\mathcal{M}$  of  $A(X)$ . Since  $H^i(X, \mathbb{Q}_\ell)$  is pure of weight  $i$ ,  $\mathcal{M}^n$  is entirely in degrees  $> n$ , in the sense of (5.3.1). Hence

$$dW_n(\mathcal{M}^n) \subset W_n(\mathcal{M}^{n+1}) = 0.$$

Since  $W_n(\mathcal{M}^n)$  consists of cycles, there is a natural projection

$$W_n(\mathcal{M}^n) \longrightarrow H^n(X, \mathbb{Q}_\ell).$$

It is surjective, because  $H^n$  is pure of weight  $n$ . Let

$$\varphi_n : \mathcal{M}^n \longrightarrow H^n(X, \mathbb{Q}_\ell)$$

be the composite of this projection with the projection of  $\mathcal{M}^n$  onto  $W_n(\mathcal{M}^n)$ , which kills the  $W_i(\mathcal{M}^n)$  for  $i > n$ . It is easy to check that

$$\varphi = (\varphi_n)$$

is a quasi-isomorphism of DG algebras. The corollary follows.

(5.3.8) For  $X$  normal, one similarly obtains a theorem analogous to Morgan's theorem for the system of quotients of  $\pi_1(X)$  which are unipotent  $\ell$ -adic Lie groups; see [9].

Let us point out some questions left aside.

(5.3.9) For  $\mathcal{M}$  a minimal model and  $(\mathcal{M}_n)$  a filtration as in (5.3.1), let  $G$  be the pro-algebraic group of automorphisms of  $\mathcal{M}$  preserving the  $\mathcal{M}_n$ 's, and let  $H$  be the subgroup of those which are homotopic to the identity. Put

$$\mathrm{Aut}_h(\mathcal{M}) = G/H.$$

It is a pro-algebraic group.

If  $X$  is an algebraic variety over an algebraically closed field  $k$ , obtained by extension of scalars from  $X_0/k_0$  with  $k_0$  countable, the natural action of  $\mathrm{Gal}(\bar{k}_0/k_0)$  on  $A(X)$  gives a homomorphism

$$\mathrm{Gal}(\bar{k}_0/k_0) \longrightarrow \mathrm{Aut}_h(\mathcal{M})(\mathbb{Q}_\ell). \quad (5.3.9.1)$$

It would be necessary to verify that this homomorphism is continuous, i.e. that for every algebraic quotient  $Q$ , of finite type, of  $\mathrm{Aut}_h(\mathcal{M})$ , the homomorphism from  $\mathrm{Gal}(\bar{k}_0/k_0)$  to  $Q(\mathbb{Q}_\ell)$  deduced from (5.3.9.1) is continuous when  $Q(\mathbb{Q}_\ell)$  is equipped with its usual topology.

One can hope for more precise results, but I do not think there is any reason to expect a reasonable action of  $\mathrm{Gal}(\bar{k}_0/k_0)$  on  $\mathcal{M}$  itself.

(5.3.10) It would be necessary to verify that the increasing filtrations attached to the gradings by weight of (5.3.3) or (5.3.6) are all conjugate to each other under the group of automorphisms of  $\mathcal{M}$ , the  $\ell$ -adic minimal model, homotopic to the identity. In the setting of (5.3.3), where  $\mathcal{M}$  is the  $\ell$ -adic completion of Sullivan's rational minimal model  $\mathcal{M}_\mathbb{Q}$ , their conjugacy class should be defined over  $\mathbb{Q}$ , equivalently should be the orbit of a filtration deduced by extension of scalars from a filtration of  $\mathcal{M}_\mathbb{Q}$ .

## 6. The formalism of mixed sheaves

### (6.1) Stability

(6.1.1) In this section we establish, in the following contexts, that the category of mixed sheaves is stable under the usual operations.

- a) Equal characteristic  $p$ : we consider only schemes of finite type over  $\mathbb{F}_p$ ; the word sheaf means a Weil sheaf.
- b) Equal characteristic 0: more precisely, we consider schemes of finite type over  $\mathbb{Z}$ , and sheaves in the sense of constructible  $\mathbb{Q}_\ell$ -sheaves, while allowing ourselves at every instant to replace  $X$  by  $X[1/n]$ .

As regards schemes, this amounts to considering schemes of finite type over  $\mathbb{Q}$ : their category is the inductive 2-limit of the categories of schemes of finite type over the  $\mathbb{Z}[1/n]$ . For sheaves the situation is not the same. If  $X$  is of finite type over  $\mathbb{Z}$ , I do not know whether every constructible sheaf on  $X_{\mathbb{Q}}$  comes from a constructible sheaf on some  $X[1/n]$ . However, if  $\mathcal{F}$  and  $\mathcal{G}$  are two constructible sheaves on  $X$ , one has

$$\mathrm{Hom}_{X_{\mathbb{Q}}}(\mathcal{F}, \mathcal{G}) = \varinjlim_n \mathrm{Hom}_{X[1/n]}(\mathcal{F}, \mathcal{G}).$$

In other words, the inductive 2-limit of the categories of sheaves on the  $X[1/n]$  identifies with a full subcategory of the category of sheaves on  $X_{\mathbb{Q}}$ .

The following terminology gives uniform statements in cases a) and b). If  $X_{\mathbb{Q}}$  is a scheme of finite type over  $\mathbb{Q}$ , obtained by base change from a scheme  $X$  of finite type over  $\mathbb{Z}$ , and if  $\mathcal{F}_{\mathbb{Q}}$  is a sheaf on  $X_{\mathbb{Q}}$ , we say that  $\mathcal{F}_{\mathbb{Q}}$  is mixed, pure, etc., if it is obtained from a sheaf  $\mathcal{F}$  on some  $X[1/n]$ , and the restriction of  $\mathcal{F}$  to  $X[1/nm]$  is mixed, pure, etc., for a suitable  $m$ . The results in case b) will be stated as variants of those in case a). One could also give potential variants, in the sense of (3.4.10); we shall not spell them out.

**Theorem 13** (6.1.2). *Let  $f : X \rightarrow Y$  be a morphism of schemes of finite type over  $\mathbb{F}_p$  (respectively over  $\mathbb{Q}$ ), and let  $\mathcal{F}$  be a mixed sheaf on  $X$ . Then the sheaves  $R^i f_* \mathcal{F}$  are mixed.*

In view of the generic base-change theorem (SGA 4 $\frac{1}{2}$ , Th. finitude, (1.9)), this follows from the following lemma.

**Lemma 24** (6.1.3). *Let  $S$  be a scheme of finite type over  $\mathbb{Z}[1/\ell]$ , let  $f : X \rightarrow Y$  be a morphism of  $S$ -schemes of finite type, and let  $\mathcal{F}$  be a mixed sheaf on  $X$ . Then there is a dense open  $U \subset S$  over which the sheaves  $R^i f_* \mathcal{F}$  are mixed.*

We repeat the proof of SGA 4 $\frac{1}{2}$ , Th. finitude, (2.1)–(2.8).

(6.1.4) First suppose that  $X$  is smooth over  $Y = S$ , that  $\mathcal{F}$  is lisse, and that the  $R^i f_* \mathcal{F}$  are lisse. The sheaves  $R^i f_* \mathcal{F}$  are then obtained, by Poincaré duality, from the  $R^i f_! \mathcal{F}$ , to which (3.3.1) applies.

(6.1.5) If  $X$  is smooth over  $Y = S$  and  $\mathcal{F}$  is lisse, we reduce to (6.1.4) after replacing  $S$  by a suitable dense open.

(6.1.6) We prove by induction on  $n$  the assertion  $(*)_n$ : the conclusion of (6.1.3) holds when  $S$  is integral, with generic point  $\eta$ , when  $f$  is an open immersion with dense image, and when  $\dim X_{\eta} < n$ . For  $n = 0$ , after shrinking  $S$ , one has  $X = Y$ , and  $(*)_0$  is evident. Suppose  $(*)_{n-1}$ , and prove  $(*)_n$ . In  $(*)_{n-1}$ , one may replace “open immersion” by “immersion”, by factoring an immersion as an open immersion followed by a closed immersion.

**Lemma 25** (6.1.7). *After shrinking  $S$ , the conclusion of (6.1.3) holds over an open  $Y' \subset Y$ , whose complement  $Y_1$  is finite over  $S$ .*

The assertion is local on  $Y$ , and we may suppose  $Y$  affine, say  $Y \subset \mathbb{A}_S^n$ . The induction hypothesis applies to the projections. Thus, for each  $i$ , there is a dense open  $U_i$  of  $\mathbb{A}_S^{n-1}$  such that the conclusion of (6.1.3) holds over  $\text{pr}_i^{-1}(U_i)$ . It therefore holds over their union, which gives (6.1.7).

**(6.1.8)** We prove  $(*)_n$  when  $X$  is smooth over  $S$  and  $\mathcal{F}$  is lisse. Since the problem is local on  $Y$ , we may suppose  $Y$  affine, and then projective, by replacing  $Y$  by its closure in a projective space. Let

$$i : Y_1 \hookrightarrow Y, \quad j : Y' \hookrightarrow Y$$

be as in (6.1.7), after shrinking  $S$ . Formally applying  $Rb_*$  to the triangle defined by

$$0 \longrightarrow j_* j^* Rf_* \mathcal{F} \longrightarrow Rf_* \mathcal{F} \longrightarrow i_* i^* Rf_* \mathcal{F} \longrightarrow 0,$$

one obtains a triangle

$$Rb_* j_* j^* Rf_* \mathcal{F} \longrightarrow Ra_* \mathcal{F} \longrightarrow Rb_* i_* i^* Rf_* \mathcal{F} \longrightarrow .$$

The cohomology sheaves of the first two terms are mixed on a dense open of  $S$ , by (3.3.1) for the first term and by (6.1.5) for the second. Hence the cohomology sheaves of  $b_* i^* Rf_* \mathcal{F}$ , then those of  $i^* Rf_* \mathcal{F}$ , and finally those of  $Rf_* \mathcal{F}$ , are mixed. To justify this derived-category argument, one writes  $\mathcal{F}$  as coming from the projective limit of a projective system of locally constant sheaves of free  $R/m^n$ -modules, compatible by reduction, as in (1.1.1), and repeats the preceding argument for that system.

**(6.1.9)** We prove  $(*)_n$  in general. The use of a derived category is justified as above. We first reduce to the case where  $X$  contains a dense open  $V$  smooth over  $S$ . When  $S$  is the spectrum of a perfect field, it suffices to replace  $X$  by  $X_{\text{red}}$ . In general, after shrinking  $S$ , making a finite radicial surjective base change  $S' \rightarrow S$ , and replacing  $X$  and  $Y$  by their reductions, nothing is changed for the étale topology. Shrinking  $V$ , we may also suppose  $\mathcal{F}$  lisse on  $V$ . Apply (6.1.8) to

$$V \xrightarrow{j} X \xrightarrow{f} Y.$$

Define  $A$  by the triangle

$$\mathcal{F} \longrightarrow Rj_* j^* \mathcal{F} \longrightarrow A \longrightarrow .$$

The cohomology sheaves of  $A$  have support in  $X - V$ , and  $\dim(X - V)_\eta < n$ . By induction we may suppose  $Rf_* A$  mixed. Applying  $Rf_*$  to the triangle above, one obtains

$$Rf_* \mathcal{F} \longrightarrow R(fj)_* j^* \mathcal{F} \longrightarrow Rf_* A \longrightarrow ,$$

and two of the three terms are mixed. The third is therefore mixed.

**(6.1.10)** To prove (6.1.2), the problem is local on  $Y$ , which we may suppose affine. Taking an affine covering of  $X$  and using the Leray spectral sequence, we reduce to the case where  $X$  is also affine. We then factor  $f$  as an open immersion followed by a proper morphism,  $f = gj$ . Thus  $Rf_* = Rg_* Rj_*$ . The open immersion is covered by some  $(*)_n$ , and the proper morphism by (3.3.1).

**Corollary 13** (6.1.11). *The four operations  $R\mathcal{H}om$ ,  $Rf_*$ ,  $f^*$ , and  $Rf^!$  transform mixed sheaves into mixed sheaves. The same stability holds for local Ext sheaves.*

**(6.1.12)** We next deduce from (6.1.2) an analogous stability for sheaves of vanishing cycles. Let  $S$  be a smooth curve over  $\mathbb{F}_q$  (respectively over  $\mathbb{Q}$ ), let  $s$  be a closed point of  $S$ , let  $X$  be an  $S$ -scheme of finite type, and let  $\mathcal{F}$  be a sheaf on  $X$ . After the base change  $S(s) \rightarrow S$ , one obtains  $(X', \mathcal{F}')$  over the henselian trait  $S(s)$ . Let  $\eta$  be the generic point of  $S(s)$ , let  $\bar{s}$  be a geometric point above  $s$ , and let  $\bar{\eta}$  be a geometric generic point of the strict localization  $S_{(\bar{s})}$ . The sheaves of vanishing cycles  $R^i\Psi(\mathcal{F})$  are sheaves on  $X_{\bar{s}}$ , endowed with a continuous action of  $\text{Gal}(\bar{\eta}/\eta)$  above its action on  $X_{\bar{s}}$ , via  $\text{Gal}(\bar{s}/s)$ . The restriction of this action to inertia is automatically quasi-unipotent.

For a sheaf  $\mathcal{G}$  with such an action, and with quasi-unipotent inertia, we define purity and mixedness by passing to the unipotent part of the inertia action and to the corresponding nilpotent logarithm. With this convention, the vanishing-cycle sheaves of a mixed sheaf are mixed. This follows from (6.1.2), by the standard description of nearby cycles after replacing the trait by finite extensions and using the stability of mixed sheaves under direct image.

## (6.2) Pure complexes

In this section we place ourselves in the situation of (6.1.1) a).

**(6.2.1)** For every scheme  $X$  of finite type over  $\mathbb{F}_q$ , let

$$a : X \longrightarrow \text{Spec}(\mathbb{F}_q)$$

and put

$$K_X = Ra^!\mathbb{Q}_\ell.$$

This is the dualizing complex. If  $X$  is smooth and purely of dimension  $n$ , then

$$K_X \simeq \mathbb{Q}_\ell(n)[2n].$$

For  $K \in D_c^b(X)$ , define

$$DK = R\mathcal{H}om(K, K_X).$$

One has the usual formalism of duality:  $D$  exchanges  $Rf_!$  and  $Rf_*$ , and for proper  $f$ ,

$$DRf_*K \simeq Rf_*DK.$$

**(6.2.2)** Let  $K \in D_c^b(X)$ . We say that  $K$  is *punctually pure of weight  $n$*  if, for every  $i$ , the cohomology sheaf  $\mathcal{H}^i K$  is punctually pure of weight  $n + i$ . We say that  $K$  is *pure of weight  $n$*  if both  $K$  and  $DK$  are punctually pure of weight  $n$ .

**(6.2.3)** If  $K$  is punctually pure of weight  $n$ , then the usual truncations show that the spectral sequence of cohomology sheaves is compatible with weights. In particular, extensions between pieces of distinct forced weights split in the cases needed below, and the duality condition in the definition of purity is stable under the operations already considered.

**(6.2.4)** This definition extends the one for lisse sheaves. If  $X$  is smooth of pure dimension  $n$ , and if  $\mathcal{F}$  is a lisse sheaf punctually pure of weight  $m$ , then  $\mathcal{F}$  placed in degree 0 is pure of weight  $m$ , for

$$D\mathcal{F} \simeq \mathcal{F}^\vee(n)[2n],$$

and the shift and Tate twist have the expected effect on weights.

**(6.2.5)** Examples and elementary consequences are as follows.

- a) If  $K$  is pure of weight  $n$ , then  $K[r](s)$  is pure of weight  $n + r - 2s$ .
- b) If  $j : U \hookrightarrow X$  is an open immersion and  $\mathcal{F}$  is lisse, punctually pure of weight  $n$  on  $U$ , then in favorable cases  $j_!\mathcal{F}$  and  $j_*\mathcal{F}$  are pure of the expected weight, by duality.
- c) For  $X$  smooth,  $j : U \hookrightarrow X$  the complement of a smooth divisor  $D$ , and  $\mathcal{F}$  lisse, punctually pure of weight  $n$  on  $U$ , tamely ramified along  $D$ , the sheaf  $j_*\mathcal{F}$  is again pure of weight  $n$ .

**Proposition 8** (6.2.6). *If  $f : X \rightarrow Y$  is proper and  $K \in D_c^b(X)$  is pure, then  $Rf_*K$  is pure of the same weight.*

Apply (6.2.3) to  $K$  and to  $DK$ , using

$$DRf_*K = Rf_*DK$$

because  $f$  is proper.

**Variant 3** (6.2.7). *For every scheme  $X$  of finite type over  $\mathbb{Z}[1/\ell]$ , with*

$$a : X \longrightarrow \mathrm{Spec}(\mathbb{Z}[1/\ell]),$$

*put*

$$K_X = Ra^!\mathbb{Q}_\ell.$$

*Again one has a duality functor*

$$D'K = R\mathcal{H}om(K, K_X)$$

*and a definition of purity parallel to (6.2.4). If  $X$  is of finite type over  $\mathbb{F}_p$ , the two duality functors differ only by a shift and a twist, and the preceding definition of purity generalizes (6.2.4). Proposition (6.2.6) remains valid.*

**(6.2.8)** Let  $k$  be an algebraically closed field, let  $S$  be the spectrum of the henselization of  $k[T]$  at  $(T)$ , let  $s$  be its closed point,  $\eta$  its generic point, and  $\bar{\eta}$  a geometric generic point. Let  $f : X \rightarrow S$  be proper.

A complex  $K \in D_c^b(X)$  is called *potentially pure* if  $(X/S, K)$  comes from an arithmetic situation as follows:

- a)  $\mathrm{Spec}(k)$  is a geometric generic point of an integral scheme  $A_0$  of finite type over  $\mathbb{Z}[1/\ell]$ ;
- b)  $S$  comes, by extension of scalars to  $k$  and henselization, from a smooth curve  $S_0$  over  $A_0$  with a section;
- c)  $f$  comes from a proper morphism  $f_0 : X_0 \rightarrow S_0$ ;
- d)  $K$  comes from a pure complex  $K_0 \in D_c^b(X_0)$ .

The following theorem generalizes (3.6.1).

**Theorem 14** (6.2.9, local invariant-cycle theorem for pure complexes). *With the preceding notation, let  $K$  be a potentially pure complex on  $X$ . Then the specialization morphism*

$$\mathrm{sp}^* : H^*(X_{\bar{s}}, K) \longrightarrow H^*(X_{\bar{\eta}}, K)^{\mathrm{Gal}(\bar{\eta}/\eta)}$$

*is surjective.*

The proof is parallel to that of (3.6.1). One has the same cross diagram, with terms

$$H^{i-1}(X_{\bar{s}}, K)(-1), \quad H^i(X_{\bar{\eta}}, K), \quad H^i(X_{\bar{s}}, K),$$

and duality identifies the relevant upper and lower terms. The analogues of (3.6.2) and (3.6.3) are proved by replacing the references to (3.3.9) and (3.3.4) by references to (6.2.6) and (6.2.3). One concludes as in (3.6).

**(6.2.10)** In the remainder of this section the arguments are geometric, except for references to (6.2.9) and to (3.4). Let  $k$  be algebraically closed. For every scheme  $X$  of finite type over  $k$ , with  $a : X \rightarrow \operatorname{Spec}(k)$ , define

$$K_X = Ra^!\mathbb{Q}_\ell, \quad D(K) = R\mathcal{H}om(K, K_X).$$

Let  $X \subset P$  be projective. Hyperplane sections of  $X$  are parametrized by the dual projective space  $\check{P}$ , and each line in  $\check{P}$  parametrizes a pencil of hyperplane sections:

$$\begin{array}{ccc} & Z & \\ q \swarrow & & \searrow f \\ X & & \check{P}. \end{array}$$

For  $K \in D_c^b(X)$ , the sheaves  $R^i f_* q^* K$  are lisse on a dense open  $U \subset \check{P}$ . A hyperplane section  $Y_u$ , for  $u \in U$ , will be called general. The group  $\pi_1(U, u)$  acts on  $H^*(Y_u, K)$ . If a line  $D$  is sufficiently general and  $u \in V = U \cap D$ , the image of  $\pi_1(V, u)$  in  $\operatorname{GL} H^*(Y_u, K)$  coincides with that of  $\pi_1(U, u)$ . In particular,

$$H^*(Y_u, K)^{\pi_1(U, u)} = H^*(Y_u, K)^{\pi_1(V, u)}. \quad (6.2.10.2)$$

**Proposition 9** (6.2.11). *With the preceding notation, let  $n$  be an integer, and suppose that for every  $i$ , the support of  $\mathcal{H}^i(DK[-2n-2])$  has dimension  $< n+1-i$ . Then:*

i) *For every hyperplane section  $Y$ , the map*

$$H^i(X, K) \longrightarrow H^i(Y, K)$$

*is an isomorphism for  $i < n$ , and*

$$H^n(X, K) \longrightarrow H^n(Y, K)$$

*is injective.*

ii) *For a general pencil and for  $u \in D$ , the image of  $H^n(X, K)$  in  $H^n(Y_u, K)$  coincides with the image of the compactly supported cohomology of the base of the pencil with coefficients in the sheaf  $R^n f_*(i^* q^* K)$ .*

The proof is parallel to that of (4.1.6). One first checks weak Lefschetz,

$$H_Y^i(X, K) = 0 \quad (i < n).$$

By duality this vanishing is reduced to the assumed dimension condition on the supports of the cohomology sheaves of  $DK[-2n-2]$ . Assertion i) follows from the long exact cohomology sequence. For a sufficiently general hyperplane section  $Y$ , the restriction of  $K$  to  $Y$  again satisfies the hypothesis, with  $n$  replaced by  $n-1$ . This follows from generic acyclicity applied to  $Z/\check{P}$  and  $q^* K$ , using

$$D(K|Y) \simeq (DK)|Y(-1)[-2].$$

One then applies (4.3.7) as in (4.3.8), and concludes as in (4.3.9).

**Corollary 14** (6.2.12, global invariant-cycle theorem for pure complexes). *Under the hypotheses of (6.2.11), if  $K$  is potentially pure, then, for  $Y_u$  a general hyperplane section of  $X$ ,  $H^n(X, K)$  identifies with the subgroup*

$$H^n(Y_u, K)^{\pi_1(U, u)}$$

*of monodromy invariants.*



It suffices to verify this for a sufficiently general hyperplane section. For a sufficiently general pencil  $D$ , the complex  $i^*q^*K$  on the total space of the pencil is potentially pure. Applying (6.2.9) at each point of  $D - V$  gives the surjectivity onto the invariant part. By (6.2.11), the image of  $H^n(X, K)$  is exactly that invariant part.

**Theorem 15** (6.2.13). *Let  $X \subset P$  be a projective scheme, let  $\eta$  be the first Chern class of  $\mathcal{O}(1)$ , and let  $K \in D_c^b(X)$  be a potentially pure complex. Identify  $\mathbb{Z}_\ell$  with  $\mathbb{Z}_\ell(1)$  over  $k$ , so that  $\eta$  may be regarded as an element of  $H^2(X, \mathbb{Z}_\ell)$ . Let  $n$  be an integer, and suppose that  $K$  and  $DK[-2n]$  satisfy the following condition: for every  $i$ , the support of  $\mathcal{H}^i$  has dimension  $\leq n - i$  (with  $\dim \emptyset = -\infty$ ). Then, for every  $i \geq 0$ , the iterated cup product*

$$\eta^i : H^{n-i}(X, K) \longrightarrow H^{n+i}(X, K)$$

*is an isomorphism.*

The proof is parallel to that of (4.1.1), by induction on  $\dim X$ . If  $X$  has dimension zero and  $\mathcal{H}^i K \neq 0$ , the hypothesis forces  $i = n$ , so the assertion is immediate. If  $Y$  is a sufficiently general hyperplane section, the restriction of  $K$  to  $Y$  satisfies the same hypotheses with  $n$  replaced by  $n - 1$ . The induction reduces the problem to the morphism

$$H^{n-1}(X, K) \longrightarrow H^{n-1}(Y, K) \longrightarrow H^{n+1}(X, K).$$

The second arrow is transposed to the restriction morphism for  $DK[-2n]$ . It remains only to check that the perfect duality between the two cohomology groups on  $Y$  induces a perfect duality between the invariant parts. This follows from (4.1.4), since the representation of  $\pi_1(U, u)$  on  $H^*(Y_u, K)$  is semisimple by (6.2.6) and (3.4.1)(iii).

**Added in proof.** Let  $X$  be a complex algebraic variety, assumed irreducible and normal. The intermediate cohomology of MacPherson and Goresky can be defined as the hypercohomology of  $X$  with coefficients in a certain complex of sheaves  $I_X$ . The construction of  $I_X$  has an analogue in étale cohomology, over an arbitrary base field. O. Gabber has just proved the purity of  $I_X$ . His theorem allows one to apply the results of this section to MacPherson–Goresky cohomology.

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## Sigla

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# Integration over a Vanishing Cycle

P. Deligne

## Introduction

Let  $k$  be a commutative field and let  $f$  be a formal power series in two indeterminates  $x$  and  $y$  over  $k$ :

$$f = \sum a_{n,m} x^n y^m.$$

Let  $I(f)$  be the formal power series in one indeterminate  $t$

$$I(f) = \sum a_{n,n} t^n.$$

The aim is to prove the following result.

**Theorem 1.** *If  $f$  represents an algebraic function of  $x$  and  $y$ , and if  $k$  has characteristic  $p \neq 0$ , then  $I(f)$  represents an algebraic function of  $t$ .*

For a formal series in  $N$  indeterminates

$$g = \sum a_n x^n,$$

one defines similarly

$$I_N(g) = \sum a_{n,\dots,n} t^n,$$

and one has the following consequence.

**Corollary 1.** *If  $g$  is algebraic over the field  $k(x_1, \dots, x_N)$ , and if  $k$  has characteristic  $p > 0$ , then  $I_N(g)$  is algebraic over  $k(t)$ .*

Under the stronger hypothesis  $g \in k(x_1, \dots, x_N)$  this result is due to H. Furstenberg. The first purpose here is to give a geometric explanation of Furstenberg's result.

Suppose first that  $k = \mathbb{C}$  and that the formal series  $f$  is convergent. Then  $I(f)$  is convergent and has the integral representation

$$I(f)(t) = \frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt}. \quad (1)$$

Here  $\pi : \mathbb{C}^2 \rightarrow \mathbb{C}$  is  $(x, y) \mapsto xy$ . The expression  $(dx \wedge dy)/dt$  is a relative holomorphic differential on  $\mathbb{C}^2 - \{(0, 0)\}$ , relative to  $\pi$ ; it is also a section of the relative dualizing sheaf at  $(0, 0)$ , in the sense of Grothendieck duality. In the coordinate  $y$  on the fibre  $xy = t$  one has

$$\frac{dx \wedge dy}{dt} = \frac{d(t/y) \wedge dy}{dt} = \frac{dy}{y},$$

up to the sign fixed by the chosen order of the variables.

If  $f$  converges in a polydisc of radius  $R$ , and if  $|t| < R^2$ , the cycle of integration is  $|y| = r$ ,  $|x| = |t|/r$ ,  $xy = t$ , with  $|t|/R < r < R$ . The integral is independent of  $r$ . Thus

$$\frac{1}{2\pi i} \int_{xy=t} f(x, y) \frac{dx \wedge dy}{dt} = \frac{1}{2\pi i} \int f(t/y, y) \frac{dy}{y},$$

and expansion into the series followed by the residue formula leaves only the terms with equal exponents in  $x$  and  $y$ .

The integration cycle is the Lefschetz vanishing cycle for  $\pi$  near  $(0, 0)$ : if  $r = |t|^{1/2}$  the cycle visibly collapses as  $|t| \rightarrow 0$ . This explains the title.

Formula (1) suggests that the operation  $I$  has an intrinsic meaning for a morphism of formal schemes  $\pi : X \rightarrow S$  over  $k$  with  $X \simeq \mathrm{Spf} k[[x, y]]$ ,  $S \simeq \mathrm{Spf} k[[t]]$ , and  $\pi$  having a non-degenerate quadratic singularity. The two branches of  $\pi^{-1}(0)$  are assumed to be defined over  $k$ , and an order between them is chosen. Thus we choose coordinates with  $\pi(x, y) = xy$ , with the first branch of  $xy = 0$  being  $x = 0$ .

Let  $\Omega_{X/k}^1$  be the Kähler differentials defined with respect to the adic topology. In coordinates  $\Omega_{X/k}^1$  is freely generated by  $dx, dy$ , and  $\Omega_{S/k}^1$  by  $dt$ . Let

$$\omega_{X/k} = \det \Omega_{X/k}^1, \quad \omega_{S/k} = \Omega_{S/k}^1, \quad \omega_{X/S} = \omega_{X/k} \otimes \pi^* \omega_{S/k}^{-1}.$$

In coordinates  $\omega_{X/S}$  is generated by  $(dx \wedge dy)/dt$ . The operation  $I$  will be interpreted as a morphism

$$\pi_* \omega_{X/S} \longrightarrow \mathcal{O}_S,$$

or, for  $S = \mathrm{Spf}(V)$ , as a  $V$ -linear map

$$H^0(X, \omega_{X/S}) \longrightarrow V.$$

Section 1 provides this interpretation algebraically. The integration sign in (1) is replaced by Grothendieck's local trace. For a uniformizer  $t$  of  $S$ , the reduction modulo  $t^n$  is denoted with the index  $n$ . Let  $A_n$  be the complete local ring whose formal spectrum is the reduction  $X_n$ ; in coordinates  $A_n = k[[x, y]]/((xy)^n)$ . It is convenient to replace  $X_n$  by the ordinary scheme  $X'_n = \mathrm{Spec} A_n$ . Let  $O$  be its closed point. Grothendieck's local trace is a morphism

$$\mathrm{Tr} : H_{\{O\}}^1(X'_n, \omega) \longrightarrow H^0(S_n, \mathcal{O}).$$

The role of the vanishing cycle is played by an element

$$v_n \in H_{\{O\}}^1(X'_n, \mathcal{O}).$$

In coordinates  $v_n$  comes from  $\mathrm{Ext}_{A_n}^1(A_n/(x+y)^{2n-1}, A_n)$ . In Section 2 these classes are related to an element of the Lie algebra of a Picard scheme. Section 3 globalizes the construction and proves the theorem and the corollary.

## 1 Algebraic analogue of the integral formula

Keep the notation above. The inclusion  $X'_1 \subset X'_n$  is a homeomorphism, since  $X'_1$  is defined by a nilpotent ideal in  $X'_n$ . The complement of the origin in  $X'_n$  has two connected components corresponding to the branches  $x = 0$  and  $y = 0$  of  $X'_1$ . Let  $u_n$  be the section of the structure sheaf of  $X'_n - \{O\}$  which is 1 on the first branch and 0 on the second.

Algebraically, the homomorphism

$$A_n = k[[x, y]]/((xy)^n) \rightarrow k[[x, y]]/(x^n) \times k[[x, y]]/(y^n)$$

induces an isomorphism after inverting  $x + y$ , and  $u_n$  is the element whose image is  $(1, 0)$ . Let  $v_n$  be the image of  $u_n$  under the boundary map

$$H^0(X'_n - \{O\}, \mathcal{O}) \longrightarrow H_{\{O\}}^1(X'_n, \mathcal{O}).$$

Applying  $\mathrm{Ext}^\bullet(-, A_n)$  to

$$0 \rightarrow ((x + y)^N) \rightarrow A_n \rightarrow A_n/((x + y)^N) \rightarrow 0$$

gives a diagram showing that, for  $N = 2n - 1$ ,  $v_n$  lies in the image of

$$\mathrm{Ext}_{A_n}^1(A_n/((x + y)^{2n-1}), A_n).$$

**1.2. Proposition.** For  $f \in k[[x, y]]$  and for every  $n$ ,

$$\mathrm{Tr} \left( f \frac{dx \wedge dy}{dt}, v_n \right) = -I(f) \pmod{t^n}. \quad (1.2.1)$$

The trace is Grothendieck's local trace. We prove the formula when  $k$  is infinite; the general case follows by extension of scalars. Let  $I'(f)$  denote the left-hand side of (1.2.1). It is a continuous  $k[[t]]/(t^n)$ -linear map  $A_n \rightarrow k[[t]]/(t^n)$ . For  $\lambda, \mu \in k^*$  let  $\sigma(\lambda, \mu)$  be the automorphism

$$(x, y, t) \mapsto (\lambda x, \mu y, \lambda \mu t).$$

It preserves  $(dx \wedge dy)/dt$  and  $v_n$ , hence

$$I'(\sigma(\lambda, \mu) f) = \sigma(\lambda, \mu)^* I'(f). \quad (1.2.2)$$

Applying this to monomials gives  $I'(x^a y^b) = 0$  unless  $a = b$ . By continuity,

$$I'(f) = I(f)I'(1).$$

The scalar  $I'(1)$  is independent of  $n$ , so it remains to compute it for  $n = 1$ .

For  $n = 1$ ,  $X'_1 = \mathrm{Spec} k[[x, y]]/(xy)$  is the union of two smooth formal arcs meeting transversely. Sections of  $\omega$  are pairs of meromorphic differentials  $(\alpha, \beta)$  on the two branches with at most simple poles at  $O$  and residues summing to zero. On the punctured curve arbitrary pole orders are allowed. The composite

$$H^0(X'_1 - \{O\}, \omega) \rightarrow H^1_{\{O\}}(X'_1, \omega) \xrightarrow{\mathrm{Tr}} k$$

is the negative of  $(\alpha, \beta) \mapsto \mathrm{Res} \alpha + \mathrm{Res} \beta$ . In these terms  $(dx \wedge dy)/dt$  and  $u_1$  give residue sum 1, proving (1.2.1).

## 2 Relation with the Picard functor

Let  $S = \mathrm{Spec} k[[t]]$ , let  $s$  be its closed point and  $\eta$  its generic point. Let  $\pi : Y \rightarrow S$  be proper and flat, purely of relative dimension one, and let  $O \in Y_s(k)$ . Assume that the completion of  $Y$  at  $O$  is the formal singularity considered above. Thus  $Y$  is regular at  $O$ ,  $O$  is an isolated non-smooth point of  $\pi$ , the singularity is non-degenerate quadratic, and the two branches are defined over  $k$  and ordered.

We may further suppose  $Y$  regular and connected. Let  $Y_\eta$  be the generic fibre over  $k((t))$ .

**2.2. Lemma.** The curve  $Y_\eta$  is absolutely irreducible and generically smooth over  $k((t))$ .

Indeed, the special fibre is connected and has a rational point, hence is absolutely connected. It is smooth near  $O$  away from  $O$ , so after a separable extension it has smooth rational points. The corresponding base change has a section; regularity then implies irreducibility, and absolute connectedness gives absolute irreducibility. The smooth locus of the generic fibre is non-empty and therefore dense.

Let  $C$  be a complete reduced generically smooth curve over a field  $K$ , and  $D$  a Cartier divisor on  $C$ . The generalized Jacobian  $\mathrm{Pic}(C; D)$  represents invertible sheaves on  $C_T$  trivialized on  $D_T$ . It is smooth, and

$$\mathrm{Lie} \mathrm{Pic}(C; D) = H^1(C, \mathcal{O}(-D)). \quad (2.3.1)$$

If  $C$  is smooth, the map

$$j : C - D \rightarrow \text{Pic}(C; D), \quad u \mapsto [\mathcal{O}(u(T))] \quad (2.3.2)$$

relates Serre duality with the pullback of invariant differentials on the Picard scheme.

Returning to  $Y/S$ , let  $Z$  be a relative Cartier divisor disjoint from  $O$ , and let  $\text{Pic}(Y; Z)^0$  be the relative generalized Picard scheme parametrizing line bundles on  $Y$  trivialized along  $Z$  and of degree zero on each geometrically irreducible component of each fibre. For each  $n$  one constructs morphisms

$$H^0(\mathcal{O}_T, \mathcal{O}_T) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(-Z_{n,T})), \quad (2.4.6)$$

$$H^0(\mathcal{O}_T, \mathcal{O}_T^*) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(1 \text{ on } Z_{n,T})^*). \quad (2.4.7)$$

The multiplicative version gives a morphism of  $k[[t]]/(t^n)$ -group schemes

$$w_n : \mathbf{G}_m \longrightarrow \text{Pic}(Y_n, Z)^0. \quad (2.4.8)$$

For  $m < n$ , the reduction of  $w_n$  modulo  $t^m$  is  $w_m$ ; for  $n = 1$  this is the morphism associated with the two branches.

Although one cannot take a direct limit to get a morphism  $\mathbf{G}_m \rightarrow \text{Pic}(Y; Z)^0$ , one can restrict to finite subgroup schemes. For every integer  $m$  one obtains

$$w : \mu_m \longrightarrow \text{Pic}(Y, Z)^0. \quad (2.4.9)$$

Passing to Lie algebras gives

$$dw : \text{Lie } \mathbf{G}_m \longrightarrow \text{Lie } \text{Pic}(Y; Z)^0, \quad (2.4.10)$$

and hence an element

$$dw(1) \in H^1(Y, \mathcal{O}(-Z)). \quad (2.4.11)$$

Let  $\omega_{Y/S}$  be the relative dualizing sheaf and let  $\varphi$  be a section of  $\omega_{Y/S}(Z)$ . In formal coordinates near  $O$ ,

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

The restriction of  $\varphi$  to the generic fibre is paired with  $H^1(Y_\eta, \mathcal{O}(-Z_\eta))$  by Serre duality.

**2.5. Proposition.** With the notation above,

$$I(f) = -(dw(1), \varphi). \quad (2.5.1)$$

The right-hand side lies in  $k[[t]]$ , not merely in  $k((t))$ . Reducing modulo  $t^n$ , the local trace and the global Serre trace are compatible; the proposition is reduced to the following lemma.

**2.6. Lemma.** The morphism from local cohomology at  $O$  to  $H^1(Y_n, \mathcal{O}(-Z_n))$  sends  $v_n$  to  $dw_n(1)$ .

The image of  $v_n$  is also the image of 1 under (2.4.6). The assertion follows from the description of  $dw_n$  via dual numbers over  $k[[t]]/(t^n)$  and the natural exact sequence comparing the additive and multiplicative constructions.

### 3 Proof of the theorem

Let  $S_0$  be a smooth curve over  $k$ , let  $s \in S_0(k)$ , and let  $S$  be the completion at  $s$ . Let  $\pi : Y_0 \rightarrow S_0$  be proper and flat, purely of relative dimension one, with  $Y$  its base change to  $S$  and with a point  $O \in Y_s$  such that  $(Y/S, O)$  is as above. We suppose, for simplicity, that  $Y_0$  is regular and that  $\pi$  has geometrically connected fibres. Let  $\omega_{Y_0/S_0}$  be the relative dualizing sheaf.

Let  $\varphi$  be a rational section of  $\omega_{Y_0/S_0}$ , regular at  $O$ . In formal coordinates it can be written

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

**3.2. Proposition.** If  $k$  has characteristic  $p > 0$ , then  $I(f)$  is algebraic over the field  $k(S_0)$  of rational functions on  $S_0$ .

After multiplying  $\varphi$  by a rational function on  $S_0$  vanishing sufficiently at  $s$ , and after shrinking  $S_0$ , we may assume that for a suitable relative Cartier divisor  $Z_0$  disjoint from  $O$ , the form  $\varphi$  is a section of  $\omega$  on  $Y_0 - Z_0$ . Let  $\text{Pic}(Y_0; Z_0)^0$  be the relative generalized Picard scheme. Via Serre duality,  $\varphi$  corresponds to a section of the dual of the vector bundle

$$\text{Lie Pic}(Y_0; Z_0)^0 = R^1 \pi_* \mathcal{O}(-Z_0).$$

After base change to  $S$  we have  $dw : \mathcal{O} \rightarrow \text{Lie Pic}(Y; Z)^0$ , and Proposition 2.5 reduces Proposition 3.2 to the following key lemma.

**3.3. Key Lemma.** If  $p > 0$ , then  $dw$  is algebraic over  $S_0$ .

In characteristic  $p$ , the inclusion  $\mu_p \subset \mathbf{G}_m$  induces an isomorphism on Lie algebras:

$$\text{Lie}(\mu_p) \xrightarrow{\sim} \text{Lie}(\mathbf{G}_m).$$

Modulo  $t^n$ , one checks that  $dw$  is the differential of the morphism

$$w : \mu_p \rightarrow \text{Pic}(Y; Z)^0.$$

It remains to use the rigidity of  $\mu_p$ : for every flat commutative group scheme  $P$  over  $S_0$ , the functor  $\text{Hom}(\mu_p, P)$  is represented by an etale scheme over  $S_0$ . Applying this to  $P = \text{Pic}(Y_0; Z_0)^0$ , the morphism  $w$  is pulled back from a universal morphism over an etale cover of  $S_0$ . Hence its differential, and therefore  $dw$ , is algebraic.

Let  $k\{x_1, \dots, x_N\}$  be the henselization of the local ring of affine  $N$ -space at the origin. Its completion is  $k[[x_1, \dots, x_N]]$ , and an element of the completion is algebraic over  $k(x_1, \dots, x_N)$  if and only if it lies in the henselization.

### Proof of the theorem

If  $f$  is algebraic, there exists a surface  $Y_0$  with a rational point  $O$ , an etale morphism  $h : Y_0 \rightarrow \mathbf{A}^2$  sending  $O$  to the origin, and a section whose formal expansion at  $O$  is  $f$ . Let  $S_0 = \mathbf{A}^1$  with coordinate  $t$ , and let  $\pi'$  be the composite

$$Y_0 \xrightarrow{h} \mathbf{A}^2 \xrightarrow{xy} \mathbf{A}^1 = S_0.$$

The differential form  $f h^*((dx \wedge dy)/dt)$  is a rational section of the relative dualizing sheaf. After compactifying and resolving singularities, one obtains a proper flat regular model to which Proposition 3.2 applies. Therefore  $I(f)$  is algebraic.

## Proof of the corollary

Proceed by induction on  $N$ . The cases  $N = 0$  and  $N = 1$  are immediate, and  $N = 2$  is the theorem. For  $N > 2$ , write

$$g = \sum g_{n,m} x_{N-1}^n x_N^m, \quad g_{n,m} \in k[[x_1, \dots, x_{N-2}]],$$

and put

$$I(g) = \sum g_{n,n} x_{N-1}^n.$$

Then

$$I_N(g) = I_{N-1}(I(g)). \quad (3.6.1)$$

If  $g$  is algebraic, the coefficients  $g_{n,m}$  are algebraic over  $k(x_1, \dots, x_{N-2})$ . Applying the theorem over the fraction field of the henselization in the first  $N - 2$  variables shows that  $I(g)$  is algebraic, and induction proves the claim.

## Degree estimates and remarks

The proof gives an estimate. With the notation of the geometric construction, let  $\alpha$  be the degree of the curve  $S_0$  over the  $t$ -line, let  $g$  be the genus of the normalization of the geometric generic fibre, and let  $z = \max(0, z' - 1)$  where  $z'$  is the number of geometric points of the divisor  $Z$ . Then the degree  $d$  of the equation satisfied by  $I(f)$  is bounded by

$$d < \alpha p^{g+z}. \quad (3.7.1)$$

For a formal algebraic series with integer coefficients, this implies that for almost all primes  $p$ , the degree  $d_p$  of the reduction of  $I(f)$  modulo  $p$  satisfies

$$d_p < \alpha p^{g+z}. \quad (3.7.2)$$

Using  $\mu_{p^m}$  when  $p^m = 0$  on the base gives analogous information modulo powers of  $p$ , for almost all primes.

A direct proof for  $N$  variables would be preferable. Analytically, for a convergent series  $g$  one expects the formula

$$I(g)(t) = \int_{Z(t)} g \frac{dz_1 \cdots dz_N}{dt}, \quad (1')$$

where

$$Z(t) = \{(x_1, \dots, x_N) : x_1 \cdots x_N = t, |x_i| = r_i, \prod r_i = |t|\}.$$

A direct proof of this kind would be a test case for a  $p$ -adic theory of vanishing cycles.

Furstenberg's proof in the rational case shows that if  $g$  is one component of a column vector  $h$  with entries in  $k[[x_1, \dots, x_N]]$  satisfying

$$h = Ah^p$$

for a square matrix  $A$  with polynomial entries, then  $I(g)$  has the same property and is algebraic. This type of equation recalls  $F$ -crystals in crystalline cohomology. Since  $T_p \text{ Pic}$  is an avatar of  $p$ -adic  $H^1$ , the proof here and Furstenberg's proof are perhaps less distant from each other than they first appear.



## A Signs

The treatment of signs in the duality formalism for coherent algebraic sheaves is delicate. We record the conventions used here. For a flat relative complete intersection morphism  $f : X \rightarrow S$ , purely of relative dimension  $n$ , the relative dualizing sheaf  $\omega_{X/S}$  is related to the relative dualizing complex by

$$K_{X/S} = \omega_{X/S}[n]. \quad (\text{S.1})$$

When  $X$  is smooth over  $S$ ,  $\omega_{X/S} = \Omega_{X/S}^n$ . If  $X$  is the fibre product over  $S$  of morphisms  $f_i : X_i \rightarrow S$  of relative dimensions  $n_i$ , then

$$K_{X/S} = \boxtimes_i K_{X_i/S}. \quad (\text{S.2})$$

Equivalently, after choosing a total order on the index set,

$$\omega_{X/S} = \bigotimes_i \omega_{X_i/S}, \quad (\text{S.3})$$

with the sign change  $(-1)^N$  when the order is changed, where  $N$  is the sum of  $n_i n_j$  over inverted pairs.

For  $f : X \rightarrow S$  proper as above, the trace is

$$\text{Tr} : Rf_* K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.4})$$

It induces

$$\text{Tr} : R^n f_* \omega_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.5})$$

For a section  $s$  of  $f$  there is a local trace

$$\text{Tr} : Rs^! K_{X/S} \rightarrow \mathcal{O}_S. \quad (\text{S.6})$$

The choices are fixed by requiring compatibility between global and local traces and by normalizing, for a smooth curve  $X$  over an algebraically closed field and a point  $x \in X(k)$ , the local trace

$$H_{\{x\}}^1(X, \Omega^1[1]) \rightarrow k \quad (\text{S.7})$$

to be the residue map  $\text{Res}_x$ .

With these conventions, for a smooth complete curve  $X$  and  $x \in X(k)$ , the boundary map

$$H^0(X - \{x\}, \Omega^1) \rightarrow H_{\{x\}}^1(X, \Omega^1)$$

composed with the trace is  $-\text{Res}_x$ . For products, if  $X = X_1 \times X_2$  and  $u_i \in H^{n_i}(X_i, \omega_{X_i})$ , then

$$\text{Tr}(\text{pr}_1^* u_1 \smile \text{pr}_2^* u_2) = (-1)^{n_1 n_2} \text{Tr}(u_1) \text{Tr}(u_2).$$

Over  $k = \mathbf{C}$ , comparison with the transcendental trace identifies the algebraic trace with integration over  $X(\mathbf{C})$ , with the standard normalization by  $(2\pi i)^{-n}$ . For projective space  $\mathbf{P}^n$  with homogeneous coordinates  $T_0, \dots, T_n$  and  $t_i = T_i/T_0$ , the Čech class of

$$\frac{dt_1 \wedge \dots \wedge dt_n}{t_1 \dots t_n}$$

has trace  $(-1)^{n(n+1)/2}$  with these conventions.

## References

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# The Lemma of Gabber

Pierre Deligne

The principal result of this exposé, Theorem 1.4, is due to O. Gabber.

## 1 Statements

**1.1** In this exposé we systematically work with algebraic spaces, rather than only with schemes. This raises a problem of references. We shall allow ourselves to refer to an article or theorem proved for schemes and apply it to algebraic spaces whenever the same proof also proves the assertion for algebraic spaces. A proof becomes problematic when it uses the fact that every point has an affine neighbourhood and when “neighbourhood” cannot be replaced by “étale neighbourhood”.

This leads us to speak of an *abelian algebraic space*  $A$  over  $S$ , where one would ordinarily speak of an abelian scheme over  $S$ : this means a commutative  $S$ -algebraic space in groups

$$f : A \longrightarrow S$$

with  $f$  proper, flat, and of finite presentation, and with geometric fibres abelian varieties. We shall call  $A$  a *semi-abelian algebraic space* over  $S$  if it is a commutative  $S$ -algebraic space in groups  $f : A \rightarrow S$ , with  $f$  separated, flat, of finite presentation, and with geometric fibres extensions of abelian varieties by tori.

Note, however, the following points.

- (a) If  $S$  is the spectrum of a field, every algebraic space in groups over  $S$  is a scheme, and indeed is quasi-projective.
- (b) If  $S$  is a trait, every algebraic space in groups  $G$  over  $S$ , locally of finite type and separated over  $S$ , is a scheme. This is proved for the smooth case in Raynaud’s work on Néron models.
- (c) If  $S$  is normal integral, with generic point  $\eta$ , and if  $A$  is an abelian algebraic space over  $S$ , Grothendieck’s arguments show that every polarization of  $A_\eta$  extends uniquely to a polarization of  $A$  over  $S$ . It follows that  $A$  is projective over  $S$ , hence in particular that  $A$  is a scheme if  $S$  is a scheme.

An algebraic space  $f : X \rightarrow S$ , flat of finite presentation, separated over  $S$ , and with proper and connected geometric fibres, is proper over  $S$ . By localization on  $S$  and passage to limits, one reduces to the case where  $S$  is the spectrum of a noetherian local ring; by faithfully flat descent one may suppose the ring complete, and then one applies EGA III, first part, 5.5. In particular, a semi-abelian algebraic space over  $S$  all of whose fibres are abelian varieties is abelian over  $S$ .

Unless explicitly stated otherwise, all schemes and algebraic spaces considered below are quasi-compact and quasi-separated.

**1.2** Let  $U$  be a dense open subspace of a normal algebraic space  $S$ , and let  $A$  be an abelian algebraic space over  $U$ . A *semi-abelian prolongation* of  $A$  over  $S$  is a semi-abelian algebraic space  $P$  over  $S$  together with an isomorphism

$$A \xrightarrow{\sim} P|_U.$$

**Theorem 1.1** (Raynaud). *Under the preceding hypotheses, if  $A$  admits a semi-abelian prolongation, then that prolongation is unique up to unique isomorphism.*

Localization and passage to limits reduce the proof to the case where  $S$  is a noetherian scheme. If  $S$  is a trait and  $U$  is its generic point, every semi-abelian prolongation  $P$  of  $A$  coincides with the identity component  $N^0$  of the Néron model  $N$  of  $A$ : since  $P$  is smooth with connected fibres, there is a morphism  $P \rightarrow N^0$  inducing the identity on the generic fibre, and SGA 7 IX, 3.1 shows that it is an isomorphism. For normal noetherian  $S$ , two prolongations coincide on an open  $V \supset U$  whose complement has codimension at least two. Raynaud's extension theorem then gives the result. A correction to this argument is supplied in the note at the end of the exposé.

### 1.3

**Remark 1.2.** *If  $P$  and  $Q$  are two semi-abelian algebraic spaces over  $S$ , with  $S$  normal and integral and with generic fibres  $P_\eta$  and  $Q_\eta$  abelian varieties over  $k(\eta)$ , the same argument gives*

$$\mathrm{Hom}_S(P, Q) \hookrightarrow \mathrm{Hom}_\eta(P_\eta, Q_\eta).$$

**Remark 1.3.** *The normality hypothesis on  $S$  is indispensable. Let  $S$  be a nodal cubic over an algebraically closed field, obtained from  $\mathbf{P}^1$  by identifying 0 and  $\infty$  in one point  $p$ ; let  $U = S - \{p\}$ ; and let  $A$  be the constant elliptic curve  $E$  over  $U$ . To give a prolongation of  $A$  is to give a prolongation over  $\mathbf{P}^1$  and a gluing isomorphism between the fibres at 0 and at  $\infty$ . The prolongation over  $\mathbf{P}^1$  is unique, namely  $E \times \mathbf{P}^1$ , but the gluing isomorphism can be chosen in at least two ways because  $\mathrm{Aut}(E)$  contains  $\{\pm 1\}$ .*

*The theorem should nevertheless remain true when  $U$  is schematically dense in  $S$  and  $S$  is only assumed geometrically unibranch. For general  $S$ , two prolongations inducing the same prolongations after the relevant normalizations should be isomorphic.*

**1.4** To avoid references that are not available, and because this case suffices for the intended application, we state Gabber's lemma under hypotheses that are probably unnecessarily restrictive.

**Theorem 1.4** (Gabber's lemma). *Let  $U$  be a dense open subscheme of an excellent normal integral scheme  $S$ , and let  $A$  be an abelian algebraic space over  $U$ . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

*such that the inverse image of  $A$  over  $f^{-1}(U)$  extends to a semi-abelian algebraic space over  $S'$ .*

**1.5** If there existed a moduli space for semi-abelian algebraic spaces, the semi-stable reduction theorem together with the valuative criterion of properness would show that this moduli object is proper over  $\mathrm{Spec}(\mathbf{Z})$ , and Gabber's lemma would follow immediately. But we are far from such a situation: the objects considered have automorphisms, sometimes infinitely many, and worse, a semi-abelian algebraic space over a complete discrete valuation ring is not determined by the associated formal algebraic space. If the special fibre is isomorphic to  $\mathbf{G}_m$ , the reduction modulo  $\mathfrak{m}^k$  is isomorphic to that of  $\mathbf{G}_m$  for every  $k$  by rigidity of tori, without  $A$  necessarily being a torus.

The bad behaviour of semi-abelian spaces with respect to formal completion comes from the fact that they are not necessarily proper. For curves the situation is better because one has the notion of a stable curve of genus  $g$ . Such curves are proper and, for  $g \geq 2$ , have only finitely many automorphisms. Because of finite automorphism groups, stable curves of genus  $g \geq 2$  do not give a fine moduli space, but they do give a proper algebraic stack over  $\mathrm{Spec}(\mathbf{Z})$  by the valuative criterion of properness and the stable reduction theorem for curves.

## 1.6

**Lemma 1.5.** *Let  $U$  be a quasi-compact dense open subspace of an algebraic space  $S$ , and let  $X \rightarrow S$  be a family of stable curves of genus  $g \geq 2$  parametrized by  $U$ . There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

*such that the inverse image of  $X$  over  $f^{-1}(U)$  extends to a family of stable curves over  $S'$ .*

*Proof.* Fix  $g \geq 2$ , and for brevity say simply “stable curve” for stable curve of genus  $g$ . Let  $\mathcal{M}$  be the algebraic stack classifying stable curves. It is proper over  $\mathrm{Spec}(\mathbf{Z})$ . There is therefore a proper scheme  $S_0$  over  $\mathrm{Spec}(\mathbf{Z})$  and a surjective morphism  $S_0 \rightarrow \mathcal{M}$ ; equivalently,  $S_0$  is endowed with a family of stable curves  $u_0 : X_0 \rightarrow S_0$  such that every stable curve over an algebraically closed field occurs as a fibre of  $X_0/S_0$ .

Over  $U \times S_0$  we have the two stable curves  $\mathrm{pr}_1^* X$  and  $\mathrm{pr}_2^* X_0$ . Let

$$I = \mathrm{Isom}(\mathrm{pr}_1^* X, \mathrm{pr}_2^* X_0).$$

This algebraic space is finite over  $U \times S_0$  and maps onto  $U$ . Let  $\Gamma$  be its schematic image in  $U \times S_0$ , and let  $\bar{I}$  and  $\bar{\Gamma}$  be the schematic closures inside  $S \times S_0$ :

$$\begin{array}{ccccc} I & \longrightarrow & \Gamma & \hookrightarrow & U \times S_0 \\ \downarrow & & \downarrow & & \downarrow \\ \bar{I} & \longrightarrow & \bar{\Gamma} & \hookrightarrow & S \times S_0. \end{array}$$

The algebraic space  $\bar{I}$ , endowed with the proper morphism

$$\bar{I} \longrightarrow \bar{\Gamma} \hookrightarrow S \times S_0 \longrightarrow S,$$

is the desired  $S'$ . The inverse image over  $\bar{I}$  of the universal family  $X_0/S_0$  extends the inverse image over  $I$  of  $X/U$ .  $\square$

In Section 3 we shall demystify the use of algebraic stacks in this proof.

**1.7** The principle of the proof of Gabber’s lemma is to reduce to the case of curves by using Jacobians. We record some preliminary remarks.

- (a) As explained in 1.1(a),  $A/U$  is automatically an abelian scheme.
- (b) Given  $U \hookrightarrow S$ ,  $A/U$ , and  $f : S' \rightarrow S$  as in Theorem 1.4, every proper surjective  $f_1 : S'' \rightarrow S$  lying over  $S'$  again satisfies the conclusion. Applying Chow’s lemma and normalizing, if the theorem holds at all then it holds with  $f : S' \rightarrow S$  projective and surjective,  $S'$  integral and normal.
- (c) It suffices to verify the theorem locally on  $S$ . If  $S$  is covered by finitely many open subspaces  $S_i$ , and over each  $S_i$  one has a solution  $S'_i \rightarrow S_i$ , one can find a projective surjective  $S' \rightarrow S$ , with  $S'|_{S_i}$  lying over  $S'_i$ , with  $f^{-1}(U)$  dense in  $S'$ , and with  $S'$  normal. The local prolongations glue by Theorem 1.1.
- (d) It suffices to verify the theorem after replacing  $U$  by a non-empty open subspace  $V \subset U$ : over integral normal  $S'$ , a prolongation of  $A|_V$  is automatically a prolongation of  $A$ .

## 2 Proof of Gabber's lemma

Let  $U \subset S$  and  $A/U$  be as in Theorem 1.4, and let  $d$  be the relative dimension of  $A$  over  $U$ . We may and shall assume  $d \geq 1$ . We seek an integral normal  $S'$  and a projective surjective morphism  $f : S' \rightarrow S$  such that the inverse image of  $A$  over  $f^{-1}(U)$  extends to a semi-abelian algebraic space over  $S'$ . In doing so we may shrink  $U$  and replace  $S$  by an integral normal scheme  $S_1$  mapping projectively and surjectively to  $S$ , replace  $U$  by its inverse image, and replace  $A$  by its inverse image.

Let  $\eta = \text{Spec } K$  be the generic point of  $U$ , and let  $A_\eta$  be the fibre of  $A/U$  at  $\eta$ . We may assume  $K$  infinite, otherwise  $S = U = \text{Spec } K$ . Choose a projective embedding

$$A_\eta \hookrightarrow \mathbf{P}^N.$$

In the Grassmannian of linear subspaces  $H \subset \mathbf{P}^N$  of codimension  $d - 1$ , those for which  $H \cap A_\eta$  is a smooth curve form the points of a dense open subset, hence have  $K$ -points. This density follows by iteration from the fact that a general hyperplane section is transverse. Choose such an  $H$  and put

$$T_\eta = A_\eta \cap H.$$

Over an algebraic closure  $\overline{K}$  of  $K$  the weak Lefschetz theorem gives

$$H^0(A_{\overline{K}}, \mathbf{Z}_\ell) \longrightarrow H^0(T_{\overline{K}}, \mathbf{Z}_\ell), \quad H^1(A_{\overline{K}}, \mathbf{Z}_\ell) \hookrightarrow H^1(T_{\overline{K}}, \mathbf{Z}_\ell).$$

Thus  $T_{\overline{K}}$  is connected and the morphism

$$\text{Alb}(T_\eta) = \text{Pic}^0(T_\eta) \longrightarrow A_\eta$$

from the Jacobian of  $T_\eta$  to  $A_\eta$  is an epimorphism. In particular  $T_\eta$  has genus  $g \geq d$ . Temporarily exclude the case  $d = 1$ ; then  $g \geq 2$ .

After shrinking  $U$  there exists a smooth curve  $T_U$  of genus  $g \geq 2$ , embedded in  $A_U$ , such that

$$\text{Pic}^0(T_U) \twoheadrightarrow A_U.$$

Applying Lemma 1.5 to the curve  $T_U$  over  $U \subset S$ , and replacing  $S$  by the resulting modification (or by the normalization of one irreducible component), we may assume that  $T_U$  extends to a family of stable curves  $T$  over  $S$ .

By Artin's representability theorem,  $\text{Pic}^0(T)$  exists as an algebraic space over  $S$ . Let

$$P := \text{Pic}^0(T)$$

be the open algebraic subspace in groups parametrizing invertible sheaves on  $T$  whose restriction to every irreducible component of every geometric fibre has degree zero. This algebraic space is separated and smooth of finite type over  $S$ . Its geometric fibres are extensions of abelian varieties by tori; hence  $P$  is a semi-abelian algebraic space over  $S$ .

We have therefore obtained, after the allowed modification of  $S$  and shrinking of  $U$ , a semi-abelian algebraic space  $P$  over  $S$  such that  $P_U$  is abelian and carries an epimorphism

$$u : P_U \rightarrow A.$$

Let  $K_U = \text{Ker}(u)$ . It is a subgroup algebraic subspace of  $P_U$ . After shrinking  $U$  if necessary, we may suppose it flat over  $U$ . By flattening by modification, a further modification of  $S$  permits us to suppose that the schematic closure  $K$  of  $K_U$  in  $P$  is flat over  $S$ ; then  $K$  is automatically a subgroup algebraic space of  $P$ .

For algebraic spaces in groups over a base  $S$ , the quotient by a flat closed subgroup algebraic space exists, is compatible with base change, is separated over  $S$ , and is flat if the original group space is flat. Let

$$N = P/K.$$

The geometric fibres of  $N$  are quotients of extensions of abelian varieties by tori, hence again extensions of abelian varieties by tori. Thus  $N$  is semi-abelian, and since  $A = P_U/K_U$ , it extends  $A$  and proves the theorem.

It remains only to handle the case  $d = 1$ . The same proof applies if one defines a stable curve of genus one as a pointed curve  $X \rightarrow S$  with section in the smooth locus and with geometric fibres either elliptic curves or nodal cubics. Alternatively, one may define  $T_\eta$  as a hyperplane section of  $A_\eta \times A_\eta$ .

### 3 Exorcising stacks

**3.1** Recall the definition of a groupoid in schemes. It is a diagram

$$S_2 \rightrightarrows S_1 \rightrightarrows S_0 \quad (3.1.1)$$

with face maps  $d_i$ ,  $i = 0, 1$  and  $i = 0, 1, 2$  respectively, such that the square

$$\begin{array}{ccc} S_2 & \longrightarrow & S_1 \\ \downarrow & & \downarrow d_1 \\ S_1 & \xrightarrow{d_0} & S_0 \end{array}$$

is cartesian and such that, for every scheme  $T$ , the sets  $\text{Hom}(T, S_i)$  define a groupoid. The objects are the elements of  $\text{Hom}(T, S_0)$ , the arrows are those of  $\text{Hom}(T, S_1)$ , the source and target of an arrow  $f$  are  $d_1(f)$  and  $d_0(f)$ , and if the target of  $f$  is the source of  $g$ , the composition is  $g \circ f = d_1\langle g, f \rangle$ , where  $\langle g, f \rangle \in \text{Hom}(T, S_2)$  is determined by  $d_2\langle g, f \rangle = f$  and  $d_0\langle g, f \rangle = g$ . We denote this groupoid by  $S_*(T)$ . As  $T$  varies, the  $S_*(T)$  form a prestack in groupoids.

The quasi-compact separated algebraic stacks are the stacks in groupoids obtained from such a prestack, for the étale topology, when  $S_0$  is quasi-compact,  $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$  is finite, and the  $d_i : S_1 \rightarrow S_0$  are étale.

The quasi-compact separated algebraic spaces correspond to the particular case where  $(d_0, d_1) : S_1 \rightarrow S_0 \times S_0$  is a closed immersion. Then  $S_1 \subset S_0 \times S_0$  is an étale equivalence relation, and the algebraic space is the quotient  $S_0/S_1$ .

**3.2** Let  $g \geq 2$ , and say simply stable curve for stable curve of genus  $g$ . We explain how to construct a diagram (3.1.1) for the stack  $\mathcal{M}$  of stable curves. If  $T_i$  is endowed with a stable curve  $u_i : X_i \rightarrow T_i$  for  $i = 1, 2$ , write

$$T_1 \times_{\mathcal{M}} T_2 = \text{Isom}(X_1, X_2).$$

This represents the functor sending  $T$  to triples  $(a_1, a_2, \varphi)$ , where  $a_i : T \rightarrow T_i$  and  $\varphi : a_1^*X_1 \xrightarrow{\sim} a_2^*X_2$ . This fibre product is finite over  $T_1 \times T_2$ .

The Hilbert scheme  $H$  of stable curves  $X$  endowed with a tricanonical embedding

$$X \hookrightarrow \mathbf{P}^{5g-6}$$

is smooth over  $\mathcal{M}$  and maps onto  $\mathcal{M}$ . The group  $\text{PGL}(5g-5)$  acts on  $H$  with finite reduced stabilizers. If  $S$  is a disjoint sum of smooth subschemes of  $H$ , transverse to the  $\text{PGL}$ -orbits, of complementary dimension and meeting all of them, then  $S$  is étale over  $\mathcal{M}$  and maps onto it, hence supplies a diagram (3.1.1). The stack  $\mathcal{M}$  is separated of finite type over  $\text{Spec}(\mathbf{Z})$ . Since deformations of stable curves are unobstructed, it is smooth; this also follows from the smoothness of  $H$  over  $\text{Spec}(\mathbf{Z})$ .

**3.3** Let  $n \geq 1$ . Let  $\mathcal{M}^0[n]$  be the algebraic stack classifying proper smooth curves  $X$  of genus  $g$  over bases  $S$  where  $n$  is invertible, together with a level- $n$  structure

$$\alpha : \text{Pic}^0(X)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

If  $\mathcal{M}^0$  is the open part of  $\mathcal{M}$  corresponding to smooth curves, the forgetful morphism

$$\mathcal{M}^0[n] \longrightarrow \mathcal{M}^0[1/n]$$

makes  $\mathcal{M}^0[n]$  a finite étale cover of  $\mathcal{M}^0[1/n]$ .

### 3.4

**Definition 3.1.** Let  $\mathcal{M}_n$  be the normalization of  $\mathcal{M}$  in  $\mathcal{M}^0[n]$ .

In the language of 3.1 this means the following. Choose  $S_0$  étale over  $\mathcal{M}$  and mapping onto it, whence a diagram  $S_*$ . Over each  $S_i$  there is a stable curve  $X_i$ , and these form a cartesian system

$$\begin{array}{ccccc} X_2 & \rightrightarrows & X_1 & \rightrightarrows & X_0 \\ \uparrow & & \uparrow & & \uparrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0. \end{array}$$

Let  $S_i^0$  be the open subset over which  $X_i$  is smooth, and let  $S_{i,n}[1/n]$  be the finite étale cover of  $S_i^0[1/n]$  given by

$$\text{Isom}(\text{Pic}^0(X_i)_n, (\mathbf{Z}/n\mathbf{Z})^{2g}).$$

Let  $S_{i,n}$  be the normalization of  $S_i$  in  $S_{i,n}[1/n]$ . Since normalization commutes with étale base change in this situation, the  $S_{i,n}$  again form a diagram (3.1.1), defining  $\mathcal{M}_n$ . The diagram

$$\begin{array}{ccccc} S_{2,n} & \rightrightarrows & S_{1,n} & \rightrightarrows & S_{0,n} \\ \downarrow & & \downarrow & & \downarrow \\ S_2 & \rightrightarrows & S_1 & \rightrightarrows & S_0 \end{array}$$

is cartesian. In particular, for  $t \in S_{0,n}(T)$  with image  $s \in S_0(T)$  one has an injection of automorphism groups

$$\text{Aut}(t) \hookrightarrow \text{Aut}(s). \tag{3.4.1}$$

### 3.5

**Proposition 3.2.** If  $n \geq 3$ , then  $\mathcal{M}_n[1/n]$  is an algebraic space.

The essential point is the following lemma.

**Lemma 3.3.** Let  $X$  be a stable curve over an algebraically closed field  $k$ , and let  $n \geq 3$  be invertible in  $k$ . Every automorphism of  $X$  acting trivially on  $\text{Pic}^0(X)_n$  is the identity.

*Proof.* Write  $\text{Pic}^0(X)$  as an extension of an abelian variety by a torus,

$$0 \longrightarrow T \longrightarrow \text{Pic}^0(X) \longrightarrow A \longrightarrow 0,$$

and let  $L$  be the character group of  $T$ . Every automorphism  $u$  of  $X$  has finite order. If it acts trivially on  $\text{Pic}^0(X)_n$ , it acts trivially on  $A_n$  and on  $T_n$ , hence on  $L/nL$ . By Serre's rigidity for  $A$ , and by the analogous direct argument for  $L$ , it acts trivially on  $A$  and on  $L$ , hence on  $T$ . Thus  $u - 1$  sends  $\text{Pic}^0(X)$  to  $T$  and is therefore zero;  $u$  acts trivially on  $\text{Pic}^0(X)$ , and the conclusion follows from the standard faithfulness result for stable curves.  $\square$



**Lemma 3.4.** *Let  $S$  be normal, let  $j : U \hookrightarrow S$  be a dense open subset, let  $A$  be an abelian algebraic space over  $U$ , let  $N$  be a semi-abelian prolongation of  $A$  over  $S$ , and let  $n$  be invertible on  $S$ .*

- (i) *The étale sheaf  $N_n$  of points killed by multiplication by  $n$  injects into  $j_*A_n$ .*
- (ii) *Every isomorphism over  $U$  from  $A_n$  to the constant sheaf  $(\mathbf{Z}/n\mathbf{Z})^{2g}$  extends to a monomorphism*

$$N_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

*Proof.* The first assertion follows from the separatedness of  $N_n$  over  $S$ . Since  $S$  is normal, the constant sheaf satisfies

$$(\mathbf{Z}/n\mathbf{Z})^{2g} \simeq j_*(\mathbf{Z}/n\mathbf{Z})^{2g},$$

and the composite

$$N_n \longrightarrow j_*j^*N_n = j_*A_n \longrightarrow j_*(\mathbf{Z}/n\mathbf{Z})^{2g} = (\mathbf{Z}/n\mathbf{Z})^{2g}$$

is the required prolongation. □

*Proof of the proposition.* Let  $T_1 = S_{1,n}[1/n]$ . Since  $T_1$  is finite and unramified over  $T_0 \times T_0$ , it suffices to verify that for every algebraically closed field  $k$ , the map

$$T_1(k) \longrightarrow T_0(k) \times T_0(k)$$

is injective. We may suppose  $n$  invertible in  $k$ , otherwise  $T_1(k) = \emptyset$ . Because  $T_*(k)$  is a groupoid, it suffices to verify that for  $t \in T_0(k)$  there is exactly one object of  $T_1(k)$  over  $(t, t)$ , i.e. that  $\text{Aut}(t)$  is reduced to the identity.

Let  $X$  denote the stable curve obtained by pullback of the universal curve. On  $T_i^0 = S_{i,n}[1/n]$  there is by definition a cartesian system of isomorphisms

$$\text{Pic}^0(X_i)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

By Lemma 3.4 it extends over  $T_i$  to a monomorphism of étale sheaves. It follows that the morphism from the stack defined by  $T_*$  to that defined by  $S_*$  factors through the stack of stable curves  $X$  endowed with a monomorphism  $\text{Pic}^0(X)_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}$ . For  $t \in T_0(k)$ , the injection (3.4.1) has image consisting of automorphisms of the corresponding stable curve acting trivially on  $\text{Pic}^0(X)_n$ . By Lemma 3.3, such an automorphism is trivial. □

### 3.6

**Corollary 3.5.** *If  $n$  is divisible by two integers  $n_i$ ,  $i = 1, 2$ , prime to one another and each at least 3, then  $\mathcal{M}_n$  is an algebraic space.*

It suffices to verify that each  $\mathcal{M}_n[1/n_i]$  is an algebraic space, i.e. that the corresponding objects have no non-trivial automorphisms. This follows from (3.4.1) and its factorization

$$\mathcal{M}_n[1/n_i] \longrightarrow \mathcal{M}_{n_i}[1/n_i] \longrightarrow \mathcal{M},$$

together with Proposition 3.5.

**3.7** In 3.6, stack language gave an étale atlas defining an algebraic space  $\mathcal{M}_n$ . The valuative criteria of separatedness and properness, together with the semi-stable reduction theorem, show that this algebraic space is proper over  $\text{Spec}(\mathbf{Z})$ . One may use this  $\mathcal{M}_n$  as the  $S_0$  in the proof of Lemma 1.5. If one wants  $S_0$  to be projective over  $\text{Spec}(\mathbf{Z})$ , one applies Chow's lemma to  $\mathcal{M}_n$ . In fact  $\mathcal{M}_n$  is projective, by Mumford.

## 4 Obtaining schemes

In this section we show how to modify and complete the proof of Gabber's lemma so as to obtain schemes rather than algebraic spaces. Another approach was indicated to me by Michel Raynaud: he says he can prove that if  $X$  is an  $S$ -algebraic space in groups, smooth and separated over  $S$ , with connected fibres, then  $X$  becomes representable after base change to a suitable blow-up  $S'/S$ .

Let  $g \geq 2$ . For brevity again say stable curve for stable curve of genus  $g$ .

**4.1** For a smooth curve  $X/S$  of genus  $g$ , autoduality of the Jacobian

$$\varepsilon : \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(X)^*$$

gives a canonical relatively ample invertible sheaf on  $\mathrm{Pic}^0(X)$ , namely

$$(1, \varepsilon)^*(\text{Poincaré sheaf}).$$

It is symmetric and is trivialized along the zero section. Its formation is compatible with arbitrary base change. Denote it by  $\mathcal{L}(X/S)$ .

### 4.2

**Lemma 4.1.** *It is possible, uniquely up to unique isomorphism, to attach to each stable curve  $X/S$  an invertible sheaf  $\mathcal{L}(X/S)$  on  $\mathrm{Pic}^0(X)$ , trivialized along the zero section, compatible with all base change, and equal to the sheaf of 4.1 when  $X$  is smooth.*

*Proof.* It suffices to treat the universal case. With the notation of 3.4, choose  $X_*/S_*$ . We seek invertible sheaves  $\mathcal{L}(X_i/S_i)$  on the  $\mathrm{Pic}^0(X_i)$ , trivialized along the zero section, extending the sheaves of 4.1 over the open subset where  $X_i$  is smooth and forming a cartesian system under the maps  $S_i \rightarrow S_j$ . Each  $S_i$  is smooth over  $\mathrm{Spec}(\mathbf{Z})$ , hence regular, and so is  $\mathrm{Pic}^0(X_i)$ , which is smooth over  $S_i$ . An extension exists by taking the determinant of any coherent extension. Correcting by the inverse image of its restriction to the zero section gives a trivialized extension. Since  $\mathrm{Pic}^0(X_i)$  has geometrically irreducible fibres, such an extension is unique up to unique isomorphism, and these unique isomorphisms give the required compatibility.  $\square$

### 4.3

**Proposition 4.2.** *For a stable curve  $X/S$ , the invertible sheaf  $\mathcal{L}(X/S)$  of 4.2 is relatively ample. In particular,  $\mathrm{Pic}^0(X)$  is quasi-projective over  $S$  and is a scheme if  $S$  is a scheme.*

As before it suffices to verify this in a universal case; one may suppose  $S$  regular and the open subset  $S^0 \subset S$  over which  $X$  is smooth dense. If one already knew that  $\mathrm{Pic}^0(X)$  were a scheme, the assertion would follow from Raynaud's theorem. Unfortunately that proof uses the scheme hypothesis essentially in order to construct, locally on  $S$ , relatively ample divisors as complements of affine open subsets.

### 4.4

**Lemma 4.3.** *Let  $X$  be a geometrically reduced connected curve over a separably closed field  $k$ . There exists a divisor  $D$ , contained in the smooth locus of  $X$ , of degree*

$$g = \dim H^1(X, \mathcal{O}_X),$$

*such that*

$$H^0(X, \mathcal{O}_X(D)) = k, \quad H^1(X, \mathcal{O}_X(D)) = 0.$$

*Proof.* The assertion on  $H^0$  follows from the other assertion by Riemann-Roch. If  $\omega$  is the dualizing sheaf, then  $\dim H^0(X, \omega) = g$ , and by duality  $H^1(X, \mathcal{O}_X(D)) = 0$  is equivalent to  $H^0(X, \omega(-D)) = 0$ . The sheaf  $\omega$  has no embedded component. We may therefore construct by induction divisors  $D_i$  of degree  $i$ , contained in the smooth locus, with

$$\dim H^0(X, \omega(-D_i)) = g - i,$$

by taking  $D_0 = 0$  and  $D_{i+1} = D_i + P_i$ , where  $P_i$  is a smooth  $k$ -point at which not all sections of  $H^0(X, \omega(-D_i))$  vanish. Take  $D = D_g$ .  $\square$

#### 4.5

**Lemma 4.4.** *Let  $f : X \rightarrow S$  be projective, with geometrically reduced connected fibres. Locally for the étale topology on  $S$ , there is an open subset*

$$V \subset \mathrm{Pic}^0(X)$$

*meeting every fibre and which is a scheme.*

*Proof.* One reduces to the case where  $S$  is strictly local with closed point  $s$ , and it suffices to construct an open  $V \subset \mathrm{Pic}^0(X)$  which is a scheme and meets the fibre over  $s$ . By Lemma 4.3, choose a divisor  $D_s$  of degree  $g = \dim H^1(X_s, \mathcal{O}_{X_s})$  on  $X_s$ , contained in the smooth locus, and satisfying  $H^1(X_s, \mathcal{O}(D_s)) = 0$ . Extend it to a relative Cartier divisor  $D$  on  $X$ , possible because  $S$  is local henselian. Then

$$f_*\mathcal{O}(D) = \mathcal{O}_S, \quad R^1f_*\mathcal{O}(D) = 0.$$

Let  $V'$  be the open subset of  $\mathrm{Pic}^0(X)$  corresponding to invertible sheaves  $\mathcal{F}$  such that  $R^1f_*\mathcal{F}(D) = 0$ . For such  $\mathcal{F}$ ,  $f_*\mathcal{F}(D)$  is free of rank one and commutes with all base change. Let  $V \subset V'$  be the open corresponding to those  $\mathcal{F}$  for which the zero locus of a basis section of  $f_*\mathcal{F}(D)$  is a relative Cartier divisor  $D(\mathcal{F})$  contained in the smooth locus  $X^0$ . By construction  $V$  contains the zero section. The map

$$\mathcal{F} \longmapsto D(\mathcal{F})$$

identifies  $V$  with an open subset of  $\mathrm{Sym}^g(X^0)$ , and hence  $V$  is a scheme.  $\square$

#### 4.6

**Lemma 4.5.** *Under the hypotheses of Lemma 4.4,  $\mathrm{Pic}^0(X)$  is locally a scheme for the étale topology on  $S$ .*

*Proof.* Again suppose  $S$  strictly local. Put  $P = \mathrm{Pic}^0(X)$ . It is a smooth algebraic space over  $S$ , pure of relative dimension  $g$ . Let  $V$  be as in Lemma 4.4. It suffices to show that  $P$  is the union of the translates  $gV$  for  $g \in P(S)$ . Reducing to the case where  $S$  is essentially of finite type over  $\mathbf{Z}$ , one may count dimensions. Let  $P^n$  be the  $n$ -fold fibre product of  $P$  over  $S$ . For a geometric point  $t$  of  $S$  and a geometric point  $\gamma = (\gamma_1, \dots, \gamma_n)$  of  $P^n$  over  $t$ , say that  $\gamma$  is allowed if the translates  $\gamma_i V$  cover  $P_t$ . The allowed points form a constructible subset  $Y$  of  $P^n$ . If  $Z$  is its complement, then fibre by fibre  $Z$  has codimension at least  $n - g$ . For  $n - g > \dim S$ , the closure of  $Z$  cannot contain the whole special fibre, so one finds a section  $\gamma$  of  $P^n - \overline{Z}$ , and then  $P$  is the union of the  $\gamma_i V$ .  $\square$

**4.7** In Lemmas 4.4 and 4.5, the projectivity hypothesis on  $f$  may be replaced by properness: locally on  $S$  for the étale topology,  $f$  is automatically projective.

#### 4.8

*Proof of Proposition 4.2.* The problem is local on  $S$  for the étale topology. By Lemma 4.5, one can apply Raynaud's theorem giving relative ampleness in the scheme case.  $\square$

**4.9** The proof of Raynaud's theorem seems to use the scheme hypothesis only through the conclusion of Lemma 4.4. This would allow one to avoid Lemma 4.5.

#### 4.10

**Proposition 4.6.** *Gabber's lemma remains true with “semi-abelian algebraic space” in the conclusion replaced by “semi-abelian scheme.”*

*Proof.* Proceeding as in Section 2, one reduces to the case where there is a stable curve  $X$  over  $S$ , smooth over  $U$ , and an epimorphism over  $U$

$$u : \mathrm{Pic}^0(X) \longrightarrow A$$

whose kernel  $K_U$  extends to a flat subgroup algebraic space  $K$  of  $\mathrm{Pic}^0(X)$ . It remains to show that

$$P := \mathrm{Pic}^0(X)/K$$

is a scheme.

Over  $U$ , Poincaré complete reducibility gives a morphism

$$v : A \longrightarrow \mathrm{Pic}^0(X)$$

such that  $uv$  is multiplication by an integer  $n \geq 1$  on  $A$ . By the extension result of 1.3,  $v$  extends to a morphism, again denoted  $v$ , from  $P$  to  $\mathrm{Pic}^0(X)$ . By density of  $U$  in  $S$ , one still has  $uv = n$ . Since multiplication by  $n$  on  $P$  is quasi-finite,  $v$  is quasi-finite, hence quasi-affine. Since  $\mathrm{Pic}^0(X)$  is a scheme,  $P$  is a scheme.  $\square$

I thank M. Raynaud and O. Gabber for a critical reading of a first version of this text.

## Note

**(1).1** Let  $S = \mathrm{Spec} A$  be affine, and let  $U \subset S$  be an open subscheme which is a finite union of principal opens  $U_i = \mathrm{Spec} A[f_i^{-1}]$ . If  $F$  is a closed subscheme of  $U$ , defined in  $U_i$  by an ideal  $I_i \subset A[f_i^{-1}]$ , then the schematic closure  $\overline{F}$  of  $F$  in  $S$  is the closed subscheme defined by

$$I = \bigcap_i \rho_i^{-1}(I_i),$$

where  $\rho_i : A \rightarrow A[f_i^{-1}]$  is the canonical map. If  $S' = \mathrm{Spec} A'$  is flat over  $S$ , then the inverse image of  $\overline{F}$  in  $S'$  is again the schematic closure of the inverse image of  $F$ . Hence schematic closure commutes with étale localization and so makes sense for algebraic spaces.

More generally, for  $f : X \rightarrow S$  quasi-compact, the closed schematic image  $f(X)^-$  is the closed subscheme defined by the ideal of elements  $a \in A$  whose inverse image on  $X$  is zero. For  $f$  the inclusion of  $F$  as above, this recovers schematic closure. The same base-change compatibility holds for schematic images.

(1).2 Let  $S$  be a noetherian algebraic space, and let  $X$  and  $Y$  be two algebraic spaces over  $S$ , each endowed with a section  $e$ . Let  $V \subset X$  be open, put  $U = e^{-1}(V) \subset S$ , and let

$$f_V : V \longrightarrow Y$$

be an  $S$ -morphism sending  $e$  to  $e$ . For each  $n$ ,  $f_V$  induces a morphism from the  $n$ -th infinitesimal neighbourhood of  $e$  in  $V$  to the corresponding neighbourhood in  $Y_U$ . Passing to the inverse limit gives a morphism

$$f_V^\wedge : X_U^\wedge \longrightarrow Y_U^\wedge$$

between formal completions along the sections.

**Proposition 4.7.** *Assume  $X$  smooth over  $S$  and  $Y$  separated of finite type.*

- (i) *If  $U$  contains all points of  $S$  of depth at most 1, then  $f_V^\wedge$  extends uniquely to a morphism  $f^\wedge : X^\wedge \rightarrow Y^\wedge$  between the formal completions of  $X$  and  $Y$  along the sections.*
- (ii) *Let  $\Gamma$  be the schematic closure in  $X \times_S Y$  of the graph  $\Gamma_V \subset V \times_S Y$  of  $f_V$ . If  $f_V^\wedge$  extends to  $f^\wedge : X^\wedge \rightarrow Y^\wedge$  and  $U$  contains all points of  $S$  of depth 0, then  $\Gamma$  is étale over  $X$  along the section  $e$ , and its formal completion there is the graph of  $f^\wedge$ .*

The proof is local on  $S$  for the étale topology; thus one may suppose  $S$  affine. Then the infinitesimal neighbourhoods of the section in  $X$  and  $Y$  are affine schemes, and (i) is proved as in Raynaud. Smoothness of  $X/S$  ensures that the infinitesimal neighbourhoods of  $e$  in  $X$  are finite and flat over  $S$ .

We use the following lemma for (ii).

**Lemma 4.8.** *Let  $S = \operatorname{Spec} A$  with  $A$  a complete noetherian local ring, let*

$$X = \operatorname{Spec} R[[T_1, \dots, T_n]],$$

*let  $e$  be the zero section, let  $V$  be an open subset of  $X$  such that  $U = e^{-1}(V)$  contains all associated points of  $S$ , and let  $F \subset V$  be a closed subscheme containing all infinitesimal neighbourhoods of  $e(U)$  in  $V$ . Then  $F = V$ .*

*Proof.* It suffices to show that  $X$  is the schematic closure of  $F$  in  $X$ . If

$$f = \sum a_I T^I$$

vanishes on  $F$ , then each coefficient  $a_I$  vanishes on  $U \subset S$ , hence is zero, and  $f = 0$ .  $\square$

*Proof of Proposition 4.7(ii).* Let  $s$  be a point of  $S$ . Write  $X_s^\wedge$  for the completion of  $X$  at  $e(s)$ , and let  $V_s^\wedge$  and  $U_s$  be the inverse images of  $V$  and  $U$ . The morphism  $f^\wedge$  induces  $X_s^\wedge \rightarrow Y$ , and  $f_V$  induces  $V_s^\wedge \rightarrow Y$ . These two morphisms coincide on a closed subscheme containing all infinitesimal neighbourhoods of  $e(U_s)$ , hence coincide by Lemma 4.8. The inverse image of  $\Gamma_V$  in  $X_s^\wedge \times_S Y$  is therefore the graph of  $f^\wedge|_{V_s^\wedge}$ , whose schematic closure is the graph of  $f^\wedge : X_s^\wedge \rightarrow Y$ . Since schematic closure commutes with the flat base change  $X_s^\wedge \times_S Y \rightarrow X \times_S Y$ , it follows that after the base change  $X_s^\wedge \rightarrow X$ , the space  $\Gamma$  is étale over  $X$ . Hence  $\Gamma$  is étale over  $X$  at  $e(s)$ .  $\square$

### (1).3

**Corollary 4.9.** *Let  $G$  and  $H$  be algebraic spaces in groups over  $S$ , let  $U$  be an open subspace of  $S$ , and let*

$$f_U : G_U \longrightarrow H_U$$

*be a morphism. Assume that  $G$  is smooth over  $S$  with connected fibres, that  $H$  is of finite type and separated over  $S$ , and that  $U$  contains all points of  $S$  of depth 0. If the formal completion of  $f_U$  along the zero sections extends to a morphism of formal completions*

$$G^\wedge \longrightarrow H^\wedge,$$

*then  $f_U$  extends uniquely to a morphism  $G \rightarrow H$ .*

If  $U$  contains all points of depth at most 1, the existence of the formal extension is guaranteed by Proposition 4.7(i).

*Proof.* Uniqueness follows from the fact that  $U$  is schematically dense in  $S$ , hence  $G_U$  is schematically dense in  $G$ . For existence, let  $\Gamma_U$  be the graph of  $f_U$ , let  $\Gamma \subset G \times_S H$  be its schematic closure, and let  $\Gamma^0$  be the open subset of  $\Gamma$  where  $\Gamma$  is étale over  $G$ . By Proposition 4.7(ii),  $\Gamma^0$  contains the zero section of  $G \times_S H$ . For every section  $\gamma$  of  $\Gamma$ , left translation by the element of  $G_U \times_S H_U$  determined by  $\gamma|_U$  sends  $\Gamma_U$  to itself. By transport of structure, left translation by  $\gamma$  sends  $\Gamma^0$  to itself. After the flat base change  $\Gamma^0/S$ , this shows that  $\Gamma^0$  is a subgroup algebraic space of  $G \times_S H$ .

The image of  $\Gamma^0$  in  $G$  is an open subgroup algebraic space, hence is all of  $G$  because  $G/S$  has connected fibres. Since  $\Gamma^0/G$  is étale and separated, the function assigning to  $x$  the number of geometric points in the fibre over  $x$  is lower semicontinuous. It is at least one everywhere and equals one over  $G_U$ , hence is constantly one. Therefore  $\Gamma^0 \rightarrow G$  is étale, surjective, and radicial, hence an isomorphism. The required extension is the composite

$$G \xleftarrow{\sim} \Gamma^0 \longrightarrow H.$$

□

(1).4 The preceding proposition should remain true if the hypothesis that  $G/S$  is smooth is weakened to flatness of finite type, and if the hypothesis that  $H$  is separated over  $S$  is omitted.

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## Sigla

EGA denotes *Éléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné. SGA denotes *Séminaire de géométrie algébrique du Bois-Marie*.

# Heisenberg Group and Reality

P. Deligne

## 1

Let  $H$  be the complex Heisenberg group, an extension of  $\mathbb{C}^2$  by  $\mathbb{C}^*$ . It has coordinates  $(x, y, \lambda)$ , the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$\tau : (x, y, \lambda) \mapsto (-x, -y, \lambda) \quad (1.2)$$

is an automorphism of  $H$ .

The coordinate system  $x, y, \lambda : H \rightarrow \mathbb{C}^2 \times \mathbb{C}^*$  exhibits the section  $\mathbb{C}^2 \times \{1\}$  of the projection of  $H$  onto  $\mathbb{C}^2$ . On this section,  $\tau(g) = g^{-1}$ . Each line  $D \subset \mathbb{C}^2$  has, by means of this section, a lift to a one-parameter subgroup of  $H$ , again denoted by  $D$ . The commutator  $(u, v) = uvu^{-1}v^{-1}$  of two elements  $u = (u_1, u_2)$  and  $v = (v_1, v_2)$  of  $\mathbb{C}^2$  is  $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$ , with

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let  $H_{\mathbb{R}}$  be the real form defined by  $x, y \in \mathbb{R}$  and  $|\lambda| = 1$ . The group  $\mathrm{SL}(2, \mathbb{C})$  acts on  $H$ , and  $\mathrm{SL}(2, \mathbb{R})$  acts on  $H_{\mathbb{R}}$ , which as a set is identified with  $\mathbb{R}^2 \times U^1$ .

It is known, by the Stone-von Neumann theorem [2], that  $H_{\mathbb{R}}$  has a unique irreducible unitary representation  $V$  for which  $\lambda \in U^1$  acts by multiplication by  $\lambda$ . In the Schrödinger model of  $V$ , one has  $V = L^2(\mathbb{R})$ , and the action of  $H_{\mathbb{R}}$  is

$$\begin{aligned} (x, 0, 1) : f(z) &\mapsto f(z + x), \\ (0, y, 1) : f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of  $V$  one deduces a projective unitary action of  $\mathrm{SL}(2, \mathbb{R})$  on  $V$ , compatible with the action of  $\mathrm{SL}(2, \mathbb{R})$  on  $H_{\mathbb{R}}$  (A. Weil [3], D. Shale [1]).

For the action of  $H_{\mathbb{R}}$  on  $V = L^2(\mathbb{R})$ , the  $C^\infty$  vectors, i.e. those  $v \in V$  for which  $h \mapsto hv$  is  $C^\infty$  in  $h \in H_{\mathbb{R}}$ , form the Schwartz space  $\mathcal{S}$  of  $C^\infty$  rapidly decreasing functions on  $\mathbb{R}$ . The holomorphic vectors, i.e. those  $v \in V$  for which  $h \mapsto hv$  extends to a holomorphic map from  $H$  to  $V$ , are the  $L^2$  functions on  $\mathbb{R}$  that extend to holomorphic functions on  $\mathbb{C}$  such that, in every horizontal strip  $|\mathrm{Im} z| < A_1$ , one has, for all  $A_2 \in \mathbb{R}$ ,

$$|f(z)| \leq O(\exp(-A_2|z|)).$$

The complex group  $H$  acts on the space  $V^{\mathrm{hol}}$  of holomorphic vectors by the formulae (1.4).

For the holomorphic action of  $H$  on  $V^{\mathrm{hol}}$ , one does not have the same uniqueness as for the unitary action of  $H_{\mathbb{R}}$  on  $V$ . If one did, the projective action of  $\mathrm{SL}(2, \mathbb{R})$  on  $V$  would extend to a projective action of  $\mathrm{SL}(2, \mathbb{C})$  on  $V^{\mathrm{hol}}$ ; since  $\mathrm{SL}(2, \mathbb{C})$  is simply connected, this action would lift to an honest action of  $\mathrm{SL}(2, \mathbb{C})$ , and the projective action of  $\mathrm{SL}(2, \mathbb{R})$  would lift to an honest action. This is not the case: only the double cover of  $\mathrm{SL}(2, \mathbb{R})$  acts. Our purpose is to make explicit what happens.



## 2

Let  $S = \mathbb{P}^1(\mathbb{C})$  be the space of lines of  $\mathbb{C}^2$ . For  $D \in S$ , lifted in  $H$  as in no. 1, let  $W_D$  be the space of holomorphic functions  $\varphi$  on  $H$  satisfying

$$\varphi(\lambda dh) = \lambda \varphi(h)$$

for  $\lambda \in \mathbb{C}^*$  and  $d \in D$ , endowed with the action of  $H$  given by  $g : \varphi(h) \mapsto \varphi(hg)$ . For  $E \neq D$ , restriction to  $E$  is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \quad (2.1)$$

where  $\mathcal{O}(E)$  denotes the space of holomorphic functions on  $E$ .

If  $R$  is a holomorphic representation of  $H$ , on which each  $\lambda \in \mathbb{C}^*$  acts by  $\lambda$ , and if  $\omega$  is a  $D$ -invariant linear form on  $R$ , then

$$w \mapsto \omega(hw) \quad (2.2)$$

is a morphism from  $R$  to  $W_D$ .

## 3. Example

Let  $V$ , as in nos. 1 and 2, be the irreducible unitary representation of  $H_{\mathbb{R}}$ , let  $\mathcal{S} \subset V$  be the space of its  $C^\infty$  vectors, and let  $\mathcal{S}' \supset V$  be the dual of the space of  $C^\infty$  vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let  $D_t \in S$  be the line generated by  $(-t, 1)$ . If  $\text{Im } t > 0$ , the linear form, written in the Schrödinger model,

$$\omega_t : f \mapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t) dx$$

is defined on  $V$ , and even on  $\mathcal{S}'$ . The form  $\omega_t$  is a holomorphic vector of the dual of  $V$ , fixed by  $D_t$ . In terms of the infinitesimal generator  $-t\partial_x + 2\pi i x$  of  $D_t$ , the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2),  $\omega_t$  defines

$$\Omega_t : V^{\text{hol}} \longrightarrow W_{D_t},$$

which extends to  $V$  and even to  $\mathcal{S}'$ . Let  $E$  be the  $x$ -axis. The map  $r_E \Omega_t : V \rightarrow \mathcal{O}(E)$  sends  $f \in V$  to

$$f(x, t) = r_E \Omega_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\left( \partial_t - \frac{1}{4\pi i} \partial_x^2 \right) f(x, t) = 0.$$

For real  $t$ , the form  $\omega_t$  remains defined on  $\mathcal{S} \subset V$ . For  $t = 0$  it is  $f \mapsto f(0)$ , and for  $t = \infty$  it is  $f \mapsto \int f(x) dx$ . The homomorphism  $\Omega_t$  remains defined on  $V^{\text{hol}}$ . For  $t = 0$ ,  $r_E \Omega_0$  is, in the Schrödinger model, the identity.

Assume  $\text{Im } t > 0$ . Let  $\overline{D}_t$  be the complex conjugate of  $D_t$ . In the affine line  $S - \{\overline{D}_t\}$ , the “half-plane”  $\{D_u \mid \text{Im } u > 0\}$  is a disk with center  $D_t$ . The map  $r_{\overline{D}_t} \circ \Omega_t : V \rightarrow W_{D_t} \rightarrow \mathcal{O}(\overline{D}_t)$  gives the holomorphic model of  $V$ : the action of  $H$  is given by

$$\begin{aligned} d \in D_t : \quad & f(x) \mapsto f(x + d), \\ e \in \overline{D}_t : \quad & f(x) \mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned}$$

and the Hilbert structure for which  $(f, f)$  is the integral of

$$f(d)\overline{f(\bar{d})}\exp(2\pi i\langle d, \bar{d} \rangle)$$

is invariant under  $H_{\mathbb{R}}$ ; moreover  $r_{\bar{D}_t} \circ \Omega_t$  identifies  $V$  with the space of  $f$  of finite norm. For  $d = (-tz, z)$ , the factor  $\exp(2\pi i\langle d, \bar{d} \rangle)$ , the commutator of  $d$  and  $\bar{d}$ , is

$$\exp(2\pi i\langle d, \bar{d} \rangle) = \exp(2\pi i(t - \bar{t})|z|^2).$$

## 4

Fix  $E$ . The isomorphisms  $r_E$  of (2.1) make it possible to regard the  $W_D$  for  $D \neq E$  as a family of representations of  $H$  operating in the fixed space  $\mathcal{O}(E)$ , with the action of  $H$  depending holomorphically on  $D$ . Let us make explicit the dependence on  $E$  of  $r_E$ . Choose coordinates as in no. 1, with  $E$  the  $x$ -axis. Let  $E'$  be generated by  $(1, u)$  and  $D$  by  $(-t, 1)$ . We use  $x$  as coordinate on both  $E$  and  $E'$ . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for  $(1 + tu)x' = x$  and  $\lambda = \exp(\pi i u x x')$ . Hence  $r_{E'} r_E^{-1}$  sends  $f_E \in \mathcal{O}(E)$  to  $f_{E'} \in \mathcal{O}(E')$ , with  $f_{E'}(x') = \lambda f_E(x)$ :

$$f_{E'}\left(\frac{x}{1+tu}\right) = \exp\left(\frac{\pi i u}{1+tu}x^2\right) f_E(x). \quad (4.1)$$

This formula is holomorphic in  $t$ : the  $W_D$  form a holomorphic vector bundle  $\mathcal{W}$  of infinite dimension over  $S$ , on which  $H$  acts.

To understand (4.1), one may observe that the point of  $E$  with coordinate  $x$  has the same image in  $\mathbb{C}^2/D$  as the point of  $E'$  with coordinate  $x/(1 + tu)$ .

From (4.1) one deduces that the following growth condition on  $v \in W_D$  is independent of the choices of  $E \neq D$  and of the coordinate  $z$  on  $E$ :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O(\exp(A|z|^2)). \quad (4.2)$$

Let  $W_D^0 \subset W_D$  be defined by this condition. The  $W_D^0$  form a holomorphic sub-bundle  $\mathcal{W}^0$  of the bundle  $\mathcal{W}$  in holomorphic representations of  $H$ .

## 5. Proposition

Let  $T > 0$  and consider the following conditions on an entire function  $f$ :

- (i)<sub>T</sub> There exists a holomorphic function  $g(z, t)$ , for  $z \in \mathbb{C}$  and  $|t| < T$ , satisfying the heat equation  $(\partial_t - \partial_z^2)g = 0$ , with initial condition  $g(z, 0) = f(z)$ .
- (ii)<sub>T</sub>  $|f(z)| \leq O(\exp(|z|^2/4T))$ .
- (iii)<sub>T</sub>  $\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|dz \wedge d\bar{z}| < \infty$ .

If  $T < T_1$ , each of  $(i)_{T_1}$ ,  $(ii)_{T_1}$ ,  $(iii)_{T_1}$  implies each of  $(i)_T$ ,  $(ii)_T$ ,  $(iii)_T$ .

*Proof.*  $(i)_{T_1} \Rightarrow (ii)_T$ . The heat equation remains satisfied by the derivatives of  $g$ , and therefore

$$\partial_z^{2n} g = \partial_t^n g, \quad \partial_z^{2n+1} g = \partial_t^n (\partial_z g).$$

Since  $g(0, t)$  is holomorphic for  $|t| < T_1$ , if  $T < T_2 < T_1$ , then at  $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \quad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for  $\partial_z^{2n+1}g$ . The Taylor series of  $f$  then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!} \frac{1}{T_2^n} |z|^{2n}\right).$$

One has

$$\frac{1}{2n+1} 2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n} \frac{1}{T^n} \frac{|z|^{2n}}{n!}\right),$$

and this sum is  $\exp(|z|^2/4T)$ .

$(ii)_{T_1} \Rightarrow (i)_T$ . It suffices to take  $g$  given by the kernel of the heat equation:

$$g(z, t) = \int_{\sqrt{t}\mathbb{R}} f(z+x) \frac{1}{2\sqrt{\pi t}} \exp(-x^2/4t) dx.$$

Changing  $\sqrt{t}$  to  $-\sqrt{t}$  changes the orientation of the cycle  $\sqrt{t}\mathbb{R}$  and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for  $|t| < T$ .

The relation between (ii) and (iii) is clear:  $(ii)_{T_1} \Rightarrow (iii)_T$ , and  $(iii)_{T_1} \Rightarrow (ii)_T$  is verified by writing  $f(z)$  as the mean of its values on a disk with center  $z$  and radius  $1/|z|$ , and estimating that mean by Cauchy–Schwarz.

## 6

In homage to the Stone-von Neumann theorem, the bundle  $\mathcal{W}$  over  $S$  is, projectively, equipped with a holomorphic connection for which the action of  $H$  is horizontal.

Let us specify “projectively.” On the projective line  $S$  there is the ample invertible sheaf  $\mathcal{O}(1)$ . Let  $\mathcal{O}(-1)$  be its dual. A square root  $\mathcal{O}(-\frac{1}{2})$  of  $\mathcal{O}(-1)$  is an invertible sheaf  $L$  endowed with an isomorphism  $L^{\otimes 2} \simeq \mathcal{O}(-1)$ . Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity  $\pm 1$ . This ambiguity obstructs the global existence of  $\mathcal{O}(-\frac{1}{2})$ . It is not on  $\mathcal{W}$ , but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \quad (6.1)$$

Note that if  $\mathcal{O}(-\frac{1}{2})'$  and  $\mathcal{O}(-\frac{1}{2})''$  are two square roots of  $\mathcal{O}(-1)$ , and if  $\nabla$  is a connection on  $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$ , then its transport to  $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$  by an isomorphism  $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$  of square roots of  $\mathcal{O}(-1)$  does not depend on the choice of  $a$ , which can be changed locally only by a constant  $\pm 1$ . Thus the notion of a “connection on  $\mathcal{W}(-\frac{1}{2})$ ” is well defined independently of the global existence, or the choice, of  $\mathcal{O}(-\frac{1}{2})$ .

Let  $D \in S$  and let  $t$  be a tangent vector at  $D$ . For a local section  $s$  of  $\mathcal{W}(-\frac{1}{2})$ , the problem is to define  $\nabla_t s$  in the fiber of  $\mathcal{W}(-\frac{1}{2})$  at  $D$ . We proceed as follows: after making an auxiliary choice  $u$ , one defines  $\nabla_t^u s$  for  $s$  a local section of  $\mathcal{W}$  or of  $\mathcal{O}(-\frac{1}{2})$ . This derivation  $\nabla_t^u$  depends on  $u$ , with dependence in  $u$  of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where  $a$  takes opposite values for  $\mathcal{W}$  and for  $\mathcal{O}(-\frac{1}{2})$ . Hence, for  $s = s_1 s_2$ , with  $s_1$  a local section of  $\mathcal{W}$  and  $s_2$  a local section of  $\mathcal{O}(-\frac{1}{2})$ ,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of  $u$ .

Let  $D \in S$  be a line of  $\mathbb{C}^2$ . A tangent vector  $t$  at  $D \in S$  is identified with a linear map  $t : D \rightarrow \mathbb{C}^2/D$ , i.e. with an element of  $(\mathbb{C}^2/D)^{\otimes 2}$ : to  $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$  one associates  $t(d) = \langle u, d \rangle v'$ , where  $u$  is any representative of  $u'$ . The auxiliary datum will be a lift  $\tilde{t} : D \rightarrow \mathbb{C}^2$  of  $t$ , or equivalently a lift of  $t$  to  $\mathbb{C}^2/D \otimes \mathbb{C}^2$ . We write it  $\tilde{t} = u' \otimes u$ , with  $u \in \mathbb{C}^2$  mapping to  $u'$  in  $\mathbb{C}^2/D$ .

The tangent space at the origin of  $H$  is  $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$ . For  $z$  in this tangent space, let  $\delta_z$  be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of  $H$ . For  $x, y \in \mathbb{C}^2$ , computing  $[x, y]$  as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y \rangle \delta_{(0,0,1)}. \quad (6.2)$$

Let  $\mathcal{U}_1$  be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation  $\delta_{(0,0,1)} = 1$ . In  $\mathcal{U}_1$ , (6.2) implies that, for  $d \in D$ ,

$$\left[ -\frac{\delta_u^2}{4\pi i}, \delta_d \right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d \rangle \delta_u = \langle u, d \rangle \delta_u = \delta_{t(d)}. \quad (6.3)$$

Let us compute to first order around  $D$ . For  $\varepsilon^2 = 0$  and  $\varphi(g)$  in  $W_D$ , so  $\delta_d \varphi = 0$ , one has

$$\delta_{d+\varepsilon t(d)} \left[ \varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = \varepsilon \left[ \delta_{t(d)} \varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = 0,$$

because  $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$ , by (6.3). To first order around  $D$ , the function of  $g$  and  $D'$  defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle  $\mathcal{W}$ . If  $\varphi(g, D')$  is a local section of  $\mathcal{W}$ , this permits one to define the derivative  $\nabla_t^u \varphi$  in  $\mathcal{W}_D$  by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on  $u$ . If  $u$  is replaced by  $u + x$  with  $x \in D$ , then in  $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u \delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u \delta_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For  $\varphi \in W_D$ ,  $\delta_x \varphi = 0$ , and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

whence

$$\frac{1}{4\pi i} \delta_{u+x}^2 \varphi = \frac{1}{4\pi i} \delta_u^2 \varphi + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of  $u$  also makes it possible to define the derivative  $\nabla_t^u$  with respect to  $t$  of a local section of  $\mathcal{O}(1)$ . A section of  $\mathcal{O}(1)$  on  $U \subset S$  is a homogeneous function of degree 1 on the inverse image of  $U$  in  $\mathbb{C}^2$ , and one differentiates with respect to  $\langle u, d \rangle u$  to obtain a homogeneous function of degree 1 on  $D$ . When  $u$  is changed into  $u + x$ ,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on  $\mathcal{O}(-\frac{1}{2})$ , with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2} \nabla_t^u(s) s^{-3/2},$$

and, for the dependence on  $u$ ,  $s$  now being a local section of  $\mathcal{O}(-\frac{1}{2})$ ,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x \rangle s. \quad (6.9)$$

The dependences on  $u$  in (6.8) and (6.9) cancel to provide a derivation  $\nabla_t$  for  $\mathcal{W}(-\frac{1}{2})$ . Since the connection  $\nabla$  is defined in terms of right-invariant vector fields on  $H$ , it commutes with the action of  $H$ .

## 7

Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let  $E$  be the  $x$ -axis. On  $S$ ,  $\mathcal{O}(1)$  has a section  $s$  with a simple zero at  $E$ . It is unique up to a factor and trivializes  $\mathcal{O}(1)$  on  $S - E$ . Thus there is a square root  $\mathcal{O}(-\frac{1}{2})$  on  $S - E$ , trivialized by a square root  $s^{-1/2}$  of  $s^{-1}$ . By such a trivialization, the desired connection is transported to a connection on the bundle  $\mathcal{W}$ . We calculate it.

Let  $D_t$  be the line generated by  $(-t, 1)$ . By  $r_E$  of (2.1), a local section of  $\mathcal{W}$  is identified with a function  $f(z, t)$  holomorphic in  $z$  and  $t$ . We compute its derivative with respect to  $t$ . In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1, 0), d \rangle (1, 0).$$

The corresponding derivation of  $\mathcal{O}(1)$  annihilates  $s$ , so that simply

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through  $s \in W_D^0(-\frac{1}{2})$  there passes a horizontal local section of  $\mathcal{W}(-\frac{1}{2})$ . It is automatically a section of  $\mathcal{W}^0(-\frac{1}{2})$ . It exists in a neighborhood of  $D$  that is larger the slower the growth of  $s$ . Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection  $\nabla$  on  $\mathcal{W}^0(-\frac{1}{2})$  defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\frac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of  $\mathcal{W}^0(-\frac{1}{2})$ . Although the sphere  $S$  is simply connected, one has the following.

## 8. Surprise

The projective-space bundle  $\mathbb{P}(\mathcal{W}^0)$  on  $S$  has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of  $\mathrm{SL}(2, \mathbb{C})$  given by the action of  $\mathrm{SL}(2, \mathbb{C})$  on the projective space of global horizontal sections of  $\mathbb{P}(\mathcal{W}^0)$ .

*First proof.* Divide  $S$  into two hemispheres  $S_1$  and  $S_2$ , and choose  $\mathcal{O}(-\frac{1}{2})_i$  on  $S_i$ . On  $S_1 \cap S_2$  one has locally an isomorphism  $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$ , defined up to sign, but  $\alpha$  can be defined globally only on the double cover of the circle  $S_1 \cap S_2$ : the monodromy is  $-1$ . If there existed global horizontal projective sections of  $\mathbb{P}(\mathcal{W}^0)$ , there would exist horizontal sections  $f_1, f_2$  of  $\mathcal{W}^0(-\frac{1}{2})$  on  $S_1$  and  $S_2$  which, on  $S_1 \cap S_2$ , would coincide up to sign. On  $S_1 \cap S_2$

this would make it possible to normalize  $\alpha : \mathcal{O}(-\frac{1}{2})_1 \rightarrow \mathcal{O}(-\frac{1}{2})_2$  by  $\alpha(f_1) = f_2$ , contradicting the nonexistence of  $\alpha$  on  $S_1 \cap S_2$ .

*Second proof.* Choose coordinates as in no. 1. Let  $E$  be the  $x$ -axis and  $D$  the  $y$ -axis. As in no. 2, identify  $W_D$  with the space of entire functions  $f(z)$ . Let  $f \neq 0$  in  $W_D$ . By the proposition, for there to exist on  $S - E$  a horizontal section of  $\mathbb{P}(\mathcal{W}^0)$  passing through  $f$ , it is necessary and sufficient that  $f$  satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon|z|^2)). \quad (8.1)$$

If  $E'$  is another line, then by (4.1) the existence of a horizontal section on  $S - E'$  passing through  $f$  is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon|z|^2)) \quad (8.2)$$

with suitable  $a \neq 0$ .

Put  $g(z) = f(z) \exp(az^2/2)$ . By (8.1) and (8.2),  $|g(z)|$  is bounded by a multiple both of  $|\exp(\varepsilon|z|^2 + az^2/2)|$  and of  $|\exp(\varepsilon|z|^2 - az^2/2)|$ ; hence in all directions except those for which  $az^2$  is purely imaginary,  $g$  decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf,  $g$  decreases exponentially quadratically in all directions, hence is zero, and therefore  $f = 0$ .

## 9

In the coordinates of no. 1, let  $D_t$  be the line generated by  $(-t, 1)$ . Let  $d = (-tz, z)$ . The commutator of  $d$  and  $\bar{d}$  is

$$\langle d, \bar{d} \rangle = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \bar{d} \rangle > 1 \iff \text{Im}(t) > 0.$$

The lines  $D$  such that, for  $d$  generating  $D$ , one has  $\langle d, \bar{d} \rangle > 1$ , therefore form a disk in  $S$ , called the disk attached to the real structure  $H_{\mathbb{R}}$ .

The real structures  $L \subset \mathbb{C}^2$  for which  $\langle \cdot, \cdot \rangle$  is real on  $L$  form a homogeneous space under  $\text{SL}(2, \mathbb{C})$ , with  $\text{SL}(2, \mathbb{R})$  as the stabilizer of  $\mathbb{R}^2$ . Each defines a real structure  $H_L = L \cdot U^1$  on  $H$ , hence a complex conjugation  $\sigma_L$  with fixed points  $LU^1$ .

The disks in  $S$  likewise form a homogeneous space under  $\text{SL}(2, \mathbb{C})$ , the stabilizer of the one associated to  $H_{\mathbb{R}}$  being  $\text{SL}(2, \mathbb{R})$ . The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in  $S$ .

Passing from a disk to the complementary disk corresponds, on real structures, to  $L \mapsto iL$ .

## 10

The calculations of no. 3 admit the following interpretation. Let  $X$  be an open disk of  $S$ , corresponding to a real structure  $L$ , whence real forms  $H_L$  of  $H$  and  $\text{SL}(L)$  of  $\text{SL}(2, \mathbb{C})$ . Then  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$  has an  $H_L$ -invariant pre-Hilbert structure; its completion  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$  is the irreducible unitary representation of  $H_L$ . One has

$$\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\frac{1}{2})^{\nabla=0}).$$

The double cover  $\mathrm{SL}(L)^\sim$  of  $\mathrm{SL}(L)$  acts on  $\mathcal{O}(-\frac{1}{2})|_X$  compatibly with its action on  $\mathcal{O}(-1)$ ; hence  $\mathrm{SL}(L)^\sim$  acts on  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ . This is the action that normalizes the action of  $H_L$  and induces the natural action of  $\mathrm{SL}(L)$  on  $H_L$ .

The subrepresentation  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$  may be interpreted as the space of real analytic vectors for the action of  $\mathrm{SL}(L)^\sim$ .

This description shows that the projective representation of  $\mathrm{SL}(L)$  extends to the monoid of  $g \in \mathrm{SL}(2, \mathbb{C})$  such that  $gX \supset X$ .

## 11

Everything preceding generalizes from  $\mathrm{SL}(2, \mathbb{C})$  to  $\mathrm{Sp}(2n, \mathbb{C})$ : replace  $\mathbb{C}^2$  by  $\mathbb{C}^{2n}$ , endowed with a nondegenerate alternating form  $\langle \cdot, \cdot \rangle$  real on  $\mathbb{R}^{2n}$ , replace  $H$  by  $\mathbb{C}^{2n} \times \mathbb{C}^*$  with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle)),$$

and replace  $S$  by the space of Lagrangian, i.e. maximal isotropic, subspaces  $D$  of  $\mathbb{C}^{2n}$ . The disk attached to a real structure becomes the Siegel space of those  $D$  for which  $|\langle d, \bar{d} \rangle| > 1$  for  $d \neq 0$  in  $D$ . The invertible sheaf  $\mathcal{O}(-\frac{1}{2})$  must be replaced by  $\det(D)^{1/2}$ .

## 12. Remark

Beside representations of  $H$  in which  $\mathbb{C}^*$  acts by multiplication by  $\lambda$ , consider likewise representations in which the action is by  $\lambda^{-1}$ . One defines, as in no. 2, a holomorphic bundle  $\mathcal{W}_-^0$  of such representations over  $S$ , and on  $\mathcal{W}_-^0(-\frac{1}{2})$  one again has a connection, given by a heat equation.

On a connected simply connected open subset  $U$  of  $S$ , consider  $W = \mathcal{W}^0(-\frac{1}{2})$ , its connection, and the action of  $H$ . In  $\Gamma(U, W)^{\nabla=0}$ , for each  $x \in U$  there is a linear form fixed by the corresponding line, defined up to a factor:  $f \mapsto f(e)$  for  $f \in \mathcal{W}_x^0$ . For each  $x \notin U$ , there exists a vector  $v(x)$  fixed by  $D_x$ ; in the Schrödinger model, and for  $x$  corresponding to  $E =$  the  $x$ -axis, it is the constant 1. It is unique up to a factor.

Fix the real structure  $H_{\mathbb{R}}$ , whence a real structure  $S_{\mathbb{R}} \subset S$  (a circle), and a decomposition of  $S - S_{\mathbb{R}}$  into two disks  $U_1, U_2$ . Then:

- (a) The complex conjugate of  $\Gamma(U_1, W)^{\nabla=0}$  is  $\Gamma(U_2, W_-)^{\nabla=0}$ .
- (b) The Hermitian form on  $\Gamma(U_1, W)^{\nabla=0}$ , invariant under  $H_{\mathbb{R}}$ , is reinterpreted as a bilinear pairing between  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ , invariant under  $H$ .

**Question.** Let  $C$  be a Jordan curve in  $S$ , separating  $S - C$  into two open sets  $U_1$  and  $U_2$ . Does there exist a unique separating  $H$ -invariant bilinear form pairing  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ , induced by a perfect duality between  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ ?

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# The Tame Symbol

Pierre Deligne

**Working English translation.** This is a faithful mathematical working translation preserving the article's numbered structure, formulas, and constructions. The accompanying French TeX layer is a cleaned working alignment layer.

## 1. Introduction

Let  $f$  and  $g$  be two meromorphic functions on a Riemann surface  $S$ . The *tame symbol* of  $f$  and  $g$  at  $x \in S$  is

$$(f, g)_x = (-1)^{v_x(f)v_x(g)} \left( \frac{f^{v_x(g)}}{g^{v_x(f)}} \right) (x), \quad (1.1)$$

where  $v_x(f)$  denotes the valuation of  $f$  at  $x$ , namely the order of the zero, or the negative of the order of the pole. The quotient appearing in (1.1) has valuation zero and hence can be evaluated at  $x$ . The definition is due to J. Tate; in the language of local symbols for the multiplicative group it gives, for compact  $S$ , the product formula

$$\prod_{x \in S} (f, g)_x = 1, \quad (1.2)$$

where almost all factors are equal to 1. This generalizes Weil reciprocity

$$f(\operatorname{div} g) = g(\operatorname{div} f)$$

for functions with disjoint divisors. If  $K$  is the fraction field of a discrete valuation ring with residue field  $k$ , formula (1.1) still defines a pairing

$$K^* \otimes K^* \longrightarrow k^*,$$

and, up to sign, this map can be interpreted in  $K$ -theory as the composite of the product  $K_1(K) \otimes K_1(K) \rightarrow K_2(K)$  with the boundary  $K_2(K) \rightarrow K_1(k)$ .

Let  $D$  be the region in  $\mathbf{C}$  bounded by a  $C^\infty$  curve  $\Gamma$ , and let  $f, g$  be meromorphic functions on a neighbourhood of  $D$ , without zeros or poles on  $\Gamma$ . The starting point of this article was a formula expressing

$$\prod_{x \in D} (f, g)_x$$

as the monodromy along  $\Gamma$  of a flat line bundle on  $\Gamma$  defined by the restrictions of  $f$  and  $g$  to  $\Gamma$ . Concretely, this is an integral formula for that product.

Following a suggestion of O. Gabber, the essential point is the construction, in the derived category of sheaves of abelian groups on a Riemann surface  $S$ , of a morphism of degree one

$$\mathcal{O}^* \otimes^L \mathcal{O}^* \longrightarrow \mathbf{C}^*[1]. \quad (1.3)$$

This morphism is given by an explicit formula at the level of complexes resolving  $\mathcal{O}^*$  and  $\mathbf{C}^*$ . It therefore has a local interpretation: to two invertible holomorphic functions  $f$  and  $g$  it attaches a  $\mathbf{C}^*$ -torsor, equivalently a flat line bundle, denoted  $(f, g)$ .



In arbitrary dimension there is an analogous morphism, with values in the complex

$$\mathcal{O}^* \xrightarrow{d\log} \Omega^1,$$

which is quasi-isomorphic to  $\mathbf{C}^*$  in dimension one but not in higher dimension. It attaches to two invertible holomorphic functions  $f, g$  a line bundle  $(f, g)$  with connection, not necessarily integrable. In fact this morphism is the one defining the product in Beilinson-Deligne cohomology,  $\mathbf{Z}(1) \otimes \mathbf{Z}(1) \rightarrow \mathbf{Z}(2)$ ; the corresponding interpretation in terms of Hodge theory is described in Section 3.

The same constructions exist on  $C^\infty$  manifolds, where  $\mathcal{O}^*$  is replaced by the sheaf of invertible  $C^\infty$  functions and  $\mathcal{O}^* \rightarrow \Omega^1$  by its differentiable analogue. Less regularity is also possible.

For a commutative complex analytic group  $G$ , (1.3) admits the generalization

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1], \quad (1.4)$$

which attaches a flat  $G$ -torsor  $(f, g]$  to an invertible holomorphic function  $f$  and a holomorphic map  $g$  with values in  $G$  on a Riemann surface. Its monodromy is analogous to the local symbols of class field theory. In particular, if  $\mathcal{L}$  is an invertible sheaf trivialized at infinity and  $g$  is a holomorphic map to  $G$ , one obtains a symbol  $(\mathcal{L}, g] \in G$  with  $(\mathcal{O}(x), g] = g(x)$ .

For the circle  $S^1$ , the map

$$(f, g) \longmapsto \text{monodromy}(f, g)$$

identifies, in a sense made precise below, the multiplicative group of invertible functions on  $S^1$  with its own Cartier dual. This autoduality is skew-symmetric. It is an analytic analogue of the Contou-Carrere result that the functor sending a commutative ring  $A$  to  $(A[[T]][T^{-1}])^*$  is its own Cartier dual. A twisted version with coefficients in a local system of free finite-rank  $\mathbf{Z}$ -modules on  $S^1$  is also proved.

If  $S$  is a Riemann surface with boundary a circle, the boundary values of invertible holomorphic functions form a maximal isotropic subspace of the group of invertible smooth functions on the boundary. Variants are proved for several boundary circles and for local systems. These results are parallel to the autoduality of the Jacobian, and of Prym varieties.

Throughout, Tate twists such as  $\mathbf{Z}(1) = 2\pi i \mathbf{Z}$  are needed to make the statements natural and independent of the choice of a square root  $i$  of  $-1$  in  $\mathbf{C}$ .

The last section applies these tools to a preliminary construction needed for holomorphic conformal field theories attached to certain orbifolds of the form “torus modulo a finite group”. Let  $B$  be an even positive definite bilinear form on  $\mathbf{Z}^n$ , and let  $S$  be an oriented circle. From  $B$  and an auxiliary datum one obtains a central extension of the group of smooth maps  $S \rightarrow \mathbf{R}^n/\mathbf{Z}^n$  by  $\mathbf{C}^*$ , and a compatibility with gluing along surfaces with boundary.

## 1 The main construction

Let  $X$  be a complex analytic variety. We write  $\mathcal{O}$  for the sheaf of holomorphic functions,  $\mathcal{O}^*$  for the invertible functions, and  $G(\mathcal{O})$  for the sheaf of holomorphic maps to a commutative complex Lie group  $G$ . The exponential sequence for  $G$  is written in the form

$$0 \longrightarrow G(\mathbf{C}) \longrightarrow G(\mathcal{O}) \xrightarrow{d\log} \Omega^1 \otimes \text{Lie}(G) \longrightarrow 0 \quad (2.1.1)$$

when  $X$  is one-dimensional and  $G$  is locally represented in logarithmic coordinates.

For a complex  $[A \rightarrow B]$  placed in degrees  $n, n+1$  we write  $[A \rightarrow B]_n$ . The fundamental morphism is obtained from the elementary formula

$$[\mathcal{O}^* \rightarrow \Omega^1]_0 \otimes^L G(\mathcal{O}) \longrightarrow G[1]. \quad (2.2.1)$$

At the level of local logarithms, it is given by the expression

$$(f, g) \longmapsto \exp \left( \frac{1}{2\pi i} \int \log(f) d \log(g) \right), \quad (2.2.2)$$

with the usual correction terms on overlaps. Passing to cohomology, (2.2.1) gives a product

$$H^i(X, \mathcal{O}^*) \times H^j(X, G(\mathcal{O})) \longrightarrow H^{i+j+1}(X, G). \quad (2.3.1)$$

For  $i = j = 0$  this attaches to local functions  $f, g$  the torsor  $(f, g]$ .

The construction may be written using Čech cocycles. Let  $\{U_i\}$  be an open cover, choose branches  $\log f_i$  of  $f$ , and represent  $g$  locally. The change of branch  $\log f_i - \log f_j \in 2\pi i \mathbf{Z}$  supplies the transition functions of a  $G$ -torsor by applying these integral changes to the periods of  $g$ . This Čech description is often more explicit than the derived-category one, but it is the same construction.

On a  $C^\infty$  manifold the construction is identical after replacing holomorphic functions by smooth invertible functions and holomorphic differentials by smooth one-forms. For a Riemann surface  $S$ , since  $[\mathcal{O}^* \rightarrow \Omega^1]$  is quasi-isomorphic to  $\mathbf{C}^*$ , the construction becomes a pairing

$$\mathcal{O}^* \otimes^L G(\mathcal{O}) \longrightarrow G[1] \quad (2.6.1)$$

which yields flat torsors.

Let  $x \in S$ , let  $f$  be invertible on  $S - \{x\}$  with prescribed divisor at  $x$ , and let  $g : S - \{x\} \rightarrow G$ . The monodromy of  $(f, g]$  around  $x$  is the local symbol. For  $G = \mathbf{G}_m$  it is the tame symbol (1.1). For  $G = \mathbf{G}_a$  one obtains the residue pairing

$$(f, g]_x = \text{res}_x(g d \log f). \quad (2.8.1)$$

For  $G = \mathbf{G}_m$  we write  $(f, g)$  instead of  $(f, g]$ . The symbol is bimultiplicative and satisfies the product formula on compact Riemann surfaces.

Let  $\mathcal{L}$  be an invertible sheaf on  $S$  trivialized at infinity. Applying the construction to a divisor representing  $\mathcal{L}$  gives a global symbol

$$(\mathcal{L}, g] \in G, \quad (2.10.1)$$

characterized by  $(\mathcal{O}(x), g] = g(x)$  and by multiplicativity in divisors. The case of cohomology with compact support gives a parallel construction when  $i = j = 1$ .

There is also a relative version. For a family of Riemann surfaces and holomorphic functions varying over a base, the torsors  $(f, g]$  vary functorially and are compatible with base change. The local monodromy formula survives in families.

## 2 The case of the multiplicative group

Take  $G = \mathbf{G}_m$ . The product (2.2.1) becomes the product in Deligne-Beilinson cohomology

$$\mathbf{Z}(1)_{\mathcal{D}} \otimes^L \mathbf{Z}(1)_{\mathcal{D}} \longrightarrow \mathbf{Z}(2)_{\mathcal{D}}, \quad (3.1.1)$$

where  $\mathbf{Z}(1) = 2\pi i\mathbf{Z}$  and  $\mathbf{Z}(2) = (2\pi i)^2\mathbf{Z}$ . Explicitly, the line bundle with connection  $(f, g)$  is represented locally by logarithms of  $f$  and  $g$  and by the connection one-form

$$\nabla = d + \frac{1}{2\pi i} \log(f) d \log(g). \quad (3.1.2)$$

When  $X$  is a Riemann surface the connection is flat and its monodromy is the tame symbol.

The Čech formula reads as follows. Choose logarithms  $u_i = \log f$  and  $v_i = \log g$  on  $U_i$ . On intersections the differences  $u_i - u_j$  and  $v_i - v_j$  lie in  $2\pi i\mathbf{Z}$ . The transition functions of  $(f, g)$  and the local connection forms are assembled from

$$\frac{1}{2\pi i} u_i dv_i, \quad \exp\left(\frac{1}{2\pi i} (u_i - u_j) v_j\right), \quad (3.2.1)$$

with the standard cocycle correction on triple intersections. The same formula with parameters gives the relative torsor.

The construction is bimultiplicative. There are canonical isomorphisms

$$(f_1 f_2, g) \simeq (f_1, g) \otimes (f_2, g), \quad (3.7.1)$$

$$(f, g_1 g_2) \simeq (f, g_1) \otimes (f, g_2). \quad (1)$$

It is skew-symmetric in the sense that there is a canonical isomorphism

$$(f, g) \otimes (g, f) \simeq \mathbf{1}. \quad (3.8.1)$$

For  $f = -1$  the canonical trivialization in (3.8.1) must be treated with the sign convention encoded in the Tate twist; otherwise one misses the contribution of the chosen square root of  $-1$ .

By bilinearity, one may extend  $(f, g)$  to sections of a local system. Let  $A$  be a local system of free finite-rank  $\mathbf{Z}$ -modules,  $B : A \otimes A \rightarrow \mathbf{Z}$  a bilinear form, and let  $f$  be a section of  $\mathcal{O}^* \otimes A$ . The expression

$$(f, f)_B \quad (3.10.1)$$

is defined by combining the pairings of the components with coefficients given by  $B$ . If  $B$  is symmetric and even, the construction is independent of ordering. If  $B$  is symmetric but not necessarily even, the parity term

$$B(x, x) \pmod{2} \quad (3.11.1)$$

controls the additional sign.

Let  $S$  be an oriented circle and  $A$  a local system of free finite-rank  $\mathbf{Z}$ -modules on  $S$ . Choosing  $i \in \mathbf{C}$  identifies  $\mathbf{Z}$  with  $\mathbf{Z}(1)$  by  $1 \mapsto 2\pi i$ . The exponential sequence gives exact sequences of sheaves on  $S$ :

$$0 \longrightarrow A \longrightarrow \mathcal{C}^\infty \otimes A \xrightarrow{\exp(2\pi i \cdot)} (\mathcal{C}^\infty)^* \otimes A \longrightarrow 0, \quad (3.13.1)$$

$$0 \longrightarrow A \longrightarrow \mathcal{C} \otimes A \longrightarrow \mathbf{C}^* \otimes A \longrightarrow 0. \quad (2)$$

These sequences will be used in the duality theorem of Section 4.

### 3 Duality

This section gives analytic analogues of the autoduality of the Jacobian and of the Contou-Carrere symbol. For an oriented  $C^\infty$  circle  $S$  the exponential sequence

$$0 \longrightarrow 2\pi i \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0 \quad (4.1.1)$$

and the complex

$$0 \longrightarrow \mathbf{C}^* \longrightarrow \mathcal{O}^* \xrightarrow{d\log} \Omega^1 \longrightarrow 0 \quad (4.1.2)$$

identify the group of smooth invertible functions, up to the correct completion, with its Cartier dual.

More generally, for a local system  $A$  of free  $\mathbf{Z}$ -modules, the pairing of Section 3 gives a perfect pairing between the group of invertible smooth functions with values in  $A$  and the corresponding dual group with values in  $A^\vee(1)$ . The duality is skew-symmetric and functorial in  $A$ .

One treats the relevant cohomology groups as topological abelian groups. For a locally convex topological vector space the dual means the continuous character group. Exactness is understood in the topological sense. The pairings are computed by integrating logarithms and logarithmic differentials along the circle; in a local coordinate  $x$  one encounters expressions of the form

$$\int_S a(x) d\log f(x) \quad (4.6.1)$$

and, for the skew form, expressions of the form

$$\int_S \log f d\log g. \quad (4.6.2)$$

If  $A$  is a local system on  $S$ , the exact sequence

$$0 \rightarrow H^0(S, A) \rightarrow H^0(S, \mathcal{O} \otimes A) \rightarrow H^0(S, \Omega^1 \otimes A) \rightarrow H^1(S, A) \rightarrow 0 \quad (4.7.1)$$

serves to write the pairing explicitly and to show its non-degeneracy. In the basic example  $S = U(1)$  and  $A = \mathbf{Z}$ , the group of invertible smooth functions decomposes into winding number and exponentials of smooth functions; the duality exchanges these two parts as expected.

Let  $S$  now be a compact Riemann surface with boundary. Boundary values of holomorphic invertible functions form an isotropic subgroup for the boundary pairing. The image is maximal isotropic. This is the analytic counterpart of the assertion that the Jacobian of a compact Riemann surface is principally polarized.

With coefficients in a local system  $A$ , and a positive definite bilinear form  $B$  on  $A$ , the boundary pairing is written

$$\langle f, g \rangle_B = (f, g)_B. \quad (4.16.1)$$

The proof shows that  $\log(f, f)_B$  is the  $L^2$ -norm of a harmonic representative. Hence for positive definite  $B$  the corresponding isotropic condition is also a positivity statement. For the annulus one obtains a Fourier-expansion model: if the two boundary components are related by the modulus of the annulus, the pairing is expressed as a convergent sum over Fourier modes.

## 4 Construction of central extensions

Let  $S$  be a compact Riemann surface with boundary and  $A$  a local system on  $S$  endowed with an even positive definite symmetric bilinear form  $B$ . The objective is to construct central extensions of groups of boundary values by  $\mathbf{C}^*$ , using the torsors and gerbes associated with the symbol.

We use the language of torsors and gerbes. A  $G$ -torsor is a sheaf locally isomorphic to  $G$  with its simply transitive  $G$ -action. A gerbe banded by  $G$  is a stack locally non-empty and locally connected, with automorphism group  $G$ ; it represents classes in  $H^2(-, G)$ . The torsor associated with  $(f, g)$  is thus the  $H^1$  incarnation of the symbol, and the gerbe construction is its  $H^2$  analogue.

A central extension of a sheaf of groups  $L$  by  $G$  is a  $G$ -torsor over  $L$  equipped with a multiplication lifting that of  $L$ , with  $G$  central. Equivalently, it is given locally by a two-cocycle, with the usual coboundary equivalence. In categorical language, the same data are encoded by a gerbe with a compatible group law.

Following an idea of Grothendieck, the auxiliary data needed to construct the extension are kept in a sheaf  $\mathcal{H}^*$ . A section consists of a lattice with bilinear form and of the choices required to make the signs and Tate twists coherent. If  $k = (\mathbf{C}, s)$  is such an auxiliary datum, one defines an extension

$$1 \longrightarrow \mathbf{C}^* \longrightarrow \tilde{L}_k \longrightarrow L \longrightarrow 1. \quad (5.12.1)$$

The construction is local on the boundary and compatible with restriction to boundary components.

If  $k' = (A', B')$  and  $k'' = (A'', B'')$  are two data, the extension associated with  $k' + k''$  is obtained from those associated with  $k'$  and  $k''$  by the contracted product. The commutator in the extension is the symbol obtained from the boundary pairing:

$$[\tilde{f}, \tilde{g}] = (f, g)_B. \quad (5.16.1)$$

For a circle lying on the boundary of a surface, the data of Section 5.1 therefore provide a central extension of the boundary loop group. These extensions are the analytic input for the construction of holomorphic conformal field theories attached to torus orbifolds.

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# Real Mixed Hodge Structures

Pierre Deligne

English working translation; formulas, filtrations, and numbering retained.

The real mixed Hodge structures of §2 form a rigid abelian tensor category, and the functor “underlying real vector space” is a fiber functor. The functor  $\mathrm{Gr}^W$  is another one. We compute what the general theory of Tannakian categories gives in this particular case.

## 1 Three opposite filtrations

### 1.1

This paragraph is a variation on §1.2 of P. Deligne, *Théorie de Hodge II*, Publ. Math. IHES 40, pp. 5–58. Let  $\mathcal{A}$  be an abelian category, and let  $A$  be an object of  $\mathcal{A}$  endowed with three opposite filtrations  $W, F, \bar{F}$ . Here “filtration” means finite filtration;  $W$  is increasing, and  $F$  and  $\bar{F}$  are decreasing. The condition of opposition is

$$\mathrm{Gr}_n^W \mathrm{Gr}_F^p \mathrm{Gr}_{\bar{F}}^q A = 0 \quad \text{for } n \neq p + q.$$

For a bigraded object  $B = \bigoplus B^{p,q}$ , we write  $W, F_W, \bar{F}_W$  for the three opposite filtrations

$$W_n B = \bigoplus_{p+q \leq n} B^{p,q}, \quad F_W^i B = \bigoplus_{p \geq i} B^{p,q}, \quad \bar{F}_W^i B = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, saying that  $W, F, \bar{F}$  are opposite on  $A$  is equivalent to saying that, for each  $n$ ,  $\mathrm{Gr}_n^W A$  has a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations induced by  $F$  and  $\bar{F}$  on  $\mathrm{Gr}_n^W A$  are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, p \geq i} A_W^{p,q}, \quad \bar{F}^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n, q \geq i} A_W^{p,q}.$$

Put

$$A_F^{p,q} := W_{p+q} A \cap F^p A \cap \left( \bar{F}^q A + \sum_{i>0} W_{p+q-i} A \cap \bar{F}^{q-i+1} A \right),$$

and, interchanging the roles of  $F, \bar{F}$  and of  $p, q$ ,

$$A_{\bar{F}}^{p,q} := W_{p+q} A \cap \bar{F}^q A \cap \left( F^p A + \sum_{i>0} W_{p+q-i} A \cap F^{p-i+1} A \right).$$

For  $p + q = n$ , the projection from  $W_n A$  to  $\mathrm{Gr}_n^W A$  induces isomorphisms

$$A_F^{p,q} \simeq A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \simeq A_W^{p,q}.$$

It follows that the  $A_F^{p,q}$ , respectively the  $A_{\bar{F}}^{p,q}$ , form a bigrading of  $A$ . The sums of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \quad A_{\bar{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

are bifiltered isomorphisms

$$a_F : (A; W, F) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, F_W), \quad a_{\bar{F}} : (A; W, \bar{F}) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, \bar{F}_W).$$

**Lemma 1.1.** *The automorphism  $d = a_F a_{\overline{F}}^{-1}$  of  $\mathrm{Gr}^W A$  satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r < p \\ s < q}} A_W^{r,s}. \quad (1.1.1)$$

*Proof.* Passing to the associated graded, the isomorphisms  $a_F$  and  $a_{\overline{F}}$  induce the identity on  $\mathrm{Gr}^W A$ . Thus  $\mathrm{Gr}^W(d) = 1$ . For a subset  $I \subset \mathbb{Z}^2$  and a bigraded object  $B$ , write  $B_I = \bigoplus_{(r,s) \in I} B^{r,s}$ . Let

$$I(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } r < p)\},$$

$$J(p,q) = \{(r,s) : r+s \leq p+q \text{ and } ((r,s) = (p,q) \text{ or } s < q)\},$$

$$K(p,q) = \{(r,s) : r+s \leq p+q \text{ and } r < p \text{ and } s < q\}.$$

In the definition of  $A_F^{p,q}$  as an intersection of two subobjects, the first subobject is  $A_{FI(p,q)}$  and the second is  $A_{\overline{F}J(p,q)}$ . Hence  $dA_W^{p,q}$  lies in the intersection corresponding to  $I(p,q) \cap J(p,q) = K(p,q) \cup \{(p,q)\}$ . Since  $\mathrm{Gr}^W(d) = 1$ , this gives exactly (1.1.1).  $\square$

**Proposition 1.1.** *The functor  $A \mapsto \mathrm{Gr}^W A$  is an equivalence from the category of objects of  $\mathcal{A}$  endowed with three opposite filtrations  $W, F, \overline{F}$  to the category of objects of  $\mathcal{A}$  endowed with a finite bigrading and with an automorphism  $d$  satisfying (1.1.1).*

The isomorphism  $a_F : A \rightarrow \mathrm{Gr}^W A$  respects  $W$ , sends  $F$  to  $F_W$ , and sends  $\overline{F}$  to  $d^{-1}\overline{F}_W$ . Thus the functor  $\mathrm{Gr}^W$  has the inverse

$$s : (B = \bigoplus B^{p,q}, d) \longmapsto (B, W, F_W, d^{-1}\overline{F}_W),$$

with the functorial isomorphism  $a_F : A \xrightarrow{\sim} s \mathrm{Gr}^W A$ . Conversely, starting from a bigraded object  $B$  with an automorphism  $d$ , define  $W, F, \overline{F}$  as above. If  $d$  satisfies (1.1.1), the evident map  $B \rightarrow \mathrm{Gr}^W B$  is an isomorphism of functors  $\mathrm{Id} \xrightarrow{\sim} \mathrm{Gr}^W \circ s$ .

**Remark 1.1.** *If 2 is invertible in the ground field, then  $d$  has a unique square root  $d^{1/2}$  still satisfying (1.1.1). Put*

$$a := d^{1/2} a_F = d^{-1/2} a_{\overline{F}}.$$

*Then  $a_{\overline{F}} = d^{1/2} a$  and  $a_F = d^{-1/2} a$ ; thus  $a$  sends  $F$ , respectively  $\overline{F}$ , to*

$$d^{1/2}(F_W), \quad d^{-1/2}(\overline{F}_W)$$

*on  $\mathrm{Gr}^W A$ . The isomorphism  $a$  is the same for  $(A; W, F, \overline{F})$  and for  $(A; W, \overline{F}, F)$ .*

## 1.4

The constructions are compatible with tensor products. If

$$\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{A}$$

is a biadditive exact functor, it is extended to filtered objects by

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a A_1 \otimes F^b A_2 \subset A_1 \otimes A_2.$$

The associated-graded functor is compatible with  $\otimes$ , and, for bifiltered objects, the canonical isomorphism

$$\mathrm{Gr}^W \mathrm{Gr}^F(A_1 \otimes A_2) \simeq \mathrm{Gr}^W \mathrm{Gr}^F(A_1) \otimes \mathrm{Gr}^W \mathrm{Gr}^F(A_2)$$

sends the filtration induced by  $F_1 \otimes F_2$  to the tensor product of the filtrations induced by  $F_1$  and  $F_2$ . For vector spaces this follows from the fact that every bifiltration underlies a bigrading. Hence, if  $(W_i, F_i, \overline{F}_i)$  are three opposite filtrations on  $A_i$ , then  $(W_1 \otimes W_2, F_1 \otimes F_2, \overline{F}_1 \otimes \overline{F}_2)$  are again opposite on  $A_1 \otimes A_2$ . The bigradings, the isomorphisms  $a_F, a_{\overline{F}}$ , and the automorphism  $d$  are all compatible with  $\otimes$ .

## 1.5

Let  $k$  be a commutative field, let  $\mathcal{A}$  be the category of finite-dimensional vector spaces over  $k$ , and let  $\mathcal{C}$  be the category of vector spaces endowed with three opposite filtrations  $W, F, \overline{F}$ . This is a rigid abelian tensor category with  $\text{End}(1) = k$ . It has the fiber functor  $\omega_0$ , the underlying vector space, hence it is neutral Tannakian. It also has the fiber functor  $\text{Gr}^W =: \omega_W$ . The isomorphisms  $a_F$  and  $a_{\overline{F}}$  are isomorphisms of fiber functors  $\omega_0 \rightarrow \omega_W$ . Put

$$\mathcal{G} = \text{Aut}(\omega_W).$$

The functorial automorphism  $d$  defines a  $k$ -point of the pro-algebraic group  $\mathcal{G}$ . The functorial bigrading of  $\text{Gr}^W(V)$  by the  $V^{p,q}$  corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow \mathcal{G}.$$

It is a section of the projection

$$\text{pr} : \mathcal{G} \longrightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion of the category of bigraded vector spaces in  $\mathcal{C}$ . We normalize  $\alpha$  by

$$\alpha(\lambda, \mu)v = \lambda^p \mu^q v \quad (v \in V^{p,q}).$$

By Proposition 1.1,  $\mathcal{G}$  is also the algebraic group of automorphisms of the fiber functor on the category of bigraded vector spaces endowed with an automorphism  $d$  satisfying (1.1.1). We now compute it when  $k$  has characteristic zero.

Let  $\mathfrak{l}$  be the Lie algebra over  $k$  freely generated by elements

$$D^{i,j} \quad (i, j < 0),$$

with the unique bigrading in which  $D^{i,j}$  has bidegree  $(i, j)$ . The  $W_n \mathfrak{l}$  are ideals, and the quotients  $\mathfrak{l}/W_n \mathfrak{l}$  are nilpotent Lie algebras. Let  $\mathcal{U}$  be the pro-algebraic group which is the projective limit of the unipotent groups

$$\exp(\mathfrak{l}/W_n \mathfrak{l}).$$

Let  $\mathbb{G}_m \times \mathbb{G}_m$  act on  $\mathfrak{l}$  so that  $(\lambda, \mu)$  acts on bidegree  $(i, j)$  by multiplication by  $\lambda^i \mu^j$ . This action respects  $W$  and therefore gives an action of  $\mathbb{G}_m \times \mathbb{G}_m$  on  $\mathcal{U}$ .

**Construction 1.1.** *The group  $\mathcal{G}$  is the semidirect product*

$$\mathcal{G} = (\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U}.$$

*Indeed, giving an action of this semidirect product on a finite-dimensional vector space  $V$  is equivalent to giving an action of  $\mathbb{G}_m \times \mathbb{G}_m$ , hence a bigrading, together with a compatible nilpotent action of  $\mathfrak{l}$ . Equivalently, it is the datum of endomorphisms  $D^{i,j}$  of bidegree  $(i, j)$ , or of their sum*

$$D = \sum D^{i,j},$$

*with*

$$D(V^{p,q}) \subset \bigoplus_{r < p, s < q} V^{r,s}.$$

*Writing  $d = \exp(D)$ , this is precisely the datum of a bigrading and of an automorphism satisfying (1.1.1). The construction is compatible with tensor products, giving the required isomorphism*

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U} \xrightarrow{\sim} \mathcal{G} = \text{Aut}(\omega_W).$$



## 2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space endowed with an increasing filtration  $W$ , whose complexification  $V_{\mathbb{C}}$  is endowed with a decreasing filtration  $F$ , such that the filtration  $W_{\mathbb{C}}$ , the filtration  $F$ , and its complex conjugate  $\overline{F}$  form a system of three opposite filtrations.

Via the dictionary between real vector spaces and complex vector spaces with a real structure, giving a real mixed Hodge structure is equivalently giving a complex vector space with three opposite filtrations  $W, F, \overline{F}$  and with a complex conjugation respecting  $W$  and interchanging  $F$  and  $\overline{F}$ . By Proposition 1.1, the functor  $\mathrm{Gr}^W$  identifies this category with the category of bigraded complex vector spaces

$$V = \bigoplus V^{p,q}$$

equipped with an automorphism  $d$  satisfying (1.1.1) and a complex conjugation  $\sigma$  interchanging  $V^{p,q}$  and  $V^{q,p}$ , for which the complex conjugate of  $d$  is  $d^{-1}$ :

$$\sigma d \sigma^{-1} = d^{-1}.$$

Let  $\mathcal{H}$  be the category of real mixed Hodge structures. It is a rigid abelian tensor category with  $\mathrm{End}(1) = \mathbb{R}$ . It has the fiber functor  $\omega_0$ , the underlying real vector space, hence is neutral Tannakian. It also has the fiber functor  $\mathrm{Gr}^W =: \omega_W$ . With the notation of Remark 1.1, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \mathrm{Gr}^W(V_{\mathbb{C}})$$

respects the real structures on both sides, since it depends symmetrically on  $F$  and  $\overline{F}$ . It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, put  $k = \mathbb{C}$ , and let  $\mathcal{G}_{\mathbb{R}}$  be the real form of  $\mathcal{G}$  defined by the following complex conjugation:

- (a) on  $\mathbb{G}_m \times \mathbb{G}_m$ , it is  $(\lambda, \mu) \mapsto (\overline{\mu}, \overline{\lambda})$ ;
- (b) on  $\mathcal{U}$ , it is the one compatible with the complex conjugation of  $\mathfrak{l}$  given by

$$D^{i,j} \longmapsto -D^{j,i}.$$

As in §1.1, giving an action of  $\mathcal{G}_{\mathbb{R}}$  on a real vector space  $V$  is equivalent to giving:

- (a) a bigrading  $V_{\mathbb{C}} = \bigoplus V^{p,q}$  such that  $V^{p,q}$  and  $V^{q,p}$  are complex conjugates of one another;
- (b) an automorphism  $d$  of  $V_{\mathbb{C}}$  such that  $\overline{d} = d^{-1}$  and

$$(d-1)V^{p,q} \subset \bigoplus_{i < p, j < q} V^{i,j}.$$

**Proposition 2.1.** *The fiber functor  $\omega_W$  induces an equivalence between  $\mathcal{H}$  and the category of representations of  $\mathcal{G}_{\mathbb{R}}$ . Thus*

$$\mathcal{G}_{\mathbb{R}} = \mathrm{Aut}(\omega_W).$$

*The inverse functor is*

$$(V, \text{ action of } \mathcal{G}_{\mathbb{R}}) \longmapsto (V, \text{ bigrading of } V_{\mathbb{C}}, d) \longmapsto (V, W, F),$$

*where  $W$  is the real filtration whose complexification is the weight filtration attached to the bigrading of  $V_{\mathbb{C}}$ , and*

$$F = d^{-1/2}(F_W)$$

*on the complexification. The isomorphism  $a$  gives the functorial identification*

$$\phi \circ \mathrm{Gr}^W \xrightarrow{\sim} \mathrm{Id}.$$

# Action of the Braid Group on a Category

Full working English translation

P. Deligne

## Abstract

We describe how to define, by generators and relations, an action of the braid group on a category. More generally, we describe how to define a functor from a generalized positive braid monoid, corresponding to a finite Coxeter group, to a monoidal category. Applications are given to constructions of Bondal (exceptional systems) and of Broué–Michel (correspondences on flag manifolds).

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## 0. Introduction

An action of a monoid  $M$  (not necessarily having a unit) on a category  $C$  is the data of functors  $T(f)$ , for  $f \in M$ , and isomorphisms of functors

$$c_{f,g} : T(f) \circ T(g) \longrightarrow T(fg)$$

which make the diagrams

$$\begin{array}{ccc} T(f) \circ T(g) \circ T(h) & \longrightarrow & T(fg) \circ T(h) \\ \downarrow & & \downarrow \\ T(f) \circ T(gh) & \longrightarrow & T(fgh) \end{array} \quad (0.1)$$

commutative.

The actions of a monoid  $M$  on a category  $C$  form a category: a morphism  $(T, c) \rightarrow (T', c')$  is the data of morphisms of functors  $T(f) \rightarrow T'(f)$  which make the diagrams

$$\begin{array}{ccc} T(f)T(g) & \xrightarrow{c_{f,g}} & T(fg) \\ \downarrow & & \downarrow \\ T'(f)T'(g) & \xrightarrow{c'_{f,g}} & T'(fg) \end{array}$$

commutative.

Suppose that  $M$  is given by generators and relations,  $M = \langle G \mid R \rangle$ . If to each generator  $g \in G$  one assigns a functor  $T(g) : C \rightarrow C$ , and if each relation  $r_1 = r_2$  in  $R$  gives an equality of composite functors, then by composing the  $T(g)$ ,  $g \in G$ , one obtains an action of  $M$  on  $C$  for which the  $c_{f,g}$  are identity isomorphisms. This construction is of no interest. The useful

question is the following. Consider the category of systems of the following type: (a) functors  $T(g) : C \rightarrow C$  for  $g \in G$ ; (b) isomorphisms between composite functors for each relation  $r_1 = r_2$  in  $R$ ; (c) compatibilities of the type (0.1), to be specified. How should (c) be specified so that the restriction functor from the category of actions of  $M$  on  $C$  to the category of systems (a),(b) satisfying (c) is an equivalence of categories? In other words, when is it “the same thing” to give an action of  $M$  on  $C$  as to give a system (a),(b) satisfying (c)?

In this article we treat the case in which  $M$  is the monoid  $B_n^+$  of strictly positive braids (1.2). We use a system of generators indexed by the non-trivial elements of the symmetric group  $S_n$ . For  $b \in B_n^+$ , let  $E(b)$  be the set of decompositions of  $b$  as a product of generators. This set has a natural order, and our main tool is the theorem that the geometric realization of the ordered set  $E(b)$  is contractible. One may think of this theorem as a uniqueness result, up to homotopy, for the decomposition of  $b$  as a product of generators. In proving the contractibility of  $E(b)$ , our methods are those of Deligne (1972) and hence, indirectly, those of Garside (1969).

Our methods apply equally well to generalized strictly positive braid monoids (1.4), corresponding to finite Coxeter groups. Moreover,  $f \mapsto T(f)$  may take values in an arbitrary monoidal category  $\mathcal{M}$ , rather than in  $\text{Hom}(C, C)$ .

At the end of this introduction we give the two applications which motivated us. The reader may prefer to read them after §2.

**Application 1.** Let  $T$  be a triangulated category and let

$$W_1 \subset \cdots \subset W_N = T$$

be a filtration of  $T$  by triangulated subcategories, with  $W_n$  thick (Verdier (1963), p. 13) in  $W_{n+1}$  for  $1 \leq n < N$ . Put  $W_0 := 0$ .

Following Bondal–Kapranov (1989), 4.1, say that the filtration  $W$  is right admissible if the following equivalent conditions are satisfied for  $1 \leq n < N$ : (i) the inclusion of  $W_n$  into  $W_{n+1}$  has a right adjoint; (ii) the quotient functor  $W_{n+1} \rightarrow W_{n+1}/W_n$  has a right adjoint; (iii) every object  $Y$  of  $W_{n+1}$  occurs in a distinguished triangle  $(X, Y, Z)$  with  $X$  in  $W_n$  and  $Z$  in the right orthogonal of  $W_n$  in  $W_{n+1}$ . For the equivalence of these conditions, see Bondal–Kapranov (1989), §1, or Verdier (1963), Ch. 1, §2, no. 6. The filtration  $W$  is left admissible if it is right admissible in the opposite triangulated category: in (i),(ii), replace “right adjoint” by “left adjoint”. It is admissible if it is both left and right admissible.

Write

$$\text{Gr}_n^W(T) := W_n/W_{n-1} \quad (1 \leq n \leq N).$$

For  $W$  right admissible, write

$$\begin{aligned} i_{n*} : W_n &\rightarrow T && \text{for the inclusion functor,} \\ i_n^! : T &\rightarrow W_n && \text{for its right adjoint,} \\ j_{n*} : \text{Gr}_n^W(T) &\rightarrow T && \text{for the composite of } i_{n*} \text{ and the right adjoint} \\ &&& \text{of } W_n \rightarrow W_n/W_{n-1} = \text{Gr}_n^W(T), \\ j_n^! : T &\rightarrow \text{Gr}_n^W(T) && \text{for the composite of the quotient functor and } i_n^!. \end{aligned}$$

As explained in Bondal–Kapranov (1989), 4.3, giving a right admissible filtration  $W$  is equivalent to giving a sequence  $(B_i)_{1 \leq i \leq N}$  of strictly full triangulated subcategories of  $T$  which generate  $T$ , with  $B_j$  right orthogonal to  $B_i$  for  $i < j$ , i.e.  $\text{Hom}(X, Y) = 0$  for  $X \in B_i$  and  $Y \in B_j$ . Passing from  $W$  to  $(B_i)$ :  $B_n$  is the essential image of  $\text{Gr}_n^W(T)$  by the functor  $j_{n!}$ . Passing from  $(B_i)$  to  $W$ :  $W_n$  is the triangulated subcategory of  $T$  generated by the  $B_i$  with  $i \leq n$ , i.e. the smallest strictly full triangulated subcategory containing these  $B_i$ .

For a permutation  $\sigma$  of  $\{1, \dots, N\}$ , the right mutation  $\sigma_d W$ , or simply  $\sigma W$ , of  $W$  is defined by

$$(\sigma W)_n := \text{the triangulated subcategory generated by the } B_i \text{ with } \sigma i \leq n.$$

Equivalently, it is the category of objects  $K$  of  $T$  for which  $j_i^! K = 0$  whenever  $\sigma i > n$ . The equivalences

$$\mathrm{Gr}_n^W T \xleftarrow{\sim} B_n \xrightarrow{\sim} \mathrm{Gr}_{\sigma n}^{\sigma W} T \quad (0.2)$$

give an equivalence of  $\mathrm{Gr}_n^W T$  with  $\mathrm{Gr}_{\sigma n}^{\sigma W} T$ .

For  $W$  left admissible, the left mutation  $\sigma_s W$  is the right mutation in the opposite triangulated category. If  $W$  is admissible, the right mutation  $\sigma_d W$  is still left admissible, and  $(\sigma^{-1})_s \sigma_d W = W$ .

A filtration  $W$  is  $\infty$ -admissible if it is admissible, as are all its iterated left or right mutations. Bondal–Kapranov (1989), §§3,4 give convenient criteria for verifying  $\infty$ -admissibility.

The following example explains the notation chosen above.

**Example 1.** Let  $X$  be a topological space and let

$$F_1 \subset \cdots \subset F_N = X$$

be a filtration of  $X$  by closed subspaces. Put  $F_0 := \emptyset$  and  $S_n := F_n - F_{n-1}$ ,  $n \in I := \{1, \dots, N\}$ . The filtration  $W$  of  $T := D^+(X)$  by the thick subcategories  $D^+(F_n)$  is admissible. The functors  $i_{n*}$  and  $i_n^!$  (respectively  $j_{n*}$  and  $j_n^!$ ) identify with the functors of the same name attached to the inclusion  $i_n : F_n \hookrightarrow X$  (respectively  $j_n : S_n \hookrightarrow X$ ).

Factor the inclusion  $j_n$  of  $S_n$  into  $X$  as the inclusion maps

$$g_{n,i} : F_n - F_i \longrightarrow F_n - F_{i-1} \quad (n \geq i > 0)$$

and  $i_n$ . Let  $\sigma$  be a permutation of  $\{1, \dots, N\}$ , and let  $\sigma W$  be the corresponding right mutation of  $W$ . The subcategory  $(\sigma W)_n$  of  $D^+(X)$  is the category of objects  $K$  such that  $j_i^! K = 0$  for  $\sigma i > n$ . The filtration  $\sigma W$  is left admissible; the right orthogonal of  $(\sigma W)_{\sigma n-1}$  in  $(\sigma W)_{\sigma n}$  is the essential image of  $D^+(S_n)$  by the composite functor of the  $g_{n,i}!$  for  $\sigma i < \sigma n$ , or the  $g_{n,i*}$  for  $\sigma i > \sigma n$ , followed by  $i_{n*} = i_n!$ . This composite functor induces the already considered equivalence from  $D^+(S_n) = \mathrm{Gr}_n^W(T)$  to  $\mathrm{Gr}_{\sigma n}^{\sigma W}(T)$ .

Let  $T$  be a triangulated category and let  $(T_i)_{i \in I}$  be a family with  $N$  elements of triangulated categories. Consider exceptional systems  $E$  of the following type (a),(b),(c):

- (a) a total order on  $I$ , identified with a bijection  $\tau : I \rightarrow \{1, \dots, N\}$ ;
- (b) an  $\infty$ -admissible filtration  $W : W_1 \subset \cdots \subset W_N = T$  of  $T$ ;
- (c) equivalences of triangulated categories  $\varphi_i : \mathrm{Gr}_{\tau(i)}^W(T) \rightarrow T_i$  for  $i \in I$ .

Exceptional systems form a category: there is a morphism from  $E$  to  $E'$  only if the corresponding orders coincide; in that case a morphism is a system of isomorphisms of functors  $\varphi_i \rightarrow \varphi'_i$ .

Let  $\sigma$  be a permutation of  $\{1, \dots, N\}$ . The right mutation  $\sigma E$  of  $E = (\tau, W, (\varphi_i))$  is the exceptional system defined by  $\sigma \circ \tau$ , by  $\sigma W$ , and by the equivalences

$$\mathrm{Gr}_{\sigma\tau(i)}^{\sigma W}(T) \xleftarrow{(0.2)} \mathrm{Gr}_{\tau(i)}^W(T) \xrightarrow{\varphi_i} T_i.$$

The construction  $E \mapsto \sigma E$  is an autoequivalence of the category of exceptional systems; its inverse is given by a left mutation.

Let  $\ell(\sigma)$  be the number of pairs  $(i, j)$  such that  $i < j$  and  $\sigma(j) < \sigma(i)$ . If  $\ell(\sigma_1 \sigma_2) = \ell(\sigma_1) + \ell(\sigma_2)$ , one checks that  $\sigma_1 \sigma_2 E$  is canonically isomorphic to  $\sigma_1(\sigma_2 E)$ , with the compatibility for a triple product  $\sigma_1 \sigma_2 \sigma_3$  if the lengths add. Applying 1.5 and 1.10, one obtains an action of the braid group on the category of exceptional systems.

This makes precise Bondal–Kapranov (1989), 4.10, which defines by mutations the action of the braid group on the set of  $\infty$ -admissible filtrations and proves the equivalence of  $\mathrm{Gr}_n^W(T)$

with  $\mathrm{Gr}_{\bar{b}n}^{bW}(T)$ , where  $b \mapsto \bar{b}$  is the projection from the braid group to the symmetric group. We obtain an equivalence

$$[b] : \mathrm{Gr}_n^W(T) \longrightarrow \mathrm{Gr}_{\bar{b}n}^{bW}(T)$$

well-defined up to a unique isomorphism, an isomorphism  $[b_1 b_2] \xrightarrow{\sim} [b_1] \circ [b_2]$ , and a compatibility for a triple product.

**Variant.** Suppose the triangulated category  $T$  is  $k$ -linear, for  $k$  a commutative field, and that the  $k$ -vector spaces

$$\mathrm{Hom}^*(X, Y) := \bigoplus_n \mathrm{Hom}(X, Y[n])$$

are finite-dimensional. If each category  $T_i$  is the bounded derived category of finite-dimensional vector spaces over  $k$ , i.e. the category of finite-dimensional graded vector spaces, the notion of exceptional system reduces to that of an exceptional sequence: a sequence  $X_1, \dots, X_N$  of objects of  $T$  such that  $\mathrm{Hom}^*(X_i, X_j) = 0$  for  $i < j$  and  $\mathrm{Hom}^*(X_i, X_i) = k$ , which generate  $T$ . More precisely, one should consider a finite family  $(X_i)_{i \in I}$ , a total order  $\leq$  on  $I$ , and require that the  $X_i$  generate  $T$ , that  $\mathrm{Hom}^*(X_i, X_j) = 0$  for  $i < j$ , and that  $\mathrm{Hom}^*(X_i, X_i) = k$ . The corresponding exceptional system is defined by the order  $\leq$ , identified with a bijection  $\tau$  as in (a), and by the filtration  $W$ , where  $W_n$  is generated by the  $X_i$  for  $\tau(i) \leq n$ . The equivalence  $\varphi_i$  is characterized by sending  $X_i$  to  $k$  in degree 0. According to Bondal–Kapranov (1989), 3.2 and 4.11, the filtration  $W$  is automatically  $\infty$ -admissible.

The action of the braid group on the category of exceptional systems therefore gives an action on the category of exceptional sequences.

**Application 2.** Let  $G$  be a split reductive group over a field  $k$  and let  $\mathcal{B}$  be the variety of its Borel subgroups. Let  $\mathcal{M}$  be the category of schemes over  $\mathcal{B} \times \mathcal{B}$ , viewed as correspondences from  $\mathcal{B}$  to  $\mathcal{B}$ . Composition of correspondences makes  $\mathcal{M}$  a monoidal category. Let  $W$  be the Weyl group, identified with the set of relative positions of Borel subgroups, i.e. with the orbits of  $G(k)$  on  $\mathcal{B}(k) \times \mathcal{B}(k)$ . For  $w \in W$ , write  $X(w)$  for the variety of pairs  $(B', B'')$  of Borel subgroups in relative position  $w$ . In this description of  $W$ , for  $e \in W$  the identity element,  $X(e)$  is the diagonal. The distinguished generators  $s_i$  correspond to those orbits  $X$  whose closure is  $X$  union the diagonal, and the group law is characterized by

$$X(w) = X(w') \circ X(w'') \quad \text{if } w = w'w'' \text{ and } \ell(w) = \ell(w') + \ell(w''). \quad (0.3)$$

The distinguished generators of  $W$  make it a finite Coxeter group. Let  $B^+$  be the corresponding generalized positive braid monoid. Broué and Michel (1995) deduce from (0.3) that one may extend  $w \mapsto X(w)$  to a construction which attaches to  $b \in B^+$  an isomorphism class of schemes  $X(b)$  over  $\mathcal{B} \times \mathcal{B}$ , with  $X(b) \simeq X(b') \circ X(b'')$  for  $b = b'b''$ . Our results show that  $X(b)$  is in fact defined up to a unique isomorphism.

## 1 Actions

**1.1.** As explained in the introduction, the actions of a monoid  $M$  on a category  $C$  form a category. To say that it is “the same thing” to give an action of  $M$  on  $C$  as to give something else means the construction of an equivalence of the category of actions with the category of that something else.

**1.2.** The monoid without unit  $B_n^+$  of strictly positive braids on  $n$  strands,  $n \geq 2$ , is the monoid generated by generators  $s_i$  ( $1 \leq i \leq n-1$ ) subject to the relations

$$s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \quad (1.2.1)$$

$$s_i s_j = s_j s_i \quad \text{for } j \geq i+2. \quad (1)$$

It is known that it injects into the braid group  $B_n$ , defined by the same generators and relations.

**1.3.** To define an action of  $B_n^+$  on a set  $E$ , it suffices to give maps  $T(s_i) : E \rightarrow E$  satisfying (1.2.1).

To define an action of  $B_n^+$  on a category  $C$ , it does not suffice to give functors  $T(s_i) : C \rightarrow C$  and isomorphisms of functors

$$T(s_i)T(s_{i+1})T(s_i) \xrightarrow{\sim} T(s_{i+1})T(s_i)T(s_{i+1}), \quad (1.3.1)$$

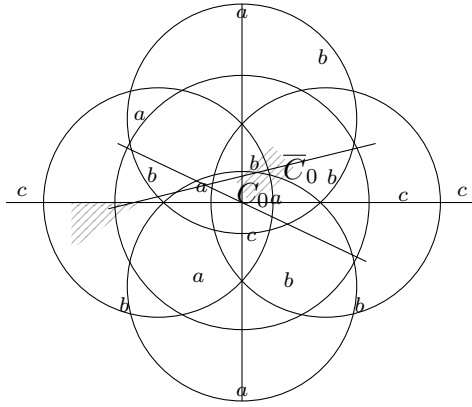
$$T(s_i)T(s_j) \xrightarrow{\sim} T(s_j)T(s_i) \quad \text{for } j \geq i + 2. \quad (2)$$

In the case of  $B_4^+$ , one must moreover require the following condition. Write  $a, b, c$  for  $T(s_1), T(s_2), T(s_3)$ . In the following cycle of twelve functors, each is linked to the next by the isomorphisms (1.3.1) or their inverses, applied to the indicated subcomposites in the original notation:

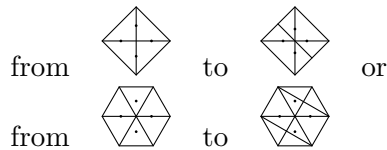
$$\begin{array}{cccc} acbcab & abcbab & abcaba & abacba \\ babcba & bacbca & bcabac & bcbabc \\ cbcabc & cbacbc & cbabcb & cabacb. \end{array} \quad (1.3.2)$$

The composite of these isomorphisms must be the identity.

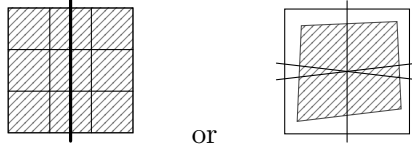
Here is the geometric meaning of (1.3.2). Consider the sphere  $S^2$ , cut into spherical triangles (chambers) by the root hyperplanes of the root system  $A_3$ . A regular tetrahedron inscribed in  $S^2$  gives a decomposition of  $S^2$  into four spherical triangles; the decomposition considered is its barycentric subdivision. Call one of the triangles the fundamental chamber, and attach to each edge a label  $a, b$ , or  $c$ , invariantly under the Weyl group, with the edges of a triangle having distinct labels and with  $b$  opposite the right angle. In stereographic projection, sending one vertex to infinity:



The shaded chamber is the fundamental chamber  $C_0$ , and the other shaded chamber is its opposite  $\overline{C}_0$ . Each word in (1.3.2) corresponds to a sequence  $G$  of adjacent chambers, a gallery, going from the fundamental chamber  $C_0$  to the opposite chamber  $\overline{C}_0$  and crossing exactly once each of the root hyperplanes: in the figure, the two lines and the four circles. The word gives the sequence of labels of the edges crossed. The cycle of words (1.3.2) gives a cycle of galleries. One passes from a gallery  $G$  to the next by a modification of one of the following types, or by two disjoint modifications of the first type:



Attach to  $G$  the path from the center of  $C_0$  to the center of  $\overline{C}_0$  obtained as the sequence of geodesics going from the center of one triangle to the center of the next. When passing from one gallery to the next, one has a homotopy between the corresponding paths:



For the sequence of galleries (1.3.2), these successive homotopies sweep out  $S^2$ , giving rise to the generator of  $\pi_1(\Omega S^2) = \pi_2(S^2)$ .

Theorem 1.5 below makes it possible to verify that the compatibility condition (1.3.2) is sufficient. More precisely, one has a restriction functor from the category of actions of  $B_4^+$  on  $C$  to the category of triples of functors  $T(s_1), T(s_2), T(s_3)$  endowed with isomorphisms (1.3.1) satisfying (1.3.2): to  $T$  one attaches the  $T(s_i)$  and the isomorphisms

$$\begin{aligned} T(s_i)T(s_{i+1})T(s_i) &\xrightarrow{\sim} T(s_i s_{i+1} s_i) = T(s_{i+1} s_i s_{i+1}) \xleftarrow{\sim} T(s_{i+1})T(s_i)T(s_{i+1}), \\ T(s_i)T(s_j) &\xrightarrow{\sim} T(s_i s_j) = T(s_j s_i) \xleftarrow{\sim} T(s_j)T(s_i). \end{aligned}$$

This restriction functor is an equivalence of categories.

**1.4.** A more convenient description is obtained by replacing (1.2.1) by a redundant system of generators and relations. Consider the more general case of the generalized strictly positive braid monoid  $B^+$  corresponding to a finite Coxeter group  $W$  with distinguished generating set  $R$ . The usual braids correspond to  $W = S_n$ , with  $R$  the set of transpositions  $(i, i+1)$ . Let  $\ell$  be the length function  $W \rightarrow \mathbb{N}$  defined by the generating system  $R$ .

Write  $\text{prod}(m, a, b)$  for a product of  $m$  alternating factors  $abab \dots$ . The group  $W$  is generated by the  $i \in R$ , subject to relations

$$\text{prod}(m_{ij}, i, j) = \text{prod}(m_{ij}, j, i) \quad (i, j \in R, i \neq j), \quad (1.4.1)$$

$$i^2 = e \quad (i \in R). \quad (1.4.2)$$

The corresponding monoid  $B^+$  is generated by generators  $\tilde{i}$ ,  $i \in R$ , subject only to the relations (1.4.1). It injects into the generalized braid group  $B$  defined by the same generators and relations (1.4.1) (Deligne (1972), 4.14(ii)). The projection  $\tilde{i} \mapsto i$  from  $B^+$  to  $W$  admits, on the complement of the identity element  $e$ , a canonical set-theoretic section  $\tau$ , characterized by

$$\tau(i) = \tilde{i} \quad (i \in R), \quad (1.4.3)$$

$$\tau(w'w'') = \tau(w')\tau(w'') \quad \text{if } \ell(w'w'') = \ell(w') + \ell(w''). \quad (1.4.4)$$

(Bourbaki, Lie IV, 1.5, Proposition 5). This section provides a new presentation of  $B^+$ :

$$\text{generators: } \tau(w) \quad (w \in W, w \neq e), \quad (1.4.5)$$

with relations (1.4.4). Note that (1.4.5), i.e. (1.4.4), makes it possible to write each  $\tau(w)$  as a product of  $\tau(i)$ ,  $i \in R$ , and implies (1.4.1) for the  $\tau(i)$ : the two sides of (1.4.1) are minimal decompositions of the same element of  $W$ .

Our main result is the following.

**Theorem 1.1** (1.5). *It is the same thing (1.1) to give an action of  $B^+$  on  $C$  as to give:*

(1.5.1) *for  $w \in W$ ,  $w \neq e$ , a functor  $T(w) : C \rightarrow C$ ;*

(1.5.2) *for  $w', w''$  with  $\ell(w'w'') = \ell(w') + \ell(w'')$ , an isomorphism of functors*

$$(1.5.2)_{w', w''} : T(w')T(w'') \longrightarrow T(w'w'');$$

(1.5.3) these isomorphisms satisfying, for  $\ell(w'w''w''') = \ell(w') + \ell(w'') + \ell(w''')$ , the commutativity of the diagram

$$\begin{array}{ccc} T(w')T(w'')T(w''') & \longrightarrow & T(w'w'')T(w''') \\ \downarrow & & \downarrow \\ T(w')T(w''w''') & \longrightarrow & T(w'w''w'''). \end{array} \quad (1.5.3)$$

In this statement, “it is the same thing” presupposes that the data (1.5.1),(1.5.2) satisfying (1.5.3) form a category. A morphism  $T_1 \rightarrow T_2$  is the data of morphisms of functors  $T_1(w) \rightarrow T_2(w)$ ,  $w \in W - \{e\}$ , which make the diagrams

$$\begin{array}{ccc} T_1(w')T_1(w'') & \longrightarrow & T_1(w'w'') \\ \downarrow & & \downarrow \\ T_2(w')T_2(w'') & \longrightarrow & T_2(w'w'') \end{array}$$

commute whenever  $w', w''$  are as in (1.5.2). We shall only construct the quasi-inverse to the restriction functor, and leave to the reader the verification that it is indeed a quasi-inverse. The difficult part of the proof is the contractibility theorem 1.7, proved in §2.

Let data (1.5.1),(1.5.2) satisfying (1.5.3) be given, and construct an action of  $B^+$  on  $C$  which extends (1.5.1),(1.5.2).

Consider the set  $E(b)$  of ways of writing  $b \in B^+$  as a product of  $\tau(w)$ . More precisely, let  $L(W - \{e\})$  be the free monoid generated by the  $w \in W - \{e\}$ . It maps to  $B^+$  by  $w \mapsto \tau(w)$ , and  $E(b) \subset L(W - \{e\})$  is the inverse image of  $b$ . The datum (1.5.1) gives, for each  $A \in L(W - \{e\})$ , a composite functor  $T(A)$ , extending (1.5.1) and such that  $T(AB) = T(A)T(B)$ .

If  $A_1$  and  $A_2$  are of the form

$$B : w' : w'' : C \quad \text{and} \quad B : (w'w'') : C \quad (1.5.4)$$

with  $\ell(w'w'') = \ell(w') + \ell(w'')$ , the datum (1.5.2) gives an isomorphism

$$T(B) : (1.5.2)_{w', w''} : T(C) : T(A_1) \longrightarrow T(A_2). \quad (1.5.5)$$

The essential point is to verify the following.

**Lemma 1.2** (1.6). *The isomorphisms (1.5.5) generate a transitive system of isomorphisms among the  $T(A)$ ,  $A \in E(b)$ .*

**Proof of 1.6  $\Rightarrow$  1.5.** Define  $T(b)$  to be the common value of the  $T(A)$ ,  $A \in E(b)$ . More precisely, it is the projective limit of the system of the  $T(A)$ : one has isomorphisms  $T(b) \xrightarrow{\sim} T(A)$ ,  $A \in E(b)$ , which make the diagrams

$$\begin{array}{ccc} & & T(A_1) \\ & \nearrow & \downarrow (1.5.5) \\ T(b) & & \\ & \searrow & T(A_2) \end{array}$$

commute.

For  $b, c \in B^+$ ,  $B \in E(b)$  and  $C \in E(c)$ , the composite word  $BC$  belongs to  $E(bc)$ . Moreover, if  $u : T(B_1) \rightarrow T(B_2)$  is an isomorphism (1.5.5), then  $u T(C)$  is an isomorphism (1.5.5) for  $bc$ ;



likewise after interchanging the roles of  $B$  and  $C$ . This allows one to define the isomorphism  $T(b)T(c) \rightarrow T(bc)$  as the one making the diagrams

$$\begin{array}{ccc} T(b)T(c) & \longrightarrow & T(bc) \\ \downarrow & & \downarrow \\ T(B)T(C) & \longrightarrow & T(BC) \end{array}$$

commutative. This construction, functorial in the data (1.5.1),(1.5.2) satisfying (1.5.3), is the desired quasi-inverse to the restriction functor.

**Proof of 1.6.** Let  $\leq$  be the order relation on  $E(b)$  generated by  $A_1 \leq A_2$  for  $A_1, A_2$  as in (1.5.4). In other words,  $A \leq B$  if  $B$  is deduced from  $A$  by factoring  $A$  as  $A_1 \cdots A_k$ , each  $A_i$  lying in  $L(W - \{e\})$  and being of the form  $w_{i,1} : \cdots : w_{i,m}$  with  $\ell(w_{i,1} \cdots w_{i,m}) = \sum_j \ell(w_{i,j})$ , and by replacing  $A_i$  by the product  $w_{i,1} \cdots w_{i,m}$  in  $W$ . The following theorem is proved in 2.3 and 2.4.

**Theorem 1.3** (1.7). *The geometric realization  $|E(b)|$  of the ordered set  $E(b)$  is contractible.*

The geometric realization of  $E(b)$  is that of the category attached in the usual way to  $E(b)$ : its set of objects is  $E(b)$ , and  $\text{Hom}(x, y)$  is reduced to a single element if  $x \leq y$  and is empty otherwise. See Quillen (1973), §1, p. 89.

Let us show that 1.6 follows from the connectedness and simple connectedness of  $|E(b)|$ . The condition (1.5.3) assures that, for  $A \leq B$ , the composites of isomorphisms (1.5.5) give a unique isomorphism

$$\varphi_{B,A} : T(A) \longrightarrow T(B)$$

with  $\varphi_{C,B}\varphi_{B,A} = \varphi_{C,A}$  for  $A \leq B \leq C$ .

By Quillen (1973), §1, Proposition 1, p. 90, the category of coverings of  $|E(b)|$  is equivalent to the category of functors from the category  $E(b)$  to the category (Sets, is) of sets and bijections. If  $|E(b)|$  is connected and simply connected, every functor from  $E(b)$  to (Sets, is) is therefore isomorphic to a constant functor: given sets  $T(x)$  for  $x \in E(b)$  and a system of bijections  $\varphi_{y,x} : T(x) \rightarrow T(y)$  for  $x \leq y$  satisfying  $\varphi_{z,y}\varphi_{y,x} = \varphi_{z,x}$ , these bijections generate a transitive system of bijections. The same result follows for  $T$  with values in a category  $\mathcal{X}$  and the  $\varphi_{x,y}$  isomorphisms: apply the preceding result to  $\text{Hom}(A, T(x))$  for  $A$  in  $\mathcal{X}$ .

**1.8. Variant: units.** A unital action of a monoid with unit  $(M, e)$  on a category  $C$  is an action  $T$  of  $M$  such that  $T(e)$  is an autoequivalence. There is then a unique isomorphism

$$T(e) \xrightarrow{\sim} \text{Id} \tag{1.8.1}$$

such that  $c_{e,f} : T(e)T(f) \rightarrow T(f)$  and  $c_{f,e} : T(f)T(e) \rightarrow T(f)$  are induced by (1.8.1). If  $M^+$  is obtained from a monoid  $M$  by adjoining a unit, it is the same thing to give an action of  $M$  or a unital action of  $M^+$ .

An action of a group  $G$  is a unital action of  $G$ .

**Proposition 1.4** (1.9). *Let  $M$  be a monoid with unit satisfying the left or right Ore condition, and let  $G$  be its group of fractions. It is the same thing (1.1) to give an action of  $G$  on a category  $C$  as to give a unital action of  $M$  on  $C$  by autoequivalences.*

*Proof.* For definiteness, assume the left Ore condition.

For  $f \in G$ , let  $L_f$  be the category whose objects are pairs  $(x, y) \in M \times M$  such that  $f = x^{-1}y$ , and whose arrows  $a : (x, y) \rightarrow (x', y')$  are elements  $a$  of  $M$  such that  $x' = ax$  and  $y' = ay$ . The Ore condition assures that  $L_f$  is filtered. For  $a : (x, y) \rightarrow (x', y')$ , one has an isomorphism

$$\varphi_a : T(x)^{-1}T(y) = T(x)^{-1}T(a)^{-1}T(a)T(y) = T(x')^{-1}T(y'),$$

and  $\varphi_{ba} = \varphi_b \varphi_a$ . Since the  $\varphi_a$  are isomorphisms and  $L_f$  is filtered, the inductive limit  $T(f)$  of the  $T(x)^{-1}T(y)$  exists, and  $T(x)^{-1}T(y) \xrightarrow{\sim} T(f)$ .

For  $f, g \in G$ , let  $L_{f,g}$  be the category of triples  $(x, y, z)$  such that  $f = x^{-1}y$  and  $g = y^{-1}z$ ; an arrow  $a : (x, y, z) \rightarrow (x', y', z')$  is an element  $a$  of  $M$  such that  $x' = ax$ ,  $y' = ay$ , and  $z' = az$ . The Ore condition again assures that this category is filtered and that the functors  $(x, y)$ ,  $(y, z)$ ,  $(x, z)$  from  $L_{f,g}$  to  $L_f$ ,  $L_g$ , and  $L_{fg}$  are cofinal. Define  $c_{f,g} : T(f)T(g) \rightarrow T(fg)$  as the inductive limit of the isomorphisms

$$T(x)^{-1}T(y)T(y)^{-1}T(z) \longrightarrow T(x)^{-1}T(z).$$

To verify (0.1), write  $f, g, h$  in the forms  $x^{-1}y$ ,  $y^{-1}z$ , and  $z^{-1}t$ . □

**1.10.** By Deligne (1972), the generalized positive braid monoid obtained from  $B^+$  (1.4) by adjoining an identity satisfies the Ore condition both on the left and on the right. By 1.8, 1.9, and 1.5, it is the same thing to give an action on a category of the corresponding generalized braid group as to give data (1.5.1), (1.5.2), (1.5.3) for which the  $T(w)$  are autoequivalences.

**1.11. Variant.** Let  $\mathcal{M}$  be a monoidal category and consider the following categories (A) and (B).

(A) The category of systems consisting of:

- (A.1) a function  $B^+ \rightarrow \text{Ob } \mathcal{M}$ ,  $b \mapsto T(b)$ ;
- (A.2) isomorphisms  $T(b')T(b'') \xrightarrow{\sim} T(b'b'')$  satisfying
- (A.3) the commutativity of the square

$$\begin{array}{ccc} T(b')T(b'')T(b''') & \longrightarrow & T(b'b'')T(b''') \\ \downarrow & & \downarrow \\ T(b')T(b''b''') & \longrightarrow & T(b'b''b'''). \end{array}$$

(B) The category of systems consisting of:

- (B.1) a function  $W - \{e\} \rightarrow \text{Ob}(\mathcal{M})$ ,  $w \mapsto T(w)$ ;
- (B.2) isomorphisms  $T(w')T(w'') \xrightarrow{\sim} T(w'w'')$  for  $\ell(w'w'') = \ell(w') + \ell(w'')$ , satisfying
- (B.3) the commutativity of the square

$$\begin{array}{ccc} T(w')T(w'')T(w''') & \longrightarrow & T(w'w'')T(w''') \\ \downarrow & & \downarrow \\ T(w')T(w''w''') & \longrightarrow & T(w'w''w''') \end{array}$$

whenever  $\ell(w'w''w''') = \ell(w') + \ell(w'') + \ell(w''')$ .

The restriction functor from (A) to (B) is an equivalence. The proof is the same as that of 1.5.

The same conclusion holds if  $\mathcal{M}$  is a Picard category, i.e. a monoidal category with unit in which every object is invertible, and if in (A) one replaces  $B^+$  by the generalized braid group  $B$ . The proof is the same as in 1.9 and 1.10.

## 2 Contractibility

**2.1.** Let  $V$  be a finite-dimensional real vector space and let  $\mathcal{M}$  be a finite set of homogeneous hyperplanes of  $V$ , called walls. Assume that the connected components of  $V - \bigcup_{M \in \mathcal{M}} M$ , called chambers, are open simplicial cones. We shall use the terminology of Bourbaki (Lie V, §1), namely walls, chambers, faces, facets, and the terminology of Deligne (1972) recalled below.

Two chambers  $A$  and  $B$  are called adjacent if  $A \neq B$  and  $A$  and  $B$  have a face in common. A gallery of length  $n$ ,  $n \geq 0$ , with source  $A$  and target  $B$ , is a sequence of chambers  $C_i$ ,  $0 \leq i \leq n$ , with  $A = C_0$ ,  $C_i$  adjacent to  $C_{i+1}$  for  $0 \leq i < n$ , and  $C_n = B$ . The composite  $GG'$  of two galleries  $G = (C_0, \dots, C_n)$  and  $G' = (C'_0, \dots, C'_m)$ , defined if  $C_n = C'_0$ , is  $(C_0, \dots, C_n = C'_0, \dots, C'_m)$ . For two chambers  $A$  and  $B$ , the minimal galleries from  $A$  to  $B$  are those of minimal length, i.e. length the number  $d(A, B)$  of walls separating  $A$  and  $B$ . They cross exactly once each wall separating  $A$  from  $B$ . Equivalence of galleries is the finest equivalence relation compatible with composition for which the minimal galleries from a chamber  $A$  to a chamber  $B$  are equivalent. Equivalent galleries have the same length. More precisely, they cross each wall the same number of times. The minimal galleries from  $A$  to  $B$  form an equivalence class, denoted  $u(A, B)$ , and if

$$d(A, B) + d(B, C) = d(A, C),$$

then  $u(A, B)u(B, C) = u(A, C)$  for composition of equivalence classes of galleries.

Let  $A, B, C$  be three chambers. The following conditions are equivalent (Deligne (1972), 1.4): (a)  $d(A, B) + d(B, C) = d(A, C)$ ; (b) there exists a minimal gallery from  $A$  to  $C$  passing through  $B$ ; (c)  $B$  lies in the intersection of the half-spaces bounded by a wall and containing  $A$  and  $C$ . Write  $D(A, C)$  for the set of chambers  $B$  satisfying these conditions, the chambers between  $A$  and  $C$ . For fixed  $A$ , the relation “ $B$  is between  $A$  and  $C$ ”, equivalently  $D(A, B) \subset D(A, C)$ , is an order relation among chambers.

By the central technical result 1.9 of Deligne (1972), for every equivalence class  $g$  of galleries with source  $A$ , if  $g$  begins with  $u(A, C)$ , i.e. if there exists  $g'$  such that  $g = u(A, C)g'$ , then the class  $g'$  is unique. Moreover, there is an optimal chamber  $C$  such that  $g$  begins with  $u(A, C)$ : for  $g$  to begin with  $u(A, B)$  it is necessary and sufficient that  $B$  be between  $A$  and  $C$ .

**2.2.** Let  $g$  be an equivalence class of galleries from  $A$  to  $B$ , let  $\ell(g)$  be their length, and let  $E(g)$  be the set of ways of writing  $g$  as a composite of classes  $u(C, D)$ . More precisely,  $E(g)$  is the set of sequences of chambers  $(C_i)$ ,  $0 \leq i \leq n$ , such that  $C_0 = A$ ,  $C_n = B$ ,  $C_i \neq C_{i+1}$  for all  $i$ , and

$$g = \text{composite of the } u(C_i, C_{i+1}). \quad (2.2.1)$$

The condition (2.2.1) implies

$$\sum d(C_i, C_{i+1}) = \ell(g). \quad (2.2.2)$$

Order  $E(g)$  as follows:

$$(C_0, \dots, C_n) \leq (D_0, \dots, D_m)$$

if there is a strictly increasing map  $\varphi : \{0, \dots, m\} \rightarrow \{0, \dots, n\}$  such that  $\varphi(0) = 0$ ,  $\varphi(m) = n$ ,  $D_j = C_{\varphi(j)}$ , and

$$d(C_{\varphi(j)}, C_{\varphi(j+1)}) = \sum_{\varphi(j) \leq i < \varphi(j+1)} d(C_i, C_{i+1}).$$

The map  $\varphi$  is then unique. If  $(C_0, \dots, C_n) \leq (D_0, \dots, D_m)$ , one passes from the decomposition  $g = \prod u(C_i, C_{i+1})$  to the decomposition  $g = \prod u(D_j, D_{j+1})$  by replacing the consecutive factors  $u(C_i, C_{i+1})$ , for  $\varphi(j) \leq i < \varphi(j+1)$ , by their composite  $u(D_j, D_{j+1})$ .

**2.3. Example.** Let  $W \subset GL(V)$  be a finite group generated by reflections. Assume that the invariant subspace  $V^W$  is 0. Take as walls the mirrors, i.e. fixed hyperplanes, of the reflections in  $W$ . It is known that the chambers of  $V$  are simplicial cones, permuted simply transitively by  $W$  (Bourbaki, Lie V, 3.9, Prop. 7 and 3.2, Th. 1).

Choose a chamber  $C_0$  and call it the fundamental chamber. For every chamber  $A$ , let  $g(A)$  be the unique element of  $W$  which sends  $C_0$  to  $A$ . For two chambers  $A$  and  $B$ , the relative position  $w(A, B)$  of  $A$  and  $B$  is the element  $w$  of  $W$  such that  $g(A)$  sends  $(C_0, wC_0)$  to  $(A, B)$ : one has  $w(A, B) = g(A)^{-1}g(B)$  and  $w(A, B)w(B, C) = w(A, C)$ .

The group  $W$ , equipped with the set  $R$  of reflections with respect to the mirrors containing the faces of  $C_0$ , is a Coxeter group (Bourbaki, Lie V, 3.2, Theorem 1).

Let  $G = (A_0, \dots, A_n)$  be a gallery of length  $n > 0$ . For  $0 \leq i < n$ , let  $r(i) \in R$  be the relative position of  $A_i$  and  $A_{i+1}$ . Attach to  $G$  the element

$$b(G) = \widetilde{r(0)} \cdots \widetilde{r(n-1)} \quad (2.3.1)$$

of  $B^+$  (1.4). The image of  $b(G)$  in  $W$  is the relative position of  $A_0$  and  $A_n$ . For a composite gallery, one has

$$b(G'G'') = b(G')b(G''). \quad (2.3.2)$$

The definition (2.1) of equivalence of galleries mimics the presentation (1.4.5) of  $B^+$ , and one verifies that (2.3.1) induces a bijection, still denoted  $b$ :

$$\left\{ \begin{array}{c} \text{equivalence classes of galleries of fixed source} \\ \text{and of positive length} \end{array} \right\} \xrightarrow{\sim} B^+. \quad (2.3.3)$$

Moreover, the map

$$E(g) \longrightarrow E(b(g)), \quad (C_0, \dots, C_n) \longmapsto w_0 : \dots : w_{n-1},$$

where  $w_i$  is the relative position of  $C_i$  and  $C_{i+1}$ , is an isomorphism of ordered sets. Theorem 1.7 is therefore a particular case of the following statement.

**Theorem 2.1** (2.4). *With the notation of 2.2, the geometric realization  $|E(g)|$  of the ordered set  $E(g)$  is contractible.*

*Proof.* We prove 2.4 by induction on the length  $\ell(g)$  of  $g$ . If  $\ell(g) = 0$ , then  $A = B$  and the set of sequences  $E(g)$  is reduced to the sequence  $(A)$ . Thus  $|E(g)|$  is contractible. Let  $\ell$  be an integer  $> 0$ , suppose 2.4 known for  $\ell(g) < \ell$ , and prove 2.4 for  $\ell(g) = \ell$ .

Let  $g$  be an equivalence class of galleries of length  $\ell$  from  $A$  to  $B$ ; let  $C$  be the optimal chamber (2.1) such that  $g$  begins with  $u(A, C)$ ; and let  $S$  be the set of walls of  $A$  which separate  $A$  from  $C$ . Since  $\ell > 0$ ,  $A \neq C$  and  $S$  is non-empty.

For  $s \in S$ , let  $E_s(g) \subset E(g)$  be the set of sequences  $(C_i)$  in  $E(g)$  such that the wall  $s$  separates  $A$  from  $C_1$ . The set  $E(g)$  is the union of the  $E_s(g)$ . Each  $E_s(g)$  is a final segment of  $E(g)$ . With the induced order, one therefore has, for geometric realizations,

$$|E(g)| = \bigcup |E_s(g)|. \quad (2.4.1)$$

For a non-empty subset  $T$  of  $S$ , let  $E_T(g)$  be the intersection of the  $E_t(g)$  for  $t \in T$ , equipped with the induced order. Then

$$|E_T(g)| = \bigcap |E_t(g)|. \quad (2.4.2)$$

**Lemma 2.2** (2.5). *Each  $|E_T(g)|$  is contractible.*

*Proof.* Let  $F_T$  be the facet of  $A$  open in the intersection  $P_T$  of the walls  $t \in T$ , and let  $A_T$  be the chamber opposite to  $A$  in the star of  $F_T$ . In the notation 1.6(ii) of Deligne (1972),  $A_T := A \cdot \Delta(P_T)$ . The chamber  $A_T$  is the least (2.1) of the chambers separated from  $A$  by the walls  $t \in T$ . In particular,  $A_T$  lies between  $A$  and  $C$ . Let  $h$  be the unique equivalence class of galleries from  $A_T$  to  $B$  such that  $g = u(A, A_T)h$ . One has  $\ell(h) < \ell(g)$ .

Let  $f : E(h) \rightarrow E_T(g)$  be the increasing map

$$(C_0 = A_T, \dots, C_n) \mapsto (A, C_0, \dots, C_n). \quad (2.5.1)$$

If  $(C_i)$  lies in  $E(g)$ , then  $C_1$  lies in  $D(A, C)$ , and for  $(C_i)$  to lie in  $E_T(g)$  it is necessary and sufficient that  $C_1$  lie in  $D(A_T, C)$ . The map  $f$  therefore induces an isomorphism of  $E(h)$  with an initial segment of  $E_T(g)$ .

Let  $f_* : E_T(g) \rightarrow E(h)$  be the increasing map

$$(C_0 = A, C_1, \dots, C_n) \mapsto \begin{cases} (C_1, \dots, C_n), & C_1 = A_T, \\ (A_T, C_1, \dots, C_n), & C_1 \neq A_T. \end{cases} \quad (2.5.2)$$

For  $x \in E(h)$  and  $y \in E_T(g)$ , one has

$$f(x) \leq y \iff x \leq f_*(y). \quad (2.5.3)$$

Viewing  $E(h)$  and  $E_T(g)$  as categories, and  $f, f_*$  as functors (cf. 1.7), (2.5.3) says that  $(f, f_*)$  is an adjoint pair of functors. By Quillen (1973), §1, Corollary 1 to Proposition 2, p. 92, the map

$$|f| : |E(h)| \longrightarrow |E_T(g)|$$

is a homotopy equivalence; 2.5 follows from the induction hypothesis.  $\square$

The non-empty intersections of the  $|E_s(g)|$  being contractible ((2.4.2) and 2.5),  $|E(g)|$  has the same homotopy type as the nerve of its finite closed covering by the  $|E_s(g)|$ . Since all intersections are non-empty, this nerve is a simplex and  $|E(g)|$  is contractible.  $\square$

**Remark 2.6.** Let  $\mathcal{C}$  be a 2-category whose objects are indexed by the chambers. Just as 1.7 gave 1.5, Theorem 2.4 gives, by the same proof, that it is the same thing (1.1) to give (A) or (B) as follows.

(A) (i) For  $g$  an equivalence class of galleries of positive length from  $A$  to  $B$ , a 1-arrow  $T(g) : X_A \rightarrow X_B$ ;

(ii) for  $g = g'g''$ , an isomorphism  $T(g'')T(g') \rightarrow T(g)$ , these isomorphisms making the diagrams

$$\begin{array}{ccc} T(g''')T(g'')T(g') & \longrightarrow & T(g''g''')T(g') \\ \downarrow & & \downarrow \\ T(g''')T(g'g'') & \longrightarrow & T(g'g''g''') \end{array}$$

commutative.

(B) (i) For two distinct chambers  $A$  and  $B$ , a 1-arrow  $T_{B,A} : X_A \rightarrow X_B$ ;

(ii) for three distinct chambers  $A, B, C$ , with  $C$  between  $A$  and  $B$ , an isomorphism

$$T_{C,B}T_{B,A} \xrightarrow{\sim} T_{C,A},$$

these isomorphisms making the diagrams

$$\begin{array}{ccc} T_{D,C}T_{C,B}T_{B,A} & \longrightarrow & T_{D,B}T_{B,A} \\ \downarrow & & \downarrow \\ T_{D,C}T_{C,A} & \longrightarrow & T_{D,A} \end{array}$$

commutative when they are defined, i.e. when  $d(A, D) = d(A, B) + d(B, C) + d(C, D)$ .

Application 1 from the introduction is more conveniently expressed in this language: the set of total orders on  $I$  is identified with the set of chambers of a root system of type  $A_n$ , for  $n = |I| - 1$ . For each total order  $\leq$ , corresponding to a chamber  $C$ , let  $X_C$  be the category of corresponding exceptional systems, and let  $\text{Hom}(X_C, X_D)$  be the category of functors from  $X_C$  to  $X_D$ . What Bondal constructs is a datum of type (B).

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# Some Master Ideas in the Work of A. Grothendieck

Pierre Deligne

## Abstract

This article attempts to explain four fundamental mathematical concepts created by Grothendieck: schemes, topoi, the six operations, and motives.

In *Récoltes et semailles* (third part), Grothendieck writes:

Take, for example, the task of proving a theorem which still remains hypothetical (to which, for some, mathematical work would seem to reduce). I see two extreme approaches to going about it. One is that of the hammer and chisel, when the problem posed is seen as a large nut, hard and smooth, whose inside one must reach, the nourishing flesh protected by the shell. The principle is simple: one places the edge of the chisel against the shell, and one strikes hard. If need be, one begins again at several different places, until the shell breaks — and one is satisfied. [...]

I could illustrate the second approach while keeping the image of the nut which has to be opened. The first parable which came to my mind just now is that one plunges the nut into an emollient liquid, simply water, why not; from time to time one rubs it so that the liquid penetrates better, and for the rest one lets time do its work. The shell softens over the weeks and months; when the time is ripe, a pressure of the hand suffices, the shell opens like that of an avocado ripened to the point! Or again, one lets the nut ripen under the sun and the rain and perhaps also under the winter frosts. When the time is ripe, it is a delicate shoot issuing from the substantial flesh which will have pierced the shell as if in play; or, to speak more exactly, the shell will have opened of itself to let it pass. [...]

The reader who is at all familiar with some of my work will have no difficulty recognizing which of these two modes of approach is “mine.”

A little farther on [ibid., p. 554, note (\*\*\*)], Grothendieck puts forward four examples: Riemann–Roch, the structure of the prime-to-the-characteristic  $\pi_1$  for curves, rationality of  $L$ -functions for schemes of finite type over a finite field, and the semi-stable reduction theorem for abelian varieties. I remember my amazement, in 1965–66, after Grothendieck’s lecture [SGA 5] proving the base-change theorem for  $Rf_!$ : dévissages, dévissages, nothing seems to happen and yet at the end of the lecture a clearly non-trivial theorem is there.

Many of Grothendieck’s ideas have become so familiar to us, and are so perfectly adapted to their object, that we forget that they were far from obvious at their birth; we even forget their author. My aim in this article is to describe four of these ideas: schemes, topoi, the six operations, and motives.

# 1 Schemes

The invention of schemes is the first of Grothendieck's ideas that comes to mind, perhaps because it was the most quickly accepted. Serre's address in Stockholm in 1962 begins: "I would like to describe here some of the recent developments in algebraic geometry. I should specify that I take this last term in the sense which it has acquired over the last few years: that of the theory of schemes." This acceptance was facilitated by the rapid publication, thanks to the collaboration of Dieudonné, of EGA.

The boldness of Grothendieck's definition is to accept that every commutative ring (with unit)  $A$  defines an affine scheme  $\text{Spec}(A)$ , that is, not to try to restrict oneself to a category of "good" rings (domains, reduced rings, noetherian rings, ...). This has a price. The points of  $\text{Spec}(A)$  (prime ideals of  $A$ ) do not have a manageable geometric meaning, and the structural sheaf  $\mathcal{O}$  is not a sheaf of functions. When one has to construct a scheme, one does not in general begin by constructing its set of points.

Perhaps more important is the decision to build a relative theory, to which the omnipresent

$$X \longrightarrow S$$

in Grothendieck's lectures bears witness. The classical case of a variety defined over a field  $k$  becomes the special case  $S = \text{Spec}(k)$ . In a relative theory, with an  $S$ -scheme  $X$ , that is, with a morphism of schemes  $f : X \rightarrow S$ , one systematically considers the scheme  $X'$  deduced from  $X$  by a base change  $u : S' \rightarrow S$ , that is, the fiber product  $X' := S' \times_S X$  and its projection  $f'$  onto  $S'$ :

$$\begin{array}{ccc} X' & \longrightarrow & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{u} & S. \end{array}$$

In the category of schemes, fiber products always exist: if allowing every commutative ring to define an affine scheme gives citizenship to strange schemes, allowing it supplies a category of schemes with good properties.

A property of  $X$  over  $S$  will be called geometric if it has good invariance properties under base change. Classical analogue: for a variety  $X$  defined over a field  $k$ , the set of points of  $X$  over an algebraically closed extension  $\Omega$  of  $k$  (for example, Weil's universal domain) is regarded as "geometric", while the set of  $k$ -points is "arithmetic".

If  $X$  is a scheme over  $S$ , and  $u : S' \rightarrow S$  is a morphism of schemes, an  $S'$ -point  $p$  of  $X$  is a morphism of  $S$ -schemes:

$$\begin{array}{ccc} & & X \\ & \nearrow p & \downarrow \\ S' & \xrightarrow{u} & S. \end{array}$$

One denotes by  $X(S')$  the set of  $S'$ -points of  $X$ . It identifies with the set of sections of  $X' \rightarrow S'$ .

**Example 1.1.** Let  $X$  be the affine scheme over a field  $k$  defined by equations

$$P_\alpha(X_1, \dots, X_n) = 0 : \quad X = \text{Spec}(k[X_1, \dots, X_n]/(P_\alpha)).$$

Let  $k'$  be a  $k$ -algebra. The set  $X(k') := X(\text{Spec}(k'))$  is the set of solutions in  $k'^n$  of the equations  $P_\alpha = 0$ . For  $k'$ , an algebraically closed extension  $\Omega$  of  $k$ , this is the set  $X(\Omega)$  that Weil regards as underlying  $X$ .



**Example 1.2.** The scheme  $GL_N$  over  $\text{Spec}(\mathbb{Z})$  is such that for every commutative ring  $A$  (automatically a  $\mathbb{Z}$ -algebra:  $\text{Spec}(A)$  is a scheme over  $\text{Spec}(\mathbb{Z})$ ),  $GL_N(\text{Spec}(A))$  “is”  $GL_N(A)$ .

In these examples, the geometric intuition that one has of  $X$ , a scheme over  $S$ , is well reflected in  $X(S')$ . Better than in the underlying set of  $X$ . In the case of Example 1.2, this underlying set is not a group, whereas each  $GL_N(S')$  is one. More precisely, it is useful to attach to the scheme  $X$  over  $S$  the contravariant functor from  $S$ -schemes to sets

$$S' \longmapsto \text{the set } X(S') \text{ of } S'\text{-points of } X.$$

In the category of  $S$ -schemes, this is simply the representable functor

$$h_X : S' \longmapsto \text{Hom}(S', X)$$

attached to  $X$ . By the Yoneda lemma, the functor  $X \mapsto h_X$  is fully faithful. Rather than thinking of an  $S$ -scheme  $X$  as a ringed space, equipped with  $X \rightarrow S$ , with suitable properties, it is often convenient to think of it as a functor

$$S\text{-points} : (\text{Schemes}/S)^\circ \longrightarrow (\text{Sets}),$$

which has the virtue of being representable. When one wants to define a “fine moduli space”, the first step is to define the corresponding functor. Typically, defining this functor requires a relative theory.

**Example 1.3.** Let  $X$  be projective over a field  $k$ . Question: what does “moduli space of closed subschemes of  $X$ ” mean? For  $u : S' \rightarrow \text{Spec}(k)$ , let  $X'$  over  $S'$  be deduced from  $X$  by base change. Let  $\mathcal{H}(S')$  be the set of closed subschemes  $Z'$  of  $X'$  which are flat over  $S'$  (flat of finite presentation, if one does not want to assume  $S'$  noetherian). Answer: it is a scheme  $\text{Hilb}(X)$  representing the functor  $S' \mapsto \mathcal{H}(S')$ .

For this point of view to be viable, one must have methods for verifying when a functor is representable. Most of the lectures of FGA are devoted to this problem. A definitive solution, extending this work, was obtained by M. Artin [1969], at least if one agrees to replace the category of schemes by the more natural category of algebraic spaces. More natural: in the same sense in which the étale topology is more natural than the Zariski topology.

## 2 Topoi

Let  $u : S' \rightarrow S$  be a morphism of schemes, the “base-change morphism”. Descent theory [Grothendieck FGA, Séminaire Bourbaki 190, 1959–60] considers problems of the following kinds. Descent of properties: let  $X$  be a scheme over  $S$ , and  $X'$  over  $S'$  deduced from  $X$  by base change. Suppose that  $X'/S'$  has a property  $P$ . Can one conclude that  $X/S$  satisfies  $P$ ? Descent of morphisms: let  $X, Y$  be over  $S$ , let  $X', Y'$  be deduced by base change, and let  $g' : X' \rightarrow Y'$  be a morphism of  $S'$ -schemes. When does  $g'$  arise by base change from  $g : X \rightarrow Y$ ? Descent of objects: let  $X'$  be over  $S'$ . What descent datum on  $S'$  is required in order to construct  $X$  over  $S$  from which  $X'$  is deduced by base change?

If  $S'$  is the disjoint union of the open subsets  $(U_i)_{i \in I}$  of a covering of  $S$ , the base change to  $S'$  is essentially the restriction to each  $U_i$ , and the preceding problems are problems of localization on  $S$  and gluing. Gluing typically requires considering the pairwise intersections

$U_i \cap U_j$ , and in terms of  $u : S' \rightarrow S$ , the disjoint union of the  $U_i \cap U_j$  is simply the fiber product  $S' \times_S S'$ .

The theory of topoi makes it possible to transpose topological intuition into descent theory. For  $u : S' \rightarrow S$ , a base-change morphism of a type considered in descent theory, for example faithfully flat and quasi-compact (fpqc), base change from  $S$  to  $S'$  becomes a localization. A descent datum is the analogue of a gluing datum.

An antecedent of descent theory is Galois descent, corresponding to  $S$ , the spectrum of a field  $k$ , and  $S'$ , the spectrum of a Galois extension. Here,  $S' \times_S S'$  is a sum of copies of  $S'$  indexed by  $\text{Gal}(k'/k)$ . The proofs, and in particular Galois theory, are however simpler in the more general framework of descent theory. In Cartier's phrase: Grothendieck proves Galois theory, and Galois descent, by Galois descent.

The tool that is topos theory enabled the construction of étale cohomology of schemes, and this is its best-known success. A sheaf on  $X$  for the Zariski topology is a contravariant functor from the category of Zariski open subsets of  $X$  to the category of sets, with a gluing condition for  $(U_i)_{i \in I}$ , a Zariski open covering of  $U$ . In the étale topology, the Zariski site considered above is replaced by the étale site: the category of Zariski open subsets is replaced by that of  $f : U \rightarrow X$  étale over  $X$ , and coverings by the covering families  $(U_i)_{i \in I}$ , that is, a surjective morphism of schemes over  $X$  from the disjoint union of the  $U_i$  to  $U$ . An antecedent: Serre's introduction of the notion of an isotrivial principal homogeneous space (= locally trivial for a topology close to the étale topology). In his Kansas [1955] and Tôhoku [1957] papers, Grothendieck had shown that, once a category of sheaves is given, a notion of cohomology groups results. The étale topology thus supplies étale cohomology. That the  $H^1$  groups obtained are reasonable is not surprising. The miracle is that the higher  $H^i$  are reasonable too.

For Grothendieck, the importance of topos theory goes far beyond the single case of the étale topology. The title given to SGA 4 bears witness to this: "Theory of topoi and étale cohomology of schemes."

Other applications:

- (A) *Crystalline cohomology*. Crystalline cohomology is the cohomology of the crystalline topos, and this definition makes its functoriality clear. The crystalline topos is nevertheless delicate to use, and it is often necessary to pass to an interpretation in terms of de Rham complexes.
- (B) *Rigid analytic spaces*. Tate's rigid analytic sheaves are coherent sheaves on a suitable ringed topos, and their cohomology is the corresponding cohomology.
- (C) *Foliations*. A variety  $X$  equipped with a foliation  $\mathcal{F}$  defines a "quotient"  $X/\mathcal{F}$ , which is a topos locally isomorphic to that of sheaves on a variety. This point of view seems to have been eclipsed by that of Connes, who instead associates to  $\mathcal{F}$  a non-commutative  $C^*$ -algebra.

I would also like to mention Grothendieck's use of "big" sites (topoi), notably for interpreting classifying spaces.

### 3 The six operations

The formalism of the six operations presupposes that of derived categories. For a history of the latter, I refer to Illusie's text [1990]. Basic idea: for all sorts of cohomology groups,

their definition supplies not only these groups, but also a complex  $K$  whose cohomology groups they are. Typically, this complex is not uniquely determined, but it is so up to quasi-isomorphism: for two variants  $K', K''$  of  $K$ , one has  $K'''$  and morphisms  $K' \leftarrow K''' \rightarrow K''$  inducing isomorphisms on cohomology. In the derived category,  $K$  thereby becomes unique up to unique isomorphism.

The six operations are functors between derived categories. They are the derived tensor product  $\otimes^L$ , the internal Hom  $R\mathcal{H}om$  (giving rise to the local  $\mathcal{E}xt^i$ ), and, for  $f : X \rightarrow Y$  a morphism of schemes, two direct-image functors:  $Rf_*$  and  $Rf_!$ , and two inverse-image functors:  $Lf^*$  and  $Rf^!$ .

The formalism of these operations was first isolated in the cohomology of coherent sheaves [Grothendieck 1963]. Here, for direct images, one should consider only proper morphisms, for which  $Rf_* = Rf_!$ . The formalism provides a relative version of Serre duality, in the form of an adjunction between the functors  $Rf_*$  and  $Rf^!$ . It also provides a fixed-point formula more general than the “Woodshole fixed point formula”, which is contemporary with it (Summer Institute on Algebraic Geometry, Whitney estate, Woodshole, Massachusetts, 1964).

Miraculously, the same formalism applies in étale cohomology — with very different proofs. It again supplies a formalism of duality (Poincaré duality) and fixed-point formulae (Lefschetz formulae).

## 4 Motives

Let  $X$  be an algebraic variety over an algebraically closed field  $k$ . For each prime number  $\ell$  prime to the characteristic, the étale topology supplies  $\ell$ -adic cohomology groups  $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$ . If  $k$  is a subfield of  $\mathbb{C}$ , one has comparison isomorphisms

$$H^i(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} H_{\text{ét}}^i(X, \mathbb{Z}_\ell).$$

For  $k$  of characteristic  $> 0$ , there is no functorial integral cohomology giving rise to such isomorphisms. Nevertheless, the  $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$  have, as  $\ell$  varies, a “family resemblance”. For  $i = 1$ , and  $X$  projective and smooth,  $\text{Pic}^0(X)$  plays the role of the nonexistent integral theory: it recovers the  $H_{\text{ét}}^1(X, \mathbb{Z}_\ell)$  and is an object “over  $\mathbb{Z}$ ”, in that groups of homomorphisms between abelian schemes are of finite type.

The theory of motives is first of all an attempt to find a substitute for the nonexistent integral cohomology, explaining the family resemblance among the  $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$ , especially for  $X$  projective and smooth. One recognizes the Master’s hand in the idea that the problem is not to define what a motive is: the problem is to define the category of motives, and to bring out the structures it carries. These structures should make it possible to prove the Weil conjecture on the model of Serre [1960]. See Grothendieck [1969].

It was on this occasion that Grothendieck invented the notion of a Tannakian category, developed in Saavedra’s thesis [1972]. The aim was to formalize the notion of tensor product of motives, corresponding by the Künneth formula to the product of varieties.

Even if the theory of motives has not achieved its original aim, its influence has been great. I refer to the Seattle conference on motives [Jannsen et al., 1994] for a panorama of its applications.

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# The Cohomology Algebra of the Complement, in an Affine Space, of a Finite Family of Affine Subspaces

P. Deligne, M. Goresky, and R. MacPherson

This paper is dedicated to W. Fulton in admiration for his mathematical achievements and his broad influence on the field of algebraic geometry.

## 0. Introduction

Let  $X$  be the complement, in a real affine space  $V$ , of a locally finite set  $E$  of proper affine subspaces of  $V$ . The flats of  $E$  are  $V$  and the non-empty intersections of elements of  $E$ . In [GM], Part III, Goresky and MacPherson apply Morse theory to a function “distance to  $v_0 \in V$ ” in order (a) to decompose the homology of  $X$  into factors indexed by the flats, and (b) to give a combinatorial description of each factor. In Section 1 we recover these results by sheaf-theoretic techniques. These techniques make clear what the decomposition of [GM] depends on, and they also apply to the determination of the  $\ell$ -adic cohomology of the complement of a finite family of affine subspaces in an affine space over an algebraically closed field.

By duality, these homology results are transported to cohomology. For  $E$  finite, one has a decomposition indexed by the flats

$$H^*(X) = \bigoplus h^*(A). \quad (0.1)$$

Let  $\text{coor}(A)$  be the free rank-one  $\mathbb{Z}$ -module with bases  $\pm e$  the coorientations of  $A$  in  $V$ , placed in degree  $\text{codim}(A)$ ; then

$$\text{coor}(A) = H_A^*(V, \mathbb{Z}). \quad (0.2)$$

The direct factor  $h^*(A)$  is the tensor product of  $\text{coor}(A)$  with a group depending only on the combinatorics of the arrangement, that is, only on the set  $\mathcal{A}^+$  of flats ordered by inclusion. The decomposition (0.1) depends on the choice, for each flat  $A$ , of a point

$$x_A \in A^0 := A - \bigcup \{\text{smaller flats}\}.$$

Only the connected component of  $A^0$  containing  $x_A$  matters. In particular, if the following connectedness hypothesis (CONN) is satisfied, then the decomposition (0.1) depends on no choice:

$$\text{if a flat } B \text{ is contained in a flat } A, \quad A = B \text{ or } \dim(A) - \dim(B) \geq 2. \quad (\text{CONN})$$

In Section 4.2, under the hypothesis (CONN), we compute the cup product of  $H^*(X)$  in terms of the decomposition (0.1). The essential point is that the component

$$h(A) \otimes h(B) \longrightarrow h(C)$$

of the cup product can be non-zero only for  $C = A \cap B$ , the intersection being transverse. We prove this by interpreting the cup product as the exterior product followed by restriction to the diagonal  $X \rightarrow X \times X$ , and by treating this restriction as a special case of a restriction morphism

$$H^*(X) \longrightarrow H^*(X'),$$

where  $X' = V' \cap X$  is the trace of  $X$  on an affine subspace.

Under (CONN), it follows from this calculation that the cohomology algebra  $H^*(X)$  depends, up to unique isomorphism, only on the following data:

- (a) the set  $\mathcal{A}^+$  of flats, ordered by inclusion;
- (b) the function  $A \mapsto \text{codim}(A)$ ;
- (c) for each flat  $A$ , the free rank-one  $\mathbb{Z}$ -module  $\text{coor}(A)$ ;
- (d) for  $C = \inf(A, B)$  and  $\text{codim}(C) = \text{codim}(A) + \text{codim}(B)$ , that is, for a transverse intersection  $C = A \cap B$ , the natural isomorphism from  $\text{coor}(A) \otimes \text{coor}(B)$  to  $\text{coor}(C)$ .

M. de Longueville and C. Schultz independently determined the integral cohomology algebra of the complement of a central arrangement satisfying (CONN) [dS]. See also [FZ].

**Acknowledgments.** Our interest in these questions was drawn by the preprint [Y] of Yuzvinsky. In the case of complex arrangements, Yuzvinsky determines, up to isomorphism and using the results of Morgan [M], the rational cohomology algebra of the complement of an arrangement.

## 1 Decomposition of Cohomology

### 1.1

We shall use the following weak sense of stratification. A stratification of a topological space  $T$ , indexed by a finite ordered set  $I$ , is a decomposition indexed by  $I$  of  $T$  into non-empty disjoint subsets  $S_i$ , the strata, such that for every  $i \in I$  the union

$$T_i := \bigcup_{j \leq i} S_j$$

is closed. The  $T_i$  are the closed strata. Note that  $i \leq j$  if and only if  $T_i \subset T_j$ , and that the  $T_i$  determine the  $S_i$  by

$$S_i = T_i - \bigcup_{j < i} T_j. \quad (1.1.1)$$

Let  $\varphi : T \rightarrow I$  be the map equal to  $i$  on  $S_i$ . If  $I$  is given the topology for which the closure of a point  $i$  is the interval

$$[\cdot, i] := \{j \mid j \leq i\},$$

then saying that the  $T_i$  are closed is equivalent to saying that  $\varphi$  is continuous.

### 1.2

We shall also write  $I$  for the category whose objects are the elements of  $I$ , with a unique arrow from  $i$  to  $j$  for  $i \leq j$ .

Each point  $i$  of the topological space  $I$  has a smallest neighborhood, namely the interval

$$[i, \cdot] := \{j \mid i \leq j\}.$$

For a sheaf  $F$  on  $I$ , the fiber  $F_i$  of  $F$  at  $i$  coincides with  $F([i, \cdot])$ . For  $i \leq j$ , restriction from  $[i, \cdot]$  to  $[j, \cdot]$  is a morphism from  $F_i$  to  $F_j$ , and this construction identifies sheaves of abelian groups on  $I$  with functors from  $I$  to  $(\text{Ab})$ .

If  $I$  has a least element 0, the functor  $\Gamma(I, F) = F_0$  is exact and the higher  $H^i$  vanish. If  $I$  has a greatest element 1, every constant sheaf is flasque and its higher  $H^i$  therefore vanish: the space  $I$  is acyclic, i.e. its cohomology with constant coefficients is that of a point.

Another proof is the following. Computing cohomology by the canonical flasque resolution, one verifies that the cohomology of  $I$  with constant coefficients coincides with that of the classifying space  $|I|$  of the category  $I$ . In particular, it is the same for  $I$  and for  $I$  with the opposite order, and we have already seen that every  $I$  having a least element is acyclic.

**Lemma 1.1** (Devissage, 1.3). *Let  $\mathcal{C}$  be a class of sheaves on  $I$ . Suppose that  $\mathcal{C}$  contains the zero sheaf, contains every constant sheaf on a closed subset  $[\cdot, i]$  extended by zero to  $I$ , and has the following two-out-of-three property: for every short exact sequence*

$$0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$$

*of sheaves on  $I$ , if two of the  $F_i$  belong to  $\mathcal{C}$ , then the third does as well. Then  $\mathcal{C}$  contains all sheaves on  $I$ .*

*Proof.* Proceed by induction on the number of elements of  $I$ . For a closed subset  $i : J \hookrightarrow I$  different from  $I$ , applying the induction hypothesis to the class of sheaves on  $J$  such that  $i_!F$  is in  $\mathcal{C}$ , one sees that every sheaf with support not equal to  $I$  is in  $\mathcal{C}$ . If  $I$  is empty, the lemma is clear. Otherwise, let  $x$  be a maximal element of  $I$  and let  $j$  be the inclusion of  $\{x\}$  in  $I$ . Taking  $F_1, F_2, F_3$  to be respectively a constant sheaf on  $\{x\}$ , on  $[\cdot, x]$ , and on  $[\cdot, x[$ , each extended by zero, one sees that for every abelian group  $\Lambda$ ,  $j_!\Lambda$  belongs to  $\mathcal{C}$ . For a sheaf  $F$  on  $I$ , conclude by observing that the support of  $F/j_!j^*F$  is contained in the closed set  $I - \{x\}$ .  $\square$

**Proposition 1.2** (1.4). *If the closed strata are acyclic, then for every sheaf  $F$  on  $I$  one has*

$$H^*(I, F) \xrightarrow{\sim} H^*(T, \varphi^*F). \quad (1.4.1)$$

*Proof.* If  $F$  is a constant sheaf on a closed set  $[\cdot, i]$ , extended by zero to  $I$ , then  $\varphi^*F$  is constant on  $T_i$ , extended by zero to  $T$ , and (1.4.1) follows from the acyclicity of  $[\cdot, i]$  and of  $T_i$ .

The general case follows by devissage: for a short exact sequence  $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$  of sheaves on  $I$ , the two sides of (1.4.1) give long exact cohomology sequences, and if (1.4.1) is an isomorphism for two of the  $F_i$ , it is so for the third. Conclude by 1.3.  $\square$

**Remark 1.3** (1.5). If  $|I|$  is the classifying space of the category  $I$ , define a continuous map  $|I| \rightarrow I$  by sending the interior of the simplex  $[i_0, \dots, i_n]$  with  $i_0 < i_1 < \dots < i_n$  to  $i_n$ . Under this map the inverse image of  $[\cdot, i]$  is  $||[\cdot, i]||$ , a cone with vertex  $i$ . The hypothesis of 1.4 is therefore satisfied, and taking  $F$  to be a constant sheaf one recovers the fact that  $I$  and  $|I|$  have the same cohomology.

## 1.6

Assume that  $T$  is locally compact, separated, and countable at infinity, and that the closed strata  $T_i$  are acyclic manifolds without boundary. Being acyclic,  $T_i$  is orientable and connected.

For every connected orientable manifold  $V$ , write  $\text{or}(V)$  for the free rank-one graded  $\mathbb{Z}$ -module, pure of degree  $\dim(V)$ , whose generators  $\pm e$  correspond to the two orientations of  $V$ . For every abelian group  $\Lambda$  and every point  $x \in V$ ,

$$\Lambda \otimes \text{or}(V) = H_{\{x\}}^*(V, \Lambda) \xrightarrow{\sim} H_c^{\dim(V)}(V, \Lambda), \quad (1.6.1)$$

and, if  $V$  is acyclic, Poincaré duality gives

$$\Lambda \otimes \text{or}(V) \xrightarrow{\sim} H_c^*(V, \Lambda). \quad (1.6.2)$$

Write simply  $\text{or}_i$  for  $\text{or}(T_i)$ .



Choose, in each stratum  $S_i$ , a point  $x_i$ . For every sheaf  $F$  on  $I$  there is an inverse-image map in cohomology with support

$$\varphi^* : H_{\{i\}}^*([i, \cdot], F) \longrightarrow H_{S_i}^* \left( \bigcup_{i \leq j} S_j, \varphi^* F \right). \quad (1.6.3)$$

Note that  $U_i := \bigcup_{i \leq j} S_j$  is an open subset of  $T$ , and that  $S_i$  is closed in  $U_i$ . There is a cup product

$$H_{S_i}^*(U_i, \varphi^* F) \otimes H_{\{x_i\}}^*(S_i, \mathbb{Z}) \longrightarrow H_{\{x_i\}}^*(U_i, \varphi^* F). \quad (1.6.4)$$

In the language of derived categories,  $H_{S_i}^*(U_i, \varphi^* F)$  is a  $\text{Hom}^*$  in  $D(U_i)$  from  $\mathbb{Z}_{S_i}$  to  $\varphi^* F$ ,  $H_{\{x_i\}}^*(S_i, \mathbb{Z})$  is a  $\text{Hom}^*$  from  $\mathbb{Z}_{\{x_i\}}$  to  $\mathbb{Z}_{S_i}$ , and the cup product is simply composition. The composite

$$H_{S_i}^*(U_i, \varphi^* F) \otimes_{\text{or}_i} \xrightarrow{1.6.4} H_{\{x_i\}}^*(U_i, \varphi^* F) \longrightarrow H_c^*(U_i, \varphi^* F) \quad (1.6.5)$$

depends only on the connected component of  $S_i$  in which  $x_i$  lies. Indeed, it factors through the tensor product by  $H_{S_i}^*(U_i, \varphi^* F)$  of the morphism

$$\text{or}_i = H_{\{x_i\}}^*(S_i, \mathbb{Z}) \longrightarrow H_c^*(S_i, \mathbb{Z}).$$

Composing (1.6.3) and (1.6.5), one obtains morphisms

$$H_{\{i\}}^*([i, \cdot], F) \otimes \text{or}_i \longrightarrow H_c^*(U_i, \varphi^* F), \quad (1.6.6)$$

which, composed with  $H_c^*(U_i, \varphi^* F) \rightarrow H_c^*(T, \varphi^* F)$ , give

$$\bigoplus_i H_{\{i\}}^*([i, \cdot], F) \otimes \text{or}_i \longrightarrow H_c^*(T, \varphi^* F). \quad (1.6.7)$$

**Theorem 1.4** (1.7). *Under the hypotheses and with the notation of 1.6, the morphism (1.6.7) is an isomorphism.*

*Proof.* For a short exact sequence  $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$ , the two sides of (1.6.7) give long exact cohomology sequences. Applying 1.3, we reduce to verifying the assertion for  $F = \Lambda_{[\cdot, i]}$ , the constant sheaf  $\Lambda$  on a closed set  $[\cdot, i]$ , extended by zero to  $I$ .

Compute  $H_{\{j\}}^*([j, \cdot], \Lambda_{[\cdot, i]})$ . If  $j \not\leq i$ , then  $\Lambda_{[\cdot, i]}$  is zero on  $[j, \cdot]$ , and one gets 0. Suppose that  $j \leq i$ . There is a long exact sequence

$$\cdots \rightarrow H_{\{j\}}^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow H^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow H^*([j, \cdot], \Lambda_{[\cdot, i]}) \rightarrow \cdots$$

which identifies with

$$\cdots \rightarrow H_{\{j\}}^*([j, i], \Lambda) \rightarrow H^*([j, i], \Lambda) \rightarrow H^*([j, i], \Lambda) \rightarrow \cdots.$$

For  $j < i$ , both  $[j, i]$  and  $]j, i]$  are acyclic, and  $H_{\{j\}}^*([j, i], \Lambda) = 0$ . For  $i = j$ , one has  $]j, i] = \emptyset$ ,  $[j, i] = \{j\}$ , and  $H_{\{j\}}^*(\{j\}, \Lambda) = \Lambda$  in degree 0. The map (1.6.7) is therefore reduced to the isomorphism (1.6.2) for  $T_i$ .  $\square$

**Corollary 1.5** (1.8). *With the hypotheses and notation of 1.6, suppose that  $I$  has a greatest element 1, and let  $S_1$  be the corresponding stratum. Let  $j_1$  be the inclusion of  $\{1\}$  in  $I$ , and let  $j_{1!}\Lambda$  be the constant sheaf  $\Lambda$  on the open set  $\{1\}$ , extended by zero to  $I$ . Then the choice of a connected component of each  $S_i$  defines an isomorphism*

$$\bigoplus_i H_{\{i\}}^*([i, 1], j_{1!}\Lambda) \otimes_{\text{or}_i} \xrightarrow{\sim} H_c^*(S_1, \Lambda). \quad (1.8.1)$$

*Proof.* Let  $j$  be the inclusion of  $S_1$  in  $T$ . The inverse image of  $j_{1!}\Lambda$  on  $T$  is  $j_!\Lambda$ , the constant sheaf  $\Lambda$  on  $S_1$  extended by zero to  $T$ . We have

$$H_c^*(S_1, \Lambda) \xrightarrow{\sim} H_c^*(T, j_!\Lambda),$$

and 1.8 follows from 1.7 applied to  $j_{1!}\Lambda$ .  $\square$

**Construction 1.6** (1.9). Let  $I$  be a finite ordered set with greatest element 1, and let  $j_1$  be the inclusion of the open set  $\{1\}$  in  $I$ . Then

$$H_{\{i\}}^*([i, 1], j_{1!}\Lambda) = \begin{cases} \Lambda & \text{in degree 0, if } i = 1, \\ \tilde{H}^{*-2}([i, 1[, \Lambda) & \text{otherwise.} \end{cases}$$

If  $]i, 1[$  is empty, the reduced cohomology  $\tilde{H}^*$  is  $\Lambda$  in degree  $-1$ .

*Construction.* The case  $i = 1$  is clear. Suppose  $i < 1$  and put  $F := j_{1!}\Lambda$ . The groups  $H^p([i, 1], F)$  are then all zero: for  $p > 0$  because  $[i, 1]$  has a least element, and for  $p = 0$  because  $F([i, 1]) = F_i = 0$ . The long exact sequence

$$\cdots \rightarrow H_{\{i\}}^*([i, 1], \cdot) \rightarrow H^*([i, 1], \cdot) \rightarrow H^*([i, 1], \cdot) \rightarrow \cdots$$

therefore reduces, for  $F$ , to an isomorphism

$$\partial : H^p([i, 1], F) \xrightarrow{\sim} H_{\{i\}}^{p+1}([i, 1], F). \quad (1.9.1)$$

Let  $\Lambda_{]i, 1[}$  be the constant sheaf  $\Lambda$  on the closed subset  $]i, 1[$  of  $[i, 1]$ , extended by zero to  $]i, 1]$ . Since  $]i, 1]$  is acyclic, because it has a greatest element, the long exact cohomology sequence defined by

$$0 \rightarrow F \rightarrow \Lambda \rightarrow \Lambda_{]i, 1[} \rightarrow 0$$

reduces to an isomorphism

$$\partial : \tilde{H}^p([i, 1[, \Lambda) \xrightarrow{\sim} H^{p+1}([i, 1], F). \quad (1.9.2)$$

The isomorphism of 1.9 is the composite of (1.9.1) and (1.9.2).  $\square$

## 1.10. Variants: homology and cohomology of $S_1$

If  $V$  is an oriented pure  $n$ -dimensional manifold, cap product with the fundamental class is an isomorphism from compactly supported cohomology of  $V$  to homology of  $V$ :

$$H_c^i(V, \Lambda) \xrightarrow{\sim} H_{n-i}(V, \Lambda).$$

Under the hypotheses of 1.8, and up to a renumbering and a twist by  $\text{or}(T)$ , (1.8.1) therefore also computes the homology of  $S_1$ .

The results obtained in compactly supported cohomology, or equivalently here in homology, give by duality—between homology and cohomology, or by Poincaré duality between cohomology and compactly supported cohomology—analogous results in cohomology.

For cohomology with coefficients in a field  $k$ , one simply has

$$H^i = \text{Hom}_k(H_i, k)$$

respectively  $H^i = \text{Hom}_k(H_c^{n-i}, k)$ , and in the present case these vector spaces are finite-dimensional. The same statement holds for  $k = \mathbb{Z}/N\mathbb{Z}$ . For  $\Lambda$  finite and  $\Lambda'$  its  $\mathbb{Q}/\mathbb{Z}$ -dual, one similarly has

$$H^i(\cdot, \Lambda') = \text{Hom}(H_i(\cdot, \Lambda), \mathbb{Q}/\mathbb{Z}),$$

respectively

$$H^i(\cdot, \Lambda') = \text{Hom}(H_c^{n-i}(\cdot, \Lambda), \mathbb{Q}/\mathbb{Z}).$$

More generally, one may use Pontryagin duality, with values in  $\mathbb{R}/\mathbb{Z}$ .

Another method, for  $\Lambda$  finitely generated over  $\mathbb{Z}$ , is to work systematically in derived categories, taking as coefficient  $\Lambda$  an object of  $D^b(\mathbb{Z})$  with finitely generated cohomology over  $\mathbb{Z}$ . If  $\Lambda'$  is the dual  $R\text{Hom}(\Lambda, \mathbb{Z})$  of  $\Lambda$ , then one still has

$$R\Gamma(S_1, \Lambda') = R\text{Hom}(R\Gamma_c(S_1, \Lambda), \mathbb{Z}) \otimes \text{or}(T).$$

**Notation.** We shall write  $h_c^*(i)$  for the groups

$$H_{\{i\}}^*([i, 1], F) \otimes \text{or}_i$$

which occur in the decomposition (1.8.1) of  $H_c^*(S_1, \Lambda)$ . The factors in the corresponding decompositions of  $H_*(S_1, \Lambda)$  and  $H^*(S_1, \Lambda)$  will be denoted by  $h_*(i)$  and  $h^*(i)$ .

The free rank-one graded  $\mathbb{Z}$ -module of coorientations of  $T_i$  in  $T$ , denoted  $\text{coor}(T_i)$ , or simply  $\text{coor}_i$ , is defined by

$$\text{or}(T_i) \otimes \text{coor}(T_i) \xrightarrow{\sim} \text{or}(T).$$

By duality, one deduces from 1.8 and 1.9 that

$$h^*(i) = \begin{cases} \Lambda & \text{in degree zero for } i = 1, \\ \tilde{H}_*([i, 1], \Lambda)[2] \otimes \text{coor}_i & \text{otherwise.} \end{cases} \quad (1.10.1)$$

In the second line, homological degree  $p$  is identified with cohomological degree  $-p$ : for  $i \neq 1$ ,

$$h^p(i) = \tilde{H}_{\text{codim}(T_i)-p-2}([i, 1], \Lambda) \otimes \text{coor}_i.$$

**Example 1.7** (1.11). Let  $V$  be a real affine space and let  $\mathcal{A}$  be a finite set of affine subspaces  $A \neq \emptyset, V$  of  $V$ , such that if the intersection of two elements of  $\mathcal{A}$  is non-empty, then that intersection belongs to  $\mathcal{A}$ . Put

$$\mathcal{A}^+ := \mathcal{A} \cup \{V\}, \quad X := V - \bigcup \mathcal{A}.$$

We call  $\mathcal{A}^+$ , respectively  $\mathcal{A}$ , the set of flats, respectively of proper flats, of the arrangement  $\mathcal{A}$ . The flats are the closed strata of a stratification of  $V$ , indexed by the set  $\mathcal{A}^+$  ordered by inclusion. Let  $\varphi : V \rightarrow \mathcal{A}^+$  be the corresponding map. We shall sometimes write  $A^0$  for the stratum corresponding to the flat  $A$ :

$$A^0 := \varphi^{-1}(A) = A - \bigcup \{\text{smaller flats}\}. \quad (1.11.1)$$

The flats are acyclic varieties. Applying 1.8, the choice in each stratum  $A^0$  of a point  $x_A$  defines an isomorphism

$$H_c^*(X, \Lambda) = \bigoplus_A h_c^*(A) \quad (\text{sum over the flats}), \quad (1.11.2)$$

where

$$h_c^*(A) = H_{\{A\}}^*([A, V], j_{V!}\Lambda) \otimes \text{or}(A),$$

and  $j_V$  is the inclusion of  $\{V\}$  in  $\mathcal{A}^+$ .

In [GM], Goresky and MacPherson obtain such a decomposition of the homology of  $X$  by applying Morse theory to a function “distance to  $v_0 \in V$ ”. The Euclidean structure on  $V$  and the point  $v_0$  are chosen generically, in the sense that:

- (a) for each flat  $A$ , the critical point on  $A$  of the distance function  $d(v_0, \cdot)$ , that is, the foot  $x_A$  of the perpendicular dropped from  $v_0$  to  $A$ , lies in  $A^0$ ;
- (b) the critical values  $d(v_0, x_A)$  are distinct.

For the flat  $V$ , one has  $x_V = v_0$ .

Let  $B(r)$  be the open ball with center  $v_0$  and radius  $r$ . Endow it with the stratification induced by that of  $V$ , whose closed strata are the traces on  $B(r)$  of the flats which meet  $B(r)$ . The hypotheses of 1.8 are again satisfied. If a flat  $A$  meets  $B(r)$ , then every flat  $B \supset A$  also meets  $B(r)$ , and the group  $h_c^*(A)$  is the same for  $B(r)$  and for  $V$ . The diagram

$$\begin{array}{ccc}
\bigoplus_{A \cap B(r) \neq \emptyset} h_c(A) & \longrightarrow & \bigoplus_A h_c(A) \\
\sim \downarrow & & \downarrow \sim \\
H_c(X \cap B(r), \Lambda) & \longrightarrow & H_c(X, \Lambda)
\end{array} \tag{1.11.3}$$

is commutative. Orient  $V$ , and identify homology and compactly supported cohomology as in 1.10. When  $r$ , increasing, passes a critical value  $r_A := d(v_0, x_A)$ , the Morse group, in homology, which measures the change from  $H_*(B(r_A - \epsilon), \Lambda)$  to  $H_*(B(r_A + \epsilon), \Lambda)$  is, by (1.11.3), the group  $h(A)$ . Our definition of the map

$$h(A) \longrightarrow H_c^*(X \cap B(r_A + \epsilon), \Lambda) \longrightarrow H_c^*(X, \Lambda)$$

corresponding to the choice of  $x_A$  as marked point in  $A^0$  is inspired by the proof in [GM] that the Morse function  $d(v_0, \cdot)$  is perfect. The decomposition by flats of  $H_*(X, \Lambda)$  implicit in [GM] therefore coincides with the decomposition (1.11.2) of  $H_c^*(X, \Lambda)$ , for the choice of the  $x_A$  as marked points.

**Example 1.8** (1.12). If  $T$  is a convex open subset of  $\mathbb{R}^n$  and  $E$  is a finite set of traces on  $T$  of affine subspaces, then the intersections  $T \cap F$  for  $F \in E$  which are non-empty are the closed strata of a stratification of  $T$  satisfying the hypotheses of 1.8. More generally, one may take  $T$  to be a convex subset of a space of negative curvature and  $E$  a set of totally geodesic subspaces.

**Example 1.9** (1.13). In Examples 1.11 and 1.12, the ordered set  $I$  indexing the strata has the property that if  $i$  and  $j$  in  $I$  have a lower bound, then they have a greatest lower bound  $\inf(i, j)$ . This is not necessarily the case in applications of 1.8. For example, one may have a disk, two curves  $A$  and  $B$ , and the two points of intersection as closed strata; the two intersection points have the same two curve strata above them without a single greatest lower bound.

**Example 1.10** (1.14). Our proof of 1.8 is sheaf-theoretic. It does not use Morse theory. It therefore applies in  $\ell$ -adic cohomology. The only difference is that for  $V$  a connected smooth algebraic variety of dimension  $n$  over an algebraically closed field, the group  $\text{or}(V)$  of 1.6 must be replaced by the Tate module  $\mathbb{Z}_\ell(-n)$  placed in degree  $2n$ .

Let  $V$  be an affine space over  $k$ , let  $\mathcal{A}$  be a finite set of affine subspaces, and let  $\mathcal{A}^+$  and  $X$  be as in 1.11. One has

$$\bigoplus_A H_{\{A\}}^*([A, V], j_{V!} \mathbb{Z}_\ell)(-\dim A)[-2 \dim A] \xrightarrow{\sim} H_c^*(X, \mathbb{Z}_\ell). \tag{1.14.1}$$

Since the  $A^0$  are connected here, (1.14.1) does not depend on the choice of marked points in the strata.

If  $k = \mathbb{C}$ , the same decomposition holds in mixed Hodge theory: in the decomposition (1.11.2) of  $H_c^*(X, \mathbb{Z})$ , the factor  $h_c^*(A)$  is pure of Hodge type  $(\dim_{\mathbb{C}} A, \dim_{\mathbb{C}} A)$ . Dually,  $h^*(A)$  is pure of Hodge type  $(\text{codim}(A), \text{codim}(A))$ .

In the complex case, cup product is compatible with the mixed Hodge structure. In algebraic geometry the arrangement  $\mathcal{A}$  may be defined over a subfield  $k_0$  of  $k$  of finite type over its prime subfield; the Galois group  $\text{Gal}(\bar{k}_0/k_0)$  acts on cohomology, and cup product is compatible with the Galois action. Under the above hypotheses and in rational cohomology, respectively  $\mathbb{Q}_\ell$ -cohomology, the component

$$h^*(A) \otimes h^*(B) \longrightarrow h^*(C)$$

of the cup product can therefore be non-zero only if

$$\text{codim}(C) = \text{codim}(A) + \text{codim}(B).$$

## 2 Functoriality in the Arrangement

### 2.1

Let  $V, \mathcal{A}, \mathcal{A}^+ := \mathcal{A} \cup \{V\}$ ,  $X := V - \bigcup \mathcal{A}$ , and  $\varphi : V \rightarrow \mathcal{A}^+$  be as in 1.11. Let  $\mathcal{A}' \subset \mathcal{A}$  satisfy the same hypotheses as  $\mathcal{A}$ , and denote the corresponding objects by primes. The stratification of  $V$  defined by  $\mathcal{A}$  refines that defined by  $\mathcal{A}'$ . If

$$\alpha : \mathcal{A}^+ \longrightarrow \mathcal{A}'^+$$

assigns to a flat of  $\mathcal{A}$  the smallest flat of  $\mathcal{A}'$  containing it, then the diagram

$$\begin{array}{ccc} & V & \\ \varphi \swarrow & & \searrow \varphi' \\ \mathcal{A}^+ & \xrightarrow{\alpha} & \mathcal{A}'^+ \end{array}$$

is commutative. Note that if the ordered sets  $\mathcal{A}^+$  and  $\mathcal{A}'^+$  are viewed as categories, then  $\alpha$  is left adjoint to the inclusion  $\text{incl}$  of  $\mathcal{A}'^+$  in  $\mathcal{A}^+$ . Indeed,

$$\alpha(A) \subset A' \iff A \subset \text{incl}(A').$$

Mark each flat  $A$  of  $\mathcal{A}^+$  by a point  $x_A \in A^0$ , and mark  $A'$  in  $\mathcal{A}'^+$  by  $x_{A'} := x_{\text{incl}(A')}$ . These points define decompositions (1.11.2) of  $H_c^*(X, \Lambda)$  and  $H_c^*(X', \Lambda)$ .

**Notation.** Below, if a flat  $A$  of  $\mathcal{A}^+$  lies in  $\mathcal{A}'^+$ , we shall write it as  $A'$  when it is to be considered as an element of  $\mathcal{A}'^+$ . Thus the interval  $[A, V]$  in  $\mathcal{A}'^+$  will be denoted  $[A', V]$ , and  $A'^0$  is  $A - \bigcup \{\text{smaller flats of } \mathcal{A}'\}$ .

The inclusion  $X \subset X'$  induces a morphism

$$H_c^*(X, \Lambda) \longrightarrow H_c^*(X', \Lambda). \quad (2.1.1)$$

**Proposition 2.1** (2.2). *Let  $A$  be a flat of  $\mathcal{A}^+$ .*

- (i) *If  $A$  is not a flat of  $\mathcal{A}'^+$ , then (2.1.1) is zero on  $h_c^*(A)$ .*
- (ii) *If  $A$  is a flat of  $\mathcal{A}'^+$ , say  $A = \text{incl}(A')$ , then (2.1.1) sends  $h_c^*(A)$  into  $h_c^*(A')$ . The morphism induced by (2.1.1) from  $h_c^*(A)$  to  $h_c^*(A')$  is the twist by  $\text{or}(A)$  of*

$$\text{incl}^* : H_{\{A\}}^*([A, V], j_{V!}\Lambda) \longrightarrow H_{\{A'\}}^*([A', V], j_{V!}'\Lambda). \quad (2.2.1)$$

Note that  $\text{incl}^{-1}(A)$  is reduced to  $\{A'\}$ , and that  $j_{V!}'\Lambda$  is the inverse image on  $[A', V]$  of  $j_{V!}\Lambda$ , so that (2.2.1) is defined. The inclusion of  $X$ , respectively  $X'$ , in  $V$  will be denoted by  $j_X$ , respectively  $j_{X'}$ .

*Proof of (ii).* In the definition of the morphism  $h_c^*(A') \rightarrow H_c^*(X', \Lambda)$ , we interpret  $H_c^*(X', \Lambda)$  as  $H_c^*(V, j_{X'}! \Lambda)$  and take the cup product of an element of

$$\text{or}_A = H_{\{x_A\}}^*(\varphi'^{-1}(A'), \mathbb{Z})$$

with a class in  $H_{A'}^*(\varphi'^{-1}([A', V]), j_{X'}! \Lambda)$ . Since we take the product with a class supported at  $x_A$ , it matters only through its restriction to a neighborhood of  $x_A$ . Take as neighborhood the open set  $U := \varphi^{-1}([A, V])$ . The morphism from  $h_c^*(A')$  to  $H_c^*(X', \Lambda)$  is deduced from

$$\varphi'^* : H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \rightarrow H_{A^0}(U, j_{X'}! \Lambda).$$

Again write  $\alpha$  for the restriction  $[A, V] \rightarrow [A', V]$  of  $\alpha$  to  $[A, V]$ . The inverse image of  $A'$  in  $[A, V]$  is reduced to  $A$ , and  $\alpha$  induces

$$\alpha^* : H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \rightarrow H_{\{A\}}^*([A, V], \alpha^* j_{V'}! \Lambda). \quad (2.2.2)$$

The morphism  $\alpha^*$  is an isomorphism. By the long exact sequence for cohomology with support, it is enough to verify that for every sheaf  $F$  on  $[A', V]$  one has

$$H^*([A', V], F) \xrightarrow{\sim} H^*([A, V], \alpha^* F), \quad H^*(]A', V], F) \xrightarrow{\sim} H^*(]A, V], \alpha^* F).$$

This is an application of 1.4, with  $\alpha$  viewed as a stratification of  $[A, V]$  or  $]A, V]$ : the inverse image of  $[\cdot, x]$  is  $[\cdot, \text{incl}(x)]$ , acyclic because it has a greatest element.

Consider the commutative diagram

$$\begin{array}{ccc} H_{\{A\}}^*([A, V], j_{V'}! \Lambda) & \rightarrow & H_{\{A\}}^*([A, V], \alpha^* j_{V'}! \Lambda) \xleftarrow{\sim} H_{\{A'\}}^*([A', V], j_{V'}! \Lambda) \\ \downarrow & & \swarrow \\ H_{\varphi^{-1}(A)}(U, j_X! \Lambda) & \longrightarrow & H_{\varphi^{-1}(A)}(U, j_{X'}! \Lambda) \end{array} \quad (2.2.3)$$

It defines a commutative diagram

$$\begin{array}{ccc} h_c^*(A) & \longrightarrow & h_c^*(A') \\ \downarrow & & \downarrow \\ H_c^*(X, \Lambda) & \longrightarrow & H_c^*(X', \Lambda), \end{array}$$

whose first line is the twist by  $\text{or}(A)$  of the first line of (2.2.3). Since  $\alpha \circ \text{incl}$  is the identity of  $\mathcal{A}'^+$ , the corresponding commutative diagram shows that this first line coincides with the inverse image by  $\text{incl}$  used in the statement of (ii).  $\square$

*Proof of (i) if  $\bigcup \mathcal{A} = \bigcup \mathcal{A}'$ .* Under the hypothesis  $\bigcup \mathcal{A} = \bigcup \mathcal{A}'$ , one has  $X = X'$ . We prove more precisely that  $h_c^*(A) = 0$  if the flat  $A$  of  $\mathcal{A}^+$  is not in  $\mathcal{A}'^+$ . This is clear from the interpretation (1.10) of  $h_c^*(A)$  as a Morse group: the “critical point” corresponding to  $A$  is superfluous, and its passage does not correspond to a change in the topology of  $X' \cap B(r)$ . Another proof is as follows. The inverse image of  $V$  by  $\alpha : [A, V] \rightarrow [A', V]$  is reduced to  $V$ , and by (2.2.2),  $h_c^*(A) \rightarrow h_c^*(A')$  is an isomorphism. From the commutative diagram

$$\begin{array}{ccc} \bigoplus_A h_c^*(A) & \longrightarrow & \bigoplus_{A'} h_c^*(A') \\ & \searrow \sim & \swarrow \sim \\ & H_c^*(X, \Lambda) & \end{array}$$

it follows that  $h_c^*(A) = 0$  for  $A$  in  $\mathcal{A}$  and not in  $\mathcal{A}'$ .  $\square$

*Proof of (i) if  $A \neq V$  and  $A \notin \bigcup \mathcal{A}'$ .* One has  $\varphi^{-1}(A) \subset X'$ . The diagram

$$\begin{array}{ccc} H_{\{A\}}^*([A, V], j_{V!}\Lambda) \otimes \text{or}(A) & \longrightarrow & H_{\{A\}}^*([A, V], \Lambda) \otimes \text{or}(A) \\ \downarrow & & \downarrow \\ H_c(X, \Lambda) & \longrightarrow & H_c^*(X', \Lambda) \end{array}$$

is commutative, and  $H_{\{A\}}^*([A, V], \Lambda) = 0$ , because  $[A, V]$  and  $]A, V]$  are both acyclic.  $\square$

*Proof of (i), general case.* Let  $\mathcal{A}''$  be the set of flats of  $\mathcal{A}$  contained in  $\bigcup \mathcal{A}'$ . Then  $\mathcal{A}' \subset \mathcal{A}'' \subset \mathcal{A}$ , and it is enough to treat separately the inclusions of  $\mathcal{A}'$  in  $\mathcal{A}''$  and of  $\mathcal{A}''$  in  $\mathcal{A}$ , both consequences of the cases already treated.  $\square$

## 2.3

Assume that the arrangement  $\mathcal{A}$  satisfies the connectedness hypothesis (CONN) of the introduction. Let  $A$  be a flat, let  $x_A$  be a point of the corresponding stratum  $A^0$ , and let  $U$  be an open ball with center  $x_A$ , small enough to meet no flats other than those which contain  $A$ . As observed in 1.12, one may apply 1.8 to  $U$ , endowed with the stratification induced by  $\mathcal{A}$ , indexed by  $[A, V]$ . The inclusion of  $U - \bigcup [A, V]$  in  $V - \bigcup [A, V]$  is a homotopy equivalence, and by 2.2 one has a commutative diagram

$$\begin{array}{ccccc} \bigoplus_{B \in [A, V]} h_c(B) & \longrightarrow & \bigoplus h_c(B) & \longrightarrow & \bigoplus_{B \in [A, V]} h_c(B) \\ \downarrow & & \downarrow & & \downarrow \\ H_c^*(U - \bigcup [A, V], \Lambda) & \longrightarrow & H_c^*(X, \Lambda) & \longrightarrow & H_c^*(V - \bigcup [A, V], \Lambda). \end{array} \quad (2.3.1)$$

The composite in the second line is an isomorphism. It makes  $H_c^*(U - \bigcup [A, V])$  a direct factor of  $H_c^*(X, \Lambda)$ , and the corresponding idempotent  $p_A$  is identified by 2.2 with the projection of  $\bigoplus_{B \in \mathcal{A}^+} h(B)$  onto the partial sum  $\bigoplus_{B \in [A, V]} h(B)$ . The idempotents  $p_A$  for  $A \in \mathcal{A}^+$  defined by the second line of (2.3.1) therefore commute with one another. They allow one to recover the decomposition of  $H_c^*(X, \Lambda)$  indexed by the flats. The dual decomposition of  $H^*(X, \Lambda)$  may similarly be reconstructed from the idempotents defined by

$$H^*(V - \bigcup [A, V], \Lambda) \longrightarrow H^*(X, \Lambda) \longrightarrow H^*(U - \bigcup [A, V], \Lambda),$$

whose composite is an isomorphism.

## 3 Restriction to a Linear Subspace

### 3.1

Resume the notation  $V, \mathcal{A}, \mathcal{A}^+, X$ , and  $\varphi : V \rightarrow \mathcal{A}^+$  of 1.11. Let  $V'$  be an affine subspace of  $V$  not contained in  $\bigcup \mathcal{A}$ , let  $\mathcal{A}'$  be the set of non-empty intersections of  $V'$  with a flat  $A \in \mathcal{A}$ , and let

$$X' := V' - \bigcup \mathcal{A}' = X \cap V'.$$

Choose a coorientation of  $V'$  in  $V$ . It defines a Gysin morphism

$$H_c^*(X', \Lambda) \longrightarrow H_c^*(X, \Lambda). \quad (3.1.1)$$

**Theorem 3.1** (3.2). *Assume that  $(V, \mathcal{A})$  and  $(V', \mathcal{A}')$  satisfy the connectedness hypothesis (CONN). Then, for  $A$  in  $\mathcal{A}$  and  $A'$  in  $\mathcal{A}'$ , the component*

$$h_c^*(A') \longrightarrow h_c^*(A) \quad (3.2.1)$$

*of the morphism (3.1.1), in the decompositions (1.11.2), can be non-zero only if  $A' = A \cap V'$  and the intersection of  $A$  and  $V'$  is transverse. In that case, the choice of the coorientation of  $V'$  in  $V$  defines an isomorphism  $\epsilon : \text{or}(A') \xrightarrow{\sim} \text{or}(A)$ ; the map “trace on  $V'$ ”,*

$$\alpha : [A, V] \longrightarrow [A', V'],$$

*is injective, and (3.2.1) is the tensor product of  $\epsilon$  with*

$$\alpha^* : H_{\{A'\}}^*([A', V'], j_{V'!}\Lambda) \longrightarrow H_{\{A\}}^*([A, V], j_{V!}\Lambda). \quad (3.2.2)$$

Let  $A \in \mathcal{A}$ , let  $\mathcal{A}_A := [A, V] \subset \mathcal{A}$ , let  $X_A := V - \bigcup \mathcal{A}_A$ , let  $\mathcal{A}'_A$  be the set of non-empty intersections of  $V'$  with a flat  $B \in \mathcal{A}_A$ , and let

$$X'_A = V' - \bigcup \mathcal{A}'_A = V' \cap X_A.$$

The group  $h_c^*(A)$  is the same for  $(V, \mathcal{A})$  and for  $(V, \mathcal{A}_A)$ . The diagram

$$\begin{array}{ccccc} H_c^*(X', \Lambda) & \longrightarrow & H_c^*(X, \Lambda) & \longrightarrow & h_c^*(A) \\ \downarrow & & \downarrow & & \parallel \\ H_c^*(X'_A, \Lambda) & \longrightarrow & H_c^*(X_A, \Lambda) & \longrightarrow & h_c^*(A) \end{array}$$

is commutative. For the square on the right, apply 2.2. The first vertical arrow was computed in 2.2. It remains to compute the composite in the second line. In other words, in order to compute  $H_c^*(X', \Lambda) \rightarrow h_c^*(A)$ , we have reduced to the case in which  $A$  is the least element of  $\mathcal{A}$ .

**Lemma 3.2** (3.3). *If  $A$  is the least element of  $\mathcal{A}$  and the intersection of  $A$  and  $V'$  is empty or is not transverse, then the composite*

$$H_c^*(X') \longrightarrow H_c^*(X) \longrightarrow h(A)$$

*is zero.*

*Proof.* Choose a Euclidean structure. If  $A$  and  $V'$  are disjoint, then for  $\epsilon$  small enough the set  $U$  of points of  $V$  at distance  $< \epsilon$  from  $V'$  is disjoint from  $A$ . The Gysin morphism factors through  $H_c^*(U \cap X', \Lambda)$ . Applying (1.8) to  $U$ , with the stratification induced by that of  $V$ , one sees that  $H_c^*(U \cap X, \Lambda)$  maps into

$$\bigoplus_{B \cap U \neq \emptyset} h(B) \subset H_c^*(X, \Lambda),$$

and therefore has zero projection to  $h(A)$ .

If  $A$  meets  $V'$  and the intersection is non-transverse, there exists a hyperplane  $H$  containing  $A$  and  $V'$ . Let  $U$  be one of the two half-spaces defined by  $H$ . The map from  $H_c^*(X', \Lambda)$  to  $H_c^*(X, \Lambda)$  then factors through  $H_c^*(U \cap X)$ . This is clear if, applying 1.10, one replaces  $H_c^*$  by homology  $H_*$ , the Gysin morphism becoming a direct image in homology. Indeed,  $X'$  lies in the boundary of  $U$ , in a part of the boundary of  $U$ , namely  $H \cap X$ , where the closure  $\overline{U}$  of  $U$  is a manifold with boundary. This permits any cycle traced on  $X'$  to be pushed into the interior of  $U$ .

As before, the image of  $H_c^*(U \cap X)$  in  $H_c^*(X)$  is contained in the sum of the  $h_c^*(B)$  for  $B \cap U \neq \emptyset$ , and hence has zero projection to  $h(A)$ .  $\square$



### 3.4

It remains to treat the case in which  $A$  is the least element of  $V$ ,  $A \cap V' \neq \emptyset$ , and the intersection of  $A$  and  $V'$  is transverse. For affine spaces  $V_1$  and  $V_2$ , for a system  $\mathcal{A}_1$  of flats of  $V_1$ , and for a point  $0$  of  $V_2$ , the triple  $(V, V', \mathcal{A})$  is isomorphic to

$$(V_1 \times V_2, V_1 \times \{0\}, \{B \times V_2 \mid B \in \mathcal{A}_1\}).$$

If  $X_1 = V_1 - \bigcup \mathcal{A}_1$ , then  $X = X_1 \times V_2$  and  $X' = X_1 \times \{0\}$ ; the Gysin morphism is the Kunneth isomorphism

$$H_c^*(X_1, \Lambda) \otimes H_c^{\dim V_2}(V_2, \mathbb{Z}) \xrightarrow{\sim} H_c^*(X, \Lambda).$$

It sends each  $h_c^*(B_1)$  isomorphically to  $h_c^*(B_1 \times V_2)$ , by the map indicated in 3.2.

### 3.5

Dually, consider the restriction morphism

$$H^*(X, \Lambda) \longrightarrow H^*(X', \Lambda). \quad (3.5.1)$$

In terms of the decompositions of 1.10 indexed by  $\mathcal{A}$  and  $\mathcal{A}'$ , respectively

$$H^*(X, \Lambda) = \bigoplus h^*(A), \quad H^*(X', \Lambda) = \bigoplus h^*(A'),$$

3.2 implies by duality that the only non-zero components of the restriction morphism (3.5.1) are the

$$h_A \longrightarrow h_{A'}$$

for  $A' = A \cap X'$ , the intersection being transverse, and they are deduced by duality from (3.2.2).

## 4 Application to the Cup Product

### 4.1

Let  $(V_1, \mathcal{A}_1)$  and  $(V_2, \mathcal{A}_2)$  satisfy the hypotheses of (1.11), and let  $V$  be the affine space  $V_1 \times V_2$ . Use the notation of 1.11 for  $V_1$  and  $V_2$ , and also for  $V$  endowed with the product stratification, whose strata, respectively closed strata, are products of strata, respectively closed strata, of  $V_1$  and  $V_2$ . The notation for  $V_1$ , respectively  $V_2$ , will carry subscript 1, respectively 2. The ordered set  $\mathcal{A}^+$  is naturally identified with the product of the ordered sets  $\mathcal{A}_1^+$  and  $\mathcal{A}_2^+$ , and the diagram

$$\begin{array}{ccc} V & \xlongequal{\quad} & V_1 \times V_2 \\ \varphi \downarrow & & \downarrow \varphi_1 \times \varphi_2 \\ \mathcal{A}^+ & \xlongequal{\quad} & \mathcal{A}_1^+ \times \mathcal{A}_2^+ \end{array}$$

is commutative. Suppose that each stratum  $A^0$  of  $V_i$  is equipped with a marked point  $x_A$ , and mark the stratum  $A^0 \times B^0$  of  $V_1 \times V_2$  by the point  $(x_A, x_B)$ .

Take as coefficients  $\Lambda$  a commutative ring. The cohomology of  $X = X_1 \times X_2$  is given by the Kunneth formula, which it is convenient for us to write in the derived category. Only the case  $\Lambda = \mathbb{Z}$  matters; the general case follows by extension of scalars  $\otimes_{\mathbb{Z}}^L \Lambda$ . With the notation and hypotheses of 1.6, the isomorphism (1.6.7) refines to an isomorphism in the derived category

$$\bigoplus_i R\Gamma_{\{i\}}([i, \cdot], F) \otimes_{\text{or}_i} \xrightarrow{\sim} R\Gamma_c(T, \varphi^* F). \quad (4.1.1)$$

The definition of the map is the same as in 1.6, and the fact that it is an isomorphism follows from 1.7.

Special case: for  $V_i$  with  $i = 1, 2$ ,

$$\bigoplus_A R\Gamma_{\{A\}}([A, V_i], j_{V_i!}\Lambda) \otimes \text{or}(A) \xrightarrow{\sim} R\Gamma_c(X_i, \Lambda), \quad (4.1.2)$$

and similarly for  $V$ . One advantage of the derived category is that the Kunneth formulas take the simple form that the cohomology of a product is the derived tensor product of the cohomologies of the factors. The morphisms (4.1.2) are compatible with the following three Kunneth formulas: first, for the intervals  $[A_i, V_i]$ ,

$$R\Gamma_{\{A_1\}}([A_1, V_1], j_{V_1!}\Lambda) \otimes^L R\Gamma_{\{A_2\}}([A_2, V_2], j_{V_2!}\Lambda) \xrightarrow{\sim} R\Gamma_{\{A_1 \times A_2\}}([A_1 \times A_2, V], j_{V!}\Lambda); \quad (4.1.3)$$

second, for the  $A_i$ ,  $\text{or}(A_1) \otimes \text{or}(A_2) \xrightarrow{\sim} \text{or}(A_1 \times A_2)$ ; and third, for  $X_1 \times X_2$ . The diagram

$$\begin{array}{ccc} (R\Gamma_{\{A_1\}}([A_1, V_1], j_{V_1!}\Lambda) \otimes \text{or}(A_1)) \otimes_{\Lambda}^L (R\Gamma_{\{A_2\}}([A_2, V_2], j_{V_2!}\Lambda) \otimes \text{or}(A_2)) & \longrightarrow & R\Gamma_c(X_1, \Lambda) \otimes_{\Lambda}^L R\Gamma_c(X_2, \Lambda) \\ \sim \downarrow & & \downarrow \sim \\ R\Gamma_{\{A_1 \times A_2\}}([A_1 \times A_2, V], j_{V!}\Lambda) \otimes \text{or}(A_1 \times A_2) & \longrightarrow & R\Gamma_c(X, \Lambda) \end{array} \quad (4.1.4)$$

is commutative. Thus the decomposition (4.1.2) of  $R\Gamma_c(X, \Lambda)$  is simply the derived tensor product of the decompositions (4.1.2) of the factors  $R\Gamma_c(X_i, \Lambda)$  for  $i = 1, 2$ .

Pass to the dual, denoted

$$D(\cdots) := R\text{Hom}_{\Lambda}(\cdots, \Lambda).$$

Under the finiteness hypotheses verified here, this is an involutive operation in the derived category and it commutes with tensor product. Poincare duality says that  $R\Gamma(X_i, \Lambda)$  is the dual of  $R\Gamma_c(X_i, \Lambda)$ , twisted by  $\text{or}(V_i)$ :

$$R\Gamma(X_i, \Lambda) = D(R\Gamma_c(X_i, \Lambda)) \otimes \text{or}(V_i).$$

The same holds for  $X$ .

The isomorphism (4.1.2) therefore dualizes to an isomorphism

$$R\Gamma(X_i, \Lambda) \xrightarrow{\sim} \bigoplus_A D(R\Gamma_{\{A\}}([A_i, V_i], j_{V_i!}\Lambda)) \otimes \text{coor}(A),$$

and, by compatibility between the Kunneth formula and Poincare duality, the corresponding diagram for  $R\Gamma(X_1, \Lambda) \otimes_{\Lambda}^L R\Gamma(X_2, \Lambda)$  and  $R\Gamma(X, \Lambda)$  is commutative. The isomorphism in the second column is the tensor product of an isomorphism

$$\text{coor}(A_1 \times A_2) = \text{coor}(A_1) \otimes \text{coor}(A_2)$$

with the isomorphism deduced by duality from (4.1.3).

## 4.2

Now let  $(V, \mathcal{A})$  be as in 1.11, and suppose that the connectedness hypothesis (CONN) of the introduction is satisfied. We apply 4.1 to  $V \times V$ . The cup product in cohomology is the composite of the exterior product deduced from the isomorphism studied in 4.1,

$$R\Gamma(X, \Lambda) \otimes^L R\Gamma(X, \Lambda) \xrightarrow{\sim} R\Gamma(X \times X, \Lambda),$$

and the restriction to the diagonal

$$\Delta^* : H^*(X \times X, \Lambda) \longrightarrow H^*(X, \Lambda).$$

This is the restriction to the diagonal subspace  $V$  of  $V \times V$ ; it is a special case of (3.5.1). By (4.1.4), the exterior product sends  $h_c^*(A_1) \otimes h_c^*(A_2)$  into  $h_c^*(A_1 \times A_2)$ . The condition that  $A_1 \times A_2$  meet the diagonal transversely is equivalent to the condition that  $A_1$  and  $A_2$  meet transversely in  $V$ . By 3.5, the only non-zero components of the cup product are therefore the components

$$h^*(A) \otimes h^*(B) \longrightarrow h^*(C) \quad (4.2.1)$$

for  $C = A \cap B$ , this intersection being transverse. In that case, (4.2.1) is deduced from the dual of the isomorphism (4.1.3) and from the dual of

$$\alpha^* : R\Gamma_{\{C\}}([C, V], j_{V!}\Lambda) \rightarrow R\Gamma_{\{(A, B)\}}([A, V] \times [B, V], j_{(V, V)!}\Lambda),$$

where  $\alpha : [A, V] \times [B, V] \rightarrow [C, V]$  is intersection.

**Remark 4.1** (4.3). Let  $V$  be an affine space over an algebraically closed field  $k$ , and let  $\mathcal{A}$ ,  $\mathcal{A}^+$ , and  $X$  be as in 1.14. The arguments used in 2.2 carry over unchanged to the case of  $\ell$ -adic cohomology. By contrast, 3.3 creates a difficulty. Consequently, our arguments do not determine the cup product in  $\ell$ -adic cohomology. We conjecture that it remains given by the same formulas as in 4.2.

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# The Nagata Embedding Theorem

Pierre Deligne

## Abstract

Except for a few corrections added in 2010, this article reproduces notes sent to Nagata in the 1970s, translating into the language of schemes his papers [2] and [3]. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to avoid the use of valuations. A more detailed exposition and additional results are given in Conrad [1]. Unless the contrary is expressly stated, all schemes considered below are assumed noetherian.

## 1 Recalls on blow-ups

### 0.1

Let  $\mathfrak{a}$  be an ideal of the ring  $A$ . The scheme obtained from  $X = \text{Spec}(A)$  by blowing up  $\mathfrak{a}$  is the union of the open subschemes  $D(x)$ , where  $x$  runs through a system of generators of  $\mathfrak{a}$ , and  $D(x)$  is the spectrum of the subring  $A[\mathfrak{a}x^{-1}]$  of  $A[x^{-1}]$ . Moreover,

$$D(x) \cap (X - V(\mathfrak{a})) = \text{Spec}(A[x^{-1}]).$$

If  $\mathfrak{c}$  is a second ideal of  $A$ , defining a closed subscheme  $V(\mathfrak{c})$  of  $X$ , the trace on  $D(x)$  of the strict transform of  $V(\mathfrak{c})$  is defined by the ideal of  $A[\mathfrak{a}x^{-1}]$  which is the trace of the ideal of  $A[x^{-1}]$  generated by  $\mathfrak{c}$ .

**Lemma (0.2).** *Let  $X$  be a scheme and let  $\mathfrak{a}$  and  $\mathfrak{b}$  be two sheaves of ideals on  $X$ . Let  $\widehat{X}$  be the blow-up of  $X$  along  $\mathfrak{a} + \mathfrak{b}$ . Then the open subset of  $\widehat{X}$  which is the union of the  $D(x)$  for  $x \in \mathfrak{b}$  contains the strict transform of  $V(\mathfrak{a})$ .*

*Proof.* The scheme  $\widehat{X}$  is the union of the  $D(x)$  for  $x \in \mathfrak{a}$  or  $x \in \mathfrak{b}$ , and, for  $x \in \mathfrak{a}$ , the open subset  $D(x)$  is disjoint from the strict transform of  $V(\mathfrak{a})$ .  $\square$

The lemma 0.3 of the original text was false. The corrected version of the lemma which follows dates from March 1996. It forced a change of notation. The original lemma used the following notation, to which the rest of the text refers: instead of  $F$  and  $G$ , one considered their complements  $U$  and  $V$ , which, with  $Y' := U \cap p^{-1}(Y)$ , formed a commutative diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \longrightarrow & X, \end{array}$$

where  $Y'$  is closed in  $p^{-1}(V)$ .

**Lemma (0.3).** *Let  $p : Z \rightarrow X$  be a morphism, let  $F \supset G$  be closed subschemes of  $X$ , and let  $Y$  be a closed subscheme of  $Z$ . Suppose that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G)$$

*scheme-theoretically. Blow up  $X$  along  $G$ , obtaining  $\widehat{X}$ , and let  $\tilde{p} : \widehat{Z} \rightarrow \widehat{X}$  be obtained from  $p : Z \rightarrow X$  by base change. Then the closure of  $Y - p^{-1}(G)$  in  $\widehat{Z}$  and  $\tilde{p}^{-1}(F - G)$ , taken in  $\widehat{X}$ , are disjoint closed subschemes of  $\widehat{Z}$ .*

*Proof.* We may suppose that  $Z$  and  $X$  are affine. Let  $\mathfrak{G}$  be the ideal of  $G$ . Cover  $\widehat{X}$  by the open subsets  $D(g)$ , for  $g \in \mathfrak{G}$  (a system of generators of  $\mathfrak{G}$  suffices), and cover  $X - G$  by

$$U(g) = D(g) - (\text{inverse image of } G).$$

Thus

$$\begin{array}{ccc} X - G & \hookrightarrow & \widehat{X} \\ \uparrow & & \uparrow \\ U(g) & \hookrightarrow & D(g). \end{array}$$

It is enough to see that, for each  $g$ , the traces on  $\tilde{p}^{-1}D(g)$  of  $\tilde{p}^{-1}(F - G)$  and of the closure of  $Y - p^{-1}G$  are disjoint. They are respectively the inverse image of the closure of  $F \cap U(g)$  in  $D(g)$ , and the closure of  $Y \cap p^{-1}(U(g))$  in  $\tilde{p}^{-1}D(g)$ .

Let a tilde denote inverse image, either from  $X$  to  $Z$ , or from  $D(g)$  to  $\tilde{p}^{-1}D(g)$ . Since  $p^{-1}G \supset Y \cap p^{-1}(F)$ , the inverse image  $\tilde{g}$  of  $g$  on  $Z$  can be written

$$\tilde{g} = y + f,$$

where  $y$  vanishes on  $Y$  and  $f$  vanishes on  $p^{-1}(F)$ . There are functions  $a_i$  on  $Z$  and elements  $f_i$  in the ideal of  $F$  such that

$$f = \sum_i a_i \tilde{f}_i.$$

Since  $F \supset G$ , the  $f_i$  are in the ideal of  $G$ , and the functions  $f_i/g$  are regular on  $D(g)$ . They vanish on  $F \cap U(g)$ , hence on the closure of  $F \cap U(g)$  in  $D(g)$ , and therefore

$$\sum_i a_i (f_i/g)^\sim$$

vanishes on  $\tilde{p}^{-1}\overline{F \cap U(g)}$ . On  $p^{-1}U(g)$  one has

$$1 = y/g + f/g = y/g + \sum_i a_i (f_i/g)^\sim,$$

and the same sum is therefore equal to 1 on  $Y \cap p^{-1}U(g)$ , hence on its closure in  $\tilde{p}^{-1}D(g)$ . This proves the required disjointness.  $\square$

**Remark.** Suppose instead that one is given closed subschemes  $F \supset G_0$  of  $X$ , with  $Y \cap p^{-1}(F)$  contained set-theoretically in  $p^{-1}(G_0)$ . Then there exists an  $r$  such that the  $r$ -th infinitesimal neighbourhood  $G$  of  $G_0$  in  $F$  satisfies the hypotheses of the lemma. If  $J$  is the ideal defining  $(Y \cap p^{-1}(F))_{\text{red}}$  in  $Y \cap p^{-1}(F)$ , it is enough to take  $r$  such that  $J^{r+1} = 0$ . Thus the lemma permits, by a blow-up which does not touch the complement  $V$  of  $G_0$ , the separation of the strict transform of  $F$  from the closure of  $Y - p^{-1}(G_0)$  in  $\widehat{Z}$ .

**Lemma (0.4).** Let

$$\begin{array}{ccc} X' & \xrightarrow{p} & X' \\ q \downarrow & & \downarrow \\ U & \hookrightarrow & X \end{array}$$

be a commutative diagram of schemes, with  $U$  a dense open subset of  $X$  and of  $X'$ , with  $p$  and  $q$  separated of finite type, and with  $q^{-1}(X) = X'$ . Let  $F$  be a closed subscheme of  $X$ , and let  $G$  be a closed subscheme of  $X'$  containing  $p^{-1}(F)$ . Then there exists an ideal  $\mathfrak{a}$ , with  $V(\mathfrak{a}) \subset F - \overline{F}$ , such that, after base change by the blow-up morphism defined by  $\mathfrak{a}$ ,  $\widehat{X} \rightarrow X$ , and after replacing the new  $X'$  by the closure of  $U$ , one has

$$q^{-1}(F) \subset G.$$

*Proof.* In Nagata's language, to take the closure of  $F$  in the Zariski–Riemann space of  $X$ , it is enough to take the “intersection” of the closures of  $F$  in the blow-ups of  $X$  whose centres are contained in  $F - \overline{F}$ .

Apply Lemma 0.3 to

$$\begin{array}{ccc} X' - q^*F & \longrightarrow & X' - G \\ \uparrow & & \uparrow \\ X - F & \longrightarrow & X - (F - \overline{F}). \end{array}$$

One obtains  $\mathfrak{a}$ , and, after the base change  $\widehat{X} \rightarrow X$ , the new closure of  $F$  is disjoint from the image of the closure of the old  $X' - q^*F$  in  $X' - G$ . In particular, after replacing  $\widehat{X}$  by  $U$ , this new closure is disjoint from the image of  $X' - G$ , which is what was required.  $\square$

**Lemma (0.5).** *Let*

$$\begin{array}{ccc} X_i & \longrightarrow & X_i \quad (1 \leq i \leq n) \\ \uparrow & & \uparrow \\ U & \longrightarrow & X \end{array}$$

*be a diagram of separated  $S$ -schemes of finite type, with  $X_i$  proper over  $S$ , and with  $U$  a dense open subscheme of  $X$  and of the  $X_i$ . Let  $X^*$  be the closure of  $U$  in the product of the  $X_i$ . Finally, let  $F_i$  be a closed subscheme of  $X_i$ ; suppose that the intersection of the inverse images of the  $F_i$  in  $X^*$  is empty. Then, after replacing the  $X_i$  by blow-ups whose centres lie in  $F_i - \overline{F_i}$ , one obtains an analogous situation in which the intersection of the inverse images of the  $\overline{F_i}$  is empty.*

*Proof.* Let  $X^{**}$  be a blow-up of  $X^*$ , with projections  $p_i : X^{**} \rightarrow X_i$ , such that in  $X^{**}$  one has

$$\bigcap_i p_i^{-1}(F_i) = \emptyset;$$

for example, blow up  $\bigcap_i p_i^{-1}(F_i)$ . Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccc} p_i^{-1}(X_i) & \longrightarrow & X^{**} \\ \uparrow & & \uparrow \\ U & \longrightarrow & X_i, \end{array}$$

and to  $F_i \subset X_i$ ,  $G_i = p_i^{-1}(F_i)$ . This provides the desired blow-ups of the  $X_i$ . Indeed, in the new situation, in the closure of  $U$  in  $X^{**}$ , one has

$$\bigcap_i p_i^{-1}(\overline{F_i}) \subset \bigcap_i G_i = \emptyset.$$

Since  $X^{**} \rightarrow X^*$  is surjective, this remains true in  $X^*$ .  $\square$

## 2 The theorems

**Definition (1.1).** *Let  $X$  and  $Y$  be two  $S$ -schemes, with  $Y$  separated over  $S$ . A quasi-domination  $f : X \dashrightarrow Y$  is a pair consisting of a dense open subscheme  $U$  of  $X$  and a morphism of  $S$ -schemes  $f : U \rightarrow Y$  whose graph is closed in  $X \times_S Y$ .*

*Let  $X$  and  $Y$  be two  $S$ -schemes, with  $Y$  separated over  $S$ , let  $U$  be a quasi-compact open subscheme of  $X$ , and let  $f : U \rightarrow Y$ . We say that  $X$  is quasi-dominant over  $Y$ , relative to  $U$  and  $f$ , if there exists a quasi-domination of the schematic closure  $\overline{U}$  of  $U$ , with values in  $Y$ , which extends  $f$ . There is at most one such quasi-domination; its graph is the closure of the graph of  $f$ .*

*A quasi-domination  $f$  is called proper or complete if the morphism  $f : U \rightarrow Y$  is proper.*

**Theorem (1.2).** *Let  $X$  and  $Y$  be two  $S$ -schemes, and let  $U \subset V \subset X$  be two dense open subschemes of  $X$ . Suppose that  $Y$  is separated of finite type over  $S$ . Let  $f : U \rightarrow Y$  be a quasi-dominion of  $V$  in  $Y$ . Then there exists an ideal  $\mathfrak{a}$ , with  $V(\mathfrak{a}) \subset X - V$ , such that the blow-up  $\widehat{X}$  of  $X$  with centre  $V(\mathfrak{a})$  quasi-dominates  $Y$ .*

*Proof.* By the base change  $X \rightarrow S$  one is reduced to the case  $X = S$ . Then  $Y$  is a scheme over  $X$ ,  $f$  is a section of  $Y$  over  $U$ , and by hypothesis  $f(U)$  is closed over  $V$ .

*Case 1:*  $U = V$ ,  $p : Y \rightarrow X$  quasi-affine, and  $X$  affine. By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & E_X \\ & \searrow & \downarrow \\ & & X, \end{array}$$

where  $E_X$  is an affine space over  $X$ . If  $\widehat{X}$  quasi-dominates  $E_X$  relative to  $f$ , then a fortiori  $\widehat{X}$  quasi-dominates  $Y$ . We may therefore suppose  $Y = E_X$ . A preliminary blow-up allows us to suppose that  $X - U$  is a divisor  $D$ . Locally on  $X$ , the section  $f$  has coordinates  $(f_i u^{-n})$ , where  $u$  is a local equation of  $D$ , where the  $f_i$  are regular, and where  $n$  is chosen once and for all sufficiently large. Let  $\mathfrak{a}$  be the ideal generated by  $u^n$  and the  $f_i$ . After blowing up  $V(\mathfrak{a})$ , the section  $f$  extends to a section of the projective completion of  $E_X$ , and we are done.

*Case 2:*  $U = V$  and  $p$  quasi-affine. By Case 1, if  $(U_i)_{0 \leq i \leq n}$  is an affine open covering of  $X$ , there exists on  $U_i$  an ideal  $\mathfrak{a}_i$  which works. This ideal admits an extension  $\tilde{\mathfrak{a}}_i$  to  $X$ , with  $V(\tilde{\mathfrak{a}}_i) \subset X - V$ , and it is enough to blow up the product of the  $\tilde{\mathfrak{a}}_i$ .

*Case 3:*  $p$  quasi-affine. By Lemma 0.3, applied to

$$\begin{array}{ccc} f(U) & \longrightarrow & Y \\ & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X, \end{array}$$

a preliminary blow-up allows us to suppose that

$$p(f(U)) \cap (V - U) = \emptyset.$$

There is therefore a quasi-compact open subscheme  $X'$  of  $X$  with  $X' \cap V = U$  and  $p(f(U)) \subset X'$ . By Case 2, there exists an ideal  $\mathfrak{a}$  on  $X'$ , with  $V(\mathfrak{a}) \subset X' - U$ , such that the blow-up  $\widehat{X'}$  quasi-dominates  $Y$ , that is,  $f(U)$ . It remains only to extend  $\mathfrak{a}$  to an ideal on  $X$ , with  $V(\mathfrak{a}) \subset X - V$ .

*General case.* Let  $(Y_i)$  be a finite covering of  $Y$  by affine open subschemes, and put

$$U_i = f^{-1}(Y_i), \quad X_i = \overline{U_i}, \quad V_i = X_i \cap V.$$

Case 3 applies to

$$Y_i \cap p^{-1}(X_i), \quad U_i \subset V_i \subset X_i,$$

and gives an ideal  $\mathfrak{a}_i$  on  $X_i$ . This ideal defines an ideal on  $X$ , still denoted  $\mathfrak{a}_i$ , with  $V(\mathfrak{a}_i) \subset X - V$ . Blow up the product of the  $\mathfrak{a}_i$ . The blow-up  $q : \widehat{X} \rightarrow X$  thus obtained is such that  $q^{-1}(X_i)$  dominates the blow-up  $\widehat{X}_i$ . If  $X'_i = q^{-1}(X_i)$ , there exists an open  $U'_i$  of  $X'_i$ , with  $U'_i \supset q^{-1}(U_i)$ , such that, above  $X_i$ ,  $f(U_i)$  is a section of  $U'_i \rightarrow Y$ . One concludes that, after this blow-up,

$$f(U) = \bigcup_i f_i(U'_i).$$

For the proof of Theorem 1.2, we may replace  $Y$  by  $f(U)$ , and conclude by Case 3 and the following lemma.  $\square$

**Lemma (1.3).** *The scheme  $f(U)$  is quasi-affine over  $\widehat{X}$ .*

*Proof.* Indeed,  $f(U)$  is quasi-finite over  $\widehat{X}$ .  $\square$

**Corollary** (1.4, a variant of Chow's lemma). *Let  $f : X \rightarrow S$  be a separated morphism of finite type. Then for every open subscheme  $U$  which is quasi-projective over  $S$  and dense in  $X$ , there exists a diagram of  $S$ -schemes*

$$\begin{array}{ccc} & X' & \\ q \swarrow & & \searrow j \\ X & & \overline{X}, \end{array}$$

where  $q$  is the blow-up morphism defined by an ideal  $\mathfrak{a}$  satisfying  $V(\mathfrak{a}) \subset X - U$ , and where  $\overline{X}$  is projective over  $S$ . In particular,  $q$  is projective and surjective and  $q^{-1}(U) \simeq U$ .

*Proof.* Let  $U^*$  be a projective  $S$ -scheme containing  $U$  as a schematically dense open subscheme. By Theorem 1.2, applied to the injection of  $U$  into  $X$ , one may choose  $U^*$  quasi-dominant over  $X$ . Thus one has a diagram

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ U^* & \longrightarrow & X, \end{array} \quad (1.4.1)$$

and here, since  $U^*$  is proper over  $S$ ,  $\varphi$  is proper.  $\square$

**Lemma** (1.5). *Let*

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ Y & \longrightarrow & X \end{array}$$

be a diagram of separated  $S$ -schemes of finite type, with  $U$  dense in  $X$ , dense in  $Y$ , and schematically dense in  $V$ , and with  $\varphi$  a quasi-domination of  $Y$  in  $X$ . If  $\mathfrak{a}$  is an ideal of  $X$ , with  $V(\mathfrak{a}) \subset X - U$ , such that the blow-up  $\widehat{X}$  of  $X$  along  $\mathfrak{a}$  quasi-dominates  $V$ , relative to  $U$ , and such that  $U$  is schematically dense in  $\widehat{X}$ , giving a diagram

$$\begin{array}{ccc} U & \hookrightarrow & W \\ \downarrow & & \downarrow \psi \\ \widehat{X} & \longrightarrow & V, \end{array}$$

then:

(a)  $\psi$  is the blow-up morphism of  $V$  along  $\varphi^*\mathfrak{a}$ ;

(b) the graph of  $\psi$  is closed in  $\widehat{X} \times_S Y$ .

*Proof.* Let  $\widehat{V}$  be the blow-up of  $V$  along  $\varphi^*\mathfrak{a}$ . By the universal property of  $\widehat{X}$ , there is a unique  $X$ -morphism  $\alpha : \widehat{V} \rightarrow \widehat{X}$ . By the universal property of  $\widehat{V}$ , there is a unique  $V$ -morphism  $\beta : W \rightarrow \widehat{V}$ . The relevant diagram is

$$\begin{array}{ccccc} U & \hookrightarrow & W & \xrightarrow{\beta} & \widehat{V} \\ \downarrow & & \downarrow \psi & & \downarrow \alpha \\ V & \xleftarrow{\varphi} & Y & \longrightarrow & \widehat{X} \end{array}$$

The graph of  $\varphi$  is, trivially, proper over  $V$ ; the same is true of its inverse image on  $V \times_S \widehat{X}$ , and it follows that the graph of  $\psi$  is proper over  $V$ , hence that  $\psi$  is proper. Thus  $\beta$  is proper,



and its image is schematically dense, because already  $U$  is schematically dense in  $\widehat{V}$ . On the other hand,  $\beta$  is locally a section of  $\alpha$  on  $W$  (because  $\beta|_U$  is such a section over  $U$ ); hence  $\beta$  is an isomorphism. By hypothesis the graph of  $\psi$  is closed in  $V \times_S \widehat{X}$ , and is contained in the inverse image of the graph of  $\varphi$ , which is closed in  $Y \times X$ . One concludes that  $\Gamma(\psi)$  is closed in  $Y \times \widehat{X}$ .  $\square$

**Lemma (1.7).** *Let*

$$X_2 \leftarrow U \rightarrow X_1$$

*be a diagram of separated  $S$ -schemes of finite type, where  $U$  is a dense open subscheme of each  $X_i$ . Then there exists  $j : U \hookrightarrow X$ , with  $X$  separated of finite type over  $S$  and  $U$  schematically dense in  $X$ , and proper quasi-dominations*

$$p_i : X \dashrightarrow X_i.$$

*Proof.* By Theorem 1.2, after blowing up  $X_1$  without touching  $U$ , one reduces to the case where  $X_1$  quasi-dominates  $X_2$ :

$$\begin{array}{ccc} & X_1 & \\ \swarrow & & \searrow \\ U & \xhookrightarrow{\quad} & X_2. \end{array}$$

Apply Lemma 1.5 to this diagram, in order to obtain

$$\begin{array}{ccc} W & \xrightarrow{\psi} & X'_2 \\ \uparrow & & \uparrow \\ U & \xhookrightarrow{\quad} & V, \\ & & \uparrow \\ & & X_1 \end{array}$$

where

- (a)  $\psi$  is the blow-up morphism of an ideal  $\mathfrak{a}$  of  $V$ , with  $V(\mathfrak{a}) \subset V - U$ ;
- (b) the graph of  $\psi$  is closed in  $X_1 \times X'_2$ ;
- (c)  $X'_2$  is a blow-up of  $X_2$ .

Let  $\mathfrak{a}'$  be an ideal of  $X_1$  extending  $\mathfrak{a}$ , with  $V(\mathfrak{a}') \subset X_1 - U$ , and let  $q : X'_1 \rightarrow X_1$  be the blow-up of  $X_1$  along  $\mathfrak{a}'$ . Then  $\psi$  induces an isomorphism between  $W$  and  $q^{-1}(V)$ , and the graph of this isomorphism is closed in  $X'_1 \times X'_2$ . Define  $X$  by gluing  $X'_1$  and  $X'_2$  along this isomorphism.  $\square$

**Remark.** *The proof shows that one may take for  $p_i : U_i \rightarrow X_i$ , with  $U_i \subset X$ , a blow-up morphism of an ideal outside  $U$ .*

**Theorem (1.6).** *Let  $f : X \rightarrow S$  be separated of finite type. There exists an immersion of  $X$  into an  $S$ -scheme proper over  $S$ .*

*Proof.* Let  $(U_i)$  be a finite open covering of  $X$  by dense quasi-projective open subschemes. For each  $U_i$ , there is, by Corollary 1.4, a commutative diagram of  $S$ -schemes

$$\begin{array}{ccc} X'_i & \xrightarrow{q_i} & X_i \\ j_i \downarrow & & \\ U_i & \xhookrightarrow{\quad} & \overline{X}_i, \end{array}$$

where  $q_i$  is proper,  $\overline{X}_i$  is proper over  $S$ , and  $U_i$  is dense in  $\overline{X}_i$ . Let  $F_i = \overline{X}_i - U_i$  and  $F'_i = q_i^{-1}(F_i)$ . Finally, let  $X^*$  be the closure of  $U$  in the product of the  $\overline{X}_i$ , with projections

$$p_i : X^* \rightarrow \overline{X}_i.$$

By Lemma 0.5, a preliminary blow-up of the  $\overline{X}_i$ , not touching  $X'_i$ , allows one to suppose that

$$\bigcap_i p_i^{-1}(F'_i) = \emptyset.$$

Let  $M_i$  be the  $S$ -scheme, separated over  $S$ , obtained by gluing  $X$  and  $\overline{X}_i - F'_i$  along  $U_i$ :

$$\begin{array}{ccc} \overline{X}_i - F'_i & & X \\ & \swarrow \quad \searrow & \\ & M_i & \end{array}$$

Apply Lemma 1.7 to  $X$  and to the  $M_i$ , to obtain  $X \hookrightarrow M$ , where  $M$  properly quasi-dominates each  $M_i$ . It remains to prove that  $M$  is proper over  $S$ .

Indeed, one has morphisms

$$X^* - p_i^{-1}(F'_i) \longrightarrow M_i$$

which glue on  $U = \bigcap_i U_i$ . Replacing  $X^*$  by a blow-up  $X^{**}$ , one obtains morphisms

$$X^{**} - p_i^{-1}(F_i) \longrightarrow M$$

which glue, and hence a morphism from the schematic closure of  $U$  in  $X^{**}$  to  $M$ . The scheme  $M$ , as the image of this morphism, is proper over  $S$ .  $\square$

**Comments.** *I hope that, by being still more constructive, one can prove Theorem 1.6 for  $f : X \rightarrow S$  separated of finite type, with  $S$  quasi-compact and quasi-separated. The rules to follow are:*

- (a) *consider only quasi-compact open subsets and closed subsets defined by ideals of finite type;*
- (b) *never take schematic closure (one would leave (a)), but take only a closed subset containing the given part and sufficiently small for the later constructions;*
- (c) *do not speak of a dense quasi-projective open subset, cf. (b). Remember that, to construct such open subsets, one could start from an affine covering  $(U_i)$ , and take*

$$U = \bigcup_i \left( U_i - \bigcup_{j < i} U_j \right).$$

*Translate the arguments in terms of the original  $U_i$ , and apply (a) and (b).*

*On the other hand, I have no idea in the case of algebraic spaces.*

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# Finiteness of the Extension of $\mathbb{Q}$ Generated by Frobenius Traces, in Finite Characteristic

Pierre Deligne

Dedicated to the memory of I. M. Gelfand

**Abstract.** Let  $Z_0$  be a scheme of finite type over  $\mathbb{F}_q$  and let  $\mathcal{F}_0$  be a  $\overline{\mathbb{Q}_\ell}$ -sheaf on  $Z_0$ . We show that there is a finite-type extension  $E \subset \overline{\mathbb{Q}_\ell}$  of  $\mathbb{Q}$  such that the local factors of the  $L$ -function of  $\mathcal{F}_0$  all have coefficients in  $E$ . If  $Z_0$  is normal and connected and  $\mathcal{F}_0$  is an irreducible  $\ell$ -adic local system whose determinant has finite order, one may take  $E$  to be a finite extension of  $\mathbb{Q}$ .

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- 0. Introduction
- 1. Irreducibility and extension of the base field
- 2. The case of curves: quantitative versions
- 3. The main theorem

## 0. Introduction

### 0.0

Fix the following notation:

- $\mathbb{F}_q$  : a finite field with  $q$  elements, of characteristic  $p$ ;
- $\mathbb{F}$  : an algebraic closure of  $\mathbb{F}_q$ ;
- $\mathbb{F}_{q^n}$  : the extension of degree  $n$  of  $\mathbb{F}_q$  inside  $\mathbb{F}$ ;
- $W(\mathbb{F}/\mathbb{F}_{q^n})$  : the subgroup  $\mathbb{Z}$  of  $\text{Gal}(\mathbb{F}/\mathbb{F}_{q^n}) = \widehat{\mathbb{Z}}$  generated by the geometric Frobenius  $F$   
(the inverse of  $x \mapsto x^{q^n}$ );
- $\ell$  : a prime number distinct from  $p$ ;
- $\overline{\mathbb{Q}_\ell}$  : an algebraic closure of  $\mathbb{Q}_\ell$ .

An object over  $\mathbb{F}_q$  will be marked by a subscript 0, and suppressing this subscript will mean extension of the base field from  $\mathbb{F}_q$  to  $\mathbb{F}$ .

For a summary of the formalism of  $\overline{\mathbb{Q}_\ell}$ -sheaves and of  $\overline{\mathbb{Q}_\ell}$ -Weil sheaves, I refer to [De2] 1.1.1 and 1.1.10.

Let  $\overline{X}_0$  be a smooth projective absolutely irreducible curve over  $\mathbb{F}_q$ , let  $S_0$  be a finite set of closed points of  $\overline{X}_0$ , and put  $X_0 := \overline{X}_0 - S_0$ . We write  $K_0$  for the field of rational functions on  $X_0$  and  $A$  for its adèle ring.

### 0.1

Lafforgue [L] provides a natural bijection between

- (A) isomorphism classes of smooth irreducible  $\overline{\mathbb{Q}_\ell}$ -Weil sheaves of rank  $r$  on  $X_0$ , and
- (B) cuspidal automorphic representations  $\pi$  of  $\text{GL}(r, A)$ , unramified outside  $S_0$ .

This bijection is characterized by equality, at each closed point  $x_0$  of  $X_0$ , of the local factors of the  $L$ -functions attached to the objects in (A) and (B).

In (B), the notion of cuspidal automorphic representation is purely algebraic: one may take as coefficient field any algebraically closed field  $k$  of characteristic zero. We take  $k = \overline{\mathbb{Q}_\ell}$ . The topology of  $\overline{\mathbb{Q}_\ell}$  plays no role.

If  $Z_0$  is a scheme of finite type over  $\mathbb{F}_q$ , the action on  $\mathbb{F}$  of  $W(\mathbb{F}/\mathbb{F}_q)$  induces an action of this Weil group on  $Z = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ . A  $\overline{\mathbb{Q}_\ell}$ -Weil sheaf  $\mathcal{F}_0$  on  $Z_0$  is a  $\overline{\mathbb{Q}_\ell}$ -sheaf  $\mathcal{F}$  on  $Z$ , endowed with an action of  $W(\mathbb{F}/\mathbb{F}_q)$  on  $(Z, \mathcal{F})$  lifting its action on  $Z$ . The notion of  $\overline{\mathbb{Q}_\ell}$ -sheaf is defined by passage to the limit from that of an étale sheaf of  $\mathcal{O}_\lambda/I$ -modules, where  $\mathcal{O}_\lambda$  is the integral closure of  $\mathbb{Z}_\ell$  in a finite extension  $E_\lambda \subset \overline{\mathbb{Q}_\ell}$  of  $\mathbb{Q}_\ell$ , and  $I$  is a power of the maximal ideal. Here the topology of  $\overline{\mathbb{Q}_\ell}$  plays a crucial role. A surprising consequence of the equivalence between (A) and (B) is that, for semisimple smooth sheaves on curves, this role is illusory.

## 0.2

One should beware that Lafforgue assumes a choice of an isomorphism  $\iota : \mathbb{C} \rightarrow \overline{\mathbb{Q}_\ell}$ , and uses it rather to compare automorphic representations in the classical sense, with coefficient field  $\mathbb{C}$ , and  $\overline{\mathbb{Q}_\ell}$ -sheaves. If one uses  $\iota$  to replace cuspidal automorphic representations with coefficients in  $\mathbb{C}$  by those with values in  $\overline{\mathbb{Q}_\ell}$ , the resulting correspondence depends, when  $r$  is even and  $q$  is an odd power of  $p$ , on the square root  $\iota(\sqrt{q})$  of  $q$  in  $\overline{\mathbb{Q}_\ell}$ . Changing  $\sqrt{q}$  to  $-\sqrt{q}$  changes the correspondence by composing it with twisting, see 0.3, by the character

$$n \mapsto (-1)^{n(r-1)}$$

of  $W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z}$ .

Instead of the correspondence used by Lafforgue, we prefer its twist by the character

$$n \mapsto (\sqrt{q})^{n(r-1)}$$

of  $W(\mathbb{F}/\mathbb{F}_q)$ . This rationalized correspondence gives a bijection between (A) and (B) independent of  $\iota$ .

## 0.3

For both the objects (A) and the objects (B), twisting by a character

$$\chi : W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \longrightarrow \overline{\mathbb{Q}_\ell}^*$$

is defined, and these twists correspond.

The set of characters  $\chi$  of  $W(\mathbb{F}/\mathbb{F}_q)$  is naturally in bijection with the set of isomorphism classes of rank-one  $\overline{\mathbb{Q}_\ell}$ -Weil sheaves  $\mathcal{L}_0$  on  $\text{Spec}(\mathbb{F}_q)$ ; for the objects (A), the twist is given by tensor product with the inverse image, under  $X_0 \rightarrow \text{Spec}(\mathbb{F}_q)$ , of such a sheaf.

Let  $v : A^* \rightarrow \mathbb{Z}$  be the homomorphism such that  $\|x\| = q^{-v(x)}$ . It factors through the idele class group  $K_0^* \backslash A^*$ . In (B), if  $\pi$  is a cuspidal automorphic representation of  $\text{GL}(r, A)$ , viewed as contained in the space of locally constant functions on  $\text{GL}(r, K_0) \backslash \text{GL}(r, A)$ , the twist  $\pi_\chi$  of  $\pi$  by  $\chi$  is the space of functions

$$f(x) \chi(v(\det x))$$

for  $f$  in  $\pi$ .

We will sometimes say “twist by  $b$ ” for “twist by the character  $\chi$  of  $\mathbb{Z}$  such that  $\chi(1) = b$ .”

## 0.4

The  $\overline{\mathbb{Q}}_\ell$ -sheaves form a full abelian subcategory of the abelian category of  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves. On  $\text{Spec}(\mathbb{F}_q)$ , the isomorphism class of a rank-one  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf  $\mathcal{L}_0$  is determined by a character

$$\chi : W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \longrightarrow \overline{\mathbb{Q}}_\ell^*,$$

and for  $\mathcal{L}_0$  to be a  $\overline{\mathbb{Q}}_\ell$ -sheaf it is necessary and sufficient that  $\chi$  extend to a continuous character of  $\text{Gal}(\mathbb{F}/\mathbb{F}_q) = \widehat{\mathbb{Z}}$ , i.e. that  $\chi(1)$  be an  $\ell$ -adic unit. This is what forces the use, in 0.1(A), of  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves. If one wanted in (A) to restrict oneself to  $\overline{\mathbb{Q}}_\ell$ -sheaves, then in (B) one would have to restrict oneself to cuspidal automorphic representations  $\pi$  whose central character  $\omega_\pi$  extends to a continuous character of the profinite completion of the idele class group. This would destroy the purely algebraic character of (B).

The following theorems allow one to pass from  $\overline{\mathbb{Q}}_\ell$ -sheaves to  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves.

- (a) Let  $Z_0$  be a connected normal variety of finite type over  $\mathbb{F}_q$ , and let  $\mathcal{F}_0$  be a smooth irreducible  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf of rank  $r$  on  $Z_0$ . Then  $\mathcal{F}_0$  is a  $\overline{\mathbb{Q}}_\ell$ -sheaf if and only if

$$\det(\mathcal{F}_0) := \bigwedge^r \mathcal{F}_0$$

is one ([De2] 1.3.14).

- (b) Let  $\mathcal{L}_0$  be a smooth rank-one  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf on  $Z_0$ . Definition:  $\mathcal{L}_0$  is of finite order if there is an  $N > 0$  such that  $\mathcal{L}_0^{\otimes N}$  is isomorphic to the constant sheaf  $\overline{\mathbb{Q}}_\ell$ . There is always an  $N > 0$  such that  $\mathcal{L}_0^{\otimes N}$  is the inverse image of a Weil sheaf on  $\text{Spec}(\mathbb{F}_q)$  ([De2] 1.3.4(i)). The group, for  $\otimes$ , of isomorphism classes of rank-one Weil sheaves on  $\text{Spec}(\mathbb{F}_q)$  is divisible. Every  $\mathcal{L}_0$  is therefore the tensor product of an  $\mathcal{L}'_0$ , of finite order, and the inverse image of a Weil sheaf  $\mathcal{L}''_0$  on  $\text{Spec}(\mathbb{F}_q)$ . For  $\mathcal{L}_0$  to be a  $\overline{\mathbb{Q}}_\ell$ -sheaf it is necessary and sufficient that  $\mathcal{L}''_0$  be one.
- (c) Every  $\mathcal{F}_0$  as in (a) is therefore obtained by twisting from a  $\overline{\mathbb{Q}}_\ell$ -sheaf  $\mathcal{F}'_0$  such that  $\det(\mathcal{F}'_0)$  has finite order; and  $\mathcal{F}_0$  is a  $\overline{\mathbb{Q}}_\ell$ -sheaf if and only if the twist is tensor product with the inverse image of a rank-one  $\overline{\mathbb{Q}}_\ell$ -sheaf on  $\text{Spec}(\mathbb{F}_q)$ .

In [L], Lafforgue states the correspondence between (A) and (B) by restricting in (A) to smooth irreducible  $\overline{\mathbb{Q}}_\ell$ -sheaves whose determinant has finite order, and by imposing in (B) the corresponding restriction on automorphic representations. The preceding discussion shows that nothing is lost thereby.

## 0.5. Notation

Let  $\mathcal{F}_0$  be a  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf on  $Z_0$  of finite type over  $\mathbb{F}_q$ , let  $x_0$  be a closed point of  $Z_0$ , and let

$$\deg(x_0) := [k(x_0) : \mathbb{F}_q]$$

be its degree. The choice of  $\bar{x}$  in  $Z$  above  $x_0$  defines an embedding of  $k(x_0)$  into  $k(\bar{x}) = \mathbb{F}$ . The Weil group  $W(\mathbb{F}/k(x_0))$  is generated by the geometric Frobenius  $F_{x_0}$ . It is of index  $\deg(x_0)$  in  $W(\mathbb{F}/\mathbb{F}_q)$ , fixes  $\bar{x}$ , and acts on the fiber  $\mathcal{F}_{\bar{x}}$  of  $\mathcal{F}_0$  at the geometric point  $\bar{x}$ . In particular  $F_{x_0}$  acts on  $\mathcal{F}_{\bar{x}}$ . The local factor at  $x_0$  of the  $L$ -function of  $\mathcal{F}_0$  is

$$\det \left( 1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_{\bar{x}} \right)^{-1}. \quad (0.5.1)$$

It does not depend on  $\bar{x}$ , only on  $x_0$ .

Let  $n \geq 1$  and let  $x$  be an  $\mathbb{F}_{q^n}$ -point of  $Z_0$ :  $x \in Z_0(\mathbb{F}_{q^n})$ . The point  $x$  is an  $\mathbb{F}_q$ -morphism

$$\text{Spec}(\mathbb{F}_{q^n}) \longrightarrow Z_0.$$

Let  $x_0$  be the closed point of  $Z_0$  which is its image, and let  $\bar{x}$  be the geometric point

$$\mathrm{Spec}(\mathbb{F}) \longrightarrow \mathrm{Spec}(\mathbb{F}_{q^n}) \xrightarrow{x} Z_0.$$

We write  $F_x$  for the “geometric Frobenius” generator of  $W(\mathbb{F}/\mathbb{F}_{q^n})$ . We have  $W(\mathbb{F}/\mathbb{F}_{q^n}) \subset W(\mathbb{F}/k(x_0))$ , and

$$F_x = F_{x_0}^{n/\deg(x_0)}. \quad (0.5.2)$$

We will say “the action of  $F_x$  (respectively  $F_{x_0}$ ) on  $\mathcal{F}_0$ ” for “the action of  $F_x$  (respectively  $F_{x_0}$ ) on  $\mathcal{F}_{\bar{x}}$ .” With this abuse of language, the local factor (0.5.1) is written

$$\det(1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_0)^{-1}.$$

It will be convenient for us to consider, rather than these local factors, the traces  $\mathrm{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in some  $Z_0(\mathbb{F}_{q^n})$ . The identity of formal series

$$\log \det(1 - ft, V) = - \sum_{n \geq 1} \mathrm{Tr}(f^n, V) \frac{t^n}{n},$$

for an endomorphism  $f$  of  $V$ , shows that these traces for  $x$  above  $x_0$  and the local factor at  $x_0$  determine one another.

Lafforgue [L] VII 6 proves the following theorem, in which (ii) and (iii) result from the equivalence between (A) and (B) of 0.1.

**Theorem 0.6 (Lafforgue).** Let  $X_0/\mathbb{F}_q$  be as in 0.0, and let  $\mathcal{F}_0$  be a smooth irreducible  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf of rank  $r$  on  $X_0$ , whose determinant  $\det(\mathcal{F}_0)$  has finite order.

- (i) At every closed point  $x_0$  of  $X_0$ , the eigenvalues of  $F_{x_0}$  acting on  $\mathcal{F}_0$  are Weil numbers of weight 0, i.e. are algebraic numbers of absolute value one at every place not dividing  $p$ . In particular  $\mathcal{F}_0$  is pure of weight 0.
- (ii) There exists a finite extension  $E \subset \overline{\mathbb{Q}}_\ell$  of  $\mathbb{Q}$  such that, for every closed point  $x_0$  of  $X_0$ , the polynomial

$$\det(1 - F_{x_0} t, \mathcal{F}_0)$$

has coefficients in  $E$ .

- (iii) For  $E$  as in (ii), for every prime number  $\ell' \neq p$  and every embedding  $\sigma$  of  $E$  into  $\overline{\mathbb{Q}}_{\ell'}$ , there exists a smooth  $\overline{\mathbb{Q}}_{\ell'}$ -Weil sheaf  $\mathcal{F}'_0$  on  $X_0$  such that, at every closed point  $x_0$  of  $X_0$ ,

$$\det(1 - F_{x_0} t, \mathcal{F}'_0) = \sigma \det(1 - F_{x_0} t, \mathcal{F}_0).$$

The minimal extension  $E$  is the field of definition of the automorphic representation  $\pi$  corresponding to  $\mathcal{F}_0$ , viewed either as an isomorphism class of representations of  $\mathrm{GL}(r, A)$  or as a subspace of the space of functions on  $\mathrm{GL}(r, K_0) \backslash \mathrm{GL}(r, A)$ . Here one must use the rationalized correspondence of 0.2.

## 0.7

Let  $Z_0$  be a connected normal scheme of finite type over  $\mathbb{F}_q$  and let  $\mathcal{F}_0$  be a smooth irreducible  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf on  $Z_0$ , whose determinant has finite order. Lafforgue [L] VII 7 shows, by reduction to the case of curves through an argument with plane sections, that assertion (i) of 0.6 remains true for  $\mathcal{F}_0$ . Beware that the reference to Hartshorne which Lafforgue gives for irreducibility of the restriction of  $\mathcal{F}_0$  to a curve  $X_0$  which is a plane section of  $Z_0$  is not adequate: Hartshorne assumes  $Z_0$  projective. We will explain in 1.5–1.9 how to correct the argument, replacing the reference to Hartshorne by a reference to Jouanolou [J].

## 0.8

Let  $\mathcal{F}_0$  and  $Z_0$  be as in 0.7. Using a method developed by Wiesend [W] and Lafforgue's theorem, Drinfeld [Dr] has shown, under the additional hypothesis that  $Z_0$  is smooth, that if  $\mathcal{F}_0$  satisfies assertion (ii) of the theorem then  $\mathcal{F}_0$  also satisfies (iii). Our aim is to show that every  $\mathcal{F}_0$  as in 0.7 satisfies (ii).

## 0.9

Let us review the sections of the present article. In Section 1, we consider smooth semisimple  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves on  $Z_0$  normal absolutely irreducible of finite type over  $\mathbb{F}_q$ . We show how to describe them in terms of smooth  $\overline{\mathbb{Q}}_\ell$ -sheaves on schemes  $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^a}$ , with determinant of finite order, and whose inverse image on  $Z = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$  is irreducible. The decomposition (1.4.1) obtained translates (0.3), together with well-known results on the relation between irreducible representations of a group  $G$  and of an extension of  $\mathbb{Z}$  by  $G$ .

In Section 2, we consider a smooth  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf  $\mathcal{F}_0$  on a smooth absolutely irreducible curve  $X_0$  over  $\mathbb{F}_q$ . For simplicity, we assume  $X_0$  affine. The “complexity” of  $X_0$  will be measured by the integer

$$b_1(X) := \dim H_c^1(X, \overline{\mathbb{Q}}_\ell).$$

We assume  $\mathcal{F}_0$  algebraic, i.e. that the traces  $\mathrm{Tr}(F_x, \mathcal{F}_0)$ , for  $x$  in some  $X_0(\mathbb{F}_{q^n})$ , are algebraic numbers. By 0.6(ii), applied to twists of irreducible subquotients of  $\mathcal{F}_0$ , we know that these traces generate a finite extension  $E$  of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}}_\ell$ . Our aim is to make explicit an integer  $N$ , depending on a “complexity” of  $(X, \mathcal{F})$ , such that  $E$  is generated by the  $\mathrm{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in  $X_0(\mathbb{F}_{q^n})$  with  $n \leq N$ . Put  $\log_q^+(a) = \sup(0, \log_q(a))$ . If  $\mathcal{F}$  is tamely ramified, one obtains  $N$  of the form

$$O(\log_q^+(b_1(X))) + O(1), \tag{0.9.1}$$

where the implicit constants in the  $O$  depend on the rank of  $\mathcal{F}_0$ . For arbitrary  $\mathcal{F}_0$ , one must replace  $b_1(X)$  by

$$b_1(X) + \sum_{s \in S} \alpha_s(\mathcal{F}),$$

where  $\alpha_s(\mathcal{F})$  measures the wildness of the ramification of  $\mathcal{F}$  at  $s$ .

Another way of measuring the wildness of  $\mathcal{F}$ , much less economical but convenient in Section 3, is in terms of a connected étale cover  $X'$  of  $X$  on which the inverse image of  $\mathcal{F}$  is tamely ramified. We will obtain another  $N$ , of the form

$$O(\log_q^+(b_1(X'))) + O(1). \tag{0.9.2}$$

Here again, the implicit constants in the  $O$  depend on the rank of  $\mathcal{F}_0$ .

The proofs make essential use of 0.6(iii), in order to reduce the posed problem to the following one. For  $\mathcal{F}_0$  and  $\mathcal{G}_0$  smooth semisimple Weil sheaves on  $X_0$ , find  $N$  such that if

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \mathcal{G}_0)$$

for all  $x$  in  $X(\mathbb{F}_{q^n})$  with  $n \leq N$ , then  $\mathcal{F}_0$  and  $\mathcal{G}_0$  are isomorphic.

## 0.10

The main theorem is proved in Section 3. After a few devissages, one is reduced to the case in which  $\mathcal{F}_0$  is a smooth  $\overline{\mathbb{Q}}_\ell$ -sheaf on a scheme  $Z_0$  which is smooth and absolutely irreducible over  $\mathbb{F}_q$ , and in which  $\mathcal{F}_0$  is algebraic, meaning algebraicity of Frobenius traces. The point is to show that the extension of  $\mathbb{Q}$  generated by Frobenius traces is a finite extension. This is done by showing that there exists an integer  $N$  such that, for every integer  $n > N$  and every point

$x \in Z_0(\mathbb{F}_{q^n})$ , the trace  $\text{Tr}(F_x, \mathcal{F}_0)$  is contained in the extension of  $\mathbb{Q}$  generated by the  $\text{Tr}(F_{x'}, \mathcal{F}_0)$  with  $x'$  in some  $Z_0(\mathbb{F}_{q^{n'}})$  and  $n' < n$ . To prove this, one passes through  $x$  a curve  $X_0$  such that  $(X_0, \mathcal{F}_0|_{X_0})$  is not too complex, and one applies to this curve the results of Section 2. More precisely, one constructs a smooth affine curve  $X_0$ , a point  $\tilde{x} \in X_0(\mathbb{F}_{q^n})$ , and an  $\mathbb{F}_q$ -morphism  $\varphi : X_0 \rightarrow Z_0$  sending  $\tilde{x}$  to  $x$ . The construction does not guarantee that the curve  $X_0$  is connected. Replace it by its connected component  $X_1$  such that  $\tilde{x} \in X_1(\mathbb{F}_{q^n})$ . The curve  $X_1$  might not be absolutely irreducible over  $\mathbb{F}_q$ , but there is a bound  $D$ , depending only on  $Z_0$ , such that  $X_1$  is absolutely irreducible over some  $\mathbb{F}_{q^d}$  with  $d \leq D$ . It is to  $X_1/\mathbb{F}_{q^d}$  that the results of Section 2 will be applied. To simplify the exposition of the strategy, we pretend here, falsely, that  $X_0$  is absolutely irreducible over  $\mathbb{F}_q$ .

Choose a connected étale cover  $Z'$  of  $Z$  on which  $\mathcal{F}$  becomes tamely ramified. This is an enormous waste, but is done only once. Let  $X'$  be a connected component of the inverse image of  $Z'$  on  $X$ . The complexity of  $(X_0, \varphi^* \mathcal{F}_0)$  will be measured by  $b_1(X')$ . We will control it by Katz's theorem [K]. Since only curves are involved here, one could instead use Bombieri [B], on which Katz bases himself, and estimate the number of connected components by the product of the degrees of equations of  $X'$ , but this would only repeat in a simple case Katz's argument. One obtains a bound for the complexity which is polynomial in  $n$ . As soon as  $n$  is large enough, the integer  $N$  of (0.9.2) satisfies  $N < n$ , and

$$\text{Tr}(F_x, \mathcal{F}_0) = \text{Tr}(F_{\tilde{x}}, \varphi^* \mathcal{F}_0)$$

is in the extension of  $\mathbb{Q}$  generated by the

$$\text{Tr}(F_{x'}, \varphi^* \mathcal{F}_0) = \text{Tr}(F_{\varphi(x')}, \mathcal{F}_0)$$

for  $x' \in X_0(\mathbb{F}_{q^{n'}})$  and  $n' \leq N < n$ .

I thank H. Esnault for a careful reading of the text and for her suggestions, which I hope have made it more readable.

## 1. Irreducibility and extensions of the base field

### 1.1

Let  $Z_0$  be a normal absolutely irreducible scheme of finite type over  $\mathbb{F}_q$ , let  $Z := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ , and let  $x$  be a geometric point of  $Z$ . It defines a geometric point, still denoted  $x$ , of  $Z_0$ . The fundamental group of  $Z_0$  is an extension

$$1 \rightarrow \pi_1(Z, x) \rightarrow \pi_1(Z_0, x) \rightarrow \text{Gal}(\mathbb{F}/\mathbb{F}_q) \rightarrow 1.$$

Taking the fiber product with  $W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \subset \text{Gal}(\mathbb{F}/\mathbb{F}_q) = \widehat{\mathbb{Z}}$ , one obtains the Weil group  $W(Z_0, x)$ :

$$1 \rightarrow \pi_1(Z, x) \rightarrow W(Z_0, x) \rightarrow \mathbb{Z} \rightarrow 1.$$

The functor “fiber at  $x$ ” is an equivalence from the category of smooth  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves on  $Z_0$  to that of continuous linear representations of  $W(Z_0, x)$  on a finite-dimensional  $\overline{\mathbb{Q}}_\ell$ -vector space.

### 1.2

If  $V$  is an irreducible  $\overline{\mathbb{Q}}_\ell$ -representation of  $W(Z_0, x)$ , its restriction to  $\pi_1(Z, x)$  is a sum of pairwise nonisomorphic irreducible representations of  $\pi_1(Z, x)$ , permuted transitively by  $W(Z_0, x)/\pi_1(Z, x) = \mathbb{Z}$ . Let  $n$  be the number of summands, and put  $Z_1 := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}$ ;  $W(Z_1, x)$  is the inverse image in  $W(Z_0, x)$  of  $n\mathbb{Z}$ . If  $S$  is one of the irreducible summands of the restriction of  $V$  to  $\pi_1(Z, x)$ , then  $S$  is a representation of  $W(Z_1, x)$  and

$$V = \text{Ind}_{W(Z_1, x)}^{W(Z_0, x)}(S).$$



If  $\mathcal{V}$  (respectively  $\mathcal{S}$ ) is the  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf corresponding to  $V$  (respectively  $S$ ), then  $\mathcal{V}$  is the direct image, under  $Z_1 \rightarrow Z_0$ , of  $\mathcal{S}$ .

### 1.3

Now let  $V$  be a semisimple representation of  $W(Z_0, x)$ . The quotient  $\mathbb{Z}$  of  $W(Z_0, x)$  permutes the isomorphism classes of simple constituents of the restriction of  $V$  to  $\pi_1(Z, x)$ . Let  $A$  be the set of orbits, and in each orbit  $a$  choose a representative  $S'_a$ ; let  $n(a)$  be the number of elements of the orbit. If  $Z_a = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^{n(a)}}$ , the representation  $S'_a$  extends to a representation  $S_a$  of  $W(Z_a, x)$ , and  $S_a$  is unique up to twist by a character of the quotient  $\mathbb{Z}$  of  $W(Z_a, x)$ . We choose the twist so that the character  $\det(S_a)$  of  $W(Z_a, x)$  has finite order.

Apply 1.2 to the irreducible constituents of  $V$ , and collect the occurrences of  $S_a$ . One obtains a decomposition

$$V = \bigoplus_a \operatorname{Ind}_{W(Z_a, x)}^{W(Z_0, x)} (S_a \otimes W_a), \quad (1.3.1)$$

where  $W_a$  is a representation of  $W(Z_a, x)$  trivial on  $\pi_1(Z, x)$ . This description of  $V$  is unique, except that one may

- a) twist  $S_a$  by a finite-order character of  $\mathbb{Z} = W(Z_a, x)/\pi_1(Z, x)$  and  $W_a$  by the inverse character;
- b) replace  $S_a$  by its conjugate  $S_a^{(i)}$  by an element  $i$  of  $W(Z_0, x)/W(Z_a, x) = \mathbb{Z}/n(a)$ .

### 1.4

Passing from the language of group representations to that of sheaves:  $S_a$  (respectively  $S_a^{(i)}$ ) corresponds to  $\mathcal{S}_{a,1}$  (respectively  $\mathcal{S}_{a,1}^{(i)}$ ) on  $Z_a$ ,  $W_a$  to  $\mathcal{W}_a$  on  $\operatorname{Spec}(\mathbb{F}_{q^{n(a)}})$ , and, if  $p_a$  is the projection from  $Z_a$  to  $Z_0$ , and  $\operatorname{pr}_a$  the projection from  $Z_a$  to  $\operatorname{Spec}(\mathbb{F}_{q^{n(a)}})$ , (1.3.1) becomes a decomposition of the  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf  $\mathcal{V}$  corresponding to  $V$ :

$$\mathcal{V} = \bigoplus_{a \in A} p_{a*} (\mathcal{S}_a \otimes \operatorname{pr}_a^* \mathcal{W}_a). \quad (1.4.1)$$

In this decomposition, the  $\det(\mathcal{S}_a)$  have finite order, and the inverse images  $\mathcal{S}_a^{(i)}$  on  $Z$  of the  $\mathcal{S}_{a,1}^{(i)}$  ( $a \in A$ ,  $i \in \mathbb{Z}/n(a)$ ) are irreducible pairwise nonisomorphic  $\overline{\mathbb{Q}}_\ell$ -sheaves.

### 1.5

Let  $\mathcal{F}_0$  be smooth irreducible of finite-order determinant on  $Z_0$  as in 1.1. By 1.2, the inverse image  $\mathcal{F}$  of  $\mathcal{F}_0$  on  $Z := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$  is a direct sum of nonisomorphic smooth irreducible sheaves  $\mathcal{S}^{(i)}$ , permuted transitively by  $\mathbb{Z} = W(\mathbb{F}/\mathbb{F}_q)$ . Let  $n$  be the number of the  $\mathcal{S}^{(i)}$ . In the remainder of this section, we give a proof of the following statement of Lafforgue [L] VII 7, cf. 0.7.

**Theorem 1.6 (Lafforgue).** At every closed point  $x_0$  of  $Z_0$ , the eigenvalues of  $F_{x_0}$  acting on  $\mathcal{F}_0$  are Weil numbers of weight 0.

If  $U_0$  is a nonempty open subset of  $Z_0$ , the restriction of  $\mathcal{F}_0$  to  $U_0$  is still irreducible. Covering  $Z_0$  by affine opens, one reduces to the case where, and we will assume that,  $Z_0$  is affine.

**Lemma 1.7.** There exists a connected Galois étale cover  $Z'$  of  $Z$  such that if the inverse image  $Y'$  of  $Z'$  by a morphism  $f : Y \rightarrow Z$  is connected, then  $\mathcal{F}$  and  $f^* \mathcal{F}$  have the same monodromy.

Fix a base point  $a$  of  $Y$ , and still denote by  $a$  its image in  $Z$ . There is a morphism

$$\pi_1(Y, a) \longrightarrow \pi_1(Z, a),$$

and the conclusion “same monodromy” means that  $\pi_1(Y, a)$  maps onto the image of  $\pi_1(Z, a)$  in  $\operatorname{GL}(\mathcal{F}_a)$ . It implies that the  $f^* \mathcal{S}^{(i)}$  remain smooth irreducible and pairwise nonisomorphic.

**Proof of 1.7.** The monodromy group  $M$ , the image of  $\pi_1(Z, a)$  in  $\mathrm{GL}(\mathcal{F}_a)$ , is a closed subgroup of a group  $\mathrm{GL}(n, \mathcal{O}_\lambda)$ , hence is a compact  $\ell$ -adic Lie group. It therefore admits an open normal pro- $\ell$  subgroup  $H$ , and the intersection  $H'$  of the kernels of the homomorphisms  $H \rightarrow \mathbb{Z}/\ell$  is again an open normal subgroup. Take  $Z'$  to be the cover corresponding to  $H'$ . Suppose  $Y'$  connected. This is equivalent to saying that  $\pi_1(Y, a)$  maps onto  $M/H'$ . Since every closed subgroup of  $H$  which maps onto  $H/H'$  coincides with  $H$ ,  $\pi_1(Y, a)$  maps onto  $M$ .

## 1.8

Choose a closed immersion of  $Z_0$  into an affine space  $E_0$  over  $\mathbb{F}_q$ , and choose  $Z'$  as in 1.7. Let  $G_0$  be the  $\mathbb{F}_q$ -scheme parametrizing affine linear subspaces of codimension  $\dim(Z_0) - 1$  in  $E_0$ . Extend the base field from  $\mathbb{F}_q$  to  $\mathbb{F}$  and apply to the composite morphism

$$f : Z' \rightarrow Z \hookrightarrow E$$

the Bertini theorem in the form given by Jouanolou [J], Th. 6.3(iv), p. 67. One obtains a nonempty open  $U$  of the smooth irreducible scheme  $G = G_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ , such that for every  $u \in U(\mathbb{F})$ , corresponding to an affine linear subspace  $L_u$  of  $E$ , the inverse image  $f^{-1}(L_u)$  of  $Z'$  on  $Z \cap L_u$  is irreducible. Note that this irreducibility implies that of  $Z \cap L_u$ . The set of  $u$  such that  $Z \cap L_u$  is not a smooth curve has dimension strictly smaller than that of  $U$ . Thus, after shrinking  $U$ , one may suppose that the  $Z \cap L_u$  are smooth curves. After further shrinking  $U$ , one may also suppose that  $U$  is of the form  $U_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ , for  $U_0$  an open subset of  $G_0$ .

The number of  $\mathbb{F}_{q^m}$ -points of  $U_0$  is, as  $m \rightarrow \infty$ , of the form

$$q^{\dim(U)m} \left( 1 + O(q^{-1/2}) \right)$$

(Lang-Weil, or the trace formula plus the Weil conjecture). As soon as  $m$  is large enough,  $U_0(\mathbb{F}_{q^m})$  is therefore nonempty. This allows one to choose  $m$  prime to  $n$  such that  $U_0(\mathbb{F}_{q^m})$  is nonempty.

For  $\mathcal{F}_0$  on  $Z_0$  to satisfy the conclusion of 1.6, it is necessary and sufficient that this conclusion be satisfied by the inverse image of  $\mathcal{F}_0$  on  $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ . Replacing  $\mathbb{F}_q$  by  $\mathbb{F}_{q^m}$ ,  $Z_0$  by  $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$ , and  $\mathcal{F}_0$  by its inverse image, one is reduced to proving 1.6 under the additional hypothesis that  $U_0$  contains an  $\mathbb{F}_q$ -point  $u_0$ . The plane section  $Z_0 \cap L_{u_0}$  of  $Z_0$  is then a smooth absolutely irreducible curve  $Y_0$ . Extend the base field to  $\mathbb{F}$ . Since  $u_0$  is in  $U_0$ , 1.7 ensures that the restrictions of the  $\mathcal{S}^{(i)}$  to  $Y$  are irreducible and pairwise nonisomorphic. Because  $m$  is prime to  $n$ , they are permuted by  $W(\mathbb{F}/\mathbb{F}_q)$ : the restriction of  $\mathcal{F}_0$  to  $Y_0$  is irreducible. Apply 0.6(i) to it: for every closed point  $y_0$  of  $Y_0 \subset Z_0$ , the eigenvalues of  $F_{y_0}$  acting on  $\mathcal{F}_0$  are Weil numbers of weight 0. One concludes by the following proposition.

**Proposition 1.9.** Let  $\mathcal{F}_0$  be a smooth  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf on a connected scheme  $X_0$  over  $\mathbb{F}_q$ . The following conditions are equivalent:

- (i) there exists a closed point  $x_0$  of  $X_0$  such that the eigenvalues of  $F_{x_0}$  acting on  $\mathcal{F}_0$  are Weil numbers of weight 0;
- (ii) the same assertion holds at all closed points of  $X_0$ .

**Proof, for  $X_0$  a smooth connected curve.** Suppose that  $\mathcal{F}_0$  satisfies (i). Every irreducible subquotient  $\mathcal{F}'_0$  still satisfies (i), and it suffices to verify (ii) for each  $\mathcal{F}'_0$ . Let  $\mathcal{F}''_0$  be a twist of  $\mathcal{F}'_0$  whose determinant has finite order. By 0.6(i),  $\mathcal{F}''_0$  satisfies (ii). Testing at  $x_0$ , one sees that the twist is by a Weil number of weight 0, and the validity of (ii) is invariant under such a twist.

**Proof, general case.** For any closed point  $x'_0$  of  $X_0$ , there is a sequence of smooth connected curves  $Y_0^j$  and morphisms  $f^j : Y_0^j \rightarrow X_0$  ( $1 \leq j \leq N$ ) such that  $x_0$  is in the image of  $Y_0^1$ ,  $x'_0$  is in the image of  $Y_0^N$ , and the images of  $Y_0^j$  and  $Y_0^{j+1}$  intersect nontrivially for  $1 \leq j < N$ . If (i) holds at  $x_0$ , the first part of the proof shows that (ii) holds for the inverse image of  $\mathcal{F}_0$  on  $Y_0^1$ , hence that (i) holds at every closed point of  $X_0$  in the image of  $Y_0$ . Similarly, if (i) holds at the closed points in  $f(Y_j)$ , and hence at at least one closed point of  $f(Y_{j+1})$ , then (i) holds at the closed points in  $f(Y_{j+1})$ , and one eventually reaches  $x'_0$ .

## 2. The case of curves: quantitative version

### 2.1

Let  $X$  be a smooth connected curve over an algebraically closed field of characteristic  $p$ , let  $\overline{X}$  be the smooth projective completion of  $X$ , and let  $S$  be the set of points at infinity:  $X = \overline{X} - S$ . Let  $\mathcal{F}$  be a smooth  $\overline{\mathbb{Q}}_\ell$ -sheaf of rank  $r$  on  $X$ . Put

$$\chi_c(X, \mathcal{F}) := \sum (-1)^i \dim H_c^i(X, \mathcal{F}).$$

This Euler-Poincare characteristic is given by the Grothendieck-Ogg-Shafarevich formula

$$\chi_c(X, \mathcal{F}) = r\chi_c(X) - \sum_{s \in S} \text{Sw}_s(\mathcal{F}). \quad (2.1.1)$$

If the curve  $\overline{X}$  has genus  $g$ , then

$$\chi_c(X) = 2 - 2g - |S|. \quad (2.1.2)$$

The term  $\text{Sw}_s(\mathcal{F})$  is the Swan conductor of  $\mathcal{F}$  at  $s$ .

Let  $K_s$  be the fraction field of the henselization of  $X$  at  $s$ . The reader who prefers may replace “henselization” by “completion.” This does not change  $\text{Gal}(\overline{K}_s/K_s)$ , for  $\overline{K}_s$  an algebraic closure of  $K_s$ . The  $\overline{\mathbb{Q}}_\ell$ -sheaves on  $\text{Spec}(K_s)$  are identified with continuous  $\overline{\mathbb{Q}}_\ell$ -linear representations  $V$  of  $\text{Gal}(\overline{K}_s/K_s)$ . The conductor  $\text{Sw}_s(\mathcal{F})$  is the Swan conductor  $\text{Sw}(V)$  of the linear representation  $V$  of  $\text{Gal}(\overline{K}_s/K_s)$  corresponding to the inverse image of  $\mathcal{F}$  on  $\text{Spec}(K_s)$ .

The conductor  $\text{Sw}(V)$  is additive in  $V$ , for short exact sequences. Let  $\text{Gal}(\overline{K}_s/K_s)^\beta$  denote the ramification subgroups of  $\text{Gal}(\overline{K}_s/K_s)$  in upper numbering. Define  $\alpha(V)$  as the infimum of the  $\beta > 0$  such that  $\text{Gal}(\overline{K}_s/K_s)^\beta$  acts trivially on  $V$ . If the representation  $V$  is irreducible, then

$$\text{Sw}(V) = \dim(V) \alpha(V) \quad (V \text{ irreducible}) \quad (2.1.3)$$

(a consequence of Serre [S] 19.1 and of the definition of the Herbrand functions relating the lower numbering to the upper numbering). If  $V$  comes from  $\mathcal{F}$  on  $X$ , we put  $\alpha_s(\mathcal{F}) := \alpha(V)$ .

It follows from (2.1.3) by additivity that

$$\text{Sw}(V) \leq \dim(V) \alpha(V). \quad (2.1.4)$$

Let  $P$  be the pro- $p$  Sylow subgroup of  $\text{Gal}(\overline{K}_s/K_s)$ . Since  $V$  is  $\ell$ -adic, it acts on  $V$  through a finite quotient  $\overline{P}$ . Both  $\text{Sw}(V)$  and  $\alpha(V)$  vanish if and only if  $V$  is tame, i.e. if  $P$  acts trivially. Otherwise,  $\alpha(V)$  is the largest rational number such that  $\overline{P}^{\alpha(V)}$  is nontrivial. The importance for us of the invariant  $\alpha(V)$  comes from the observation that

$$\alpha(V \otimes W) \leq \sup(\alpha(V), \alpha(W)). \quad (2.1.5)$$

Poincare duality places  $H_c^2(X, \mathcal{F})$  and  $H^0(X, \mathcal{F}^\vee(1))$  in perfect duality. In particular,  $H_c^2(X, \mathcal{F})$  has rank at most  $r$ , and has rank  $r$  if  $\mathcal{F}$  is constant. If the curve  $X$  is affine, then  $H_c^0(X, \mathcal{F}) = 0$ , and (2.1.1), (2.1.4) give

$$\dim H_c^1(X, \mathcal{F}) \leq r \left( b_1(X) + \sum \alpha_s(\mathcal{F}) \right) \quad (X \text{ affine}), \quad (2.1.6)$$

where we have put

$$b_1(X) = \dim H_c^1(X, \overline{\mathbb{Q}}_\ell) = 2g + |S| - 1 \quad (X \text{ affine}). \quad (2.1.7)$$

## 2.2

Let  $X_0 = \overline{X}_0 - S_0$  be a curve over  $\mathbb{F}_q$  as in 0.0, and let  $\mathcal{F}_0$  be a smooth  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf of rank  $r$  on  $X_0$ . Assume  $\mathcal{F}_0$  algebraic, in the sense of 0.9. We do not assume  $\mathcal{F}_0$  irreducible. Let  $E$  be a Galois extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}}_\ell$  containing the  $\text{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in any  $X_0(\mathbb{F}_{q^n})$ ,  $n \geq 1$ . By 0.6, for every  $\sigma \in \text{Gal}(E/\mathbb{Q})$ , there is a smooth semisimple  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf  $\sigma(\mathcal{F}_0)$  such that, for every  $x$  in  $X_0(\mathbb{F}_{q^n})$ ,

$$\text{Tr}(F_x, \sigma(\mathcal{F}_0)) = \sigma(\text{Tr}(F_x, \mathcal{F}_0)).$$

We write  $\sigma(\mathcal{F})$  for the inverse image of  $\sigma(\mathcal{F}_0)$  on  $X$ . This notation is reasonable because  $\sigma(\mathcal{F})$ , which is defined up to isomorphism, depends only on  $\mathcal{F}$ ; this follows from (1.4.1) and a compatibility between “ $\sigma$ ” and twisting. For each  $s \in S$ , one has

$$\text{Sw}_s(\sigma\mathcal{F}) = \text{Sw}_s(\mathcal{F}), \quad \alpha_s(\sigma\mathcal{F}) = \alpha_s(\mathcal{F}). \quad (2.2.1)$$

**Sketch of proof.** One may suppose  $\mathcal{F}_0$  irreducible. If  $\mathcal{F}_0$  corresponds to an automorphic representation  $\pi$ , by the rationalized correspondence, then  $\pi$  is defined over  $E$  and  $\sigma\mathcal{F}_0$  corresponds to  $\sigma\pi$ . At each  $s \in S_0$ ,  $\mathcal{F}_0$  defines a representation of the local Weil group. By the local correspondence, its Frobenius-semisimplification ([De1] 8.6) corresponds to the local component  $\pi_s$  of  $\pi$ . Note that isomorphism classes of continuous  $\overline{\mathbb{Q}}_\ell$ -linear  $F$ -semisimple representations of the local Weil group are in bijection with isomorphism classes of  $\overline{\mathbb{Q}}_\ell$ -linear  $F$ -semisimple representations, loc. cit., of the Weil-Deligne group. The latter are purely algebraic objects. If  $\pi_s$  is defined over  $E$ , then the corresponding representation will be also, and conjugation by  $\sigma$  is compatible with the rationalized correspondence. In particular,  $\mathcal{F}_0$  and  $\sigma(\mathcal{F}_0)$  give rise to representations of the wild inertia group at  $s$  having the same kernel, and one is reduced to (2.1.3).

For our main result, the following more elementary lemma suffices.

**Lemma 2.3.** If  $\mathcal{F}$  is tamely ramified, then  $\sigma(\mathcal{F})$  is also tamely ramified.

**Proof.** The image under  $\sigma$  of the  $L$ -function of  $\mathcal{F}_0$ ,

$$L(X_0, \mathcal{F}_0, t) = \prod_{x \in |X_0|} \det \left( 1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_0 \right)^{-1},$$

is the  $L$ -function of  $\sigma(\mathcal{F}_0)$ . This  $L$ -function is a rational function, and by Grothendieck’s cohomological interpretation of  $L$ -functions the order of its pole at  $t = \infty$  is  $-\chi_c(\mathcal{F})$ . Thus  $\chi_c(\mathcal{F}) = \chi_c(\sigma\mathcal{F})$ . Since  $\mathcal{F}$  is tamely ramified, the  $\text{Sw}_s(\mathcal{F})$  are zero. Comparing (2.1.1) for  $\mathcal{F}$  and  $\sigma\mathcal{F}$ , one sees that the sum of the  $\text{Sw}_s(\sigma\mathcal{F})$ , hence each of them, is zero, i.e. that  $\sigma(\mathcal{F})$  is tamely ramified.

## 2.4

Suppose  $X_0$  affine, and let  $\mathcal{F}_0, \mathcal{G}_0$  be two smooth semisimple Weil sheaves of rank  $r$  on  $X_0$ . Write  $\log_q^+$  for the function  $\sup(0, \log_q)$  and  $[ ]$  for the integer-part function. Put

$$\alpha_s(\mathcal{F}, \mathcal{G}) := \alpha_s(\mathcal{F} \oplus \mathcal{G}) = \sup(\alpha_s(\mathcal{F}), \alpha_s(\mathcal{G})), \quad (2.4.1)$$

$$N_0 = 2 \log_q^+ \left( 2r^2 \left( b_1(X) + \sum_{s \in S} \alpha_s(\mathcal{F}, \mathcal{G}) \right) \right), \quad (2.4.2)$$

$$N = [N_0] + 2r. \quad (2.4.3)$$

**Proposition 2.5.** If for every integer  $n \leq N$  and every  $x \in X(\mathbb{F}_{q^n})$  one has

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \mathcal{G}_0),$$

then  $\mathcal{F}_0$  is isomorphic to  $\mathcal{G}_0$ .

**Proof.** Apply (1.3), (1.4) to  $\mathcal{F}_0 \oplus \mathcal{G}_0$ . One obtains decompositions (1.4.1)

$$\mathcal{F}_0 = \bigoplus_{a \in A} p_{a*}(\mathcal{S}_{a,1} \otimes \mathrm{pr}_a^* \mathcal{W}_a), \quad (2.5.1)$$

$$\mathcal{G}_0 = \bigoplus_{a \in A} p_{a*}(\mathcal{S}_{a,1} \otimes \mathrm{pr}_a^* \mathcal{W}'_a). \quad (1)$$

In (2.5.1), to each  $a \in A$  is attached an integer  $n(a) \geq 1$ ;  $\mathcal{S}_{a,1}$  is a smooth  $\overline{\mathbb{Q}}_\ell$ -Weil sheaf on  $X_a := X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^{n(a)}}$ ;  $\mathcal{W}_a$  and  $\mathcal{W}'_a$  are  $\overline{\mathbb{Q}}_\ell$ -Weil sheaves on  $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$ ; and  $p_a$  and  $\mathrm{pr}_a$  are the projections from  $X_a$  to  $X_0$  and to  $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$ . Let  $\mathcal{S}_{a,1}^{(i)}$  denote the image of  $\mathcal{S}_{a,1}$  by the  $i$ -th power of the Frobenius  $F \in \mathrm{Gal}(\mathbb{F}_{q^{n(a)}}/\mathbb{F}_q)$ , and omit the subscript 1 to indicate the inverse image from  $X_a$  to  $X$ . By (1.4), the  $\mathcal{S}_{a,1}$ ,  $\mathcal{W}_a$ , and  $\mathcal{W}'_a$  in (2.5.1) satisfy

- (i)  $\det(\mathcal{S}_{a,1})$  has finite order;
- (ii) the  $\mathcal{S}_a^{(i)}$  ( $a \in A$ ,  $i \in \mathbb{Z}/n(a)$ ) are smooth irreducible  $\overline{\mathbb{Q}}_\ell$ -sheaves on  $X$ , pairwise nonisomorphic;
- (iii) for each  $a$ ,  $\mathcal{W}_a$  or  $\mathcal{W}'_a$  is nonzero.

Since  $\mathcal{S}_a^{(i)}$  is a direct factor of  $\mathcal{F}$  or  $\mathcal{G}$ , one has at every point  $s \in S$

$$\alpha_s(\mathcal{S}_a^{(i)}) \leq \sup(\alpha_s(\mathcal{F}), \alpha_s(\mathcal{G})).$$

Let  $A(n)$  denote the set of  $a$  in  $A$  such that  $n(a) \mid n$ .

**Lemma 2.6.** If  $n > N_0$  (2.4.2), the functions on  $X_0(\mathbb{F}_{q^n})$

$$t_{a,i} : x \longmapsto \mathrm{Tr}(F_x, \mathcal{S}_{a,1}^{(i)}) \quad (a \in A(n), i \in \mathbb{Z}/n(a))$$

are linearly independent.

**Proof.** The functions  $t_{a,i}$  take values in a number field  $E \subset \overline{\mathbb{Q}}_\ell$ . Embed  $E$  into  $\mathbb{C}$  to treat them as complex-valued functions. The idea of the proof is to show that they are nearly orthogonal in  $L^2(X_0(\mathbb{F}_{q^n}))$ .

By 0.6(i),  $\mathcal{S}_{a,1}^{(i)}$  is pure of weight 0. The complex conjugate function of  $t_{a,i}$  is therefore

$$x \longmapsto \mathrm{Tr}(F_x, (\mathcal{S}_{a,1}^{(i)})^\vee),$$

and the scalar product  $\langle t_{b,j}, t_{a,i} \rangle = \sum \overline{t_{b,j}(x)} t_{a,i}(x)$  is

$$\langle t_{b,j}, t_{a,i} \rangle = \sum \mathrm{Tr}(F_x, \mathcal{H}om(\mathcal{S}_{a,1}^{(i)}, \mathcal{S}_{b,1}^{(j)})).$$

By the trace formula, this sum is the trace of the Frobenius  $F \in W(\mathbb{F}/\mathbb{F}_{q^n})$ :

$$\langle t_{b,j}, t_{a,i} \rangle = \sum (-1)^k \mathrm{Tr} \left( F, H_c^k \left( X, \mathcal{H}om(\mathcal{S}_a^{(i)}, \mathcal{S}_b^{(j)}) \right) \right).$$

Since  $X$  is affine,  $H_c^0$  is zero. The “dominant” term is given by  $H_c^2$ : it is  $q^n$  if  $(a,i) = (b,j)$  and zero otherwise. The promised almost orthogonality comes from the fact that the term  $k = 1$  is

$O(q^{n/2})$  for large  $n$ . More precisely, since  $\mathcal{H}om(\mathcal{S}_{a,1}^{(i)}, \mathcal{S}_{b,1}^{(j)})$  is pure of weight 0, the eigenvalues of  $F$  on its  $H_c^1$  have absolute value  $q^{n/2}$  or 1, and the term  $k = 1$  is bounded in absolute value by  $q^{n/2}$  times

$$\dim \mathcal{S}_c^1 \leq \dim \mathcal{S}_b^{(j)} \cdot \dim \mathcal{S}_a^{(i)} \cdot \left( b_1(X) + \sum \alpha_s(\mathcal{F}, \mathcal{G}) \right).$$

Suppose that there is a linear dependence relation  $\sum \lambda_{b,j} t_{b,j} = 0$ . Choose  $a, i$  such that  $|\lambda_{a,i}|$  is maximal among the  $|\lambda_{b,j}|$ . Dividing by  $\lambda_{a,i}$ , one may suppose  $\lambda_{a,i} = 1$  and  $|\lambda_{b,j}| \leq 1$  for  $b \in A(n)$  and  $j \in \mathbb{Z}/n(b)$ . Then

$$0 = \left\langle \sum \lambda_{b,j} t_{b,j}, t_{a,i} \right\rangle = \sum_{b,j} \lambda_{b,j} \langle t_{b,j}, t_{a,i} \rangle = q^n + \text{remainder},$$

where the absolute value of the remainder is bounded by

$$\sum_{b,j} |\lambda_{b,j}| \left| \text{Tr} \left( F, H_c^1(X, \mathcal{H}om(\mathcal{S}_a^{(i)}, \mathcal{S}_b^{(j)})) \right) \right| \leq q^{n/2} 2r^2 \left( b_1(X) + \sum \alpha_s(\mathcal{F}, \mathcal{G}) \right).$$

The hypothesis on  $n$  ensures  $|\text{remainder}| < q^n$ : contradiction.

**Corollary 2.7.** Let  $F$  denote the Frobenius of  $W(\mathbb{F}/\mathbb{F}_{q^n})$ . If  $n > N_0$  and  $\text{Tr}(F_x, \mathcal{F}_0) = \text{Tr}(F_x, \mathcal{G}_0)$  for  $x \in X_0(\mathbb{F}_{q^n})$ , then

$$\text{Tr}(F, \mathcal{W}_a) = \text{Tr}(F, \mathcal{W}'_a)$$

for each  $a \in A(n)$ .

**Proof.** The decompositions (2.5.1) give the identity between trace functions

$$\sum t_{a,i} \text{Tr}(F, \mathcal{W}_a) = \sum t_{a,i} \text{Tr}(F, \mathcal{W}'_a),$$

and one applies 2.6.

## 2.8. End of the proof of 2.5

It remains to show that, for each  $a$ , and for  $F$  the geometric Frobenius generator of  $W(\mathbb{F}/\mathbb{F}_{q^{n(a)}})$ ,  $F$  has the same multiset of eigenvalues on  $\mathcal{W}_a$  and  $\mathcal{W}'_a$ . By 2.7, if  $n$  is divisible by  $n(a)$  and

$$[N_0] + 1 \leq n \leq N = [N_0] + 2r,$$

then

$$\text{Tr}(F^{n/n(a)}, \mathcal{W}_a) = \text{Tr}(F^{n/n(a)}, \mathcal{W}'_a).$$

There are at least  $[2r/a]$  such values of  $n$ ;  $\mathcal{W}_a$  and  $\mathcal{W}'_a$  are of dimension at most  $[r/a]$ ; it remains to apply the following lemma.

**Lemma 2.9.** Let  $\alpha_i$  be  $k$  distinct nonzero numbers. If the  $\lambda_i$  satisfy, for a suitable  $m$ ,

$$\sum \lambda_i \alpha_i^r = 0 \quad \text{for } m \leq r < m + k, \tag{2.9.1}$$

then all the  $\lambda_i$  are zero.

**Proof.** Equation (2.9.1) can be rewritten

$$\sum (\lambda_i \alpha_i^m) \alpha_i^s = 0 \quad \text{for } 0 \leq s < k,$$

and the Vandermonde determinant  $\det(\alpha_i^s)$  is nonzero.

In the proposition which follows,  $X_0$  is assumed affine, and  $\mathcal{F}_0$  is a smooth Weil sheaf of rank  $r$  on  $X_0$ . Assume  $\mathcal{F}_0$  algebraic in the sense of 0.9. Put

$$N_0 = 2 \log_q^+ \left( 2r^2 \left( b_1(X) + \sum \alpha_s(\mathcal{F}) \right) \right), \quad N = [N_0] + 2r.$$

**Proposition 2.10.** Let  $E_0$  be the extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}_\ell}$  generated by the  $\text{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in  $X_0(\mathbb{F}_{q^n})$  with  $n \leq N$ . Then, for every  $n$  and every  $x \in X_0(\mathbb{F}_{q^n})$ ,  $\text{Tr}(F_x, \mathcal{F}_0)$  is in  $E_0$ .

**Proof.** After semisimplifying  $\mathcal{F}_0$ , one reduces to the case where  $\mathcal{F}_0$  is semisimple. Let  $E$  be as in 2.2, a sufficiently large Galois extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}_\ell}$ . We must show that for  $\sigma \in \text{Gal}(E/E_0)$ , the  $\text{Tr}(F_x, \mathcal{F}_0)$  are fixed by  $\sigma$ . This is equivalent to

$$\text{Tr}(F_x, \mathcal{F}_0) = \text{Tr}(F_x, \sigma(\mathcal{F}_0)). \quad (2.10.1)$$

By hypothesis, (2.10.1) is true if  $x$  is in some  $X(\mathbb{F}_{q^n})$  with  $n \leq N$ . By 2.5 and by the fact that  $\alpha_s(\mathcal{F}_0) = \alpha_s(\sigma(\mathcal{F}_0))$ ,  $\mathcal{F}_0$  is isomorphic to  $\sigma(\mathcal{F}_0)$ . The assertion follows.

### 2.11. Variants

Keep the hypotheses and notation of 2.5. Let  $q : X' \rightarrow X$  be a connected etale cover of  $X$  on which the inverse images of  $\mathcal{F}$  and  $\mathcal{G}$  are tamely ramified. For every  $\overline{\mathbb{Q}_\ell}$ -sheaf  $\mathcal{H}$  on  $X$  whose inverse image on  $X'$  is tamely ramified, the morphism

$$q : H^*(X, \mathcal{H}) \rightarrow H^*(X', q^* \mathcal{H})$$

is injective, because its composite with  $\text{Tr}_q$  is multiplication by the degree of the cover. Thus

$$\dim H_c^1(X, \mathcal{H}) \leq \dim H_c^1(X', \mathcal{H}) \leq \text{rank}(\mathcal{H})b_1(X').$$

If one repeats the arguments proving 2.5 using this estimate rather than (2.1.6), one obtains the following.

**Variant 2.12.** Put  $N'_0 := 2 \log_q^+(2r^2 b_1(X'))$  and  $N' := [N'_0] + 2r$ . If for every integer  $n \leq N'$  and every  $x \in X_0(\mathbb{F}_{q^n})$  one has

$$\text{Tr}(F_x, \mathcal{F}_0) = \text{Tr}(F_x, \mathcal{G}_0),$$

then  $\mathcal{F}_0$  is isomorphic to  $\mathcal{G}_0$ .

Similarly, if  $\mathcal{F}_0$  is as in 2.10 and its inverse image by  $q : X' \rightarrow X$  is tamely ramified, then  $\sigma(\mathcal{F}_0)$  has the same property: after a finite extension of the base field  $\mathbb{F}_q$ , one may suppose that  $q : X' \rightarrow X$  comes from  $q_0 : X'_0 \rightarrow X_0$  and apply 2.3. Repeating similarly the arguments which prove 2.10, one obtains the following.

**Variant 2.13.** Let  $E_0$  be the extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}_\ell}$  generated by the  $\text{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in  $X_0(\mathbb{F}_{q^n})$  with  $n \leq N'$ , where  $N'$  is as in 2.12. Then, for every  $n$  and every  $x \in X_0(\mathbb{F}_{q^n})$ ,  $\text{Tr}(F_x, \mathcal{F}_0)$  is in  $E_0$ .

**Remark 2.14.** The hypothesis that  $\mathcal{F}_0$  is algebraic ensures that  $E_0$  is a finite extension of  $\mathbb{Q}$ . It is not required in order to prove 2.13. If  $E_0 \subset E$  are extensions of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}_\ell}$ , and if  $P$  is the set of  $\sigma : E \rightarrow \overline{\mathbb{Q}_\ell}$  extending the inclusion of  $E_0$  in  $\overline{\mathbb{Q}_\ell}$ , then

$$E_0 = \{e \in E \mid \sigma(e) = e \text{ for every } \sigma \in \Sigma\},$$

and this observation can replace the Galois argument.

## 3. Main theorem

**Theorem 3.1.** Let  $Z_0$  be a scheme of finite type over  $\mathbb{F}_q$ . If  $\mathcal{F}_0$  is an algebraic  $\overline{\mathbb{Q}_\ell}$ -Weil sheaf (0.9) on  $Z_0$ , there exists a finite extension  $E \subset \overline{\mathbb{Q}_\ell}$  of  $\mathbb{Q}$  such that for every  $n$  and every  $x \in Z_0(\mathbb{F}_{q^n})$ ,  $\text{Tr}(F_x, \mathcal{F}_0)$  lies in  $E$ .

The fact that all these traces lie in  $E$  is equivalent to saying that, at every closed point  $x_0$  of  $Z_0$ , the local factor of the  $L$ -function of  $\mathcal{F}_0$  has coefficients in  $E$ ; see 0.5.

**Lemma 3.2.** Theorem 3.1 is implied by the special case in which one assumes that

- (i)  $Z_0$  is affine, smooth, irreducible, and is endowed with an étale morphism  $\varphi : Z_0 \rightarrow \mathbb{E}_0^k$  to an affine space over  $\mathbb{F}_q$ ;
- (ii)  $\mathcal{F}_0$  is smooth, irreducible, and  $\det(\mathcal{F}_0)$  has finite order.

**Proof.** The scheme  $Z_0$  of 3.1 admits a partition into irreducible locally closed parts  $F_i$  satisfying (i) and such that  $\mathcal{F}_0|_{F_i}$  is smooth. It suffices to treat separately each  $(F_i, \mathcal{F}_0|_{F_i})$ . One can then treat separately each irreducible subquotient of  $\mathcal{F}_0|_{F_i}$ , and twist it so that it satisfies (ii).

**Lemma 3.3.** There exists a connected étale cover  $q : Z' \rightarrow Z$  such that the monodromy group of  $q^*\mathcal{F}$  is a pro- $\ell$  group.

What matters for us is that the restriction of  $q^*\mathcal{F}$  to a curve drawn on  $Z'$  is tamely ramified at infinity.

**Proof.** There exists a finite extension  $E_\lambda$  of  $\mathbb{Q}_\ell$  in  $\overline{\mathbb{Q}_\ell}$ , with ring of integers  $\mathcal{O}_\lambda$ , and a torsion-free smooth  $\mathcal{O}_\lambda$ -sheaf  $\tilde{\mathcal{F}}$  such that

$$\mathcal{F} \simeq \tilde{\mathcal{F}} \otimes_{\mathcal{O}_\lambda} \overline{\mathbb{Q}_\ell}.$$

The reduction of  $\tilde{\mathcal{F}}$  modulo the maximal ideal  $\mathfrak{m}_\lambda$  of  $\mathcal{O}_\lambda$  is a locally constant étale sheaf, and it suffices to take for  $Z'$  a cover of  $Z$  on which it becomes constant. If  $\mathcal{F}$  has rank  $r$ , one can take for  $Z'$  a connected component of the cover

$$\text{Isom} \left( \tilde{\mathcal{F}}/\mathfrak{m}_\lambda \tilde{\mathcal{F}}, (\mathcal{O}_\lambda/\mathfrak{m}_\lambda)^r \right).$$

On the cover  $Z'$ , the monodromy of each  $\tilde{\mathcal{F}}/\mathfrak{m}_\lambda^n \tilde{\mathcal{F}}$  is an  $\ell$ -group.

**Proposition 3.4.** If the integer  $n$  is sufficiently large, then for every  $x \in Z_0(\mathbb{F}_{q^n})$  the trace  $\text{Tr}(F_x, \mathcal{F}_0)$  is contained in the extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}_\ell}$  generated by the traces  $\text{Tr}(F_y, \mathcal{F}_0)$  with  $y \in Z_0(\mathbb{F}_{q^m})$  and  $m < n$ .

If  $N$  is such that the integers  $n > N$  are “sufficiently large,” it follows by induction that every trace  $\text{Tr}(F_x, \mathcal{F}_0)$  is contained in the extension of  $\mathbb{Q}$  generated by the  $\text{Tr}(F_y, \mathcal{F}_0)$  with  $y$  in some  $Z_0(\mathbb{F}_{q^m})$  with  $m \leq N$ . This extension is finite, and therefore 3.4 implies 3.1.

Let  $x \in Z_0(\mathbb{F}_{q^n})$ , and try to prove the conclusion of 3.4 for  $x$ . We will succeed for  $n$  large enough.

Choose a generator of  $\mathbb{F}_{q^n}$  over  $\mathbb{F}_q$ . This choice defines an  $\mathbb{F}_{q^n}$ -point  $x'$  of the affine line  $\mathbb{E}_0^1$  over  $\mathbb{F}_q$ , whose  $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ -conjugates are all distinct.

**Lemma 3.5.** If  $y$  is an  $\mathbb{F}_{q^n}$ -point of  $\mathbb{E}_0^1$ , there exists an  $\mathbb{F}_q$ -morphism  $P : \mathbb{E}_0^1 \rightarrow \mathbb{E}_0^1$ , i.e. a polynomial  $P \in \mathbb{F}_q[T]$ , which sends  $x'$  to  $y$  and has degree  $\leq n - 1$ .

**Proof.** The  $\sigma x'$ , for  $\sigma \in \text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ , are  $n$  distinct points of the affine line. There is therefore a unique polynomial  $P \in \mathbb{F}_{q^n}[T]$ , of degree  $\leq n - 1$ , which for each  $\sigma$  takes at  $\sigma x'$  the value  $\sigma y$ . Its uniqueness ensures its Galois invariance, hence that it lies in  $\mathbb{F}_q[T]$ .

**Corollary 3.6.** There exists an  $\mathbb{F}_q$ -morphism  $P : \mathbb{E}_0^1 \rightarrow \mathbb{E}_0^k$ , whose coordinates are polynomials of degree  $\leq n - 1$ , which sends  $x' \in \mathbb{E}_0^1(\mathbb{F}_{q^n})$  to  $\varphi(x) \in \mathbb{E}_0^k(\mathbb{F}_{q^n})$ .

This is checked by applying 3.5 to each coordinate of  $\varphi(x)$ .



### 3.7

Fix  $P$  as in 3.6. Let  $X_0''$  be the fiber product of  $Z_0$  and  $\mathbb{E}_0^1$  over  $\mathbb{E}_0^k$ . Since  $\varphi(x) = P(x')$ , there exists  $\bar{x} \in X_0''(\mathbb{F}_{q^n})$  mapping to  $x$  and  $x'$ . Let  $X_0$  denote the connected component of  $X_0''$  containing  $\bar{x}$ :

$$\begin{array}{ccccc} X_0 & \hookrightarrow & X_0'' & \xrightarrow{\varphi''} & \mathbb{E}_0^1 \\ & \searrow \tilde{P} & \downarrow & & \downarrow P \\ & & Z_0 & \xrightarrow{\varphi} & \mathbb{E}_0^k \end{array} \quad \begin{array}{c} \bar{x} \in X_0(\mathbb{F}_{q^n}) \\ \downarrow \\ x \in Z_0(\mathbb{F}_{q^n}). \end{array} \quad (3.7.1)$$

The morphism  $\varphi''$  is étale, of degree over the generic point of  $\mathbb{E}_0^1$  at most equal to the degree  $D$  of  $\varphi$ . Let  $k_0$  be the field of constants of the curve  $X_0$  over  $\mathbb{F}_q$ . The degree  $d$  of  $k_0$  over  $\mathbb{F}_q$  is  $\leq D$ , because  $X_0$  is of degree  $\leq D$  over  $\mathbb{E}_0^1$ , and  $d$  divides  $n$ , because  $X_0$  has a point over  $\mathbb{F}_{q^n}$ . Extend the base field from  $\mathbb{F}_q$  to  $\mathbb{F}_{q^d}$ , and let  $X_1$  (respectively  $Z_1$ ) be the connected component of  $X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^d}$  (respectively of  $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^d}$ ) which contains  $\bar{x}$  (respectively  $x$ ):

$$\begin{array}{ccccc} \bar{x} : \text{Spec}(\mathbb{F}_{q^n}) & \longrightarrow & X_1 & \xrightarrow{\sim} & X_0 \\ \parallel & & \downarrow & & \downarrow \tilde{P} \\ x : \text{Spec}(\mathbb{F}_{q^n}) & \longrightarrow & Z_1 & \longrightarrow & Z_0 \\ & & \downarrow & & \downarrow \\ & & \text{Spec}(\mathbb{F}_{q^d}) & \longrightarrow & \text{Spec}(\mathbb{F}_q). \end{array} \quad (3.7.2)$$

The curve  $X_1$  over  $\mathbb{F}_{q^d}$  is smooth and absolutely irreducible. Finally extend the base field from  $\mathbb{F}_{q^d}$  to  $\mathbb{F}$ , to obtain  $X$  and  $Z$ , and let  $X'$  be a connected component of the inverse image  $X''$  of the étale cover  $Z'$  of  $Z$ :

$$\begin{array}{ccccc} X' & \hookrightarrow & X'' & \longrightarrow & X \\ \downarrow & & \downarrow & & \\ Z' & \longrightarrow & Z. & & \end{array} \quad (3.7.3)$$

### 3.8. Proof of 3.4

The inverse image of  $\mathcal{F}$  on  $X'$  is tamely ramified. We propose to apply 2.13 to the curve  $X_1$  over  $\mathbb{F}_{q^d}$ , to  $X' \rightarrow X$ , and to the inverse image  $\mathcal{F}_1$  of  $\mathcal{F}_0$  on  $X_1$ . Let us estimate  $b_1(X')$ .

The composite  $\mathbb{F}$ -morphism

$$Z' \rightarrow Z \xrightarrow{\varphi} \mathbb{E}^k$$

is affine. It therefore factors through a closed immersion  $Z' \hookrightarrow \mathbb{E}^{k+k'}$ , and there is a family of  $A'$  equations of degree  $\leq B$  defining  $Z'$  in  $\mathbb{E}^{k+k'}$ .

The graph  $\Gamma_P$  of  $P : \mathbb{E}^1 \rightarrow \mathbb{E}^k$  is defined in  $\mathbb{E}^{1+k}$  by  $k$  equations of degree  $\leq n-1$ . The fiber product, calculated over  $\mathbb{F}$ , of  $Z' \rightarrow \mathbb{E}^k$  and  $P : \mathbb{E}^1 \rightarrow \mathbb{E}^k$  is the intersection, in  $\mathbb{E}^{1+k+k'}$ , of the inverse images of  $Z' \subset \mathbb{E}^{k+k'}$  and of  $\Gamma_P \subset \mathbb{E}^{1+k}$ . It is therefore defined, in  $\mathbb{E}^{1+k+k'}$ , by  $A := A' + k$  equations of degree  $\leq \sup(B, n-1)$ . By Katz [K], this implies an upper bound for the sum of the Betti numbers of this fiber product, and a fortiori of  $b_1(X')$ , since  $X'$  is a connected component of the fiber product.

For  $n > B$ , one has  $A$  equations of degree  $\leq n-1$  in an affine space of dimension  $k+k'+1$ , and Katz gives

$$b_1(X') \leq 6 \cdot 2^A (An + 3)^{k+k'+1}. \quad (3.8.1)$$

Put, as in 2.12,

$$N'_0 = 2 \log_{q^d}^+(2r^2 b_1(X')) \quad \text{and} \quad N' = [N'_0] + 2r.$$

The bound (3.8.1) is polynomial in  $n$ , and  $d$  is bounded by  $D$ . As soon as  $n$  is large enough, one therefore has

$$\frac{n}{d} > N'.$$

Assume  $n$  large enough for this inequality to hold.

By (3.7.2), one has  $\text{Tr}(F_{\bar{x}}, \mathcal{F}_1) = \text{Tr}(F_x, \mathcal{F}_0)$ . By 2.13, this trace is contained in the field generated by the  $\text{Tr}(F_y, \mathcal{F}_1)$  for  $y$  in  $X_1(\mathbb{F}_{q^m})$  with  $d \mid m$  and with  $m/d$ , the degree of  $\mathbb{F}_{q^m}$  over  $\mathbb{F}_{q^d}$ , at most  $N'$ . Again by (3.7.2), these traces are also the  $\text{Tr}(F_y, \tilde{P}^* \mathcal{F}_0)$  for  $y$  in  $X_0(\mathbb{F}_{q^m})$  with  $m/d \leq N'$ . We have

$$\text{Tr}(F_y, \tilde{P}^* \mathcal{F}_0) = \text{Tr}(F_{\tilde{P}(y)}, \mathcal{F}_0),$$

and  $\text{Tr}(F_x, \mathcal{F}_0)$  is in the field generated by the  $\text{Tr}(F_y, \mathcal{F}_0)$  with  $y \in Z_0(\mathbb{F}_{q^m})$  and

$$\frac{m}{d} \leq N' < \frac{n}{d},$$

and hence  $m < n$ . This proves 3.4.

**Remark 3.9.** If one does not assume  $\mathcal{F}_0$  algebraic, it remains true that the extension of  $\mathbb{Q}$  in  $\overline{\mathbb{Q}}_\ell$  generated by the  $\text{Tr}(F_x, \mathcal{F}_0)$  for  $x$  in any  $Z_0(\mathbb{F}_{q^n})$  is an extension of finite type. One may deduce this by twisting from the algebraic case, or repeat the preceding arguments while applying 2.14.

**Remark 3.10.** Except in the proof of 3.3, our arguments have used the sheaf  $\mathcal{F}_0$  only through its inverse images on smooth curves. Theorem 3.1 therefore remains valid if, instead of  $\mathcal{F}_0$ , one is given “trace” functions  $t_n$  on the  $X_0(\mathbb{F}_{q^n})$ , with values in  $\overline{\mathbb{Q}}_\ell$ , having the following two properties:

- a. For every smooth curve  $X_0$  and every morphism  $\varphi : X_0 \rightarrow Z_0$ , there is a smooth  $\overline{\mathbb{Q}}_\ell$ -sheaf  $\mathcal{F}_0[\varphi]$  on  $X_0$  such that, for every  $n$  and every  $x \in X_0(\mathbb{F}_{q^n})$ ,

$$\text{Tr}(F_x, \mathcal{F}_0[\varphi]) = t_n(\varphi(x)).$$

- b. There exists a connected etale cover  $q : Z' \rightarrow Z$  such that for every  $\varphi : X_0 \rightarrow Z_0$  as in a., if  $X'$  denotes the etale cover of  $X$  obtained as the inverse image of  $Z'$ , then the inverse image of  $\mathcal{F}_0[\varphi]$  on  $X'$  is tamely ramified at infinity.

Hypothesis b. replaces 3.3. The conclusion is that the values of the  $t_n$  generate a finite-type extension of  $\mathbb{Q}$ ; a finite extension if the values of the  $t_n$  are algebraic.

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# Multizetas, after Francis Brown

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## 0. Introduction

Let  $k \geq 0$  and let  $s = (s_1, \dots, s_k)$  be a sequence of  $k$  integers  $\geq 1$ . The multizeta number  $\zeta(s)$  is the iterated sum (the name is explained in 1.1)

$$\zeta(s) := \sum_{n_1 > \dots > n_k > 0} \frac{1}{n_1^{s_1} \cdots n_k^{s_k}} \quad (0.1)$$

(the sum being over strictly decreasing sequences of  $k$  integers  $> 0$ ). For  $k = 0$ , this definition gives  $\zeta(s) = 1$ . For  $k \geq 1$ , the sum (0.1) converges if and only if  $s_1 \geq 2$ . We shall consider only convergent  $\zeta(s)$ , i.e. those for which (0.1) converges.

The product of two multizeta numbers is a linear combination, with integer coefficients, of multizeta numbers. See 3.6. It is therefore equivalent to determine the polynomial relations with rational coefficients among the multizeta numbers, or to determine the  $\mathbb{Q}$ -linear relations among them.

For  $k = 1$ , the convergent  $\zeta(s)$  are the values at the integers  $n \geq 2$  of Riemann's zeta function. Euler showed that for  $n$  even,  $\zeta(n)$  is a rational multiple of  $\pi^n$  ([3]). Computing  $\zeta(3)$  to 10 significant figures, he also checked that  $\zeta(3)$  was not the product of  $\pi^3$  by a rational number of small denominator. Following a correspondence with Goldbach, Euler studied in [4], for  $k = 2$ , the variant of the sums (0.1) in which one sums over  $n_1 \geq n_2$ :

$$\zeta_{\text{Euler}}(s_1, s_2) = \zeta(s_1, s_2) + \zeta(s_1 + s_2).$$

He proves a series of  $\mathbb{Q}$ -linear relations among multizeta numbers, among them  $\zeta(2, 1) = \zeta(3)$ . One of his motivations was the hope of obtaining information on the values of  $\zeta$  at the odd integers  $\geq 3$  (see the end of his letter to Goldbach of January 5, 1743).

The problem of determining all  $\mathbb{Q}$ -linear relations among multizeta numbers has two aspects:

- (A) What relations should one expect? Prove them.
- (B) Prove that there are no others.

The available evidence, both numerical and theoretical, suggests that every  $\mathbb{Q}$ -linear relation among the  $\zeta(s)$  is a sum of isobaric relations, i.e. relations among  $\zeta(s)$  of the same weight  $\sum s_i$ . If one restricts oneself to considering only isobaric relations, one knows nothing about problem (B). We shall speak only of (A).

F. Brown has defined a class of  $\mathbb{Q}$ -linear relations among the  $\zeta(s)$ , the motivic relations. More precisely, he defines a graded  $\mathbb{Q}$ -algebra  $M$ , homogeneous elements  $\zeta^m(s)$  of  $M$ , of degree the weight  $\sum s_i$  of  $s$ , and a homomorphism  $\text{real} : M \rightarrow \mathbb{R}$  such that  $\text{real}(\zeta^m(s)) = \zeta(s)$ . As a vector space, the algebra  $M$  is generated by the  $\zeta^m(s)$ . The motivic relations among the  $\zeta(s)$  are those that come from relations among the  $\zeta^m(s)$ .

A variant of Grothendieck's period conjecture implies that every  $\mathbb{Q}$ -linear relation among the  $\zeta(s)$  is motivic. This prediction has been verified for many of the known relations. Warning: although some proofs of relations among the  $\zeta(s)$  are easy to lift to proofs of the same relations among the  $\zeta^m(s)$ , this is far from always being the case.

**Theorem 0.2 (F. Brown [1]).** The  $\zeta^m(s_1, \dots, s_k)$  for which each  $s_i$  is equal to 2 or 3 form a basis of  $M$  over  $\mathbb{Q}$ .

Each  $\zeta(s)$  can therefore be expressed uniquely, in a motivic way, as a linear combination with rational coefficients of  $\zeta(s_1, \dots, s_k)$  with  $k \geq 0$  and each  $s_i \in \{2, 3\}$ . Unfortunately, Brown's proof does not provide an algorithm, or at least not a usable algorithm, for finding what this linear combination is. From Theorem 0.2, F. Brown deduces that the category of mixed Tate motives over  $\mathbb{Z}$  is generated by the fundamental group of  $\mathbb{P}^1 - \{0, 1, \infty\}$ . See 7.18.

Some of our notations differ from those of [1]. The correspondence between the notations is involutive:  $\zeta(s_1, \dots, s_k)$  becomes  $\zeta(s_k, \dots, s_1)$ , the composition of paths  $\alpha\beta$  becomes  $\beta\alpha$ , a monomial in the  $e_t$  (respectively a sequence of 0 and 1) is to be replaced by the same one read from right to left, and  $e_1$  is to be replaced by  $-e_1$ .

## 1. Integral formula

### 1.1

Let  $\omega_1, \dots, \omega_N$  be 1-forms on the interval  $[0, 1]$ . The iterated integral  $\text{It} \int_0^1 \omega_1, \dots, \omega_N$  is the integral, over the simplex

$$\sigma_N := \{(t_1, \dots, t_N) \mid 1 \geq t_1 \geq \dots \geq t_N \geq 0\}, \quad (1.1.1)$$

contained in the cube  $[0, 1]^N$ , of

$$\text{pr}_1^* \omega_1 \wedge \text{pr}_2^* \omega_2 \wedge \dots \wedge \text{pr}_N^* \omega_N.$$

Let  $S$  be the operation that attaches to a 1-form  $\omega$  the function

$$S[\omega](t) := \int_0^t \omega.$$

The iterated integral  $\text{It} \int_0^1 \omega_1, \dots, \omega_N$  can be constructed as follows: apply  $S$  to  $\omega_N$ , multiply the function obtained by  $\omega_{N-1}$ , apply  $S$ , multiply by  $\omega_{N-2}$ , and so on; at the end, evaluate at 1:

$$\text{It} \int_0^1 \omega_1, \dots, \omega_N = S[\omega_1.S[\omega_2.\dots S[\omega_N]\dots]](1). \quad (1.1.2)$$

It is this construction that explains the name “iterated integral”. A sum such as (0.1) admits an analogous description in terms of the operator  $P$  which, to a function  $f(n)$  ( $n \geq 1$ ), attaches the function

$$P[f](n) := \sum_{m=1}^{n-1} f(m);$$

this explains the name “iterated sum” given to (0.1).

Let us compute in this way the iterated integral

$$\text{It} \int_0^1 \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_1-1) \text{ times}}, \frac{dt}{1-t}, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_2-1) \text{ times}}, \frac{dt}{1-t}, \dots, \underbrace{\frac{dt}{t}, \dots, \frac{dt}{t}}_{(s_k-1) \text{ times}}, \frac{dt}{1-t}. \quad (1.1.3)$$

We have

$$\frac{dt}{1-t} = \sum_{n \geq 0} t^n dt.$$

Applying  $S$  term by term, one obtains  $\sum_{n \geq 1} t^n/n$ . Multiplying by  $dt/t$  and applying  $S$ , one obtains  $\sum_{n \geq 1} t^n/n^2$ . Iterating  $(s_k - 1)$  times, one obtains  $\sum_{n \geq 1} t^n/n^{s_k}$ . Multiplying by  $dt/(1 - t) = \sum_{m \geq 0} t^m dt$  and applying  $S$ , one obtains

$$S \left( \sum_{m \geq 0, n \geq 1} \frac{t^{m+n}}{n^{s_k}} dt \right) = \sum_{m, n \geq 1} \frac{t^{m+n}}{(m+n)n^{s_k}} = \sum_{m > n > 0} \frac{t^m}{mn^{s_k}}.$$

Continuing in this way, one finally obtains:

**Proposition 1.2.** The iterated integral (1.1.3) is equal to  $\zeta(s_1, \dots, s_k)$ .

### 1.3

Whereas the notion of an infinite sum is foreign to algebraic geometry, the study of integrals of algebraic quantities is one of its sources. It is thanks to Proposition 1.2 that algebraic geometry, more precisely the theory of mixed Tate motives, is useful in the study of multizeta numbers.

## 2. Pro-unipotent fundamental group and cohomology

### 2.1

We shall need a variant  $Z'$  of the descending central series  $Z$  of a group  $\Gamma$ . For each  $i \geq 1$ ,  $\Gamma/Z_i$  is a nilpotent group. The set  $T_i$  of its torsion elements is therefore a subgroup. We define  $Z'_i$  as the inverse image of  $T_i$  in  $\Gamma$ . Below, we shall always assume  $\Gamma$  to be finitely generated. This hypothesis implies that the  $Z'_i/Z'_{i+1}$  are finitely generated free abelian groups.

First suppose  $\Gamma$  nilpotent and torsion-free, i.e. that there is an  $A$  such that  $Z'_A = 0$ . For each  $i$  ( $1 \leq i < A$ ), choose a basis of  $Z'_i/Z'_{i+1}$  and lift it to  $Z'_i$ . Let  $\gamma_1, \dots, \gamma_M$  be the elements of  $\Gamma$  obtained, arranged in an arbitrary order, and let  $w(j)$  be the integer  $i$  such that  $\gamma_j$  lifts an element of a basis of  $Z'_i/Z'_{i+1}$ . One checks by induction on  $A$  that the map

$$\mathbb{Z}^M \longrightarrow \Gamma : n \longmapsto \gamma_1^{n_1} \cdots \gamma_M^{n_M}$$

is bijective. In such a “system of coordinates”, the group law of  $\Gamma$  is given by polynomials with rational coefficients and integer values on the integers. More precisely, if the coordinate of index  $j$  is viewed as having weight  $w(j)$ , the  $j$ -th coordinate of  $m \circ n$  is a polynomial of weight  $\leq w(j)$  in the  $m_k$  and  $n_l$ .

The same polynomials define a unipotent algebraic group over  $\mathbb{Q}$ , the unipotent envelope  $\Gamma^{\text{un}}$  of  $\Gamma$ . The group  $\Gamma^{\text{un}}(\mathbb{Q})$  of its rational points is  $\mathbb{Q}^N$ , endowed with the group law given by the same formulae as that of  $\Gamma$ . The group  $\Gamma^{\text{un}}(\mathbb{Q})$  is the Malcev completion  $\Gamma \otimes \mathbb{Q}$  of  $\Gamma$ .

Let  $\mathbb{Q}[\Gamma]$  be the group algebra of  $\Gamma$ , and let  $I$  be its augmentation ideal. Quillen gave an elegant description of the affine algebra  $O(\Gamma^{\text{un}})$  of  $\Gamma^{\text{un}}$ : it is the inductive limit of the duals of the  $\mathbb{Q}[\Gamma]/I^N$ . More precisely, the inclusion of  $\Gamma$  in  $\mathbb{Q}[\Gamma]$  identifies the dual of  $\mathbb{Q}[\Gamma]$  with the space of functions  $\Gamma \rightarrow \mathbb{Q}$ , and under this identification the dual of  $\mathbb{Q}[\Gamma]/I^{N+1}$  becomes the space of polynomials in the  $n_j$  of weight  $\leq N$ .

For an arbitrary finitely generated  $\Gamma$ , the preceding applies to the  $\Gamma/Z'_i$ , and one defines  $\Gamma^{\text{un}}$  as the projective limit group scheme of the  $(\Gamma/Z'_i)^{\text{un}}$ . The affine algebra of  $\Gamma^{\text{un}}$  is still the inductive limit of the duals of the  $\mathbb{Q}[\Gamma]/I^N$ .

### 2.2

Let  $E$  be a reasonable topological space, for example a complex algebraic variety. Assume  $E$  connected. A path  $\gamma$  from  $a$  to  $b$ ,  $[0, 1] \rightarrow E$ , gives

$$\gamma_N : [0, 1]^N \rightarrow E^N$$

and, by restriction to the simplex  $\sigma_N$  of (1.1.1), a singular chain of  $E^N$  which is a cycle modulo the union of  $b = x_1, x_1 = x_2, \dots, x_N = a$ . The homology class of this cycle depends only on the homotopy class of  $\gamma$ .

Make  $a = b$ . Elaborating the preceding construction (cf. [2] §3), one obtains an isomorphism of  $\mathbb{Q}[\pi_1(E, a)]/I^{N+1}$  with a relative homology group of  $E^N$  and, for its dual  $(\mathbb{Q}[\pi_1(E, a)]/I^{N+1})^\vee$ , with a relative cohomology group of  $E^N$ . From this, by passage to the inductive limit, one obtains a cohomological description of the affine algebra of  $\pi_1(E, a)^{\text{un}}$ .

No longer suppose that  $a = b$ . Let  $\pi_1(E; b, a)$  be the set of homotopy classes of paths from  $a$  to  $b$ . The group  $\pi_1(E, a)$  acts on the right, by composition of paths, on  $\pi_1(E; b, a)$ . This action makes  $\pi_1(E; b, a)$  a principal homogeneous space, we prefer to say torsor, under  $\pi_1(E, a)$ . We write  $\pi_1(E; b, a)^{\text{un}}$  for the  $\pi_1(E, a)^{\text{un}}$ -torsor deduced from it by pushing forward by  $\pi_1(E, a) \rightarrow \pi_1(E, a)^{\text{un}}$ . Let  $\mathbb{Q}[\pi_1(E; b, a)]$  be the space of formal linear combinations of elements of  $\pi_1(E; b, a)$ . It is a right  $\mathbb{Q}[\pi_1(E, a)]$ -module, free of rank one. Set

$$\mathbb{Q}[\pi_1(E; b, a)]/I^N := \mathbb{Q}[\pi_1(E; b, a)]/\mathbb{Q}[\pi_1(E; b, a)]I^N.$$

The preceding construction again provides a cohomological description of the  $(\mathbb{Q}[\pi_1(E; b, a)]/I^N)^\vee$ , and of the affine algebra of  $\pi_1(E; b, a)^{\text{un}}$  which is its inductive limit.

## 2.3

Let  $X$  be an algebraic variety defined over a subfield  $K$  of  $\mathbb{C}$ . The Betti cohomology  $H_B^*(X)$  of  $X$  is the rational cohomology of the space  $X(\mathbb{C})$  of complex points of  $X$ , endowed with its classical topology. It is endowed with structures reflecting the algebraic structure of  $X$ , in particular with a mixed Hodge structure. One also has the algebraic de Rham cohomology  $H_{\text{dR}}^*(X)$ . If  $X$  is nonsingular,  $H_{\text{dR}}^*(X)$  is the hypercohomology of the de Rham complex of sheaves of algebraic differential forms. It is a  $K$ -vector space, one has a comparison isomorphism

$$\text{comp}_{B, \text{dR}} : H_{\text{dR}}^i(X) \otimes_K \mathbb{C} \xrightarrow{\sim} H_B^i(X) \otimes_{\mathbb{Q}} \mathbb{C}, \quad (2.3.1)$$

and the filtrations  $W$  and  $F$  of the mixed Hodge structure of  $H_B^i(X)$  come from filtrations of  $H_{\text{dR}}^i(X)$ . The coefficients of the matrix of the isomorphism (2.3.1), in a basis of  $H_{\text{dR}}^i(X)$  and one of  $H_B^i(X)$ , are called periods.

Thanks to 2.2, all this applies to the affine algebras of the group schemes  $\pi_1(X, a)^{\text{un}}$  or of their homogeneous spaces  $\pi_1(X; b, a)^{\text{un}}$ . We shall write  $\pi_1(X, a)_B$  for  $\pi_1(X, a)^{\text{un}}$ ,  $\pi_1(X, a)_{\text{dR}}$  for its de Rham variant, and similarly for the  $\pi_1(X; b, a)$ . We shall put  $B/\text{dR}$  in the subscript when the statement is valid equally well in Betti and in de Rham.

## 2.4. Example

Take  $K = \mathbb{Q}$  and let  $X$  be the multiplicative group  $\mathbb{G}_m$ , whose group of complex points is  $\mathbb{C}^*$ . Here the fundamental group, which is commutative, is independent of the base point and is identified with the homology group  $H_1$ . In de Rham, the dual  $H_{\text{dR}}^1$  of  $H_1^{\text{dR}}$  is  $\mathbb{Q} dz/z$ . In Betti,  $H_1(\mathbb{C}^*, \mathbb{Z})$  is  $\mathbb{Z}$ , generated by a positive turn around 0. If one does not want to speak of a “positive turn”, i.e. to choose which of  $i$  and  $-i$  is  $i$ , it is better to say that  $H_1 = \pi_1 = 2\pi i\mathbb{Z}$ , with  $\ell \in 2\pi i\mathbb{Z}$  corresponding to the loop  $\exp(\ell t)$  ( $0 \leq t \leq 1$ ), on which the integral of  $dz/z$  is  $\ell$ .

The  $H_1^B(\mathbb{G}_m) = H_1(\mathbb{C}^*, \mathbb{Q})$  and  $H_1^{\text{dR}}(\mathbb{G}_m)$  are the Betti and de Rham realizations of the Tate motive  $\mathbb{Q}(1)$  (see §5). We have  $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$  (1 being dual to  $dz/z \in H_{\text{dR}}^1(\mathbb{G}_m)$ ), and  $\mathbb{Q}(1)_B$ , included in  $\mathbb{Q}(1)_{\text{dR}} \otimes \mathbb{C}$  by  $\text{comp}_{\text{dR}, B}$ , is  $2\pi i\mathbb{Q}$ ;  $\mathbb{Q}(1)$  is purely of Hodge type  $(-1, -1)$ .

The algebraic group  $\pi_1(\mathbb{G}_m, 1)_{\text{dR}}$  is the additive group  $\mathbb{G}_a$ , whose group of rational points is  $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$ , and whose affine algebra is  $\mathbb{Q}[T]$  (more precisely:  $\text{Sym}^*(H_{\text{dR}}^1(\mathbb{G}_m))$ ). The algebraic group  $\pi_1(\mathbb{G}_m, 1)_B$  is the additive group whose group of rational points is  $2\pi i\mathbb{Q}$ . Affine algebra:  $\text{Sym}^*(H_B^1(\mathbb{G}_m))$ . It is the algebra of functions  $f$  on  $2\pi i\mathbb{Z}$ , with rational values, and such that,

if one identifies  $2\pi i\mathbb{Z}$  with  $\mathbb{Z}$  by  $n \mapsto 2\pi in$ , the following equivalent conditions are verified for  $N$  large enough: (a)  $f$  is a polynomial of degree  $< N$ ; (b) denoting by  $\Delta$  the operator  $g \mapsto (g(n+1) - g(n))$ , one has  $\Delta^N f = 0$ ; (c) the linear form on the group algebra  $\mathbb{Q}[\mathbb{Z}]$  defined by  $f$  is zero on  $I^N$ .

By abuse of language, we shall sometimes write  $\mathbb{Q}(1)_{B/\mathrm{dR}}$  for these algebraic groups.

### 3. The case of $\mathbb{P}^1 - \{0, 1, \infty\}$

#### 3.1

If  $X$  is the complement in  $\mathbb{P}^1$  of a finite set  $T$  of rational points, the variety  $X$  is defined over  $\mathbb{Q}$  and the mixed Hodge structures of 2.2, 2.3 are Hodge-Tate:  $\mathrm{Gr}_n^W$  is zero for  $n$  odd, and for  $n = 2m$  is purely of Hodge type  $(m, m)$ . For each of the groups  $M$  considered in 2.2, let  $M_{\mathrm{dR}}$  be its de Rham variant (2.3). After renumbering, the weight and Hodge filtrations  $W$  and  $F$  of  $M_{\mathrm{dR}}$  are opposite. They define the grading of  $M_{\mathrm{dR}}$  for which

$$\begin{aligned} W_{-2n}(M_{\mathrm{dR}}) &= \bigoplus_{m \geq n} M_{\mathrm{dR}}^m, \\ F^{-p}(M_{\mathrm{dR}}) &= \bigoplus_{m \leq p} M_{\mathrm{dR}}^m. \end{aligned} \tag{3.1.1}$$

The convention (3.1.1) ensures that  $\mathbb{Q}(1)_{\mathrm{dR}}$  is of degree one. It is sometimes convenient to grade by the opposite of the degree, the codegree, and to set  $(M_{\mathrm{dR}})_m := (M_{\mathrm{dR}})^{-m}$ .

Special case: the affine algebra of  $\pi_1(X; b, a)_{\mathrm{dR}}$  is graded. It is in degrees  $\leq 0$  and is reduced to  $\mathbb{Q}$  in degree 0. The augmentation morphism  $O(\pi_1(X; b, a)_{\mathrm{dR}}) \rightarrow \mathbb{Q}$  is a point  ${}_b 1_a$  of  $\pi_1(X; b, a)_{\mathrm{dR}}$ . In de Rham, one thus has a canonical path  ${}_b 1_a$  from one base point to another: for a certain pro-unipotent group scheme  $\Pi$ , each  $\pi_1(X; b, a)_{\mathrm{dR}}$  is identified with a copy  ${}_b \Pi_a$  of  $\Pi$ , and composition of paths is the group law.

The group  $\Pi$  has the following description. Let  $L$  be the Lie algebra over  $\mathbb{Q}$  generated by elements  $e_t$  ( $t \in T$ ) subject only to the relation  $\sum e_t = 0$ . It is endowed with the grading for which each  $e_t$  is homogeneous of degree 1. The descending central series  $Z$  is the filtration by “degrees  $\geq i$ ”. Let  $L^\wedge$  be the completion of  $L$  for this filtration, and let  $\exp(L/Z_i)$  be the unipotent algebraic group with Lie algebra  $L/Z_i$ . We have

$$\Pi = \exp(L^\wedge) := \varprojlim \exp(L/Z_i). \tag{3.1.2}$$

Another description: the enveloping algebra  $A$  of  $L$  is the associative algebra generated by elements  $e_t$  ( $t \in T$ ) subject only to the relation  $\sum e_t = 0$ . It is graded, each  $e_t$  being homogeneous of degree one, and the  $N$ -th power of the augmentation ideal  $I$  is the part of  $A$  of degree  $\geq N$ . The completion  $A^\wedge = \varprojlim A/I^N$  of  $A$  is endowed with a coproduct

$$\Delta : A^\wedge \rightarrow A^\wedge \hat{\otimes} A^\wedge : e_t \mapsto e_t \otimes 1 + 1 \otimes e_t,$$

$L^\wedge$  is the Lie algebra of primitive elements, and for every  $\mathbb{Q}$ -algebra  $R$ , if one sets

$$A^\wedge \hat{\otimes} R := \varprojlim_N (A/I^N) \otimes_{\mathbb{Q}} R,$$

the group  $\Pi(R)$  of  $R$ -points of  $\Pi$  is the group of group-like elements of  $(A^\wedge \hat{\otimes} R)^*$ :

$$\Pi(R) = \{g \in (A^\wedge \hat{\otimes} R)^* \mid \Delta g = g \otimes g\}. \tag{3.1.3}$$

The affine algebra  $O(\Pi)$  of  $\Pi$  (with the discrete topology) and the completed enveloping algebra  $A^\wedge$  (with its projective-limit topology) are dual to one another. The topological dual of  $A^\wedge$  is



identified with the graded dual  $\bigoplus (A_n)^\vee$  of  $A$ , and this gives the grading (in degrees  $\leq 0$ ) of  $O(\Pi)$ .

We shall write  $\omega$  for the logarithmic 1-form on  $X$  with values in  $L$ :

$$\omega := \sum_{t \neq \infty} e_t \frac{dz}{z - t}. \quad (3.1.4)$$

At each  $t \in T$ , it has residue  $e_t$ .

Suppose that  $\infty \in T$  and put  $T' := T - \{\infty\}$ . Then  $A = \mathbb{Q}\langle (e_t)_{t \in T'} \rangle$  and

$$A^\wedge = \mathbb{Q}\langle\langle (e_t)_{t \in T'} \rangle\rangle$$

(associative formal series in the  $e_t$ ). For every word  $w = t_1 \cdots t_k$  in the alphabet  $T'$ , let  $c(w)$  be the linear form on  $A^\wedge$  “coefficient of the monomial  $e_{t_1} \cdots e_{t_k}$ ”. We again write  $c(w)$  for the restriction of  $c(w)$  to  $\Pi \subset A^{\wedge*}$ , and for a formal linear combination of words we put  $c(\sum \lambda_w w) = \sum \lambda_w c(w)$ . The  $c(w)$  form a vector-space basis of  $O(\Pi)$ . The product in  $O(\Pi)$  corresponds to the shuffle product of words:

$$c(w')c(w'') = c(w' \amalg w''), \quad (3.1.5)$$

where, for  $w' = t_1 \cdots t_k$  and  $w'' = t_{k+1} \cdots t_{k+\ell}$ ,  $w' \amalg w''$  is the sum of the  $t_{\sigma^{-1}(1)} \cdots t_{\sigma^{-1}(k+\ell)}$  for  $\sigma$  running through the set  $S_{k,\ell}$  of permutations of  $\{1, \dots, k+\ell\}$  increasing on  $\{1, \dots, k\}$  and on  $\{k+1, \dots, k+\ell\}$ .

### 3.2

A path  $\gamma$  from  $a$  to  $b$  defines a rational point of  $\pi_1(X; b, a)_B = \pi_1(X(\mathbb{C}); b, a)^{\text{un}}$ . Let  $\text{comp}_{\text{dR}, B}(\gamma)$  be the image of this rational point by the comparison isomorphism

$$\text{comp}_{\text{dR}, B} : \pi_1(X; b, a)_B(\mathbb{C}) \xrightarrow{\sim} \pi_1(X; b, a)_{\text{dR}}(\mathbb{C}) = \Pi(\mathbb{C}).$$

This image is obtained by “integration” along  $\gamma$ , in the noncommutative group  $\Pi$ , of the form  $\omega$ : it is the value at 1 of the solution  $g : [0, 1] \rightarrow \Pi(\mathbb{C})$ , equal to 1 at 0, of

$$dg(t) \cdot g(t)^{-1} = \omega \left( \frac{d\gamma}{dt} \right) dt. \quad (3.2.1)$$

### 3.3

The same formalism applies when one allows the base points  $a, b$  to be tangential base points (a nonzero tangent vector at a “rational point at infinity”  $t \in T$  of  $X$ ). Let  $\Theta_t$  be the tangent space to  $\mathbb{P}^1$  at  $t$ ,  $\Theta_t^* := \Theta_t - \{0\}$ , and  $a \in \Theta_t^*$  a tangential base point at  $t$ . Everything happens as if one had a morphism from  $\Theta_t^*$  to  $X$ : one has morphisms

$$\pi_1(\Theta_t^*, a)_{B/\text{dR}} = \mathbb{Q}(1)_{B/\text{dR}} \longrightarrow \pi_1(X, a)_{B/\text{dR}} \quad (3.3.1)$$

compatible with the comparison isomorphisms. On the Betti side, this morphism is induced by the germs of maps  $\varphi$  from  $(\Theta_t, 0)$  to  $(\mathbb{P}^1, t)$  with differential the identity. The space of these germs is contractible, and each  $\varphi$  induces a map from a small punctured disk around 0 (homotopic to  $\Theta_t^*$ ) to  $X$ . In de Rham, and after passage to Lie algebras, (3.3.1) is the morphism from  $\mathbb{Q}(1)_{\text{dR}} = \mathbb{Q}$  to  $L^\wedge$  which sends 1 to  $e_t$ .

The  $\mathbb{Q}$ -form  $\pi_1(X; b, a)_B$  of  $\Pi(\mathbb{C})$  is characterized by the property that the images of paths from  $a$  to  $b$  are rational points.

Warning:  $\pi_1(X; b, a)_{\text{dR}}$  is independent of  $a$  and  $b$ , and  $\pi_1(X; b, a)_B$  is locally constant in  $a$  and  $b$ , but the comparison isomorphism between their complexifications is not locally constant. The independence of  $\pi_1(X; b, a)_{\text{dR}}$  from  $a$  and  $b$  comes from the fact that one has the canonical de Rham path  ${}_b 1_a$  between any two base points. The image of this path by  $\text{comp}_{B, \text{dR}}$  is not locally constant.

### 3.4

The case that interests us is that in which  $X = \mathbb{P}^1 - \{0, 1, \infty\}$  and in which, as base points, one takes the tangential base points 1 at 0 and  $-1$  at 1. They will be denoted 0 and 1. Here

$$\omega = e_0 \frac{dz}{z} + e_1 \frac{dz}{z-1}$$

and for every  $\mathbb{Q}$ -algebra  $R$ ,

$$\Pi(R) = \{\text{group-like elements of } R\langle\langle e_0, e_1 \rangle\rangle^*\}$$

(3.1.3). In Betti, one has the “straight path”  $dch$  from 0 to 1. We shall again write  $dch$  for its image by  $\text{comp}_{\text{dR}, B}$  in  $\Pi(\mathbb{C})$ . This image is the following limit for  $\epsilon, \eta \rightarrow 0$  through values  $> 0$ :

$$\text{comp}_{\text{dR}, B}(dch) = \lim \exp(-\log \eta e_1) \text{comp}_{\text{dR}, B}([\epsilon, 1 - \eta]) \exp(\log \epsilon e_0). \quad (3.4.1)$$

The third (respectively first) factor is to be viewed as computing, in the punctured tangent space at 0 (respectively at 1),  $\text{comp}_{\text{dR}, B}([1, \epsilon])$  (respectively  $\text{comp}_{\text{dR}, B}([- \eta, -1])$ ), the form  $\omega$  being replaced by  $(du/u)e_0$  (respectively  $(du/u)e_1$ ).

In general one can express the solution of (3.2.1) as a sum of iterated integrals. Applying 1.2, one obtains that, for convergent  $\zeta(s_1, \dots, s_k)$ , one has:

**Proposition 3.5.** The coefficient of  $e_0^{s_1-1} e_1 \cdots e_0^{s_k-1} e_1$  in  $dch \in \Pi(\mathbb{C}) \subset \mathbb{C}\langle\langle e_0, e_1 \rangle\rangle^*$  is  $(-1)^k \zeta(s_1, \dots, s_k)$ .

This proposition determines the coefficients  $c(w)$  of monomials of  $dch$  only for  $w$  a word in the alphabet  $\{0, 1\}$  which does not begin with 1 and does not end with 0. The coefficients  $c(0)$  of  $e_0$  and  $c(1)$  of  $e_1$  in  $dch$  are zero. This, together with the fact that  $dch$  is group-like, determines the missing coefficients. More precisely, the vanishing of  $c(0)$  and  $c(1)$  means that  $dch$  belongs to the derived group  $\Pi'$  of  $\Pi$ , whose graded Lie algebra is the part of degree  $\geq 2$  of that of  $\Pi$ . On  $\Pi'$ , writing  $\{p\}$  for a power in the monoid of words,

$$c(0^{\{p\}}) = c(1^{\{p\}}) = 0 \quad \text{for } p > 0, \quad (3.5.1)$$

and, applying (3.1.5) to  $w10^{\{p-1\}}$  and 0 (respectively to 1 and  $1^{\{p-1\}}0w$ ), one checks by induction on  $p \geq 0$  that

$$c(w10^{\{p\}}) = (-1)^p c((w \amalg 0^{\{p\}})1), \quad (3.5.2)$$

respectively

$$c(1^{\{p\}}0w) = (-1)^p c(0(1^{\{p\}} \amalg w))$$

on  $\Pi'$ .

### 3.6

The identity (3.1.5), evaluated at  $dch$ , expresses the product of two multizeta numbers as a linear combination with integer coefficients of multizeta numbers.

## 4. Motivic multizetas

### 4.1

Continue to put  $X := \mathbb{P}^1 - \{0, 1, \infty\}$  and to take as base points the tangential base points 0 or 1 (see 3.4). In 5.6 we shall give a precise meaning to the expression “motivic structure on the affine algebras of the  $\pi_1(X; b, a)_{B/\text{dR}}$ ”. We shall also say “motivic structure on the  $\pi_1(X; b, a)_{B/\text{dR}}$ ”. Heuristically, these are structures coming from geometry, and hence having a meaning both for

Betti and for de Rham, and respected by the comparison isomorphisms. Examples: the product giving the algebra structure, the coproducts coming from composition of paths, the weight filtration, the isomorphism  $\pi_1(X; b, a)_{B/\mathrm{dR}} \rightarrow \pi_1(X; \sigma(b), \sigma(a))_{B/\mathrm{dR}}$  induced by the involution  $\sigma : x \mapsto 1 - x$  of  $\mathbb{P}^1$ . Likewise, the morphisms (3.3.1) are a motivic structure on  $\mathbb{Q}(1)_{B/\mathrm{dR}}$  and  $\pi_1(X, a)_{B/\mathrm{dR}}$ . By contrast, neither the Hodge filtration of the affine algebra of  $\pi_1(X; b, a)_{\mathrm{dR}}$ , nor therefore its grading, nor the independence of  $\pi_1(X; b, a)_{\mathrm{dR}}$  from  $a$  and  $b$ , is motivic. They are available only for de Rham. The structures in question will be structures of multilinear algebra. Let  $H_B$  (respectively  $H_{\mathrm{dR}}$ ) be the group scheme over  $\mathbb{Q}$  of automorphisms of  $\pi_1(X; 1, 0)_B$  (respectively  $\pi_1(X; 1, 0)_{\mathrm{dR}}$ ) respecting all its motivic structures. The weight filtration  $W$  of the affine algebra is a filtration by finite-dimensional subspaces, and  $H_B$  (respectively  $H_{\mathrm{dR}}$ ) is a projective limit of algebraic subgroups of the  $\mathrm{GL}(W_i)$ .

## 4.2

Let  $P_{\mathrm{dR}, B}$  be the scheme of isomorphisms from  $\pi_1(X; 1, 0)_B$  to  $\pi_1(X; 1, 0)_{\mathrm{dR}}$  respecting all motivic structures. The group  $H_B$  acts on the right on  $P_{\mathrm{dR}, B}$ . The comparison isomorphism  $\mathrm{comp}_{\mathrm{dR}, B}$  is a complex point of  $P_{\mathrm{dR}, B}$ . The scheme  $P_{\mathrm{dR}, B}$  is therefore nonempty. It is an  $H_B$ -torsor (terminology of 2.2).

If  $Z_B$  is a subscheme of  $\pi_1(X; 1, 0)_B$  stable under  $H_B$ , twisting by the  $H_B$ -torsor  $P_{\mathrm{dR}, B}$  transforms  $\pi_1(X; 1, 0)_B$  into  $\pi_1(X; 1, 0)_{\mathrm{dR}} = {}_1\Pi_0$ , and  $Z_B$  into a subscheme  $Z_{\mathrm{dR}}$  of  ${}_1\Pi_0$ . Over  $\mathbb{C}$ ,  $Z_{\mathrm{dR}}(\mathbb{C})$  is the image of  $Z_B(\mathbb{C})$  by  $\mathrm{comp}_{\mathrm{dR}, B}$ .

I like to think of the  $H_B$ -orbit of  $dch$  as the space of points  $ch$  of  $\pi_1(X; 1, 0)_B$  having all the properties of  $dch$  expressible in motivic terms. For example: the morphism  $\pi_1(X; 1, 0)_B \rightarrow \pi_1(X; 0, 1)_B$  induced by the involution  $x \mapsto 1 - x$  of  $\mathbb{P}^1$  must transform  $ch$  into its inverse.

Let  $M_B \subset \pi_1(X; 1, 0)_B$  be the closure of the  $H_B$ -orbit of  $dch$ . The twist of  $M_B$  by  $P_{\mathrm{dR}, B}$  is a closed subscheme  $M_{\mathrm{dR}}$  of  ${}_1\Pi_0$ . We again write  $dch$  for the image of  $dch$  by  $\mathrm{comp}_{\mathrm{dR}, B}$ . It is a complex point of  $M_{\mathrm{dR}}$ , and  $M_{\mathrm{dR}}(\mathbb{C}) \subset {}_1\Pi_0(\mathbb{C})$  is the closure of its  $H_{\mathrm{dR}}(\mathbb{C})$ -orbit.

**Definition 4.3.** Let  $s = (s_1, \dots, s_k)$ . If  $k \neq 0$ , we assume  $s_1 \geq 2$ . Let  $w$  be the word  $0^{\{s_1-1\}}1 \dots 0^{\{s_k-1\}}1$  in the alphabet  $\{0, 1\}$ , and let  $c(w) \in O(\Pi)$  be the function “coefficient of the monomial  $e_0^{s_1-1}e_1 \dots e_0^{s_k-1}e_1$ ” (3.1). We denote by  $\zeta^m(s)$  the restriction to  $M_{\mathrm{dR}}$  of  $(-1)^k c(w)$ .

The  $\zeta^m(s)$  are the motivic multizetas (F. Brown [1] 2.3). The algebra denoted  $M$  in the introduction is  $O(M_{\mathrm{dR}})$ . Before [1], only the restriction of the  $\zeta^m(s)$  to the orbit of 1 was considered. This gave rise to painful contortions. By 3.5, the value of  $\zeta^m(s)$  at the complex point  $dch$  of  $M_{\mathrm{dR}}$  is  $\zeta(s)$ . Every polynomial identity with rational coefficients in the  $\zeta^m(s)$  therefore gives, by evaluation at  $dch$ , a polynomial relation among the  $\zeta(s)$ . Grothendieck’s yoga of periods predicts that every polynomial relation among the  $\zeta(s)$  comes from geometry; more precisely, that it is obtained as above.

For every word  $w$  in 0 and 1, similarly write  $c^m(w)$  for the restriction of  $c(w)$  to  $M_{\mathrm{dR}}$ . Relying on the remark following 3.5, one can show that the  $c^m(w)$  are linear combinations with integer coefficients of the  $\zeta^m(s)$ .

By construction,  $H_{\mathrm{dR}}$  acts on  $M_{\mathrm{dR}}$ . Its Lie algebra therefore acts by derivations on the affine algebra of  $M_{\mathrm{dR}}$ , a quotient of that of  $\Pi$ . These derivations, which have a meaning only once one has lifted from the  $\zeta(s)$  to the  $\zeta^m(s)$ , will play a key role in F. Brown’s arguments.

## 5. Mixed Tate motives

### 5.1

We are here in one of the cases in which the philosophy of motives is not only a precious guide, but allows proofs. Over every field  $K$ , Voevodsky has defined a motivic triangulated category  $V_K$ , endowed with a tensor product, and containing Tate objects  $\mathbb{Z}(n)$  ( $n \in \mathbb{Z}$ ). If  $K$  is a

number field, after tensoring with  $\mathbb{Q}$  one can construct a subcategory  $MT(K)$  of  $V_K \otimes \mathbb{Q}$ , the category of mixed Tate motives over  $K$ , and this category is Tannakian. This rests on our knowledge of the  $K$ -theory groups of a number field, due to the transcendental determination of  $H_i(BGL(n, K), \mathbb{C})$  for  $n \gg i$ .

All this is for me a black box. It is difficult to open, and this is why the identities promised by 0.2 are not explicit.

## 5.2

Let us review some properties of the category  $MT(K)$ . It is a Tannakian category over  $\mathbb{Q}$ . It contains a rank-one object  $\mathbb{Q}(1)$ , the Tate motive. Since this motive has rank one,  $\mathbb{Q}(n) := \mathbb{Q}(1)^{\otimes n}$  is defined for  $n \in \mathbb{Z}$ . The objects of  $MT(K)$  are endowed with a finite increasing functorial filtration, compatible with the tensor product, the weight filtration  $W$ . It is indexed by even integers. The functor  $M \mapsto \text{Gr}^W(M)$  is exact, and  $\text{Gr}_{-2n}^W(M)$  is a sum of copies of  $\mathbb{Q}(n)$ .

Suppose, for simplicity, that  $K = \mathbb{Q}$ . Then one has Betti and de Rham fiber functors

$$\omega_B, \omega_{\text{dR}} : MT(K) \longrightarrow \mathbb{Q}\text{-vector spaces}$$

and a comparison isomorphism

$$\text{comp}_{\text{dR}, B} : \omega_B \otimes \mathbb{C} \longrightarrow \omega_{\text{dR}} \otimes \mathbb{C}.$$

Note that, because  $K = \mathbb{Q}$ , the fiber functor  $\omega_{\text{dR}}$  coincides with the fiber functor  $\omega$  of [2]. For the motive  $\mathbb{Q}(1)$ ,  $\omega_B(\mathbb{Q}(1))$ ,  $\omega_{\text{dR}}(\mathbb{Q}(1))$ , and  $\text{comp}_{\text{dR}, B}$  are as in 2.4.

The functor  $\omega_{\text{dR}}$  is endowed with a Hodge filtration  $F$ , opposite, up to a renumbering, to the weight filtration of  $\omega_{\text{dR}}(M)$  by the  $\omega_{\text{dR}}(W_{2n}M)$ . This makes it into a graded functor, as in 3.1.

## 5.3

If  $T$  is a Tannakian category, we shall call a  $T$ -algebra an Ind-object  $A$  of  $T$ , endowed with a product  $A \otimes A \rightarrow A$ . One defines the category of affine  $T$ -schemes as the opposite of that of commutative  $T$ -algebras with unit, and an affine  $T$ -group scheme as a group in this category. For  $T$  a category of motives, one replaces the prefix  $T$  by the adjective motivic.

## 5.4

The  $\pi_1(X; b, a)_B$ ,  $\pi_1(X; b, a)_{\text{dR}}$  of 3.1, 3.3, and the comparison isomorphism between their complexifications are motivic: they are images of a motivic group scheme  $\pi_1(X; b, a)_{\text{mot}}$  ([2] §4). The morphisms “composition of paths” and (3.3.1) are motivic as well.

If  $X = \mathbb{P}^1 - \{0, 1, \infty\}$  and  $a, b$  are chosen among the tangential base points 0, 1 of 3.4, the points 0, 1,  $\infty$  remain distinct after reduction modulo  $p$ , and the tangent vectors chosen at 0 and 1 remain nonzero. One can deduce that the affine algebras  $O(\pi_1(X; b, a)_{\text{mot}})$  are Ind-objects of a Tannakian subcategory  $MT(\mathbb{Z})$  of  $MT(\mathbb{Q})$ , that of mixed Tate motives with good reduction over  $\text{Spec}(\mathbb{Z})$  ([2] 1.7, 1.4, 1.8). In  $MT(\mathbb{Z})$ ,

$$\text{Ext}^1(\mathbb{Q}(0), \mathbb{Q}(n)) \text{ is } \begin{cases} \text{one-dimensional over } \mathbb{Q}, & n \text{ odd and } n \geq 3, \\ 0, & \text{otherwise,} \end{cases} \quad (5.4.1)$$

$$\text{Ext}^2(\mathbb{Q}(0), \mathbb{Q}(n)) = 0. \quad (5.4.2)$$

## 5.5

Let  $T$  be a Tannakian category over a field  $K$ , and let  $\omega$  be a fiber functor over an extension  $K'$  of  $K$ . The family of the  $\omega(Y)$ , for  $Y$  in  $T$ , endowed with the  $\omega(f) : \otimes \omega(Y_i) \rightarrow \otimes \omega(Z_j)$  for

$f : \otimes Y_i \rightarrow \otimes Z_j$  a morphism of  $T$ , is a family of  $K'$ -vector spaces endowed with a structure of multilinear algebra. The subgroup scheme of  $\prod \mathrm{GL}(\omega(Y))$  (product over  $\mathrm{Ob}(T)$ ) that respects this structure is the group scheme  $\mathrm{Aut}(\omega)$  of automorphisms of the fiber functor  $\omega$ . It is denoted  $\pi(T, \omega)$  and is called the fundamental group of  $T$  at  $\omega$ . One will see in examples that it is of the “same nature” as the  $\omega(Y)$ , for example that its affine algebra is graded if the functor  $\omega$  is graded. The explanation of this fact is that there exists a  $T$ -group scheme  $\pi(T)$ , acting functorially on each object of  $T$ , such that for every fiber functor  $\omega$ ,  $\pi(T, \omega)$  is  $\omega(\pi(T))$ .

Suppose  $K' = K$ . In this case,  $\omega$  induces an equivalence of  $T$  with the category of finite-dimensional linear representations of  $\pi(T, \omega)$ . If  $T$  is a category of motives, the fundamental groups are also called motivic Galois groups.

## 5.6

Let  $\langle(Y_i)\rangle$  be the Tannakian subcategory of  $T$  generated (by subquotients, sums, duals, and tensor products) by a family  $(Y_i)$  of objects of  $T$ . A  $T$ -structure on the family of the  $\omega(Y_i)$  is the image by  $\omega$  of a morphism of  $\langle(Y_i)\rangle$ . More precisely, it is the datum of the image by  $\omega$  of subquotients of sums of tensor products of the  $Y_i$  and  $Y_i^\vee$ , and of morphisms between such subquotients. If  $T$  is a category of motives, one will rather say motivic structure. The subgroup of  $\prod \mathrm{GL}(\omega(Y_i))$  which respects all  $T$ -structures is therefore the fundamental group of  $\langle(Y_i)\rangle$  at  $\omega$ . Every algebraic subgroup of a group  $\mathrm{GL}(V)$  is the subgroup respecting a few subspaces and elements in tensor spaces on  $V$ . The equivalence 5.5 therefore ensures that this fundamental group of  $\langle(Y_i)\rangle$  at  $\omega$  is the quotient of  $\pi(T, \omega)$  which acts faithfully on the family of the  $\omega(Y_i)$ . Same terminology if one starts from a family of ind-objects of  $T$ : one replaces it by the family of subobjects in  $T$ .

Take  $T := MT(\mathbb{Z})$ . With the notation  $B/\mathrm{dR}$  of 2.3, set

$$G_{B/\mathrm{dR}} := \pi(MT(\mathbb{Z}), \omega_{B/\mathrm{dR}}). \quad (5.6.1)$$

The group scheme  $H_{B/\mathrm{dR}}$  of automorphisms of  $\pi_1(X; 1, 0)_{B/\mathrm{dR}}$  “respecting all motivic structures” considered in 4.1 is the quotient of  $G_{B/\mathrm{dR}}$  acting faithfully on

$$\pi_1(X; 1, 0)_{B/\mathrm{dR}} = \omega_{B/\mathrm{dR}}(\pi_1(X; 1, 0)_{\mathrm{mot}}).$$

By the equivalence 5.5, a closed subscheme of  $\pi_1(X; 1, 0)_B$  stable under  $H_B$  is the Betti realization of a closed subscheme of  $\pi_1(X; 1, 0)_{\mathrm{mot}}$ . Example: there is a unique motivic subscheme  $M_{\mathrm{mot}}$  of  $\pi_1(X; 1, 0)_{\mathrm{mot}}$  whose Betti realization is the orbit closure  $M_B$ ; the subscheme  $M_{\mathrm{dR}}$  (4.2) of  $\pi_1(X; 1, 0)_{\mathrm{dR}}$  is its de Rham realization.

## 5.7

The action of  $G_{B/\mathrm{dR}}$  on  $\mathbb{Q}(1)_{B/\mathrm{dR}}$  defines a morphism  $t$  from  $G_{B/\mathrm{dR}}$  to the multiplicative group  $\mathbb{G}_m$ . For every  $M$  in  $MT(\mathbb{Z})$ ,  $G_{B/\mathrm{dR}}$  respects the weight filtration of  $\omega_{B/\mathrm{dR}}M$ , and since  $\mathrm{Gr}_{-2n}^W(M)$  is a sum of copies of  $\mathbb{Q}(n)$ , the action of  $g$  in  $G_{B/\mathrm{dR}}$  on  $\mathrm{Gr}_{-2n}^W(M)$  is multiplication by  $t(g)^n$ . The kernel  $U_{B/\mathrm{dR}}$  of  $t$  therefore acts trivially on the  $\mathrm{Gr}_{-2n}^W(M)$ : it is a pro-unipotent group scheme.

Let  $\tau(\lambda)$  be multiplication by  $\lambda^n$  on  $\omega_{\mathrm{dR}}^n(M)$ . The  $\tau(\lambda)$  define a section  $\tau : \mathbb{G}_m \rightarrow G_{\mathrm{dR}}$  of  $t$ , which makes  $G_{\mathrm{dR}}$  a semidirect product  $\mathbb{G}_m \ltimes U_{\mathrm{dR}}$ . The action of  $\mathbb{G}_m$  on  $U_{\mathrm{dR}}$  (by inner automorphisms in  $G_{\mathrm{dR}}$ ) will again be denoted  $\tau$ . It permits one to define the graded Lie algebra  $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$ . The vanishing (5.4.2) implies that this graded Lie algebra is free, and (5.4.1) that it is freely generated by one generator in each odd degree  $\geq 3$ . The functor  $\omega_{\mathrm{dR}}$  induces an equivalence of categories from  $MT(\mathbb{Z})$  with the category of finite-dimensional graded vector spaces, endowed with an action, compatible with the gradings and therefore nilpotent, of  $\mathrm{Lie}^{\mathrm{gr}}(U_{\mathrm{dR}})$ .

## 5.8

Let  $P$  be the scheme of isomorphisms of fiber functors from  $\omega_{\text{dR}}$  to  $\omega_B$ . The comparison isomorphism  $\text{comp}_{B,\text{dR}}$  is a complex point of  $P$ . The scheme  $P$  is therefore nonempty, and is a  $G_{\text{dR}}$ -torsor. Since  $G_{\text{dR}}$  is an extension of  $\mathbb{G}_m$  by  $U_{\text{dR}}$ , a Galois cohomology argument shows that every  $G_{\text{dR}}$ -torsor has a rational point: there exists an isomorphism of fiber functors  $p : \omega_{\text{dR}} \rightarrow \omega_B$ . Write

$$p = \text{comp}_{B,\text{dR}} \cdot a \quad (5.8.1)$$

with  $a \in G_{\text{dR}}(\mathbb{C})$ . It follows from (5.8.1) that for every  $M$  in  $MT(\mathbb{Z})$ , if one identifies  $\omega_{\text{dR}}(M) \otimes \mathbb{C}$  with  $\omega_B(M) \otimes \mathbb{C}$  by  $\text{comp}_{B,\text{dR}}$ ,  $\omega_B(M) \subset \omega_B(M) \otimes \mathbb{C} = \omega_{\text{dR}}(M) \otimes \mathbb{C}$  verifies

$$\omega_B(M) = a(\omega_{\text{dR}}(M)). \quad (5.8.2)$$

Since  $\omega_B(\mathbb{Q}(1)) = 2\pi i\mathbb{Q}$ , one has  $t(a) \in 2\pi i\mathbb{Q}^*$ , and  $a = u\tau(\lambda)$  with  $u \in U_{\text{dR}}(\mathbb{C})$  and  $\lambda \in 2\pi i\mathbb{Q}^*$ . A compatibility with complex conjugation makes it possible to show that one can choose ([2] 2.12)

$$a \in U_{\text{dR}}(\mathbb{R}) \cdot \tau(2\pi i). \quad (5.8.3)$$

## 6. The space $\tilde{H}(2, 3)$ of functions on $\pi_1(X; 1, 0)_{\text{dR}}$

From now on,  $X := \mathbb{P}^1 - \{0, 1, \infty\}$ , and the notation of 3.4 is in force. By abuse of language, write  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$  (respectively  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$ ) for the group scheme over  $\mathbb{Q}$  such that, for every  $\mathbb{Q}$ -algebra  $R$ , the set of its  $R$ -points is  $R\langle\langle e_0, e_1 \rangle\rangle$  (respectively  $R\langle\langle e_0, e_1 \rangle\rangle^*$ ). Recall that  $\pi_1(X; b, a)_{\text{dR}}$  is a copy  ${}_b\Pi_a$  of the subscheme  $\Pi$  of group-like elements (3.1.3) of  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$ . We shall regard it as contained in a copy  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle_{b,a}$  of  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ . A point  $x$  of  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$  will sometimes be denoted  ${}_bx_a$  when regarded as belonging to this copy.

### 6.1

As a  $\mathbb{Q}$ -vector space, the affine algebra  $O(\Pi)$  of  $\Pi$  has for basis the “coefficients of monomials”  $c(w)$  of 3.1. It is graded. If the word  $w$  in the alphabet  $\{0, 1\}$  has length  $d$ , the element  $c(w)$  of  $O(\Pi)$  is of degree  $-d$ , of codegree  $d$ . When we write  $B'$ , with decorations attached, for a set of words,  $B$ , with the same decorations attached, will be the corresponding set of  $c(w)$ .

**Definition 6.2.** (i)  $B'(2, 3)$  is the set of words that are concatenations of the words 01 and 001;

(ii)  $\tilde{H}(2, 3)$  is the vector subspace of  $O(\Pi)$  with basis  $B(2, 3)$ .

The restrictions to  $M_{\text{dR}}$  of the  $c(w)$  in  $B(2, 3)$  are those of the  $(-1)^k \zeta^m(s_1, \dots, s_k)$  where each  $s_i$  is 2 or 3.

For every graded vector space  $Z$ , with finite-dimensional homogeneous components, the Poincaré series of  $Z$  is

$$P(Z, t) = \sum \dim(Z_d) t^d. \quad (6.2.1)$$

If  $\tilde{H}(2, 3)$  is graded by the codegree, one has

$$P(\tilde{H}(2, 3), t) = (1 - t^2 - t^3)^{-1}. \quad (6.2.2)$$

This is verified by expanding  $(1 - t^2 - t^3)^{-1} = \sum (t^2 + t^3)^n$ .

### 6.3

The words  $w \in B'(2, 3)$  are characterized by:

- a. they do not begin with 1 and do not end with 0;
- b. they do not contain two consecutive 1's;
- c. they do not contain three consecutive 0's.

For a function  $f \in O(\Pi)$  to be a linear combination of  $c(w)$  where  $w$  satisfies a, b, or c is equivalent respectively to:

- a'. for  $\gamma$  in  $\Pi$ ,  $f(\exp(ue_1)\gamma \exp(te_0))$  is independent of  $t$  and  $u$ ;
- b'. for  $\gamma, \delta$  in  $\Pi$ ,  $f(\gamma \exp(te_1)\delta)$  is of degree  $\leq 1$  in  $t$ ;
- c'. for  $\gamma, \delta$  in  $\Pi$ ,  $f(\gamma \exp(te_0)\delta)$  is of degree  $\leq 2$  in  $t$ .

To verify this, note that the linear combinations of elements of  $\Pi(\mathbb{Q})$  are dense in  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ , and that if  $f$  is interpreted as a continuous linear form on  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ , condition a' (respectively b', respectively c') is equivalent to the same condition, where one allows  $\gamma, \delta$  to be arbitrary in  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ . The conditions a', b', c' are motivic, because they are expressible in terms of composition of paths and of (3.3.1). Here one must interpret  $\gamma$  (respectively  $\gamma, \delta$ ) as being in  ${}_1\Pi_0$  (respectively in  ${}_1\Pi_1$  and  ${}_1\Pi_0$ , respectively in  ${}_1\Pi_0$  and  ${}_0\Pi_0$ ).

To be convinced of the motivic character of a', b', c', one can observe that these conditions admit a ‘‘Betti’’ formulation. One obtains it by regarding  $f$  as a function on  $\pi_1(X; 1, 0)$ , and replacing  $\exp(te_0)$  (respectively  $\exp(te_1)$ ) by  $\sigma_0^t$  (respectively  $\sigma_1^t$ ) with  $t \in \mathbb{Z}$  and  $\sigma_0$  (respectively  $\sigma_1$ ) a positive turn around 0 in the punctured tangent space to  $\mathbb{P}^1$  at 0 (respectively at 1) (cf. 3.3).

We leave to the reader the verification of the following lemma:

**Lemma 6.4.** Let  $f = \sum \lambda_w c(w)$  in  $O(\Pi)$ . In order that, for all  $\gamma_0, \dots, \gamma_r$  in  $\Pi$ ,

$$f(\gamma_0 \exp(te_0)\gamma_1 \exp(te_0) \cdots \exp(te_0)\gamma_r) \quad (6.4.1)$$

be of degree  $\leq d$  in  $t$ , it is necessary and sufficient that for every  $w$  such that  $\lambda_w \neq 0$ , the total length of at most  $r$  disjoint strings of consecutive 0's in  $w$  be  $\leq d$ .

### 6.5

Let  $B'_\ell(2, 3)$  (respectively  $B'_{\leq \ell}(2, 3)$ ) be the set of words  $w$  in  $B'(2, 3)$  having exactly (respectively at most)  $\ell$  factors 001. With the convention of 6.1, let  $N_\ell$  be the subspace of  $\tilde{H}(2, 3)$  with basis  $B_{\leq \ell}(2, 3)$ . It follows from 6.4 that for  $f$  in  $\tilde{H}(2, 3)$  to be in  $N_\ell$ , it is necessary and sufficient that for  $r = \ell + 1$ , and every choice of the  $\gamma_i$ , (6.4.1) be of degree  $\leq 2\ell + 1$  in  $t$ . This property is motivic. The action of  $G_{\text{dR}}$  on  $O({}_1\Pi_0)$  therefore respects  $\tilde{H}(2, 3)$  and its increasing filtration  $N$ .

**Proposition 6.6.** The action of  $U_{\text{dR}}$  on  $\text{Gr}^N \tilde{H}(2, 3)$  is trivial.

**Proof.** The graded Lie algebra of  $U_{\text{dR}}$  is generated by elements  $\nu_d$  of odd degrees  $d \geq 3$ , and, if  $O({}_1\Pi_0)$  is graded by the degree (6.1), the action is compatible with the gradings. It suffices to show that each  $\nu_d$  acts trivially. Let  $w$  be a word of length  $n$ , a concatenation of words 01 and of  $\ell$  words 001. We have  $n \equiv \ell \pmod{2}$ . The image of  $c(w)$  by  $\nu_d$  is a linear combination of  $c(w')$  with  $w'$  of length  $n - d$  and a concatenation of words 01 and of  $\ell' \leq \ell$  words 001. We have  $n - d \equiv \ell' \pmod{2}$ , and,  $d$  being odd,  $\ell$  and  $\ell'$  do not have the same parity. That  $\ell' < \ell$  proves 6.6.

## 6.7

It follows from 6.6 that the action of  $\mathrm{Lie}^{\mathrm{gr}} U_{\mathrm{dR}}$  on  $N_\ell/N_{\ell-2}$  factors through the abelianization  $\mathrm{Lie}^{\mathrm{gr}} U_{\mathrm{dR}}^{\mathrm{ab}}$ . The action of  $\nu_d$  on  $N_\ell/N_{\ell-2}$  therefore does not depend, up to a factor, on the choice of the generators  $\nu_d$ . It is given by a map

$$\mathrm{Gr}^N(\nu_d) : \mathrm{Gr}_\ell^N(\tilde{H}(2, 3)) \longrightarrow \mathrm{Gr}_{\ell-1}^N(\tilde{H}(2, 3)). \quad (6.7.1)$$

The vector space  $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))$  has for basis the image of  $B_\ell(2, 3)$  (6.5). It is graded by the codegree. Let  $B'_\ell(2, 3)_n$  be the part of  $B'_\ell(2, 3)$  consisting of words of length  $n$ . Write  $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))_n$  for the direct factor of codegree  $n$  of  $\mathrm{Gr}_\ell^N(\tilde{H}(2, 3))$ . It has for basis the image of  $B_\ell(2, 3)_n$ . It is nonzero only for  $n \equiv \ell \pmod{2}$  and  $n - 3\ell \geq 0$ .

The morphisms (6.7.1) induce

$$\sum_d \mathrm{Gr}^N(\nu_d) : \mathrm{Gr}_\ell^N(\tilde{H}(2, 3))_n \longrightarrow \bigoplus_d \mathrm{Gr}_{\ell-1}^N(\tilde{H}(2, 3))_{n-d}, \quad (6.7.2)$$

where it suffices to sum over odd  $d$  satisfying  $3 \leq d \leq 3 + n - 3\ell$ . The left-hand side of (6.7.2) has a basis indexed by  $B'_\ell(2, 3)_n$ . The right-hand side has a basis indexed by the union of the  $B'_{\ell-1}(2, 3)_{n-d}$ . If  $\ell \geq 1$ , one obtains a bijection between these sets of words by associating to the word  $w \in B'_\ell(2, 3)_n$  the word  $\bar{w}$  deduced from it by removing from  $w$  its final  $(001)(01) \cdots (01)$ .

F. Brown computes the morphisms (6.7.2) and proves:

**Theorem 6.8.** For  $\ell \geq 1$ , the morphisms (6.7.2) are isomorphisms.

The strategy that he uses to prove 6.8 will be explained in §8.

**Corollary 6.9.** The restriction morphism to  $M_{\mathrm{dR}}$

$$\mathrm{restr} : \tilde{H}(2, 3) \longrightarrow O(M_{\mathrm{dR}}) \quad (6.9.1)$$

is injective.

Applying (6.2.2), one deduces from 6.9:

**Corollary 6.10.** The Poincaré series of  $O(M_{\mathrm{dR}})$  satisfies

$$P(O(M_{\mathrm{dR}}), t) \geq (1 - t^2 - t^3)^{-1} \quad (6.10.1)$$

(coefficientwise inequality), with equality if and only if (6.9.1) is bijective.

Assume 6.8, and prove 6.9.

**Lemma 6.11.** The morphism (6.9.1) is injective on  $N_0$ .

**Proof.** The morphism induced by (6.9.1) from  $N_0$  to  $O(M_{\mathrm{dR}})$  is compatible with the gradings by the codegree. It therefore suffices to verify its injectivity separately in each codegree. The  $c((01)^n)$  form a basis of  $N_0$ , and  $c((01)^n)$  is of codegree  $2n$ . Since these codegrees are distinct, it suffices to verify that for each  $n$  the restriction of  $c((01)^n)$  to  $M_{\mathrm{dR}}$  is nonzero. The value of  $c((01)^n)$  at  $dch \in M_{\mathrm{dR}}(\mathbb{C})$  is  $(-1)^n \zeta(2^{\{n\}})$ , where  $\zeta(2^{\{n\}}) = \zeta(2, \dots, 2)$  with  $n$  copies of “2”. One concludes by observing that every multizeta number is  $> 0$ , hence nonzero.

**Remark 6.12.** Expanding the identities due to Euler

$$\prod_{n>0} (1 - z^2/n^2) = \sin(\pi z)/\pi z = \sum (-1)^n (\pi z)^{2n}/(2n+1)!,$$

one sees that  $\zeta(2^{\{n\}}) = \pi^{2n}/(2n+1)!$ . That  $\zeta(2^{\{n\}})$  is a rational multiple of  $\pi^{2n}$  already followed from the fact that the motivic subspace  $N_0$  of  $O(1\Pi_0)$  is fixed under  $U_{\mathrm{dR}}$ , hence is the de Rham realization of a sum of Tate motives. This forces  $\zeta(2^{\{n\}})$  to be a period of  $\mathbb{Q}(-2n)$ .



### 6.13. Proof of 6.8 $\Rightarrow$ 6.9

Filter the image  $\text{restr}(\tilde{H}(2, 3))$  of  $\tilde{H}(2, 3)$  in  $O(M_{\text{dR}})$  by the images of the  $N_\ell$ . It suffices to show the injectivity of

$$\tilde{H}(2, 3) \longrightarrow \text{restr}(\tilde{H}(2, 3)) \quad (6.13.1)$$

after passage to the associated graded. We prove by induction on  $\ell \geq 0$  the injectivity on  $\text{Gr}_\ell^N(\tilde{H}(2, 3))$ . Lemma 6.11 proves this injectivity for  $\ell = 0$ . If  $\ell > 0$ , consider the morphism

$$\sum \nu_d : \text{Gr}_\ell^N \tilde{H}(2, 3) \longrightarrow \bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2, 3),$$

where the target is a sum of copies of  $\bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2, 3)$  indexed by the odd integers  $d \geq 3$ . Theorem 6.8 ensures that this morphism is bijective. The morphism (6.9.1) commutes with the derivations  $\nu_d$ . The diagram

$$\begin{array}{ccc} \text{Gr}_\ell^N \tilde{H}(2, 3) & \xrightarrow{\text{Gr}_\ell^N(6.13.1)} & \text{Gr}_\ell^N \text{restr}(\tilde{H}(2, 3)) \\ \sum \nu_d \downarrow \sim & & \downarrow \sum \nu_d \\ \bigoplus \text{Gr}_{\ell-1}^N \tilde{H}(2, 3) & \xrightarrow{\text{Gr}_{\ell-1}^N(6.13.1)} & \bigoplus \text{Gr}_{\ell-1}^N \text{restr}(\tilde{H}(2, 3)) \end{array}$$

is therefore commutative. The induction hypothesis ensures that  $\text{Gr}_{\ell-1}^N(6.13.1)$  is injective. Since the first vertical arrow is injective,  $\text{Gr}_\ell^N(6.13.1)$  is injective too.

## 7. Structure of the orbit closure $M_{\text{dR}}$

### 7.1

Consider the groupoid of the  ${}_b\Pi_a$  for  $a, b \in \{0, 1\}$ , endowed with  $e_0 \in \text{Lie}({}_0\Pi_0)$  and  $e_1 \in \text{Lie}({}_1\Pi_1)$ . Let  $V_{\text{dR}}$  be the group scheme of automorphisms of this structure. The action of  $U_{\text{dR}}$  on the groupoid of the  ${}_b\Pi_a$  factors through a morphism

$$I : U_{\text{dR}} \longrightarrow V_{\text{dR}}. \quad (7.1.1)$$

We write  $IU_{\text{dR}}$  for the image of  $U_{\text{dR}}$  in  $V_{\text{dR}}$ .

The group of automorphisms that respect  $e_0 \in \text{Lie}({}_0\Pi_0)$  and  $e_1 \in \text{Lie}({}_1\Pi_1)$  only up to a factor, the same for  $e_0$  and  $e_1$ , is a semidirect product  $\mathbb{G}_m \ltimes V_{\text{dR}}$ . We write  $\circ$  for its group law, and  $\tau$  for the inclusion of  $\mathbb{G}_m$ . The action of  $\mathbb{G}_m$  on the  ${}_b\Pi_a$  is that of the subgroup  $\mathbb{G}_m$  of  $G_{\text{dR}}$ . It corresponds to the gradings.

By [2] 5.9,

$$v \longmapsto v({}_110) : V_{\text{dR}} \longrightarrow {}_1\Pi_0 \quad (7.1.2)$$

is an isomorphism of schemes. This isomorphism identifies  $V_{\text{dR}}$  with  $\Pi$ , endowed with a new group law, denoted  $\circ$ . The action  $\tau$  of  $\mathbb{G}_m$  which defines the grading of  $O({}_1\Pi_0)$  fixes  ${}_110$ . It follows that (7.1.2) is compatible with the gradings. Since it sends identity element to identity element, it defines an isomorphism of graded vector spaces

$$\text{Lie}^{\text{gr}} V_{\text{dR}} \xrightarrow{\sim} \text{Lie}^{\text{gr}} \Pi. \quad (7.1.3)$$

The graded Lie algebra  $\text{Lie}^{\text{gr}} V_{\text{dR}}$  is thus identified with  $\text{Lie}^{\text{gr}} \Pi$ , endowed with a new bracket, the Ihara bracket  $\{, \}$ .

## 7.2

Write  $x \mapsto \gamma_{b,a}\langle x \rangle$  for the action on  ${}_b\Pi_a$  of  $\gamma$  in  $(\Pi, \circ)$ . It is computed as follows. The action on  ${}_a\Pi_a$  respects the group structure. For  $a = 0$ , it is induced by the automorphism

$$\gamma_{00} : e_0 \mapsto e_0, \quad e_1 \mapsto \gamma^{-1}e_1\gamma \quad (7.2.1)$$

of the completed enveloping algebra  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$  of  $\text{Lie } \Pi$ . To verify this, note that  $e_0 \in \text{Lie}({}_0\Pi_0)$  and  $e_1 \in \text{Lie}({}_1\Pi_1)$  are fixed by definition of  $V_{\text{dR}}$ . Writing  ${}_0(e_1)_0 = ({}_110)^{-1}{}_1(e_1)_1({}_110)$ , one obtains as promised

$${}_0(e_1)_0 \mapsto \gamma^{-1}{}_1(e_1)_1\gamma.$$

The automorphism  $\gamma_{10}$  of  ${}_1\Pi_0$  is induced by the  $\gamma_{00}$ -semilinear automorphism

$$\gamma_{10} : x \mapsto \gamma \gamma_{00}\langle x \rangle \quad (7.2.2)$$

of the right  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ -module  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ . Verify this as above, starting from  ${}_1x_0 = {}_110{}_0x_0$ . Since  $\gamma \circ x$  sends  ${}_110$  to  $\gamma_{10}\langle {}_1x_0 \rangle$ , this is also

$$\gamma_{10} : x \mapsto \gamma \circ x, \quad (7.2.3)$$

and therefore

$$x \circ y = x x_{00}\langle y \rangle. \quad (7.2.4)$$

## 7.3

Via  $\text{comp}_{\text{dR}, B}$ , regard  $\pi_1(X; b, a)_B$  as a  $\mathbb{Q}$ -structure on the complexification of  $\pi_1(X; b, a)_{\text{dR}} = {}_b\Pi_a$ , the Betti  $\mathbb{Q}$ -structure, denoted  $({}_b\Pi_a)_B$ . For this new  $\mathbb{Q}$ -structure,  $2\pi i e_0$  in  $\text{Lie}({}_0\Pi_0)_{\mathbb{C}}$  and  $2\pi i e_1$  in  $\text{Lie}({}_1\Pi_1)_{\mathbb{C}}$  are rational: their exponentials are the  $\sigma_0$  and  $\sigma_1$  of 6.3. One deduces:

**Lemma 7.4.** Let  $ch$  be a rational point of  $({}_1\Pi_0)_B$ . Regard its image by  $\text{comp}_{\text{dR}, B}$ , still denoted  $ch$ , as a complex point of  $(\Pi, \circ)$ . Then the complex point  $ch \circ \tau(2\pi i)$  of  $\mathbb{G}_m \ltimes V_{\text{dR}} = \mathbb{G}_m \ltimes (\Pi, \circ)$  transforms the  $\mathbb{Q}$ -structure of  ${}_b\Pi_a$  into its Betti  $\mathbb{Q}$ -structure.

## 7.5

Let  $a = u\tau(2\pi i) \in U_{\text{dR}}(\mathbb{R})\tau(2\pi i) \subset G_{\text{dR}}(\mathbb{C})$  be as in (5.8.3). If  $ch$  is as in 7.4, both  $ch\tau(2\pi i)$  and  $I(u)\tau(2\pi i)$  transform the  $\mathbb{Q}$ -structure of the  ${}_b\Pi_a$  into their Betti  $\mathbb{Q}$ -structure. If  $\gamma \in V_{\text{dR}}(\mathbb{C}) = \Pi(\mathbb{C})$  is defined by

$$ch \circ \tau(2\pi i) = I(u) \circ \tau(2\pi i) \circ \gamma, \quad (7.5.1)$$

then  $\gamma$  respects the  $\mathbb{Q}$ -structure, and hence is in  $V_{\text{dR}}(\mathbb{Q}) = \Pi(\mathbb{Q})$ . Writing again  $\tau$  for the action of  $\mathbb{G}_m$  on  $\Pi$ , one can rewrite (7.5.1) in the form

$$ch = I(u) \circ \tau(2\pi i)(\gamma).$$

Apply this to the straight path  $dch$ :

**Lemma 7.6.** For  $u$  as above, there exists  $\gamma$  in  $\Pi(\mathbb{Q})$  such that

$$dch = I(u) \circ \tau(2\pi i)(\gamma). \quad (7.6.1)$$

Every such  $\gamma$  verifies

$$\tau(-1)(\gamma) = \gamma. \quad (7.6.2)$$

**Proof.** Equation (7.6.2) follows from the fact that  $u$  and  $dch$ , and hence also  $\tau(2\pi i)(\gamma)$ , are real.

## 7.7

Apply (7.2.3) to (7.6.1). One obtains that the orbit of  $dch$  under  $U_{\text{dR}}(\mathbb{C})$  is

$$IU_{\text{dR}}(\mathbb{C}) \circ \tau(2\pi i)(\gamma) \subset \Pi_{\text{dR}}(\mathbb{C}).$$

To obtain the orbit under  $G_{\text{dR}} = \mathbb{G}_m \ltimes U_{\text{dR}}$ , it suffices to replace  $\tau(2\pi i)(\gamma)$  by its orbit  $\tau(\mathbb{C}^*)(\gamma)$  under  $\mathbb{G}_m$ :

$$G_{\text{dR}}(\mathbb{C})(dch) = IU_{\text{dR}}(\mathbb{C}) \circ \tau(\mathbb{C}^*)(\gamma). \quad (7.7.1)$$

Indeed,  $IU_{\text{dR}}$  is already stable under  $\mathbb{G}_m$ . As we already knew, the orbit (7.7.1) is, in the de Rham sense, defined over  $\mathbb{Q}$ . It is the set of complex points of

$$IU_{\text{dR}} \circ \tau(\mathbb{G}_m)(\gamma) \subset \Pi. \quad (7.7.2)$$

In (7.7.2), the product is that of  $(\Pi, \circ) = V_{\text{dR}}$ , and the result is regarded as a subscheme of  ${}_1\Pi_0$ .

In  $\mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle$ , write  $\gamma = \sum c_w w$ . By (7.6.2), the sum runs only over monomials  $w$  of even degree. If

$$\rho(\lambda) := \sum \lambda^{\deg(w)/2} c_w w,$$

then  $\rho(\lambda)$  is defined for  $\lambda \in \mathbb{G}_a$  and verifies

$$\rho(\lambda^2) = \tau(\lambda)(\gamma). \quad (7.7.3)$$

**Proposition 7.8.** The morphism of schemes

$$IU_{\text{dR}} \times \mathbb{G}_a \longrightarrow \Pi : (u, \lambda) \longmapsto u \circ \rho(\lambda) \quad (7.8.1)$$

induces an isomorphism from  $IU_{\text{dR}} \times \mathbb{G}_a$  to  $M_{\text{dR}} \subset {}_1\Pi_0$  defined in 4.2.

**Proof.** If  $u$  is in  $IU_{\text{dR}} \subset V_{\text{dR}} = \Pi \subset \mathbb{Q}\langle\langle e_0, e_1 \rangle\rangle^*$ , it can be written  $u = 1 + \sum e_w w$ , where the sum runs only over monomials of degree  $\geq 3$ . This is a consequence of the fact that  $\text{Lie}^{\text{gr}} U_{\text{dR}}$  is zero in degree  $< 3$ . The coefficient of  $e_0 e_1$  in  $u \circ \rho(\lambda)$  therefore coincides with that in  $\rho(\lambda)$ . The coefficient of  $e_0 e_1$  in  $dch$  is  $-\pi^2/6 = (2\pi i)^2/24$ . By (7.7.3), that in  $\gamma$  is therefore  $1/24$ , and that in  $\rho(\lambda)$  is  $\lambda/24$ . Starting from  $x = u \circ \rho(\lambda)$ , one recovers  $\lambda$  as  $24c(e_0 e_1)(x)$ . That  $x$  be of the form  $u \circ \rho(\lambda)$  is equivalent to saying that  $u$  defined by  $x = u \circ \rho(24c(e_0 e_1)(x))$  is in  $IU_{\text{dR}}$ . This is a closed condition, and (7.8.1) is an isomorphism onto a closed subscheme of  $\Pi$ , in which the orbit of  $dch$  is dense. This proves 7.8.

## 7.9

We shall write  $D$  for the image of  $\mathbb{G}_a$  by  $\rho$ . Thus  $\rho : \mathbb{G}_a \xrightarrow{\sim} D$ , and 7.8 says that the group law  $\circ$  induces an isomorphism

$$\circ : IU_{\text{dR}} \times D \xrightarrow{\sim} M_{\text{dR}}. \quad (7.9.1)$$

This isomorphism is compatible with the gradings of affine algebras, if  $O(D) = \mathbb{Q}[\lambda]$  is graded by giving  $\lambda$  degree  $-2$  (codegree 2).

## 7.10

By 6.6 applied to  $N_0$ , the  $(-1)^n c((01)^n)$ , and therefore their restrictions  $\zeta^m(2^{\{n\}})$  to  $M_{\text{dR}}$ , are  $U_{\text{dR}}$ -invariant. Transported from  $M_{\text{dR}}$  to  $IU_{\text{dR}} \times D$  by (7.9.1), the action of  $u \in U_{\text{dR}}$  is  $(a, \lambda) \mapsto (I(u)a, \lambda)$ . Viewed as functions on  $IU_{\text{dR}} \times D$ , the  $\zeta^m(2^{\{n\}})$  are therefore inverse images of functions on  $D$ . Up to a factor, there is only one such function in each even degree. To determine which one is  $\zeta^m(2^{\{n\}})$ , it suffices to test on  $dch$ . Set  $\pi_m^2 := 6\zeta^m(2)$ , a definition suggested by the identity  $\zeta(2) = \pi^2/6$ , and  $w_m^{2n} := (\pi_m^2)^n$ . By 6.12, one has:

**Lemma 7.11.** (i) For each  $n$ ,

$$\zeta^m(2^{\{n\}}) = \pi_m^{2n} / (2n+1)!.$$

(ii)  $IU_{\text{dR}}$  is the subscheme of  $M_{\text{dR}}$  defined by  $\pi_m^2 = 0$ .

**Proposition 7.12.** For each integer  $n \geq 1$ :

- (i)  $\zeta^m(2n)$  is a rational multiple of  $\pi_m^{2n}$  (which multiple is seen by evaluating on  $dch$ );
- (ii) as a function on  $IU_{\text{dR}} \times D$ ,  $\zeta^m(2n+1)$  is the inverse image of a function on  $IU_{\text{dR}}$ , and this function is a nonzero homomorphism from  $IU_{\text{dR}}$  to  $\mathbb{G}_a$ ;
- (iii) on abelianized groups, the  $\zeta^m(2n+1)$  for  $n \geq 1$  induce an isomorphism

$$U_{\text{dR}}^{\text{ab}} \xrightarrow{\sim} IU_{\text{dR}}^{\text{ab}} \xrightarrow{\sim} (\text{product of copies of } \mathbb{G}_a).$$

**Proof (sketch).** The depth filtration of  $\text{Lie } \Pi$  is the filtration by the subalgebras “degree in  $e_1 \geq i$ ”. Let  $D_i\Pi$  be the corresponding subgroups. If a subgroup  $Q$  of  $\Pi$ , viewed as a subgroup of  ${}_0\Pi_0$ , is stable under  $V_{\text{dR}}$ , it follows from (7.2.4) that it is also a subgroup of  $(\Pi, \circ)$ . This applies to the  $D_i\Pi$ .

The subgroup  $D_1\Pi$  is defined by the equation  $c(0) = 0$ . It contains the derived group  $\Pi'$ , which is defined by  $c(0) = c(1) = 0$ , hence contains both  $IU_{\text{dR}}$  and  $dch$ . The quotient  $D_1\Pi/D_2\Pi$  is also the quotient  $D_1\Pi/D_2\Pi$  for the group law  $\circ$ . It is abelian, and the  $c(0^{\{n-1\}}1)$  define an isomorphism of  $D_1\Pi/D_2\Pi$  with a product of copies of  $\mathbb{G}_a$ . Because  $IU_{\text{dR}} \subset (\Pi, \circ)$  has a Lie algebra generated in odd degrees, if  $n$  is even,  $c(0^{\{n-1\}}1)$  is zero on  $IU_{\text{dR}}$ ;  $\zeta^m(n)$ , transported by (7.9.1) into a function on  $IU_{\text{dR}} \times D$ , is the inverse image of a function on  $D$ , and (i) is proved by the arguments of 7.10. By (7.6.2), the  $c(0^{\{n-1\}}1)$  are zero on  $D$  for  $n$  odd. That  $\zeta(n) \neq 0$  ensures that, for  $n$  odd and  $n \geq 3$ , they are not zero on  $IU_{\text{dR}}$ . This proves (ii). Assertion (iii) follows from (ii) and from the fact that the graded Lie algebra of  $U_{\text{dR}}$  has one generator in each odd degree  $\geq 3$ .

**Corollary 7.13.** For all integers  $a, b \geq 0$ , there exist rational numbers  $\alpha_{a,b,r}$  ( $1 \leq r \leq a+b+1$ ), unique, such that

$$\zeta^m(2^{\{b\}}32^{\{a\}}) = \sum_{r=1}^{a+b+1} \alpha_{a,b,r} \zeta^m(2r+1) \zeta^m(2^{\{a+b+1-r\}}). \quad (7.13.1)$$

**Proof.** Regard, by (7.9.1),  $\zeta^m(2^{\{b\}}32^{\{a\}})$  as a function on  $IU_{\text{dR}} \times D$ . Expand it according to powers of a coordinate on  $D$ :

$$\zeta^m(2^{\{b\}}32^{\{a\}}) = \sum u_{2r+1} \pi_m^{2(a+b+1-r)},$$

with  $u_{2r+1}$  of codegree  $2r+1$  on  $IU_{\text{dR}}$ . By 6.6 and 7.9, a left translate of  $u_{2r+1}$  differs from  $u_{2r+1}$  only by addition of a constant. It is therefore a homomorphism from  $IU_{\text{dR}}$  to the additive group. Being of weight  $2r+1$ , it is by 7.12 a rational multiple of  $\zeta^m(2r+1)$ , and 7.13 follows from 7.11.

**Deus ex Machina 7.14 (D. Zagier [6]).** Set

$$A_{a,b}^r = \binom{2r}{2a+2}, \quad B_{a,b}^r = (1-2^{-2r}) \binom{2r}{2b+1}.$$

Then

$$\zeta(2^{\{b\}}32^{\{a\}}) = 2 \sum (-1)^r (A_{ab}^r - B_{ab}^r) \zeta(2r+1) \zeta(2^{\{a+b+1-r\}}). \quad (7.14.1)$$

This theorem suggests that the  $\alpha_{a,b,r}$  of (7.13.1) are the  $2(-1)^r (A_{a,b}^r - B_{a,b}^r)$ . How to prove this starting from 7.14 will be explained in §8.

## 7.15

The affine algebra  $O(U_{\text{dR}})$  of  $U_{\text{dR}}$  is in duality, in the graded sense, with the enveloping algebra of  $\text{Lie}^{\text{gr}}(U_{\text{dR}})$ . This enveloping algebra is a free associative algebra in generators of degrees the odd integers  $\geq 3$ . The Poincare series of  $O(U_{\text{dR}})$ , graded by the codegree, is therefore

$$P(O(U_{\text{dR}}), t) = \sum_{a \geq 0} \left( \sum_{n \geq 1} t^{2n+1} \right)^a = \left( 1 - \sum_{n \geq 1} t^{2n+1} \right)^{-1} = (1 - (t^3/(1-t^2)))^{-1} = \frac{1-t^2}{1-t^2-t^3}.$$

If the affine algebra of the additive group  $\mathbb{G}_a$ , with coordinate  $\lambda$ , is graded by giving  $\lambda$  codegree 2, one has  $P(O(\mathbb{G}_a), t) = (1-t^2)^{-1}$  and

$$P(O(U_{\text{dR}} \times \mathbb{G}_a), t) = (1-t^2-t^3)^{-1}.$$

The affine algebra  $O(IU_{\text{dR}})$  is a graded subalgebra of  $O(U_{\text{dR}})$ . By 7.9, one therefore has:

**Proposition 7.16.** The Poincare series of  $O(M_{\text{dR}})$ , graded by the codegree, satisfies

$$P(O(M_{\text{dR}}), t) \leq (1-t^2-t^3)^{-1} \quad (7.16.1)$$

(coefficientwise inequality), with equality if and only if the morphism  $I : U_{\text{dR}} \rightarrow IU_{\text{dR}}$  is an isomorphism, i.e. if  $U_{\text{dR}}$  acts faithfully on  $\pi_1(X; 1, 0)_{\text{dR}}$ .

Comparing with 6.10, one obtains:

**Theorem 7.17.** (i)  $U_{\text{dR}}$  acts faithfully on  ${}_1\Pi_0$ .

(ii) The morphism of  $\mathbb{Q}$ -vector spaces  $\tilde{H}(2, 3) \rightarrow O(M_{\text{dR}})$  is an isomorphism.

The affine algebra of  $\pi_1(X; 1, 0)_{\text{mot}}$  contains a  $\mathbb{Q}(-1)$  as a subquotient. It follows therefore from (i) that the action of  $G_{\text{dR}}$  on the affine algebra of  $\pi_1(X; 1, 0)_{\text{dR}}$  is faithful. The theory of Tannakian categories gives, as a consequence of (i):

**Corollary 7.18.** The affine algebra  $O(\pi_1(X; 1, 0)_{\text{mot}})$ , which is an Ind-object of  $MT(\mathbb{Z})$ , generates the Tannakian category  $MT(\mathbb{Z})$ .

More precisely, every mixed Tate motive with good reduction over  $\text{Spec}(\mathbb{Z})$  is deduced from subobjects in  $MT(\mathbb{Z})$  of  $O(\pi_1(X; 1, 0)_{\text{mot}})$  by the operations of tensor products, direct sums, duals, and subquotients.

Assertion (ii) of 7.17 is equivalent to saying that the  $\zeta^m(s)$  with  $s_i \in \{2, 3\}$  form a basis of  $O(M_{\text{dR}})$ .

## 8. Sketches of proofs

### 8.1

The group scheme  $V_{\text{dR}} = (\Pi, \circ)$  acts on  ${}_1\Pi_0$  (7.1), and therefore on its affine algebra  $O({}_1\Pi_0)$ . This action is the action of  $(\Pi, \circ)$  on  $O(\Pi)$  by left translations (7.2.3):  $x$  transforms  $f$  into  $y \mapsto f(x^{-1} \circ y)$ . These actions are given by (7.2.1), (7.2.2). Differentiating these formulae, one obtains the action of  $\text{Lie}(V_{\text{dR}}) = (\text{Lie } \Pi, \{, \})$  on  $O({}_1\Pi_0)$ , and dually the coaction of the Lie coalgebra. The latter is  $m/m^2$ , for  $m$  the ideal of functions vanishing at the identity element 1, and the coaction is the projection onto  $m/m^2 \otimes O(\Pi)$  of

$$f \longmapsto f(x^{-1} \circ y) - f(y)$$

in  $m \otimes O(\Pi) \subset O(\Pi \times \Pi)$ .

Goncharov gave a convenient formula for this coaction (see [5] Theorem 1.2). For  $w$  a word in the alphabet  $\{0, 1\}$ , consider the subwords  $cId$  of  $1w0$ , with  $I$  nonempty: one considers all decompositions  $w = w'Iw''$  of  $w$ , with  $I$  nonempty, and  $c, d$  are the letters adjacent to  $I$  in  $1w0$ . For each decomposition, put  $w \setminus I := w'w''$ . Formula for the coaction:

$$c(w) \mapsto - \sum p_{cd}(I) \otimes c(w \setminus I) \quad (8.1.1)$$

where  $p_{cd}(I)$  is the projection in  $m/m^2$  of the following element of  $m \subset O(\Pi)$ :

$$p_{cd}(I) = \text{projection of } \begin{cases} 0, & c = d, \\ c(I), & c = 1, d = 0, \\ \text{the antipode } c(I)^* \text{ of } c(I), & c = 0, d = 1, \end{cases} \quad (8.1.2)$$

where, if  $I = a_1 \cdots a_k$ , the antipode is  $(-1)^k c(a_k \cdots a_1)$ .

## 8.2

The action of  $U_{\text{dR}}$  on  $O(\Pi)$  factors through that of  $IU_{\text{dR}} \subset (\Pi, \circ)$ , and to obtain the coaction of the Lie coalgebra it remains to compose (8.1.1) with

$$\text{coLie}(\Pi, \circ) \longrightarrow \text{coLie}(IU_{\text{dR}}) \longrightarrow \text{coLie}(U_{\text{dR}}).$$

The  $c(I)$  and  $c(I)^*$  of (8.1.2) matter no longer except by their restriction to  $M_{\text{dR}}$ , taken modulo  $\pi_m^2$  and the square of the maximal ideal at 1.

We shall choose the generators  $\nu_{2r+1}$  of the Lie algebra  $\text{Lie}^{\text{gr}}(U_{\text{dR}})$  so that, writing again  $\nu_{2r+1}$  for the image of  $\nu_{2r+1}$  in the tangent space of  $IU_{\text{dR}} \subset \Pi$  at 1, one has

$$\langle \nu_{2r+1}, d\zeta^m(2r+1) \rangle = 1$$

(possible by 7.12(ii)). This determines the image of  $\nu_{2r+1}$  in  $\text{Lie}^{\text{gr}}(U_{\text{dR}}^{\text{ab}})$ . If  $r = a + b + 1$  and  $\alpha_{a,b}$  is the coefficient of  $\zeta^m(2r+1)$  in (7.13.1), then

$$\langle \nu_{2r+1}, d\zeta^m(2^{\{b\}} 3 2^{\{a\}}) \rangle = \alpha_{a,b}.$$

**Proposition 8.3.** Formula (7.14.1) remains true with  $\zeta$  replaced by  $\zeta^m$ .

**Proof.** Proceed by induction on  $a + b$ . If  $a + b = 0$ , the identity (7.14.1)<sup>m</sup> to be proved reduces to  $\zeta^m(3) = \zeta^m(3)$ . Suppose (7.14.1)<sup>m</sup> true for  $a + b < N$ . This ensures that, for  $a + b < N$  and  $r = a + b + 1$ , one has, with the notation of 7.14,

$$\zeta^m(2^{\{b\}} 3 2^{\{a\}}) = 2(-1)^r (A_{ab}^r - B_{ab}^r) \zeta^m(2r+1) \quad \text{on } IU_{\text{dR}}. \quad (8.3.1)$$

For  $a + b = N$ , if the  $\alpha_{a,b,r}$  are defined as in 7.13, one has, for the action of  $\nu_{2r+1}$ ,

$$\nu_{2r+1} \zeta^m(2^{\{b\}} 3 2^{\{a\}}) = -\alpha_{a,b,r} \zeta^m(2^{\{a+b+1-r\}}) : \quad (8.3.2)$$

apply 7.12. For  $r \leq N$ , compute the left-hand side by (8.1.1), (8.1.2). One must sum over the subwords  $I$  of  $w = (01)^b(001)(01)^a$ , of length  $2r + 1$ , and surrounded by 0, 1 or 1, 0 in  $1w0$ . The  $c(w \setminus I)$ , restricted to  $M_{\text{dR}}$ , are  $\pm \zeta^m(2^{\{a+b+1-r\}})$ . The  $c(I)$  or  $c(I)^*$  that appear in (8.1.2), restricted to  $M_{\text{dR}}$ , are  $\pm \zeta^m(2^{\{b'\}} 3 2^{\{a'\}})$  to which (8.3.1) applies by the induction hypothesis, or a  $c((01)^m 0)$  which, on  $\Pi'$ , is expressed by (3.5.2) as a sum of  $\zeta(2^{\{b'\}} 3 2^{\{a'\}})$  to which (8.3.1) applies. This permits the computation of the  $\alpha_{a,b,r}$  for  $r \leq N$ , and one finds that they have the expected value. This leaves unknown in (7.13.1) only the coefficient  $\alpha_{a,b,a+b+1}$ . It is determined by evaluating at  $dch$  and applying (7.14.1).

## 8.4

Armed with (8.3.1), one can now compute the morphisms (6.7.1) by (8.1.1), (8.1.2). To prove 6.8, the idea is that, in the matrix obtained for (6.7.2), the dominant terms, in the 2-adic sense, come from the  $2^{-2r}$  which appears in  $B_{ab}^r$  (7.14). If, in the target and source of (6.7.2), one orders the basis vectors in suitable orders which correspond by the bijection defined in 6.7, and if in the target one multiplies the basis vectors by suitable powers of 2, the matrix of (6.7.2) is a square matrix with integer coefficients whose reduction modulo 2 is triangular with 1's on the diagonal. This ensures that it is invertible.

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# Semisimplicity of Tensor Products in Characteristic $p$

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English translation, literal mathematical layer

**Abstract.** Let  $G$  be an affine group scheme over a field  $k$  of characteristic  $p \neq 0$ , and let  $(V_i)$  be a finite family of semisimple representations of  $G$ . We show that, if

$$\sum_i (\dim V_i - 1) < p,$$

then the tensor-product representation of  $G$  on  $\bigotimes_i V_i$  is again semisimple. Under the additional hypothesis that  $G$  is smooth, this theorem was proved by J.-P. Serre in 1994. We reduce to that case. More generally, the  $V_i$  may be taken to be objects of a Tannakian category over  $k$ .

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## 1 Introduction

Let  $T$  be a Tannakian category over a field  $k$  of characteristic  $p \neq 0$ . For the definition of Tannakian categories we refer to [3], p. 193, or to [1], 2.1 and 2.8. Recall that among the axioms are the existence of duals, and the existence of a fiber functor  $\omega$  with values in the vector spaces over a suitable extension  $K$  of  $k$ . The dimension  $\dim(V)$  of an object  $V$  of  $T$  is  $\dim_K \omega(V)$ . Since two fiber functors become isomorphic over a suitable larger extension ([1] 1.12, 1.13), this dimension does not depend on the choice of the fiber functor  $\omega$ .

**Theorem 0.1.** Let  $(V_i)_{i \in I}$  be a finite family of objects of  $T$ . If the  $V_i$  are semisimple and

$$\sum_{i \in I} (\dim(V_i) - 1) < p,$$

then the tensor product  $\bigotimes_{i \in I} V_i$  is a semisimple object of  $T$ .

**Example 0.2.** If  $T$  is the category of finite-dimensional linear representations of an affine group scheme over  $k$ , one obtains the theorem announced in the abstract.

Other examples are the category of finite-dimensional representations of a Lie algebra over  $k$ , or that of finite-dimensional  $p$ -representations of a  $p$ -Lie algebra over  $k$ . By passage to an algebraic envelope, these cases also reduce to the preceding one.



**Example 0.3.** Suppose that a group  $\Gamma$  acts on a commutative field  $K$  of characteristic  $p$ , and let  $k := K^\Gamma$  be the subfield of invariants. Let  $T$  be the category of finite-dimensional  $K$ -vector spaces, endowed with a semilinear action of  $\Gamma$ . With the tensor product over  $K$ , this category is Tannakian over  $k$ : it admits the fiber functor over  $K$  “underlying vector space.” If semilinear representations  $V_i$  of  $\Gamma$  on  $K$  are semisimple and

$$\sum_i (\dim_K(V_i) - 1) < p,$$

then the semilinear tensor-product representation of the  $V_i$  is therefore semisimple.

**0.4.** Let  $V$  be a finite-dimensional vector space over  $k$  of characteristic  $p$ . As explained in [4], §4, an automorphism  $g$  of  $V$  such that  $g^p = 1$  defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto g^t,$$

where  $g^t$  is defined by the binomial formula

$$g^t := \sum_{n < p} \binom{t}{n} (g - 1)^n. \quad (0.4.1)$$

Similarly, an endomorphism  $N$  of  $V$  such that  $N^p = 0$  defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto \exp(tN),$$

where  $\exp(tN)$  is defined by

$$\exp(tN) := \sum_{n < p} \frac{(tN)^n}{n!}. \quad (0.4.2)$$

Indeed, after any extension of scalars to a commutative  $k$ -algebra  $\Lambda$ , if for  $t \in \Lambda$  one defines  $\exp(tN)$  by (0.4.2), the hypothesis  $N^p = 0$  ensures that

$$\exp((t' + t'')N) = \exp(t'N) \exp(t''N).$$

**Definition 0.5.** Suppose that  $k$  is algebraically closed of characteristic  $p$ , and let  $G$  be a subgroup scheme of  $\mathrm{GL}(V)$ .

- (i)  $G$  is *saturated* if, for every  $g \in G(k)$  such that  $g^p = 1$ , the morphism  $t \mapsto g^t$  from  $\mathbb{G}_a$  to  $\mathrm{GL}(V)$  factors through  $G$ .
- (ii)  $G$  is *infinitesimally saturated* if, for every  $N \in \mathrm{Lie}(G)$  such that  $N^p = 0$  (in the sense of the  $p$ -Lie-algebra structure of  $\mathrm{Lie}(G)$ , or as an endomorphism of  $V$ , which amounts to the same thing), the morphism  $t \mapsto \exp(tN)$  from  $\mathbb{G}_a$  to  $\mathrm{GL}(V)$  factors through  $G$ .

We shall say that  $G$  is *doubly saturated* if it is saturated and infinitesimally saturated.

**0.6.** Let  $G$  be an affine group scheme of finite type over an algebraically closed field  $k$  of characteristic  $p$ . In the statement 0.1, take  $T$  to be the category of finite-dimensional representations of  $G$ , suppose that the family  $(V_i)$  of representations is faithful, and put  $V := \bigoplus V_i$ . To prove 0.1 when  $G$  is reduced, Serre [4] begins by reducing to the case in which  $G$  is saturated in  $\mathrm{GL}(V)$ . If that is the case, the finite group  $\pi_0(G)$  of connected components of  $G$  has order prime to  $p$ , and this permits a reduction to the case in which  $G$  is connected. For  $G$  not necessarily reduced, we use an analogous argument to reduce to the case in which  $G$  is infinitesimally saturated in  $\mathrm{GL}(V)$ , and our key result in that case is the following structure theorem. This theorem will provide a reduction to the case in which  $G$  is smooth.

Let  $G^0$  be the identity component of  $G$ , let  $G_{\mathrm{red}}^0$  be the reduced subgroup scheme of  $G^0$ , and let  $R_{\mathrm{u}} G_{\mathrm{red}}^0$  be the unipotent radical of  $G_{\mathrm{red}}^0$ . Both  $G_{\mathrm{red}}^0$  and  $R_{\mathrm{u}} G_{\mathrm{red}}^0$  are smooth, i.e. linear algebraic groups in the sense of the Chevalley seminar of 1956/58.

**Theorem 0.7.** Let  $(V_i)_{i \in I}$  be a finite faithful family of representations of  $G$ . Let  $V$  be the sum of the  $V_i$ . Suppose that each representation  $V_i$  has dimension  $< p$  and that, as a subgroup of  $\mathrm{GL}(V)$ ,  $G$  is infinitesimally saturated. Under these hypotheses:

- (i)  $G_{\mathrm{red}}^0$  and  $R_{\mathrm{u}} G_{\mathrm{red}}^0$  are invariant subgroup schemes of  $G$ , and  $G^0/G_{\mathrm{red}}^0$  is of multiplicative type.
- (ii) If the representations  $V_i$  are semisimple,  $G_{\mathrm{red}}^0$  is reductive.
- (iii) If  $G_{\mathrm{red}}^0$  is reductive, there exists in  $G^0$  a central connected subgroup scheme  $M$  of multiplicative type such that the morphism

$$M \times G_{\mathrm{red}}^0 \longrightarrow G^0$$

realizes  $G^0$  as a quotient of  $M \times G_{\mathrm{red}}^0$ .

## 2 Proof of Theorem 0.7

**Proposition 1.1.** In an affine group scheme  $G$  of finite type over a field, every connected subgroup scheme of multiplicative type is contained in a maximal connected subgroup scheme of multiplicative type.

**Proof, after a letter from M. Raynaud to the author.** It is enough to show that, in the set of connected subgroup schemes of multiplicative type of  $G$ , ordered by inclusion, every totally ordered subset  $\mathcal{M}$  is bounded above (Bourbaki, Ens. III, §2, no. 4, Theorem 2). For  $M$  in  $\mathcal{M}$ , the centralizer  $Z_G(M)$  of  $M$  in  $G$  is a subgroup scheme, automatically closed, and the map  $M \mapsto Z_G(M)$  is decreasing. The ordered set of closed subgroup schemes of  $G$  is noetherian. The set of the  $Z_G(M)$  for  $M$  in  $\mathcal{M}$  therefore has a least element  $C$ . Every  $M$  in  $\mathcal{M}$  is contained in the center  $Z(C)$  of  $C$ . This center is affine and commutative. The identity component of its largest subgroup scheme of multiplicative type (SGA3 XVII 7.2.1) bounds  $\mathcal{M}$  from above. This is the required upper bound.

**1.2.** Fix  $k$ ,  $G$ , and the finite family of representations  $(V_i)_{i \in I}$ , with sum  $V$ , satisfying the hypotheses of 0.7:  $k$  is algebraically closed of characteristic  $p \neq 0$ ,  $\dim(V_i) < p$ , and, as a subgroup scheme of  $\mathrm{GL}(V)$ ,  $G$  is infinitesimally saturated.

**Lemma 1.3.**

- (i) The identity component  $G^0$  of  $G$  is infinitesimally saturated in  $\mathrm{GL}(V)$ .
- (ii) If a representation  $W$  of  $G$  is semisimple, its restriction to  $G^0$  is also semisimple.
- (iii) If the reduced subgroup  $G_{\mathrm{red}}^0$  of  $G^0$  (respectively its unipotent radical  $R_{\mathrm{u}} G_{\mathrm{red}}^0$ ) is an invariant subgroup scheme of  $G^0$ , then it is one of  $G$ .

**Proof.** Assertion (i) follows from  $\mathrm{Lie} G = \mathrm{Lie} G^0$  and from the connectedness of  $\mathbb{G}_a$ .

It is enough to prove (ii) for  $W$  irreducible. Let  $S \subset W$  be the sum of the irreducible  $G^0$ -subrepresentations of  $W$ . We must show that  $S = W$ . Each  $g \in G(k)$  normalizes  $G^0$ . By transport of structure,  $S$  is therefore stable under  $g$ . Since  $k$  is algebraically closed,  $G$  is the union of the  $gG^0$  for  $g \in G(k)$ , and  $S$  is stable under  $G$ . Since  $W$  is irreducible and  $S \neq 0$ ,  $S$  is all of  $W$ .

Similarly, by transport of structure, each  $g \in G(k)$  normalizes  $G_{\mathrm{red}}^0$  and  $R_{\mathrm{u}} G_{\mathrm{red}}^0$ , and (iii) follows because  $G$  is the union of the  $gG^0$ .

**1.4.** Lemma 1.3 shows that, to prove 0.7 for  $G$ , it suffices to prove it for  $G^0$ . Moreover, if  $p = 2$ , each  $V_i$  has dimension one,  $G$  is of multiplicative type, and 0.7 is trivial.

In the rest of this section, besides the hypotheses of 1.2, we shall suppose that  $p \neq 2$  and that  $G$  is connected:  $G = G^0$ . We shall prove that  $G_{\text{red}}$  is an invariant subgroup scheme of  $G$  in 1.18, that  $G/G_{\text{red}}$  is of multiplicative type in 1.19, the invariance in  $G$  of  $R_u G_{\text{red}}$  in 1.20–1.23, 0.7(ii) in 1.24, and 0.7(iii) in 1.25.

**1.5.** Let  $H$  be a maximal connected subgroup of multiplicative type of  $G$  (1.1), and let  $Z_G^0(H)$  be the identity component of its centralizer  $Z_G(H)$ . The quotient

$$U := Z_G^0(H)/H$$

contains no subgroup  $\mu_p$ , since the inverse image in  $Z_G^0(H)$  of such a subgroup would be an extension of  $\mu_p$  by  $H$ , hence of multiplicative type (SGA3 XVII 7.1.1 b)). Thus  $U$  is unipotent (SGA3 XVII 4.3.1(v)).

**1.6.** Let  $X$  be the group of characters of  $H$ . The connectedness of  $H$  is equivalent to  $X$  having no torsion prime to  $p$ . Giving an action of  $H$  on a vector space  $W$  is equivalent to giving an  $X$ -grading

$$W = \bigoplus_{\alpha \in X} W^\alpha$$

of  $W$ , and this construction is functorial and compatible with tensor product. In particular,  $V$  and the  $V_i$  are  $X$ -graded, and the action by inner automorphisms of  $H$  on  $G$  defines an  $X$ -grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha$$

of  $\text{Lie}(G)$ .

**Lemma 1.7.** The central extension  $Z_G^0(H)$  of  $U$  by  $H$  is split.

Since  $U$  is unipotent and  $H$  is of multiplicative type, there is no nontrivial morphism from  $U$  to  $H$  (SGA3 XVII 2.4(ii)). The splitting guaranteed by Lemma 1.7 is therefore unique. We shall use the corresponding isomorphism of  $H \times U$  with  $Z_G^0(H)$  to regard  $U$  as a subgroup of  $Z_G^0(H)$ .

**Proof.** Let  $V_i^\alpha$  ( $\alpha \in X$ ) be the homogeneous components of the  $V_i$ . The group  $Z_G^0(H)$  acts on  $V_i^\alpha$ . Let  $\chi_i^\alpha$  be the determinant character of this representation. Its restriction to  $H$  is  $\dim(V_i^\alpha)$  times the character  $\alpha$  of  $H$ .

Let  $J \subset I \times X$  be the set of pairs  $(i, \alpha)$  such that  $V_i^\alpha \neq 0$ , and for  $j = (i, \alpha)$  put

$$V^j := V_i^\alpha, \quad \chi_j := \chi_i^\alpha.$$

The family  $(V_i)$  of representations of  $H$  is faithful: the  $\alpha$  for  $j = (i, \alpha)$  in  $J$  generate  $X$ . If  $j = (i, \alpha)$  is in  $J$ ,  $\dim V_i^\alpha \leq \dim V_i < p$  is prime to  $p$ . The restrictions to  $H$  of the  $\chi_j$  ( $j \in J$ ) therefore generate a subgroup of  $X$  of finite index prime to  $p$ . Let  $F$  be the kernel of

$$(\chi_j) : H \longrightarrow \mathbb{G}_m^J$$

and let  $Q$  be the cokernel. The group  $F$  is finite, reduced, and of order prime to  $p$ . In the commutative diagram with exact rows

$$\begin{array}{ccccccc} 1 & \longrightarrow & H/F & \longrightarrow & Z_G^0(H)/F & \longrightarrow & U \longrightarrow 1 \\ & & \parallel & & \downarrow & & \downarrow \\ 1 & \longrightarrow & H/F & \longrightarrow & \mathbb{G}_m^J & \longrightarrow & Q \longrightarrow 1, \end{array}$$

the morphism from the unipotent group  $U$  to the group of multiplicative type  $Q$  is trivial. The kernel of  $(\chi_j)$  therefore lifts  $U$  into  $Z_G^0(H)/F$ . Its inverse image in  $Z_G^0(H)$  is a central extension  $E$  of  $U$  by  $F$ . By Lemma 1.8 below, this extension is trivial. A lifting of  $U$  into  $E$  splits the central extension  $Z_G^0(H)$  of  $U$  by  $H$ .

**Lemma 1.8.** A central extension of a unipotent group by a finite reduced group of order prime to  $p$  is trivial.

We give two proofs of the lemma, based on different ideas. The second assumes that  $U$  is connected, which is sufficient for the application above.

**First proof.** Let

$$1 \longrightarrow F \longrightarrow E \longrightarrow U \longrightarrow 1$$

be a central extension as in the lemma, and let  $a$  be an integer prime to  $p$  that kills  $F$ . View  $E$  as a central extension in the category of fppf sheaves of groups on  $\mathrm{Spec}(k)$ . For any  $S/k$ ,  $U(S)$  admits a finite central filtration with successive quotients killed by  $p$  (apply SGA3 XV 3.5(v)). Hence every finite subset  $X$  of  $U(S)$  generates a finite  $p$ -group  $P_X$ . Let  $Q_X$  be its intersection with the image of  $E(S)$ . The inverse image of  $Q_X$  in  $E(S)$  is a central extension  $E_X$  of  $Q_X$  by  $F(S) = F^{\pi_0(S)}$ . The group  $H^2(Q_X, F(S))$  classifying these extensions is trivial, since it is killed both by  $a$  and by the power  $|Q_X|$  of  $p$ . The extension  $E_X$  is therefore trivial. Moreover, the lifting of  $Q_X$  into  $E_X$  is unique: it is the set of the  $a$ -th powers of elements of  $E_X$ .

Passing to the inductive limit over  $X$  and sheafifying, one deduces that the subsheaf of sets of  $E$  which is the image of

$$x \longmapsto x^a : E \longrightarrow E$$

is a subsheaf in groups lifting  $U$  into  $E$ .

**Second proof.** The action by right translations of  $F$  on  $E$  makes  $E$  an  $F$ -torsor over  $U$ . It is trivialized over the identity element, and therefore corresponds to a morphism

$$\pi_1(U, e) \longrightarrow F.$$

The reduced scheme  $U_{\mathrm{red}}$  is isomorphic to an affine space  $\mathbb{A}^N$ . Since

$$\pi_1(U, e) = \pi_1(U_{\mathrm{red}}, e) \simeq \pi_1(\mathbb{A}^N, 0)$$

has no nontrivial quotient of order prime to  $p$  (SGA4 XV 2.2), the  $F$ -torsor  $E$  is trivial: the identity component of  $E$  lifts  $U$  into  $E$ .

**Lemma 1.9.** Let  $f : V \rightarrow W$  be a morphism of schemes of finite type over  $k$ , let  $v \in V(k)$ , and let  $(df)_v$  be the morphism induced by  $f$  from the Zariski tangent space of  $V$  at  $v$  to the Zariski tangent space of  $W$  at  $f(v)$ . If  $V$  is smooth over  $k$  and  $(df)_v$  is an isomorphism, then  $f$  is etale at  $v$  and  $W$  is smooth over  $k$  in a neighborhood of  $f(v)$ .

**Proof.** Let  $A$  (respectively  $B$ ) be the completion of the local ring of  $V$  at  $v$  (respectively of  $W$  at  $f(v)$ ). Let  $m$  be the maximal ideal of  $A$  and  $n$  that of  $B$ . The injectivity of  $(df)_v$  ensures that  $A$  is a quotient of  $B$ . We must show that  $B \xrightarrow{\sim} A$  (SGA1 I 4.4). Let  $t_1, \dots, t_N$  in  $n$  lift a basis of  $n/n^2$ . The bijectivity of  $(df)_v$  ensures that the images of the  $t_i$  in  $m/m^2$  form a basis. The  $t_i$  define an epimorphism

$$g : T_i \mapsto t_i : k[[T_1, \dots, T_N]] \longrightarrow B.$$

In the commutative diagram

$$\begin{array}{ccc} & k[[T_1, \dots, T_N]] & \\ \swarrow & & \searrow g \\ A & \xleftarrow{\hspace{1.5cm}} & B \end{array}$$

the smoothness of  $V$  at  $v$  ensures that  $g$  is an isomorphism. It follows that  $B \xrightarrow{\sim} A$ .

In characteristic  $p$ , the Lie algebra of a group scheme is a  $p$ -Lie algebra. We shall write  $\gamma \mapsto \gamma^p$  for the  $p$ -th power operation, often denoted  $\gamma^{[p]}$ . For every linear representation  $\rho$  of the group, one has

$$\rho(\gamma^p) = \rho(\gamma)^p.$$

**Lemma 1.10.** For every  $\gamma \in \text{Lie } U$ , one has  $\gamma^p = 0$ .

As explained under 1.7, we regard here  $U$  as a subgroup of  $Z_G^0(H) \subset G$ .

**Proof.** Since  $U$  is a unipotent group scheme, for each  $i$  the image  $\gamma_i$  of  $\gamma$  in  $\text{End}(V_i)$  is nilpotent. Since  $\dim(V_i) < p$ , one has  $\gamma_i^p = 0$ . Since the family  $(V_i)$  of representations is faithful,  $\gamma^p = 0$ .

**Corollary 1.11.** The unipotent group  $U$  is smooth.

**Proof.** Let  $(\gamma_a)_{1 \leq a \leq A}$  be a basis of  $\text{Lie } U$ . By 1.10, for each  $a$ ,

$$\rho_a : \lambda \mapsto \exp(\lambda \gamma_a) : \mathbb{G}_a \longrightarrow \text{GL}(V)$$

is defined by 0.4. The hypothesis that  $G$  is infinitesimally saturated ensures that this morphism factors through  $G$ . Since  $U$  centralizes  $H$ , the action of  $H$  by inner automorphisms fixes  $\text{Lie } U$ , hence each  $\rho_a$ , and  $\rho_a$  factors through  $Z_G(H)$ , and even through  $Z_G^0(H) = H \times U$  because  $\mathbb{G}_a$  is connected. Every morphism from  $\mathbb{G}_a$  to  $H$  being trivial,  $\rho_a$  factors through  $U$ . Let  $\mathbb{E}^A$  be affine space of dimension  $A$  and set

$$f : \mathbb{E}^A \longrightarrow U, (t_a) \mapsto \prod \rho_a(t_a),$$

the product being taken with respect to the group law of  $U$ . By construction,  $df$  is an isomorphism at 0. Applying 1.9, one deduces that  $U$  is smooth in a neighborhood of the identity element, hence everywhere by homogeneity.

Recall that the action by inner automorphisms of  $H$  on  $\text{Lie } G$  defines a grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha.$$

**Lemma 1.12.** If  $\alpha \in X$  is nonzero and  $\gamma \in \text{Lie}(G)^\alpha$ , then  $\gamma^p = 0$ .

**Proof.** Regard  $\gamma$  as an endomorphism of  $V$ . If  $\gamma^p$  were nonzero, in the sense of the  $p$ -Lie algebra structure of  $\text{Lie } G$  or as an endomorphism of  $V$  (which amounts to the same thing), there would exist  $i, \beta \in X$ , and  $v \in V_i^\beta$  such that  $\gamma^p v \neq 0$ . This implies the nonvanishing of the  $\gamma^l v$  for  $0 \leq l \leq p$ . The element  $\gamma^l v$  of  $V_i$  lies in  $V_i^{\beta + l\alpha}$ . For  $0 \leq l < p$ , the  $\beta + l\alpha$  are distinct, because  $X$  has no torsion prime to  $p$ . The  $\gamma^l v$  for  $0 \leq l < p$  are therefore linearly independent. This contradicts the hypothesis  $\dim V_i < p$ .

**1.13.** Let  $(\gamma_b)_{1 \leq b \leq B}$  be a family of elements of  $\text{Lie } G$ , the disjoint union of bases of the  $(\text{Lie } G)^\alpha$  for  $\alpha \neq 0$ . Let  $\alpha_b$  be the character  $\alpha$  of  $H$  such that  $\gamma_b \in (\text{Lie } G)^\alpha$ . Since  $G$  is infinitesimally saturated, the morphism  $\lambda \mapsto \exp(\lambda \gamma_b) : \mathbb{G}_a \rightarrow \text{GL}(V)$  provided by 0.4 and 1.12 factors through a morphism

$$\rho_b : \mathbb{G}_a \longrightarrow G.$$

This morphism is equivariant for the action of  $H$  on  $\mathbb{G}_a$  through  $\alpha_b$  and on  $G$  by inner automorphisms, because the same equivariance holds after composition with the inclusion of  $G$  in  $\text{GL}(V)$ .

Let  $q$  (respectively  $q_0$ ) be the projection of  $G$  onto  $G/Z_G(H)$  (respectively  $G/Z_G^0(H)$ ). Consider the morphism

$$f : \mathbb{E}^B \longrightarrow G, \quad (t_b) \longmapsto \prod_{b=1}^B \rho_b(t_b),$$

the product being taken with respect to the group law of  $G$ .

**Lemma 1.14.** The composite morphism  $qf : \mathbb{E}^B \rightarrow G/Z_G(H)$  is étale at 0.

Since  $G/Z_G^0(H)$  is étale over  $G/Z_G(H)$ , the same statement holds for  $q_0f$ .

**Proof.** Let  $e$  be the identity element of  $G$ , and let  $T_{q(e)}$  denote the tangent space of  $G/Z_G(H)$  at  $q(e)$ . By 1.9, it is enough to check that  $qf$  induces an isomorphism from the tangent space at 0 onto  $T_{q(e)}$ , and for that it is enough to check that

$$dq : \text{Lie}(G) \longrightarrow T_{q(e)}$$

induces an isomorphism

$$\bigoplus_{\alpha \neq 0} \text{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

If  $G$  were smooth,  $Z_G(H)$  would be smooth as the fixed locus of an action on  $G$  of the group of multiplicative type  $H$ ; one would deduce an exact sequence

$$0 \longrightarrow \text{Lie } Z_G^0(H) \longrightarrow \text{Lie } G \longrightarrow T_{q(e)} \longrightarrow 0,$$

and one would conclude by noting that  $\text{Lie } Z_G^0(H) = \text{Lie}(G)^H$ .

We shall deduce 1.14 from the following lemma.

**Lemma 1.15.** Suppose that a group  $H$  of multiplicative type, with character group  $X$ , acts on an affine  $k$ -scheme of finite type  $S = \text{Spec}(A)$ . Let  $S^H$  be the fixed locus and let  $s \in S^H(k)$ . Let  $I$  be the ideal defining  $S^H$  in  $S$ . View  $I/I^2$  as a coherent sheaf on  $S^H$ , and let

$$(I/I^2)_s = s^*(I/I^2)$$

be its fiber at  $s$ . Let  $m$  be the maximal ideal at  $s$ . The group  $H$  acts on  $m/m^2$ , hence  $m/m^2$  is  $X$ -graded. The inclusion of  $I$  in  $m$  induces an isomorphism

$$(I/I^2)_s \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha. \quad (1.15.1)$$

The hypothesis “ $S$  affine” is probably unnecessary.

**Proof.** The action of  $H$  gives an  $X$ -grading of  $A$ . The quotient  $A/I$  is the largest graded quotient of  $A$  purely of degree 0:  $I$  is the ideal generated by the  $A^\alpha$  for  $\alpha \neq 0$ . The two sides of (1.15.1) are unchanged when  $A$  is replaced by  $A/I^2$ . We may therefore suppose that  $I^2 = 0$ . Then  $A^\alpha A^\beta = 0$  for  $\alpha, \beta \neq 0$ ,  $I$  is the sum of the  $A^\alpha$  for  $\alpha \neq 0$ , and  $A^0 \xrightarrow{\sim} A/I$ . Let  $m_0$  be the maximal ideal of  $s$  in  $A/I \xleftarrow{\sim} A^0$ . We have

$$m = m_0 \oplus \bigoplus_{\alpha \neq 0} A^\alpha.$$

Thus

$$m/m^2 = m_0/m_0^2 \oplus \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

and

$$I/I^2 \otimes_{A^0} A^0/m_0 = \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

which gives (1.15.1).

**End of the proof of 1.14.** The morphism  $G \rightarrow G/Z_G(H)$  is flat. If  $I$  is the ideal defining  $Z_G(H)$  in  $G$ , and  $n$  is the maximal ideal at the point  $q(e)$  of  $G/Z_G(H)$ , then the vector bundle  $I/I^2$  on  $Z_G(H)$  is the inverse image of  $n/n^2$  on  $q(e)$ . Its fiber at the identity element  $e$  of  $G$  is  $n/n^2$ , and therefore by (1.15.1)

$$n/n^2 \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha.$$

Passing to duals gives

$$\bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

**Corollary 1.16.** The morphism

$$F : \mathbb{E}^B \times Z_G^0(H) \longrightarrow G, \quad (t_b), z \longmapsto \prod \rho_b(t_b) \cdot z$$

is etale at  $(0, e)$ .

**Proof.** The square

$$\begin{array}{ccc} \mathbb{E}^B \times Z_G(H) & \xrightarrow{F} & G \\ \downarrow & & \downarrow \\ \mathbb{E}^B & \xrightarrow{q_0 f} & G/Z_G^0(H) \end{array}$$

is cartesian. Hence  $F$  is etale at every point lying above  $0 \in \mathbb{E}^B$ .

**1.17.** If  $f : T \rightarrow U$  is a quasi-compact morphism of schemes, the closed scheme-theoretic image  $\overline{f(T)}$  of  $f$  is the smallest closed subscheme of  $U$  through which  $f$  factors. If  $f : \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$  is affine, it is defined by the ideal kernel of  $f^* : B \rightarrow A$ . Its formation is compatible with every flat base change  $U' \rightarrow U$ . If  $f$  is a morphism of schemes over  $S$ , if a group scheme  $H$  flat over  $S$  acts on  $T$  and  $U$ , and if  $f$  is equivariant, this compatibility with base change implies that the action of  $H$  on  $U$  induces an action of  $H$  on the closed scheme-theoretic image  $\overline{f(T)}$ .

**1.18. Proof of the invariance of  $G_{\text{red}}$  in  $G$ .** The morphism  $F$  of 1.16 is equivariant for the action of  $H$  on  $\mathbb{E}^B$  by the  $\alpha_b$ , and on  $G$  by inner automorphisms. Since  $U$  is smooth (1.11), the reduced subscheme of

$$Z_G^0(H) = H \times U$$

is  $H_{\text{red}} \times U$ . An etale morphism induces an etale morphism on reduced subschemes. The morphism induced by  $F$  from  $\mathbb{E}^B \times H_{\text{red}} \times U$  to  $G_{\text{red}}$  is therefore etale at 0, and its image contains a nonempty open of  $G_{\text{red}}$ . The closed scheme-theoretic image of the morphism

$$F_1 : \mathbb{E}^B \times H_{\text{red}} \times U \longrightarrow G$$

induced by  $F$  therefore coincides with  $G_{\text{red}}$ . Since this morphism is  $H$ -equivariant, 1.17 ensures that  $G_{\text{red}}$  is stable under the action of  $H$  on  $G$  by inner automorphisms.

Because  $F$  is etale at the origin,  $G$  is generated as an fppf sheaf by the image of  $F$ , hence by  $H$  and  $G_{\text{red}}$ , since  $U \subset G_{\text{red}}$ . Since  $H$  normalizes  $G_{\text{red}}$ , it follows that  $G_{\text{red}}$  is an invariant subgroup scheme of  $G$ .

**Proposition 1.19.** One has

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}.$$

Consequently  $G/G_{\text{red}}$  is of multiplicative type.

**Proof.** If  $Y_1$  and  $Y_2$  are two closed subschemes of  $Z$ , then for any morphism  $u : Z' \rightarrow Z$  one has

$$u^{-1}(Y_1 \cap Y_2) = u^{-1}(Y_1) \cap u^{-1}(Y_2).$$

If  $u$  is etale, and  $Y_2 = Z_{\text{red}}$ , one obtains

$$u^{-1}(Y_1 \cap Z_{\text{red}}) = u^{-1}(Y_1) \cap Z'_{\text{red}}.$$

Apply this to  $F$ , which is etale in a neighborhood of  $\{0\} \times H \subset \{0\} \times Z_G^0(H)$ . One obtains that, in a neighborhood of  $\{0\} \times H$ , the inverse image of  $H \cap G_{\text{red}}$  is  $H_{\text{red}}$ ; hence  $H_{\text{red}} \rightarrow H \cap G_{\text{red}}$  is etale, therefore an isomorphism. If  $H$ ,  $H_{\text{red}}$ ,  $G$ ,  $G_{\text{red}}$  and the quotients are interpreted as fppf sheaves, the isomorphism

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}$$

follows from  $H$  normalizing  $G_{\text{red}}$ , from  $HG_{\text{red}} = G$ , and from  $H_{\text{red}} \xrightarrow{\sim} H \cap G_{\text{red}}$ .

**1.20. Proof of the invariance in  $G$  of  $R_u G_{\text{red}}$ .** Let  $R$  denote the unipotent radical  $R_u G_{\text{red}}$  of the smooth group  $G_{\text{red}}$ , and let

$$L := G_{\text{red}}/R.$$

Let  $T_L$  be a maximal torus of the reductive group  $L$ . It is a maximal connected subgroup of multiplicative type. The inverse image of  $T_L$  in  $G_{\text{red}}$  is an extension of the torus  $T_L$  by the smooth connected unipotent group  $R$ . By SGA3 XVII 5.1.1(i), such an extension is trivial: the torus  $T_L$  lifts to a torus  $T$ , automatically maximal, of  $G_{\text{red}}$ . We shall take  $H$  to be a maximal connected subgroup of multiplicative type of  $G$  containing  $T$ . Then

$$T = H_{\text{red}} = H \cap G_{\text{red}}.$$

**Lemma 1.21.** The connected centralizer  $Z_{G_{\text{red}}}^0(T)$  of  $T$  in  $G_{\text{red}}$  is a product  $T \times W$ , with  $W$  unipotent. The group scheme  $H$  normalizes  $W$ .



**Proof.** Since  $H$  normalizes  $T$ ,  $H$  normalizes

$$Z := Z_{G_{\text{red}}}^0(T).$$

As with  $G$ , the subgroup  $G_{\text{red}}$  of  $\text{GL}(V)$  is infinitesimally saturated. Applying 1.7 to  $G_{\text{red}}$  and  $T$  (instead of  $G$  and  $H$ ) gives the decomposition

$$Z = T \times W.$$

The proof of 1.7 describes  $W \subset Z$  as the identity component of the intersection of the kernels of the determinant characters of the representations of  $Z$  in the  $V_i^\beta$ , for  $\beta$  a weight of  $T$  in  $V$ . This description shows that  $H$  normalizes  $W$ .

**Lemma 1.22.** The action of  $H$  on  $\text{Lie}(G_{\text{red}})$  preserves the subspace  $\text{Lie}(R)$ .

**Proof.** Decompose

$$l = \text{Lie } L, \quad g = \text{Lie } G_{\text{red}}, \quad r = \text{Lie } R$$

according to the weights of  $T$ . Since  $H$  normalizes  $T$ , the decomposition obtained for  $g$  is respected by  $H$ . We shall show that each  $r^\alpha$ , hence their sum  $r$ , is stable under  $H$ .

The connected centralizer of  $T_L$  in  $L$  is reduced to  $T_L$ . Hence we have a morphism of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & W & \longrightarrow & Z = T \times W & \longrightarrow & T_L \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & R & \longrightarrow & G_{\text{red}} & \longrightarrow & L. \end{array}$$

The part of weight 0 of  $g$  is  $\text{Lie}(Z)$ . Its intersection with  $\text{Lie } R$  is  $\text{Lie } W$ , stable under  $H$  by 1.21.

If  $\alpha$  is not a weight of  $T$  on  $l$ , then  $r^\alpha = g^\alpha$ , which is stable under  $H$ .

If  $\alpha$  is a nonzero weight of  $T$  on  $l$ , then  $l^\alpha$  and  $l^{-\alpha}$  have dimension one. Under our hypothesis that  $p \neq 2$ , the bracket induces a nondegenerate pairing of  $l^\alpha$  and  $l^{-\alpha}$  with values in a line  $h_\alpha \subset \text{Lie } T_L$ . The bracket of  $g$  induces a pairing of  $g^\alpha$  and  $g^{-\alpha}$  with values in

$$g^0 = \text{Lie } Z.$$

Its composite with  $g^0 \rightarrow g^0 / \text{Lie } W \simeq \text{Lie } T_L$  factors through  $g^{\pm\alpha} \rightarrow l^{\pm\alpha}$  and the preceding pairing. It therefore has values in a line  $d$  of  $g^0 / \text{Lie } W$ , and

$$r^{\pm\alpha} = \text{Ker}(g^{\pm\alpha} \rightarrow \text{Hom}(g^{\mp\alpha}, d)) :$$

the  $r^{\pm\alpha}$  are the kernels of the pairing of  $g^\alpha$  and  $g^{-\alpha}$  with values in  $d$ . The decomposition of  $g$  according to the weights of  $T$ , and  $W \subset g^0$ , are respected by  $H$ ; this description shows that the  $r^\alpha$ , and hence  $r$ , are subrepresentations of  $H$  acting on  $g$ .

**1.23. End of the proof of the invariance of  $R$ .** Let  $\text{Fil}$  denote the finite increasing filtration of  $\bigoplus V_i$  such that  $\text{Fil}_0 = 0$  and  $\text{Fil}_{n+1} / \text{Fil}_n$  is the space of invariants of  $R$  in  $\bigoplus V_i / \text{Fil}_n$ . It is the sum of filtrations of the  $V_i$ . It is respected by  $G_{\text{red}}$ , since  $R$  is an invariant subgroup of  $G_{\text{red}}$ .

The reduced subgroup of the largest subgroup of  $G_{\text{red}}$  which acts trivially on

$$\text{Gr}^{\text{Fil}}(\bigoplus V_i)$$

is a smooth invariant unipotent subgroup containing  $R$ . It therefore coincides with  $R$ .

If  $\gamma \in \text{Lie } R$ , then  $\gamma$  acts nilpotently on  $V_i$ . Since the family of representations  $V_i$  of dimension  $< p$  is faithful, one has  $\gamma^p = 0$ . Since  $G_{\text{red}}$  is infinitesimally saturated, the corresponding morphism

$$\rho : \mathbb{G}_a \longrightarrow \prod \text{GL}(V_i)$$

factors through  $G_{\text{red}}$ . Since this representation of  $\mathbb{G}_a$  is trivial on

$$\text{Gr}^{\text{Fil}}\left(\bigoplus V_i\right),$$

the morphism  $\rho$  even factors through  $R$ .

Choose a basis  $(\gamma_c)_{1 \leq c \leq C}$  of  $r = \text{Lie } R$ , the disjoint union of bases of the  $r^\beta$  for  $\beta$  a weight of  $H$  acting on  $r$ . Let  $\rho_c : \mathbb{G}_a \rightarrow R$  be the morphism defined by  $\gamma_c$ . As in 5.6, the morphism of schemes

$$f : \mathbb{E}^C \longrightarrow R, \quad (t_c) \longmapsto \prod_{c=1}^C \rho_c(t_c)$$

is etale at 0. Its composite with the inclusion of  $R$  in  $G_{\text{red}}$  is  $H$ -equivariant, for a suitable action of  $H$  on  $\mathbb{E}^C$ . By the same argument as in 1.17 and 1.18, it follows that  $R$  is stable under the action of  $H$  on  $G_{\text{red}}$ .

**Corollary 1.24.** If the representations  $V_i$  are semisimple, the group scheme  $G_{\text{red}}$  is reductive.

**Proof.** Replacing each  $V_i$  by a family of irreducible representations of which  $V_i$  is the sum, we reduce to supposing the  $V_i$  irreducible. We must prove that the unipotent radical  $R = R_u G_{\text{red}}$  is trivial.

Since  $R$  is unipotent, the subspace  $V_i^R$  of  $R$ -invariants in  $V_i$  is nonzero. Because  $R$  is invariant in  $G$ ,  $V_i^R$  is stable under  $G$ , hence, by irreducibility of  $V_i$ , coincides with  $V_i$ . Since the family of the  $V_i$  is faithful,  $R$  is trivial.

**1.25. Proof of 0.7(iii).** Suppose that  $G_{\text{red}}$  is reductive, let  $T$  be a maximal torus of  $G_{\text{red}}$ , and take  $H$  to be a maximal connected subgroup of multiplicative type of  $G$  containing  $T$ . It normalizes  $G_{\text{red}}$  and its action on  $G_{\text{red}}$  fixes  $T$ . The group scheme of automorphisms of the reductive group  $G_{\text{red}}$  which fix  $T$  is the image  $T^{\text{ad}}$  of  $T$  in the adjoint group. If  $Z$  is the subgroup of  $H$  which centralizes  $G_{\text{red}}$ , the morphism

$$Z \times T \longrightarrow H$$

is a quotient by  $T$ : the image  $T^{\text{ad}}$  of  $T$  in the adjoint group. View all these group schemes as fppf sheaves. Since  $H$  and  $G_{\text{red}}$  generate  $G$ ,  $Z$  is central. Since the action of  $H$  on  $G_{\text{red}}$  factors through a quotient of  $T$ ,  $Z$  and  $T$  generate  $H$ . Since  $Z$  is generated by  $Z^0$  and  $Z_{\text{red}}$  and  $Z_{\text{red}} \subset G_{\text{red}}$ ,  $Z^0$  and  $G_{\text{red}}$  generate  $G$ : the morphism

$$Z^0 \times G_{\text{red}} \longrightarrow G$$

is an epimorphism, and this verifies 0.7(iii).

### 3 Saturation (cf. [4] §4)

**2.1.** Let  $V$  be a finite-dimensional vector space over an algebraically closed field  $k$  of characteristic  $p$ , and let  $G$  be a subgroup scheme of  $\text{GL}(V)$ . The intersection  $G^*$  of the doubly saturated subgroup schemes (0.5) of  $\text{GL}(V)$  containing  $G$  is doubly saturated. We call  $G^*$  the double saturation of  $G$ .

Let  $(V_i)_{i \in I}$  be a finite family of finite-dimensional vector spaces over  $k$ . For each  $i$ , let  $N_i$  be an endomorphism of  $V_i$  such that  $N_i^p = 0$ . It defines an action  $t \mapsto \exp(tN_i)$  of  $\mathbb{G}_a$  on  $V_i$  (0.4). For  $a \in I$ , again denote by  $N_a$  the endomorphism of  $\bigotimes V_i$  which is the tensor product of  $N_a \in \text{End}(V_a)$  and of the identities  $1 \in \text{End}(V_j)$  for  $j \neq a$ . Let  $N$  be the endomorphism  $\sum N_a$  of  $\bigotimes V_i$ .

**Proposition 2.2.** Suppose there exist integers  $v_i$  such that  $N_i^{v_i} = 0$  and

$$\sum_i (v_i - 1) < p.$$

Then  $N^p = 0$ , and the action  $t \mapsto \exp(tN)$  of  $\mathbb{G}_a$  on  $\bigotimes V_i$  is the tensor product of the actions of  $\mathbb{G}_a$  on the  $V_i$  defined by the  $N_i$ .

**Proof.** The endomorphisms  $N_a$  of  $\bigotimes V_i$  commute with one another, and every monomial of degree  $p$  in the  $N_a$  has degree at least  $v_a$  in  $N_a$  for some  $a \in I$ , hence is zero. It follows that

$$N^p = \left( \sum N_a \right)^p = 0,$$

and also that the truncated exponentials satisfy

$$\exp(tN) = \exp\left(\sum tN_a\right) = \prod \exp(tN_a) = \bigotimes \exp(tN_a).$$

**Corollary 2.3.** If  $\sum_i (\dim(V_i) - 1) < p$ , the action  $\exp(tN)$  of  $\mathbb{G}_a$  on  $\bigotimes V_i$  is the tensor product of the actions of  $\mathbb{G}_a$  on the  $V_i$  defined by the  $N_i$ .

**Proof.** Take  $v_i = \dim(V_i)$  in 2.2.

By the same arguments as those of [4], §4, one deduces from [4], §4, Proposition 10, and from 2.3 the following proposition 2.4. Let  $(V_i)_{i \in I}$  be a finite family of finite-dimensional vector spaces over  $k$  and let  $G$  be a subgroup scheme of  $\prod \text{GL}(V_i)$ . The subgroup scheme  $\prod \text{GL}(V_i)$  of  $\text{GL}(\bigoplus V_i)$  is saturated and infinitesimally saturated. The double saturation  $G^*$  of  $G$ , viewed as a subgroup of  $\text{GL}(\bigoplus V_i)$ , is therefore contained in  $\prod \text{GL}(V_i)$ .

**Proposition 2.4.** If  $\sum_i (\dim(V_i) - 1) < p$ , then a subspace  $W$  of  $\bigotimes V_i$  is stable under the action of  $G$  if and only if it is stable under that of  $G^*$ . Thus the representation  $\bigotimes V_i$  of  $G$  is semisimple if and only if it is semisimple as a representation of  $G^*$ .

## 4 The case in which $k$ is algebraically closed

Let  $G$  be an affine group scheme over an algebraically closed field  $k$  of characteristic  $p$ , and let  $(V_i)_{i \in I}$  be a finite family of representations of  $G$ .

**Proposition 3.1.** If the representations  $V_i$  are semisimple and  $\sum_i (\dim V_i - 1) < p$ , then the representation  $\bigotimes V_i$  of  $G$  is again semisimple.

**Proof.** We leave to the reader the verification that this statement is implied by the special case in which one makes the additional hypotheses that each  $V_i$  is simple of dimension  $\geq 2$ , and that  $|I| \geq 2$ . These hypotheses imply that  $p \neq 2$ , and that each  $V_i$  has dimension  $< p$ .

Replacing  $G$  by its image in  $\prod \text{GL}(V_i)$ , one may suppose that  $G$  is a subgroup scheme of  $\prod \text{GL}(V_i)$ . By 2.4, one may moreover suppose that, as a subgroup of  $\text{GL}(\bigoplus V_i)$ ,  $G$  is doubly saturated. In the rest of the proof we shall suppose all these properties satisfied. That  $G$  is

saturated is equivalent to  $G_{\text{red}}$  being so and, by [4], §4, Proposition 11, implies that the finite group

$$G/G^0 = G_{\text{red}}/G_{\text{red}}^0$$

has order prime to  $p$ .

By 0.7(ii)(iii), the group  $G_{\text{red}}^0$  is reductive and  $G^0$  contains a central connected group  $M$  of multiplicative type such that

$$M \times G_{\text{red}}^0 \longrightarrow G^0$$

is an epimorphism.

By Lemma 1.3(ii), the  $V_i$  are semisimple as representations of  $G^0$ . If  $W$  is an irreducible representation of  $G^0$ , then  $M$ , being of multiplicative type and central, acts on  $W$  by a character. The representation  $W$  is therefore still irreducible as a representation of  $G_{\text{red}}^0$ . Thus the  $V_i$  are semisimple as representations of  $G_{\text{red}}^0$ , and by [4] the tensor product of these representations is again semisimple as a representation of  $G_{\text{red}}^0$ . The following lemma shows that it is semisimple as a representation of  $G$ .

**Lemma 3.2.** Suppose that a group scheme  $G$  is an extension of a linearly reductive group scheme  $A$  by a group scheme  $B$ . Then every representation  $V$  of  $G$  which is semisimple as a representation of  $B$  is semisimple.

That  $A$  is linearly reductive means that all its linear representations are semisimple. A group of multiplicative type, and a finite group of order prime to  $p$ , are linearly reductive.

**Proof.** Let  $S$  be the sum of the irreducible subrepresentations of  $V$ . We must show that  $S = V$ , and to verify this it is enough to show that  $S$  has a complement  $T$  in  $V$  stable under  $G$ . Consider the exact sequence

$$1 \longrightarrow \text{Hom}_B(V/S, S) \longrightarrow \text{Hom}_B(V, S) \longrightarrow \text{Hom}_B(S, S) \longrightarrow 1.$$

It is an exact sequence of representations of  $A$ . Since  $A$  is linearly reductive, the identity  $1_S$  of  $S$ , fixed by  $A$ , lifts to a homomorphism  $f : V \rightarrow S$  of representations of  $B$  which is fixed by  $A$ , i.e. is a morphism of representations of  $G$ . Its kernel is the required complement.

**Corollary 3.3.** The statement 0.1 is true when  $k$  is algebraically closed.

**Proof.** Let  $T'$  be the Tannakian subcategory of  $T$  generated from the  $V_i$  by tensor product, duality, and passage to a subquotient. By [3] 3.3.1.1,  $T'$  admits a fiber functor over  $k$ , hence is equivalent to the category of representations of an affine group scheme over  $k$ . Another reference is [1] 6.20.

**Remark 3.4.** The reduction from  $T$  to  $T'$  is in fact unnecessary. I have checked (unpublished) that every Tannakian category over an algebraically closed field  $k$  is equivalent to the Tannakian category of representations of an affine group scheme over  $k$ .

## 5 Extension of scalars in Tannakian categories and proof of 0.1

**4.1.** Let  $k$  be a commutative field and let  $T$  be a  $k$ -linear abelian category. Define the tensor product of an object  $X$  of  $T$  and a finite-dimensional  $k$ -vector space  $V$  to be the object of  $T$  which represents the functor

$$Y \longmapsto V \otimes \text{Hom}(Y, X).$$

The choice of a basis  $(e_i)_{i \in I}$  of  $V$  identifies  $V \otimes X$  with the sum of a family of copies of  $X$  indexed by  $I$ .

Let  $k'$  be a finite extension of  $k$ . We define as follows the  $k'$ -linear abelian category  $T_{k'}$  obtained from  $T$  by extension of scalars to  $k'$ , the scalar-extension functor

$$e_{k'/k} : T_k \longrightarrow T_{k'},$$

and the scalar-restriction functor

$$r_{k \setminus k'} : T_{k'} \longrightarrow T_k.$$

The category  $T_{k'}$  is the category of objects  $X$  of  $T$  endowed with a  $k'$ -module structure  $k' \otimes X \rightarrow X$ . For  $X$  in  $T$ ,

$$e_{k'/k}(X) := k' \otimes X,$$

endowed with its natural  $k'$ -module structure. The functor  $r_{k \setminus k'}$  forgets the  $k'$ -module structure. The functors  $e_{k'/k}$  and  $r_{k \setminus k'}$  are exact, and  $r_{k \setminus k'}$  is right adjoint to  $e_{k'/k}$ . Example: if  $T$  is the category of left modules over a  $k$ -algebra  $A$ , then  $T_{k'}$  identifies with the category of left modules over the  $k'$ -algebra

$$A' := k' \otimes_k A,$$

and  $e_{k'/k}$  and  $r_{k \setminus k'}$  are the usual extension- and restriction-of-scalars functors.

Suppose that the objects of  $T$  have finite length and that the Hom groups are finite-dimensional over  $k$ . If  $X$  is a simple object of  $T$ , the division algebra  $\text{End}(X)$ , being finite-dimensional over  $k$ , is central simple over its center, which is a finite extension of  $k$ .

**Lemma 4.2.** Under these hypotheses:

- (i) Let  $X$  be a simple object of  $T$ ,  $D := \text{End}(X)$ , and  $F$  the center of  $D$ . The object  $e_{k'/k}(X)$  of  $T_{k'}$  is semisimple if and only if  $F \otimes_k k'$  is a product of fields.
- (ii) If  $Y$  is a semisimple object of  $T_{k'}$ , then  $r_{k \setminus k'}(Y)$  is semisimple.
- (iii) If  $X$  is in  $T$  and  $e_{k'/k}(X)$  is semisimple, then  $X$  is semisimple.

**Proof.** (i) The functor  $Z \mapsto \text{Hom}(X, Z)$  is an equivalence from the category of objects of  $T$  which are sums of copies of  $X$  to the category of finite-dimensional right  $D$ -vector spaces. The category of sums of copies of  $X$  endowed with a  $k'$ -module structure identifies with the category of finite-type right  $D \otimes_k k'$ -modules. The object  $e_{k'/k}(X)$  of  $T_{k'}$  thus corresponds to the  $D \otimes_k k'$ -module  $D \otimes_k k'$ . It is known to be semisimple if and only if  $F \otimes_k k'$  is a product of fields.

(ii) Reduce to the case in which  $Y$  is simple. The sum  $S$  of the simple subobjects of  $r_{k \setminus k'}(Y)$  is nonzero and is a  $k'$ -submodule. By the simplicity of  $Y$ , it coincides with  $r_{k \setminus k'}(Y)$ .

(iii) The object  $X$  is a direct factor of  $r_{k \setminus k'} e_{k'/k}(X)$ , and one applies (ii).

**4.3.** Now suppose that  $T$  is a Tannakian category over  $k$ , and equip  $T_{k'}$  with the tensor product over  $k'$ :

$$X \otimes_{k'} Y = \text{coker}(X \otimes k' \otimes Y \rightrightarrows X \otimes Y).$$

The scalar-extension functor  $e_{k'/k}$  is compatible with tensor products and with their associativity and commutativity data. It sends the unit object to the unit object, and duals, in the sense of [1] 2.2, to duals. The category  $T_{k'}$  admits internal Homs: if  $X$  and  $Y$  in  $T$  are endowed with  $k'$ -module structures, their internal Hom in  $T_{k'}$  is the kernel of the two morphisms

$$\text{Hom}(X, Y) \rightrightarrows \text{Hom}(k' \otimes X, Y) = k'^{\vee} \otimes \text{Hom}(X, Y),$$

where  $k'^{\vee}$  is the  $k$ -dual of the  $k$ -vector space  $k'$ . We shall denote it by  $\text{Hom}_{k'}(X, Y)$ .

**Theorem 4.4.** If  $T$  is a Tannakian category over  $k$  and  $k'$  is a finite extension of  $k$ , then the category  $T_{k'}$ , endowed with the tensor product over  $k'$ , is Tannakian over  $k'$ .

It is possible to deduce this theorem from Theorem 1.12 of [1], which characterizes Tannakian categories as categories of representations of affine groupoids  $G$  acting transitively on a  $k$ -scheme  $S$  ( $G \rightarrow S \times_k S$  affine and faithfully flat): if  $T$  is such a category of representations, then  $T_{k'}$  identifies with the category of representations of the groupoid obtained from it by extension of scalars to  $k'$ .

Conversely, and according to a strategy indicated to me by Grothendieck about thirty years ago, the more elementary proof of 4.4 that follows should make it possible to prove [1] 1.12 more simply.

If  $k' \supset k'' \supset k$ , then  $T_{k'}$  identifies with  $(T_{k''})_{k'}$ . This observation permits one to reduce the proof of 4.4 to the following two special cases:  $k'/k$  separable, and  $k'/k$  inseparable of degree  $p$ . The delicate point is to show the existence of duals, in the sense of [1] 2.2, in  $T_{k'}$ . Once this is acquired, if  $\omega$  is a fiber functor over the  $k$ -algebra  $K$ , then  $\omega$  provides a fiber functor  $\omega'$  of  $T_{k'}$  over the  $k'$ -algebra

$$K' := k' \otimes_k K :$$

to  $X$  in  $T$ , endowed with a  $k'$ -module structure, attach  $\omega(X)$ , endowed with its  $k'$ -module structure. This structure makes  $\omega(X)$  a  $k' \otimes K$ -module.

**Proof if  $k'$  is a finite separable extension of  $k$ .** If  $X$  is in  $T$  and has  $X^\vee$  as dual, in the sense of [1] 2.2, the object  $e_{k'/k}(X)$  of  $T_{k'}$  admits  $e_{k'/k}(X^\vee)$  as dual. One concludes by observing that every object  $Y$  of  $T_{k'}$  is a direct factor of an object of this form. More precisely, the adjunction morphism

$$e_{k'/k} r_{k \setminus k'}(Y) \longrightarrow Y$$

is split: if one regards  $Y$  as an object of  $T$ ,  $k' \otimes Y$  has two natural  $k'$ -module structures, one coming from  $k'$  (this is that of  $e_{k'/k} r_{k \setminus k'}(Y)$ ), the other from  $Y$ . The adjunction morphism is the natural morphism

$$k' \otimes Y \longrightarrow (k' \otimes Y) \otimes_{k' \otimes k} k'.$$

It is split because the morphism  $k' \otimes_k k' \rightarrow k'$  is the projection onto a direct factor.

**Proof if  $k' = k(a^{1/p})$ .** Let  $\omega$  be a fiber functor over an extension  $K$  of  $k$ . The key point is the following lemma.

**Lemma 4.5.** If an object  $X$  of  $T$  is endowed with a  $k'$ -module structure, this structure makes  $\omega(X)$  a  $k' \otimes_k K$ -module. This module is free.

**Proof.** If  $K' := k' \otimes_k K$  is a field, this is clear. Otherwise,  $a$  has a  $p$ -th root  $\alpha$  in  $K$ , and the  $K$ -algebra  $K'$  is isomorphic to  $K[\varepsilon]/(\varepsilon^p)$ : take

$$\varepsilon = \sqrt[p]{a} - \alpha.$$

Let  $d$  be the dimension over  $K$  of  $\omega(X)/\varepsilon\omega(X)$ . The  $K'$ -module  $\omega(X)$  is isomorphic to a sum of  $d$  monogenic modules  $K[\varepsilon]/(\varepsilon^j)$ ,  $1 \leq j \leq p$ . One checks that it is free if and only if the exterior power over  $K'$

$$\bigwedge \text{olimits}_{K'}^d \omega(X)$$

is free of rank one. By the exactness of  $\omega$ , this exterior power over  $K'$  is the image under  $\omega$  of the exterior power over  $k'$  of  $X$ , defined as the image of the antisymmetrization

$$a : \bigotimes \text{olimits}_{k'}^d X \longrightarrow \bigotimes \text{olimits}_{k'}^d X.$$

Replacing  $X$  by this exterior power, we are reduced to the case in which  $X$  is such that  $\omega(X)$  is a nonzero monogenic  $K'$ -module.

The  $k'$ -module structure of  $X$  gives a morphism

$$k' \otimes \mathbf{1} \longrightarrow \text{Hom}(X, X). \quad (4.5.1)$$

One knows that the unit object  $\mathbf{1}$  is simple ([2] 1.17). The kernel of (4.5.1) is therefore of the form  $A \otimes \mathbf{1}$ , for  $A$  a vector subspace of  $k'$ . This subspace is a proper ideal of  $k'$ , so  $A = 0$  and (4.5.1) is a monomorphism. Applying  $\omega$ , one deduces that  $\omega(X)$  is a faithful  $K'$ -module, and one concludes by observing that a faithful monogenic  $K'$ -module is free.

**4.6. End of the proof if  $k' = k(a^{1/p})$ .** Let  $\mathbf{1}'$  denote the unit object  $e_{k'/k}(\mathbf{1})$  of  $T_{k'}$ . For  $X$  in  $T_{k'}$ , put

$$X^\vee := \text{Hom}_{k'}(X, \mathbf{1}').$$

We show that  $X^\vee$  is a dual of  $X$  in the sense of [1] 2.2. In  $T_{k'}$  there is an evaluation morphism

$$\text{ev} : X^\vee \otimes_{k'} X \longrightarrow \mathbf{1}' \quad (4.6.1)$$

and, for every  $Y$  in  $T_{k'}$ , a morphism

$$Y \otimes_{k'} X^\vee \longrightarrow \text{Hom}_{k'}(X, Y). \quad (4.6.2)$$

The functor  $\omega'$ , with values in  $K'$ -modules, transforms this morphism into

$$\omega(Y) \otimes \omega(X)^\vee \longrightarrow \text{Hom}_{K'}(\omega(X), \omega(Y)). \quad (4.6.3)$$

Lemma 4.5 ensures that (4.6.3) is an isomorphism. Since  $\omega'$  is exact and satisfies  $\omega(Z) = 0 \Rightarrow Z = 0$ , because  $\omega$  has these properties, it is conservative and (4.6.2) is an isomorphism. Taking  $Y = X$ , and composing the evident morphism  $\mathbf{1}' \rightarrow \text{Hom}_{k'}(X, X)$  with the inverse of (4.6.2), we obtain

$$\delta : \mathbf{1}' \longrightarrow X \otimes_{k'} X^\vee.$$

The morphisms  $\text{ev}$  and  $\delta$  satisfy the identities [1] (2.1.2) making  $X^\vee$  a dual of  $X$ , since these identities become true after application of  $\omega'$ .

**Corollary 4.7.** Let  $X$  be an object of a Tannakian category  $T$  over  $k$ . If  $X$  admits a module structure over an extension  $k'$  of  $k$ , then  $[k' : k]$  divides  $\dim X$ .

**Proof.** Let  $\omega$  be a fiber functor of  $T$  over an extension  $K$  of  $k$ , put  $K' := k' \otimes K$ , and regard  $\omega(X)$ , with its  $k'$ -module structure deduced from that of  $X$ , as a  $K'$ -module. If  $X$  is regarded as an object of the Tannakian category  $T_{k'}$ , this  $K'$ -module is the image of  $X$  under the fiber functor  $\omega'$  of  $T_{k'}$  deduced from  $\omega$ . As the image of the object  $X$  of the Tannakian category  $T_{k'}$  under the fiber functor  $\omega'$ , it is a locally free  $K'$ -module of constant rank, equal to the dimension of  $X$  viewed as an object of  $T_{k'}$ . Thus

$$\begin{aligned} \dim(X) &:= \dim_K \omega(X) \\ &= [K' : K] \cdot \text{rank}_{K'} \omega(X) \\ &= [k' : k] \cdot \dim(X \text{ viewed as an object of } T_{k'}). \end{aligned}$$

**4.8. Warning.** If  $D$  is a division algebra over  $k$  and a nonzero object  $X$  admits a  $D$ -module structure, one does not necessarily have  $\dim X \geq [D : k]$ . Example: take  $T$  to be the category of  $\mathbb{Z}/2$ -graded complex vector spaces  $V$  endowed with a homogeneous antilinear automorphism  $j$  whose square is 1 on  $V^0$  and  $-1$  on  $V^1$ . Tensor product is taken over  $\mathbb{C}$ . This category is Tannakian over  $\mathbb{R}$ , and over  $\mathbb{C}$  admits the fiber functor “underlying complex vector space.” An odd object identifies with a vector space over the quaternion field  $\mathbb{H}$ . A simple odd object  $S$  has dimension 2 and  $\text{End}(S)$  is isomorphic to  $\mathbb{H}$ .

**Corollary 4.9.** If  $X$  is a simple object of a Tannakian category  $T$  over  $k$  of characteristic  $p$  and  $\dim X < p$ , then the center of  $\text{End}(X)$  is a separable extension of  $k$ .

**Proof.** By 4.7 it is an extension of degree  $< p$ .

**4.10.** Let  $k_1$  be an algebraic extension of  $k$ . If  $T$  is a  $k$ -linear abelian category, the categories  $T_{k'}$ , for  $k'$  a finite extension of  $k$  in  $k_1$ , form a 2-inductive system of abelian categories. I write “2-inductive” rather than “inductive,” because the scalar-extension functors

$$e_{k''/k'} : T_{k'} \longrightarrow (T_{k'})_{k''} = T_{k''}$$

for  $k' \subset k''$  satisfy

$$e_{k'''/k''} e_{k''/k'} = e_{k'''/k'}$$

only up to a canonical isomorphism; these isomorphisms satisfy a cocycle condition for  $k' \subset k'' \subset k''' \subset k''''$ . Define  $T_{k_1}$  to be the 2-inductive limit of the  $T_{k'}$ , for  $k'$  a finite extension of  $k$  in  $k_1$ . Since the functors  $e_{k''/k'}$  are exact, the category  $T_{k_1}$  is again abelian.

If  $T$  is Tannakian, with fiber functor  $\omega$  over  $K$ , the categories  $T_{k'}$  for  $k' \subset k_1$  are Tannakian, and  $\omega$  provides a fiber functor of  $T_{k'}$  over  $k' \otimes K$ . Passing to the inductive limit, one sees that  $T_{k_1}$  is Tannakian, that  $\omega$  provides a fiber functor  $\omega_1$  of  $T_{k_1}$  over  $k_1 \otimes K$ , and that the  $e_{k'/k}$  provide a scalar-extension functor

$$e_{k_1/k} : T \longrightarrow T_{k_1},$$

compatible with tensor product.

**Corollary 4.11.** Let  $X$  be an object of  $T$ .

- (i) If the object  $e_{k_1/k}(X)$  of  $T_{k_1}$  is semisimple, then  $X$  is semisimple.
- (ii) If  $\dim(X) < p$  and  $X$  is semisimple, then  $e_{k_1/k}(X)$  is semisimple.

**Proof.** For  $k'$  a finite extension of  $k$  in  $k_1$ , the length of  $e_{k'/k}(X)$  increases with  $k'$  and is bounded by  $\dim(X)$ . There exists therefore a finite extension  $k'$  such that the length of  $e_{k''/k}(X)$  is constant for  $k'' \supset k'$ .

Prove (i) for  $X$ . By 4.2(iii), it is enough to show that for a suitable finite extension  $k''$  of  $k'$  in  $k_1$ ,  $e_{k''/k'}(X)$  is semisimple. Let

$$0 = X_0 \subset X_1 \subset X_2 \subset \cdots \subset X_n = e_{k'/k}(X)$$

be a filtration of  $e_{k'/k}(X)$  with simple successive quotients  $S_i$ . The extension of  $S_i$  by  $X_{i-1}$  defining  $X_i$  becomes split over  $k_1$ . There is therefore a finite extension  $k''$  of  $k'$  in  $k_1$  over which these extensions split. The  $S_i$  remain simple because

$$\lg e_{k''/k}(X) = \lg e_{k'/k}(X).$$

It follows that  $e_{k''/k'}(X)$  is semisimple.

For (ii), let  $X$  be semisimple with  $\dim(X) < p$ . It is enough to treat  $X$  simple. By 4.9, the center of  $\text{End}(X)$  is separable over  $k$ ; hence, by 4.2(i), all finite scalar extensions occurring in the construction preserve semisimplicity, and so  $e_{k_1/k}(X)$  is semisimple.

**4.12. Proof of 0.1.** Let  $T$  be a Tannakian category over  $k$  of characteristic  $p$ , and take for  $k_1$  an algebraic closure  $\bar{k}$  of  $k$ . Corollary 4.11 reduces 0.1 for  $T$  to 0.1 for  $T_{\bar{k}}$ , proved in 3.3.



## 6 Tensor products of exterior powers

The results that follow were suggested by the referee.

Theorem 0.7 makes it possible to generalize to objects of a Tannakian category Serre's theorem [5] on the semisimplicity of tensor products of exterior powers of representations.

**Theorem 5.1.** Let  $T$  be a Tannakian category over a field  $k$  of characteristic  $p > 0$ , let  $(V_i)_{i \in I}$  be a finite family of semisimple objects of  $T$ , and let  $(m_i)_{i \in I}$  be a family of integers  $\geq 0$ . If

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.1.1)$$

then the tensor product of the  $\bigwedge^{m_i} V_i$  is semisimple.

Recall that the  $m$ -th exterior power of an object  $V$  of  $T$  is the image of the antisymmetrization

$$a : \bigotimes \text{olimits}^m V \longrightarrow \bigotimes \text{olimits}^m V.$$

For every fiber functor  $\omega$ , one has

$$\omega(\bigwedge \text{olimits}^m V) = \bigwedge \text{olimits}^m \omega(V),$$

and the dual of  $\bigwedge^m V$  is  $\bigwedge^m (V^\vee)$ .

We shall reduce the proof of 5.1 to the special case in which, for each  $i$ ,  $\dim(V_i) < p$ . The arguments of 3.3, 4.11 and 4.12 then reduce further to supposing that  $k$  is algebraically closed, that  $T$  is the category of representations of an affine group scheme  $G$  over  $k$ , and that the family of representations  $(V_i)$  is faithful, i.e. that  $G$  is a subgroup scheme of  $\prod \text{GL}(V_i)$ , and hence of  $\text{GL}(V)$ , for  $V := \bigoplus V_i$ .

After a few preliminaries (5.3 to 5.10), we shall reduce in 5.11 to the case in which  $G$  is doubly saturated in  $\text{GL}(V)$ , and treat that case in 5.12.

**5.2. Reduction to the case  $\dim(V_i) < p$ .** We successively reduce to supposing that:

- (1)  $m_i \leq \dim V_i$ ; otherwise  $\bigwedge^{m_i} V_i = 0$ .
- (2)  $0 < m_i < \dim V_i$ ; otherwise  $\bigwedge^{m_i} V_i$  has dimension one and the factor  $\bigwedge^{m_i} V_i$  can be omitted. This changes neither hypothesis (5.1.1) nor the conclusion, since if  $L$  has dimension one, semisimplicity of  $X$  is equivalent to that of  $X \otimes L$ , the lattices of subobjects of  $X$  and  $X \otimes L$  being isomorphic.
- (3)  $m_i \leq (\dim V_i)/2$ ; otherwise replace  $\bigwedge^{m_i} V_i$  by  $\bigwedge^{\dim V_i - m_i} V_i^\vee$ , which differs from it only by tensoring with the rank-one object  $\bigwedge^{\dim V_i} V_i$ .
- (4)  $\sum m_i \geq 2$ ; otherwise the tensor product is 1 (if  $\sum m_i = 0$ ) or is reduced to a  $V_i$  (if  $\sum m_i = 1$ ).

If these conditions are satisfied, then  $\dim V_i < p$ . Indeed, if  $|I| \geq 2$ , (5.1.1) implies

$$\dim V_i - 1 = 1(\dim V_i - 1) \leq m_i(\dim V_i - m_i) < p - 1,$$

and if  $|I| = 1$ , then  $m_i \geq 2$ ,  $\dim V_i \geq 4$ , and

$$\dim V_i \leq 2(\dim V_i - 2) \leq m_i(\dim V_i - m_i) < p.$$

**5.3.** Let  $V$  be a representation of  $\mathrm{SL}(2)$ . The action of the multiplicative group of diagonal matrices  $(a, a^{-1})$  defines a grading

$$V = \bigoplus V^j$$

of  $V$ . The action of

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SL}(2)$$

exchanges  $V^j$  and  $V^{-j}$ . Hence  $\dim V^j = \dim V^{-j}$ . Let  $E$  be the action of the element

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

of the Lie algebra. This endomorphism of  $V$  has degree 2. If the weights of  $V$  are  $< v$ , i.e. if  $V^j = 0$  for  $|j| \geq v$ , then  $E^v = 0$ .

If the weights of  $V$  are  $< p$ , one knows that the representation  $V$  is semisimple, a sum of  $\mathrm{Sym}^i$  of the fundamental representation with  $i < p$ , and that the  $j$ -th iterate  $E^j$  of  $E$  induces an isomorphism

$$E^j : V^{-j} \xrightarrow{\sim} V^j. \quad (5.3.1)$$

Conversely, if  $V$  is a graded vector space, with  $V^j = 0$  for  $|j| \geq p$ , and if  $E$  is an endomorphism of degree 2 of  $V$  satisfying (5.3.1), then there exists a unique action of  $\mathrm{SL}(2)$  on  $V$  defining this grading and this endomorphism; one has  $E^p = 0$ , and the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

in  $\mathrm{SL}(2)$  is given by the truncated exponential

$$\exp(tE) := \sum_{n < p} \frac{(tE)^n}{n!}. \quad (5.3.2)$$

**5.4.** Let  $E$  be a nilpotent endomorphism of a vector space  $V$  of dimension  $\leq p$ . Decomposing  $E$  into Jordan blocks, one checks that  $E$  comes from a representation of  $\mathrm{SL}(2)$  as in 5.3, with weights  $< p$ . If  $E$  has only one Jordan block, the dominant weight is  $\dim(V) - 1$ .

**Lemma 5.5.** With the hypotheses and notation of 5.4, the weights of the representation  $\bigwedge^m V$  of  $\mathrm{SL}(2)$  are  $\leq m(\dim V - m)$ .

Lifting the representation  $V$  of  $\mathrm{SL}(2)$  to characteristic 0, one reduces to checking in characteristic 0 that, for a representation  $W$  of  $\mathrm{SL}(2)$  of dimension  $d$ , one has

$$\text{the weights of the representation } \rho_m \text{ of } \mathrm{SL}(2) \text{ on } \bigwedge^m W \text{ are } \leq m(d - m). \quad (5.5.1)$$

Since, in characteristic 0, every representation of  $\mathrm{SL}(2)$  is a direct sum of  $\mathrm{Sym}^j$  of the fundamental representation, (5.5.1) is equivalent to

$$(d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix})^{m(d-m)+1} = 0. \quad (5.5.2)$$

Let  $E$  be the nilpotent endomorphism of  $W$  giving the action of  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$ . If one identifies endomorphisms with actions of the one-dimensional Lie algebra, then

$$d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

is the natural extension  $E_m$  of  $E$  to  $\bigwedge^m W$ . Nilpotent endomorphisms with a single Jordan block are dense among nilpotent endomorphisms. Hence it is enough to check that

$$(E_m)^{m(d-m)+1} = 0$$

when  $E$  has a single Jordan block. In that case the weights of the representation  $W$  are

$$(d-1), (d-3), \dots, -(d-1),$$

each with multiplicity one. The largest weight of the representation  $\bigwedge^m V$  is therefore

$$(d-1) + (d-3) + \dots + (d-(2m-1)) = md - m^2 = m(d-m),$$

and (5.5.2) follows.

**5.6.** Return to characteristic  $p$ . Let  $N$  be a nilpotent endomorphism of a vector space  $V$  of dimension  $d \leq p$ , and let  $N_m$  be its extension, in the sense of Lie algebras as in 5.5, to  $\bigwedge^m V$ . Since  $d \leq p$ , one has  $N^p = 0$  and  $N$  defines an action

$$\rho(t) = \exp(tN)$$

of  $\mathbb{G}_a$  on  $V$ . This action extends to an action of  $\mathbb{G}_a$  on  $\bigwedge^m V$ .

**Proposition 5.7.** With the notation of 5.6, one has

$$(N_m)^{m(d-m)+1} = 0, \tag{5.7.1}$$

and, if  $m(d-m) < p$ , the action of  $\mathbb{G}_a$  on  $\bigwedge^m V$  is given by the truncated exponential  $\exp(tN_m)$ .

**Proof.** As in 5.4, there exists an action  $\rho$  of  $\mathrm{SL}(2)$  on  $V$ , with weights  $< p$ , such that

$$d\rho \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N.$$

It extends to an action  $\rho_m$  of  $\mathrm{SL}(2)$  on  $\bigwedge^m V$ . By 5.5,  $\bigwedge^m V$  has weights  $\leq m(d-m)$ . The vanishing (5.7.1) follows.

If  $m(d-m) < p$ , then by 5.4 the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}(2)$$

on  $\bigwedge^m V$  is given by the truncated exponential  $\exp(tN_m)$ .

**5.8.** Let  $(V_i)$  be a finite family of vector spaces of dimension  $< p$ , and let  $(m_i)$  be a family of integers  $\geq 0$ . For each  $i$ , let  $N_i$  be a nilpotent endomorphism of  $V_i$ . The truncated exponential  $\exp(tN_i)$  is an action of  $\mathbb{G}_a$  on  $V_i$ . These actions define an action of  $\mathbb{G}_a$  on

$$\bigotimes_i \bigwedge^{m_i} V_i$$

and, by passage to the Lie algebra, an endomorphism  $N$  of this tensor product. It is deduced from the  $N_i$ , viewed as actions on the  $V_i$  of the one-dimensional Lie algebra.

**Corollary 5.9.** With the notation of 5.8, if

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.9.1)$$

then  $N^p = 0$  and the action of  $\mathbb{G}_a$  on  $\bigotimes_i \bigwedge^{m_i} V_i$  is given by the truncated exponential  $\exp(tN)$ .

**Proof.** As in 5.4, choose an action  $\rho_i$  of  $\mathrm{SL}(2)$  on  $V_i$ , with weights  $< p$ , such that

$$d\rho_i \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N_i.$$

By 5.5, the weights of  $\mathrm{SL}(2)$  acting on  $\bigotimes_i \bigwedge^{m_i} V_i$  are

$$\leq \sum_i m_i(\dim V_i - m_i) < p,$$

and one applies 5.4 as in 5.7. One could equally well apply 5.7 and 2.2.

**5.10.** The truncated exponential induces a bijection between nilpotent endomorphisms  $N$  with  $N^p = 0$  and automorphisms  $g$  with  $g^p = 1$ , and the corresponding actions of  $\mathbb{G}_a$  coincide:

$$\exp(N)^t = \exp(tN).$$

With the notation of 5.8, if the automorphism  $g_i$  of  $V_i$  satisfies  $g_i^p = 1$ , and if  $g$  is the automorphism of  $\bigotimes_i \bigwedge^{m_i} V_i$  induced by the  $g_i$ , then condition (5.9.1) ensures that  $g^t$  is induced by the  $g_i^t$ .

Recall that, by the remarks following 5.1, 5.2 reduces the proof of 5.1 to the case in which  $k$  is algebraically closed and  $T$  is the category of representations of

$$G \subset \prod \mathrm{GL}(V_i) \subset \mathrm{GL}(V),$$

with  $V := \bigoplus_i V_i$ .

**5.11. Reduction to the case in which  $G$  is doubly saturated in  $\mathrm{GL}(V)$ .** Let  $G^*$  be the double saturation of  $G$  (2.1). As in 2.4, 5.9 and 5.10 ensure that, if (5.1.1) is satisfied, a subspace of  $\bigotimes_i \bigwedge^{m_i} V_i$  stable under  $G$  is automatically stable under  $G^*$ . Semisimplicity as a representation of  $G^*$  therefore implies semisimplicity as a representation of  $G$ .

**5.12. End of the proof of 5.1.** As in 3.1, one reduces to supposing  $G$  reduced. One can then apply [5], p. 26, consequence of the “Main Theorem” on p. 20, to the representations  $V_i$  of  $G(k)$ . One could instead invoke [6].

**5.13.** In 5.1, we use the Schur functors

$$V_i \mapsto \bigwedge^{m_i} V_i,$$

associated with the representations of  $\mathrm{GL}(\dim V_i)$  of dominant weight

$$(1, \dots, 1, 0, \dots, 0)$$

( $m_i$  entries equal to 1). More generally one can consider Schur functors  $S_{\lambda(i)}$ , where  $\lambda(i)$  is a dominant weight

$$(\lambda(i)_1, \dots, \lambda(i)_{\dim V_i})$$

of  $\mathrm{GL}(\dim V_i)$ . Regard  $\lambda(i)$  as a partition. Let  $\mu(i)$  be its dual and put

$$n(\lambda(i)) = \sum_j \mu(i)_j (\dim V_i - \mu(i)_j)$$

(cf. [6] 5.2). Condition (5.5.1) is to be replaced by

$$\sum_i n(\lambda(i)) < p. \quad (5.13.1)$$

That (5.13.1) and the semisimplicity of the  $V_i$  suffice to ensure the semisimplicity of

$$\bigotimes_i S_{\lambda(i)} V_i$$

can be deduced from 5.1 and from the fact that  $S_{\lambda(i)}$  is a subquotient, and even, under the hypotheses made, a direct factor, of the functor

$$V \mapsto \bigotimes_j \bigwedge^{\lambda(i)_j} V$$

for  $V$  of dimension  $\dim(V_i)$ .

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# Counting $\ell$ -adic Sheaves

Pierre Deligne  
English working translation

To Gerard Laumon, on the occasion of his sixtieth birthday.

## 1 Introduction

### 1.1 1.1

Let  $X_0$  be a smooth, projective, absolutely connected curve of genus  $g$  over a finite field  $\mathbf{F}_q$  of characteristic  $p$ , and let  $X$  be obtained from it by extension of scalars to an algebraic closure  $\mathbf{F}$  of  $\mathbf{F}_q$ :

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbf{F}) & \longrightarrow & \mathrm{Spec}(\mathbf{F}_q). \end{array} \quad (1.1.1)$$

The endomorphism of  $X_0$  which is the identity on the underlying topological space and sends  $f \mapsto f^q$  on the structure sheaf is an endomorphism of the  $\mathbf{F}_q$ -scheme. We write  $\mathrm{Frob}$  for the endomorphism of the  $\mathbf{F}$ -scheme  $X$  obtained from it by extension of scalars. This is the Frobenius endomorphism of  $X$ . Fix a prime number  $\ell \neq p$ , an algebraic closure  $\overline{\mathbf{Q}}_\ell$  of  $\mathbf{Q}_\ell$ , and let  $E$  be the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X$ . Inverse image by  $\mathrm{Frob}$  induces a permutation of  $E$ . We denote it by  $V$ .

### 1.2 1.2

In the article [Dr], which remains as mysterious to me as it was thirty-one years ago, Drinfeld computes the number of fixed points of  $V : E \rightarrow E$ . A lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $L_0$  of rank one on  $\mathrm{Spec}(\mathbf{F}_q)$  is determined up to isomorphism by the unit  $\lambda \in \overline{\mathbf{Q}}_\ell^*$  by which the geometric Frobenius  $\mathrm{Fr} \in \mathrm{Gal}(\mathbf{F}/\mathbf{F}_q)$  acts on the fibre of  $L_0$  at the geometric point  $\mathrm{Spec}(\mathbf{F})$ . The  $\mathbf{F}_q$ -twist of  $F_0$  on  $X_0$  by  $L_0$  is the tensor product with the inverse image of  $L_0$  on  $X_0$ . By abuse of language, we shall also say “ $\mathbf{F}_q$ -twist by  $\lambda$ .”

Drinfeld uses the fact that the isomorphism class of a lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F$  on  $X$  is fixed by  $\mathrm{Frob}^*$  if and only if  $F$  is the inverse image of a  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F_0$  on  $X_0$ , and that, if  $F$  is irreducible,  $F_0$  is unique up to  $\mathbf{F}_q$ -twist. The problem solved in [Dr] is thus to count lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X_0$ , taken up to  $\mathbf{F}_q$ -twist, considering only those which are irreducible and remain so after inverse image on  $X$ .

In 1981 one almost had the correspondence between irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X_0$  and everywhere unramified cuspidal automorphic representations for  $\mathrm{GL}(2, k(X_0))$ . Thanks to this, Drinfeld reduced the problem to an application of the trace formula for  $\mathrm{GL}(2)$ . This reduction shows that the desired number is independent of  $\ell$ . The formula obtained by Drinfeld has the following miraculous properties (A) and (B).

For  $n \geq 1$ , let  $N_n$  be the number of fixed points of the  $n$ -th iterate  $V^n : E \rightarrow E$  of  $V$ . Computing  $N_n$  is the above problem with  $X_0/\mathbf{F}_q$  replaced by  $(X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n})/\mathbf{F}_{q^n}$ .

(A) The function  $n \mapsto N_n$  has the form

$$N_n = \sum a_i \beta_i^n, \quad (1.2.1)$$

for integers  $a_i$  and suitable Weil numbers  $\beta_i$ . The  $\beta_i$  are monomials in  $q$  and in the eigenvalues of the endomorphism  $\text{Frob}^*$  of  $H^1(X, \overline{\mathbf{Q}}_\ell)$ . Once the genus  $g$  is fixed, which monomials  $\beta_i$  occur, and with what multiplicities  $a_i$ , is independent of the curve under consideration.

The trace formula expresses  $N_n$  as the sum of several terms. Taken separately, these terms do not all have the form (1.2.1).

Let  $P$  be a compact Riemann surface of genus  $g$ . Consider complex local systems of vector spaces on  $P$ . Let  $M$  be the moduli space of those which are irreducible of rank 2. The space  $M$  and the set  $E$  are empty if  $g = 0$  or  $1$ . Otherwise  $M$  is a connected complex symplectic variety. Its complex dimension is therefore an even integer  $2N$ .

(B) In (1.2.1), the dominant term is  $q^N$ : one of the  $\beta_i$  is  $q^N$ , its multiplicity  $a_i$  is 1, and the other  $\beta_i$  satisfy  $|\beta_i| < q^N$ .

### 1.3 1.3

Formula (1.2.1) is reminiscent of a Lefschetz trace formula in which, in good cases, the number of fixed points of the iterates  $T^n$  ( $n \geq 1$ ) of an endomorphism  $T$  of a space  $S$  is

$$\text{Tr}(T^{*n}, H_{\mathbb{Q}}^*(S)) := \sum (-1)^i \text{Tr}(T^{*n}, H_{\mathbb{Q}}^i(S)). \quad (1.3.1)$$

The question mark is there to recall that one must consider a cohomology with support conditions. The support conditions to be imposed depend on the behaviour of  $T$  at infinity. For example, if for an exhaustion function  $f$  on  $S$  one has  $f(T(x)) > f(x)$  (respectively  $f(T(x)) < f(x)$ ) for  $f(x)$  sufficiently large, the cohomology to consider is compactly supported cohomology (respectively ordinary cohomology).

Assume that  $H_{\mathbb{Q}}^*$  is finite-dimensional so that (1.3.1) makes sense. The right hand side of (1.3.1) is then of the form (1.2.1): it is  $\sum a(\beta)\beta^n$ , where  $\beta$  ranges over the eigenvalues of  $T^*$  and the integer  $a(\beta)$  is the alternating sum of the multiplicities of  $\beta$  as an eigenvalue of the endomorphisms  $T^*$  on the  $H_{\mathbb{Q}}^i(S)$ .

Unfortunately I have no idea how to regard the set  $E$  of isomorphism classes of irreducible lisse  $\mathbf{Q}_\ell$ -sheaves of rank 2 on  $X$  as a “space” having a cohomology  $H_{\mathbb{Q}}^*(E)$  for which one could hope that the number of fixed points of  $V^n$  is given by a formula of type (1.3.1).

### 1.4 1.4

Even though one does not know how to think geometrically of  $E$ , if  $f \in E$  is the isomorphism class of  $F$ , one knows what the formal completion  $\widehat{E}_f$  of  $E$  at  $f$  should be. A finite-dimensional local  $\mathbf{Q}_\ell$ -algebra  $\Lambda$  is augmented over  $\mathbf{Q}_\ell$ . A deformation of  $F$  over  $\text{Spec}(\Lambda)$  is a lisse  $\mathbf{Q}_\ell$ -sheaf  $F_\Lambda$  on  $X$ , endowed with a  $\Lambda$ -module structure and an isomorphism

$$F_\Lambda \otimes_\Lambda \overline{\mathbf{Q}}_\ell \xrightarrow{\sim} F,$$

such that at one point  $x$  (and therefore at every point  $x$  of  $X$ ),  $(F_\Lambda)_x$  is a free  $\Lambda$ -module.

The functor on  $\text{Spec}(\Lambda)$  of isomorphism classes of deformations of  $F$  over  $\text{Spec}(\Lambda)$  is pro-representable. Let  $R_F$  be the complete local  $\overline{\mathbf{Q}}_\ell$ -algebra with residue field  $\overline{\mathbf{Q}}_\ell$  such that  $\text{Spf}(R_F)$  pro-represents it. There are no obstructions to deformations, and  $R_F$  is therefore isomorphic to a formal power series algebra  $\overline{\mathbf{Q}}_\ell[[t_1, \dots, t_m]]$ .

Because  $F$  is irreducible, its only automorphisms are multiplications by  $\lambda \in \overline{\mathbf{Q}}_\ell^*$ . They extend to every deformation. If  $F$  and  $F'$  are two representatives of the isomorphism class  $f$ , two isomorphisms from  $F$  to  $F'$  therefore induce the same isomorphism between  $R_F$  and  $R_{F'}$ . Thus  $\widehat{E}_f := \text{Spf}(R_F)$  depends, up to unique isomorphism, only on  $f$ .

The Zariski tangent space of  $\widehat{E}_f$  at  $f$  is  $H^1(X, \text{End}(F))$ . The symmetric bilinear form  $\text{Tr}(fg)$  on  $\text{End}(F)$  is a self-duality. It induces on  $H^1(X, \text{End}(F))$  a symplectic form with values

in  $H^2(X, \overline{\mathbf{Q}}_\ell) = \overline{\mathbf{Q}}_\ell(-1)$ . The same construction over  $\mathbf{C}$  gives the symplectic structure of  $M$ . The formal completion  $\widehat{E}_f$  is therefore smooth of even dimension, the same as that of  $M$ .

The inverse-image functor by  $\text{Frob} : X \rightarrow X$  transforms deformations of  $F$  into deformations of  $\text{Frob}^* F$ . It therefore induces a morphism

$$\widehat{V}_f : \widehat{E}_f \longrightarrow \widehat{E}_{V(f)}.$$

The argument showing that, because  $\text{Frob}$  induces an equivalence of étale sites,  $V$  is bijective, also shows that the  $\widehat{V}_f$  are isomorphisms. Another proof is as follows. Let  $\omega$  be the symplectic form defined above on the tangent space at  $f$ . If  $[t] \in H^1(X, \text{End}(F))$  is the class of the tangent vector  $t$  at  $F$ , the class of  $d\widehat{V}_f(t)$  in  $H^1(X, \text{End}(\text{Frob}^* F))$  is  $\text{Frob}^*([t])$ . For two tangent vectors  $t_1, t_2$ ,  $\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2))$  is thus the inverse image, by  $\text{Frob}^*$ , of  $\omega(t_1, t_2) \in \overline{\mathbf{Q}}_\ell(-1)$ , identified with  $H^2(X, \overline{\mathbf{Q}}_\ell)$ . Since  $\text{Frob} : X \rightarrow X$  has degree  $q$ , the endomorphism  $\text{Frob}^*$  of  $H^2(X, \overline{\mathbf{Q}}_\ell)$  is multiplication by  $q$ , and

$$\omega(d\widehat{V}_f(t_1), d\widehat{V}_f(t_2)) = q \omega(t_1, t_2),$$

which is also written  $(d\widehat{V}_f)^* \omega = q\omega$ . The differential  $d\widehat{V}_f$  is therefore an isomorphism.

The map  $d\widehat{V}_f$  between Zariski tangent spaces is identified with

$$\text{Frob}^* : H^1(X, \text{End}(F)) \longrightarrow H^1(X, \text{End}(\text{Frob}^* F)).$$

If  $f$  is a fixed point of  $V$ , then  $F$  is the inverse image of  $F_0$  on  $X_0$ . One knows that  $F_0$  may be chosen pure, and  $\text{End}(F_0)$  is therefore pure of weight 0. The eigenvalues of  $d\widehat{V}_f$  on the Zariski tangent space of  $\widehat{E}_f$  at  $f$  are therefore Weil numbers of weight 1. In particular no eigenvalue of  $d\widehat{V}_f$  at  $f$  is equal to 1. This is what makes it reasonable that a Lefschetz-type formula (1.3.1) might count the fixed points of  $V^n$ , all taken with multiplicity one.

## 1.5 1.5

If  $Y_0$  is a scheme of finite type over  $\mathbf{F}_q$ , the Frobenius endomorphism  $\text{Frob}$  of  $Y := Y_0 \otimes_{\mathbf{F}_q} \mathbf{F}$  induces a permutation  $\text{Frob}$  of  $Y(\mathbf{F})$ . The preceding discussion shows that the nature of  $(E, V)$  is different from that of  $(Y, \text{Frob})$ :

- (i)  $E$  is “over  $\overline{\mathbf{Q}}_\ell$ ,” hence an object of characteristic 0.
- (ii) The  $\widehat{V}_f$  are isomorphisms. One would like to say that  $V$  has “degree one.”
- (iii) The fixed points of  $V$  are isolated because the eigenvalues of  $dV$  at a fixed point are Weil numbers of weight 1. The fixed points of  $\text{Frob}$  are isolated because  $d\text{Frob} = 0$ .
- (iv) If  $E$  has dimension  $2N$ , in the sense that the  $\widehat{E}_f$  do, the order of magnitude of the number of fixed points of  $V^n$  is  $(q^n)^N$ , rather than  $(q^n)^{2N}$ .

## 1.6 1.6

The moduli space  $M$  is a complex algebraic variety. Contrary to what a hasty analogy might suggest,  $E$  is not naturally the space of  $\overline{\mathbf{Q}}_\ell$ -points of an algebraic variety over  $\overline{\mathbf{Q}}_\ell$ . Let  $x \in X(\mathbf{F})$ . The set  $E$  is identified with the set of isomorphism classes of continuous irreducible representations  $V$  of  $\pi_1(X, x)$ , of dimension 2 over  $\overline{\mathbf{Q}}_\ell$ . As coordinates on  $E$ , one would like to take the  $\text{Tr}(\gamma, V)$  for  $\gamma \in \pi_1(X, x)$ . In the complex case this construction gives  $M$  its “Betti” structure as a complex algebraic variety. Here, since  $\pi_1(X, x)$  is a profinite group, every representation  $V$  admits an invariant lattice, and the  $\text{Tr}(\gamma, V)$  take their values in the valuation ring  $\mathbf{Z}_\ell$  of the valued field  $\overline{\mathbf{Q}}_\ell$ .

The point of view “ $E$  as a space of representations of  $\pi_1(X, x)$ ” also does not help one understand that the number of fixed points of  $V$  is independent of  $\ell$ . Another mystery: why,



once the genus  $g$  is fixed, does the number of fixed points of  $V$  admit a uniform description in terms of  $H^1(X, \overline{\mathbf{Q}}_\ell)$ , of its symplectic structure with values in  $\overline{\mathbf{Q}}_\ell(-1)$ , and of its endomorphism  $\text{Frob}^*$ , whereas  $\pi_1(X, x)$  is a quotient of the profinite completion of the topological  $\pi_1$  of  $P$ , which quotient depends on  $X$ ?

## 1.7 1.7

Let  $E_r$  be the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank  $r$  on  $X$ . Thanks to Lafforgue [L], the number of fixed points of the bijection  $\text{Frob}^* : E_r \rightarrow E_r$  has an automorphic interpretation. For  $S_0 \subset X_0$  a finite set of closed points of  $X$ , and  $S = S_0 \otimes_{\mathbf{F}_q} \mathbf{F}$ , one may also consider irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves on  $X - S$ , with prescribed ramification at each  $s \in S$ . In all cases that have been computed, the miracles (A) and (B) discovered by Drinfeld persist, mutatis mutandis. The aim of the talk is to give a panorama of what is known.

## 2 What is to be counted?

### 2.1 2.1

Fix an integer  $r \geq 1$ , and let  $X_0, S_0/\mathbf{F}_q$  and  $X, S/\mathbf{F}$  be as in 1.1 and 1.7. For  $s \in S$ , let  $X_{(s)}$  be the henselization of  $X$  at  $s$  and  $X_{(s)}^* := X_{(s)} - \{s\}$ . Suppose given, for each  $s \in S$ , a lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf of rank  $r$ , denoted  $R(s)$ , on  $X_{(s)}^*$ . Only its isomorphism class matters. The morphism  $\text{Frob} : X \rightarrow X$  induces a morphism, again denoted  $\text{Frob}$ , from  $X_{(s)}^*$  to  $X_{(\text{Frob}(s))}^*$ . We assume that, for every  $s \in S$ ,  $R(s)$  is isomorphic to the inverse image of  $R(\text{Frob}(s))$  by  $\text{Frob}$ :

$$R(s) \simeq \text{Frob}^*(R(\text{Frob}(s))). \quad (2.1.1)$$

Let  $R := (R(s))_{s \in S}$  be the family of the  $R(s)$ , and write  $E(R)$  for the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves  $F$  of rank  $r$  on  $X - S$  such that, for every  $s \in S$ , the inverse image  $F|_{X_{(s)}^*}$  of  $F$  on  $X_{(s)}^*$  is isomorphic to  $R(s)$ . Hypothesis (2.1.1) ensures that inverse image by  $\text{Frob}$  induces a permutation of  $E(R)$ . We denote this permutation by  $V$ .

Our choice of the letter  $V$ , the first letter of “Verschiebung,” is due to an analogy with the case of certain abelian varieties, explained in 2.2 below. In that analogy  $q$  is classically a prime number, rather than a power of a prime number.

### 2.2 2.2

For any scheme  $Y$  over  $\mathbf{F}_q$ , let  $\text{Frob}_Y$  be the endomorphism of the  $\mathbf{F}_q$ -scheme  $Y$  which is the identity on the underlying topological space and which on the structure sheaf is given by  $\text{Frob}_Y^*(f) = f^q$ . This is the Frobenius endomorphism of  $Y$ .

Take for  $Y$  the abelian variety  $\text{Pic}^0(X_0)$  over  $\mathbf{F}_q$ . We obtain

$$\text{Frob}_{\text{Pic}^0(X_0)} : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(X_0). \quad (2.2.1)$$

The Jacobian  $\text{Pic}^0(X_0)$  represents the following functor on  $\mathbf{F}_q$ -schemes: the sheaf (for the Zariski topology, or for the étale topology; it is the same) associated to the presheaf attaching to  $S$  the group  $\text{Pic}^0(S \times_{\mathbf{F}_q} X_0)$  of invertible sheaves on  $S \times_{\mathbf{F}_q} X_0$ , of degree zero on each fibre of the projection to  $S$ . If  $f : Y_0 \rightarrow X_0$  is a morphism, inverse image of invertible sheaves by  $f$  defines a morphism  $f^* : \text{Pic}^0(X_0) \rightarrow \text{Pic}^0(Y_0)$ .

Take for  $f$  the endomorphism  $\text{Frob}_{X_0}$  of  $X_0$ . The endomorphism  $\text{Frob}_{X_0}^*$  of  $\text{Pic}^0(X_0)$  deserves to be called  $V$ , for it satisfies

$$\text{Frob}_{X_0}^* \circ \text{Frob}_{\text{Pic}^0(X_0)} = \text{Frob}_{\text{Pic}^0(X_0)} \circ \text{Frob}_{X_0}^* = q. \quad (2.2.2)$$

Proof of (2.2.2). Interpret (2.2.2) as two equalities between endomorphisms of the functor represented by  $\text{Pic}^0(X_0)$ . For any contravariant functor  $G$  on the category of  $\mathbf{F}_q$ -schemes, the Frobenius endomorphism  $\text{Frob}_G$  of  $G$  is defined by

$$\text{Frob}_G(S) : G(S) \rightarrow G(S), \quad \text{namely } G(\text{Frob}_S).$$

If  $G$  is represented by  $Y$ ,  $\text{Frob}_G$  is induced by  $\text{Frob}_Y$ .

Take for  $G$  the functor  $S \mapsto \text{Pic}(S \times_{\mathbf{F}_q} X_0)$  of isomorphism classes of invertible sheaves on  $S \times_{\mathbf{F}_q} X_0$ . The endomorphism  $\text{Frob}_G(S)$  of  $\text{Pic}(S \times_{\mathbf{F}_q} X_0)$  is inverse image of isomorphism classes of invertible sheaves by  $\text{Frob}_S \times \text{id}_{X_0} : S \times_{\mathbf{F}_q} X_0 \rightarrow S \times_{\mathbf{F}_q} X_0$ . If one composes this inverse image with the inverse image by  $\text{Frob}_{X_0}$ , one obtains inverse image by  $\text{Frob}_{S \times_{\mathbf{F}_q} X_0}$ , which is nothing other than raising to the  $q$ -th tensor power. Restricting to invertible sheaves of degree zero on the fibres of the projection to  $S$ , and passing from the presheaf obtained to the associated sheaf, gives (2.2.2).

If now one extends scalars from  $\mathbf{F}_q$  to  $\mathbf{F}$ ,  $\text{Frob}_{X_0}$  becomes the endomorphism  $\text{Frob}$  of  $X$ ,  $\text{Pic}^0(X_0)$  becomes  $\text{Pic}^0(X)$ , and the endomorphism  $\text{Frob}_{X_0}^*$  of  $\text{Pic}^0(X_0)$  becomes the endomorphism of  $\text{Pic}^0(X)$  “inverse image by  $\text{Frob}$ .”

### 2.3 2.3–2.6

The questions we ask are the following.

- (i) Compute, for  $n \geq 1$ , the number  $N_n(R)$  of fixed points of the  $n$ -th iterate  $V^n$  of the permutation  $V$  of  $E(R)$ . Replacing  $(X_0, S_0)$  by  $(X_0, S_0) \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$  replaces  $V$  by  $V^n$ .
- (ii) Determine whether the miracles 1.2 (A) and (B) persist.
- (iii) If so, why?

Let  $F$  be a lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf on  $X - S$ . As in 1.2, one has:

**Lemma 2.4.** (i) For the isomorphism class of  $F$  to be fixed under  $\text{Frob}^*$ , it is necessary and sufficient that  $F$  be isomorphic to the inverse image on  $X$  of a  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F_0$  on  $X_0$ . Moreover, if  $F$  is irreducible,

- (ii) such an  $F_0$  is unique up to  $\mathbf{F}_q$ -twist, and
- (iii) the  $\mathbf{F}_q$ -twists of  $F_0$  by  $\lambda \in \overline{\mathbf{Z}}_\ell^*$  are pairwise non-isomorphic.

According to 2.4, the number of fixed points of the permutation  $V$  of  $E(R)$  is the number of classes, modulo  $\mathbf{F}_q$ -twist, of lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves  $F_0$  of rank  $r$  on  $X_0 - S_0$ , such that

- (a) the inverse image  $F$  of  $F_0$  on  $X - S$  is irreducible;
- (b) for each  $s \in S$ ,  $F|_{X_{(s)}^*}$  is isomorphic to  $R(s)$ .

Let  $D_0 = \sum n_i x_i$  be a divisor of degree one on  $X_0 - S_0$ . Regard  $F_0$  as a Galois representation (see 2.7), let  $F_{x_i}$  be a Frobenius at  $x_i$ , and put

$$\langle D_0, F_0 \rangle := \prod \det(F_{x_i}, F_0)^{n_i}.$$

If  $F'_0$  is the  $\mathbf{F}_q$ -twist of  $F_0$  by  $\lambda$ , then

$$\langle D_0, F'_0 \rangle = \lambda^r \langle D_0, F_0 \rangle.$$

By 2.4 (ii) and (iii), it is therefore equivalent either to count the  $F_0$  satisfying (a) and (b) up to  $\mathbf{F}_q$ -twist, or to count, up to isomorphism, only those which also satisfy  $\langle D_0, F \rangle = 1$ , and then divide by  $r$ . For an automorphic counterpart, see the remark following 2.6.

Put  $K := k(X_0)$ . For each closed point  $x$  of  $X_0$ , let  $k(x)$  be the residue field of  $x$ ,  $\deg(x) := [k(x) : \mathbf{F}_q]$  its degree,  $K_x$  the completion of  $K$  at  $x$ , and  $O_x$  the valuation ring of  $K_x$ . Let  $\mathbf{A}$  be the adèle ring of  $K$ . The sum  $v := \sum \deg(x)v_x : \mathbf{A}^* \rightarrow \mathbf{Z}$  is trivial on  $K^*$ . One has  $\|a\| = q^{-v(a)}$ .

By Lafforgue,  $F_0$  corresponds to a cuspidal automorphic representation  $\pi$  of  $\mathrm{GL}(r, \mathbf{A})$ . If  $F'_0$  is the  $\mathbf{F}_q$ -twist of  $F_0$  by  $\lambda$ , the corresponding representation  $\pi'$  is the  $\mathbf{F}_q$ -twist of  $\pi$  by  $\lambda$ , defined as follows. As a representation, it is the tensor product of  $\pi$  by the one-dimensional representation  $g \mapsto \lambda^{v \det g}$ . As a space of functions on  $\mathrm{GL}(r, \mathbf{A})$ , it is the space of products of  $f \in \pi$  with the character  $g \mapsto \lambda^{v \det g}$ .

Translated into automorphic language, problem 2.3 (i) is to count, up to  $\mathbf{F}_q$ -twist, the cuspidal automorphic representations  $\pi$  of  $\mathrm{GL}(r, \mathbf{A})$ , unramified outside  $S_0$ , and such that

- (a') for every  $n \geq 1$ ,  $\pi$  remains cuspidal after base change from  $K$  to  $K \otimes \mathbf{F}_{q^n}$ . This condition expresses the irreducibility 2.4 (a) of  $F$ . It suffices to check it for  $n$  dividing  $r$ .
- (b') for each  $s_0 \in S_0$  and  $s \in S$  above  $s_0$ , the local component  $\pi_{s_0}$  of  $\pi$  satisfies a condition dictated by  $R(s)$  and by the local Langlands correspondence.

**Lemma 2.6.** If the cuspidal automorphic representation  $\pi$  satisfies (a'), then its  $\mathbf{F}_q$ -twists are pairwise distinct. This is the translation of 2.4 (iii). If  $d$  is an idele satisfying  $v(d) = 1$ , it is therefore equivalent to count up to  $\mathbf{F}_q$ -twist the  $\pi$  satisfying (a') and (b'), or to count only those whose central character  $\omega_\pi$  satisfies  $\omega_\pi(d) = 1$ , and then divide by  $r$ .

In finite characteristic the notion of automorphic representation is purely algebraic. This makes it possible to take the automorphic representations  $\pi$  to have coefficients in  $\overline{\mathbf{Q}}_\ell$ , rather than in  $\mathbf{C}$ . If one does this,  $\pi_{s_0}$  is an irreducible admissible  $\overline{\mathbf{Q}}_\ell$ -representation of  $\mathrm{GL}(r, K_{s_0})$ , and the local Langlands correspondence associates to it an isomorphism class of continuous  $F$ -semisimple  $\overline{\mathbf{Q}}_\ell$ -representations of dimension  $r$  of the local Weil group  $W(\overline{K}_{s_0}/K_{s_0})$ . Condition (b') is that its restriction to the inertia subgroup be given by  $R(s)$ .

Almost all known results on question 2.3 (i) are obtained through the dictionary 2.5.

## 2.4 2.7–2.16

In 2.1 we used the language of  $\overline{\mathbf{Q}}_\ell$ -sheaves. This saved us from having to choose base points. An equivalent language is that of Galois representations. Here is the translation. Let  $\overline{k(X)}$  be an algebraic closure of the field of rational functions  $k(X)$  of  $X$ , and let  $k(X)^{\mathrm{nr}}$  be the largest subextension unramified outside  $S$ . The Galois group  $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$  is the fundamental group of  $X$  at the geometric point  $\bar{\eta} := \mathrm{Spec}(k(X))$  of  $X$ : the fibre functor at  $\bar{\eta}$  is an equivalence from lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves (respectively irreducible ones of rank  $r$ ) on  $X - S$  to linear  $\overline{\mathbf{Q}}_\ell$ -representations of  $\mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X))$  (respectively irreducible representations of rank  $r$ ).

The scheme  $X_{(s)}^*$  is the spectrum of an extension  $k(X_{(s)}^*)$  of  $k(X)$ . If one chooses an embedding of this extension into  $\overline{k(X)}$ , the corresponding inertia group  $I_s := \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*))$  is likewise the fundamental group of  $X_{(s)}^*$  at  $\bar{\eta}$ . It maps to  $\pi_1(X, \bar{\eta})$ :

$$\pi_1(X_{(s)}^*, \bar{\eta}) = I_s = \mathrm{Gal}(\overline{k(X)}/k(X_{(s)}^*)) \rightarrow \mathrm{Gal}(k(X)^{\mathrm{nr}}/k(X)) = \pi_1(X, \bar{\eta}). \quad (2.7.1)$$

The fibre at  $\bar{\eta}$  of  $R(s)$  is a representation  $\rho_s$  of  $I_s$ . The condition that  $F|_{X_{(s)}^*}$  be isomorphic to  $R(s)$  is equivalent to the restriction to  $I_s$  being isomorphic to  $\rho_s$ . Condition (2.1.1) is equivalent to saying that the  $\rho_s$ , for  $s$  over  $s_0$ , come from a representation of a decomposition group  $D_{s_0} \subset \mathrm{Gal}(k(X)/k(X_0))$  at  $s_0$ . It implies that the representations  $\rho_s$  are quasi-unipotent.

I hope problem 2.3 (i) is more manageable when a lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F$  such that  $F|_{X_{(s)}^*} \simeq R(s)$  for every  $s \in S$  is automatically irreducible. This is the case if the following condition holds. Consider an integer  $0 < r' < r$  and, for each  $s \in S$ , a lisse sub- $\overline{\mathbf{Q}}_\ell$ -sheaf  $R'(s)$  of rank  $r'$  of  $R(s)$ . Put  $R''(s) := R(s)/R'(s)$ . The condition is:

For all  $r'$  and all  $R'(s)$ , either there exists no lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F'$  of rank  $r'$  on  $X - S$  with  $F'|_{X_{(s)}^*} \simeq R'(s)$  for all  $s$ , or there exists no lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F''$  of rank  $r - r'$  on  $X - S$  with  $F''|_{X_{(s)}^*} \simeq R''(s)$  for all  $s$ . (2.8.1)

Condition (2.8.1) is satisfied if one of the  $R(s)$  is irreducible. For other cases, see 2.10.4, 2.11, and 2.12.

For any field  $k$  of characteristic  $p$  or 0, containing all roots of unity of order prime to  $p$ , set

$$\widehat{\mathbf{Z}}_{p'}(1)(k) := \varprojlim \mu_n(k). \quad (2.9.1)$$

The projective limit is taken over the set of subgroups  $\mu_n(k) \subset k^*$  ( $n$  prime to  $p$ ), ordered by inclusion. The transition morphism from  $\mu_m(k)$  to  $\mu_n(k)$  is  $x \mapsto x^{m/n}$ . If no ambiguity is created, we omit “ $p'$ ” and the mention of  $k$ .

For  $k = \mathbf{F}$ , and  $\mathbf{F}_{q^k}$  the subfield of  $\mathbf{F}$  with  $q^k$  elements, the subgroups  $\mathbf{F}_{q^k}^*$  of  $\mathbf{F}^*$  are cofinal among the  $\mu_n(\mathbf{F})$ . Since  $(q^{kd} - 1)/(q^k - 1) = 1 + q^k + \dots + q^{k(d-1)}$ , the transition morphism from  $\mu_{q^{kd-1}} = \mathbf{F}_{q^{kd}}^*$  to  $\mu_{q^{k-1}} = \mathbf{F}_{q^k}^*$  is the norm morphism, and

$$\widehat{\mathbf{Z}}(1) \xrightarrow{\sim} \varprojlim \mathbf{F}_{q^k}^* \quad (2.9.2)$$

(the projective limit being taken with respect to the norm morphisms between the  $\mathbf{F}_{q^k}^*$ ). For  $k$  a product  $\prod k_\alpha$  of finite extensions of  $\mathbf{F}_q$  and  $\iota : k \rightarrow k_\alpha \xrightarrow{\sim} \mathbf{F}_{q^k} \hookrightarrow \mathbf{F}$ , write  $\text{pr}_\iota$  for the composite

$$\text{pr}_\iota : \widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_{q^k}^* \xleftarrow{\sim} k_\alpha^* \hookrightarrow \bigoplus k_\alpha^* = k^*. \quad (2.9.3)$$

For  $k = \mathbf{F}_{q^k}$  and  $\iota$  the identical inclusion,  $\text{Gal}(\mathbf{F}/\mathbf{F}_{q^k})$  acts on  $\widehat{\mathbf{Z}}(1)$ , and (2.9.3) induces an isomorphism of coinvariants with  $\mathbf{F}_{q^k}^*$ .

The local fundamental group  $\pi_1(X_{(s)}^*, \bar{\eta}) = I_s$  is an extension of  $\widehat{\mathbf{Z}}(1)$  by a pro- $p$ -group  $P_s$ . The quotient  $I_s/P_s \simeq \widehat{\mathbf{Z}}(1)$  is the tame fundamental group of  $X_{(s)}^*$ . The morphism  $\text{Frob} : X_{(s)}^* \rightarrow X_{(\text{Frob}(s))}^*$  induces a morphism between tame fundamental groups. Via the identifications of these groups with  $\widehat{\mathbf{Z}}(1)$ , this morphism is multiplication by  $q$ .

Let  $\mathbf{Z}_\ell(1)$  denote the quotient  $\varprojlim \mu_{\ell^n}(\mathbf{F})$  of  $\widehat{\mathbf{Z}}(1)$ . Every quasi-unipotent  $\overline{\mathbf{Q}}_\ell$ -representation  $\rho$  of  $\widehat{\mathbf{Z}}(1)$  is isomorphic to a sum over finite-order characters  $\chi : \widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$ :

$$\rho \simeq \bigoplus_\chi \chi \otimes \nu_\rho(\chi), \quad (2.9.4)$$

where  $\nu_\rho(\chi)$  is a unipotent representation of the quotient  $\mathbf{Z}_\ell(1)$  of  $\widehat{\mathbf{Z}}(1)$ . Let  $\lambda_\rho(\chi)$  be the following partition of the integer  $\dim \nu_\rho(\chi)$ : the decreasing sequence of sizes of the Jordan blocks of the image of a generator of  $\mathbf{Z}_\ell(1)$  in the representation  $\nu_\rho(\chi)$ . The isomorphism class of  $\rho$  is determined by the function  $\chi \mapsto \lambda_\rho(\chi)$ .

Assume that the  $R(s)$  are tame, i.e. that the corresponding representations  $\rho_s$  of the inertia groups  $I_s$  (2.7) factor through a representation of  $\widehat{\mathbf{Z}}(1)$ . We write  $\chi \mapsto \lambda_s(\chi)$  for the function, constructed above, attached to this representation. With this notation, condition (2.1.1) is equivalent to

$$\lambda_{\text{Frob}(s)}(\chi) = \lambda_s(\chi^q). \quad (2.9.5)$$

If  $L$  is a  $\overline{\mathbf{Q}}_\ell$ -sheaf of rank one on  $X - S$ , the restriction of  $L$  to  $X_{(s)}^*$  defines a character  $l_s$  of the fundamental group  $I_s$  of  $X_{(s)}^*$ . Since  $\ell \neq p$ , the restriction of  $l_s$  to  $P_s$  is of finite order: there is a power  $P$  of  $p$  such that the  $l_s^P$  factor through characters, again denoted  $l_s^P$ , of  $\widehat{\mathbf{Z}}(1) = I_s/P_s$ . One has

$$\prod l_s^P = 1. \quad (2.10.1)$$

The analogue over  $\mathbf{C}$  of (2.10.1) is that, for  $P$  as in 1.2 and  $S$  a finite subset of  $P$ , the sum over  $S$  of a small positive circle around each  $s \in S$  is a boundary in  $P - S$ .

In rank  $r$ , (2.10.1) applied to  $\bigwedge^r F$  provides a necessary condition for the existence of  $F$  on  $X - S$  satisfying  $F|_{X_{(s)}^*} \simeq R(s)$ , and interesting cases in which the hypothesis of the irreducibility criterion (2.8.1) is satisfied. Let  $\rho_s$  be the representation of  $I_s$  corresponding to  $R(s)$ . Assume, as in 2.9, that the representations  $\rho_s$  are tame. Let  $\mathcal{R}(s)$  be the multiset of characters of  $\widehat{\mathbf{Z}}(1)$  forming the semisimplification of  $\rho_s$ . With the notation of 2.9, the multiplicity in  $\mathcal{R}(s)$  of a character  $\chi$  is  $|\lambda_s(\chi)| = \dim \nu_{\rho_s}(\chi)$ . Assume that

$$\prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (2.10.2)$$

By (2.10.1), this is a necessary condition for the existence of an  $F$  on  $X - S$  such that  $F|_{X_{(s)}^*} \simeq R_s$  for all  $s \in S$ . Consider the condition:

$$\text{For every } 0 < r' < r \text{ and every family } (\mathcal{R}'(s))_{s \in S} \text{ of sub-multisets with } r' \text{ elements} \\ \text{of the multisets } \mathcal{R}(s), \text{ one has } \prod_{s \in S} \prod_{\varepsilon \in \mathcal{R}'(s)} \varepsilon \neq 1. \quad (2.10.3)$$

**Lemma 2.11.** Condition (2.10.3) implies (2.8.1), and hence the irreducibility of every lisse sheaf  $F$  of rank  $r$  on  $X - S$  satisfying  $F|_{X_{(s)}^*} \simeq R_s$  for all  $s \in S$ .

**Remark 2.12.** There exist  $R(s)$  as above, giving rise to multisets  $\mathcal{R}(s)$  satisfying (2.10.2), for which (2.10.3) is false but where condition (2.8.1) is nevertheless satisfied. Here is an example. Take  $X = \mathbf{P}^1$ ,  $r = 4$ ,  $|S| = 3$ , and semisimple  $R(s)$ , sums of multisets of characters  $\mathcal{R}(s)$  of the following type

$$\{a_1, a_1, a_3, a_4\}, \quad \{b_1, b_1, b_3, b_4\}, \quad \{c_1, c_2, c_3, c_4\}.$$

Assume that the product of all these characters is 1 (2.10.2), and that  $a_4 b_4 c_4 = 1$ , so that (2.10.3) is false. If the  $a_i, b_i, c_i$  are “general,” condition (2.8.1) is satisfied, because there is no  $F$  of rank 3 with local monodromy given by

$$\{a_1, a_1, a_3\}, \quad \{b_1, b_1, b_3\}, \quad \{c_1, c_2, c_3\}.$$

The verification is by reduction to the complex case, by lifting  $(X, S)$  to characteristic 0. This reduction brings us to the following lemma.

**Lemma 2.13.** Let  $u_1, u_3, v_1, v_3, w_1, w_2, w_3 \in \overline{\mathbf{Q}}_\ell^*$ . Assume that  $u_1 v_1 w_i \neq 1$  for  $i = 1, 2, 3$ . Under this hypothesis there do not exist elements  $U, V, W$  of  $\text{GL}(3, \overline{\mathbf{Q}}_\ell)$ , conjugate respectively to the diagonal matrices  $(u_1, u_1, u_3)$ ,  $(v_1, v_1, v_3)$ ,  $(w_1, w_2, w_3)$ , with  $UVW = 1$ .

*Proof.* Let  $L$  be the intersection of the kernels of  $U - u_1$  and  $V - v_1$ . These kernels have dimension at least 2, so  $L$  is nonzero. On  $L$ ,  $U$  is multiplication by  $u_1$ ,  $V$  is multiplication by  $v_1$ , and, since  $UVW = 1$ ,  $W$  is multiplication by  $1/(u_1 v_1)$ . Since  $1/(u_1 v_1)$  is not an eigenvalue of  $W$ , this is impossible.

Let  $P$  be a compact Riemann surface of genus  $g$ . Suppose an injection of  $S$  into  $P$  has been chosen; we identify  $S$  with a finite set of points of  $P$ . If  $P_{(s)}$  is a small disk around  $s$ , the fundamental group of  $P_{(s)}^* := P_{(s)} - \{0\}$  is  $\mathbf{Z}$ , generated by one positive turn around  $s$ . More intrinsically, this local fundamental group is  $\mathbf{Z}(1) := 2\pi i \mathbf{Z}$ , with  $2\pi i$  corresponding to the positive turn around  $s$ . For each  $n \geq 1$ , identify  $\mathbf{Z}(1)/n$  with the group of  $n$ -th roots of unity in  $\mathbf{C}$  by  $z \mapsto \exp(z/n)$ . Passing to the limit gives an isomorphism of  $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$  with the projective limit of the  $\mathbf{Z}(1)/n$  for  $n$  prime to  $p$ .

A complex representation  $\rho$  of  $\mathbf{Z}(1)$  is isomorphic to a sum over characters  $\chi : \mathbf{Z}(1) \rightarrow \mathbf{C}^*$ :

$$\rho \simeq \bigoplus_{\chi} \chi \otimes \nu(\chi), \quad (2.14.1)$$

where  $\nu(\chi)$  is a unipotent representation of  $\mathbf{Z}(1)$ . Define the partition  $\lambda(\chi)$  of  $\dim \nu(\chi)$  as in 2.9. The isomorphism class of  $\rho$  is determined by the function  $\chi \mapsto \nu(\chi)$ . We shall only have

to consider the case where the characters  $\chi$  occurring in (2.14.1) are of finite order prime to  $p$ , and hence may be viewed as complex characters of  $\widehat{\mathbf{Z}}_{p'}(1)(\mathbf{C})$ .

If  $T$  is a family  $(T_s)_{s \in S}$  of isomorphism classes of complex representations of  $\mathbf{Z}(1)$ , we write  $M_{\mathbf{C}}(T)$  for the moduli space of irreducible complex local systems of rank  $r$  on  $P - S$ , with local monodromy  $T_s$  at  $s$ . This moduli space has two natural structures of algebraic variety, under the names Betti and de Rham; only the underlying topological space will matter to us.

Choose

an isomorphism between the groups of roots of unity of order prime to  $p$  in  $\overline{\mathbf{Q}}_{\ell}$  and in  $\mathbf{C}$ ,

(2.14.2)

an isomorphism between the groups of roots of unity of order prime to  $p$  in  $\mathbf{F}$  and in  $\mathbf{C}$ .

(2.14.3)

These data are exactly what is needed to attach to an isomorphism class of quasi-unipotent  $\ell$ -adic representations of  $\widehat{\mathbf{Z}}(1)$  an isomorphism class of complex representations of  $\mathbf{Z}(1)$ , as one checks by comparing 2.9 and 2.14.

If the  $R(s)$  are tame, giving rise to representations  $\rho_s$  of  $\widehat{\mathbf{Z}}(1)$ , we write  $R^*$  for the corresponding family of complex representations  $R^*(s)$  of  $\mathbf{Z}(1)$ , and we say that the moduli space  $M_{\mathbf{C}}(R^*)$  corresponds to  $E(R)$ , or is the complex analogue of  $E(R)$ .

**Conjecture 2.15.** Under the hypotheses and with the notation of 2.14:

- (i) As a function of  $n \geq 1$ , the number of fixed points  $N_n(R)$  of  $V^n$  acting on  $E(R)$  is of the form

$$N_n(R) = \sum a_i \beta_i^n \quad (2.11.1)$$

for integers  $a_i$  and suitable numbers  $\beta_i$ .

- (ii) The sum of the  $a_i$  is equal to the Euler-Poincaré characteristic of the moduli space  $M_{\mathbf{C}}(R^*)$  corresponding to  $E(R)$ .

- (iii) As in 1.2 (B), the dominant term in (2.15.1) is  $q$  to the power  $\dim_{\mathbf{C}}(M_{\mathbf{C}}(R^*))/2$ .

As will be explained in 6.6, if  $g > 0$  then  $\chi(M_{\mathbf{C}}(R^*)) = 0$ . For a refinement of 2.15 (i) and (ii), see 6.3 and 6.7.

Using the analogy between the exponential and Artin-Schreier sheaves, it is possible to define complex analogues for certain wild representations  $\rho_s$ . Rather than moduli spaces of local systems on  $P - S$ , the complex analogues are moduli spaces of algebraic vector bundles with connection on  $P - S$ , with prescribed formal completions at each  $s \in S$ . I conjecture that 2.11 remains valid in this more general setting.

**Variant 2.16.** Let  $\rho$  and  $\rho'$  be two quasi-unipotent  $\ell$ -adic representations  $\rho, \rho' : \widehat{\mathbf{Z}}(1) \rightarrow \mathrm{GL}(r, \overline{\mathbf{Q}}_{\ell})$  (respectively two complex representations  $\rho, \rho' : \mathbf{Z}(1) \rightarrow \mathrm{GL}(r, \mathbf{C})$ ). We say that the isomorphism class of  $\rho'$  is in the closure of the isomorphism class of  $\rho$  if  $\rho'$  is a limit of conjugates of  $\rho$ . With the notation of 2.9 (respectively 2.14), this means that, for each character  $\chi$ ,  $\nu_{\rho}(\chi)$  and  $\nu_{\rho'}(\chi)$  have the same dimension  $r(\chi)$ , and that the unipotent conjugacy class of  $\mathrm{GL}(r(\chi))$  defined by the partition  $\lambda_{\rho'}(\chi)$  of  $r(\chi)$  is in the closure of that defined by  $\lambda_{\rho}(\chi)$ .

Transporting this definition to tame  $\overline{\mathbf{Q}}_{\ell}$ -sheaves on  $X_{(s)}^*$ , one defines  $\bar{E}(R)$  as the set of classes, up to semisimplification, of lisse  $\overline{\mathbf{Q}}_{\ell}$ -sheaves  $F$  of rank  $r$  on  $X - S$  such that for each  $s \in S$ ,  $F|_{X_{(s)}^*}$  lies in the closure of the isomorphism class of  $R(s)$ . One defines  $\overline{M}_{\mathbf{C}}(R^*)$  as the moduli space of local systems of rank  $r$  on  $P - S$ , whose local monodromy at  $s$  lies in the closure of  $R^*(s)$ . These local systems are taken up to semisimplification.

One still has  $V : \bar{E}(R) \rightarrow \bar{E}(R)$ , and one can repeat Conjecture 2.15 for  $\bar{E}(R)$  and for the corresponding  $\overline{M}_{\mathbf{C}}(R^*)$ . In some cases, the formulas giving the number of fixed points of  $V$  become simpler if  $E(R)$  is replaced by  $\bar{E}(R)$ . See 4.5.

The precisions 2.17 to 2.22 that follow, added in July 2013, owe much to the talks I gave at IHES in March and April 2013.

## 2.5 2.17–2.22

Let us first place ourselves in the everywhere unramified case:  $S$  is empty, and  $E_r$  is the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank  $r$  on  $X$ . Let  $W(\mathbf{F})$  be the ring of Witt vectors of  $\mathbf{F}$ , let  $K$  be its fraction field, and let  $\overline{K}$  be an algebraic closure of  $K$ . Let  $X_W$  be a lifting of  $X$  over  $W$ , and denote by  $X_K$  (respectively  $X_{\overline{K}}$ ) the curve over  $K$  (respectively  $\overline{K}$ ) obtained from it. Let  $M_K$  (respectively  $M_{\overline{K}}$ ) be the scheme over  $K$  (respectively  $\overline{K}$ ) which is the moduli space of geometrically irreducible vector bundles with connection of rank  $r$  on  $X_K$  (respectively  $X_{\overline{K}}$ ). By extension of scalars from  $K$  to  $\overline{K}$ ,  $M_K$  becomes  $M_{\overline{K}}$ .

Let  $\iota$  be an embedding of  $\overline{K}$  into  $\mathbf{C}$ , and in 2.14 take for  $P$  the Riemann surface deduced from  $X_{\overline{K}}$  by extension of scalars from  $\overline{K}$  to  $\mathbf{C}$ . For this choice of  $P$ , the corresponding space  $M_{\mathbf{C}}$  defined in 2.14 is deduced from  $M_{\overline{K}}$  by extension of scalars from  $K$  to  $\mathbf{C}$ , and, for every prime number  $\ell'$ , the  $\ell'$ -adic cohomology of  $M_{\overline{K}}$  is therefore identified with the cohomology of  $M_{\mathbf{C}}$  with coefficients in  $\mathbf{Q}_{\ell'}$ . The conjectures 2.15 (i) and (ii) (for  $S$  empty) are therefore consequences of the following.

**Conjecture 2.18.** The cohomology of  $M_{\overline{K}}$  admits an endomorphism “which deserves to be called  $V^*$ ” such that, for each  $n \geq 1$ , the number  $N_n$  of fixed points of  $V^n$  acting on  $E_r$  is given by

$$N_n = \sum (-1)^i \text{Tr}(V^{*n}, H^i(M_{\overline{K}})). \quad (2.18.1)$$

In (2.18.1),  $H^i$  is  $\ell'$ -adic cohomology for any prime number  $\ell'$ . One should also be able to take de Rham cohomology of  $M_K/K$ . The clause in quotation marks has no precise sense, but suggests compatibilities. For instance,  $V^*$  should be an endomorphism of the cohomology algebra.

There will not generally exist an endomorphism  $V : M_{\overline{K}} \rightarrow M_{\overline{K}}$  inducing  $V^*$  in cohomology, but I hope that in the analytic space  $M_{\overline{K}}^{\text{an}}$  (in the sense of Berkovich) associated to  $M_{\overline{K}}$  there is an open  $M_{\overline{K}}^0$  such that

- (a) the restriction morphism  $H^*(M_{\overline{K}}) = H^*(M_{\overline{K}}^{\text{an}}) \rightarrow H^*(M_{\overline{K}}^0)$  is an isomorphism; and
- (b) a crystalline interpretation defines  $V = \text{Frob}^* : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$ , inducing  $V^*$  in cohomology.

The conjectural morphism  $V : M_{\overline{K}}^0 \rightarrow M_{\overline{K}}^0$  should send  $M_{\overline{K}}^0$  into a proper open subset of  $M_{\overline{K}}^0$ , corresponding to vector bundles with connection for which parallel transport has a larger radius of convergence; this is what makes ordinary cohomology, rather than compactly supported cohomology, the natural choice in (2.18.1). A variant of Conjecture 2.18, for the completions 2.16 of  $E_r$  and  $M_{\overline{K}}$ , might be more natural.

**Example 2.19.** Suppose  $r = 1$ . In this case,  $M_{\overline{K}}$  is the universal additive extension of the Jacobian  $\text{Pic}^0(X_{\overline{K}})$ , and

$$H^* \text{Pic}^0(X_{\overline{K}}) \xrightarrow{\sim} H^*(M_{\overline{K}}) \quad (2.19.1)$$

is an isomorphism. There is also a natural isomorphism

$$H^*(\text{Pic}^0(X)) \xrightarrow{\sim} H^*(\text{Pic}^0(X_{\overline{K}})). \quad (2.19.2)$$

Over  $\mathbf{F}$ , one has  $V : \text{Pic}^0(X) \rightarrow \text{Pic}^0(X)$ , deduced by functoriality from  $\text{Frob} : X \rightarrow X$  (cf. 2.2). This endomorphism  $V$  of  $\text{Pic}^0(X)$  gives

$$V^* : H^*(\text{Pic}^0(X)) \rightarrow H^*(\text{Pic}^0(X)). \quad (2.19.3)$$

Define the endomorphism

$$V^* : H^*(M_{\overline{K}}) \rightarrow H^*(M_{\overline{K}}) \quad (2.19.4)$$

to be transported from (2.19.3) by (2.19.2) and (2.19.1).

**Proposition 2.20.** If  $r = 1$  and  $V^*$  is defined as above, then (2.18.1) is true.

*Proof for  $n = 1$ .* One reduces to this case by an extension of scalars from  $\mathbf{F}_q$  to  $\mathbf{F}_{q^n}$ . The endomorphisms  $\text{Frob}$  and  $V$  of  $\text{Pic}^0(X)$  are transposes of one another by the autoduality of the Jacobian  $\text{Pic}^0(X)$  of  $X$ . Hence

$$\text{Tr}(V^*, H^i(\text{Pic}^0(X))) = \text{Tr}(F^*, H^i(\text{Pic}^0(X)))$$

and

$$\sum (-1)^i \text{Tr}(V^*, H^i(\text{Pic}^0(X))) = |\text{Pic}^0(X_0)(\mathbf{F}_q)|. \quad (2.20.1)$$

Class field theory identifies the set of fixed points of  $V$  acting on  $E_1$  with the Pontryagin dual of the finite group  $\text{Pic}^0(X_0)(\mathbf{F}_q)$ , with values in the roots of unity of  $\overline{\mathbf{Q}}_\ell$  (see 6.1). Thus

$$N_1 = |\text{Pic}^0(X_0)(\mathbf{F}_q)|, \quad (2.20.2)$$

and, by (2.20.1), this verifies (2.18.1). Notice that, again by the autoduality of the Jacobian, the subscheme  $\text{Pic}^0(X)^V$  of  $\text{Pic}^0(X)$  fixed by  $V$  is the Cartier dual of the finite abelian group  $\text{Pic}^0(X)^{\text{Frob}=\text{Pic}^0(X_0)(\mathbf{F}_q)}$ .

Now pass to the tamely ramified case. Choose over  $W$  a lifting  $(X_W, S_W)$  of  $X$  and of its étale divisor  $S$ . Let  $(X_K, S_K)$  and  $(X_{\overline{K}}, S_{\overline{K}})$  be deduced from  $(X_W, S_W)$  by extension of scalars from  $W$  to  $K$  and to  $\overline{K}$ . To define in 2.14 a complex analogue  $M_{\mathbf{C}}(R^*)$  of  $E(R)$ , we had to make the choices (2.14.2) and (2.14.3). To define a  $p$ -adic analogue we choose

$$\text{an isomorphism between the groups of roots of unity of order prime to } p \text{ in } \overline{\mathbf{Q}}_\ell \text{ and in } K. \quad (2.21.1)$$

There is no need to give an analogue of (2.14.3), because reduction modulo  $p$  supplies a canonical isomorphism between the groups of roots of unity of order prime to  $p$  in  $K$  or  $\overline{K}$  and in  $\mathbf{F}$ .

Let  $X_{K(s)}$  be the completion of  $X_K$  at (the lifting of)  $s$ , and  $X_{K(s)}^* := X_{K(s)} - \{s\}$ . The datum (2.21.1) gives a natural isomorphism between the finite-order characters  $\widehat{\mathbf{Z}}(1) \rightarrow \overline{\mathbf{Q}}_\ell^*$  and the prime-to- $p$  part of  $/\mathbf{Z}$ . It is exactly what is needed to attach to  $R(s)$  an isomorphism class  $R(s)_K^*$  of vector bundles with connection on  $X_{K(s)}^*$ . Define  $M_K(R^*)$  to be the scheme over  $K$  which is the moduli space of geometrically irreducible vector bundles with connection of rank  $r$  on  $X_K - S_K$ , with completion at  $s \in S$  isomorphic to  $R(s)_K$ .

Let  $\iota$  be an embedding of  $K$  into  $\mathbf{C}$ . If in 2.14 one takes  $P$  and  $S \rightarrow P$  deduced from  $(X_K, S_K)$  by extension of scalars from  $K$  to  $\mathbf{C}$ , if (2.14.2) is deduced from (2.21.1) by  $\iota$ , and if (2.14.3) is induced by  $\iota$ , then  $M_{\mathbf{C}}(R^*)$  defined in 2.14 is deduced from  $M_K(R^*)$  by extension of scalars from  $K$  to  $\mathbf{C}$ .

Let  $M_{\overline{K}}(R^*)$  be obtained from  $M_K(R^*)$  by extension of scalars from  $K$  to  $\overline{K}$ . As in 2.18, I conjecture that the cohomology of  $M_{\overline{K}}(R^*)$  admits an endomorphism “which deserves to be called  $V^*$ ” such that for  $n \geq 1$ , the number of fixed points of  $V$  acting on  $E(R)$  is given by

$$N_n = \sum (-1)^i \text{Tr}(V^*, H^i(M_{\overline{K}}(R^*))), \quad (2.21.2)$$

and this conjecture implies 2.15 (i) and (ii).

Continue to assume that the  $R(s)$  are tamely ramified, and assume condition (2.10.2). Write  $\det$  for the functor “maximal exterior power”  $\bigwedge^r$ . It induces a map

$$\det : E(R) \rightarrow E(\det R). \quad (2.22.1)$$

Hypothesis (2.10.2) ensures that  $\det R$  satisfies (2.10.1), and that  $E(\det R)$  is nonempty, hence a torsor under the group  $E_1$  of isomorphism classes of lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank one on  $X$ . The action is tensor product.

In 2.16, for  $r = 1$  and  $S$  empty, we defined a moduli space  $M_K$ . Here we denote it by  $M_{1K}$ . The functor  $\bigwedge^r$  induces

$$\det : M_K(R^*) \rightarrow M_K(\det R^*), \quad (2.22.2)$$



and  $M_K(\det R^*)$  is a principal homogeneous space under  $M_{1K}$ . Extend scalars from  $K$  to  $\overline{K}$  and consider the Leray spectral sequence for the morphism  $\det$ :

$$E_2^{pq} = H^p(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \Rightarrow H^{p+q}(M_{\overline{K}}(R^*), \mathbf{Q}'_\ell).$$

I conjecture the existence of an endomorphism  $V^*$  of this spectral sequence, respecting its multiplicative structure. The  $E_2$ -term is given by

$$H^p(M_{\overline{K}}(\det R^*)) \otimes H^0(M_{\overline{K}}(\det R^*), R^q \det \mathbf{Q}'_\ell) \xrightarrow{\sim} E_2^{pq},$$

and the cohomologies of  $M_{\overline{K}}(\det R^*)$  and of  $M_{1\overline{K}}$  are canonically isomorphic. The conjectured existence of  $V^*$  therefore implies 6.7.

### 3 Arinkin's Method

#### 3.1 3.1–3.3

Let  $s_0$  be a closed point of  $X_0$ . One may identify the closed points  $s$  of  $X$  both with the  $\mathbf{F}$ -points of the  $\mathbf{F}$ -scheme  $X$ , and with the  $\mathbf{F}$ -points of the  $\mathbf{F}_q$ -scheme  $X_0$ . In doing so, the  $s$  above  $s_0$  are identified with the  $\mathbf{F}$ -points of  $\mathrm{Spec}(k(s_0))$ , i.e. with the embeddings of  $k(s_0)$  into  $\mathbf{F}$  extending the embedding of  $\mathbf{F}_q$  into  $\mathbf{F}$ . If one chooses such an embedding  $s$ , a character  $\varepsilon_0 : k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$  gives, by 2.9, a character  $\varepsilon := \varepsilon_0 \mathrm{pr}_s : \widehat{\mathbf{Z}}(1) \rightarrow k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$ . If  $s = \mathrm{Frob}(s')$ , the character  $\varepsilon'$  of  $\widehat{\mathbf{Z}}(1)$  defined by  $\varepsilon_0$  and  $s'$  is  $\varepsilon^q$ .

At each  $s_0 \in S_0$ , give a multiset  $R(s_0)$  of  $r$  characters  $k(s_0)^* \rightarrow \overline{\mathbf{Q}}_\ell^*$ . By 3.1 one deduces, for every  $s \in S$ , a multiset of  $r$  characters  $\mathcal{R}(s)$  of  $\widehat{\mathbf{Z}}(1)$ . Let  $\rho_s$  be the representation of  $\widehat{\mathbf{Z}}(1)$  that is the sum of the one-dimensional representations defined by these characters. The corresponding  $\overline{\mathbf{Q}}_\ell$ -sheaves  $R(s)$  on the  $X_{(s)}^*$  satisfy (2.1.1). Write  $R$  for the family  $(R(s))_{s \in S}$ . We assume that

$$\prod_s \prod_{\varepsilon \in \mathcal{R}(s)} \varepsilon = 1. \quad (3.2.1)$$

By (2.10.1), this is a necessary condition for the existence of a lisse  $\overline{\mathbf{Q}}_\ell$ -sheaf  $F$  on  $X - S$  with  $F|_{X_{(s)}^*} \simeq R(s)$ . If  $s_0$  has degree  $d$ , if  $\varepsilon_0$  is a character of  $k(s_0)^*$ , and if for every  $s$  above  $s_0$ ,  $\varepsilon[s] := \varepsilon_0 \mathrm{pr}_s$  is the corresponding character of  $\widehat{\mathbf{Z}}(1)$ , then the product of these  $\varepsilon[s]$  is the character that 3.1 attaches to  $\varepsilon_0^{(q^d-1)/(q-1)} = \varepsilon_0 \mathrm{N}_{k(s_0)/\mathbf{F}_q}$  and to any  $s$  over  $s_0$ . It is also the character that 3.1 attaches to  $\varepsilon_0|_{\mathbf{F}_q^*}$  and to  $\mathbf{F}_q \hookrightarrow \mathbf{F}$ . Hypothesis (3.2.1) can therefore be rewritten as

$$\prod_{s_0 \in S_0} \prod_{\varepsilon_0 \in R(s_0)} \varepsilon_0|_{\mathbf{F}_q^*} = 1. \quad (3.2.2)$$

Assume further that the  $\mathcal{R}(s)$  satisfy the general-position condition (2.10.3). I learned from Arinkin how, under this hypothesis, to construct an algebraic variety  $Z$  over  $\mathbf{F}_q$  such that, for every  $n \geq 1$ , the number  $N_n(R)$  of fixed points of the permutation  $V^n$  of  $E(R)$  is equal to the number of  $\mathbf{F}_{q^n}$ -points of  $Z$ :

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.3.1)$$

Under the hypotheses of 3.3, Conjecture 2.15 (i) is therefore true. Beware that  $Z$  will only be defined after cutting into locally closed pieces.

#### 3.2 3.4–3.6

For each  $s_0 \in S_0$ , let  $\varepsilon[s_0](i)$  ( $0 \leq i \leq a(s_0)$ ) be the characters of  $k(s_0)^*$  occurring in  $R(s_0)$ , arranged in any order. Let  $n[s_0](i)$  be the multiplicity of  $\varepsilon[s_0](i)$  in the multiset  $R(s_0)$ , and

let  $n[s_0]$  be the family of integers  $(n[s_0](i))_{0 \leq i \leq a(s_0)}$ . For  $s \in S$  above  $s_0$ , write  $\varepsilon[s](i)$  for the character  $\varepsilon[s_0](i) \circ \text{pr}_s$  of  $\widehat{\mathbf{Z}}(1)$  (3.1), and set  $n[s] := n[s_0]$ . The multiset  $\mathcal{R}(s)$  is

$$\mathcal{R}(s) := \{\text{the } \varepsilon[s](i), \text{ with multiplicities } n[s](i)\}. \quad (3.4.1)$$

Let  $E_0$  be a vector bundle of rank  $r$  on  $X_0$ . A parabolic structure  $\alpha_0$  of type  $(n(s_0))_{s_0 \in S_0}$  on  $E_0$  is the datum, for each  $s_0 \in S_0$ , of a finite filtration  $F(s_0)$  of the  $k(s_0)$ -vector space  $(E_0)_{s_0}$ , such that  $\text{Gr}_{F(s_0)}^i((E_0)_{s_0})$  has dimension  $n[s_0](i)$ . Parabolic structures for  $E$  on  $X$  are defined similarly. If  $E_0$  on  $X_0$  is endowed with a parabolic structure  $\alpha_0$  of type  $(n(s_0))_{s_0 \in S_0}$ , its inverse image  $E$  on  $X$  inherits a parabolic structure  $\alpha$  of type  $(n(s))_{s \in S}$ . We say that  $(E_0, \alpha_0)$  is geometrically indecomposable if its inverse image  $E$  on  $X$  admits no nontrivial decomposition  $E = E' \oplus E''$  compatible with the parabolic structure  $\alpha$ , i.e. inducing decompositions of the  $F^i E_s$ . Let  $\text{End}(E_0, \alpha_0)$  (respectively  $\text{End}(E, \alpha)$ ) be the algebra of endomorphisms of  $E_0$  (respectively  $E$ ) respecting the filtrations  $F(s_0)$  of the  $(E_0)_{s_0}$  (respectively  $F(s)$  of the  $E_s$ ). Each of the following conditions is equivalent to geometric indecomposability:

$$\text{End}(E, \alpha) = \text{End}(E_0, \alpha_0) \otimes_{\mathbf{F}_q} \mathbf{F} \text{ has no idempotent other than 0 and 1.} \quad (3.4.2)$$

$$\text{End}(E_0, \alpha_0) / \text{rad End}(E_0, \alpha_0) = \mathbf{F}_q. \quad (3.4.3)$$

Fix an integer  $d$ . Consider vector bundles  $E_0$  of rank  $r$  and degree  $d$  on  $X_0$ , endowed with a parabolic structure  $\alpha$  of type  $(n[s_0])_{s_0 \in S_0}$ . Let  $Z(\mathbf{F}_q)$  be the set of isomorphism classes of such  $(E_0, \alpha)$  that are geometrically indecomposable. Define  $Z(\mathbf{F}_{q^n})$  similarly, replacing  $X_0/\mathbf{F}_q$  by  $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}/\mathbf{F}_{q^n}$ . I learned from Arinkin the following theorem.

**Theorem 3.5.** Under the hypotheses of 3.2 and 3.3 and with the notation of 3.4, the number  $N_n(R)$  of fixed points of  $V^n$  (2.1) is

$$N_n(R) = |Z(\mathbf{F}_{q^n})|. \quad (3.5.1)$$

Replacing  $X_0$  by  $X_0 \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$ , one reduces to the case  $n = 1$ . Beware that the proof does not suggest a natural bijection between  $E(R)^V$  and  $Z(\mathbf{F}_q)$ . It gives a family of one-dimensional vector spaces indexed by  $E(R)^V$ , with direct sum  $V$ , another family, with direct sum  $V'$ , indexed by  $Z(\mathbf{F}_q)$ , and an isomorphism of vector spaces  $V \simeq V'$ .

Sketch of proof of 3.5  $\Rightarrow$  (3.3.1). For every  $\mathbf{F}_q$ -scheme  $T$ , let  $X_T := X_0 \times_{\mathbf{F}_q} T$  and  $S_T := S_0 \times_{\mathbf{F}_q} T$ . Define  $Z(T)$  as the set of isomorphism classes of vector bundles of rank  $r$  on  $X_T$ , endowed with a parabolic structure along  $S_T$ , and such that the conditions of 3.2 and 3.3 are satisfied on each geometric fibre of  $X_T \rightarrow T$ : degree  $d$ , type of the parabolic structure, indecomposability.

Naively one would like to take for the scheme  $Z$  of (3.3.1) a scheme representing the functor  $Z$ . For several reasons this functor is not representable:

- (a) The objects considered have nontrivial automorphisms, for instance homotheties; thus this functor is not a sheaf, even for the Zariski topology. For the functor  $\text{Pic}$ , the same problem is bypassed by passing to the associated sheaf. Here the situation is worse, because, for  $(E_T, \alpha_T)$  a vector bundle with parabolic structure on  $X_T$ , and for  $(E_t, \alpha_t)$  the induced bundle on the fibre  $X_t$  of  $X_T \rightarrow T$  at  $t \in T$ ,
- (b) the dimension of the automorphism group of  $(E_t, \alpha_t)$  (equal to  $\dim \text{End}(E_t, \alpha_t)$ ) is not always locally constant in  $t$ .
- (c) The set of  $t$  for which  $(E_t, \alpha_t)$  is geometrically indecomposable is not always locally closed.

Because of (c), one cannot even hope that  $Z$  comes from an algebraic stack  $C$  (i.e. that  $Z(T)$  be the set of isomorphism classes in  $C(T)$ ). Nevertheless:

- (d) The automorphism groups are connected algebraic groups. If  $k' \subset k''$  are finite extensions of  $\mathbf{F}_q$ , this ensures that

$$Z(k') \xrightarrow{\sim} Z(k'')^{\text{Gal}(k''/k')}.$$

- (e) In (b) above, the function  $t \mapsto \dim \text{End}(E_t, \alpha_t)$  is constructible; in (c), the set of  $t$  such that  $(E_t, \alpha_t)$  is geometrically indecomposable is constructible.

Properties (d) and (e) make it possible, by “cutting  $Z$  into pieces” and taking the associated (representable) sheaves, to obtain  $Z$  as promised by (3.3.1).

### 3.3 3.7–3.16

Sketch of proof of 3.5 (for  $n = 1$ ). One applies the strategy 2.5. We must show that  $|Z(\mathbf{F}_q)|$  is the number of cuspidal automorphic representations  $\pi$  of  $\text{GL}(r, \mathbf{A})$ , counted up to  $\mathbf{F}_q$ -twist, that are unramified outside  $S_0$  and whose local component  $\pi_{s_0}$  at  $s_0 \in S_0$  is of the following type. By the local Langlands correspondence, the representation  $\pi_{s_0}$  of  $\text{GL}(r, K_{s_0})$  corresponds to an  $F$ -semisimple  $\overline{\mathbf{Q}}_\ell$ -representation  $\rho$  of the Weil group  $W(\overline{K}_{s_0}/K_{s_0})$ . We first require  $\rho$  to be semisimple (in the language of the Weil–Deligne group:  $N = 0$ ; in Arthur’s language: trivial action of the local  $\text{SL}(2)$  or  $\text{SU}(2)$ ). The abelianization  $W(\overline{K}_{s_0}/K_{s_0})^{\text{ab}}$  is  $K_{s_0}^*$ . Its inertia group  $O_{s_0}^*$  has quotient  $k(s_0)^*$ . We require the restriction of  $\rho$  to inertia to factor through the representation of  $k(s_0)^*$  which is the sum of the  $\varepsilon_0$  in  $R(s_0)$ .

Let  $P[s_0] \subset \text{GL}(r, O_{s_0})$  be the parahoric subgroup inverse image of the standard parabolic  $\overline{P}[s_0]$  of  $\text{GL}(r, k(s_0))$  of type  $n(s_0)$ . The reductive quotient  $L[s_0]$  of  $\overline{P}[s_0]$  is  $\prod \text{GL}(n[s_0](i), k(s_0))$ . Let  $U[s_0]$  be the kernel of  $P[s_0] \rightarrow L[s_0]$ , and let  $\varepsilon[s_0]$  be the character

$$\varepsilon[s_0] = \prod \varepsilon[s_0](i)(\det g_i) : P[s_0] \rightarrow \overline{P}[s_0] \rightarrow L[s_0] \rightarrow \overline{\mathbf{Q}}_\ell^*$$

of the parahoric  $P[s_0]$ . By construction, the  $\varepsilon[s_0](i)$  are pairwise distinct. We shall admit the following assertion.

**Assertion 3.8.** The irreducible admissible representations of  $\text{GL}(r, K_{s_0})$  of the type defined in 3.7 are those which admit a line stable under  $P$  on which  $P$  acts through the character  $\varepsilon[s_0]$ . This line is unique.

Let  $A'$  be the set of cuspidal automorphic representations  $\pi$  of  $\text{GL}(r, \mathbf{A})$  admitting a line  $L(\pi)$  fixed by the  $\text{GL}(r, O_x)$  for  $x \notin S_0$ , and stable by  $P[s_0]$  for  $s_0 \in S$ , with the action of  $P[s_0]$  given by  $\varepsilon[s_0]$ . The line  $L(\pi)$  is unique if it exists. By Lafforgue [L] and 3.8,  $N_1(R)$  is the number of classes modulo  $\mathbf{F}_q$ -twist in  $A'$ . Fix an idele  $d$  of valuation 1, and let  $A$  be the set of  $\pi$  in  $A'$  whose central character  $\omega_\pi$  is equal to 1 at  $d$ . As observed after 2.3.1, one has

$$N_1(R) = |A' \bmod \mathbf{F}_q\text{-twist}| = |A \bmod \mathbf{F}_q\text{-twist by } \zeta \in \mu_r(\overline{\mathbf{Q}}_\ell)| = \frac{1}{r} |A|. \quad (3.9.1)$$

Let  $\mathcal{A}$  be the vector space of functions on  $\text{GL}(r, \mathbf{A})$  which are

- (a) invariant on the left by  $\text{GL}(r, K) \cdot d^{\mathbf{Z}}$ ;
- (b) invariant on the right under  $\text{GL}(r, O_x)$ , for  $x \notin S_0$ ;
- (c) transformed by  $\varepsilon[s_0]$  for the right action of  $P[s_0]$ , for  $s_0 \in S_0$ .

A key point of the proof of 3.5 is the following lemma.

**Lemma 3.11.** The vector space  $\mathcal{A}$  is the direct sum of the lines  $L(\pi)$ , for  $\pi \in A$ . It is therefore of dimension  $rN_1(R)$ .

This lemma is the automorphic counterpart, and a consequence, of the fact that irreducibility of the  $\overline{\mathbf{Q}}_\ell$ -sheaves  $F_0$  under consideration follows from the local hypotheses made. One must check that the functions in  $\mathcal{A}$  are automatically cuspidal. This done, if one writes the space

of cuspidal functions on  $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$  as a direct sum of irreducible representations  $\pi$  of  $\mathrm{GL}(r, \mathbf{A})$ , the  $\pi$  in  $A$  contribute  $L(\pi)$  to  $\mathcal{A}$ , and the others contribute nothing. That the functions in  $\mathcal{A}$  are cuspidal follows from the fact that, in the spectral decomposition of functions on  $\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A})$ , the induced representations giving rise to Eisenstein series admit no nonzero vector satisfying 3.10 (b) and (c).

The space  $\mathcal{A}$ , the sum of the  $L(\pi)$  for  $\pi \in A$ , is identified with the space of functions  $f$  on

$$\mathrm{GL}(r, K)d^{\mathbf{Z}} \backslash \mathrm{GL}(r, \mathbf{A}) / \prod_{x \notin S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.1)$$

such that, for every  $s_0 \in S_0$ , the right action of  $P[s_0]$  satisfies  $f(gp) = \varepsilon[s_0](p)f(g)$ . Let  $v$  be the map from (3.12.1) to  $\mathbf{Z}/r$  induced by  $v \det : \mathrm{GL}(r, \mathbf{A}) \rightarrow \mathbf{Z}$ . Let  $\bar{A}$  be the quotient of  $A$  by the  $\mathbf{F}_q$ -twist action of  $\mu_r(\mathbf{Q}_\ell)$ , and for each orbit  $B$ , let  $V_B$  be the subspace of  $\mathcal{A}$  which is the sum of the  $L(\pi)$  for  $\pi \in B$ . By (2.3.1),  $\dim(V_B) = r$ . For  $m \in \mathbf{Z}/r$ , write  $(3.12.1)_m$  for the part of (3.12.1) where  $v = m$ , let  $\varphi_m$  be its characteristic function, and put  $L(\pi, m) = \{\varphi_m f \mid f \in L(\pi)\}$ . The dimension of  $L(\pi, m)$  is at most one. For each character  $\eta$  of  $\mathbf{Z}/r$ , if  $\pi'$  is the  $\mathbf{F}_q$ -twist of  $\pi$  by  $\eta(1)$ , multiplication by  $\eta^v$  transforms  $L(\pi)$  into  $L(\pi')$ . For  $\pi \in B$ ,  $L(\pi, m)$  is therefore independent of  $\pi$ ; write  $L(B, m) := L(\pi, m)$ . The space  $V_B$  is stable under the multiplications by the  $\eta^v$ , hence under multiplication by the  $\varphi_m$ : for  $\pi \in B$ ,

$$V_B = \bigoplus_{m \in \mathbf{Z}/r} L(B, m).$$

Each  $L(B, m)$  is therefore one-dimensional. The sum  $\mathcal{A}_m$  of the  $L(B, m)$  is of dimension  $|\bar{A}| = N_1(R)$ . It is identified with the space of functions  $f$  on  $(3.12.1)_m$  such that, for each  $s_0 \in S_0$ , the right action of  $P[s_0]$  satisfies  $f(gp) = \varepsilon[s_0](p)f(g)$ .

Let  $\mathrm{GL}(r, \mathbf{A})^{(d)}$  be the set of  $g \in \mathrm{GL}(r, \mathbf{A})$  such that  $v \det(g) = d$ . If  $\bar{d}$  is the class of  $d$  modulo  $r$ , the set

$$\mathrm{GL}(r, K) \backslash \mathrm{GL}(r, \mathbf{A})^{(d)} / \prod_{x \in S_0} \mathrm{GL}(r, O_x) \prod_{s_0 \in S_0} U[s_0] \quad (3.12.2)$$

maps bijectively onto  $(3.12.1)_{\bar{d}}$ . Hence:

**Lemma 3.13.**  $N_1(R)$  is the dimension of the space, again denoted  $\mathcal{A}_d$ , of functions  $f$  on (3.12.2) which, for the right action of the  $P[s_0]$  ( $s_0 \in S_0$ ), satisfy  $f(gp) = \varepsilon[s_0](p)f(g)$ .

The set (3.12.2) is interpreted as the set of isomorphism classes of triples  $(E_0, \alpha, \beta)$ , where  $E_0$  is a vector bundle of rank  $r$  and degree  $d$  on  $X_0$ , and where  $\alpha$  and  $\beta$  are structures of the following types on  $E_0$ :

- (a)  $\alpha$  is a parabolic structure of type  $(n[s_0])_{s_0 \in S_0}$ ;
- (b)  $\beta$  is the datum of bases of the vector spaces  $\mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0})$ .

We identify a basis as in (b) with an isomorphism

$$\beta_{s_0, i} : k(s_0)^{n[s_0](i)} \xrightarrow{\sim} \mathrm{Gr}_{F(s_0)}^i((E_0)_{s_0}).$$

With this interpretation, the space  $\mathcal{A}_d$  of 3.13 becomes the space of functions  $f(E_0, \alpha, \beta)$ , for  $E_0, \alpha, \beta$  as above, such that

- (i)  $f(E_0, \alpha, \beta)$  depends only on the isomorphism class of  $(E_0, \alpha, \beta)$ ;
- (ii) for  $s_0 \in S_0$  and  $g \in L[s_0] = \prod \mathrm{GL}(n_i, k(s_0))$ , one has

$$f(E_0, \alpha, \beta g) = \varepsilon[s_0](g)f(E_0, \alpha, \beta).$$

For an automorphism  $T$  of  $(E_0, \alpha)$ , let  $\text{Gr}_F^i(s_0)(T_{s_0})$  be the automorphism of  $\text{Gr}_F^i(s_0)((E_0)_{s_0})$  induced by  $T$ , and put

$$\chi(T) := \prod_{s_0} \prod_i \varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})). \quad (3.14.1)$$

For any  $\beta$ , hypothesis (i) gives  $f(E_0, \alpha, T(\beta)) = f(E_0, \alpha, \beta)$ . Hypothesis (ii) gives  $f(E_0, \alpha, T(\beta)) = \chi(T)f(E_0, \alpha, \beta)$ . If there exists an automorphism  $T$  with  $\chi(T) \neq 1$ , then  $f(E_0, \alpha, \beta) = 0$ . Conversely, if  $\chi(T) = 1$  for every automorphism  $T$  of  $(E_0, \alpha)$ , then  $f$  may be prescribed freely at one  $(E_0, \alpha, \beta)$ , and its value there determines its value at all  $(E_0, \alpha, \beta')$ . Thus the dimension of  $\mathcal{A}_d$  is the number of isomorphism classes of  $(E_0, \alpha)$  as above such that for every automorphism  $T$ , one has  $\chi(T) = 1$ . Theorem 3.5 follows from the characterization (3.4.3) of geometric indecomposability and the following proposition.

**Proposition 3.15.** Let  $E_0$  be a vector bundle of rank  $r$  on  $X_0$ , endowed with a parabolic structure of type  $(n(s_0))_{s_0 \in S_0}$ . The following conditions are equivalent:

- (i) The quotient of the  $\mathbf{F}_q$ -algebra  $\text{End}(E_0, \alpha)$  by its radical is reduced to  $\mathbf{F}_q$ .
- (ii) For every automorphism  $T$  of  $(E_0, \alpha)$ ,  $\chi(T)$ , defined by (3.14.1), is equal to 1.

We shall deduce that (i) implies (ii) from (3.2.2), and that the negation of (i) implies the negation of (ii) from the general-position hypothesis (2.10.3).

*Proof of (i)  $\Rightarrow$  (ii).* If  $\lambda$  is the image in  $\mathbf{F}_q^*$  of the automorphism  $T \in \text{End}(E_0, \alpha)^*$ , then

$$\varepsilon[s_0](i)(\det \text{Gr}_F^i(s_0)(T_{s_0})) = \varepsilon(s_0)(i)^{n[s_0](i)}(\lambda),$$

so that (3.2.2) implies  $\chi(T) = 1$ .

*Proof of (ii)  $\Rightarrow$  (i).* The quotient of the  $\mathbf{F}_q$ -algebra  $\text{End}(E_0, \alpha)$  by its radical is a product of matrix algebras over finite extensions of  $\mathbf{F}_q$ . If it is not reduced to  $\mathbf{F}_q$ , it contains a commutative separable  $\mathbf{F}_q$ -algebra of dimension  $> 1$ . This algebra admits a lifting into  $\text{End}(E_0, \alpha)$ . Suppose then that  $\text{End}(E_0, \alpha)$  contains a commutative separable subalgebra  $k$  of dimension  $> 1$ . We must show that the character (3.14.1) of  $\text{End}(E_0, \alpha)^*$  is nontrivial. We shall prove that its restriction  $\chi$  to  $k^*$  is nontrivial.

Split case. Suppose first that  $k$  is a product of copies of  $\mathbf{F}_q$ , say  $k \simeq \mathbf{F}_q^I$ , and that the closed points  $s_0 \in S_0$  have degree 1, so that  $\mathbf{F}_q \xrightarrow{\sim} k(s_0)$ . The  $\mathbf{F}_q^I$ -module structure of  $E_0$  gives a decomposition

$$E_0 = \bigoplus_{\iota \in I} E_0^\iota \quad (3.15.1)$$

such that  $(\lambda_\iota)_{\iota \in I} \in \mathbf{F}_q^I$  acts on  $E_0^\iota$  by multiplication by  $\lambda_\iota$ . The decomposition (3.15.1) is compatible with the parabolic structure. For  $\iota \in I$ , put

$$n^\iota[s_0](i) = \dim \text{Gr}_{F(s_0)}^i((E_0^\iota)_{s_0}).$$

The sum over  $i$  of the  $n^\iota[s_0](i)$  is the rank of the bundle  $E_0^\iota$ . The restriction of  $\chi$  to the factor with index  $\iota$  of  $(\mathbf{F}_q^I)^* = \bigoplus_{\iota \in I} \mathbf{F}_q^*$  is

$$\lambda \longmapsto \prod_{s_0 \in S_0} \prod_i \varepsilon[s_0](i)^{n^\iota[s_0](i)}(\lambda). \quad (3.15.2)$$

Its composite with the morphism  $\widehat{\mathbf{Z}}(1) \rightarrow \mathbf{F}_q^*$  of 3.1 is the character

$$\prod_{s \in S} \prod_i \varepsilon[s](i)^{n^\iota[s](i)}$$

of  $\widehat{\mathbf{Z}}(1)$ . Hypothesis (2.10.3) ensures that this character is nontrivial.

Reduction of the general case to the split case. It suffices to show that nontriviality of  $\chi$  is invariant under extension of scalars from  $\mathbf{F}_q$  to a finite extension  $\mathbf{F}_{q^n}$ . Under such an extension of scalars,  $X_0$  becomes a curve  $X'$  over  $\mathbf{F}_{q^n}$ ,  $S_0$  becomes a divisor  $S'$  of  $X'$ , and  $k$  becomes  $k' := k \otimes_{\mathbf{F}_q} \mathbf{F}_{q^n}$ . For  $s' \in S'$  above  $s_0 \in S_0$ , put  $n[s'] := n[s_0]$ , and let  $\varepsilon[s'](i)$  be the composite of  $\varepsilon[s_0](i)$  with the norm  $N_{k(s')/k(s_0)}$ . The bundle  $E'$  on  $X'$  inverse image of  $E_0$  inherits a parabolic structure  $\alpha'$  of type  $(n(s'))_{s' \in S'}$ . Repeating for  $E'$  on  $X'/\mathbf{F}_{q^n}$  the construction (3.14.1), one obtains a character  $\chi'$  of  $k'^*$ . The curve  $X/\mathbf{F}$ , the integers  $n[s](i)$ , and the characters  $\varepsilon[s](i)$  of  $\hat{\mathbf{Z}}(1)$ , come equally from  $X_0/\mathbf{F}_q$  and from  $X'/\mathbf{F}_{q^n}$ .

To prove that triviality of  $\chi$  is equivalent to triviality of  $\chi'$ , it suffices to prove that

$$\chi' = \chi \circ N_{k'/k}. \quad (3.15.3)$$

Indeed, the norm  $N_{k'/k}$  is surjective. Identity (3.15.3) follows from identity (3.16.1) below, applied to each  $\mathrm{Gr}_F^i(s_0)(E_{s_0})$ , endowed with its  $(k, k(s_0))$ -bimodule structure.

Let  $k_0$  be a finite extension of  $\mathbf{F}_q$ ,  $N_0$  a finite-dimensional  $k_0$ -vector space,  $k$  a product of finite extensions of  $\mathbf{F}_q$ , and  $\rho : k \rightarrow \mathrm{End}_{k_0}(N_0)$ . The  $(k, k_0)$ -bimodule  $N$  defines a homomorphism

$$[N] : k^* \rightarrow k_0^*, \quad \lambda \mapsto \det(\rho(\lambda)).$$

Extend scalars from  $\mathbf{F}_q$  to  $\mathbf{F}_{q^n}$ . One obtains  $k'_0$ ,  $k'$ , and a  $(k', k'_0)$ -bimodule  $N'$ . This bimodule defines

$$[N'] : k'^* \rightarrow k'_0^*, \quad \lambda \mapsto \det_{k'_0}(\rho(\lambda)).$$

**Lemma 3.16.** The diagram

$$\begin{array}{ccc} k'^* & \xrightarrow{[N']} & k'_0^* \\ N_{k'/k} \downarrow & & \downarrow N_{k'_0/k_0} \\ k^* & \xrightarrow{[N]} & k_0^* \end{array} \quad (3.16.1)$$

is commutative.

*Proof.* Extend scalars from  $\mathbf{F}_q$  to  $\mathbf{F}$ . Diagram (3.16.1) embeds in an analogous diagram (3.16.2), where  $k$  and  $k_0$  are replaced by  $k_F := k \otimes_{\mathbf{F}_q} \mathbf{F}$ ,  $k_{0F} := k_0 \otimes_{\mathbf{F}_q} \mathbf{F}$ , where  $N_0$  is replaced by a  $(k_F, k_{0F})$ -bimodule  $N$ , and where the extension  $\mathbf{F}_{q^n}$  of  $\mathbf{F}_q$  is replaced by a product  $\mathbf{F}^J$  of  $n$  copies of  $\mathbf{F}$ :

$$\begin{array}{ccc} (k_F \otimes \mathbf{F}^J)^* & \xrightarrow{[N \otimes \mathbf{F}^J]} & (k_{0F} \otimes \mathbf{F}^J)^* \\ \downarrow & & \downarrow \\ k_F^* & \xrightarrow{[N]} & k_{0F}^*. \end{array} \quad (3.16.2)$$

One has  $(k_F \otimes \mathbf{F}^J)^* = (k_F^*)^J$ ,  $(k_{0F} \otimes \mathbf{F}^J)^* \simeq (k_{0F}^*)^J$ , and the vertical “norm” maps are the product maps. The first horizontal line is the product of copies, indexed by  $J$ , of the second horizontal line. The commutativity of (3.16.2) follows.

## 4 Trace Formula

### 4.1 4.1–4.5

In [DF], the counting problem 2.3 (i) is solved when  $|S_0| \geq 2$  and the imposed local monodromies are unipotent, with a single Jordan block. In automorphic language this means asking the local components  $\pi_{s_0}$  ( $s_0 \in S_0$ ) to be of the form

$$\text{special representation} \otimes \chi \det g, \quad (4.1.1)$$

for  $\chi$  an unramified character of  $K_{s_0}^*$ .

The hypothesis  $|S_0| \geq 2$  allows one to pass from automorphic representations for  $\mathrm{GL}(r, K)$  to automorphic representations for the multiplicative group of a division algebra  $D$  of dimension  $r^2$  over  $K$ . It is reassuring to use the trace formula only in a case where the quotient is compact (modulo the centre), but it should be possible instead to use the trace formula for  $\mathrm{GL}(n)$ , for a suitable test function, with the simplification brought by the fact that one wants to detect automorphic representations which at two places are in the discrete series. After all, this is how one relates  $\mathrm{GL}(r, K)$  and  $D^*$ . The explanations that follow will be simplified by staying with  $\mathrm{GL}(r, K)$ , but admitting that the only nonzero terms in the required trace formula are those associated with conjugacy classes of elliptic elements of finite order in  $\mathrm{GL}(r, K)$ .

Choose an embedding  $\mathbf{F}_{q^r} \hookrightarrow M_r(\mathbf{F}_q)$ . It defines a morphism  $\mathbf{F}_{q^r}^* \hookrightarrow \mathrm{GL}(r, \mathbf{F}_q) \hookrightarrow \mathrm{GL}(r, K)$ , through which the set of orbits of  $\mathrm{Gal}(\mathbf{F}_{q^r}/\mathbf{F}_q)$  in  $\mathbf{F}_{q^r}^*$  maps bijectively to the set of conjugacy classes under consideration.

Let  $T(\gamma)$  be the term of the trace formula (giving the number of fixed points of  $V$ ) thus associated to  $\gamma \in \mathbf{F}_{q^r}^*$ . It depends only on the subextension  $\mathbf{F}_{q^m}$  of  $\mathbf{F}_q$  generated by  $\gamma$ . Put  $T(m) := T(\gamma)$ . The number  $c_m$  of elements of  $\mathbf{F}_{q^m}^*$  which generate the extension  $\mathbf{F}_{q^m}$  of  $\mathbf{F}_q$  is

$$c_m = \sum_{a|m} \mu(a)(q^{m/a} - 1).$$

The number of fixed points of  $V$  is

$$N_1 = \sum_{m|r} \frac{c(m)}{m} T(m) \tag{4.1.2}$$

$$= \sum_{m|r} \sum_{ab=m} \frac{1}{m} \mu(a)(q^b - 1) T(ab) \tag{1}$$

$$= \sum_b \sum_{ab|r} \frac{1}{m} \mu(a)(q^b - 1) T(ab). \tag{2}$$

The number of fixed points of  $V^n$  ( $n \geq 1$ ) is obtained similarly, after extending scalars from  $\mathbf{F}_q$  to  $\mathbf{F}_{q^n}$ .

**Surprise 4.2.** When  $n$  varies, each  $T(m)$  is generally not, as a function of  $n$ , of the form (1.2.1). On the other hand, in (4.1.2), each term of the sum over  $b$  is of this form.

I have no explanation, neither geometric nor automorphic, of 4.2, only a proof. I likewise have no explanation for the following.

**Surprise 4.3.** Under the hypotheses of 4.1, the number  $N_1$  of fixed points of  $V$  is divisible by  $q$ .

If  $r = 2$ , the trace formula is a little less frightening. Drinfeld [Dr] used it for the case where  $S_0$  is empty. Flicker (unpublished) treated the case where  $|S_0| = 1$  and where the local monodromy at each  $s \in S$  is unipotent with a single Jordan block, i.e. since  $r = 2$ , nontrivial. Gathering the information thus obtained, one gets:

**Surprise 4.5.** Let  $E$  be the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X - S$ , with unipotent local monodromy (both nontrivial and trivial). The number of fixed points of  $V := \mathrm{Frob}^*$  on  $E$  depends only on  $X_0/\mathbf{F}_q$  and on  $|S|$ .

This is the surprise to which 2.16 alludes. If, as in [DF], one requires the local monodromy at each  $s \in S$  to be unipotent with a single Jordan block, the number of fixed points depends on  $X_0$ , on  $|S|$ , and also on the action of Frobenius on  $S$  (viewed as a conjugacy class in the symmetric group  $S_{|S|}$ ).

## 5 Examples

### 5.1 5.1–5.9

Take  $X_0 = \mathbf{P}^1$ ,  $|S| = 4$ ,  $r = 2$ , and let  $E$  be the set of isomorphism classes of irreducible lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X - S$ , with local monodromy at each  $s \in S$  nontrivial unipotent. In this case, the number of fixed points of the iterates of the permutation  $V = \text{Frob}^*$  of  $E$  is given by

$$N_n = q^n. \quad (5.1.1)$$

If  $|S_0| \geq 2$ , (5.1.1) is an application of the main result of [DF]. The following curious result ([DF] §7) allows one to deduce the case where  $S_0$  is reduced to one closed point of degree 4.

**Proposition 5.2.** Each permutation  $\sigma$  of type  $(2, 2)$  of  $S$  extends to a projectivity of  $\mathbf{P}^1$ . The action of this projectivity on  $E$  is trivial.

The proof is by reduction to an analogous theorem over  $\mathbf{C}$ .

It is natural to complete the set  $E$  of 5.1 to  $\bar{E}$ , the set of isomorphism classes, up to semisimplification, of lisse  $\overline{\mathbf{Q}}_\ell$ -sheaves of rank 2 on  $X - S$ , with unipotent local monodromy at each  $s \in S$ . Since  $X_0 = \mathbf{P}^1$  and  $|S| = 4$ , this completion  $\bar{E}$  differs from  $E$  only by the adjunction of one point corresponding to the constant  $\overline{\mathbf{Q}}_\ell$ -sheaf  $\overline{\mathbf{Q}}_\ell^2$ . We again write  $V$  for the permutation of  $\bar{E}$  induced by  $\text{Frob}^*$ . The number  $\bar{N}_n$  of fixed points of  $V^n$  acting on  $\bar{E}$  is

$$\bar{N}_n = q^n + 1 \quad (\text{fixed points on } \bar{E}). \quad (5.3.1)$$

The complex analogue of  $\bar{E}$  is the moduli space  $\bar{M}$  of complex local systems of rank 2 on  $\mathbf{P}^1 - S$ , for  $|S| = 4$ , with unipotent local monodromy. These local systems are taken up to semisimplification. If one chooses a base point  $o \in \mathbf{P}^1 - S$ , the functor  $F \mapsto F_o$  is an equivalence from local systems to representations of  $\pi_1(\mathbf{P}^1 - S, o)$ , and one can take as coordinates on  $\bar{M}$  the traces of elements of  $\pi_1$ . The space  $\bar{M}$  is affine, purely of dimension 2, with a unique singular point corresponding to the constant local system  $\mathbf{C}^2$ . The singularity is of type  $D_4$ . The nonzero Betti numbers of  $\bar{M}$  are

$$b_0 = 1, \quad b_2 = 1. \quad (5.4.1)$$

I have not made a complete verification, but the situation seems exactly the same for  $X_0 = \mathbf{P}^1$ ,  $|S| = 3$ ,  $r = 3$ , with unipotent local monodromy. Again, if one of the local monodromies is not a single Jordan block, the  $\overline{\mathbf{Q}}_\ell$ -sheaf is an iterated extension of constant local systems  $\overline{\mathbf{Q}}_\ell$ . The analogue of 5.3 holds for  $\sigma$  a cyclic permutation of the three points of  $S$ . The number of fixed points of  $V^n$  is again  $q^n + 1$ , with the “1” given by reducible local systems with unipotent local monodromy, all having a constant local system as semisimplification. The unique singular point of the complex analogue is of type  $E_6$ .

Take  $X_0 = \mathbf{P}^1$ ,  $|S| = 4$ ,  $r = 2$ , and assume that, as in 3.4, the imposed local monodromy is, at each  $s_0 \in S_0$ , given by two distinct characters  $\alpha'[s_0]$ ,  $\alpha''[s_0]$  of  $k(s_0)^*$ . Assume (3.2.2) and the general-position hypothesis (2.10.3). Using (3.5.1), one obtains:

**Proposition 5.7.** Under the hypotheses of 5.6, one has

$$N_1 = q + 1 + |S_0(\mathbf{F}_q)|. \quad (5.7.1)$$

Arinkin has checked that for  $X_0 = \mathbf{P}^1$  and  $r = 2$ , if a vector bundle  $E_0$  of rank 2 on  $X_0$  with parabolic structure  $\alpha_0$  at  $S_0$  is indecomposable, then  $\text{End}(E_0, \alpha)$  is reduced to the scalars. If one defines  $Z$  as in 3.6, and passes to the associated sheaf for the Zariski topology, one obtains a functor representable by an algebraic space. In the case  $|S| = 4$ , if one takes the degree  $d = 1$ , it is represented by the sum of two copies of  $X_0$ , glued along  $X_0 - S_0$ . The number of points of this scheme over  $\mathbf{F}_q$  is given by (5.7.1).



Pass to the complex analogue. On  $\mathbf{P}^1 - S$ , with  $|S| = 4$ , consider the moduli space of complex local systems of rank 2 with prescribed local monodromy at each  $s \in S$ . At  $s \in S$ , require the local monodromy to be conjugate to  $A_s = \begin{pmatrix} a_s & 0 \\ 0 & b_s \end{pmatrix}$ . Assume  $\prod_s a_s b_s = 1$ , and that if, for every  $s$ ,  $c_s$  is one of  $a_s$  or  $b_s$ , one never has  $\prod c_s = 1$ . Put  $A := (A_s)_{s \in S}$ , and write  $M(A)$  for the moduli space, 2.10. It is smooth and purely of dimension 2. One may view  $M(A)$ , for  $A_s$  close to 1, as a deformation of  $\bar{M}$  considered in 5.4. In this deformation, the space of vanishing cycles has dimension 4, since the singularity of  $\bar{M}$  is of type  $D_4$ . At infinity the geometry does not change, and the nonzero Betti numbers are therefore

$$b_0 = 1, \quad b_2 = 5. \quad (5.8.1)$$

Take  $X_0 = \mathbf{P}^1$ ,  $|S| = 3$ ,  $r = 3$ , and assume that, as in 3.4, the imposed local monodromy is at each  $s_0 \in S_0$  given by three distinct characters of  $k(s_0)^*$ . Assume (3.2.2) and the general-position hypothesis 2.10.3. Using (3.5.1), one obtains here

$$N_1 = q + 1 + 2|S_0(\mathbf{F}_q)|. \quad (5.9.1)$$

The sheaf associated to  $Z$  as in 3.6 is indeed, for  $d = 1$ , represented by the sum of three copies of  $X_0$ , glued along  $X_0 - S_0$ . If  $S_0$  consists of three rational points over  $\mathbf{F}_q$ , (5.9.1) says that  $N_1 = (q + 1) + 6$ , as suggested by the  $E_6$  singularity of the complex analogue for unipotent local monodromy (cf. 5.5).

## 6 Rank 1

### 6.1 6.1–6.7

Let  $E(1, \emptyset)$  denote the set  $E(R)$  of 2.1 for  $r = 1$  and  $S_0 = \emptyset$ . Tensor product gives it the structure of an abelian group. Class field theory identifies  $E(1, \emptyset)^V$  with the group of  $\bar{\mathbf{Q}}_\ell^*$ -valued characters of the finite group  $\text{Pic}^0(X_0)(\mathbf{F}_q)$ . Indeed, this theory identifies the abelianized fundamental group of  $X_0$  with the profinite completion of  $\text{Pic}(X_0)$ . The group  $\text{Pic}(X_0)$  maps to  $\mathbf{Z}$  by the degree map, with kernel the finite group  $\text{Pic}^0(X_0)(\mathbf{F}_q)$  of  $\mathbf{F}_q$ -points of the Jacobian  $\text{Pic}^0(X_0)$ . The isomorphism classes of lisse  $\bar{\mathbf{Q}}_\ell^*$ -sheaves  $F_0$  of rank one on  $X_0$  are thus identified with characters  $\chi$  of  $\text{Pic}(X_0)$  with values in  $\bar{\mathbf{Z}}_\ell^*$ ;  $\mathbf{F}_q$ -twist by  $\lambda$  corresponds to multiplication by the character  $\lambda^n$  of  $\mathbf{Z}$ ; and, by attaching to  $F_0$  the restriction of  $\chi$  to  $\text{Pic}^0(X_0)(\mathbf{F}_q)$ , one obtains a bijection between  $\mathbf{F}_q$ -twist classes of lisse  $F_0$  of rank 1 on  $X_0$  and characters of  $\text{Pic}^0(X_0)(\mathbf{F}_q)$ .

For arbitrary  $r \geq 1$  and  $R$  as in 2.1, tensor product induces an action of the abelian group  $E(1, \emptyset)$  on  $E(R)$ , and of  $E(1, \emptyset)^V$  on  $E(R)^V$ . If  $r = 1$ , and  $R$  satisfies the compatibility (2.10.1), class field theory shows that  $E(R)^V$  is a principal homogeneous space under  $E(1, \emptyset)^V$ . Thus

$$|E(R)^V| = |E(1, \emptyset)^V| = |\text{Pic}^0(X_0)(\mathbf{F}_q)| \quad (r = 1 \text{ and } (2.10.1)). \quad (6.2.1)$$

If  $r > 1$ , taking the exterior power  $\wedge^r$  gives a map

$$\det : E(R) \rightarrow E(\bigwedge^r R)$$

such that, for the action of  $E(1, \emptyset)$ , denoted  $\otimes$ , one has

$$\det(l \otimes F) = l^r \otimes \det(F).$$

Similarly, after passage to the fixed points of  $V = \text{Frob}^*$ , for  $\det : E(R)^V \rightarrow E(\wedge^r R)^V$  and the action of  $E(1, \emptyset)^V$ .

The action of  $E(1, \emptyset)^V$  on  $E(R)^V$  is not necessarily free, and the fibres of

$$\det : E(R)^V \rightarrow E(\bigwedge^r R)^V$$

do not necessarily all have the same number of elements. Nevertheless, in all cases that have been computed,  $|E(1, \emptyset)^V|$  divides  $|E(R)^V|$ . As a function of  $n$ ,  $|\text{Pic}^0(X_0)(\mathbf{F}_{q^n})|$  has the form (2.15.1):

$$|\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| = \sum (-1)^i \text{Tr}(\text{Frob}^*, H^i(\text{Pic}^0(X), \mathbf{Q}_\ell)) = \sum (-1)^i \text{Tr}(\text{Frob}^*, \bigwedge^i H^1(X, \mathbf{Q}_\ell)).$$

I conjecture the following precise form of Conjecture 2.15 (i).

**Conjecture 6.3.** As a function of  $n$ , the number of fixed points  $N_n(R)$  of  $V^n$  is of the form

$$N_n(R) = |\text{Pic}^0(X_0)(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n \quad (6.3.1)$$

for integers  $c_k$  and suitable numbers  $\gamma_k$ .

Here is an analogue of 6.3 in algebraic geometry. Let  $A_0$  be an abelian variety over  $\mathbf{F}_q$ , and let  $f_0 : X_0 \rightarrow A_0$  be a proper smooth morphism. Although the fibres of  $f : X_0(\mathbf{F}_{q^n}) \rightarrow A_0(\mathbf{F}_{q^n})$  need not have the same number of elements (example:  $X_0 = A_0$  and  $f_0$  multiplication by an integer prime to  $p$ ),  $|A_0(\mathbf{F}_{q^n})|$  divides  $|X_0(\mathbf{F}_{q^n})|$ . Indeed, the  $R^i f_{0!} \mathbf{Q}_\ell$  are lisse  $\mathbf{Q}_\ell$ -sheaves on  $A$ , and the (abelian) fundamental group of  $A$  acts on these sheaves. One knows that this action is semisimple. Let  $(R^i f_{0!} \mathbf{Q}_\ell)_0$  be the subsheaf of invariants. It comes from a subsheaf  $(R^i f_{0!} \mathbf{Q}_\ell)_0$  of  $R^i f_{0!} \mathbf{Q}_\ell$  on  $A_0$ , and this last sheaf is the inverse image on  $A_0$  of a sheaf on  $\text{Spec}(\mathbf{F}_q)$ , denoted  $H^i(X_0/A_0)$ . One knows that

$$H^i(A, (R^j f_{0!} \mathbf{Q}_\ell)_0) \xrightarrow{\sim} H^i(A, R^j f_{0!} \mathbf{Q}_\ell). \quad (6.4.1)$$

Grothendieck's trace formula for the number of  $\mathbf{F}_q$ -points of  $X_0$  can therefore be rewritten as

$$|X_0(\mathbf{F}_{q^n})| = \text{Tr} \left( \text{Frob}^*, \left( \sum (-1)^i H^i(A, \mathbf{Q}_\ell) \right) \otimes \left( \sum (-1)^j H^j(X/A) \right) \right). \quad (6.4.2)$$

If the  $\gamma_k$  are the eigenvalues of Frobenius on the  $H^j(X/A)$ , and the  $c_k$  are the alternating sums of their multiplicities, one deduces from (6.4.2) that

$$|X_0(\mathbf{F}_{q^n})| = |A_0(\mathbf{F}_{q^n})| \sum c_k \gamma_k^n, \quad (6.4.3)$$

a formula analogous to (6.3.1). The promised divisibility of  $|X_0(\mathbf{F}_{q^n})|$  by  $|A_0(\mathbf{F}_{q^n})|$  follows from the fact that the Frobenius eigenvalues on  $H^j(X/A)$ , contained in the cohomology of a fibre of  $X_0 \rightarrow A_0$ , are algebraic integers.

In 6.4 the crucial point is not that  $f_0$  is proper and smooth, but that the  $R^i f_{0!} \mathbf{Q}_\ell$  are lisse. One can avoid having to assume that the  $R^i f_{0!} \mathbf{Q}_\ell$  are semisimple by considering, rather than the subsheaf of invariants for the action of  $\pi_1(A)$ , the largest subsheaf on which the action is unipotent. Assume that there is an action of  $A_0$  on  $X_0$ , and an integer  $r$ , such that  $f_0$  is equivariant for the action of  $A_0$  on  $A_0$  by  $x : y \mapsto rx + y$ . Under this hypothesis, the inverse images of the  $R^i f_{0!} \mathbf{Q}_\ell$  on  $A$  by multiplication by  $r : A \rightarrow A$  are constant sheaves. This implies both their lissity and their semisimplicity.

If  $g \geq 1$ , Conjecture 6.3 implies that in the conjectural expression (2.11.1) for  $N_n(R)$ , one has  $\sum a_i = 0$ . Pass to the complex analogues as in 2.14. Let  $M(1, \emptyset)$  be the group of isomorphism classes of complex local systems of rank 1 on  $P$ . It is isomorphic to  $(\mathbf{C}^*)^{2g}$ , and, if the compatibility  $\prod \det R^*(s) = 1$  is satisfied,  $M(\det R^*)$  is a principal homogeneous space under  $M(1, \emptyset)$ . The group  $M(1, \emptyset)$  acts on  $M(R^*)$  and on  $M(\det R^*)$ . Denote the action by  $\otimes$ . The map induced by  $\bigwedge^r$ ,

$$\det : M(R^*) \rightarrow M(\det R^*),$$

satisfies  $\det(l \otimes F) = l^r \otimes \det(F)$ . This implies that  $\det$  is a fibration, that the  $R^i \det_! \mathbf{Z}$  are locally constant, and that the action of  $\pi_1 M(\det R^*)$  on the  $R^i \det_! \mathbf{Z}$  factors through its quotient  $\pi_1 / r\pi_1 \simeq (\mathbf{Z}/r\mathbf{Z})^{2g}$ . If  $g \geq 1$ , one deduces by arguments analogous to those of 6.4 that  $\chi(M(R^*)) = 0$ , in agreement with 2.15 (ii). Let  $(R^i \det_! \mathbf{Q})_0$  be the local subsystem of  $R^i \det_! \mathbf{Q}$  of invariants under the action of  $\pi_1$ , and let  $\chi(M(R^*)/M(\det R^*))$  be the alternating sum of the ranks of the  $(R^i \det_! \mathbf{Q})_0$ .

**Conjecture 6.7.** With the notation of 6.3 and 6.6, one has

$$\sum c_k = \chi(M(R^*)/M(\det R^*)).$$

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